ELEVATOR NEUTRAL SHIFT MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Elevator Neutral Shift Mechanism Removal

(2) Elevator Neutral Shift Mechanism Installation

TASK 27-31-91-000-801

2. Elevator Neutral Shift Mechanism Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

E. Prepare for the Removal

SUBTASK 27-31-91-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-91-010-001

(2) Remove this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-91-010-002

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-91-480-001

(4) Use the trammel bar, SPL-1677 to set B dimension at 39.17 ±0.01 in. (994.92 ±0.26 mm). (Figure 401 (Sheet 2))

NOTE: Make sure to protect wire bundles when using the trammel trammel bar, SPL-1677. The sharp edges on the trammel trammel bar, SPL-1677 can cause chafing or damage to the wires.

SUBTASK 27-31-91-860-002

(5) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	4	C00456	AFCS SYS A MACH TRIM AC
D	1	C01049	AFCS SYS A WARN LIGHT (BAT)
D	2	C01045	AFCS SYS A FCC DC
D	3	C01048	AFCS SYS A ENGAGE INTLK
D	4	C00457	AFCS SYS A MACH TRIM DC
D	5	C01044	AFCS MCP DC 1

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	1	C00374	AFCS SYS B WARN LIGHT (BAT)
B	2	C01064	AFCS SYS B MACH TRIM DC
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
C	3	C01047	AFCS MCP DC 2
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-31-91-480-002

(6) Install the lock assembly, SPL-1672 on the stabilizer trim wheel at the control stand (Figure 402):

(a) Turn the stabilizer trim wheel to put the handle at the top of the wheel.

(b) Adjust the height of the lock assembly, SPL-1672 until the yoke is under the handle.

(c) Install the pin through the yoke.

(d) Install the safety pin.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-91-480-003

(7) Install the rig pin E-5 from the rigging pin kit, SPL-1585 in the aft control quadrant, (Figure 401).

F. Neutral Shift Mechanism Removal

(Figure 401)

SUBTASK 27-31-91-020-001

(1) Do these steps to remove the elevator neutral shift mechanism [1]:

(a) Remove the nut [8], washer [9], nut [10], washers [7], and bolt [6] to disconnect the elevator neutral shift mechanism [1] from the mach trim actuator [5].

(b) Remove the nuts [28], washers [26], washers [24], and bolts [25] to disconnect control rod [23] and control rod [27] from the elevator neutral shift mechanism [1].

(c) Remove the nuts [4], washers [3], and bolts [2] that attach the elevator neutral shift mechanism [1] to the support brackets [29].

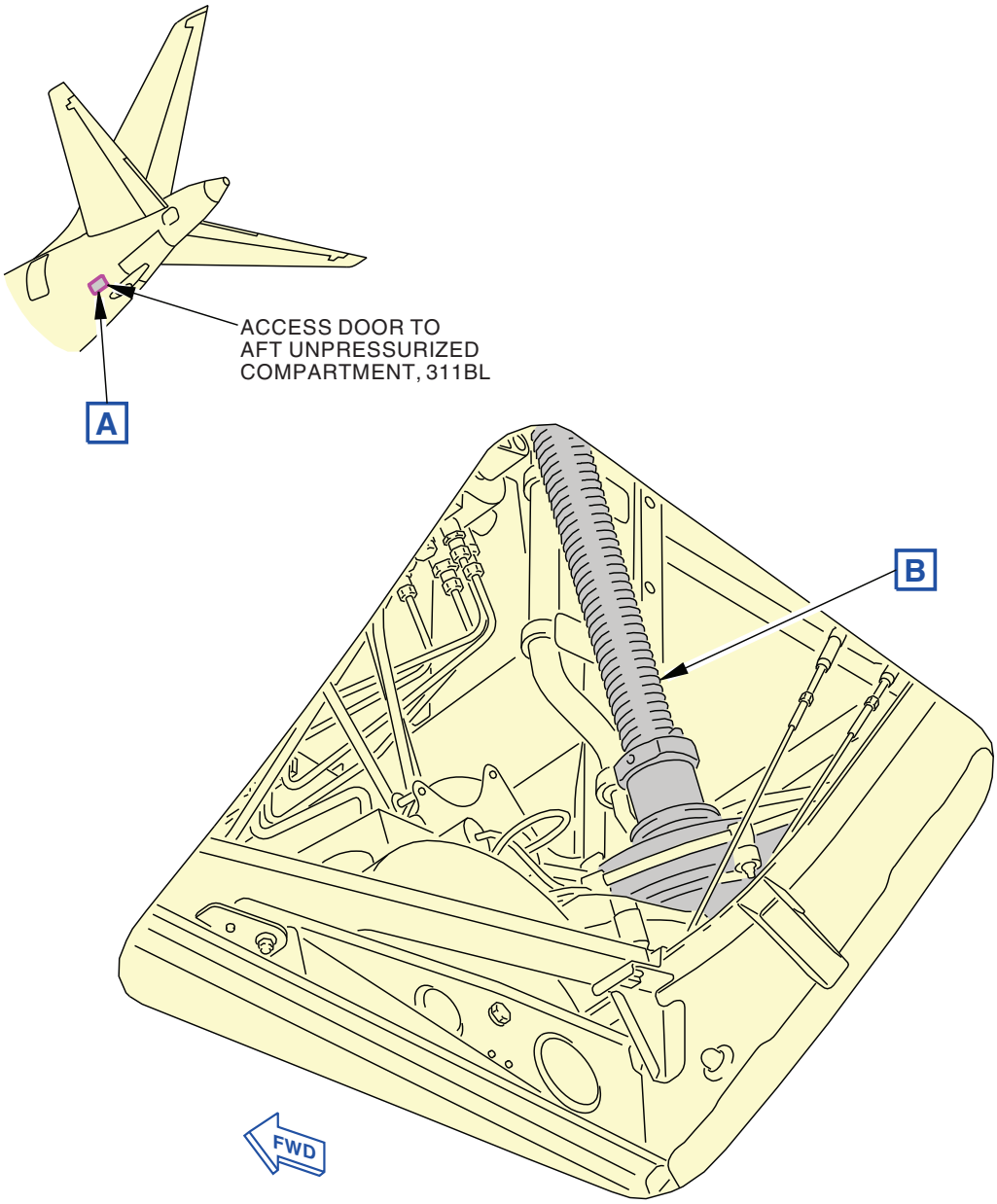
(d) Remove the nuts [13], washers [12], and bolts [11] that attach the elevator neutral shift mechanism [1] to the support brackets [29].

(e) Remove the elevator neutral shift mechanism [1] and spacers [17] from the airplane.

NOTE: Note the quantity, orientation, and location of the spacers [17] before you remove them from the elevator neutral shift mechanism [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



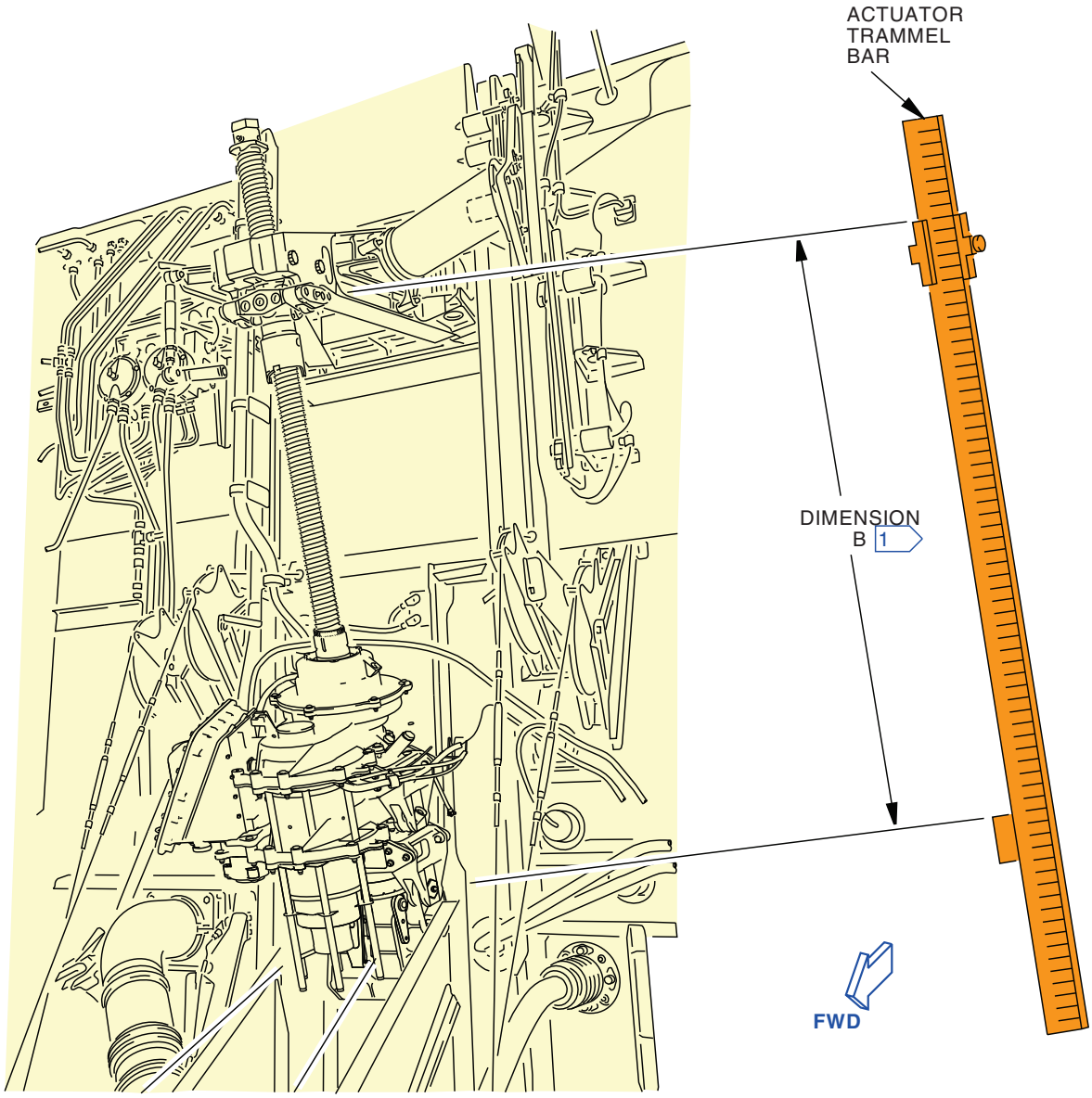
**VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL**

A

2412607 S00061530990_V1

**Elevator Neutral Shift Mechanism Installation
Figure 401/27-31-91-990-801 (Sheet 1 of 6)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

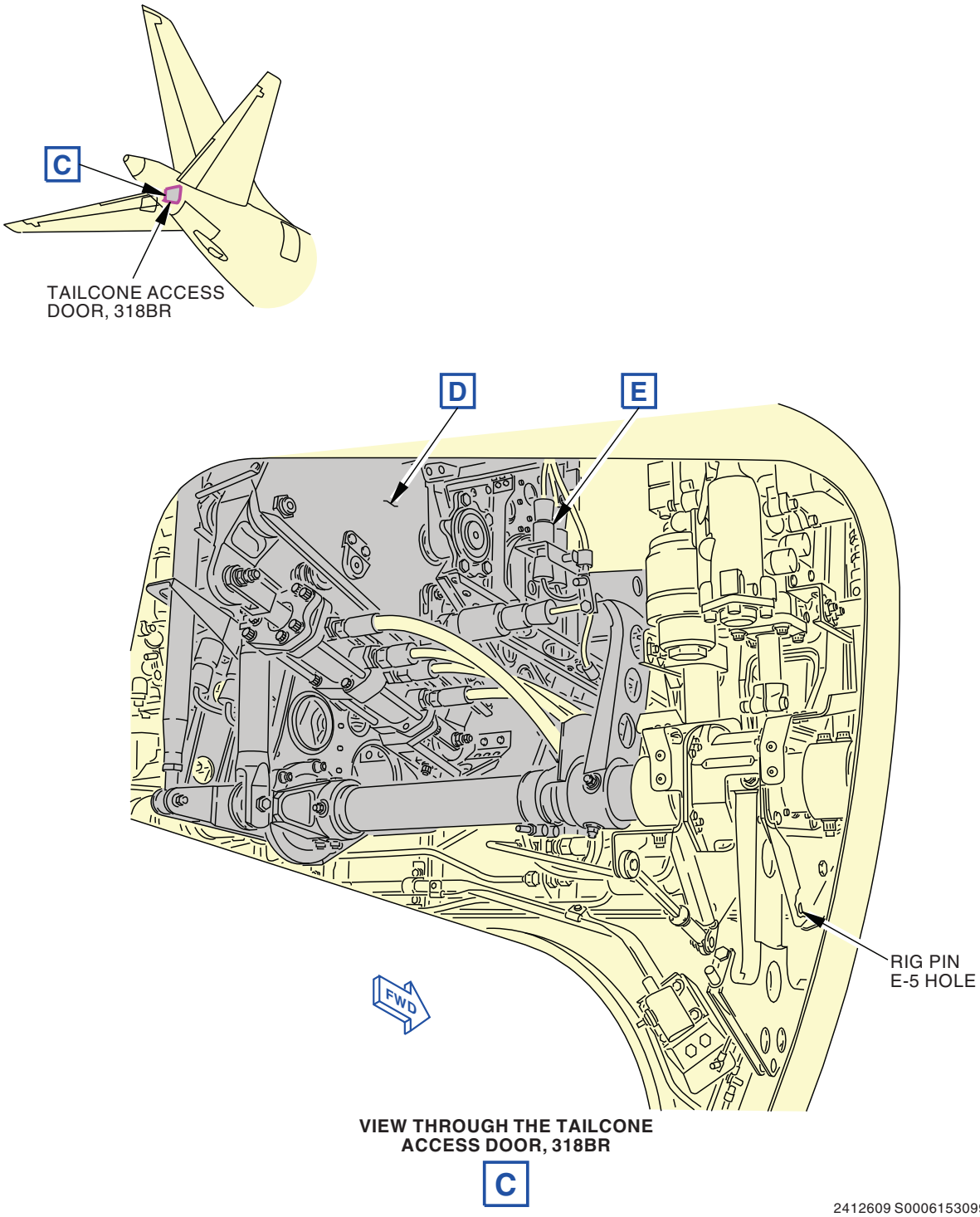
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES (NEUTRAL POSITION).

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

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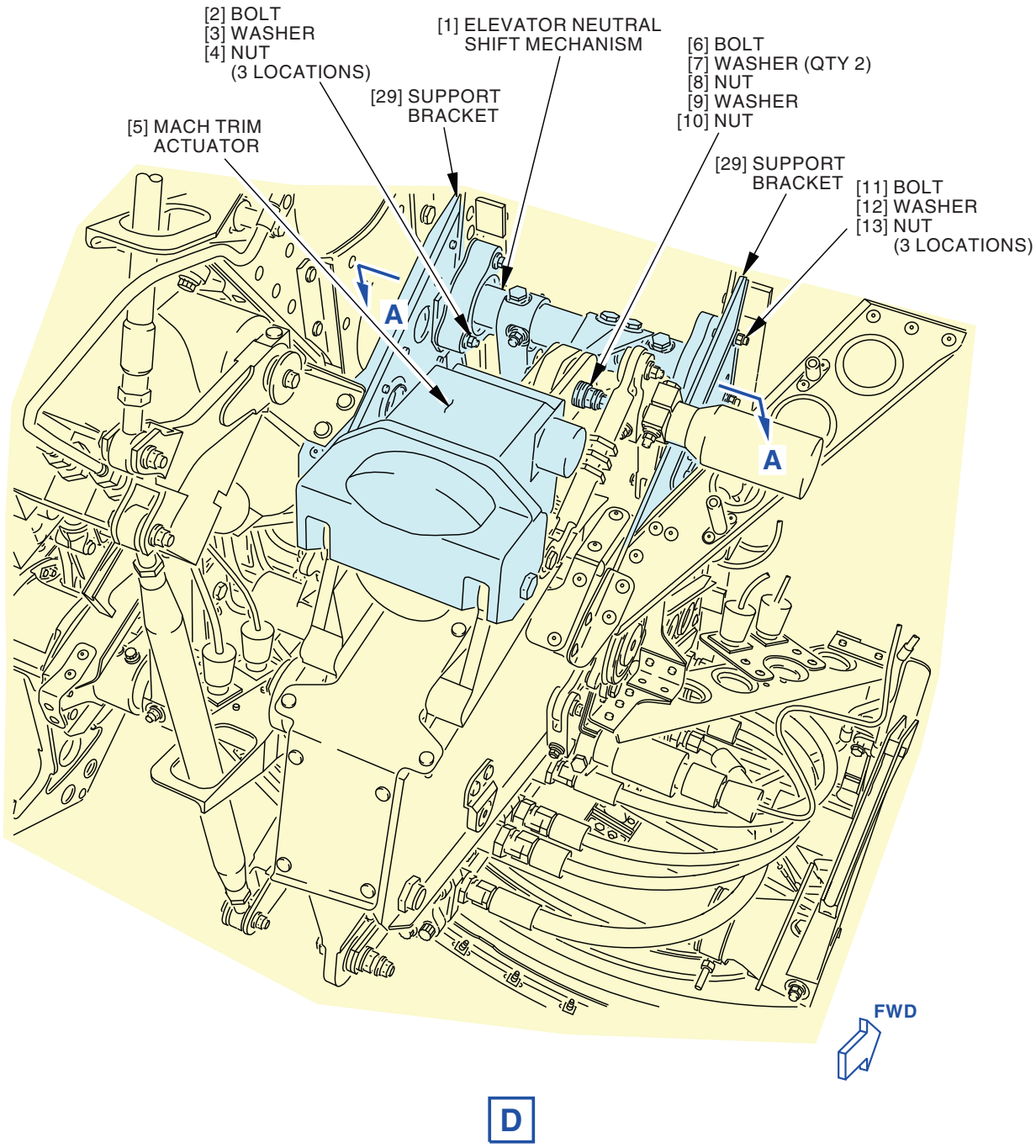
Elevator Neutral Shift Mechanism Installation
Figure 401/27-31-91-990-801 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



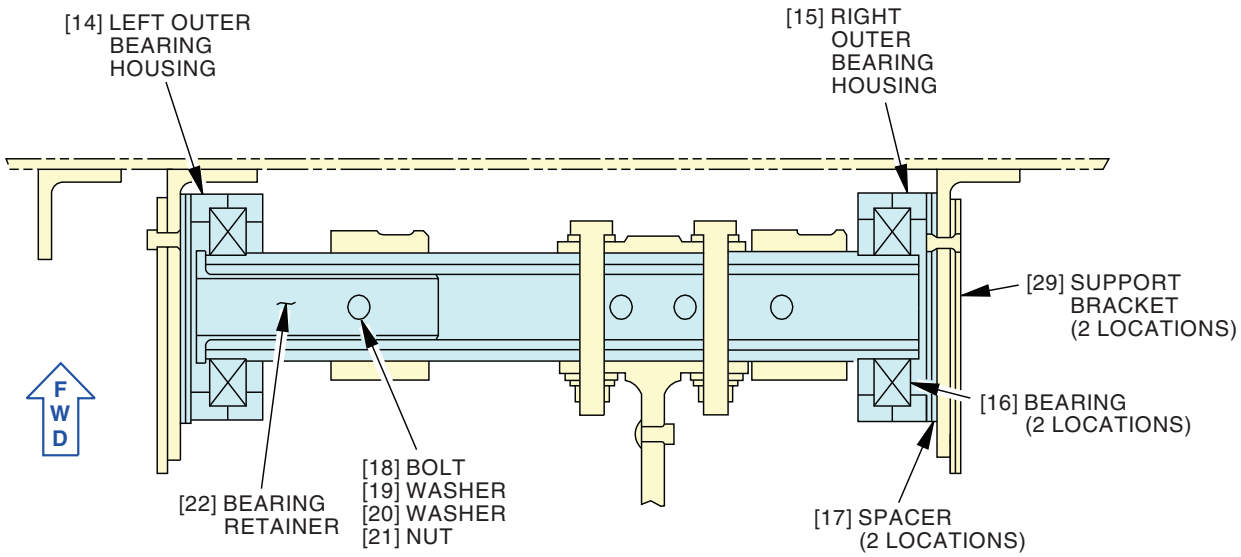
Elevator Neutral Shift Mechanism Installation
Figure 401/27-31-91-990-801 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Neutral Shift Mechanism Installation
Figure 401/27-31-91-990-801 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

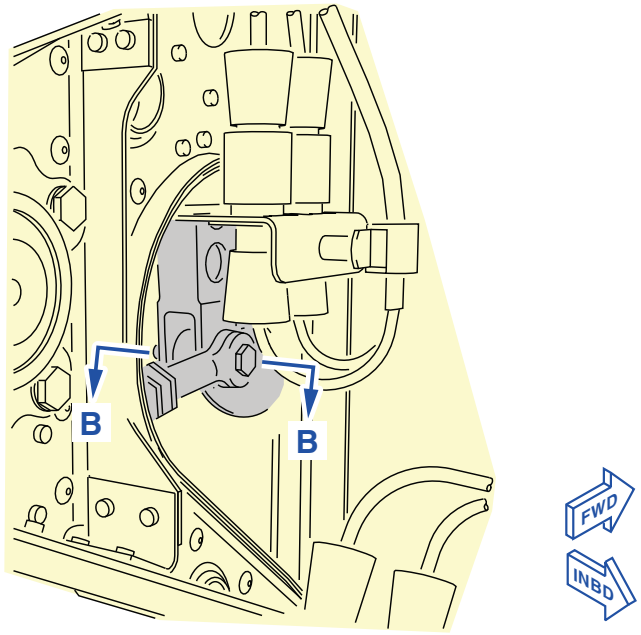


A-A

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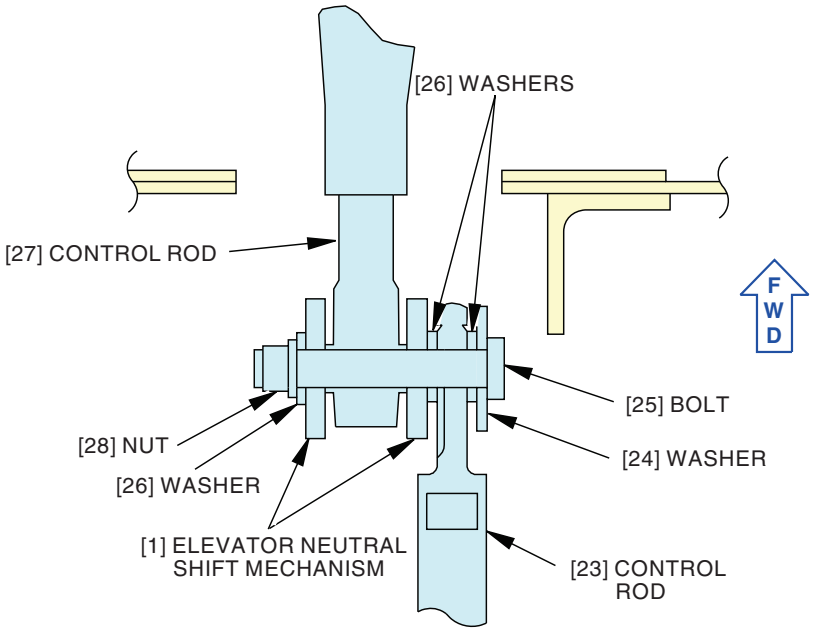
Elevator Neutral Shift Mechanism Installation
Figure 401/27-31-91-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR NEUTRAL SHIFT MECHANISM
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

E



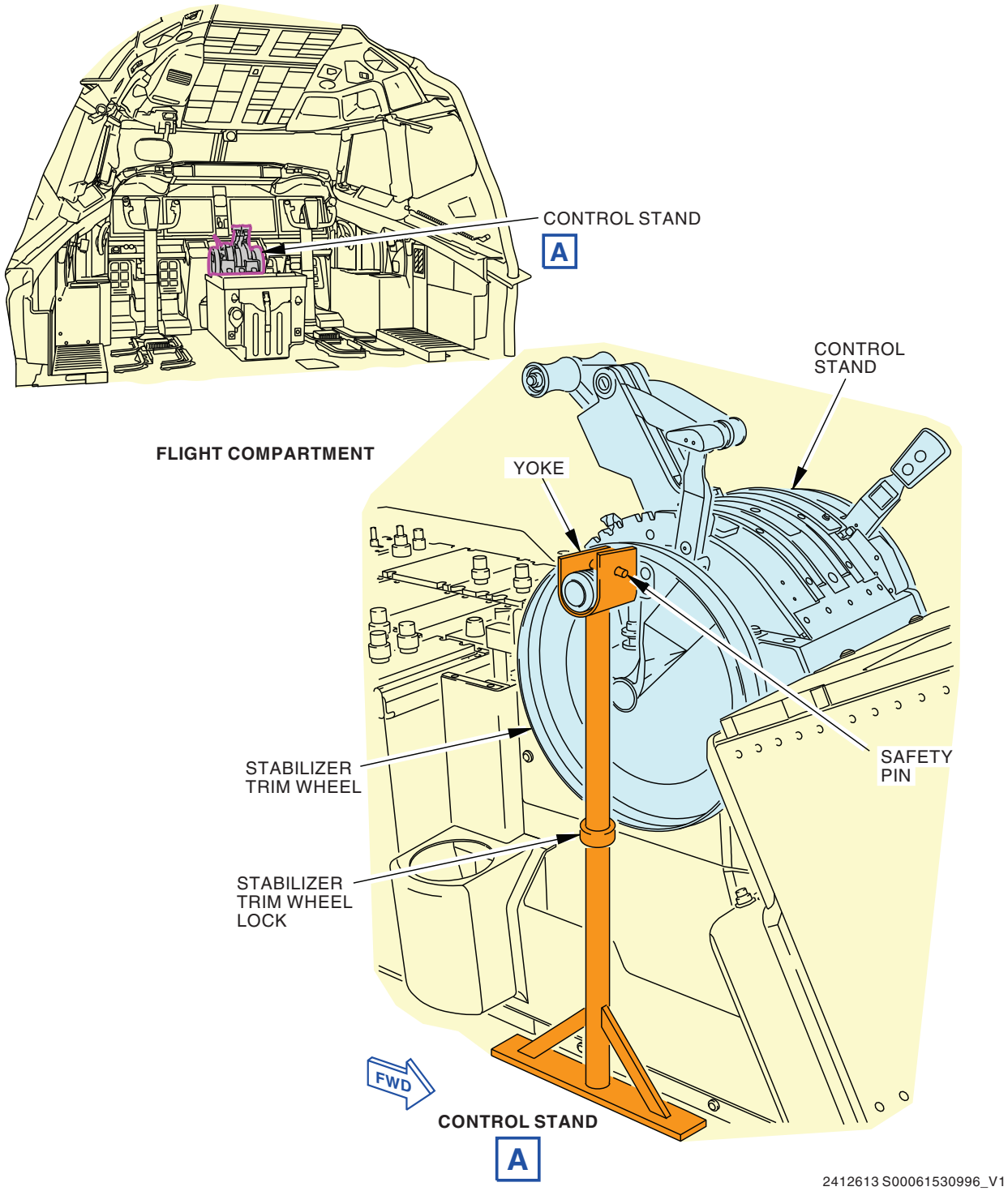
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

B-B

2412612 S00061530995_V1

Elevator Neutral Shift Mechanism Installation
Figure 401/27-31-91-990-801 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Wheel Lock Installation
Figure 402/27-31-91-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-91-400-801

3. Elevator Neutral Shift Mechanism Installation

(Figure 401)

A. References

Reference	Title
27-31-00-700-811	Neutral Shift - Test (P/B 501)
27-31-00-820-804	Input Rods of the Elevator Neutral Shift Mechanism and Output Rod of the Feel and Centering Unit - Adjustment (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Elevator neutral shift mechanism	27-31-91-01-065	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

G. Neutral Shift Mechanism Installation

(Figure 401)

SUBTASK 27-31-91-640-001

(1) Do these steps to replace the bearings [16] of the elevator neutral shift mechanism [1]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the right outer bearing housing [15] from the short end of the elevator neutral shift mechanism [1].

(b) Remove the bearing [16] from the short end of the elevator neutral shift mechanism [1].

(c) Remove the left outer bearing housing [14] from the long end of the elevator neutral shift mechanism [1].

(d) Remove the nut [21], washer [20], washer [19], and bolt [18] from the long end of the elevator neutral shift mechanism [1].

(e) Remove the bearing retainer [22] and bearing [16] from the elevator neutral shift mechanism [1].

(f) Apply sealant, A00247 between each new bearing [16], their bearing housings and bearing retainer [22].

(g) Install the bearings [16], left outer bearing housing [14], right outer bearing housing [15], and bearing retainer [22] back on the elevator neutral shift mechanism [1].

(h) Turn the bearing retainer [22] until you can install the bolt [18], washer [19] (under the bolt head), washer [20] (under the nut), and the nut [21].

SUBTASK 27-31-91-420-001

(2) Do these steps to install the elevator neutral shift mechanism [1]:

(a) Put the elevator neutral shift mechanism [1] between the support brackets [29].

1) Align the elevator neutral shift mechanism [1] with the mach trim actuator [5].

2) Install the spacers [17] between the elevator neutral shift mechanism [1] and support brackets [29].

a) Add or remove spacers [17], if necessary.

(b) Install the bolts [2], washers [3] (under the nuts), and nuts [4] to attach the left side of the elevator neutral shift mechanism [1] to the left support bracket [29].

(c) Install the bolts [11], washers [12] (under the nuts), and nuts [13] to attach the right side of the elevator neutral shift mechanism [1] to the right support bracket [29].

WARNING

MAKE SURE THAT YOU TURN THE NEUTRAL SHIFT CRANK IN THE UP DIRECTION. IF YOU INSTALL THE NEUTRAL SHIFT CRANK INCORRECTLY, THE STABILIZER CAN MOVE QUICKLY WHEN YOU ENGAGE THE AUTOPILOT. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(d) Make sure the crank of the elevator neutral shift mechanism [1] is in the up direction and connected to the mach trim actuator [5].

1) Install the bolt [6] from the left to the right side (bolt head on the left side).

2) Install the washers [7] (under the nut) and nut [8].

a) Tighten the nut [8] to 85 in-lb (10 N·m)–100 in-lb (11 N·m).

CAUTION

DO NOT TIGHTEN THE NUT [10] AGAIN AFTER YOU TIGHTEN THE NUT [8]. YOU CAN CAUSE DAMAGE TO THE BOLT ASSEMBLY.

3) Install the washer [9] and nut [10].

a) Tighten the nut [10] to 30 in-lb (3 N·m)–35 in-lb (4 N·m).

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(e) Connect the control rods [27] and control rods [23]to the elevator neutral shift mechanism [1].

1) Install the bolts [25], washers [24], washers [26], and nuts [28].

SUBTASK 27-31-91-080-001

(3) Remove the rig pin E-5 of the rigging pin kit, SPL-1585 from the aft control quadrant.

SUBTASK 27-31-91-860-003

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	4	C00456	AFCS SYS A MACH TRIM AC
D	1	C01049	AFCS SYS A WARN LIGHT (BAT)
D	2	C01045	AFCS SYS A FCC DC
D	3	C01048	AFCS SYS A ENGAGE INTLK
D	4	C00457	AFCS SYS A MACH TRIM DC
D	5	C01044	AFCS MCP DC 1

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	1	C00374	AFCS SYS B WARN LIGHT (BAT)
B	2	C01064	AFCS SYS B MACH TRIM DC
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
C	3	C01047	AFCS MCP DC 2
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-31-91-080-002

(5) Remove the lock assembly, SPL-1672 from the stabilizer trim wheel (Figure 402).

SUBTASK 27-31-91-860-004

(6) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-91-820-001

(7) Do this task to adjust the elevator neutral shift mechanism [1]: Input Rods of the Elevator Neutral Shift Mechanism and Output Rod of the Feel and Centering Unit - Adjustment, TASK 27-31-00-820-804.

SUBTASK 27-31-91-730-001

(8) Do this task: Neutral Shift - Test, TASK 27-31-00-700-811.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-91-410-001

(1) Install this access panel:

Number	Name/Location
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-91-410-002

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR NEUTRAL SHIFT MECHANISM - INSPECTION/CHECK

1. General

A. This procedure contains this task:

(1) Elevator Neutral Shift Mechanism Inspection

(a) This task supplies the data for a dimensional inspection of the elevator neutral shift mechanism [1].

(b) This task does not remove or install the elevator neutral shift mechanism [1], refer to the Elevator Neutral Shift Mechanism - Removal/Installation for procedures to remove and install the elevator neutral shift mechanism [1].

TASK 27-31-91-200-801

2. Elevator Neutral Shift Mechanism - Inspection

(Figure 601)

A. General

(1) This procedure gives wear limits and repair data for the elevator neutral shift mechanism [1].

B. References

Reference	Title
27-31-91-000-801	Elevator Neutral Shift Mechanism Removal (P/B 401)
27-31-91-400-801	Elevator Neutral Shift Mechanism Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Procedure

SUBTASK 27-31-91-020-002

(1) Do this task to remove the elevator neutral shift mechanism [1]: Elevator Neutral Shift Mechanism Removal, TASK 27-31-91-000-801.

SUBTASK 27-31-91-020-003

(2) Use these tools to check for wear on the parts of the elevator neutral shift mechanism [1] connections:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-31-91-020-004

(3) Do these steps to examine the parts of the elevator neutral shift mechanism [1] connections for worn areas:

(a) Remove the parts to be checked.

(b) Measure the parts as necessary with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Compare the dimensions you measured with the permitted wear dimensions that follow.

(d) If the parts are not in the tolerance, replace or repair the part as required.

Table 601/27-31-91-993-801

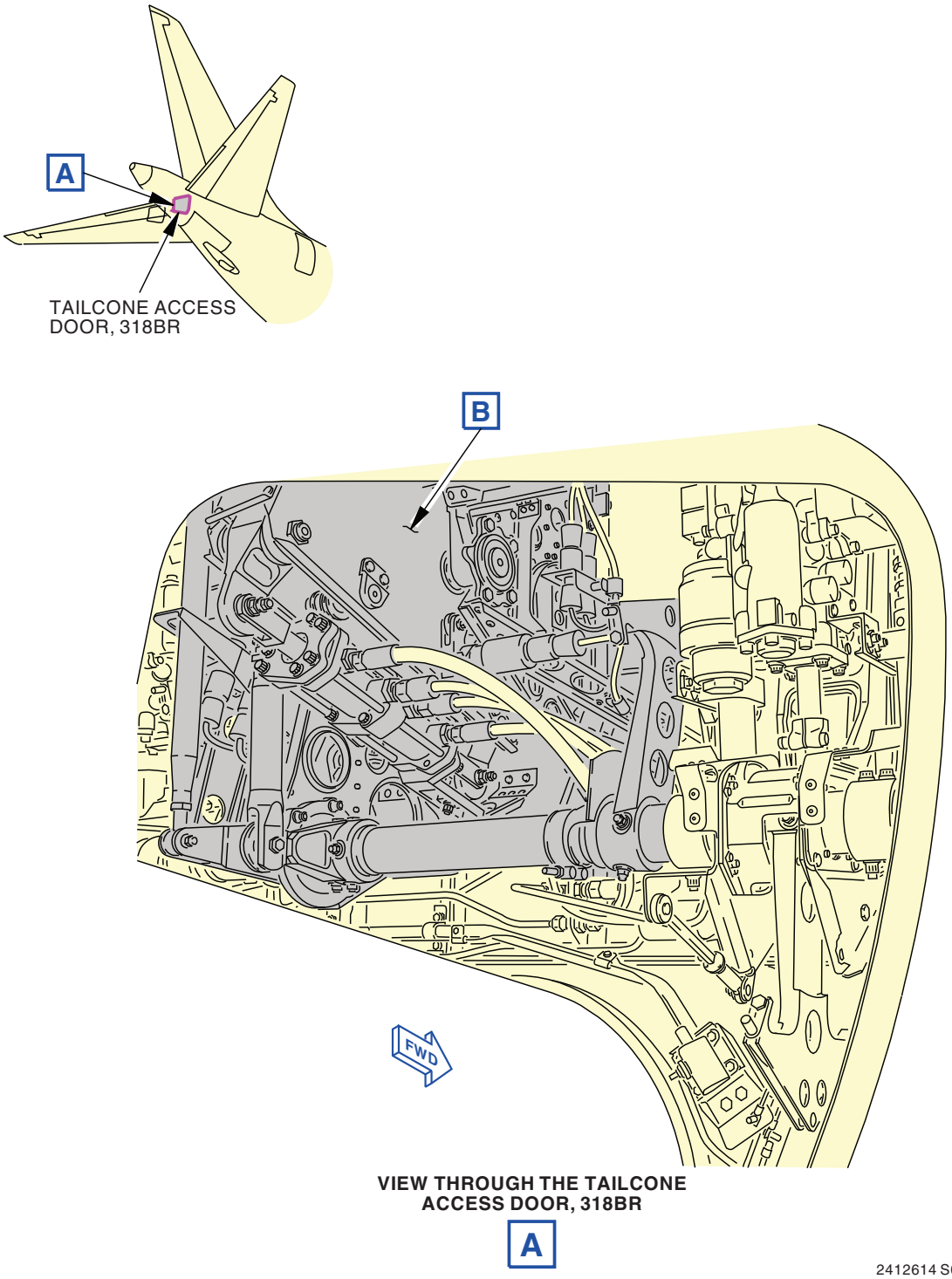
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 outer bearing housing [3]	I.D.	1.9375 in 49.212 mm	1.9385 in 49.238 mm	1.9415 in 49.314 mm	0.004 in 0.102 mm
#1 bearing [4]	O.D.	1.9365 in 49.187 mm	1.9375 in 49.212 mm	1.9360 in 49.174 mm	
#2 bearing [5]	I.D.	0.9995 in 25.387 mm	1.0000 in 25.400 mm	1.0025 in 25.463 mm	0.0035 in 0.089 mm
#2 elevator neutral shift mechanism [1]	O.D.	0.9985 in 25.362 mm	0.9990 in 25.375 mm	0.9965 in 25.311 mm	

SUBTASK 27-31-91-020-005

(4) Do this task to install the elevator neutral shift mechanism [1]: Elevator Neutral Shift Mechanism Installation, TASK 27-31-91-400-801.

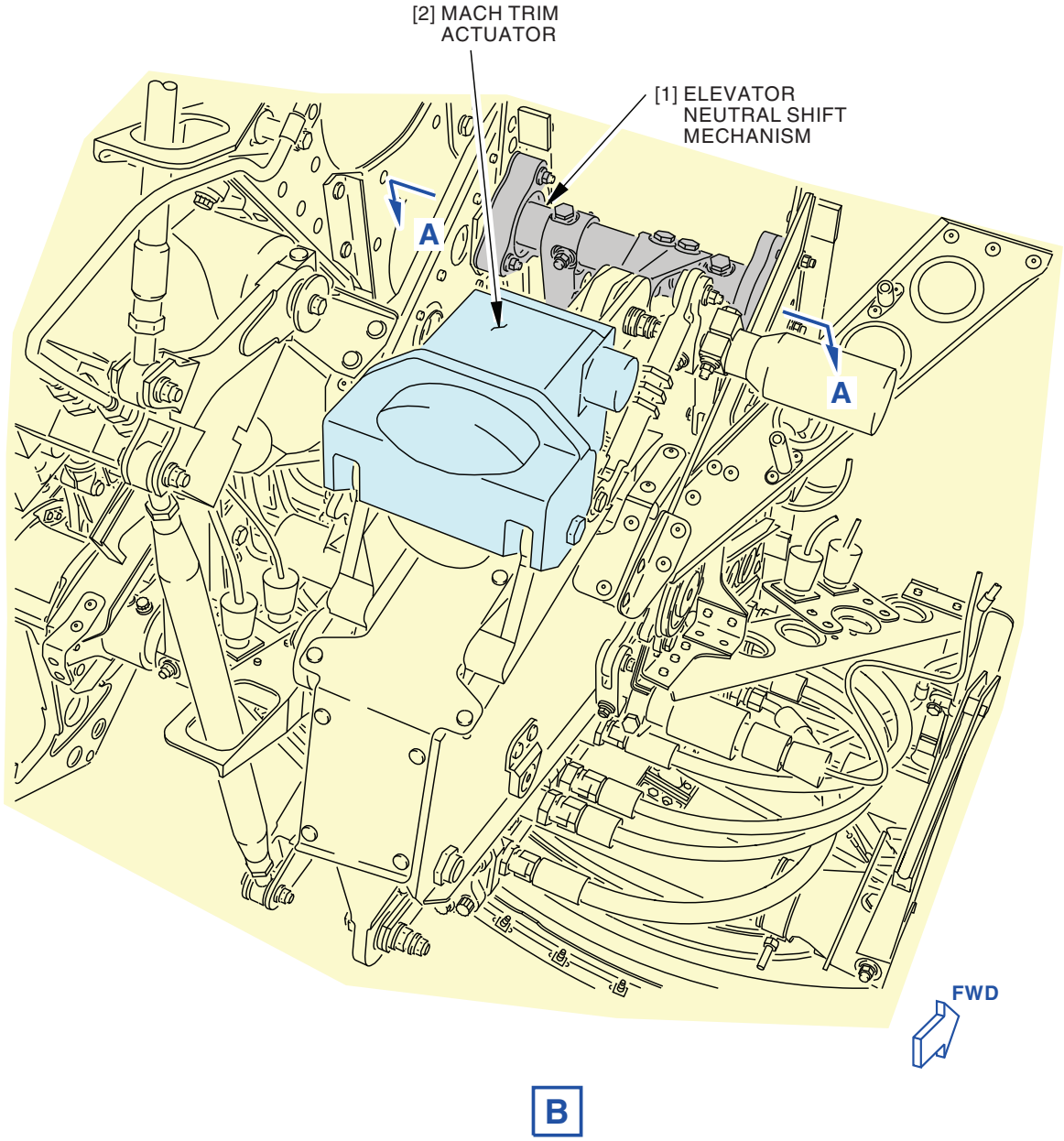
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Neutral Shift Mechanism Wear Limits
Figure 601/27-31-91-990-803 (Sheet 1 of 3)

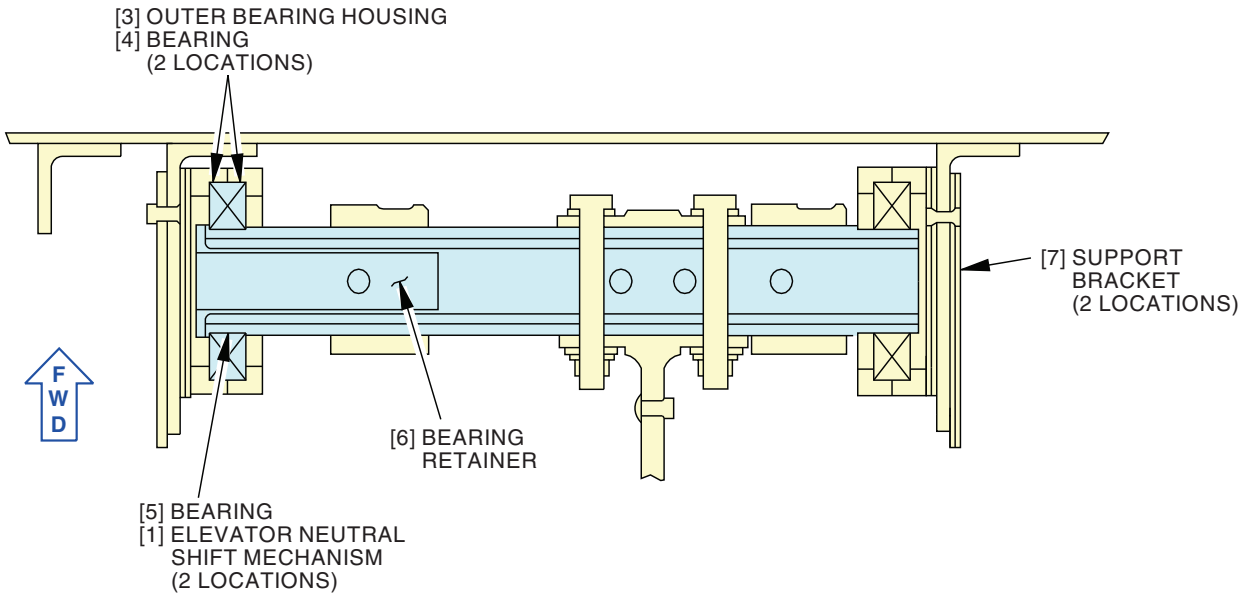
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Elevator Neutral Shift Mechanism Wear Limits
Figure 601/27-31-91-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Elevator Neutral Shift Mechanism Wear Limits
Figure 601/27-31-91-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FEEL SYSTEM PITOT PROBE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the feel system pitot probe

(2) An installation of the feel system pitot probe.

B. The feel system pitot probe is referred to as the "pitot probe [8]" in this procedure.

C. A pitot probe [8] is installed on each side of the vertical fin. This procedure is applicable for the left and right pitot probes [8].

TASK 27-31-94-000-801

2. Feel System Pitot Probe Removal

(Figure 401 and Figure 402)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975

B. Location Zones

Zone	Area
323	Vertical Fin - Front Spar To Rear Spar

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Access Panels

Number	Name/Location
323FL	Vertical Fin, Access

D. Feel System Pitot Probe Removal

SUBTASK 27-31-94-860-001

(1) For the left pitot probe [8], open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT

SUBTASK 27-31-94-860-002

(2) For the right pitot probe [8], open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-94-010-001

WARNING MAKE SURE THAT THE PITOT PROBE HEAT IS OFF. A HOT PROBE CAN CAUSE INJURIES TO PERSONNEL.

CAUTION MAKE SURE THAT THE PITOT PROBE HAS NO WEIGHT ON IT. THE PITOT PROBE CAN BEND OR TWIST OUT OF TOLERANCE.

CAUTION MAKE SURE THAT THE AREA AROUND THE PITOT PROBE IS CLEAR OF UNWANTED MATERIAL. CONTAMINATION OF THE PITOT SYSTEM CAN OCCUR.

(3) Remove this access panel:

Number	Name/Location
323FL	Vertical Fin, Access

SUBTASK 27-31-94-140-002

(4) Use the sealant removal tool, COM-2481, to remove the sealant from around the baseplate [5] of the pitot probe [8].

SUBTASK 27-31-94-020-001

(5) Remove the bolts [9] from the baseplate [5] of the pitot probe [8].

SUBTASK 27-31-94-020-002

(6) Do these steps to loosen the pitot probe [8] from the airplane skin:

(a) Hold the probe strut [7].

(b) Pull the pitot probe [8] out from the airplane skin until you can get access to the pitot hose fitting [10] and electrical connector [1] on the base of the pitot probe [8].

SUBTASK 27-31-94-020-004

(7) Remove the long screw [6] from the pitot probe side of the electrical connector [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-94-020-005

(8) Disconnect the electrical connector [1].

SUBTASK 27-31-94-420-001

(9) Install the long screw [6] in the electrical connector [1].

SUBTASK 27-31-94-020-007

(10) Temporarily attach the electrical connector [1] and pitot hose fitting [10] to the airplane structure with tape so they do not fall inside vertical fin.

SUBTASK 27-31-94-020-003

CAUTION

USE TWO WRENCHES TO APPLY COUNTER PRESSURE ON THE FITTING DURING DISASSEMBLY. USE ONE WRENCH ON EACH SIDE OF THE FITTING. IF YOU DO NOT OBEY THIS INSTRUCTION, DAMAGE TO THE TUBE OR FITTING CAN OCCUR.

(11) Disconnect the elbow [11] from the baseplate [5].

SUBTASK 27-31-94-020-008

(12) Put a cap on the elbow [11] to keep out unwanted material.

NOTE: This step is not necessary if you will replace the pitot probe immediately.

SUBTASK 27-31-94-020-009

(13) Disconnect the bonding jumper [24] from the baseplate [5].

(a) Remove the bolt [19], washer [20], and bonding jumper [24].

SUBTASK 27-31-94-020-014

(14) Remove the pitot probe [8] with the baseplate [5] as whole assembly.

(a) If the gasket [2] is damaged, discard it.

NOTE: The gasket can be reused if it is not damaged.

SUBTASK 27-31-94-020-010

(15) If it is necessary, remove the baseplate [5] from the pitot probe [8].

(a) Remove the lockwire, and discard it.

(b) Remove the bolts [3] and washers [4] from the baseplate [5].

NOTE: Hardened sealant can be softened with heat for easier removal. The baseplate and probe may be heated in an oven, 150°F (65.6°C) for 20 minutes to soften the sealant.

(c) Remove the baseplate [5].

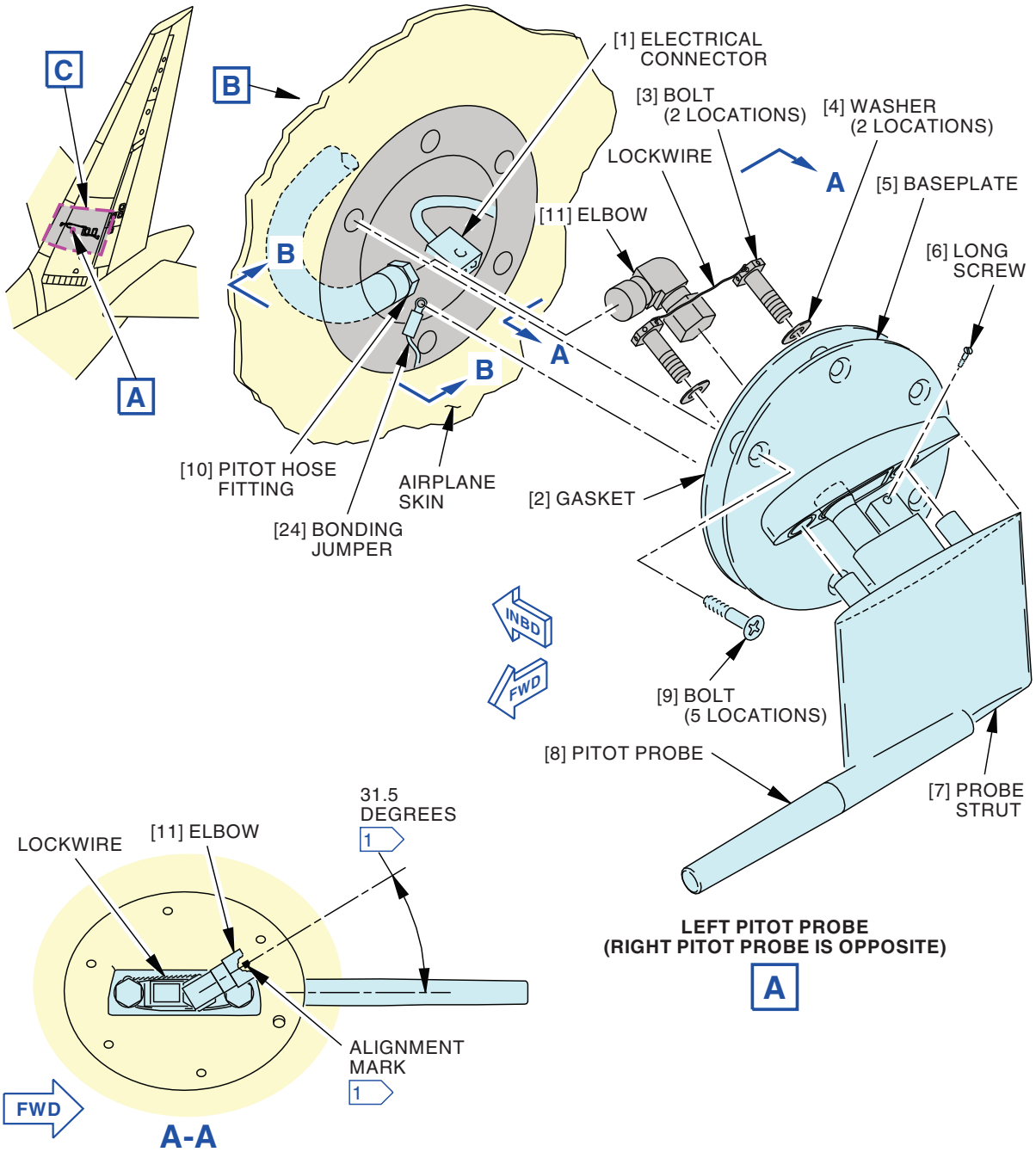
SUBTASK 27-31-94-020-011

(16) Keep the baseplate [5] for another installation.

NOTE: The baseplates are not interchangeable between positions.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

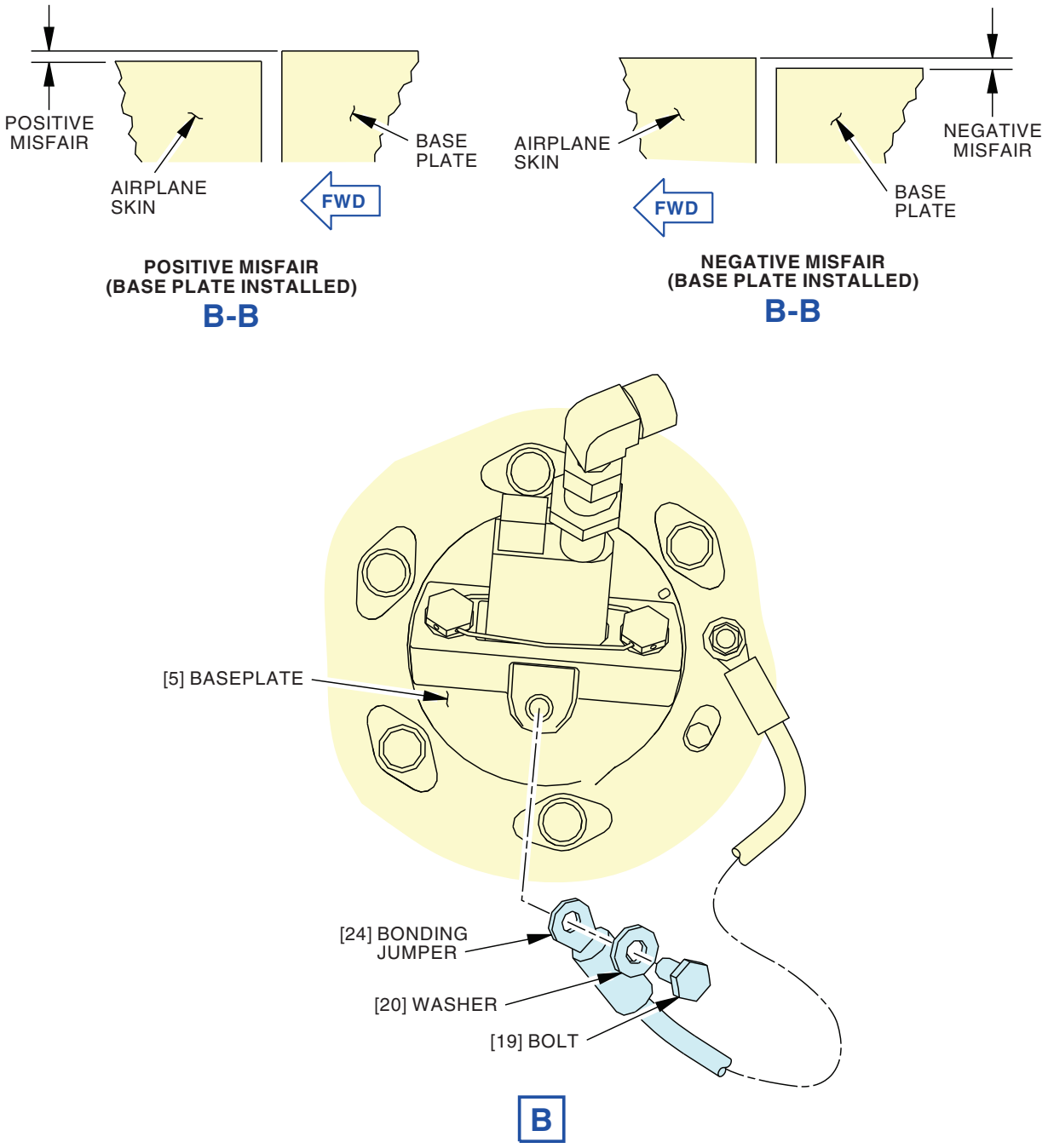


1 THE CENTER OF THE 90-DEGREE ELBOW SHOULD ALIGN 31.5 DEGREES ABOVE THE CENTER OF THE PITOT PROBE (OR ALIGNMENT MARK SHOWN). THIS WILL PREVENT MOISTURE FROM COLLECTING IN LOW SPOTS OF THE PITOT ASSEMBLY AND PITOT HOSE.

2412617 S00061531006_V3

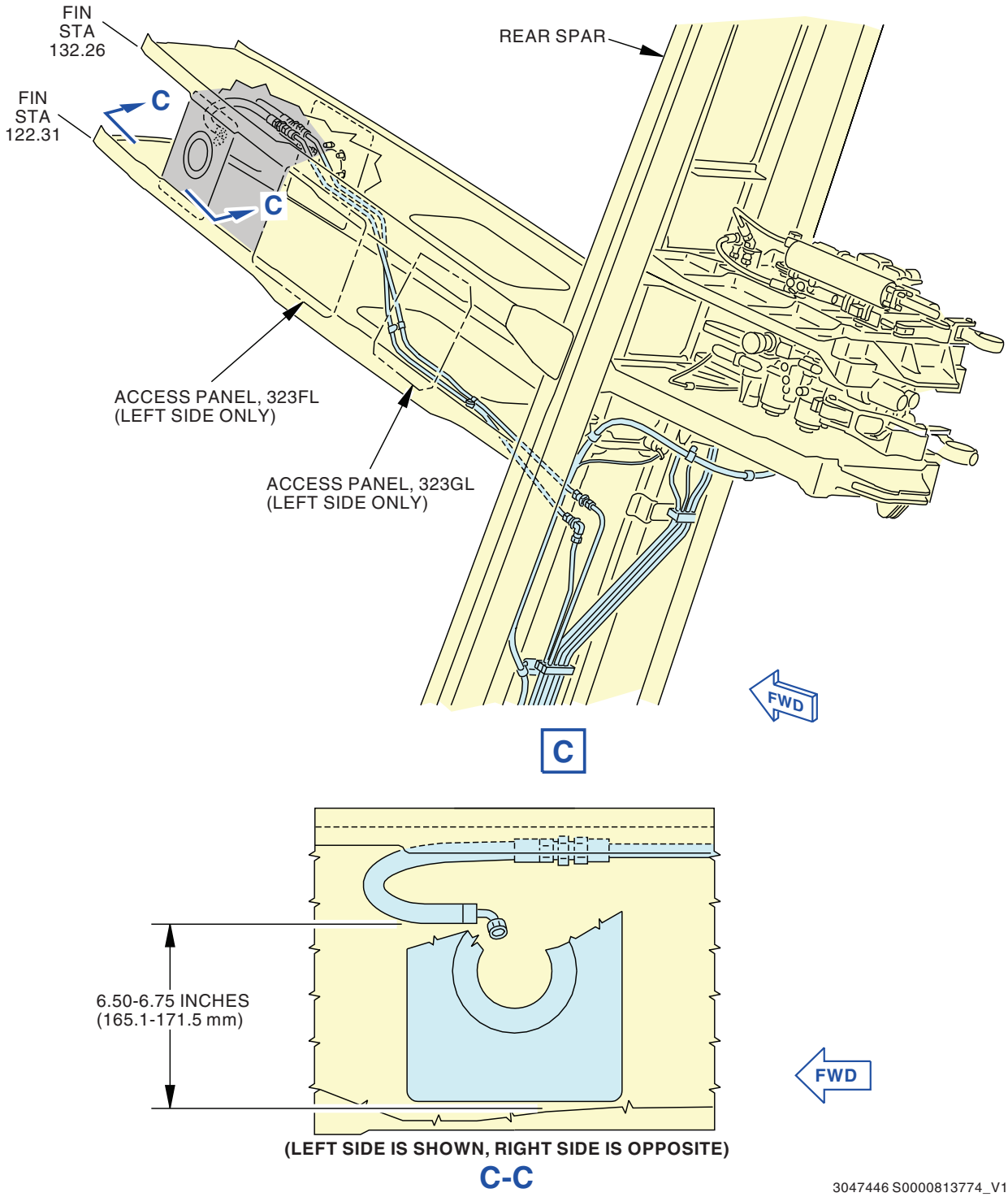
Feel System Pitot Probe Installation
Figure 401/27-31-94-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



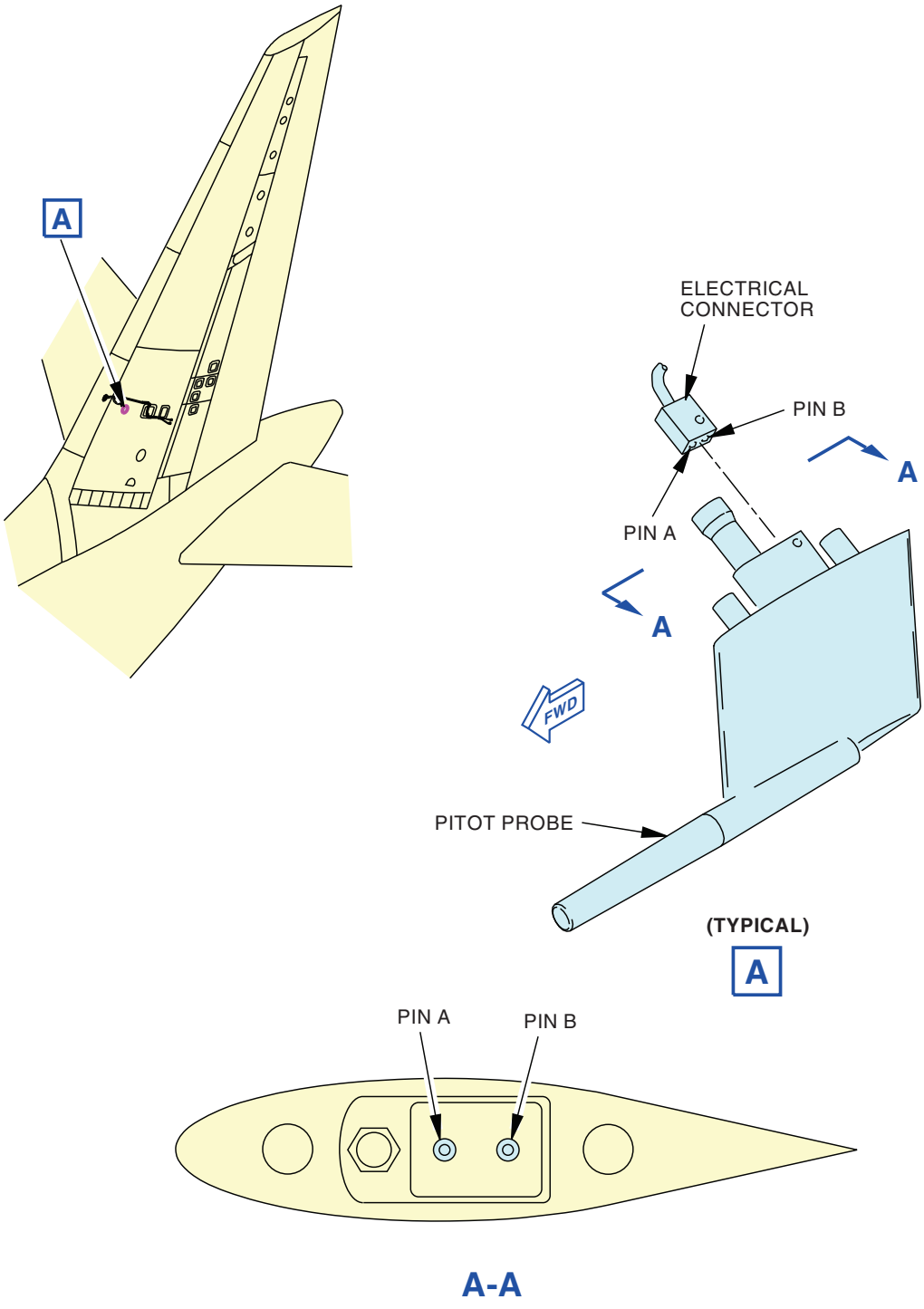
Feel System Pitot Probe Installation
Figure 401/27-31-94-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Feel System Pitot Probe Installation
Figure 401/27-31-94-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



3009755 S0000783506_V1

Feel System Pitot Probe Electrical Connector Polarity
Figure 402/27-31-94-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-94-400-801

3. Feel System Pitot Probe Installation

(Figure 401 and Figure 402)

A. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
27-31-17-790-801	Elevator Pitot System Leak - Test (P/B 201)
30-31-00-730-801	Pitot Probe, AOA Sensor, and TAT Probe Heater - System Test (P/B 501)
51-22-00 P/B 201	SEALS AND SEALING - MAINTENANCE PRACTICES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A00708	Sealant - Fast Curing, 2-Part - PR-1828	AMS 3277
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Pitot probe	27-31-94-01-060	PTF ALL

E. Location Zones

Zone	Area
323	Vertical Fin - Front Spar To Rear Spar

F. Access Panels

Number	Name/Location
323FL	Vertical Fin, Access

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Feel System Pitot Probe Installation

SUBTASK 27-31-94-110-001

(1) For the surface of the baseplate [5], do this task: Clean Bare, Clad, or Plated Metal with Solvent, TASK 20-10-34-110-802.

SUBTASK 27-31-94-420-002

(2) If removed, do these steps to install the pitot probe [8] on the baseplate [5]:

(a) Put the pitot probe [8] on the baseplate [5].

(b) Install the bolts [3] and washers [4] in the baseplate [5].

(c) Install the MS20995NC32 lockwire, G01912, on the bolts [3].

(d) Apply sealant, A00247, around the head of the two bolts [3].

SUBTASK 27-31-94-420-015

(3) Put the gasket [2] into position on the baseplate [5].

SUBTASK 27-31-94-420-005

(4) Connect the bolt [19], washer [20], and bonding jumper [24] to the baseplate [5].

(a) Measure the resistance between the terminal of the bonding jumper [24] and airplane skin.

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the resistance is not more than 2.5 milliohms (0.0025 ohm).

SUBTASK 27-31-94-420-007

(5) Make sure that the torque of the pitot hose fitting [10] is 161 in-lb (18.2 N·m) - 179 in-lb (20.2 N·m).

SUBTASK 27-31-94-420-013

(6) Do these steps to install the elbow [11] to the pitot probe [8]:

(a) If installed, remove the cap.

(b) Make sure that the center of the elbow [11] is aligned with the alignment mark on the inboard side of the pitot probe [8] base (Figure 401).

NOTE: The lowest point of the flexible hose must be the connection with the pitot probe.

(c) Install the elbow [11] to the pitot probe [8].

SUBTASK 27-31-94-420-006

(7) Tighten the elbow [11] to the base of the pitot probe [8] to 140 ±14 in-lb (15.8 ±1.6 N·m).

SUBTASK 27-31-94-420-014

CAUTION

BE CAREFUL WHEN YOU CONNECT THE ELECTRICAL CONNECTOR TO THE PITOT PROBE. IF YOU DO NOT OBEY, DAMAGE TO THE WIRE BUNDLE AT THE ELECTRICAL CONNECTOR CAN OCCUR.

(8) Do these steps to connect the electrical connector [1] to the pitot probe [8]:

(a) Examine the electrical connector [1] for loose, bent, or broken pins.

(b) Remove the long screw [6] from the electrical connector [1].

(c) Connect the electrical connector [1].

1) Make sure that the pins are connected correctly (Figure 402).

NOTE: Pin B is the pin on the side closest to the offset screw hole.

(d) Install the long screw [6] in the electrical connector [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-94-390-001

(9) Apply a thin layer of sealant, A00247, or PR-1828 sealant, A00708 (quick curing), on the mating surfaces of the baseplate [5] and pitot probe [8].

NOTE: You can use the PR-1828 sealant, A00708, quick curing sealant, in applications where the aircraft needs to go quickly back into service. The two part sealant should be mixed and then applied per SEALS AND SEALING - MAINTENANCE PRACTICES, PAGEBLOCK 51-22-00/201.

NOTE: If the aircraft needs to go quickly back into service, and you do not use the quick curing sealant, it is not necessary to apply either sealant immediately if the cure time of either of the two sealants will cause a flight delay. To prevent moisture damage to the airplane, sealant must be applied as soon as it is convenient for the operator.

SUBTASK 27-31-94-420-010

CAUTION

DO NOT LET DUST, DIRT, OR OTHER UNWANTED MATERIAL GET IN THE PROBES. CONTAMINATION CAN CAUSE DAMAGE TO THE SYSTEM.

(10) Do these steps to install the pitot probe [8]:

(a) Hold the probe strut [7].

(b) Put the pitot hose fitting [10] and electrical connector [1] in the installation hole.

(c) Install the bolts [9] in the baseplate [5] of the pitot probe [8].

1) Apply compound, C00528, to the bolts [9] and baseplate [5] hole.

2) Tighten the bolts [9] to 32 in-lb (3.6 N·m) - 39 in-lb (4.4 N·m).

(d) Put the pitot probe [8] into position on the airplane skin.

1) Measure the baseplate [5] to skin misfair at the forward and aft edges of the baseplate [5] to be within +/-0.040 in. (1.0 mm).

SUBTASK 27-31-94-390-002

(11) Fill the gap between the baseplate [5] and the airplane skin with sealant, A00247, or PR-1828 sealant, A00708 (quick curing).

NOTE: It is not necessary to apply the sealant immediately if the cure time of the sealant will cause a flight delay. To prevent moisture damage to the airplane, sealant must be applied as quickly as possible.

(a) Make sure that the baseplate [5] and sealant are flat with the airplane skin.

(b) Make sure that there is not a loop or downward oriented dip in the flexible pitot hose.

NOTE: A loop or downward oriented dip in the flexible pitot hose can cause water accumulation. Water accumulation can cause an erratic airspeed indication, resulting in a FEEL DIFF PRESS caution light.

(c) If it is necessary, loosen elbow [11] and adjust hose routing to prevent fluid pooling.

1) Tighten the elbow [11] to 140 ±14 in-lb (15.8 ±1.6 N·m).

SUBTASK 27-31-94-410-001

(12) Install this access panel:

Number	Name/Location
323FL	Vertical Fin, Access

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-94-760-001

(13) Measure the resistance between the strut of the pitot probe [8] and airplane skin.

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) If the resistance is more than 10 milliohms (0.010 ohms), do these steps:

1) Remove the pitot probe [8].

2) Clean the bonding surfaces, including the countersunk holes in the pitot probe [8] (SWPM 20-20-00).

3) Replace the bolts with new bolts.

4) Install the pitot probe [8].

5) Measure the resistance between the strut of the pitot probe [8] and airplane skin.

a) Use an intrinsically safe approved bonding meter, COM-1550.

6) If the resistance is more than 10 milliohms (0.010 ohms), do these steps:

a) Remove the pitot probe [8].

b) Replace the nutplates and rivets that attach the pitot probe [8] (SRM PROCEDURE 51-40-02-0G-0).

c) Install the pitot probe [8].

d) Make sure that the bonding resistance is not more than 10 milliohms (0.010 ohms).

SUBTASK 27-31-94-860-005

(14) For the left pitot probe [8], remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT

SUBTASK 27-31-94-860-006

(15) For the right pitot probe [8], remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	4	C00237	HEATERS ELEV PITOT RIGHT

H. Feel System Pitot Probe Installation Test

SUBTASK 27-31-94-710-001

(1) Do this task: Elevator Pitot System Leak - Test, TASK 27-31-17-790-801.

SUBTASK 27-31-94-710-002

(2) Do this task: Pitot Probe, AOA Sensor, and TAT Probe Heater - System Test, TASK 30-31-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STALL WARNING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Stall Warning System - Operational Test

(2) Stall Management Yaw Damper BITE Test - Existing Faults

(3) Stall Management Yaw Damper BITE Test - Fault History

(4) Stall Management Yaw Damper BITE Test

TASK 27-32-00-710-801

2. Stall Warning System - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Stall Warning System Operational Test

SUBTASK 27-32-00-710-001

(1) Do a test of the No. 1 Stall Warning System:

(a) Push and hold the STALL WARNING TEST NO. 1 switch on the P5-18 panel.

NOTE: The amber SPD LIM light on the on-side Primary Flight Display will come on. A YAW DAMPER light on the P5-3 panel with a related MASTER CAUTION light is possible. A SPD LIM annunciation that stays on more than 5 seconds after the stick shaker operates could be a problem.

1) Make sure that the captain's control column shaker operates for more than five seconds.

(b) Release the STALL WARNING TEST NO. 1 switch.

1) Make sure that the captain's control column shaker stops.

SUBTASK 27-32-00-710-002

(2) Do a test of the No. 2 Stall Warning System:

NOTE: SPEED BRAKE DO NOT ARM light may momentarily turn on and off for SMYD 285A1010-110.

(a) Push and hold the STALL WARNING TEST NO. 2 switch on the P5-18 panel.

NOTE: The amber SPD LIM light on the on-side Primary Flight Display will come on. A YAW DAMPER light on the P5-3 panel with a related MASTER CAUTION light is possible. A SPD LIM annunciation that stays on more than 5 seconds after the stick shaker operates could be a problem.

1) Make sure that the first officer's control column shaker operates for more than five seconds.

(b) Release the STALL WARNING TEST NO. 2 switch.

1) Make sure that the first officer's control column shaker stops.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-32-00-740-801

3. Stall Management Yaw Damper (SMYD) BITE Test - Existing Faults

(Figure 501)

A. General

(1) This procedure does the EXISTING FAULTS portion of the Stall Management Yaw Damper (SMYD) Built-In-Test Equipment (BITE).

(2) This procedure does an exchange of SMYD 1 and 2 if a maintenance message is present, and NO FAULT is found.

(3) Do this BITE test for SMYD 1 and 2 to make sure that all the faults are found.

(4) For EXISTING FAULTS, the SMYD does internal tests, discrete output checks, sensor interface checks, and checks of inputs to the SMYD. All faults that are found during the EXISTING FAULTS are active. A maintenance message identifies a specific failure which is found by the SMYD.

B. References

Reference	Title
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Stall Management Yaw Damper (SMYD) BITE Test - Existing Faults

SUBTASK 27-32-00-010-001

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-32-00-860-020

(2) To align the Inertial Reference Unit (IRU), do this task: Air Data Inertial Reference System - Alignment from the ISDU, TASK 34-21-00-820-802 or Air Data Inertial Reference System - Alignment from the FMC CDU, TASK 34-21-00-820-801 or Air Data Inertial Reference System - Automatic Alignment from the IRS MSU, TASK 34-21-00-820-803.

SUBTASK 27-32-00-200-001

(3) To make sure that the Flight Management Computer (FMC) is powered and has good weight data entered, do these steps:

NOTE: Zero Fuel Weight (ZFW) is calculated by subtracting the total fuel weight from the combined total weight of the airplane and contents. For the BITE test, any number between 70 and 150 (normal ZFW in thousands of pounds) will verify a check pass.

(a) Push the INIT REF button on the Control Display Unit (CDU).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the PERF INIT page, enter the following:

1) 3L: ZFW prompt, enter 94 lb (43 kg).

NOTE: GW/CRZ CG in 1L will auto calculate after ZFW is entered.

2) 4L: RES prompt, enter 4.0

3) 5L: CI prompt, enter 30.

SUBTASK 27-32-00-740-001

(4) Do the EXISTING FAULTS test on the SMYD BITE:

(a) Push the ON/OFF switch.

NOTE: The display will show EXISTING FAULTS? If a fault occurs, make sure that all FMC and Air Data Inertial Reference Unit (ADIRU) values are good.

(b) Push the YES switch and make sure that the display shows:

1) Either a message corresponding the first fault in the list.

2) Or the display shows NO FAULTS.

(c) If there are existing faults, refer to the IFIM and do the applicable procedure(s).

(d) If the display shows NO FAULTS, push the MENU switch to return to the EXISTING FAULTS? display.

(e) Do these steps to unlatch existing faults:

1) Push the down arrow until RESET LATCHES? shows.

2) Push the YES switch to reset the latches.

NOTE: END OF LIST will display when done.

(f) If a maintenance message is present and the SMYD shows NO FAULTS:

1) Do an exchange check of the SMYD 1 and 2 to find if a SMYD is damaged.

a) Refer to the IFIM and do the applicable procedure(s).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-32-00-860-004

(1) Push the ON/OFF switch to turn the SMYD display off.

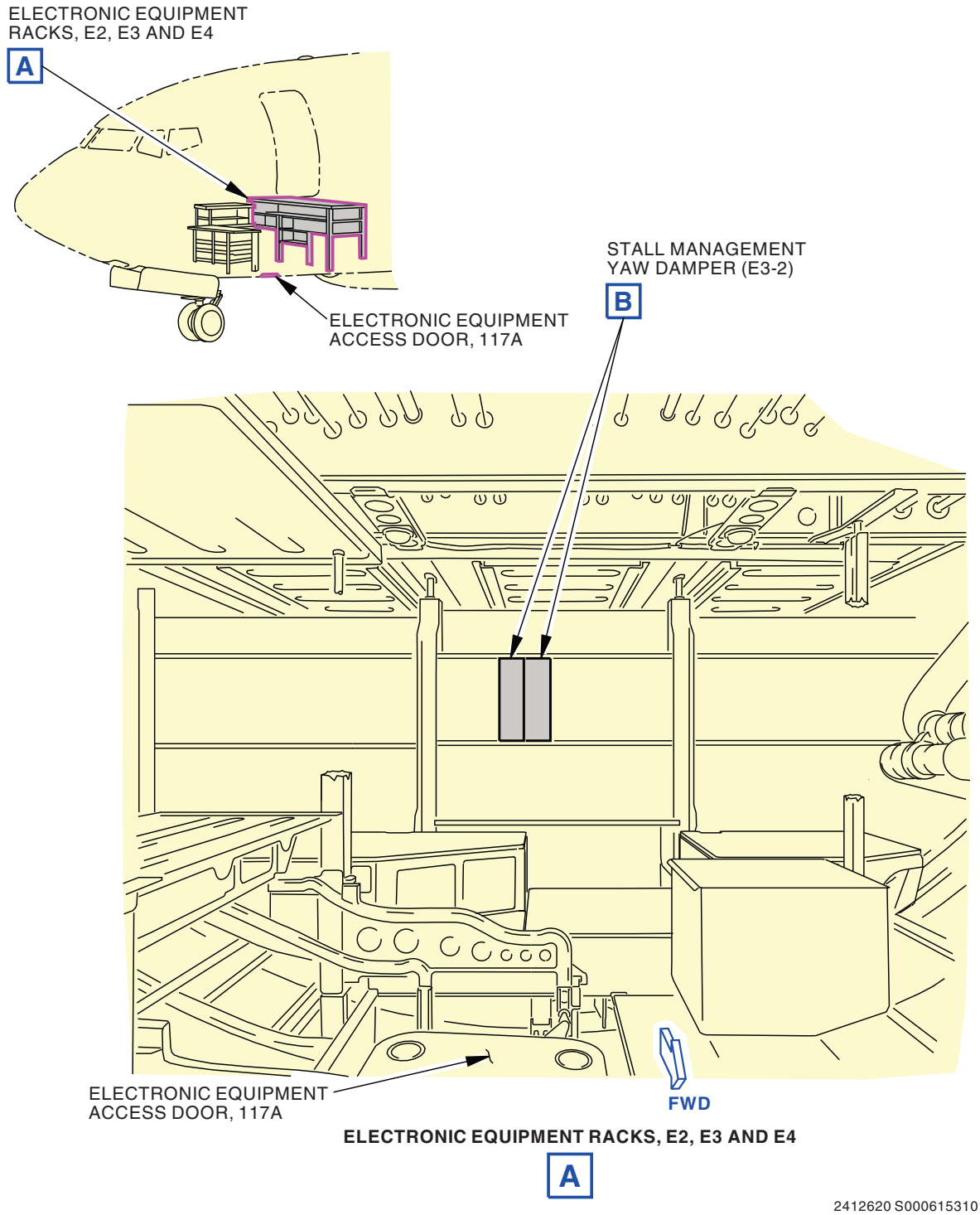
SUBTASK 27-32-00-410-001

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

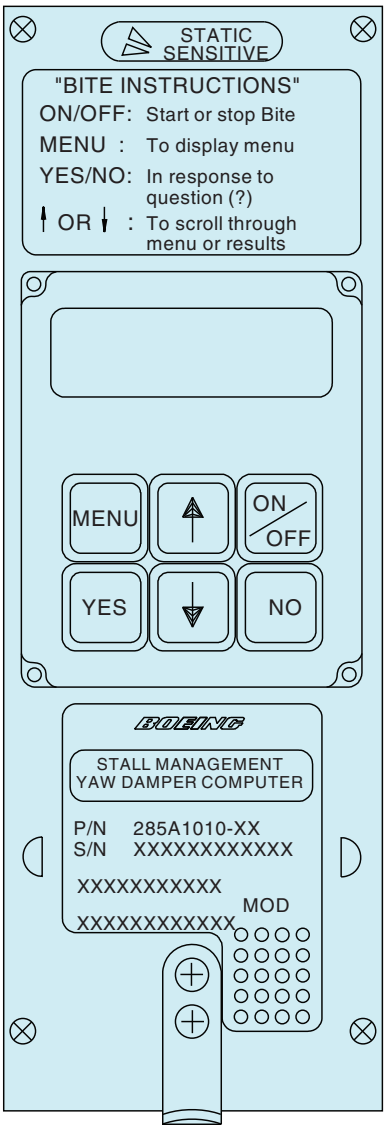
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stall Management Yaw Damper Test
Figure 501/27-32-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STALL MANAGEMENT YAW DAMPER (EXAMPLE)

B

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Stall Management Yaw Damper Test
Figure 501/27-32-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-32-00-740-802

4. Stall Management Yaw Damper (SMYD) BITE Test - Fault History

(Figure 501)

A. General

(1) This procedure does the FAULT HISTORY? portion of the Stall Management Yaw Damper (SMYD) Built-In-Test Equipment (BITE).

(2) The SMYD keeps a record of faults in FAULT HISTORY. The faults are grouped by flight leg. Flight leg 0 (LEG 0) is the most recent flight and includes faults that occurred on the ground after the most recent flight. Only flight legs with faults will be displayed.

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-32-00-010-002

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Stall Management Yaw Damper (SMYD) BITE Test - Fault History

SUBTASK 27-32-00-740-002

(1) Do the FAULT HISTORY test on the SMYD BITE.

(a) Push the ON/OFF switch on the SMYD BITE display to turn the BITE display on.

NOTE: EXISTING FAULTS? will be shown.

(b) Push the down arrow until the display shows FAULT HISTORY?.

1) Push the YES switch.

a) If there are faults in history, the display will show FLIGHT LEG -n, where n is the most recent flight with faults recorded.

b) If there are no faults, the message NO FAULT HISTORY will show.

2) Push the up arrow or down arrow to go through the flight legs.

3) If you push YES at any flight leg display the message corresponding to the first fault in the list will show.

4) To show the faults for the flight leg use the up arrow and down arrow.

NOTE: To return to the FLIGHT LEG N display, push the MENU switch.

5) To return to the FAULT HISTORY display, push the MENU switch.

6) Refer to the IFIM and do the applicable procedure(s).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-32-00-860-007

(1) Push the ON/OFF switch to turn the SMYD display off.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-32-00-410-002

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 27-32-00-740-803

5. Stall Management Yaw Damper (SMYD) BITE Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task does the GROUND TEST? portion of the Stall Management Yaw Damper (SMYD) Built-In-Test Equipment (BITE).

(2) For GROUND TEST, the SMYD has several menus:

(a) SELF TEST? - This does a self test sequence of the SMYD.

(b) DISCRETE INPUTS? - This monitors the status of each displayed discrete inputs to the SMYD.

(c) ARINC429 INPUTS? - This monitors the status of each ARINC429 input to the SMYD.

(d) ANALOG INPUTS? - This shows the status of these sensors:

1) AOA SENSOR?

2) FLAP SENSOR?

(e) SERVO TEST? - This does a test of the integrity of the interface between the SMYD and rudder power control unit.

1) YD LVDT?

NOTE: This is only for the SMYD in the number 2 position.

(f) DISPLAY TEST? This does a test of the SMYD display.

(3) This task has a procedure for each of these menu items.

(4) This task also contains the INPUT SIGNAL INTEGRITY TEST procedure from the OTHER FUNCTNS? menu.

(5) During this test, the MASTER CAUTION annunciators come on.

B. References

Reference	Title
29-00-00-800-802	Hydraulic Power System - Activation (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)
WDM 27-32-11	Wiring Diagram Manual
WDM 27-32-21	Wiring Diagram Manual

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205
SPL-1917	Fixture - Test, Angle of Attack Sensor, ROSEMOUNT AOA's Part #: J34002-19 Supplier: 81205 Opt Part #: A34012-24 Supplier: 81205 Opt Part #: J34002-18 Supplier: 81205

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Test

SUBTASK 27-32-00-860-010

(1) To align the Inertial Reference Unit (IRU), do this task: Air Data Inertial Reference System - Alignment from the ISDU, TASK 34-21-00-820-802 or Air Data Inertial Reference System - Alignment from the FMC CDU, TASK 34-21-00-820-801 or Air Data Inertial Reference System - Automatic Alignment from the IRS MSU, TASK 34-21-00-820-803.

SUBTASK 27-32-00-860-019

(2) To make sure that the Flight Management Computer (FMC) is powered and has good weight data entered, do these steps:

NOTE: Zero Fuel Weight (ZFW) is calculated by subtracting the total fuel weight from the combined total weight of the airplane and contents. For the BITE test, any number between 70 and 150 (normal ZFW in thousands of pounds) will verify a check pass.

(a) Push the INIT REF button on the Control Display Unit (CDU).

(b) On the PERF INIT page, enter the following:

1) 3L: ZFW prompt, enter 94 for lbs or 43 for kg.

NOTE: GW/CRZ CG in 1L will auto calculate after ZFW is entered.

2) 4L: RES prompt, enter 4.0

3) 5L: CI prompt, enter 30.

SUBTASK 27-32-00-010-003

(3) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

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SUBTASK 27-32-00-860-011

(4) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	9	C00109	FLIGHT RECORDER AC
C	10	C00468	FLIGHT RECORDER DC

G. SELF TEST

SUBTASK 27-32-00-740-021

(1) Self Test using the Onboard Maintenance Function (OMF) (Recommended Method):

(a) For SMYD 1 and SMYD 2, do these steps:

1) Set the GND TEST switch on the P61-4 panel, to the ENABLE position.
2) Push the SYS button in the Multi Function Display (MFD) section on the P9-82 multi function panel.
3) Use the selector on the P9-82 multi function panel to make these selections:
   - MAINT
   - ONBD MAINT
   - LINE MAINT
   - GROUND TESTS
   - 27 - Flight Controls Stall Management Yaw Damper System
   - System Test.
4) Follow the on-screen instructions and then select the rectangular CONTINUE button.
5) Use the selector on the P9-82 multi function panel, to make this selection:
   - START TEST.
6) Make sure that when the test is complete, there are no faults and you see the TEST CONDITION: PASS message.
   a) If you see TEST CONDITION: FAILED, refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).
   b) If you do additional tests on the P9-82 multi function panel, select the NEW TEST button.

SUBTASK 27-32-00-740-022

(2) Self Test using the SMYD (Alternate Method):

(a) On the SMYD 1 and SMYD 2, do these steps:

1) Push the ON/OFF button.
2) Push the down arrow button.
3) Make sure that the display shows: GROUND TESTS?.
4) Push the YES button.
5) Make sure that the display shows: SELF TEST?.
6) Push the YES button to start the self test.

   NOTE: TEST IN PROGRESS will show while the test is run.
7) When the self test is complete, make sure that the display shows: SMYD LRU OK.

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8) If the display shows a fault message, refer to the IFIM and do the applicable procedure(s).

9) Push the ON/OFF button.

H. DISCRETE INPUTS

SUBTASK 27-32-00-740-016

(1) Do these steps on the applicable SMYD:

(a) Push the ON/OFF button.

(b) Push the down arrow button.

(c) Make sure that the display shows: GROUND TESTS?.

(d) Push the YES button.

(e) Push the down arrow button until the display shows: DISCRETE INPUTS?.

(f) Push the YES button.

1) For the applicable SMYD, make sure that the program pin inputs are correct (WDM 27-32-11, WDM 27-32-21).

NOTE: For each pin, the display will show OPEN or GND. Some pins are not applicable because of the SMYD model and configuration.

NOTE: Pins that are not shown on the Wiring Diagram Manual (WDM) are OPEN.

(g) Push the down arrow button.

(h) Make sure that the display shows: asym en = gnd.

(i) Open this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

(j) Make sure that the display shows: asym en = open.

(k) Push the down arrow button.

(l) Make sure that the display shows: asymetry = open.

(m) On the Flap/Slat Electronics Unit (FSEU), do these steps:

1) Push the ON/OFF button.

2) Push the down arrow until the display shows: OTHER FUNCTNS?.

3) Push the YES button.

4) Make sure that the display shows: SET OUTPUTS?.

5) Push the YES button.

6) Push the down arrow until the display shows: LE INTRANS?.

7) Push the YES button.

8) Make sure that the display shows: SET ON?.

9) Push the YES button.

(n) Make sure that the SMYD display shows: asymetry = gnd.

NOTE: You only have 75 seconds to complete this task. If you do not finish in 75 seconds, you will have to repeat the test and restart with steps to use the FSEU to set LE INTRANS to “SET ON” state.

(o) Push the down arrow button.

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(p) Make sure that the display shows: LE UCM = open.

(q) On the FSEU, do these steps:

1) Push the MENU button.
2) Push the down arrow until the display shows: LE UCM?.
3) Push the YES button.
4) Make sure that the display shows: SET ON?.
5) Push the YES button.

(r) Make sure that the SMYD display shows: LE UCM = gnd.

NOTE: You only have 75 seconds to complete this task. If you do not finish in 75 seconds, you will have to repeat the test and restart with steps to use the FSEU to set LE UCM to “SET ON” state.

(s) On the FSEU, push the ON/OFF button.

(t) On the SMYD, push the down arrow button.

(u) Make sure that the display shows: TOGA = open.

(v) Push and hold the TOGA switch on the number 1 or number 2 throttle lever.

(w) Make sure that the display shows: TOGA = gnd.

(x) Release the TOGA switch.

(y) Make sure that the display shows: TOGA = open.

(z) Push the down arrow button.

(aa) Make sure that the display shows: nlg = gnd.



WARNING

MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES AND LANDING GEAR DOOR AREAS BEFORE YOU SIMULATE THE AIRPLANE TO THE AIR MODE. THE SIMULATED AIR MODE CAN CAUSE MOVEMENT TO CONTROL SURFACES, LANDING GEAR, AND LANDING GEAR DOORS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(ab) Install the proximity sensor test set, SPL-1690, between S1014 and S1015 nose gear air/ground sensor face and its steel target.

NOTE: The sensors are located in the aft lower nose wheel well.

(ac) Make sure that the display shows: nlg = air.

(ad) Remove the proximity sensor test set, SPL-1690, from S1014 and S1015.

(ae) Make sure that the display shows: nlg = gnd.

(af) Push the down arrow button.

(ag) Make sure that the display shows: air/gnd = gnd.

(ah) Set the GND TEST switch on the P61-4 panel, to the ENABLE position.

(ai) Push the SYS button in the MFD section on the P9-82 multi function panel.

(aj) Use the selector on the P9-82 multi function panel to make these selections:

- MAINT
- ONBD MAINT
- OTHER FUNCTIONS

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- SPECIAL FUNCTIONS
- 32 - Proximity Switch Electronic Unit
- PSEU - Air/Ground Override
- CONTINUE
- CONTINUE under the PRECONDITIONS screen
- START FUNCTION
- Set SYS 1 and 2 IN AIR.

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES AND LANDING GEAR DOOR AREAS BEFORE YOU SIMULATE THE AIRPLANE TO THE AIR MODE. THE SIMULATED AIR MODE CAN CAUSE MOVEMENT TO CONTROL SURFACES, LANDING GEAR, AND LANDING GEAR DOORS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(ak) Select CONTINUE by pushing on the center selector button SEL.

1) Make sure that these conditions are seen on the MAX Display System (MDS):
   - Air/Gnd Override is ACTIVE
   - System 1: AIR
   - System 2: AIR.

(al) Make sure that the display shows: air/gnd = air.

(am) Select Set SYS 1 and 2 ON GND.

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES AND LANDING GEAR DOOR AREAS BEFORE YOU SIMULATE THE AIRPLANE TO THE AIR MODE. THE SIMULATED AIR MODE CAN CAUSE MOVEMENT TO CONTROL SURFACES, LANDING GEAR, AND LANDING GEAR DOORS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(an) Select CONTINUE by pushing on the center selector button SEL.

1) Make sure that these conditions are seen on the MDS:
   - Air/Gnd Override is NOT ACTIVE
   - System 1: GND
   - System 2: GND.

(ao) Make sure that the SMYD display shows: air/gnd = gnd.

(ap) Push the down arrow button.

(aq) Make sure that the display shows: MLG = gnd.

(ar) Open these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

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(as) Make sure that the display shows: MLG = open.

(at) Close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

(au) Make sure that the display shows: MLG = gnd.

(av) Push the down arrow button.

(aw) Make sure that the display shows: A/S pwr present.

(ax) For SMYD 1, open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

(ay) For SMYD 2, open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2

(az) Make sure that the display shows: A/S pwr missing.

(ba) For SMYD 1, close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

(bb) For SMYD 2, close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2

(bc) Make sure that the display shows: A/S pwr present.

(bd) Push the down arrow button.

(be) Make sure that the display shows: S/S pwr present.

(bf) For SMYD 1, open this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

(bg) For SMYD 2, open this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

(bh) Make sure that the display shows: S/S pwr missing.

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(bi) For SMYD 1, close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

(bj) For SMYD 2, close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

(bk) Make sure that the display shows: S/S pwr present.

(bl) Push the down arrow button.

(bm) Make sure that the display shows: YD pwr present.

(bn) For SMYD 1, open this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	7	C00286	YAW DAMPER 1 DC

(bo) For SMYD 2, open this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	6	C01354	YAW DAMPER 2 DC

(bp) Make sure that the display shows: YD pwr missing.

(bq) For SMYD 1, close this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	7	C00286	YAW DAMPER 1 DC

(br) For SMYD 2, close this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	6	C01354	YAW DAMPER 2 DC

(bs) Make sure that the display shows: YD pwr present.

(bt) Push the down arrow button.

(bu) Make sure that the display shows: FC B sw = on.

(bv) Set the FLT CONTROL B switch, on the P5-3 overhead panel, to the OFF position.

(bw) Make sure that the display shows: FC B sw = off.

(bx) Set the FLT CONTROL B switch, on the P5-3 overhead panel, to the ON position.

(by) Make sure that the display shows: FC B sw = on.

(bz) Push the down arrow button.

(ca) Make sure that the display shows: eng rqst = false.

(cb) Push and hold the YAW DAMPER switch, on P5-3 overhead panel, to the ON position.

(cc) Make sure that the display shows: eng rqst = true.

(cd) Release the YAW DAMPER switch, on P5-3 overhead panel, to the OFF position.

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(ce) Push the down arrow button.

(cf) Make sure that the display shows: flap up = up.

WARNING

KEEP PERSONS AND EQUIPMENT AWAY FROM ALL CONTROL SURFACES AND THE NOSE GEAR WHEN HYDRAULIC POWER IS SUPPLIED. THE AILERONS, ELEVATORS, RUDDER, FLAPS, SLATS, SPOILERS, AND THE NOSE GEAR ARE SUPPLIED WITH POWER BY THE HYDRAULIC SYSTEMS. INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR WHEN HYDRAULIC POWER IS SUPPLIED.

(cg) Supply the system B hydraulic power (TASK 29-00-00-800-802).

(ch) Move the flaps to the detent 1 position.

(ci) Make sure that the display shows: flap up = not up.

(cj) Push the ON/OFF button.

(ck) Close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

(cl) Move the flaps to the detent 0 position.

(cm) Remove the system B hydraulic power (TASK 29-11-00-860-805).

I. ARINC429 TEST

SUBTASK 27-32-00-740-009

(1) Do these steps on the applicable SMYD:

(a) Push the ON/OFF button.

(b) Push the down arrow button.

(c) Make sure that the display shows: GROUND TESTS?.

(d) Push the YES button.

(e) Push the down arrow button until the display shows: ARINC429 INPUTS?.

(f) Push the YES button.

(g) Make sure that the display shows: ON SIDE IR BUS?.

(h) Push the YES button.

(i) Make sure that the display shows: BUS ACTIVE.

(j) For SMYD 1, open these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

(k) For SMYD 2, open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

(l) Make sure that the display shows: BUS INACTIVE.

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(m) For SMYD 1, close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

(n) For SMYD 2, close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

(o) Make sure that the display shows: BUS ACTIVE.

(p) Push the down arrow button.

(q) Make sure that the display shows: OFF SIDE IR BUS?.

(r) Push the YES button.

(s) Make sure that the display shows: BUS ACTIVE.

(t) For SMYD 1, open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

(u) For SMYD 2, open these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

(v) Make sure that the display shows: BUS INACTIVE.

(w) For SMYD 1, close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

(x) For SMYD 2, close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

(y) Push the down arrow button.

(z) Make sure that the display shows: ON SIDE ADR BUS?.

(aa) Push the YES button.

(ab) Make sure that the display shows: BUS ACTIVE.

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(ac) For SMYD 1, open these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

(ad) For SMYD 2, open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

(ae) Make sure that the display shows: BUS INACTIVE.

(af) For SMYD 1, close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

(ag) For SMYD 2, close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

(ah) Make sure that the display shows: BUS ACTIVE.

(ai) Push the down arrow button.

(aj) Make sure that the display shows: CDS DEU BUS?.

(ak) Push the YES button.

(al) Make sure that the display shows: BUS ACTIVE.

(am) For SMYD 1, open these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	10	C04022	DISPLAY DPC 1 HOLDUP

(an) For SMYD 2, open these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	11	C04025	DISPLAY DPC 2 PRI

(ao) Make sure that the display shows: BUS INACTIVE.

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CAUTION

MAKE SURE THAT YOU CLOSE THE CIRCUIT BREAKERS IN THE SPECIFIED SEQUENCE. IF YOU DO NOT CLOSE THEM IN THE SPECIFIED SEQUENCE, THE DPC CAN POSSIBLY BECOME CORRUPT AND A REPLACEMENT WILL BE NECESSARY.

(ap) Close these circuit breakers in the order specified.

1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	10	C04022	DISPLAY DPC 1 HOLDUP

2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	11	C04025	DISPLAY DPC 2 PRI

(aq) Wait for the DPC to turn on.

(ar) Make sure that the display shows: BUS ACTIVE.

(as) Push the down arrow button.

(at) Make sure that the display shows: FMC BUS?.

(au) Push the down arrow button.

(av) Make sure that the display shows: DFCS MCP BUS?.

(aw) Push the YES button.

(ax) Make sure that the display shows: BUS ACTIVE.

(ay) Open these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	5	C01044	AFCS MCP DC 1

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	3	C01047	AFCS MCP DC 2

(az) Make sure that the display shows: BUS INACTIVE.

(ba) Close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	5	C01044	AFCS MCP DC 1

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F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	3	C01047	AFCS MCP DC 2

(bb) Push the down arrow button.

(bc) Make sure that the display shows: CROSS CHANNEL?.

(bd) Push the YES button.

(be) Make sure that the display shows: BUS ACTIVE.

(bf) For SMYD 1, open this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	5	C01206	SMYD-2 CMPTR DC

(bg) Make sure that the SMYD 1 display shows: BUS INACTIVE.

(bh) For SMYD 1, close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	5	C01206	SMYD-2 CMPTR DC

(bi) Make sure that the SMYD 1 display shows: BUS ACTIVE.

(bj) For SMYD 2, do these steps:

1) Push the ON/OFF button.
2) Push the down arrow button.

(bk) For SMYD 1, do these steps:

1) Push the MENU button.
2) Make sure that the display shows: ARINC429 INPUTS?.
3) Push the MENU button.

(bl) Make sure that the display shows: GROUND TESTS?.

(bm) Push the down arrow button.

(bn) Make sure that the display shows: OTHER FUNCTNS?.

(bo) Push the YES button.

(bp) Make sure that the display shows: SYSTEM CONFIG?.

(bq) Push the down arrow button.

(br) Make sure that the display shows: I/O MONITOR?.

(bs) Push the YES button.

(bt) Make sure that the display shows: ARINC429 INPUTS?.

(bu) Push the YES button.

(bv) Make sure that the display shows: ON SIDE IR BUS?.

(bw) Push the down arrow button until the display shows: FMC BUS?.

(bx) Push the YES button.

(by) Push the down arrow button until the display shows: FMC label 075?.

(bz) Push the YES button.

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(ca) Open this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	6	C01017	FMCS CMPTR 1

(cb) Make sure that the SMYD 1 display shows: NO DATA.

(cc) Close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	6	C01017	FMCS CMPTR 1

(cd) On the CDU, make sure that the PERF INIT page shows 94 lb (43 kg) adjacent to Line Select Key (LSK) 3L.

(ce) Push the ON/OFF button.

J. SENSOR TEST

SUBTASK 27-32-00-860-024

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	3	C01072	HEATERS ALPHA VANE LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT

SUBTASK 27-32-00-480-001

WARNING MAKE SURE THAT THE AOA SENSORS ARE COOL BEFORE YOU TOUCH THEM. THE AOA SENSORS CAN BE SUFFICIENTLY HOT TO BURN YOU.

(2) Install the angle of attack probe test fixture, SPL-1917, on the left alpha vane.

SUBTASK 27-32-00-740-017

(3) Do this sensor test for SMYD 1:

(a) On SMYD 1, push the ON/OFF button.

(b) Push the down arrow button.

(c) Make sure that the display shows: GROUND TESTS?.

(d) Push the YES button.

(e) Push the down arrow button until the display shows: ANALOG INPUTS?.

(f) Push the YES button.

(g) Make sure that the display shows: AOA SENSOR?.

(h) Push the YES button.

NOTE: The display will show AOA and a value. Positive angles correspond to the Angle of Attack (AOA) sensor trailing edge above test fixture zero point. Negative angles correspond to the AOA sensor trailing edge below test fixture zero point. The Aerodynamic Offset of each sensor is noted on its trailing edge.

(i) Move the left AOA sensor to minus 35 degrees.

(j) Make sure that the display shows: -35 - Aerodynamic Offset, ± 1 degree.

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(k) Move the left AOA sensor to minus 14 degrees.
(l) Make sure that the display shows: -14 - Aerodynamic Offset, ± 1 degree.
(m) Move the left AOA sensor to 0 degrees.
(n) Make sure that the display shows: 0 - Aerodynamic Offset, ± 1 degree.
(o) Move the left AOA sensor to plus 14 degrees.
(p) Make sure that the display shows: 14 - Aerodynamic Offset, ± 1 degree.
(q) Move the left AOA sensor to plus 35 degrees.
(r) Make sure that the display shows: 35 - Aerodynamic Offset, ± 1 degree.
(s) Move the left AOA sensor to 0 degrees.
(t) Push the MENU button.
(u) Make sure that the display shows: AOA SENSOR?.
(v) Push the down arrow button.
(w) Make sure that the display shows: FLAP SENSOR?.



WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(x) Supply the system B hydraulic power (TASK 29-00-00-800-802).
(y) Push the YES button.
(z) Make sure that the display shows: FLAP DET 0.
(aa) Move the flaps to the detent 1 position.
(ab) Make sure that the display shows: FLAP DET 1.
(ac) Move the flaps to the detent 2 position.
(ad) Make sure that the display shows: FLAP DET 2.
(ae) Move the flaps to the detent 5 position.
(af) Make sure that the display shows: FLAP DET 5.
(ag) Move the flaps to the detent 10 position.
(ah) Make sure that the display shows: FLAP DET 10.
(ai) Move the flaps to the detent 15 position.
(aj) Make sure that the display shows: FLAP DET 15.
(ak) Move the flaps to the detent 25 position.
(al) Make sure that the display shows: FLAP DET 25.
(am) Move the flaps to the detent 30 position.
(an) Make sure that the display shows: FLAP DET 30.
(ao) Move the flaps to the detent 40 position.
(ap) Make sure that the display shows: FLAP DET 40.
(aq) Move the flaps to 0 detent.
(ar) Make sure that the display shows: FLAP DET 0.

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(as) Push the ON/OFF button.

(at) Remove the system B hydraulic power (TASK 29-11-00-860-805).

SUBTASK 27-32-00-080-001

(4) Remove the angle of attack probe test fixture, SPL-1917, from the left alpha vane.

SUBTASK 27-32-00-480-002

WARNING MAKE SURE THAT THE AOA SENSORS ARE COOL BEFORE YOU TOUCH THEM. THE AOA SENSORS CAN BE SUFFICIENTLY HOT TO BURN YOU.

(5) Install the angle of attack probe test fixture, SPL-1917, on the right alpha vane.

SUBTASK 27-32-00-740-018

(6) Do this sensor test for SMYD 2:

(a) On SMYD 2, push the ON/OFF button.

(b) Push the down arrow button.

(c) Make sure that the display shows: GROUND TESTS?.

(d) Push the YES button.

(e) Push the down arrow button until the display shows: ANALOG INPUTS?.

(f) Push the YES button.

(g) Make sure that the display shows: AOA SENSOR?.

(h) Push the YES button.

NOTE: The display will show AOA and a value. Positive angles correspond to the AOA sensor trailing edge above test fixture zero point. Negative angles correspond to the AOA sensor trailing edge below test fixture zero point. The Aerodynamic Offset of each sensor is noted on its trailing edge.

(i) Move the right AOA sensor to minus 35 degrees.

(j) Make sure that the display shows: -35 + Aerodynamic Offset, ± 1 degree.

(k) Move the right AOA sensor to minus 14 degrees.

(l) Make sure that the display shows: -14 + Aerodynamic Offset, ± 1 degree.

(m) Move the right AOA sensor to 0 degrees.

(n) Make sure that the display shows: 0 + Aerodynamic Offset, ± 1 degree.

(o) Move the right AOA sensor to plus 14 degrees.

(p) Make sure that the display shows: 14 + Aerodynamic Offset, ± 1 degree.

(q) Move the right AOA sensor to plus 35 degrees.

(r) Make sure that the display shows: 35 + Aerodynamic Offset, ± 1 degree.

(s) Move the right AOA sensor to 0 degrees.

(t) Push the MENU button.

(u) Make sure that the display shows: AOA SENSOR?.

(v) Push the down arrow button.

(w) Make sure that the display shows: FLAP SENSOR?.

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WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(x) Supply the system B hydraulic power (TASK 29-00-00-800-802).

(y) Push the YES button.

(z) Make sure that the display shows: FLAP DET 0.

(aa) Move the flaps to the detent 1 position.

(ab) Make sure that the display shows: FLAP DET 1.

(ac) Move the flaps to the detent 2 position.

(ad) Make sure that the display shows: FLAP DET 2.

(ae) Move the flaps to the detent 5 position.

(af) Make sure that the display shows: FLAP DET 5.

(ag) Move the flaps to the detent 10 position.

(ah) Make sure that the display shows: FLAP DET 10.

(ai) Move the flaps to the detent 15 position.

(aj) Make sure that the display shows: FLAP DET 15.

(ak) Move the flaps to the detent 25 position.

(al) Make sure that the display shows: FLAP DET 25.

(am) Move the flaps to the detent 30 position.

(an) Make sure that the display shows: FLAP DET 30.

(ao) Move the flaps to the detent 40 position.

(ap) Make sure that the display shows: FLAP DET 40.

(aq) Move the flaps to the detent 0 position.

(ar) Make sure that the display shows: FLAP DET 0.

(as) Push the ON/OFF button.

(at) Remove the system B hydraulic power (TASK 29-11-00-860-805).

SUBTASK 27-32-00-080-002

(7) Remove the angle of attack probe test fixture, SPL-1917, from the right alpha vane.

SUBTASK 27-32-00-860-016

(8) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	3	C01072	HEATERS ALPHA VANE LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT

K. SERVO TEST

SUBTASK 27-32-00-740-019

(1) Do this servo test for SMYD 1:

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WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Supply the system B hydraulic power (TASK 29-00-00-800-802).

1) Make sure that the FLT CONTROL A and FLT CONTROL B switches, on the P5-3 overhead panel, are set to the ON position.

2) Make sure that the SPOILER A and SPOILER B switches are set to the ON position.

(b) On SMYD 1, push the ON/OFF button.

(c) Push the down arrow button.

(d) Make sure that the display shows: GROUND TESTS?.

(e) Push the YES button.

(f) Push the down arrow button until the display shows: SERVO TEST?.

NOTE: A series of warning messages will show.

(g) Push the YES button.

(h) Make sure that the display shows these messages in this sequence:

- WARNING! WARNING!
- RUDDER MOVEMENT
- RUDDER CLEARED?

(i) Push the YES button.

(j) Make sure that the display shows: SYS B HYD ON?.

(k) Make sure that system B has hydraulic power.

(l) Push the YES button.

NOTE: If NO is pushed, then TURN ON HYD B will show on the display.

(m) Make sure that the display shows: FLT CTRL B ON.

(n) Make sure that the FLT CONTROL B switch is set to the ON position.

(o) Push the YES button.

NOTE: If NO is pushed, then FLT CTRL B TO ON will show on the display.

(p) Make sure that the display shows: YAW DMPR ENGAGED?.

(q) Set the YAW DAMPER switch to the ON position.

(r) Push the YES button.

NOTE: If NO is pushed, then ENGAGE YAW DMPR will show on the display.

(s) Make sure that the display shows: ZERO COMMAND?.

NOTE: During this test the SMYD will initiate a zero rudder command.

(t) Push the YES button.

(u) Make sure that the display shows these messages in this sequence:

NOTE: There will be approximately a 10 second delay between messages.

NOTE: The YD values are 0.0 ± 0.2.

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- RUDDER MOVING
- YD cmd value
- YD LVDT value
- END ZERO COMMAND?

(v) Push the YES button.

(w) Make sure that the display shows these messages in this sequence:

- RUDDER MOVING
- TEST EXITING
- ZERO COMMAND?

(x) Push the down arrow button.

(y) Make sure that the display shows: SWEEP TEST?.

(z) On SMYD 2, do these steps:

1) Push the ON/OFF button.
2) Push the down arrow button until the display shows: OTHER FUNCTNS?.
3) Push the YES button.
4) Push the down arrow button until the display shows: I/O MONITOR?.
5) Push the YES button.
6) Push the down arrow button until the display shows: OTHER DATA?.
7) Push the YES button.
8) Push the down arrow button until the display shows: WORD 151?.
9) Push the YES button.
10) Make sure that the display shows a value in degrees.

NOTE: The value is a number that can be disregarded.

(aa) On SMYD 1, push the YES button.

(ab) Make sure that the display shows these messages in this sequence:

NOTE: There will be approximately a 2 second or a 10 second delay between messages.

NOTE: During this test the SMYD will initiate a left rudder command, then right rudder command and then zero rudder command. If test passes, the message TEST PASSED will show.

1) Monitor the SMYD 2 display during the test.

NOTE: The value shown on the SMYD 2 will change to a number similar to the LVDT value on the SMYD 1.

NOTE: The YD LVDT values are numbers that can be disregarded.

- RUD CMD LEFT
- YD LVDT value
- RUD CMD RIGHT
- YD LVDT value
- RUD CMD ZERO
- YD LVDT value

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- TEST PASSED.

(ac) If the test failed, TEST FAILED will show followed by a message for the first fault.

1) To isolate the fault, refer to the IFIM and do the applicable procedure(s).

(ad) Set the FLT CONTROL B switch to the OFF position.

(ae) Make sure that the YAW DAMPER switch disengages.

(af) Make sure that the YAW DAMPER light, on the P5-3 overhead panel, comes on.

(ag) Set the FLT CONTROL B switch to the ON position.

(ah) Set the YAW DAMPER switch to the ON position.

(ai) Open these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

(aj) Make sure that the YAW DAMPER switch disengages.

(ak) Push the ON/OFF button on SMYD 2 and SMYD 1.

(al) Make sure that the SMYD 1 and SMYD 2 displays show: TURN OFF DISPLAY?.

(am) Push the YES button on SMYD 1 and SMYD 2.

(an) Close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

(ao) Remove the system B hydraulic power (TASK 29-11-00-860-805).

(ap) Set the FLT CONTROL A and B switches to the OFF position.

SUBTASK 27-32-00-740-020

(2) Do this servo test for SMYD 2:

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Supply the system B hydraulic power (TASK 29-00-00-800-802).

(b) Make sure that the FLT CONTROL A and B switches are set to the STBY RUD position.

(c) On SMYD 2, push the ON/OFF button.

(d) Push the down arrow button.

(e) Make sure that the display shows: GROUND TESTS?.

(f) Push the YES button.

(g) Push the down arrow button until the display shows: SERVO TEST?.

NOTE: A series of warning messages will show.

(h) Push the YES button.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Make sure that the display shows these messages in this sequence:
- WARNING! WARNING!
- RUDDER MOVEMENT
- RUDDER CLEARED?

(j) Push the YES button.

(k) Make sure that the display shows: FLT CTRL A STBY?.

(l) Make sure that the FLT CONTROL A switch is set to the STBY RUD position.

(m) Push the YES button.

NOTE: If NO is pushed, then FLT CTRL A TO STBY will show on the display.

(n) Make sure that the display shows: FLT CTRL B STBY?.

(o) Make sure that the FLT CONTROL B switch is set to the STBY RUD position.

(p) Push the YES button.

NOTE: If NO is pushed, then FLT CTRL B TO STBY will show on the display.

(q) Make sure that the display shows: YAW DMPR ENGAGED?.

(r) Set the YAW DAMPER switch to the ON position.

(s) Push the YES button.

NOTE: If NO is pushed, then ENGAGE YAW DMPR will show on the display.

(t) Make sure that the display shows: ZERO COMMAND?.

NOTE: During this test the SMYD will initiate a zero rudder command.

(u) Push the YES button.

(v) Make sure that the display shows these messages in this sequence:

NOTE: There will be approximately a 10 second delay between messages.

NOTE: The YD values are 0.0 ± 0.2.
- RUDDER MOVING
- YD cmd value
- YD LVDT value
- END ZERO COMMAND?

(w) Push the YES button.

(x) Make sure that the display shows these messages in this sequence:
- RUDDER MOVING
- TEST EXITING
- ZERO COMMAND?

(y) Push the down arrow button.

(z) Make sure that the display shows: SWEEP TEST?.

(aa) On SMYD 1, do these steps:
1) Push the ON/OFF button.
2) Push the down arrow button until the display shows: OTHER FUNCTNS?.
3) Push the YES button.
4) Push the down arrow button until the display shows: I/O MONITOR?.

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5) Push the YES button.

6) Push the down arrow button until the display shows: OTHER DATA?.

7) Push the YES button.

8) Push the down arrow button until the display shows: WORD 151?.

9) Push the YES button.

10) Make sure that the display shows a value in degrees.

NOTE: The value is a number that can be disregarded.

(ab) On SMYD 2, push the YES button.

(ac) Make sure that the display shows these messages in this sequence:

NOTE: There will be approximately a 2 second or a 10 second delay between messages.

NOTE: During this test the SMYD will initiate a left rudder command, then right rudder command and then zero rudder command. If test passes, the message TEST PASSED will show.

1) Monitor the SMYD 1 display during the test.

NOTE: The value shown on SMYD 1 will change to a number similar to the LVDT value on SMYD 2.

NOTE: The YD LVDT values are numbers that can be disregarded.

- RUD CMD LEFT
- YD LVDT value
- RUD CMD RIGHT
- YD LVDT value
- RUD CMD ZERO
- YD LVDT value
- TEST PASSED.

(ad) If the test failed, TEST FAILED shows followed by a message for the first fault.

1) To isolate the fault, refer to the IFIM and do the applicable procedure(s).

(ae) Set the FLT CONTROL A switch to the ON position.

(af) Make sure that the YAW DAMPER switch disengages.

(ag) Make sure that the YAW DAMPER light, on the P5-3 overhead panel, comes on.

(ah) Set the FLT CONTROL A switch to the STBY RUD position.

(ai) Set the YAW DAMPER switch to the ON position.

(aj) Open these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

(ak) Make sure that the YAW DAMPER switch disengages.

(al) Push the ON/OFF button on SMYD 2 and SMYD 1.

(am) Make sure that the SMYD 1 and SMYD 2 displays show: TURN OFF DISPLAY?.

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(an) Push the YES button on SMYD 1 and SMYD 2.

(ao) Close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

(ap) Remove the system B hydraulic power (TASK 29-11-00-860-805).

(aq) Set the FLT CONTROL A and B switches to the OFF position.

L. DISPLAY TEST

SUBTASK 27-32-00-740-023

(1) Do these steps on the applicable SMYD:

(a) Push the ON/OFF button.

(b) Push the down arrow button until the display shows: GROUND TESTS?.

(c) Push the YES button.

(d) Push the down arrow button until the display shows: DISPLAY TEST?.

(e) Push the YES button.

(f) Monitor the SMYD display during this test.

(g) Make sure that all the pixels on the SMYD display come on.

(h) Push the ON/OFF button.

M. INPUT SIGNAL INTEGRITY TEST

SUBTASK 27-32-00-740-015

(1) Do these steps on the applicable SMYD:

(a) Push the ON/OFF button.

(b) Push the down arrow button until the display shows: OTHER FUNCTNS?.

(c) Push the YES button.

(d) Push the down arrow button.

(e) Make sure that the display shows: I/O MONITOR?.

(f) Push the YES button.

(g) Make sure that the display shows: ARINC429 INPUTS?.

(h) Push the YES button.

(i) Push the down arrow button until the display shows: CDS DEU BUS?.

(j) Push the YES button.

(k) If the display shows WORD 017?, then do these steps:

1) Push the YES button.

2) Set the speedbrake lever to the FLIGHT DETENT position.

3) Make sure that the display shows a value between -039 and -029.

4) Set the speedbrake lever to the DOWN position.

5) Make sure that the display shows a value between -003 and +003.

(l) Push the MENU button.

(m) Push the down arrow button until the display shows: WORD 352?.

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(n) Push the YES button until the display shows: BIT 27.

(o) Set the WING ANTI-ICE switch, on the P5-11 overhead panel, to the ON position.

(p) Make sure that the display shows: 1.

(q) Set the WING ANTI-ICE switch, on the P5-11 overhead panel, to the OFF position.

(r) Make sure that the display shows: 0.

(s) Push the ON/OFF button.

N. Put the Airplane Back to Its Usual Condition

SUBTASK 27-32-00-860-023

(1) Set the FLT CONTROL A and B switches, on the P5-3 overhead panel, to the ON position.

SUBTASK 27-32-00-410-003

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-32-00-860-014

(3) Set the GND TEST switch on the P61-4 panel, to the NORM position.

——— END OF TASK ———

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STALL WARNING TEST MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the stall warning test module

(2) Installation of the stall warning test module.

TASK 27-32-21-000-801

2. Stall Warning Test Module Removal

(Figure 401)

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-32-21-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	5	C01206	SMYD-2 CMPTR DC

C. Stall Warning Test Module Removal

SUBTASK 27-32-21-020-001

(1) Remove the stall warning test module [1]:

(a) Hold the stall warning test module [1].

(b) Release the quarter turn fasteners [2] which attach the stall warning test module [1] to the aft overhead panel.

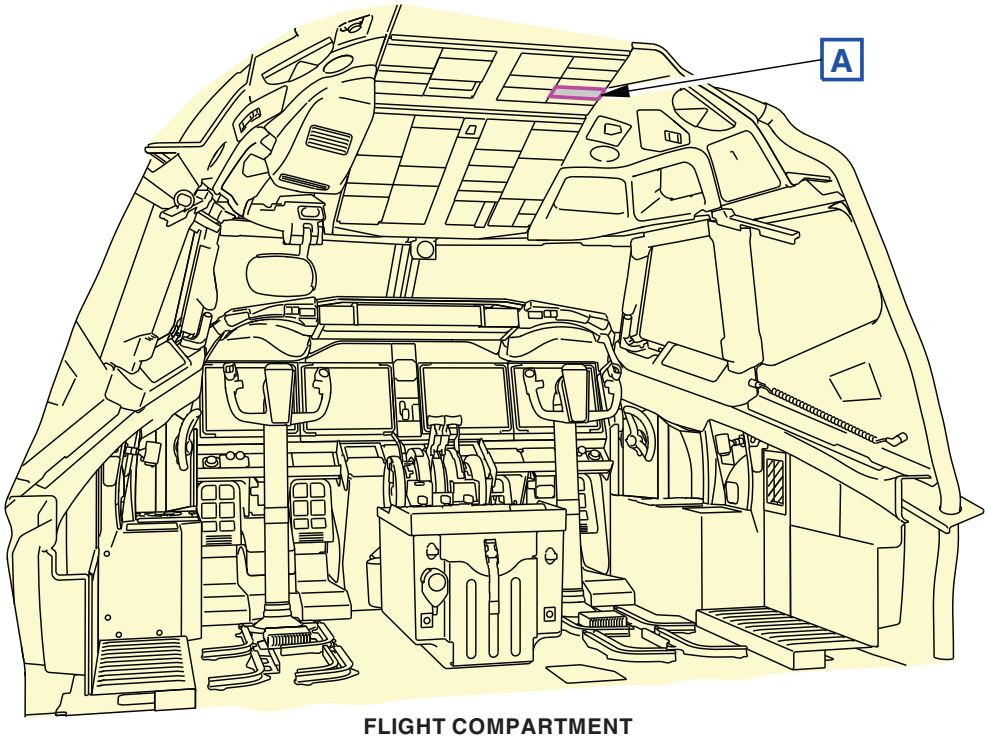
(c) Lower the stall warning test module [1] from the aft overhead panel.

(d) Disconnect the electrical connectors [3] from the back of the stall warning test module [1].

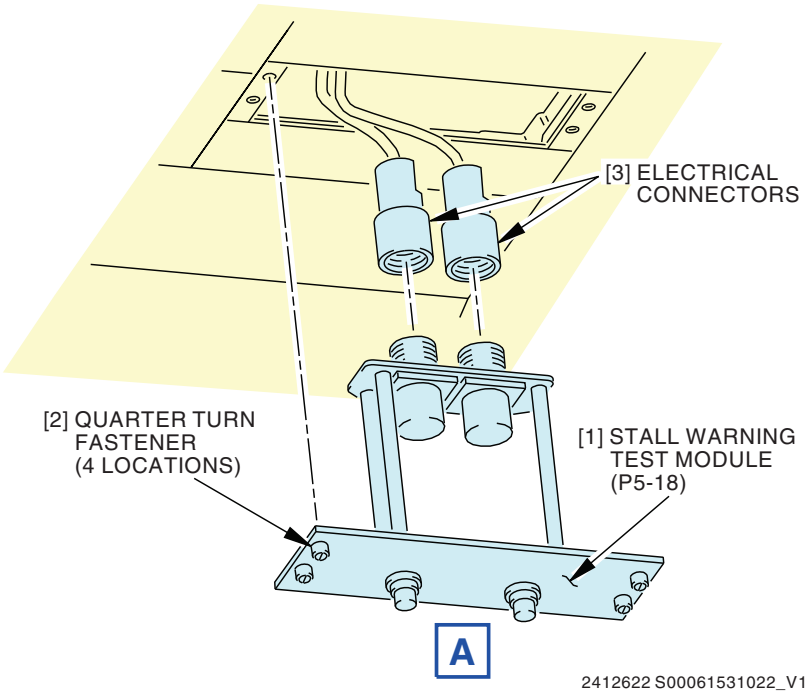
(e) Remove the stall warning test module [1].

——— END OF TASK ———

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FLIGHT COMPARTMENT



Stall Warning Test Module Installation
Figure 401/27-32-21-990-801

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TASK 27-32-21-400-801

3. Stall Warning Test Module Installation

(Figure 401)

A. References

Reference	Title
24-22-00-860-803	Supply External Power (P/B 201)

B. Stall Warning Test Module Installation

SUBTASK 27-32-21-420-001

(1) Connect the electrical connectors [3] to the stall warning test module [1].

SUBTASK 27-32-21-420-002

(2) Install the stall warning test module [1] in the aft overhead panel.

SUBTASK 27-32-21-410-001

(3) Tighten the quarter turn fasteners [2].

SUBTASK 27-32-21-860-002

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	5	C01206	SMYD-2 CMPTR DC

C. Stall Warning Test Module Installation Test

SUBTASK 27-32-21-860-003

(1) Do this task: Supply External Power, TASK 24-22-00-860-803.

SUBTASK 27-32-21-860-004

(2) Push and hold the STALL WARNING TEST NO. 1 switch on the module.

NOTE: The amber SPD LIM light on the on-side Primary Flight Display will come on. A YAW DAMPER light on the P5-3 panel with a related MASTER CAUTION light is possible. A SPD LIM annunciation that stays on more than 5 seconds after the stick shaker operates could be a problem.

(a) Make sure that the captain's control column shaker operates.

SUBTASK 27-32-21-860-005

(3) Release the STALL WARNING TEST NO. 1 switch.

(a) Make sure that the shaker stops.

SUBTASK 27-32-21-860-006

(4) Push and hold the STALL WARNING TEST NO. 2 switch on the module.

NOTE: The amber SPD LIM light on the on-side Primary Flight Display will come on. A YAW DAMPER light on the P5-3 panel with a related MASTER CAUTION light is possible. A SPD LIM annunciation that stays on more than 5 seconds after the stick shaker operates could be a problem.

(a) Make sure that the first officer's control column shaker operates.

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SUBTASK 27-32-21-860-007

(5) Release the STALL WARNING TEST NO. 2 switch.

(a) Make sure that the shaker stops.

——— END OF TASK ———

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CONTROL COLUMN SHAKER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the control column shaker

(2) Installation of the control column shaker.

B. There are two Control Column Shakers.

C. The procedure for the removal and installation of the two Control Column Shakers are the same.

TASK 27-32-31-000-801

2. Control Column Shaker Removal

(Figure 401)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-32-31-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

C. Control Column Shaker Removal

SUBTASK 27-32-31-020-001

(1) Disconnect the electrical connector from the stick shaker [1].

SUBTASK 27-32-31-020-002

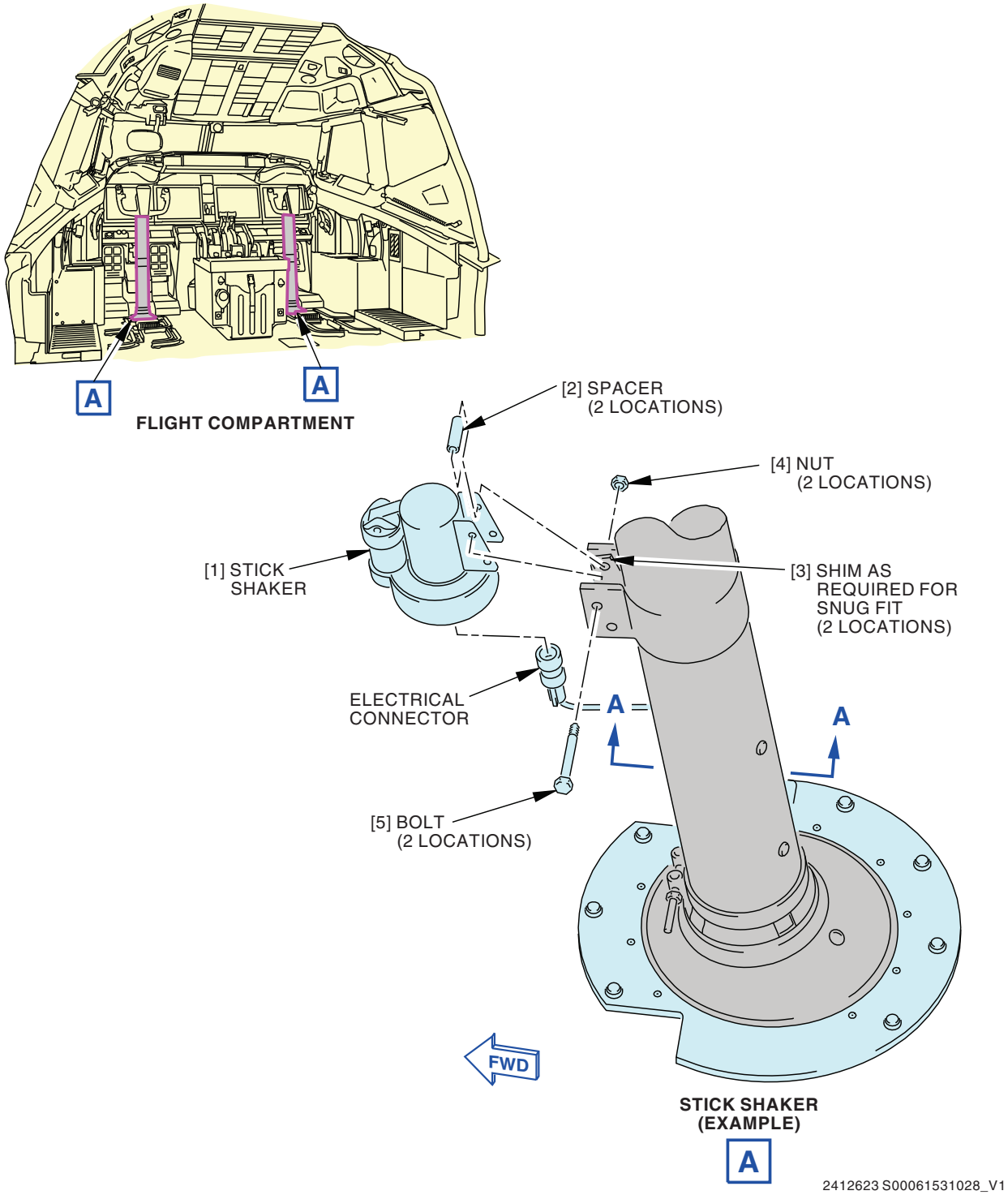
(2) Remove the two nuts [4], two bolts [5], two shims [3], and two spacers [2] which attach the stick shaker [1] to the strap on the control column.

SUBTASK 27-32-31-020-003

(3) Remove the stick shaker [1] from the control column.

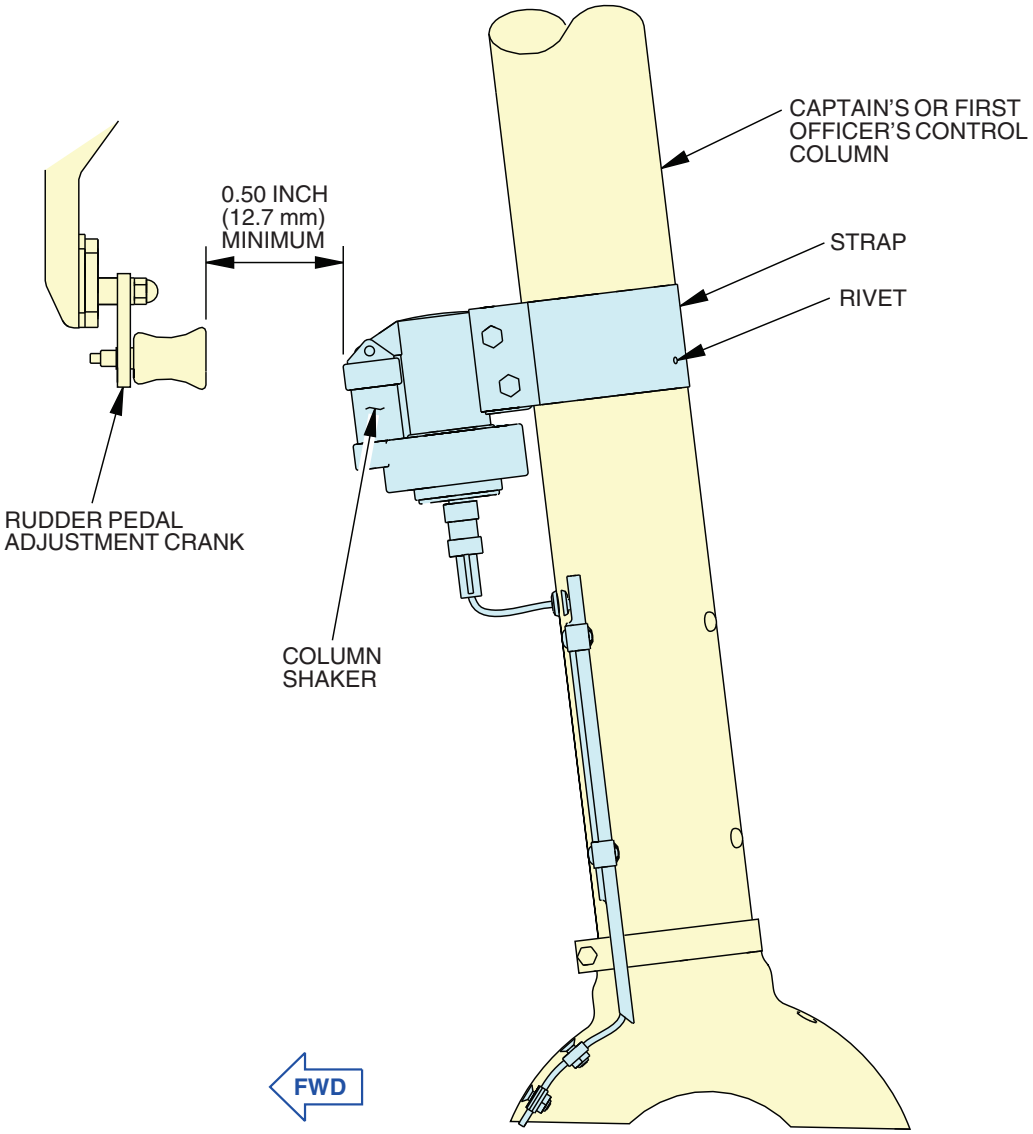
END OF TASK

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Control Column Shaker Installation
Figure 401/27-32-31-990-801 (Sheet 1 of 2)

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A-A

2412624 S00061531029_V3

Control Column Shaker Installation
Figure 401/27-32-31-990-801 (Sheet 2 of 2)

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TASK 27-32-31-400-801

3. Control Column Shaker Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Stick shaker	27-32-31-01-065	PTF ALL

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Control Column Shaker Installation

SUBTASK 27-32-31-420-001

(1) Attach the stick shaker [1] to the strap on the control column with the bolts [5], nuts [4], shims [3], and spacers [2].

(a) Make sure that the strap is attached to the control column with the rivet.

1) If necessary, slide the strap up and down along the control column to make sure that the strap rivet is aligned into the column hole.

(b) Make sure that there is a minimum of 0.5 in. (12.7 mm) clearance between the shaker and rudder pedal adjustment crank when the control column is in the full forward position.

SUBTASK 27-32-31-420-002

(2) Tighten the bolts [5] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 27-32-31-420-003

(3) Add or remove shims [3] as necessary between the stick shaker [1] and strap to get a snug fit.

SUBTASK 27-32-31-420-004

(4) Connect the electrical connector to the stick shaker [1]..

SUBTASK 27-32-31-860-002

(5) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

D. Control Column Shaker Installation Test

SUBTASK 27-32-31-710-001

(1) Do a check of the stick shaker [1]:

(a) Push and hold the STALL WARNING TEST NO. 1 switch (for the Captain's shaker) on the P5-18 panel.

NOTE: The amber SPD LIM light on the on-side Primary Flight Display will come on. A YAW DAMPER light on the P5-3 panel with a related MASTER CAUTION light is possible. A SPD LIM annunciation that stays on more than 5 seconds after the stick shaker operates could be a problem.

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(b) Make sure that the Captain's stick shaker [1] operates.

(c) Release the STALL WARNING TEST NO. 1 switch.

(d) Make sure that the Captain's stick shaker [1] stops.

(e) Push and hold the STALL WARNING TEST NO. 2 switch (for the First Officer's shaker) on the P5-18 panel.

NOTE: The amber SPD LIM light on the on-side Primary Flight Display will come on. A YAW DAMPER light on the P5-3 panel with a related MASTER CAUTION light is possible. A SPD LIM annunciation that stays on more than 5 seconds after the stick shaker operates could be a problem.

(f) Make sure that the First Officer's stick shaker [1] operates.

(g) Release the STALL WARNING TEST NO. 2 switch.

(h) Make sure that the First Officer's stick shaker [1] stops.

END OF TASK

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STALL MANAGEMENT YAW DAMPER (SMYD) - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the Stall Management Yaw Damper (SMYD)

(2) Installation of the Stall Management Yaw Damper (SMYD).

B. There are two Stall Management Yaw Damper (SMYD).

C. The two SMYD are located in the Electrical and Electronics Compartment.

D. The procedure for the removal and installation of the two SMYDs are the same.

TASK 27-32-42-000-801

2. Stall Management Yaw Damper (SMYD) Removal

(Figure 401)

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 27-32-42-860-001

(1) To remove the No. 1 Stall Management Yaw Damper (SMYD), do this step:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	7	C00286	YAW DAMPER 1 DC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT
E	5	C01204	SMYD-1 CMPTR DC
E	6	C01205	SMYD-1 SNSR EXC AC

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-32-42-860-002

(2) To remove the No. 2 SMYD, do this step:

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(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	6	C01354	YAW DAMPER 2 DC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC
B	6	C01393	STICK SHAKER RIGHT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2

SUBTASK 27-32-42-010-001

(3) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Stall Management Yaw Damper (SMYD) Removal

SUBTASK 27-32-42-910-001

CAUTION DO NOT TOUCH THE SMYD BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE SMYD.

(1) Do the necessary steps to protect the SMYD from electrostatic discharge (TASK 20-40-12-000-802).

SUBTASK 27-32-42-020-001

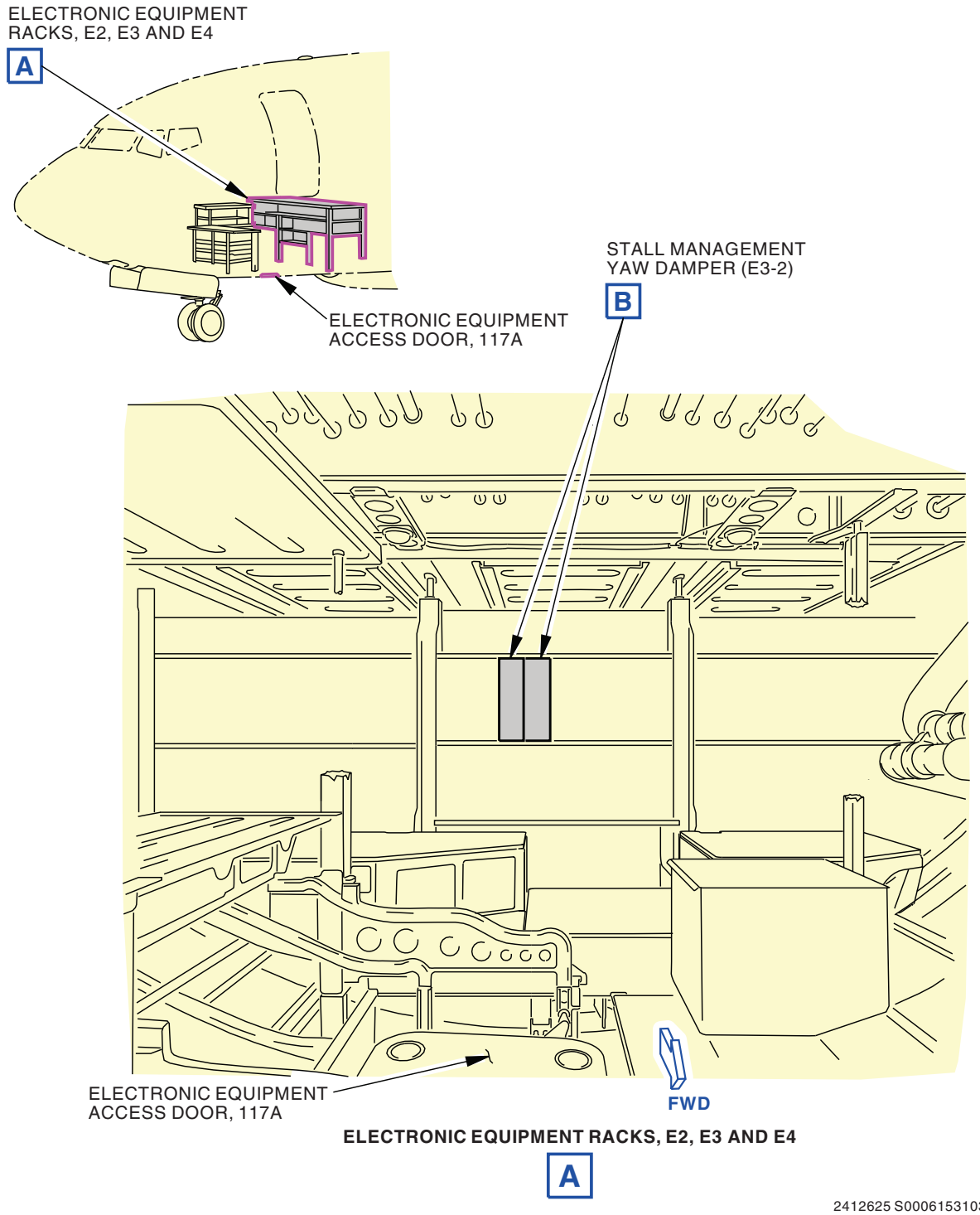
(2) Remove the SMYD (TASK 20-10-07-000-801).

SUBTASK 27-32-42-420-004

(3) Install conductive or antistatic caps on the connectors.

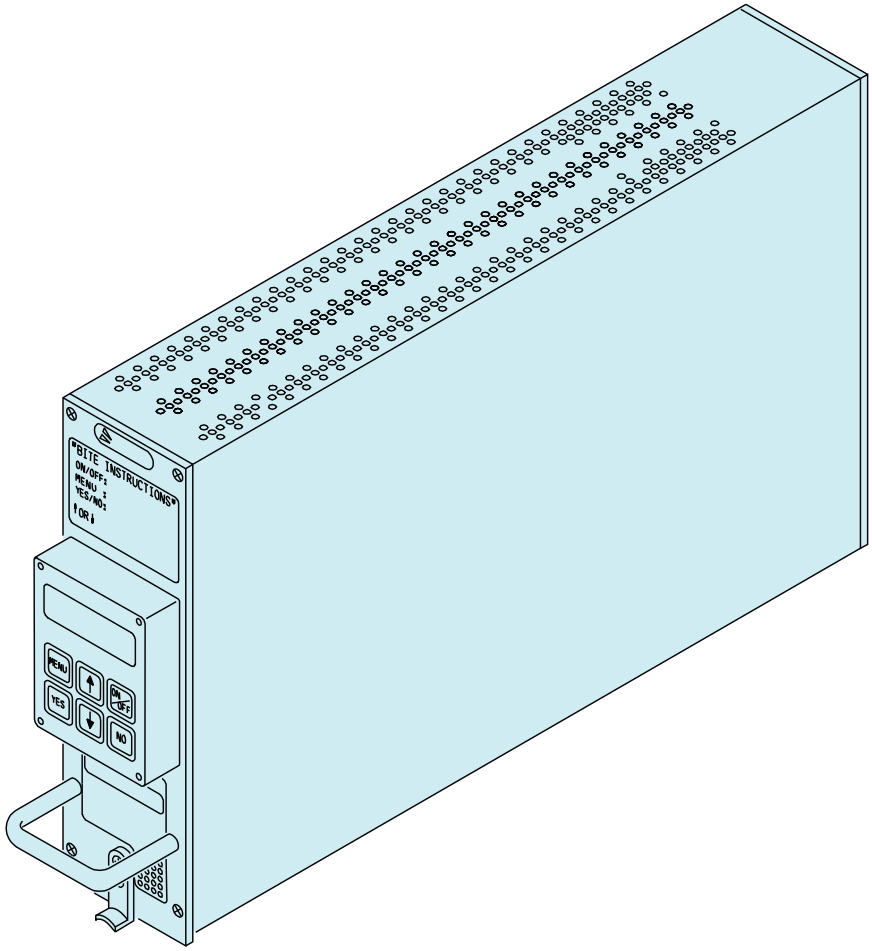
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stall Management Yaw Damper Installation
Figure 401/27-32-42-990-801 (Sheet 1 of 2)

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STALL MANAGEMENT YAW DAMPER (EXAMPLE)

B

2412626 S00061531034_V2

Stall Management Yaw Damper Installation
Figure 401/27-32-42-990-801 (Sheet 2 of 2)

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TASK 27-32-42-400-801

3. Stall Management Yaw Damper (SMYD) Installation

(Figure 401)

A. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
27-32-00-740-803	Stall Management Yaw Damper (SMYD) BITE Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Stall management yaw damper (SMYD)	27-32-42-01-010	PTF ALL

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Stall Management Yaw Damper (SMYD) Installation

SUBTASK 27-32-42-910-002

CAUTION

DO NOT TOUCH THE SMYD BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE SMYD.

(1) Do the necessary steps to protect the stall management yaw damper (SMYD) [1] from electrostatic discharge.

SUBTASK 27-32-42-420-003

(2) Install the stall management yaw damper (SMYD) [1] (TASK 20-10-07-400-801).

F. Stall Management Yaw Damper Module Installation Test

SUBTASK 27-32-42-860-010

(1) For the applicable Stall Management Yaw Damper (SMYD), do this step:

Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR
E	4	C01392	STICK SHAKER LEFT
E	5	C01204	SMYD-1 CMPTR DC
E	6	C01205	SMYD-1 SNSR EXC AC

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F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC
B	6	C01393	STICK SHAKER RIGHT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

SUBTASK 27-32-42-740-016

(2) Do the SELF test and SERVO test for the stall management yaw damper (SMYD) [1] (TASK 27-32-00-740-803).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-32-42-410-001

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HORIZONTAL STABILIZER TRIM CONTROL SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Horizontal Stabilizer Trim Control System - Deactivation

(2) Horizontal Stabilizer Trim Control System - Activation

TASK 27-41-00-800-801

2. Horizontal Stabilizer Trim Control System - Deactivation

A. General

(1) This task will deactivate the Horizontal Stabilizer Trim Control System.

B. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
330	Subzone - Left Horizontal Stabilizer and Elevator
340	Subzone - Right Horizontal Stabilizer and Elevator

E. Procedure

SUBTASK 27-41-00-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-41-00-860-002

(2) Set the STAB TRIM PRI switch, S272, on the aft of the control stand to the CUTOUT position.

SUBTASK 27-41-00-860-003

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-00-860-004

(4) Open the following circuit breaker (if installed):

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 27-41-00-860-005

(5) If necessary, use the stabilizer trim wheel on the control stand to set the horizontal stabilizer at zero degrees (4 units of trim).

SUBTASK 27-41-00-480-001

(6) Install the lock assembly, SPL-1672 on the stabilizer trim wheel at the control stand (Figure 201):

(a) Turn the trim wheel to put the handle at the top of the wheel.

(b) Adjust the height of the trim lock to put the trim wheel handle correctly on the yoke.

(c) Install the pin through the yoke.

(d) Install the safety pin.

F. Horizontal Stabilizer Trim Control System - Tryout

NOTE: This tryout is to make sure that the Horizontal Stabilizer Trim Control System is in zero state.

SUBTASK 27-41-00-210-001

(1) Make sure that the lock assembly, SPL-1672 is installed on the stabilizer trim wheel at the control stand (Figure 201).

SUBTASK 27-41-00-210-002

(2) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-00-210-003

(3) Open the following circuit breaker (if installed):

(a) Make sure that this circuit breaker is open and has safety tag:

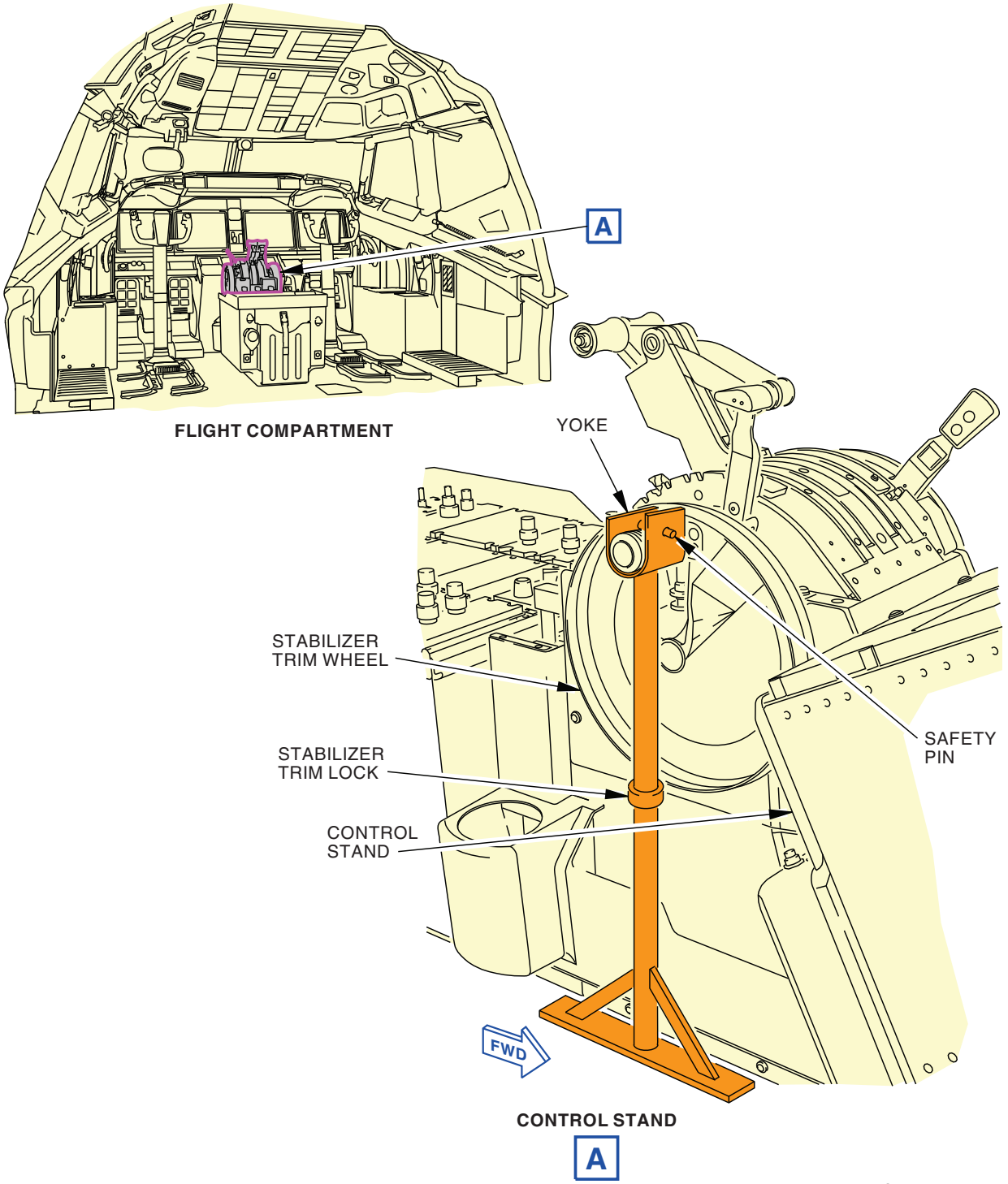
CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

ECCN 9E991 BOEING PROPRIETARY - See title page for details

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Lock Installation
Figure 201/27-41-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-00-800-802

3. Horizontal Stabilizer Trim Control System - Activation

A. General

(1) This task will activate the Horizontal Stabilizer Trim Control System.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
330	Subzone - Left Horizontal Stabilizer and Elevator
340	Subzone - Right Horizontal Stabilizer and Elevator

E. Procedure

SUBTASK 27-41-00-080-001

(1) Remove the lock assembly, SPL-1672 from the stabilizer trim wheel at the control stand (Figure 201).

SUBTASK 27-41-00-860-006

(2) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-41-00-860-007

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-00-860-008

(4) Close the following circuit breaker (if installed):

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(a) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 27-41-00-860-009

(5) Set the STAB TRIM PRI switch, S272, on the aft of the control stand to the NORMAL position.

SUBTASK 27-41-00-210-004

(6) Use the stabilizer trim wheel to make sure that the horizontal stabilizer operates through full travel between the stops.

——— END OF TASK ———

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HORIZONTAL STABILIZER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Stabilizer Control Cable and Chain Adjustment - Use this task to adjust the horizontal stabilizer control cables and chain.

(2) Stabilizer Manual Trim and Trim Indicator Test - Use this test to make sure the horizontal stabilizer moves to the correct position when it is operated from the stabilizer trim wheel. This test also makes sure that the stabilizer trim indicator operates correctly.

(3) Horizontal Stabilizer Trim Control System Friction Test - Use this test to check the force to operate the horizontal stabilizer.

(4) Stabilizer Electric Trim System Test - Use this test to make sure the horizontal stabilizer moves at the correct speed when it is operated from the STAB TRIM switches.

(5) Stabilizer Trim Wheel Free Movement Test - Use this test to test the amount of horizontal stabilizer movement after the STAB TRIM switches are released.

(6) Stabilizer Trim Limit Switches Test - Use this test to make sure the stabilizer trim limit switches operate correctly.

(7) Stabilizer Clutch Friction System Test - Use this test to make sure that the correct force will disconnect the manual clutch.

(8) Column Actuated Stabilizer Trim Cutout Switch Override Test - Use this test to make sure the horizontal stabilizer follows the commands from the STAB TRIM switches, even though the column actuated STAB TRIM cutout switches give the opposite command.

(9) Stabilizer Trim Secondary Brake System Test - Use this test to make sure the stabilizer secondary brake can prevent movement of the horizontal stabilizer.

(10) Stabilizer Trim Cutout Test - Use this procedure to make sure the stabilizer trim cutout switch operates correctly. The stabilizer trim cutout switch stops the horizontal stabilizer when the control column command is opposite from the STAB TRIM switches.

(11) Column Cutout Switches Test - Use this test to make sure the captain's and first officer's column cutout switches operate correctly.

TASK 27-41-00-820-801

2. Stabilizer Control Cable and Chain - Adjustment

(Figure 501 and Figure 502)

A. General

(1) Use this task to adjust the horizontal stabilizer control cables and chain.

(2) Before the cable adjustment is done, permit a minimum of one hour at a constant ambient temperature ± 5°F (± 2.8°C) for the airframe temperature to become stable.

(a) It is recommended to do the cable adjustments in a controlled environment such as a hangar.

NOTE: Direct sunlight on the airframe will cause the cable readings to be inaccurate.

(b) All temperature readings will be taken as ambient outside but adjacent to the applicable area of the airframe.

(c) The cable tension values will not be correct when there are temperature differences along the cable run.

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(d) It is recommended that airplanes supplied with the cables that contain higher zinc coating (Grade D or Grade K) be re-tensioned 3 to 4 months into service. This will bring the tension of the control cables within the desired range. After the cables are re-tensioned, the high zinc cables (Grade D or Grade K) will continue to relax at a rate that is equivalent to the Grade B cables. To identify the grade of the cable, refer to the table below:

Table 501/27-41-00-993-819 Control Cable Grades and Diameters

CONTROL CABLE DIAMETER (TERMINAL SIZE REF)	CONSTRUCTION FIRST DASH NUMBER *[1]									
	GRADE A					GRADE B		GRADE D		GRADE K
	ZINC			TIN OVER ZINC		ZINC	TIN OVER ZINC	ZINC	TIN OVER ZINC	ZINC
	7 X 7	7 X 7 (RT)	7 X 19	7 X 7	7 X 7 (RT)	7 X 19	7 X 19	7 X 19	7 X 19	7 X 19
.094 (3/32)	3B	3G*[2]	3A	3D	3E *[2]	- — —	- — —	- — —	- — —	- — —
.125 (1/8)	- — —	- — —	4A	- — —	- — —	4F	4C	4J*[3]	4H*[3]	4K*[3]
.188 (3/16)	- — —	- — —	- — —	- — —	- — —	- — —	- — —	6J*[4]	- — —	- — —

*[1] First dash number in the part number of the cable. Example: BACC2C-3D-01085-CE is Grade A, Tin Over Zinc, 7 x 7.

*[2] Code 3E and 3G are specified only where there is a specific requirement for REDUCED TOLERANCE (RT) percent elongation wire rope, 7 X 7 RT, per BMS7-265.

*[3] Only Grade D and Grade K .125 (1/8) cables (4J, 4H and 4K) are high zinc cables.

*[4] It is not necessary to adjust the Grade D 0.188 (3/16) cables (6J) 3 to 4 months after installation.

(e) It is recommended that cable assemblies on each side of a cable loop be replaced at the same time using the same cable grade material, in order to minimize asymmetric relaxation.

(3) Before the cable adjustment is done, allow the tensiometer to stabilize to the same temperature as the airplane.

(4) If the cables need to be replaced, do this task: Stabilizer Control Cable Removal, TASK 27-41-82-000-801.

B. References

Reference	Title
27-41-82-000-801	Stabilizer Control Cable Removal (P/B 401)
SOPM 20-50-16	Assembly of Swaged Cables

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
COM-1554	Tensiometer - Cable, High Tension (200 lbs and above) Part #: 102-03108 Supplier: 21844 Part #: 102-03120 Supplier: 21844 Part #: ACM-300 Supplier: 13331 Part #: ACM-600 Supplier: 13331 Part #: ACX-500-1 Supplier: 13331 Part #: T5-8008-106-00 Supplier: 0N8U4 Part #: T60-1001-C9-1A Supplier: 0N8U4
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-10319	Wrench - Pinion Gear (Rotate) Part #: F71252 Supplier: 81205 Part #: F71267 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

F. Access Panels

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door

G. Prepare for the Adjustment

SUBTASK 27-41-00-010-001

(1) Open these access panels:

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door

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SUBTASK 27-41-00-020-001

WARNING DO NOT OPERATE THE STABILIZER ELECTRICALLY UNTIL YOU ADJUST THE STABILIZER TRIM LIMIT SWITCHES, CABLES, AND CHAIN. THE HORIZONTAL STABILIZER CAN MOVE QUICKLY WHEN YOU OPERATE THE STABILIZER TRIM LIMIT SWITCHES. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Disconnect the flexible shaft [8] from the stabilizer forward control mechanism.

(a) Remove the lockwire from the coupling nut of the flexible shaft [8].

SUBTASK 27-41-00-020-002

(3) Remove the pins [14] from the connector link [13] to disconnect the chain [10].

SUBTASK 27-41-00-020-003

(4) To remove the upper turnbuckle [2], do these steps:

(a) Remove the lockwire from the upper turnbuckle [2].

(b) Remove the nut [6], washer [5], washer [4], and bolt [3].

(c) Remove the nut [17], washer [19], bushing [21], washer [22], washer [23], washer [24], and bolt [25].

(d) Remove the upper turnbuckle [2].

SUBTASK 27-41-00-820-001

(5) If the new cables are installed, adjust the lower turnbuckle [9] on the stabilizer forward mechanism to its maximum length 0.50 in. (12.7 mm) minimum thread engagement on each rod end.

NOTE: This will move the forward drum aft.

H. Stabilizer Control Cable and Chain Adjustment

SUBTASK 27-41-00-710-001

(1) Adjust the tension of the cable STA and cable STB.

NOTE: It is necessary to use the pinion gear wrench, SPL-10319, to operate the stabilizer system.

NOTE: It is necessary to use the control wheel switches to move the stabilizer, since the stab trim control wheel chain has been disconnected.

(a) If the new cables are installed, do these steps to stretch the cables:

NOTE: If the tension is adjusted on existing cables (cables that were already on the aircraft), do not need to do these steps. Go to the steps to adjust the cable STA and cable STB to their correct rigging tension.

1) Adjust the lower turnbuckle [9] to get the cable tension of 350 ±50 lbf (1557 ±223 N).

a) Operate the stabilizer system through a full cycle.

b) Put the cables at the center of the forward cable drum.

c) Tighten the lower turnbuckle [9].

d) Use the high tension cable tensiometer, COM-1554, to make sure that the cable tension is at 350 ±50 lbf (1557 ±223 N).

e) Operate the stabilizer system through a full cycle.

f) Put the cables at the center of the forward cable drum.

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g) Make sure that the cable tension stays at 350 ±50 lbf (1557 ±223 N).

h) Operate the stabilizer system through 5 full cycles.

i) Make sure that the cable tension stays at 350 ±50 lbf (1557 ±223 N).

(b) Do these steps to adjust cable STA and cable STB to the correct rigging tension:

1) Operate the stabilizer system through a full cycle.

2) Put the cables at the center of the forward cable drum.

3) Adjust the lower turnbuckle [9] to get the cable rigging tension (Table 502).

NOTE: For new cables, adjust the tensions close to the maximum values in the table. The cables will stretch over a period of time.

For cables that were already on the aircraft, adjust the tensions close to the minimum values in the table.

The minimum (low) cable tension listed in the table below applies only during maintenance action and must not be construed as an in-service operational limit for cable tension. There are no operational limits for cables that are in service.

Table 502/27-41-00-993-820 Cable Rigging Tension Values

Temperature	Low Tolerance	High Tolerance
110°F (43.3°C)	137 lb (62 kg)	187 lb (85 kg)
109°F (42.8°C)	136 lb (62 kg)	186 lb (84 kg)
108°F (42.2°C)	136 lb (62 kg)	186 lb (84 kg)
107°F (41.7°C)	136 lb (62 kg)	186 lb (84 kg)
106°F (41.1°C)	135 lb (61 kg)	185 lb (84 kg)
105°F (40.6°C)	135 lb (61 kg)	185 lb (84 kg)
104°F (40.0°C)	134 lb (61 kg)	184 lb (83 kg)
103°F (39.4°C)	134 lb (61 kg)	184 lb (83 kg)
102°F (38.9°C)	134 lb (61 kg)	184 lb (83 kg)
101°F (38.3°C)	133 lb (60 kg)	183 lb (83 kg)
100°F (37.8°C)	133 lb (60 kg)	183 lb (83 kg)
99°F (37.2°C)	132 lb (60 kg)	182 lb (83 kg)
98°F (36.7°C)	132 lb (60 kg)	182 lb (83 kg)
97°F (36.1°C)	131 lb (59 kg)	181 lb (82 kg)
96°F (35.6°C)	131 lb (59 kg)	181 lb (82 kg)
95°F (35.0°C)	131 lb (59 kg)	181 lb (82 kg)
94°F (34.4°C)	130 lb (59 kg)	180 lb (82 kg)
93°F (33.9°C)	130 lb (59 kg)	180 lb (82 kg)
92°F (33.3°C)	129 lb (59 kg)	179 lb (81 kg)
91°F (32.8°C)	129 lb (59 kg)	179 lb (81 kg)
90°F (32.2°C)	128 lb (58 kg)	178 lb (81 kg)

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Table 502/27-41-00-993-820 Cable Rigging Tension Values (Continued)

Temperature	Low Tolerance	High Tolerance
89°F (31.7°C)	128 lb (58 kg)	178 lb (81 kg)
88°F (31.1°C)	128 lb (58 kg)	178 lb (81 kg)
87°F (30.6°C)	127 lb (58 kg)	177 lb (80 kg)
86°F (30.0°C)	127 lb (58 kg)	177 lb (80 kg)
85°F (29.4°C)	126 lb (57 kg)	176 lb (80 kg)
84°F (28.9°C)	126 lb (57 kg)	176 lb (80 kg)
83°F (28.3°C)	126 lb (57 kg)	176 lb (80 kg)
82°F (27.8°C)	125 lb (57 kg)	175 lb (79 kg)
81°F (27.2°C)	125 lb (57 kg)	175 lb (79 kg)
80°F (26.7°C)	124 lb (56 kg)	174 lb (79 kg)
79°F (26.1°C)	124 lb (56 kg)	174 lb (79 kg)
78°F (25.6°C)	123 lb (56 kg)	173 lb (78 kg)
77°F (25.0°C)	123 lb (56 kg)	173 lb (78 kg)
76°F (24.4°C)	123 lb (56 kg)	173 lb (78 kg)
75°F (23.9°C)	122 lb (55 kg)	172 lb (78 kg)
74°F (23.3°C)	122 lb (55 kg)	172 lb (78 kg)
73°F (22.8°C)	121 lb (55 kg)	171 lb (78 kg)
72°F (22.2°C)	121 lb (55 kg)	171 lb (78 kg)
71°F (21.7°C)	120 lb (54 kg)	170 lb (77 kg)
70°F (21.1°C)	120 lb (54 kg)	170 lb (77 kg)
69°F (20.6°C)	120 lb (54 kg)	170 lb (77 kg)
68°F (20.0°C)	119 lb (54 kg)	169 lb (77 kg)
67°F (19.4°C)	119 lb (54 kg)	169 lb (77 kg)
66°F (18.9°C)	118 lb (54 kg)	168 lb (76 kg)
65°F (18.3°C)	118 lb (54 kg)	168 lb (76 kg)
64°F (17.8°C)	118 lb (54 kg)	168 lb (76 kg)
63°F (17.2°C)	117 lb (53 kg)	167 lb (76 kg)
62°F (16.7°C)	117 lb (53 kg)	167 lb (76 kg)
61°F (16.1°C)	116 lb (53 kg)	166 lb (75 kg)
60°F (15.6°C)	116 lb (53 kg)	166 lb (75 kg)
59°F (15.0°C)	115 lb (52 kg)	165 lb (75 kg)
58°F (14.4°C)	115 lb (52 kg)	165 lb (75 kg)
57°F (13.9°C)	115 lb (52 kg)	165 lb (75 kg)
56°F (13.3°C)	114 lb (52 kg)	164 lb (74 kg)

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Table 502/27-41-00-993-820 Cable Rigging Tension Values (Continued)

Temperature	Low Tolerance	High Tolerance
55°F (12.8°C)	114 lb (52 kg)	164 lb (74 kg)
54°F (12.2°C)	113 lb (51 kg)	163 lb (74 kg)
53°F (11.7°C)	113 lb (51 kg)	163 lb (74 kg)
52°F (11.1°C)	112 lb (51 kg)	162 lb (73 kg)
51°F (10.6°C)	112 lb (51 kg)	162 lb (73 kg)
50°F (10.0°C)	112 lb (51 kg)	162 lb (73 kg)
49°F (9.4°C)	111 lb (50 kg)	161 lb (73 kg)
48°F (8.9°C)	111 lb (50 kg)	161 lb (73 kg)
47°F (8.3°C)	110 lb (50 kg)	160 lb (73 kg)
46°F (7.8°C)	110 lb (50 kg)	160 lb (73 kg)
45°F (7.2°C)	109 lb (49 kg)	159 lb (72 kg)
44°F (6.7°C)	109 lb (49 kg)	159 lb (72 kg)
43°F (6.1°C)	109 lb (49 kg)	159 lb (72 kg)
42°F (5.6°C)	108 lb (49 kg)	158 lb (72 kg)
41°F (5.0°C)	108 lb (49 kg)	158 lb (72 kg)
40°F (4.4°C)	107 lb (49 kg)	157 lb (71 kg)
39°F (3.9°C)	107 lb (49 kg)	157 lb (71 kg)
38°F (3.3°C)	107 lb (49 kg)	157 lb (71 kg)
37°F (2.8°C)	106 lb (48 kg)	156 lb (71 kg)
36°F (2.2°C)	106 lb (48 kg)	156 lb (71 kg)
35°F (1.7°C)	105 lb (48 kg)	155 lb (70 kg)
34°F (1.1°C)	105 lb (48 kg)	155 lb (70 kg)
33°F (0.6°C)	104 lb (47 kg)	154 lb (70 kg)
32°F (0.0°C)	104 lb (47 kg)	154 lb (70 kg)
31°F (-0.6°C)	104 lb (47 kg)	154 lb (70 kg)
30°F (-1.1°C)	103 lb (47 kg)	153 lb (69 kg)
29°F (-1.7°C)	103 lb (47 kg)	153 lb (69 kg)
28°F (-2.2°C)	102 lb (46 kg)	152 lb (69 kg)
27°F (-2.8°C)	102 lb (46 kg)	152 lb (69 kg)
26°F (-3.3°C)	101 lb (46 kg)	151 lb (68 kg)
25°F (-3.9°C)	101 lb (46 kg)	151 lb (68 kg)
24°F (-4.4°C)	101 lb (46 kg)	151 lb (68 kg)
23°F (-5.0°C)	100 lb (45 kg)	150 lb (68 kg)
22°F (-5.6°C)	100 lb (45 kg)	150 lb (68 kg)

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Table 502/27-41-00-993-820 Cable Rigging Tension Values (Continued)

Temperature	Low Tolerance	High Tolerance
21°F (-6.1°C)	99 lb (45 kg)	149 lb (68 kg)
20°F (-6.7°C)	99 lb (45 kg)	149 lb (68 kg)
19°F (-7.2°C)	99 lb (45 kg)	149 lb (68 kg)
18°F (-7.8°C)	98 lb (44 kg)	148 lb (67 kg)
17°F (-8.3°C)	98 lb (44 kg)	148 lb (67 kg)
16°F (-8.9°C)	97 lb (44 kg)	147 lb (67 kg)
15°F (-9.4°C)	97 lb (44 kg)	147 lb (67 kg)
14°F (-10.0°C)	96 lb (44 kg)	146 lb (66 kg)
13°F (-10.6°C)	96 lb (44 kg)	146 lb (66 kg)
12°F (-11.1°C)	96 lb (44 kg)	146 lb (66 kg)
11°F (-11.7°C)	95 lb (43 kg)	145 lb (66 kg)
10°F (-12.2°C)	95 lb (43 kg)	145 lb (66 kg)
9°F (-12.8°C)	94 lb (43 kg)	144 lb (65 kg)
8°F (-13.3°C)	94 lb (43 kg)	144 lb (65 kg)
7°F (-13.9°C)	93 lb (42 kg)	143 lb (65 kg)
6°F (-14.4°C)	93 lb (42 kg)	143 lb (65 kg)
5°F (-15.0°C)	93 lb (42 kg)	143 lb (65 kg)
4°F (-15.6°C)	92 lb (42 kg)	142 lb (64 kg)
3°F (-16.1°C)	92 lb (42 kg)	142 lb (64 kg)
2°F (-16.7°C)	91 lb (41 kg)	141 lb (64 kg)
1°F (-17.2°C)	91 lb (41 kg)	141 lb (64 kg)
0°F (-17.8°C)	91 lb (41 kg)	141 lb (64 kg)
-1°F (-18.3°C)	90 lb (41 kg)	140 lb (64 kg)
-2°F (-18.9°C)	90 lb (41 kg)	140 lb (64 kg)
-3°F (-19.4°C)	89 lb (40 kg)	139 lb (63 kg)
-4°F (-20.0°C)	89 lb (40 kg)	139 lb (63 kg)
-5°F (-20.6°C)	88 lb (40 kg)	138 lb (63 kg)
-6°F (-21.1°C)	88 lb (40 kg)	138 lb (63 kg)
-7°F (-21.7°C)	88 lb (40 kg)	138 lb (63 kg)
-8°F (-22.2°C)	87 lb (39 kg)	137 lb (62 kg)
-9°F (-22.8°C)	87 lb (39 kg)	137 lb (62 kg)
-10°F (-23.3°C)	86 lb (39 kg)	136 lb (62 kg)
-11°F (-23.9°C)	86 lb (39 kg)	136 lb (62 kg)
-12°F (-24.4°C)	85 lb (39 kg)	135 lb (61 kg)

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Table 502/27-41-00-993-820 Cable Rigging Tension Values (Continued)

Temperature	Low Tolerance	High Tolerance
-13°F (-25.0°C)	85 lb (39 kg)	135 lb (61 kg)
-14°F (-25.6°C)	85 lb (39 kg)	135 lb (61 kg)
-15°F (-26.1°C)	84 lb (38 kg)	134 lb (61 kg)
-16°F (-26.7°C)	84 lb (38 kg)	134 lb (61 kg)
-17°F (-27.2°C)	83 lb (38 kg)	133 lb (60 kg)
-18°F (-27.8°C)	83 lb (38 kg)	133 lb (60 kg)
-19°F (-28.3°C)	82 lb (37 kg)	132 lb (60 kg)
-20°F (-28.9°C)	82 lb (37 kg)	132 lb (60 kg)
-21°F (-29.4°C)	82 lb (37 kg)	132 lb (60 kg)
-22°F (-30.0°C)	81 lb (37 kg)	131 lb (59 kg)
-23°F (-30.6°C)	81 lb (37 kg)	131 lb (59 kg)
-24°F (-31.1°C)	80 lb (36 kg)	130 lb (59 kg)
-25°F (-31.7°C)	80 lb (36 kg)	130 lb (59 kg)
-26°F (-32.2°C)	80 lb (36 kg)	130 lb (59 kg)
-27°F (-32.8°C)	79 lb (36 kg)	129 lb (59 kg)
-28°F (-33.3°C)	79 lb (36 kg)	129 lb (59 kg)
-29°F (-33.9°C)	78 lb (35 kg)	128 lb (58 kg)
-30°F (-34.4°C)	78 lb (35 kg)	128 lb (58 kg)
-31°F (-35.0°C)	77 lb (35 kg)	127 lb (58 kg)
-32°F (-35.6°C)	77 lb (35 kg)	127 lb (58 kg)
-33°F (-36.1°C)	77 lb (35 kg)	127 lb (58 kg)
-34°F (-36.7°C)	76 lb (34 kg)	126 lb (57 kg)
-35°F (-37.2°C)	76 lb (34 kg)	126 lb (57 kg)
-36°F (-37.8°C)	75 lb (34 kg)	125 lb (57 kg)
-37°F (-38.3°C)	75 lb (34 kg)	125 lb (57 kg)
-38°F (-38.9°C)	74 lb (34 kg)	124 lb (56 kg)
-39°F (-39.4°C)	74 lb (34 kg)	124 lb (56 kg)
-40°F (-40.0°C)	74 lb (34 kg)	124 lb (56 kg)

4) Operate the stabilizer system through 5 full cycles.

5) Put the cables at the center of the forward cable drum.

6) Do the following steps to make sure that the stabilizer trim cable tension is correct:

   a) Use the high tension cable tensiometer, COM-1554, or low tension cable tensiometer, COM-1553, to make sure that the cable tension is the same as the tension in the Table 502.

   b) If the cable tension is not correct, adjust the lower turnbuckle [9] to get the correct cable rigging tension (Table 502).

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7) Make sure that the lower turnbuckle [9] length, “C” dimension, is 11.00 in. (279.4 mm) - 15.25 in. (387.3 mm) after the adjustment.

NOTE: Boeing recommend that you adjust the lower turnbuckle length, the “C” dimension, to between 12.25 in. (311.2 mm) - 15.25 in. (387.3 mm). This is because the tension of the cable will decrease in-service.

a) If the “C” dimension value is not within limits, replace or shorten the cables (SOPM 20-50-16).

8) Make sure that the horizontal stabilizer moves freely through the full travel to the airplane nose down limit.

SUBTASK 27-41-00-020-005

(2) Connect the chain [10].

(a) Install the chain [10] over the two idler sprockets [11] and forward mechanism sprocket [12].

(b) Connect the chain [10] at the connector link [13].

(c) Install the pins [14] on the outboard side of the chain [10].

SUBTASK 27-41-00-820-002

(3) Adjust the chain [10] tension.

(a) Apply a 5 lbf (22 N) force at approximately 4.5 ±1.0 in. (114.3 ±25.4 mm) from where the chain [10] touches the forward mechanism sprocket [12].

(b) Make sure that the movement of the chain [10] is 0.25 ±0.07 in. (6.35 ±1.78 mm).

NOTE: Boeing recommend that you adjust the movement of the chain to the lower movement limit between 0.18 in. (4.57 mm) - 0.20 in. (5.08 mm). This is because the tension of the chain will decrease in-service.

(c) Do a check of the chain [10] tension.

1) Turn the stabilizer trim wheel approximately 8 turns in the airplane nose up direction.

a) Measure the chain [10] tension.

b) If necessary, adjust the chain [10] tension again.

2) Turn the stabilizer trim wheel approximately 8 turns in the airplane nose down direction.

a) Measure the chain [10] tension.

b) If necessary, adjust the chain [10] tension again.

3) Repeat this check until the chain [10] rigging tension is maintained.

(d) If it is necessary, do these steps to get the correct chain [10] movement:

1) Adjust the upper adjusting nuts [15] and upper adjusting nuts [18].

2) Measure the adjustment, Option 1.

a) Make sure that all three “A” dimensions equal ± 0.06 in. (1.52 mm) from each other.

<1> Measure the “A” dimensions of left and right aft support link [7].

NOTE: The “A” dimension is the distance of the aft support link between the center of the two connection bolts.

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<2> Measure the “A” dimension of the forward support link [1].

NOTE: The “A” dimension is the distance of the forward support link between the center of the two connection bolts.

<3> Make sure that all three “A” dimensions are ± 0.06 in. (1.52 mm) from each other.

3) Measure the adjustment, Option 2.

a) Make sure that all three “X, Y, and Z” dimensions are equal ± 0.12 in. (3.05 mm) from each other.

4) Install MS20995NC32 lockwire, G01912.

SUBTASK 27-41-00-420-001

(4) Connect the flexible shaft [8] to the forward mechanism.

CAUTION

MAKE SURE THAT YOU ARE CAREFUL WITH THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE SHARP EDGES ON THE TRAMMEL BAR, SPL-1677 CAN CAUSE CHAFING OR DAMAGE TO THE WIRES.

(a) Use the trammel bar, SPL-1677, to set the "B" dimension to the 39.17 ±0.03 in. (994.9 ±0.8 mm).

(b) Turn the flexible shaft [8] until the stabilizer position indicator is at 4 units of trim.

(c) Connect the flexible shaft [8] to the forward mechanism.

1) Install MS20995NC32 lockwire, G50347, to the coupling nut of the flexible shaft [8] with the double twist method.

SUBTASK 27-41-00-420-002

(5) Install the upper turnbuckle [2] to the forward mechanism.

(a) Install the bolt [25], washer [24], washer [23], washer [22], bushing [21], washer [19], and nut [17].

(b) Adjust the upper turnbuckle [2] until you can install the bolt [3] freely through the rod end at the forward mechanism.

(c) Install the washer [4], washer [5], nut [6], and bolt [3].

SUBTASK 27-41-00-710-014

(6) Operate the stabilizer system to the NOSE DOWN stop (stabilizer leading edge moves up) and do a check for the conditions that follow (Figure 502):

NOTE: A removal of the floor panels is not necessary. It is necessary to examine the exposed parts of the cable run for the stabilizer control cables in the forward equipment compartment and aft unpressurized compartment.

(a) The clearance between the cables and cable guard is not less than 0.25 in. (6.35 mm) for the upper position (View E, Figure 501).

NOTE: The cable guard is installed at station 237 in the bulkhead.

1) If it is necessary, adjust the cable guard.

(b) The clearance between the cables and wire raceway on top of the nose wheel well is not less than 0.35 in. (8.9 mm).

(c) The cables do not touch the pulley flanges.

(d) The cables are in ±2 degrees of the plane of the pulleys.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) The clearance between the aileron control cable and the chain at the forward mechanism sprocket [12] is not less than 0.75 in. (19.0 mm).

(f) The pulleys turn freely with no interference from the cable guards.

(g) The cables are installed through the grommets and the air pressure seals.

(h) The cables do not touch the cable guards, rub strips, or grommets.

(i) The clearance between the cables and adjacent structure is not less than 0.20 in. (5.1 mm).

SUBTASK 27-41-00-710-002

(7) Operate the stabilizer system to the NOSE UP stop (stabilizer leading edge moves down) and do a check for the conditions that follow (Figure 502):

NOTE: A removal of the floor panels is not necessary. It is necessary to examine the exposed parts of the cable run for the stabilizer control cables in the forward equipment compartment and aft unpressurized compartment.

(a) Make sure that the two STAB TRIM indicators on the control stand show 16.4±0.5 units of trim.

NOTE: It is necessary to use manual trim to position the stabilizer ball nut against the Aircraft-Nose-Up (ANU) stop limit, to get the value of 16.4±0.5 units trim.

(b) Make sure that the difference between the two STAB TRIM indicators is less than 0.5 units of trim.

(c) The clearance between the cables and cable guard is not less than 0.25 in. (6.4 mm) for the lower position (View E, Figure 501).

NOTE: The cable guard is installed at station 237 in the bulkhead.

1) If it is necessary, adjust the cable guard.

(d) The clearance between the cables and wire raceway on top of the nose wheel well is not less than 0.35 in. (8.9 mm).

(e) The cables do not touch the pulley flanges.

(f) The cables are in ±2 degrees of the plane of the pulleys.

(g) The clearance between the aileron control cable and the chain at the forward mechanism sprocket [12] is not less than 0.75 in. (19.0 mm).

(h) The pulleys turn freely with no interference from the cable guards.

(i) The cables are installed through the grommets and the air pressure seals.

(j) The cables do not touch the cable guards, rub strips, or grommets.

(k) The clearance between the cables and adjacent structure is not less than 0.20 in. (5.1 mm).

SUBTASK 27-41-00-420-003

(8) Install MS20995NC32 lockwires, G01912, to the upper turnbuckle [2] and lower turnbuckle [9] with the double twist method.

NOTE: The lockwire must be attached the tab of the washer of the upper turnbuckle.

SUBTASK 27-41-00-410-001

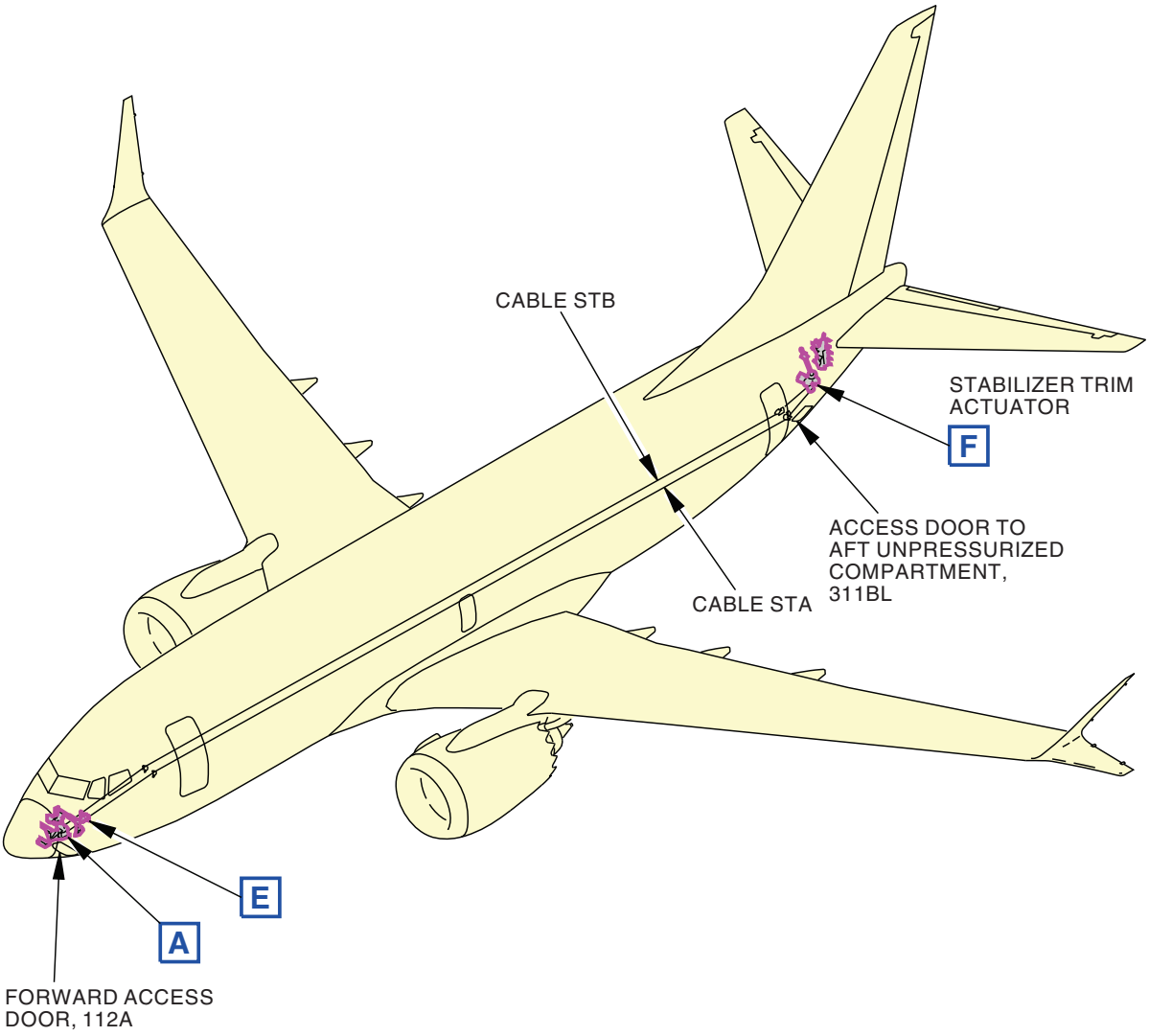
(9) Close these access panels:

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door

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END OF TASK

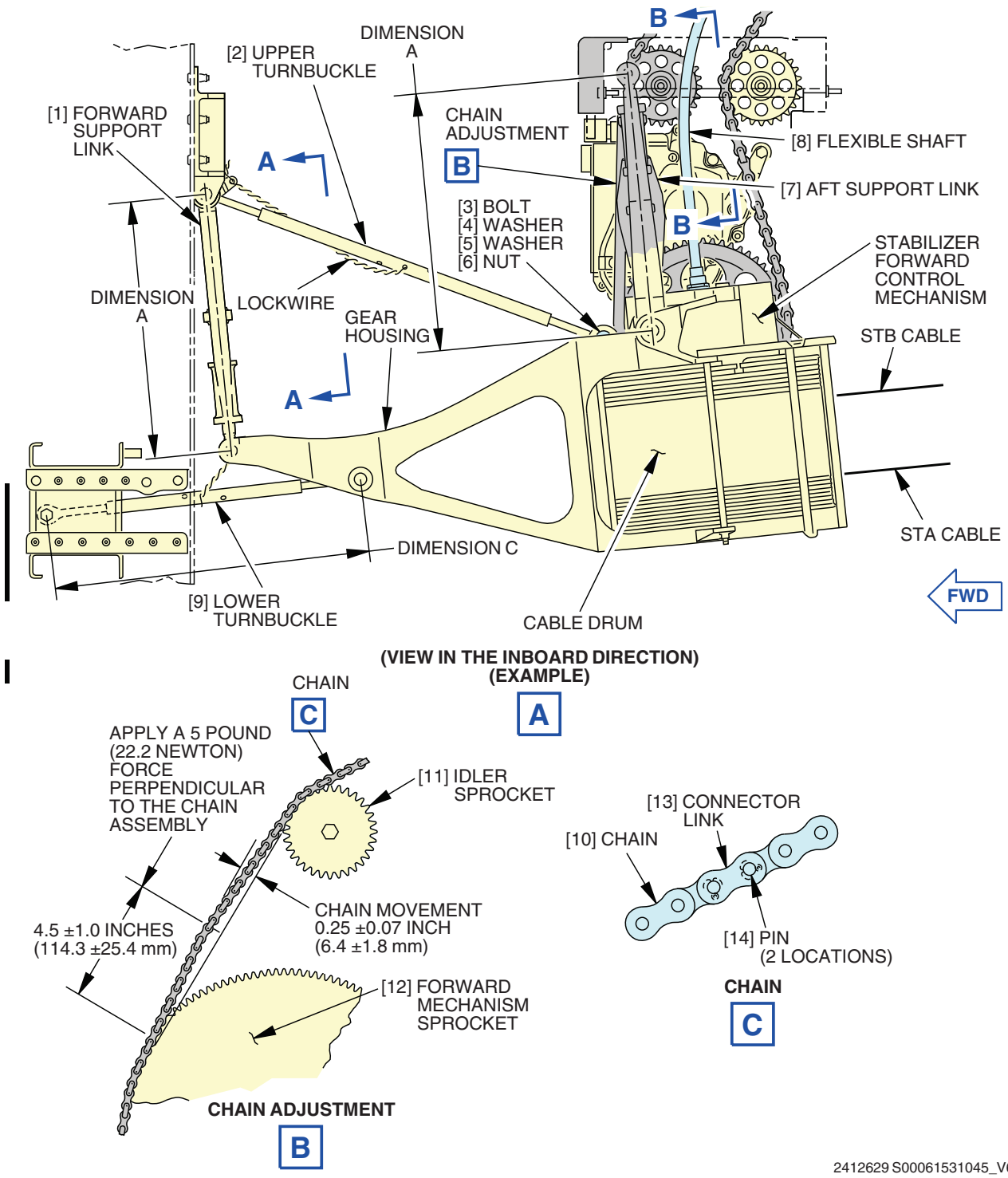
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412628 S00061531044_V2

Stabilizer Control Cable and Chain Adjustment
Figure 501/27-41-00-990-802 (Sheet 1 of 5)

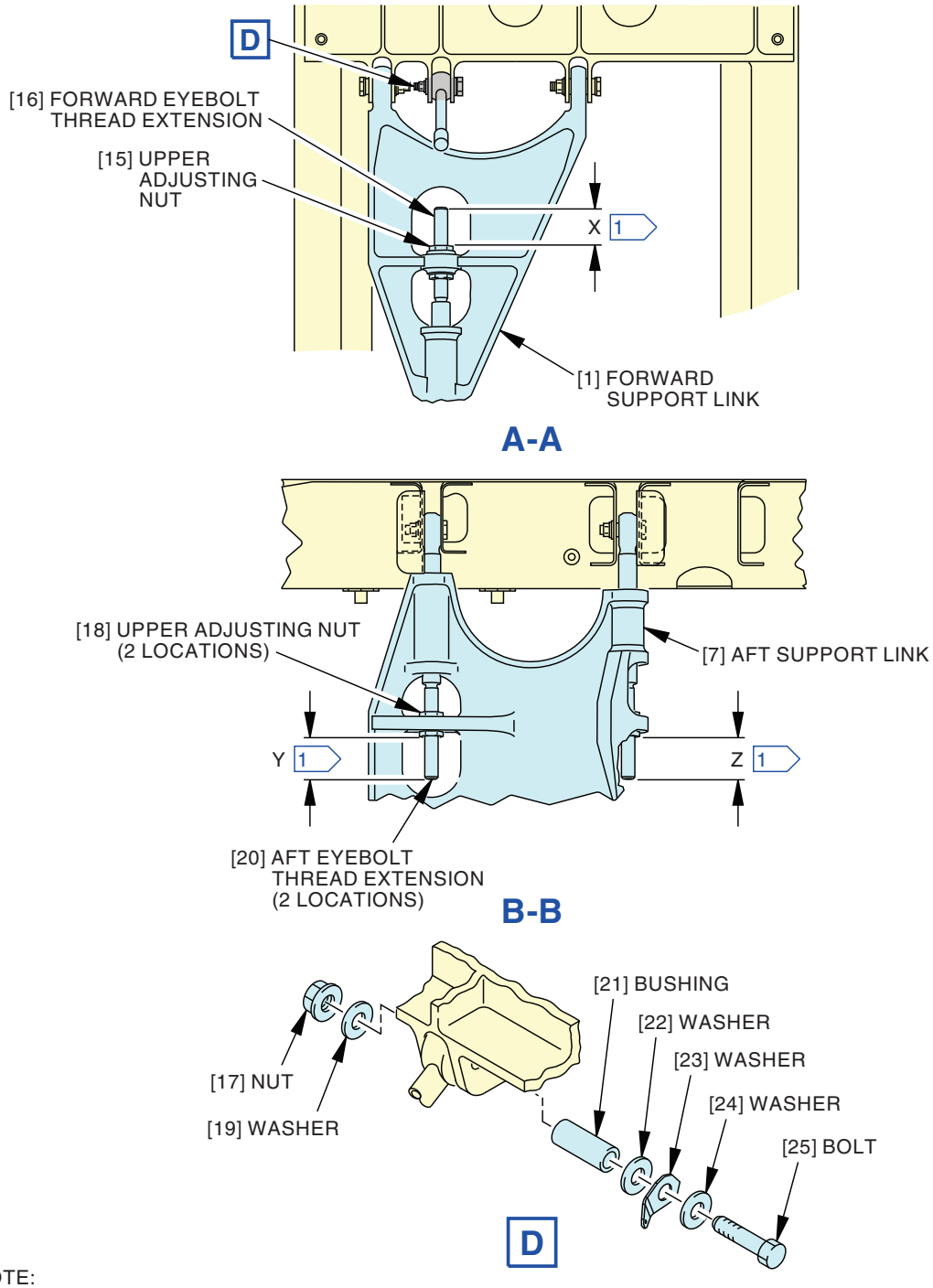
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412629 S00061531045_V6

Stabilizer Control Cable and Chain Adjustment
Figure 501/27-41-00-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



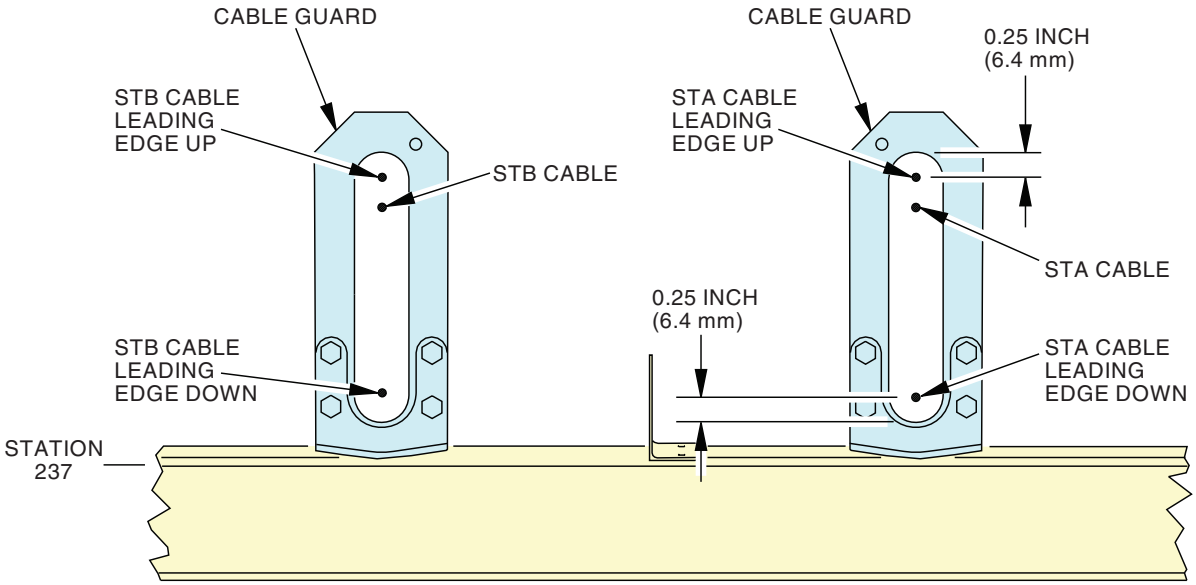
NOTE:

1 DIMENSIONS X, Y AND Z MUST BE WITHIN ±0.12 INCH (±3.1 mm) OF EACH OTHER.

2412630 S00061531046_V2

Stabilizer Control Cable and Chain Adjustment
Figure 501/27-41-00-990-802 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



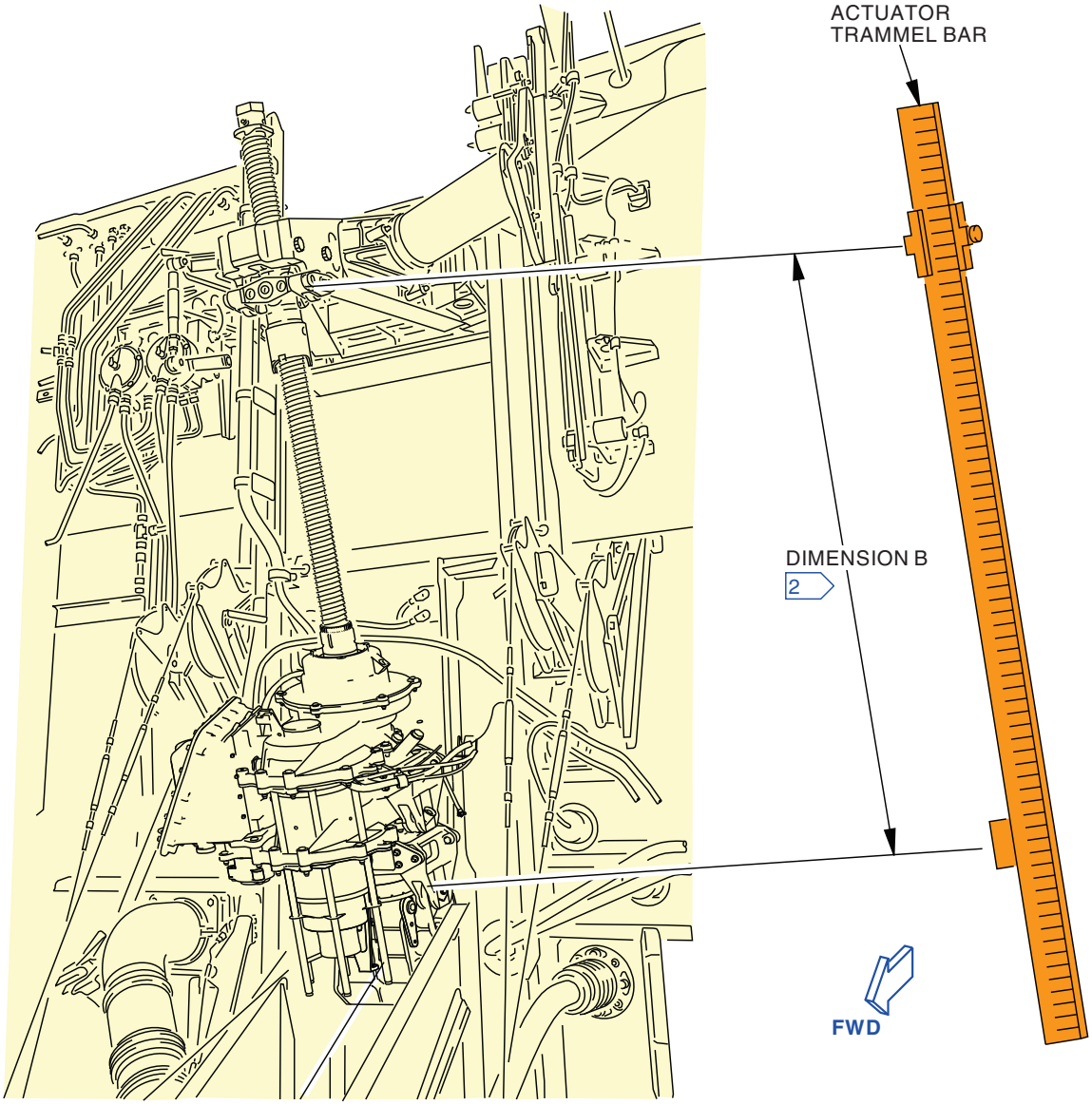
(VIEW IN THE AFT DIRECTION)

E

2412631 S00061531047_V2

Stabilizer Control Cable and Chain Adjustment
Figure 501/27-41-00-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

F

NOTE:

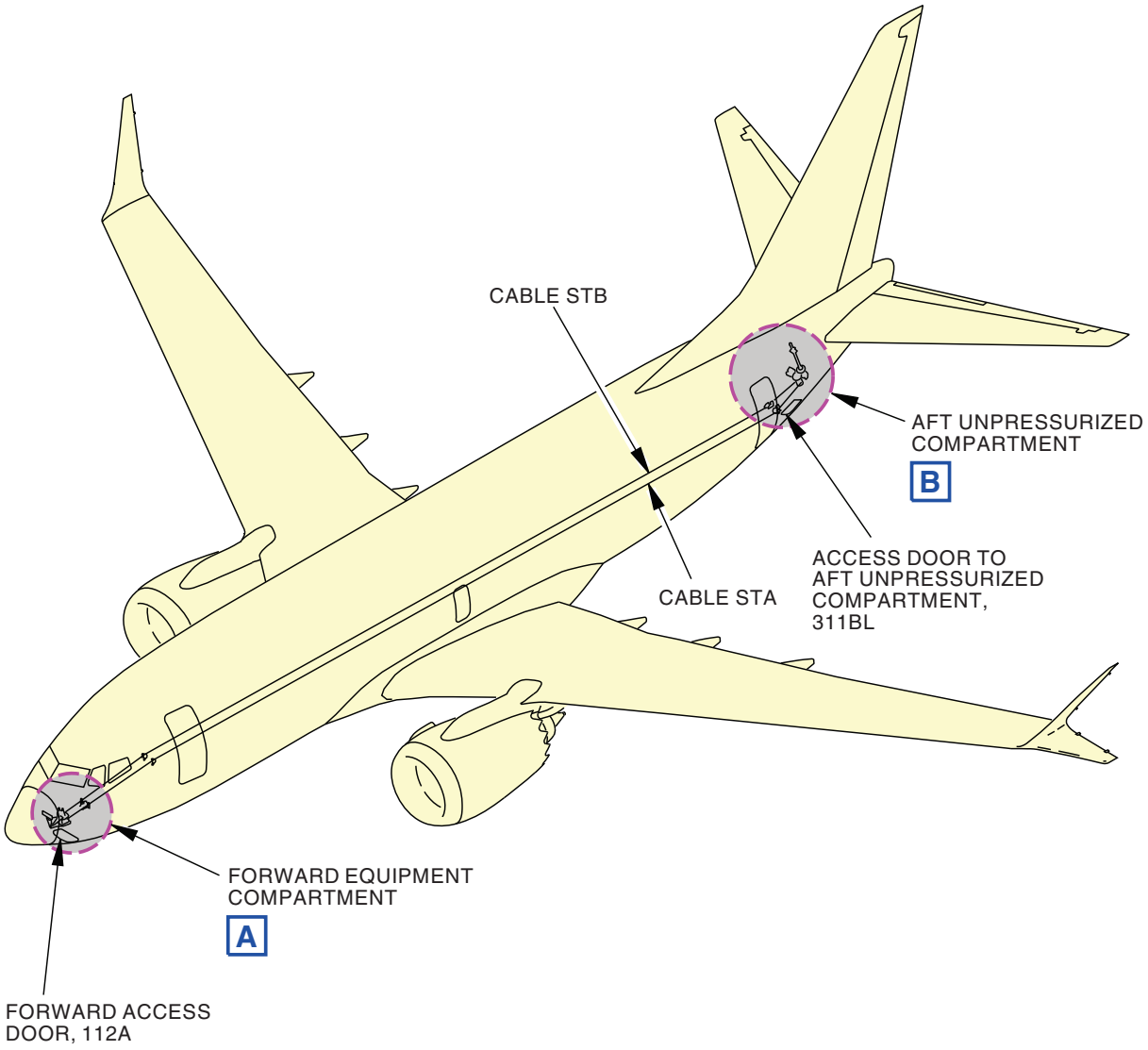
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

2 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS)

2412632 S00061531048_V4

Stabilizer Control Cable and Chain Adjustment
Figure 501/27-41-00-990-802 (Sheet 5 of 5)

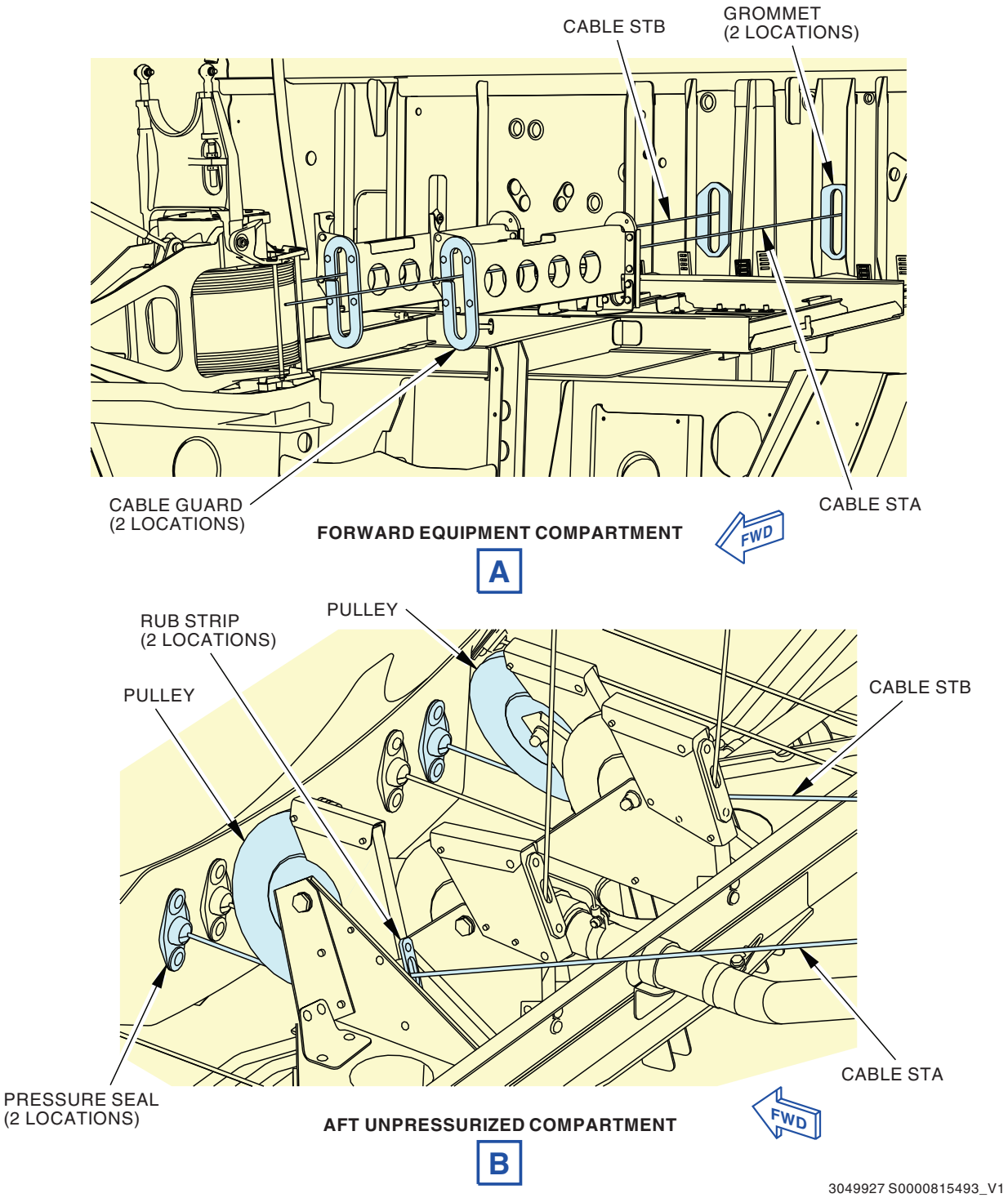
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3049890 S0000815492_V1

Stabilizer Control Cables - Inspection
Figure 502/27-41-00-990-809 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Control Cables - Inspection
Figure 502/27-41-00-990-809 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-00-700-801

3. Stabilizer Manual Trim and Trim Indicator - Test

(Figure 503, Figure 504)

A. General

(1) Use this test to make sure the horizontal stabilizer moves to the correct position when it is operated from the stabilizer trim wheel. It also make sure that the stabilizer trim indicator operates correctly.

(2) If this test is performed as part of stabilizer control cable STA/STB replacement or removal/installation procedures, make sure that the forward and aft stabilizer cable drum wraps of the control cables when the stabilizer is in the full airplane nose UP and airplane nose DOWN positions.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Prepare for the Test

SUBTASK 27-41-00-010-002

(1) To get access to the stabilizer trim actuator, do this step:

Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Stabilizer Manual Trim and Trim Indicator Test

SUBTASK 27-41-00-710-003

WARNING

MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do the test of the stabilizer manual trim and trim indicator test.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Turn the captain's control trim wheel counterclockwise (full APL NOSE DOWN) until it touches the stops.

CAUTION MAKE SURE TO PREVENT DAMAGE TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CHAF OR CAN CAUSE DAMAGE TO THE WIRES OR BUNDLE INSULATION.

1) Use the trammel bar, SPL-1677, to make sure that the "B" dimension is 45.84 ±0.10 in. (1164.3 ±2.5 mm).
2) Make sure that the two STAB TRIM indicators on the control stand show -0.7 ±0.5 units of trim.
3) Make sure that the difference between the two STAB TRIM indicators is less than 0.5 units of trim.
4) Make sure that the STB cable is a minimum of 0.75 full wraps on the forward drum.
5) Make sure that the STA cable is a minimum of 0.75 full wraps on the aft drum.

(b) Turn the captain's control trim wheel clockwise (full APL NOSE UP) until it touches the stops.

1) Make sure that the horizontal stabilizer moves down freely through the full travel.

CAUTION MAKE SURE TO PREVENT DAMAGE TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CHAF OR CAN CAUSE DAMAGE TO THE WIRES OR BUNDLE INSULATION.

2) Use the trammel bar, SPL-1677, to make sure that the "B" dimension is 21.32 ±0.10 in. (541.5 ±2.6 mm).
3) Make sure that the two STAB TRIM indicators on the control stand show 16.4 ±0.5 unit of trim.
4) Make sure that the difference between the two STAB TRIM indicators is less than 0.5 units of trim.
5) Make sure that the STA cable is a minimum of 0.75 full wraps on the forward drum.
6) Make sure that the STB cable is a minimum of 0.75 full wraps on the aft drum.

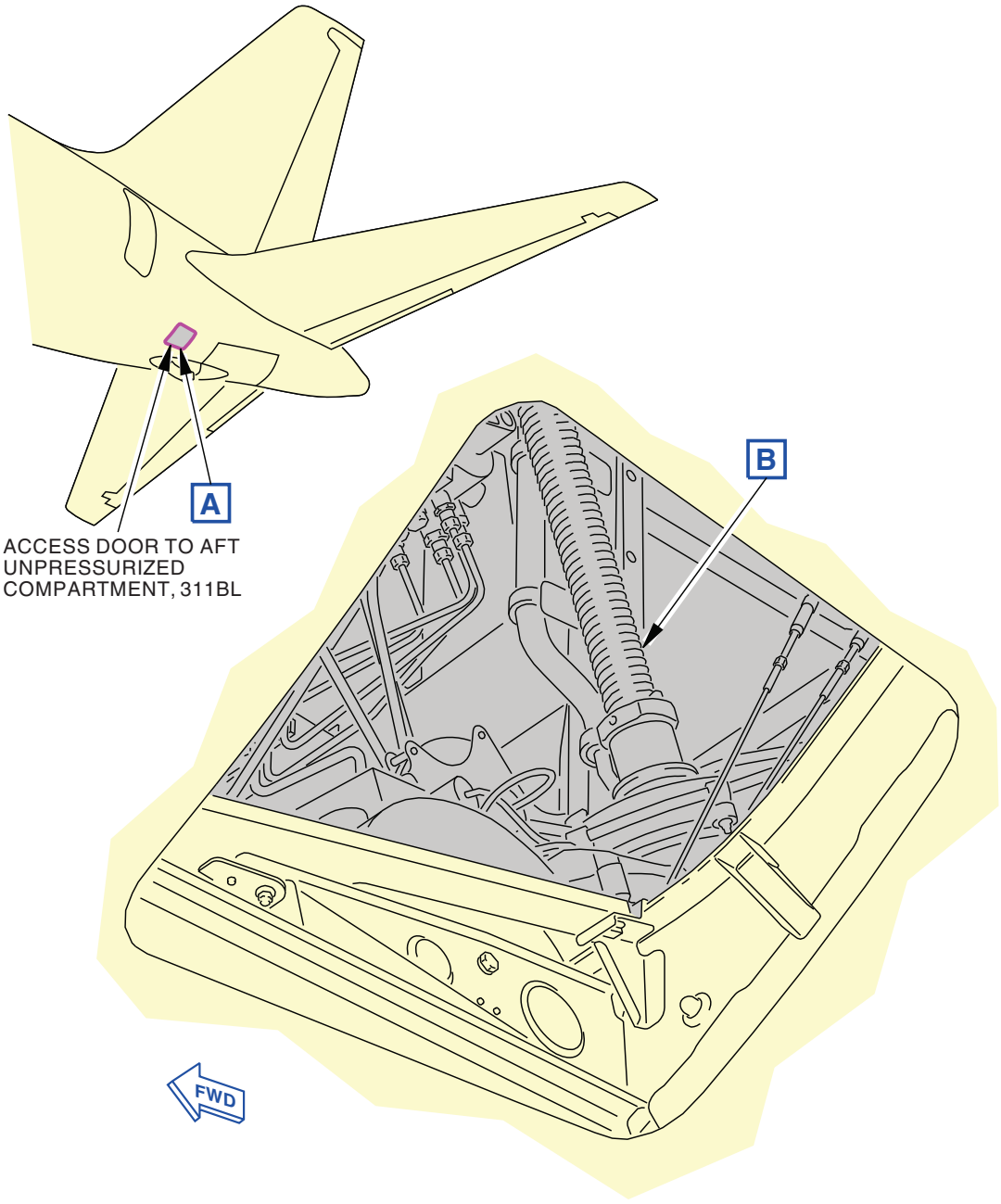
SUBTASK 27-41-00-410-002

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

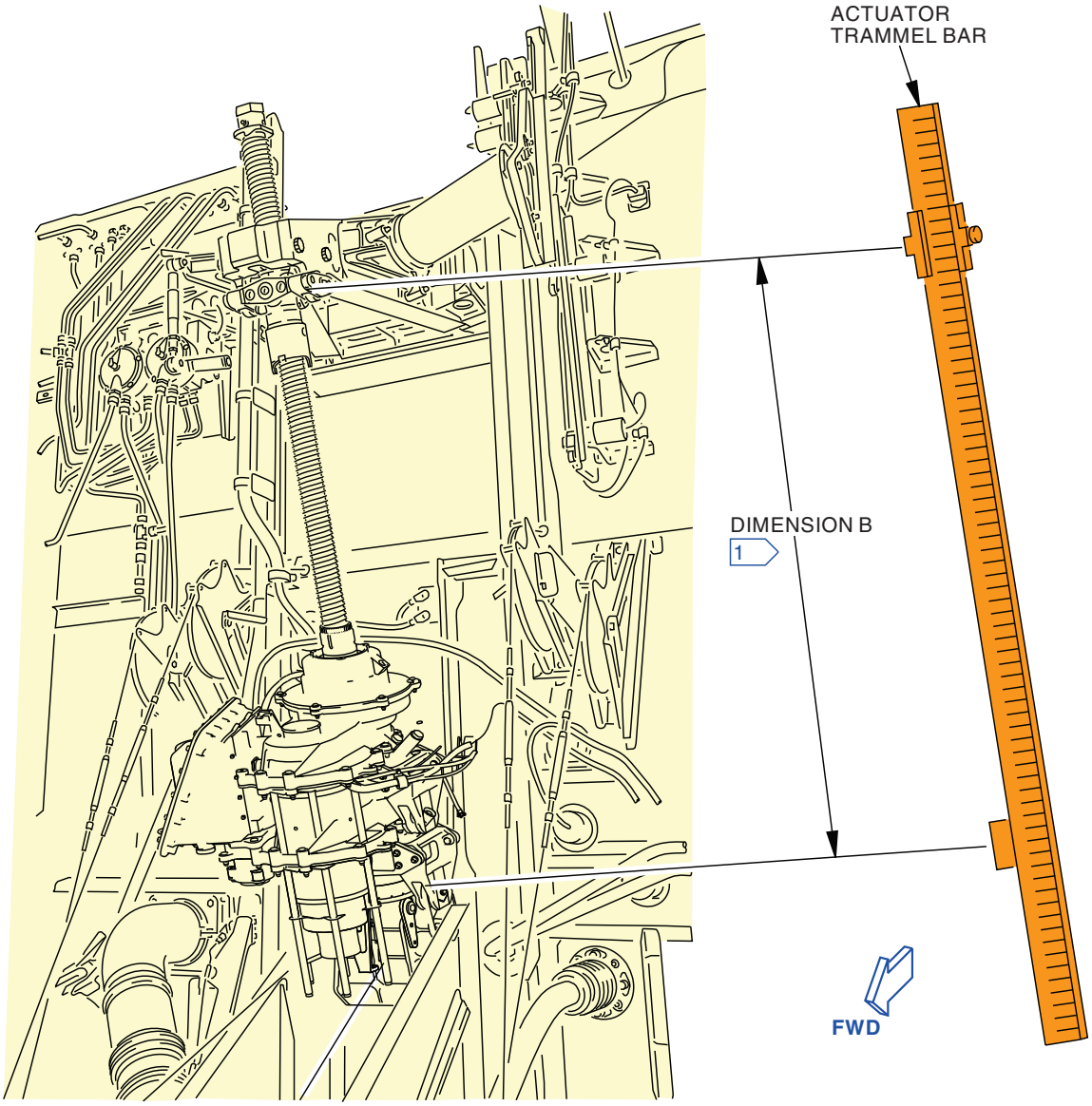


VIEW THROUGH THE ACCESS DOOR TO AFT
UNPRESSURIZED COMPARTMENT, 311BL
A

2412633 S00061531050_V1

Stabilizer Trim Jackscrew Adjustment
Figure 503/27-41-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

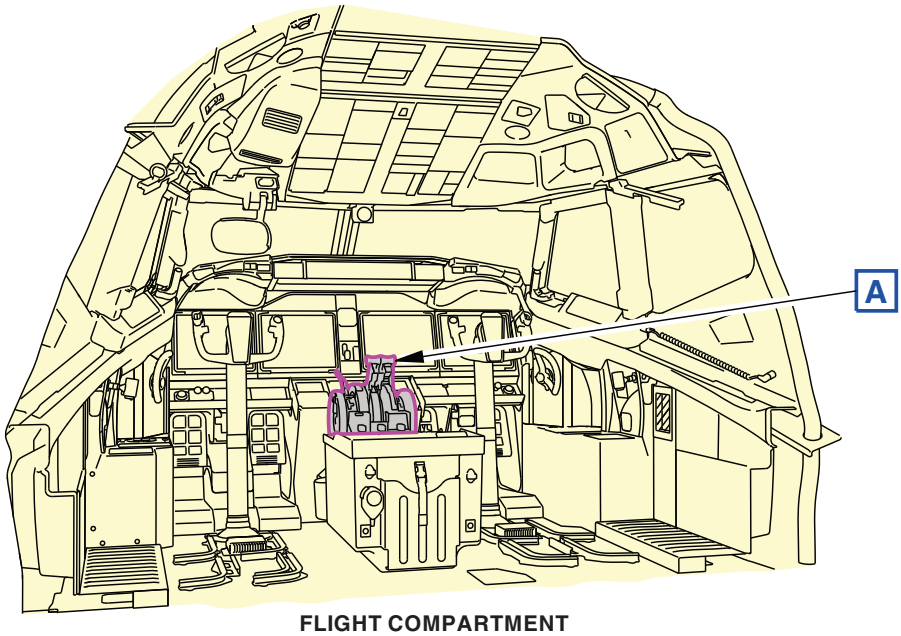
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS)

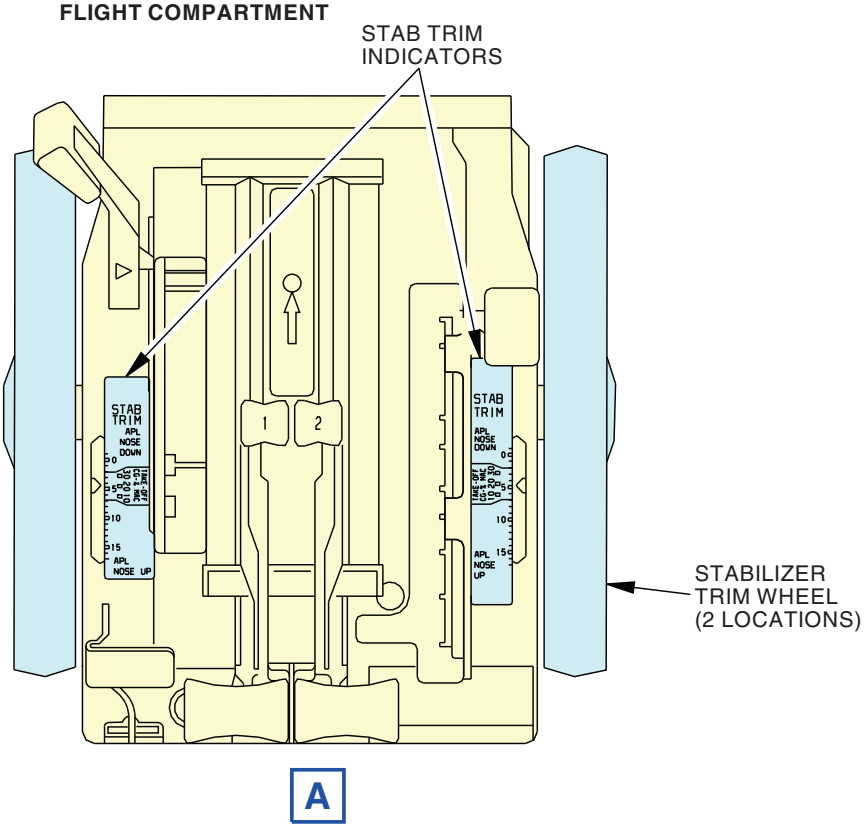
2412634 S00061531051_V4

Stabilizer Trim Jackscrew Adjustment
Figure 503/27-41-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



2412635 S00061531052_V2

Stabilizer Trim Wheel and Stabilizer Trim Indicator Locations
Figure 504/27-41-00-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-00-700-802

4. Horizontal Stabilizer Trim Control System Friction Test

(Figure 503)

A. General

(1) Use this procedure to make sure the system operates smoothly.

B. References

Reference	Title
27-41-61-010-801	Stabilizer Trim Control Wheel Removal (P/B 401)
27-41-61-410-801	Stabilizer Trim Control Wheel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1748	Adapter - Torque Wrench, Manual, Stabilizer Trim Wheel Part #: C27064-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Prepare for the Test

SUBTASK 27-41-00-860-011

(1) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the CUTOUT position.

SUBTASK 27-41-00-480-002

(2) Install the adapter, SPL-1748, on the captain's control trim wheel (Figure 505).

(a) Remove the fasteners that attach the stabilizer trim wheel to the shaft (TASK 27-41-61-010-801).

F. Procedure

SUBTASK 27-41-00-710-004

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do the friction test of the horizontal stabilizer trim control:

(a) Turn the captain's control trim wheel counterclockwise (full APL NOSE DOWN) until it touches the stops.

(b) Turn the captain's control trim wheel clockwise (full APL NOSE UP) until it touches the stops.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure the torque necessary to operate the control trim wheel is not more than 62 in-lb (7.0 N·m) or not less than 22 in-lb (2.5 N·m).

2) Make sure the system operates smoothly.

(c) Turn the captain's control trim wheel counterclockwise (full APL NOSE DOWN) until it touches the stops.

1) Make sure the torque necessary to operate the control trim wheel is not more than 62 in-lb (7.0 N·m) or not less than 22 in-lb (2.5 N·m).

2) Make sure the system operates smoothly.

SUBTASK 27-41-00-080-002

(2) Remove the adapter, SPL-1748, from the captain's control trim wheel.

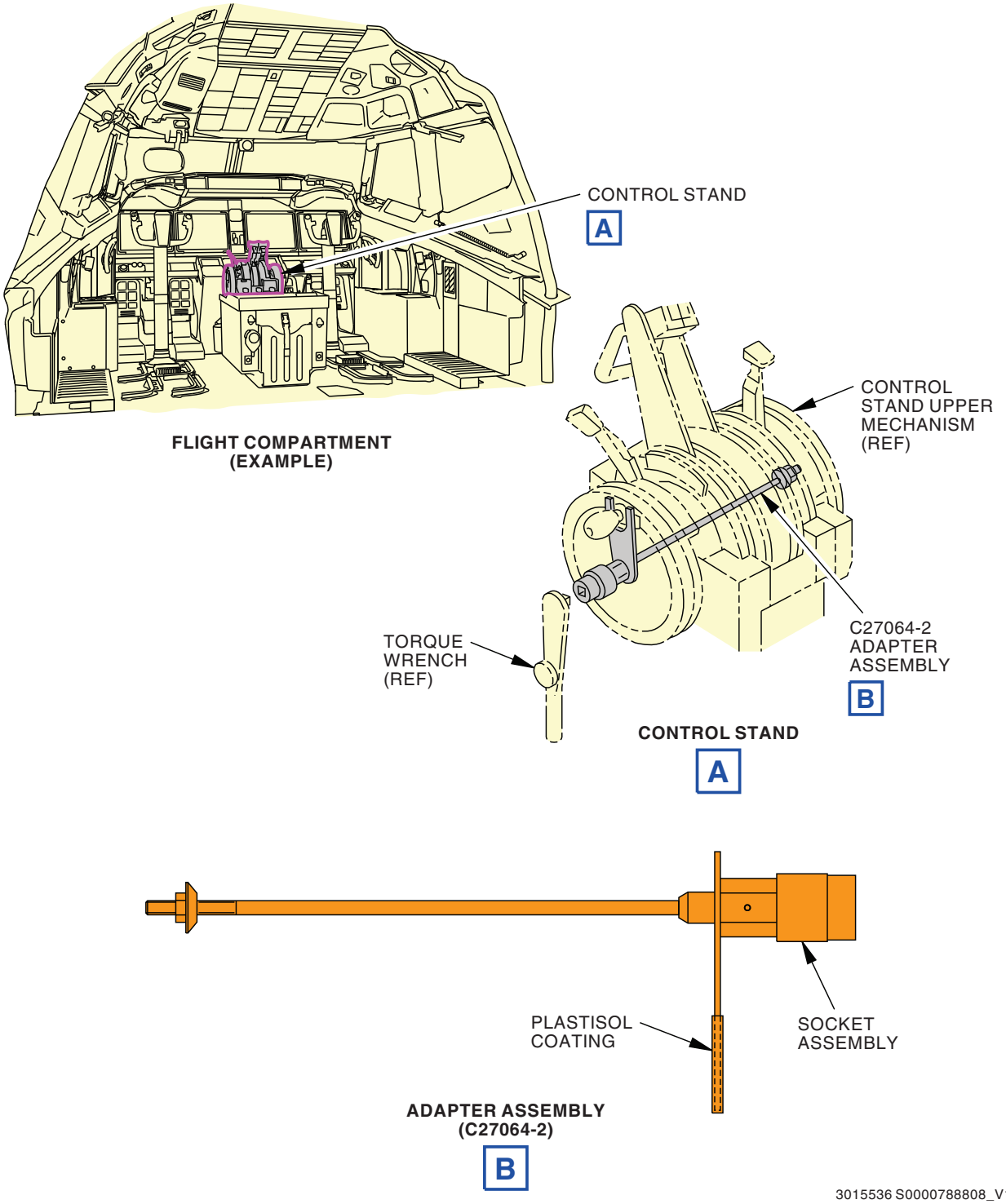
(a) Install the fasteners that attach the stabilizer trim wheel to the shaft (TASK 27-41-61-410-801).

SUBTASK 27-41-00-860-032

(3) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the NORMAL position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Manual Stabilizer Trim Wheel Torque Wrench Adapter Installation
Figure 505/27-41-00-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-00-700-803

5. Stabilizer Electric Trim System Test

A. General

(1) Use this test to make sure the horizontal stabilizer moves at the correct speed when it is operated from the STAB TRIM switches.

B. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
330	Subzone - Left Horizontal Stabilizer and Elevator
340	Subzone - Right Horizontal Stabilizer and Elevator

D. Prepare for the Test

SUBTASK 27-41-00-860-013

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-00-860-014

(2) Make sure that the STAB TRIM PRI (Cutout) switch, S272, and STAB TRIM PRIMARY switch, S149, on the aft area of the control stand is in the NORMAL position.

SUBTASK 27-41-00-860-015

(3) Make sure that the STAB TRIM override switch on the P8 panel is in the NORM position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

SUBTASK 27-41-00-860-033

(4) Supply hydraulic power for systems A and B (TASK 29-11-00-860-801).

E. Procedure

SUBTASK 27-41-00-710-005

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do the test of the electrical trim system test:

(a) Set the flaps to any position from 1-40 units, Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.
   a) Make sure the stabilizer leading edge moves up.
   b) Release the stabilizer trim switches.
2) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.
   a) Make sure the stabilizer leading edge moves down.
   b) Release the stabilizer trim switches.
3) Push and hold the stabilizer trim switches on the first officer's control wheel to the NOSE DOWN position.
   a) Make sure the stabilizer leading edge moves up.
   b) Release the stabilizer trim switches.
4) Push and hold the stabilizer trim switches on the first officer's control wheel to the NOSE UP position.
   a) Make sure the stabilizer leading edge moves down.
   b) Release the stabilizer trim switches.
5) Push the stabilizer trim switches on the captain's control wheel to the NOSE UP position until it stops and then to the NOSE DOWN position until it stops.
   a) Make sure the time for the stabilizer to move from its lower limit of travel to its upper limit of travel is between 31-39 seconds.
   b) Make sure that the operation is smooth and there is no unusual noise.
6) Use the stabilizer trim switches on the captain's control wheel to move the stabilizer many times in the two directions one after the other for 3 seconds each.
   a) Make sure that the operation is smooth and there is no unusual noise.
7) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position until the system comes to the full trim speed.
   a) Move the stabilizer trim switches momentarily to OFF and then to the NOSE DOWN position.
   b) Make sure the stabilizer leading edge moves up immediately after you move the stabilizer trim switches to the NOSE DOWN position.
8) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until the system comes to the full trim speed.
   a) Move the stabilizer trim switches momentarily to OFF and then to the NOSE UP position.
   b) Make sure the stabilizer leading edge moves down immediately after you move the stabilizer trim switches to the NOSE UP position.

(b) Set the flaps to the full up position, Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.
   1) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.
      a) Make sure the stabilizer leading edge moves up.
      b) Release the stabilizer trim switches.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.
   a) Make sure the stabilizer leading edge moves down.
   b) Release the stabilizer trim switches.
3) Push and hold the stabilizer trim switches on the first officer's control wheel to the NOSE DOWN position.
   a) Make sure the stabilizer leading edge moves up.
   b) Release the stabilizer trim switches.
4) Push and hold the stabilizer trim switches on the first officer's control wheel to the NOSE UP position.
   a) Make sure the stabilizer leading edge moves down.
   b) Release the stabilizer trim switches.
5) Push the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until it stops and then to the NOSE UP position until it stops.
   a) Make sure the time for the stabilizer to move from its upper limit of travel to its lower limit of travel is 47-58 seconds.
   b) Make sure that the operation is smooth and there is no unusual noise.
6) Use the stabilizer trim switches on the captain's control wheel to move the stabilizer many times in the two directions, one after the other, for 3 seconds each.
   a) Make sure that the operation is smooth and there is no unusual noise.
7) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position until the system comes to the full trim speed.
   a) Move the stabilizer trim switches momentarily to OFF and then to the NOSE DOWN position.
   b) Make sure the stabilizer leading edge moves up immediately after you move the stabilizer trim switches to the NOSE DOWN position.
8) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until the system comes to the full trim speed.
   a) Move the stabilizer trim switches momentarily to OFF and then to the NOSE UP position.
   b) Make sure the stabilizer leading edge moves down immediately after you move the stabilizer trim switches to the NOSE UP position.
9) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.
   a) Make sure the stabilizer leading edge moves down.
   b) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the CUTOUT position.
   c) Make sure that the movement of the stabilizer stops immediately.
   d) Set the STAB TRIM PRI switch, S272, to the NORMAL position.
   e) Make sure the stabilizer continues to move.
   f) Release the stabilizer trim switches.

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SUBTASK 27-41-00-860-034

(2) Do the test of the Stabilizer Trim Primary and Backup (Cutout) Switches (S272 and S149):

NOTE: This test specifies the use of the captain's control wheel to complete the test. If necessary, use of the first officer's control wheel instead of the captain's control wheel is acceptable, provided that this substitution is maintained throughout the test.

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

(b) Make sure that the STAB TRIM PRIMARY and BACKUP (Cutout) Switches (S272 and S149) are in the NORMAL (guarded) position.

(c) Make sure that the Column Actuated Stabilizer Trim (Override) Switch (S847 OR S1 OF P8-47 Module) is set to the NORM (guarded) position.

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

DO NOT GO MORE THAN THE DUTY CYCLE OF THE STABILIZER TRIM (MAIN ELECTRIC) MOTOR OF 2 MINUTES ON AND 13 MINUTES OFF DURING THE TEST. IF YOU DO NOT OBEY INSTRUCTIONS, DAMAGE TO THE STABILIZER TRIM MOTOR CAN OCCUR.

(d) Position the stabilizer to an intermediate position.

NOTE: Approximately 8 units of trim is satisfactory for an intermediate position.

(e) Set the STAB TRIM PRIMARY (S272) and BACKUP (S149) (Cutout) Switches to the CUTOUT (unguarded) position.

(f) Move the STAB TRIM switches on the captain's control wheel to the NOSE UP position, and hold them in this position.

(g) Make sure that the stabilizer does not move.

(h) Move the STAB TRIM switches on the captain's control wheel to the NOSE DOWN position, and hold them in this position.

(i) Make sure that the stabilizer does not move.

(j) Set the STAB TRIM PRIMARY and BACKUP (Cutout) Switches (S272 and S149) to the NORMAL (guarded) position.

(k) Make sure that the stabilizer trims in both the NOSE UP and NOSE DOWN directions using the STAB TRIM switches on the captain's control wheel.

(l) Set the STAB TRIM PRIMARY (Cutout) Switch (S272) to the CUTOUT (unguarded) position.

NOTE: The STAB TRIM BACKUP (Cutout) Switch (S149) should still be in the NORMAL (guarded) position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) Move the STAB TRIM switches on the captain's control wheel to the NOSE UP position, and hold them in this position.

(n) Make sure that the stabilizer does not move.

(o) Move the STAB TRIM switches on the captain's control wheel to the NOSE DOWN position, and hold them in this position.

(p) Make sure that the stabilizer does not move.

(q) Set the STAB TRIM PRIMARY (Cutout) Switch (S272) to the NORMAL (guarded) position.

NOTE: The STAB TRIM BACKUP (Cutout) Switch (S149) should still be in the NORMAL (guarded) position.

(r) Make sure that the stabilizer trims in both the NOSE UP and NOSE DOWN directions using the STAB TRIM switches on the captain's control wheel.

(s) Set the STAB TRIM BACKUP (Cutout) Switch (S149) to the CUTOUT (unguarded) position.

NOTE: The STAB TRIM PRIMARY (Cutout) Switch (S272) should still be in the NORMAL (guarded) position.

(t) Move the STAB TRIM switches on the captain's control wheel to the NOSE UP position, and hold them in this position.

(u) Make sure that the stabilizer does not move.

(v) Move the STAB TRIM switches on the captain's control wheel to the NOSE DOWN position, and hold them in this position.

(w) Make sure that the stabilizer does not move.

(x) Set the STAB TRIM BACKUP (Cutout) Switch (S149) to the NORMAL (guarded) position.

NOTE: The STAB TRIM PRIMARY (Cutout) Switch (S272) should still be in the NORMAL (guarded) position.

(y) Make sure that the stabilizer trims in both the NOSE UP and NOSE DOWN directions using the STAB TRIM switches on the captain's control wheel.

SUBTASK 27-41-00-710-013

(3) Remove hydraulic power from systems A and B (TASK 29-11-00-860-805).

——— END OF TASK ———

TASK 27-41-00-700-804

6. Stabilizer Trim Wheel Free Movement Test

A. General

(1) Use this test to check the freeplay between the horizontal stabilizer trim motor and the ballscrew.

B. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-41-00-860-017

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-00-860-018

(2) Make sure that the STAB TRIM PRI (Cutout) switch, S272, and STAB TRIM PRIMARY, S149, on the aft area of the control stand is in the NORMAL position.

SUBTASK 27-41-00-860-019

(3) Make sure that the STAB TRIM override switch on the P8 panel is in the NORM position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

E. Stabilizer Trim Wheel Free Movement Test

SUBTASK 27-41-00-710-006

WARNING MAKE SURE THAT ALL PERSONNEL, AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) Do the test of the stabilizer trim wheel free movement:

(a) Set the flaps to the 5-unit position (TASK 27-51-00-860-803).

1) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position until the system comes to full trim speed.

a) Release the stabilizer trim switches.

<1> Make sure that the stabilizer trim wheel continues to move not more than 2.5 turns (900 degrees).

NOTE: Reverse travel (APL NOSE DOWN) of the trim wheel is allowed if it does not exceed 27 degrees.

2) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until the system comes to full trim speed.

a) Release the stabilizer trim switches.

<1> Make sure that the stabilizer trim wheel continues to move not more than 2.5 turns (900 degrees).

NOTE: Reverse travel (APL NOSE UP) of the trim wheel is allowed if it does not exceed 100 degrees.

(b) Set the flaps to the full up position (TASK 27-51-00-860-804).

1) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until the system comes to full trim speed.

a) Release the stabilizer trim switches.

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<1> Make sure that the stabilizer trim wheel continues to move not more than 0.5 turns (180 degrees).

NOTE: Reverse travel (APL NOSE UP) of the trim wheel is allowed if it does not exceed 65 degrees.

2) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position until the system comes to full trim speed.

a) Release the stabilizer trim switches.

<1> Make sure that the stabilizer trim wheel continues to move not more than 0.5 turns (180 degrees).

NOTE: Reverse travel (APL NOSE UP) of the trim wheel is allowed if it does not exceed 65 degrees.

——— END OF TASK ———

TASK 27-41-00-700-805

7. Stabilizer Trim Limit Switches Test

(Figure 506)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to make sure that the stabilizer trim limit switches operate correctly.

(2) The trim system can move the stabilizer to a position more than the limits that were set during the adjustment of the limit switches. This condition is usual. Thus the "B" dimension tolerances for the test are larger than the tolerance for the adjustment.

(3) This test is applicable only for limit switches S145, S844, S144.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Test

SUBTASK 27-41-00-860-021

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-00-860-022

(2) Make sure that the STAB TRIM PRI switch, S272, on the aft area of the control stand is in the NORMAL position.

SUBTASK 27-41-00-860-023

(3) Make sure that the STAB TRIM override switch on the P8 is in the NORM position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

SUBTASK 27-41-00-860-024

(4) Make sure that the flaps are in the full up position.

SUBTASK 27-41-00-010-003

(5) To get access to the limit switches, do this step:

Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Procedure

SUBTASK 27-41-00-710-007

(1) Do the test of the limit switch S844:

(a) Move the stabilizer to the maximum leading edge up (APL NOSE DOWN) position (the mechanical limits). This can be done using the STAB TRIM switches and then the stabilizer trim wheel.

(b) Turn the captain's stabilizer trim wheel clockwise approximately 3 turns.

(c) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

1) Make sure that the stabilizer leading edge does not move.

2) Release the stabilizer trim switches.

(d) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

1) Make sure that the stabilizer leading edge moves down (APL NOSE UP).

2) Release the stabilizer trim switches.

(e) Move the stabilizer to approximately 5 units of trim.

(f) Move the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until the stabilizer stops automatically.

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CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

1) Use the trammel bar, SPL-1677 to measure the "B" dimension.

NOTE: Make sure to protect wire bundles when using the trammel trammel bar, SPL-1677. The sharp edges on the trammel trammel bar, SPL-1677 can cause chafing or damage to the wires.

2) Make sure the "B" dimension is 39.39 ±0.10 in. (1000.5 ±2.5 mm).

NOTE: The STAB TRIM indicator gives approximate indications. Do not use the STAB TRIM indicator indications to adjust the stabilizer trim switches ("B" dimension).

a) Make sure that the STAB TRIM indicator on the control stand reads 3.85 units of trim.

SUBTASK 27-41-00-710-008

(2) Do the test of the limit switch, S145:

(a) If you use the standby hydraulic pump motor to move the flaps, then do these steps:

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

1) Put the ALTERNATE FLAP arm switch in the ARM position.

NOTE: The standby hydraulic pump motor will operate when the ALTERNATE FLAP arm switch is in the ARM position.

2) Move the alternate flap drive control switch to extend the flaps to the 5-unit position.

(b) If you use the system B hydraulic power to move the flaps, then do these steps:

1) Pressurize hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

2) Move the flap control lever to extend the flaps to the 5-unit position.

(c) Move the stabilizer to the maximum leading edge up (APL NOSE DOWN) position (mechanical limits).

NOTE: This can be done with the stabilizer trim switches and then the stabilizer trim wheel.

(d) Turn the captain's stabilizer trim wheel clockwise approximately 2.00 ±0.25 turns.

(e) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

1) Make sure that the stabilizer leading edge does not move.

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2) Release the stabilizer trim switches.

(f) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

1) Make sure that stabilizer leading edge moves down (APL NOSE UP).

2) Release the stabilizer trim switches.

(g) Move the stabilizer to approximately 1 unit of trim.

(h) Move the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until the stabilizer stops automatically.

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

1) Use the trammel bar, SPL-1677 to measure the "B" dimension.

NOTE: Make sure to protect wire bundles when using the trammel trammel bar, SPL-1677. The sharp edges on the trammel trammel bar, SPL-1677 can cause chafing or damage to the wires.

2) Make sure that the "B" dimension is 45.48 ±0.10 in. (1155.2 ±2.5 mm).

a) Make sure that that the STAB TRIM indicator on the control stand reads -0.45 units of trim.

NOTE: The STAB TRIM indicator gives approximate indications. Do not use the STAB TRIM indicator indications to adjust the stabilizer trim switches ("B" dimension).

(i) Move the flap control lever or the alternate flap drive control switch to retract the flaps.

(j) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-41-00-710-009

(3) Do the test of the limit switch S144:

(a) Move the stabilizer to the maximum leading edge down (APL NOSE UP) position (mechanical limits).

NOTE: This can be done with the stabilizer trim switches and then the stabilizer trim wheel.

(b) Turn the captain's stabilizer trim wheel counterclockwise approximately 3 turns.

(c) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

1) Make sure that the stabilizer leading edge does not move.

2) Release the stabilizer trim switches.

(d) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

1) Make sure that stabilizer leading edge moves up (APL NOSE DOWN).

2) Release the stabilizer trim switches.

(e) Move the stabilizer to approximately 12 units of trim.

(f) Move the stabilizer trim switches on the captain's control wheel to the NOSE UP position until the stabilizer stops automatically.

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1) Use the trammel bar, SPL-1677 to measure the "B" dimension.

NOTE: Make sure to protect wire bundles when using the trammel trammel bar, SPL-1677. The sharp edges on the trammel trammel bar, SPL-1677 can cause chafing or damage to the wires.

2) Make sure that the "B" dimension is 24.80 ±0.10 in. (629.9 ±2.6 mm).

a) Make sure that the STAB TRIM indicator on the control stand reads 14.0 units of trim.

NOTE: The STAB TRIM indicator gives approximate indications. Do not use the STAB TRIM indicator indications to adjust the stabilizer trim switches ("B" dimension).

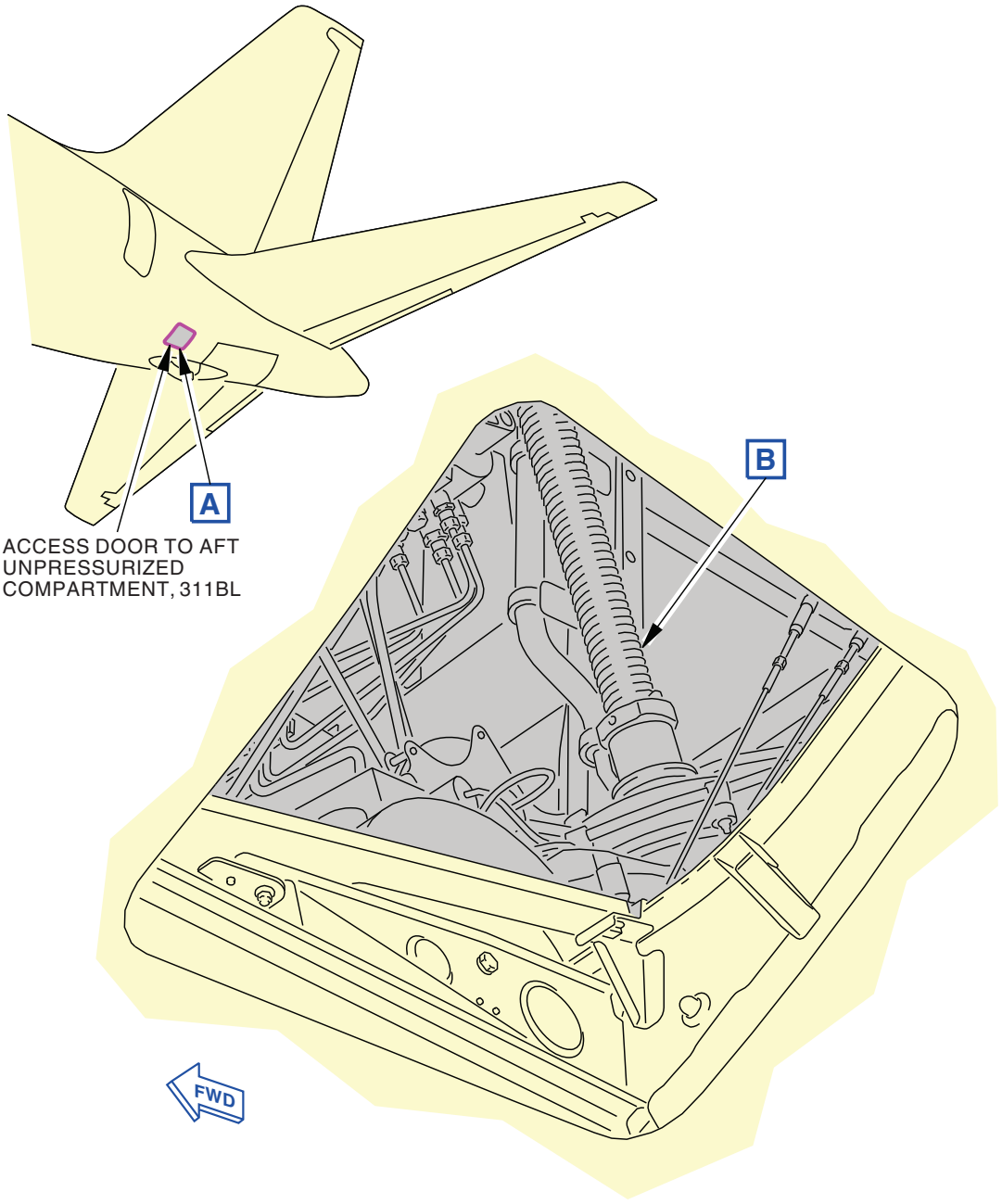
SUBTASK 27-41-00-410-003

(4) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

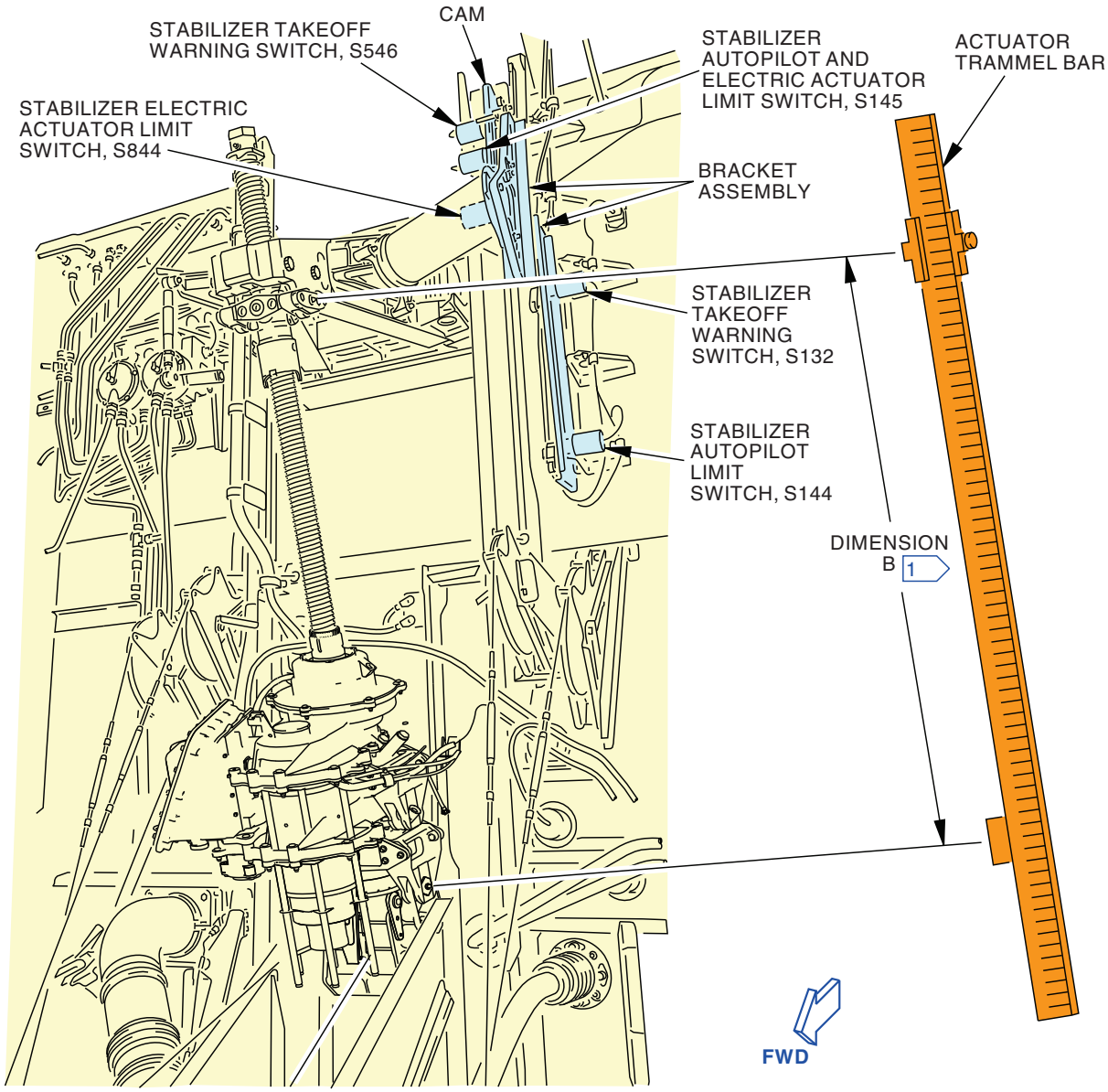


VIEW THROUGH THE ACCESS DOOR TO AFT
UNPRESSURIZED COMPARTMENT, 311BL
A

2412636 S00061531057_V1

Stabilizer Trim Limit Switches Test
Figure 506/27-41-00-990-805 (Sheet 1 of 2)

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STABILIZER TRIM ACTUATOR

B

1 **T**HE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS)

2412637 S00061531058_V3

Stabilizer Trim Limit Switches Test
Figure 506/27-41-00-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-00-700-806

8. Stabilizer Clutch Friction System Test

A. General

(1) Use this test to make sure that the correct force disconnects the manual clutch.

B. References

Reference	Title
27-41-61-010-801	Stabilizer Trim Control Wheel Removal (P/B 401)
27-41-61-410-801	Stabilizer Trim Control Wheel Installation (P/B 401)
27-41-71-000-801	Stabilizer Trim Motor Removal (P/B 401)
27-41-71-400-801	Stabilizer Trim Motor Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1684	Wrench Assy, Pinion Gear, Stabilizer Trim Actuator (Lock) Part #: F71267 Supplier: 81205
SPL-1748	Adapter - Torque Wrench, Manual, Stabilizer Trim Wheel Part #: C27064-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Prepare for the Test

SUBTASK 27-41-00-020-006

(1) Do this task: Stabilizer Trim Motor Removal, TASK 27-41-71-000-801.

SUBTASK 27-41-00-480-003

(2) Install the wrench, SPL-1684, on the internal spline on the stabilizer gearbox.

(a) Make sure the wrench does not turn.

SUBTASK 27-41-00-480-004

(3) Install the adapter, SPL-1748, on the captain's stabilizer trim wheel (Figure 505).

(a) Remove the fasteners that attach the stabilizer trim wheel to the shaft (TASK 27-41-61-010-801).

F. Procedure

SUBTASK 27-41-00-710-010

(1) Do a check of the manual clutch disconnect:

(a) Turn the captain's stabilizer trim wheel 3 turns or more counterclockwise, releasing the load each turn..

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(b) Make sure the torque necessary to operate the stabilizer trim wheel is not more than 163 in-lb (18.42 N·m).

NOTE: 163 in-lb (18.42 N·m) is equal to a tangential force at the handle of the trim wheel of approximately 46 lbf (205 N).

(c) Turn the captain's stabilizer trim wheel 3 turns or more clockwise, releasing the load each turn.

(d) Make sure the torque necessary to operate the stabilizer trim wheel is not more than 163 in-lb (18.42 N·m).

NOTE: 163 in-lb (18.42 N·m) is equal to a tangential force at the handle of the trim wheel of approximately 46 lbf (205 N).

SUBTASK 27-41-00-480-005

(2) Remove the wrench, SPL-1684 and the adapter, SPL-1748.

(a) Install the fasteners that attach the stabilizer trim wheel to the shaft (TASK 27-41-61-410-801).

SUBTASK 27-41-00-420-005

(3) Do this task: Stabilizer Trim Motor Installation, TASK 27-41-71-400-801.

——— END OF TASK ———

TASK 27-41-00-700-807

9. Column Actuated Stabilizer Trim Cutout Switch Override Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to make sure that the horizontal stabilizer follows the commands from the STAB TRIM switches, even though the column actuated STAB TRIM cutout switches give the opposite command.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
330	Subzone - Left Horizontal Stabilizer and Elevator
340	Subzone - Right Horizontal Stabilizer and Elevator

D. Prepare for the Test

SUBTASK 27-41-00-860-026

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-00-780-001

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

E. Procedure

SUBTASK 27-41-00-710-011

(1) Do the column actuated stabilizer trim cutout switch override test:

(a) Make sure that the STAB TRIM PRI switch, S272 on the aft area of the control stand is in the NORMAL position.

(b) Set the STAB TRIM override switch on the P8 panel to the OVRD position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

(c) Turn the stabilizer trim wheel to set the stabilizer to approximately 8 units of trim.

(d) Pull the captain's control column full aft.

(e) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

1) Make sure that the stabilizer leading edge moves up (STAB TRIM indicator movement in the APL NOSE DOWN direction).

2) Release the stabilizer trim switches.

(f) Put the captain's control column to the neutral detent.

(g) Set the STAB TRIM override switch on the P8 panel to the NORM position.

(h) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

1) Make sure that the stabilizer leading edge moves up (STAB TRIM indicator movement in the APL NOSE DOWN direction).

2) Pull the captain's control column full aft.

3) Make sure that the movement of the stabilizer stops.

4) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

5) Make sure that the stabilizer leading edge moves down (STAB TRIM indicator movement in the APL NOSE UP direction).

6) Release the stabilizer trim switches.

(i) Put the captain's control column to the neutral detent.

(j) Set the STAB TRIM override switch on the P8 panel, to the OVRD position.

(k) Push the captain's control column full forward.

(l) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

1) Make sure that the stabilizer leading edge moves down (STAB TRIM indicator movement in the APL NOSE UP direction).

2) Release the stabilizer trim switches.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) Put the captain's control column to the neutral detent.

(n) Set the STAB TRIM override switch on the P8 panel, to the NORM position.

(o) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

1) Make sure that the stabilizer leading edge moves down (STAB TRIM indicator movement in the APL NOSE UP direction).

2) Push the captain's control column full forward.

3) Make sure that the movement of the stabilizer stops.

4) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

5) Make sure that the stabilizer leading edge moves up (STAB TRIM indicator movement in the APL NOSE DOWN direction).

6) Release the stabilizer trim switches.

(p) Put the captain's control column to the neutral detent.

SUBTASK 27-41-00-780-002

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

——— END OF TASK ———

TASK 27-41-00-700-808

10. Stabilizer Trim Secondary Brake System Test

(Figure 507)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test is to make sure that the stabilizer secondary brake can prevent movement of the horizontal stabilizer.

B. References

Reference	Title
27-41-81-000-801	Stabilizer Ball Nut and Jackscrew Gearbox Removal (P/B 401)
27-41-81-400-801	Stabilizer Ball Nut and Jackscrew Gearbox Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-13949	Torque Socket - Upper Stop, Stabilizer Trim Jack Screw Part #: C27090-1 Supplier: 81205
STD-14434	Tape Measure - Flexible Steel

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
312	Area Aft of Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-41-00-860-027

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-00-010-004

(2) To get access to the stabilizer trim actuator, do this step:

Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Stabilizer Trim Secondary Brake System Test

SUBTASK 27-41-00-700-001

(1) Do the stabilizer trim secondary brake system test:

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE CONTROL SURFACES AND LANDING GEAR DOOR AREAS. THE CONTROL SURFACES, THE LANDING GEAR, AND THE LANDING GEAR DOORS CAN MOVE WHEN YOU DO THE AIR MODE SIMULATION. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Turn the captain's stabilizer trim wheel several turns counterclockwise (2-3 turns is satisfactory).

NOTE: If necessary, a permitted alternative is to use the first officer's control wheel (turn clockwise) to do the test. If you do this, make sure that you use the first officer's control wheel to do all the test procedure.

1) Make sure that the stabilizer is not moved to the maximum Aircraft-Nose-Up (ANU) and Aircraft-Nose-Down (AND) positions (mechanical stops).

(b) Install the torque socket, SPL-13949, onto the upstop of the jackscrew.

CAUTION IF MORE THAN 800 IN-LB (90.4 N•M) OF TORQUE IS APPLIED TO BALLSCREW ACTUATOR ASSEMBLIES, DAMAGE TO THE BALLSCREW ACTUATOR ASSEMBLY MAY OCCUR.

(c) Apply a 500 ±20 in-lb (56.5 ±2.3 N·m) counterclockwise torque to the upstop of the jackscrew and hold.

1) Make sure that the torque is not put on the safety rod nut.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) While maintaining torque on the jackscrew, use a marker to make a line on the jackscrew shield and the retainer - seal (View A-A, Figure 507).

NOTE: The arc distance should be measured on the outside radius of the jackscrew shield.

3) Release the torque.

CAUTION

IF MORE THAN 800 IN-LB (90.4 N•M) OF TORQUE IS APPLIED TO BALLSCREW ACTUATOR ASSEMBLIES, DAMAGE TO THE BALLSCREW ACTUATOR ASSEMBLY MAY OCCUR.

(d) Apply a 500 ±20 in-lb (56.5 ±2.3 N·m) clockwise torque to the upstop of the jackscrew and hold.

1) Make sure that the torque is not put on the safety rod nut.

2) While maintaining torque on the jackscrew, use a marker to make a line on the jackscrew shield and retainer - seal (View A-A, Figure 507).

NOTE: The arc distance should be measured on the outside radius of the jackscrew shield.

3) Release the torque.

(e) Use a tape measure, STD-14434, to measure the arc distance between the lines.

1) Make sure that the arc distance dimension is not more than 1.06 in. (26.9 mm) (36.0 degrees).

a) If the arc distance dimension is more than 1.06 in. (26.9 mm) (36.0 degrees), replace the horizontal stabilizer trim actuator (TASK 27-41-81-000-801 and TASK 27-41-81-400-801).

(f) Remove the torque socket, SPL-13949, from the upstop of the jackscrew.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-00-410-004

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-41-00-860-028

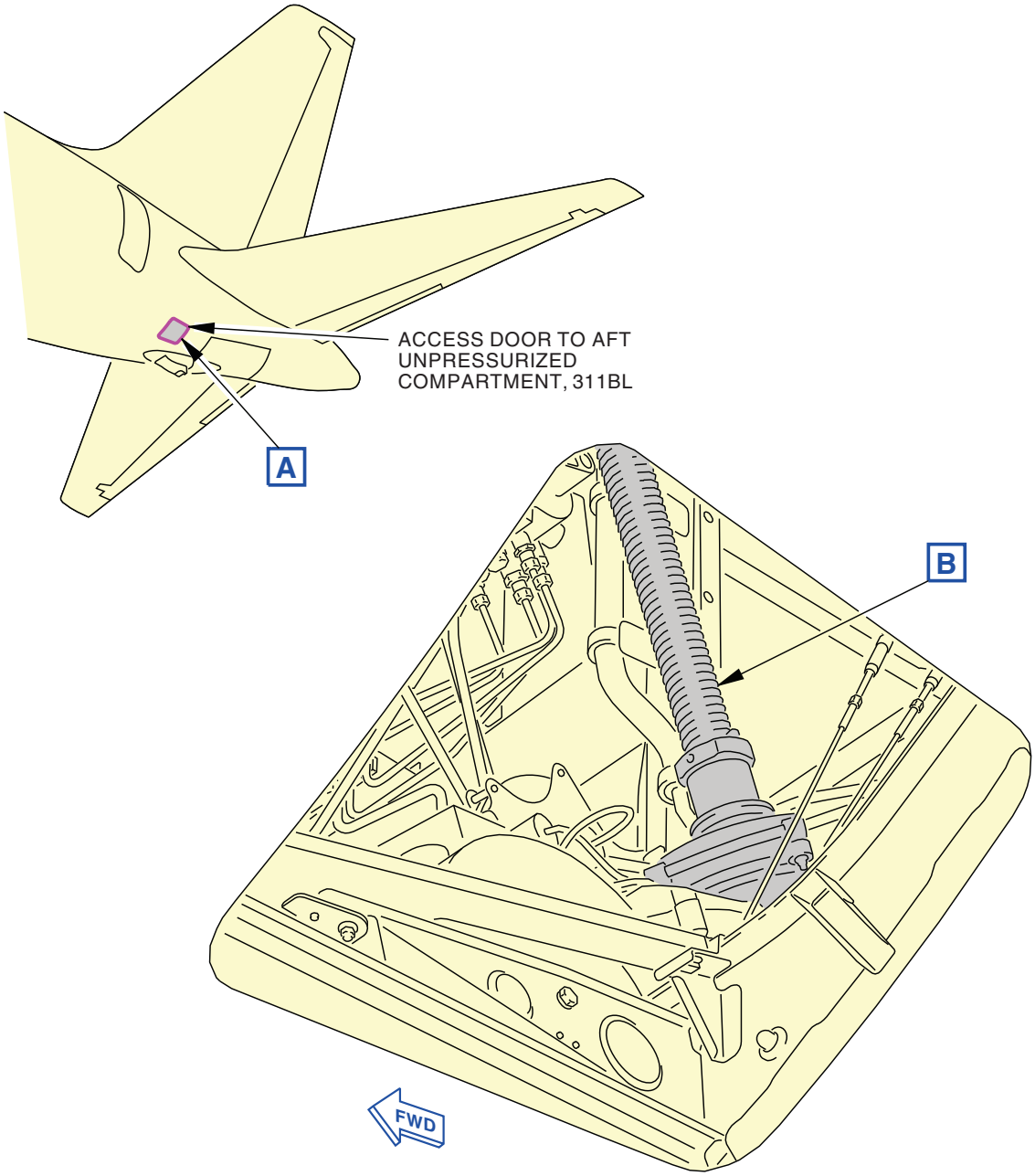
(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



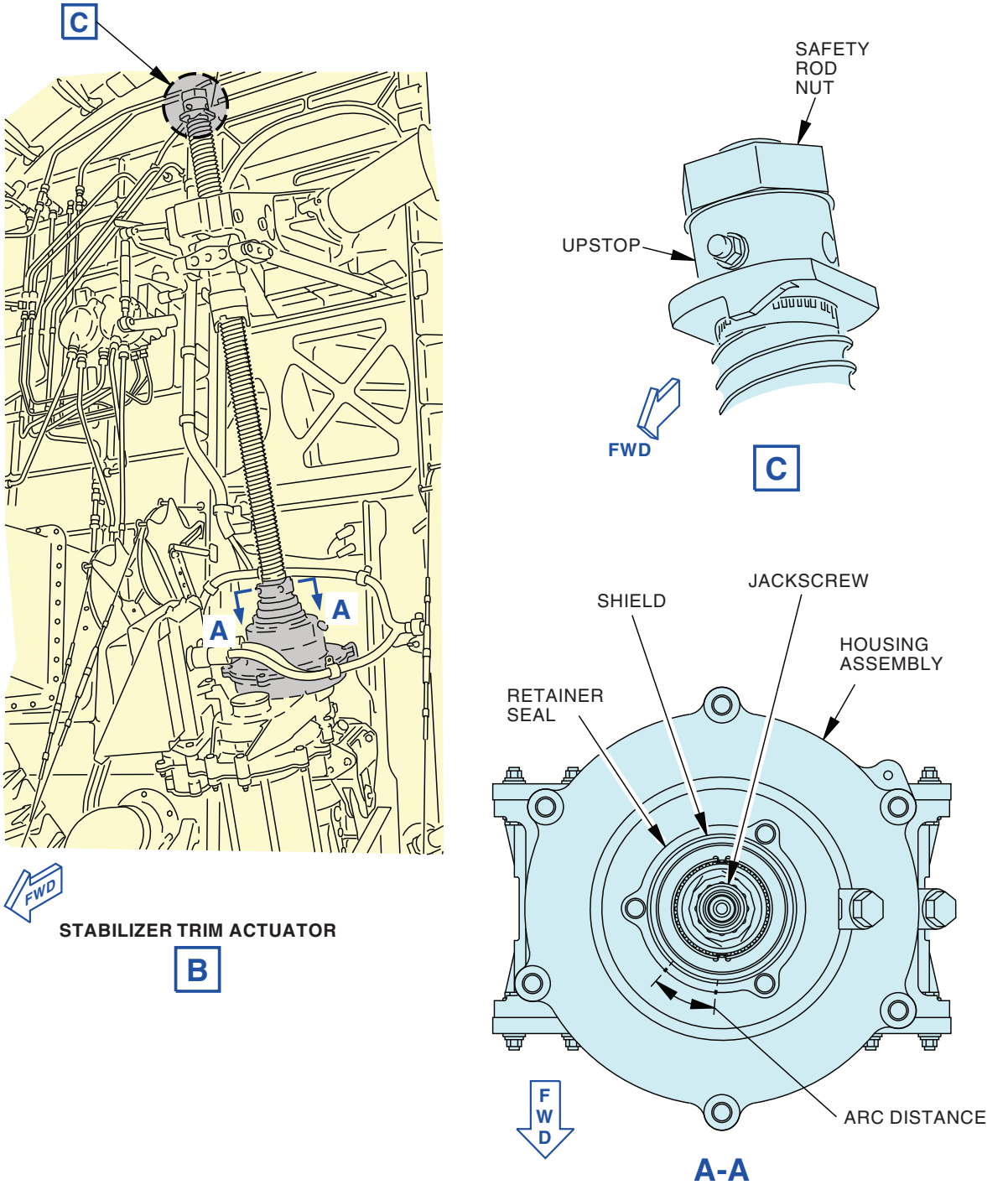
**VIEW THROUGH THE ACCESS DOOR TO AFT
UNPRESSURIZED COMPARTMENT, 311BL**

A

2412638 S00061531062_V1

**Stabilizer Trim Secondary Brake System Test
Figure 507/27-41-00-990-806 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Secondary Brake System Test
Figure 507/27-41-00-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-00-700-809

11. Stabilizer Trim Cutout Test

A. General

(1) Use this procedure to make sure the stabilizer trim cutout switch operates correctly. It stops the horizontal stabilizer when the control column command is opposite from the stabilizer trim switches.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
330	Subzone - Left Horizontal Stabilizer and Elevator
340	Subzone - Right Horizontal Stabilizer and Elevator

E. Prepare for the Test

SUBTASK 27-41-00-860-029

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-41-00-860-030

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Procedure

SUBTASK 27-41-00-710-012

(1) Do the stabilizer trim cutout test:

(a) Make sure that the STAB TRIM PRI switch, S272 and BACKUP switch, S149 on the aft area of the control stand is in the NORMAL position.

(b) Make sure that the STAB TRIM override switch on the P8 panel is in the NORM position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

(c) Position the stabilizer to an intermediate position (Approximately 8 units of trim is satisfactory).

(d) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

1) Make sure that the stabilizer leading edge moves down (APL NOSE UP).

2) While the stabilizer moves, push the captain's control column forward until the column cutout switch stops the stabilizer.

3) Immediately release the control column to the neutral position with the two stabilizer trim switches still held in the nose up position.

4) With the control column in the neutral position, make sure the stabilizer leading edge continues to move down.

5) Release the stabilizer trim switches.

(e) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

1) Make sure that the stabilizer leading edge moves up (APL NOSE DOWN).

2) While the stabilizer moves, pull the captain's control column aft until the column cutout switch stops the stabilizer.

3) Immediately release the control column to the neutral position with the two stab trim switches still held in the nose down position.

4) With the control column in the neutral position, make sure the stabilizer leading edge continues to move up.

5) Release the stabilizer trim switches.

(f) Position the stabilizer to an intermediate position (Approximately 8 units of trim is satisfactory).

(g) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

1) Make sure that the stabilizer leading edge moves down (APL NOSE UP).

2) While the stabilizer moves, push the captain's control column forward until the column cutout switch stops the stabilizer.

3) Hold the captain's control column in this position and immediately release the stabilizer trim switches.

4) Use a protractor, COM-1691 to make sure the captain's control column is between 2.90° and 3.70° forward from the neutral position.

5) Put the captain's control column back to the neutral position.

(h) Position the stabilizer to an intermediate position (Approximately 8 units of trim is satisfactory).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

1) Make sure that the stabilizer leading edge moves up (APL NOSE DOWN).

2) While the stabilizer moves, pull the captain's control column aft until the column cutout switch stops the stabilizer.

3) Hold the captain's control column in this position and immediately release the stabilizer trim switches.

4) Use a protractor, COM-1691 to make sure the captain's control column is between 3.20° and 4.21° aft from the neutral position.

5) Put the captain's control column back to the neutral position.

SUBTASK 27-41-00-860-031

(2) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

——— END OF TASK ———

TASK 27-41-00-710-801

12. Column Cutout Switches Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to make sure that the captain's and first officer's column cutout switches operate correctly.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Column Cutout Switches Test

SUBTASK 27-41-00-800-001

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF ALL FLIGHT CONTROL SURFACES, THE THRUST REVERSERS AND THE LANDING GEAR. THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS AND THE LANDING GEAR CAN MOVE QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-41-00-730-001

(2) Do the test of the captain's column cutout switch:

(a) Do one of the steps that follow:

1) Install the protractor, COM-1691, on the captain's control column.

NOTE: The neutral position of the control column is 6.9 degrees forward of the vertical position.

2) Use the MAX Display System (MDS) to view the control column position throughout this test:

a) Select MAINT.

b) Select MAINT DATA PGS.

c) Select 27 FLIGHT CONTROL - DISPLAY.

d) Make sure that COLUMN POSN shows.

NOTE: The neutral position of the control column is 0 degrees.

(b) Make sure that the STAB TRIM PRI switch is set to the NORMAL position.

(c) Make sure that the STAB TRIM override switch is set to the NORM position.

(d) Set the stabilizer to 8 units of trim.

(e) Move and hold the STAB TRIM switches on the captain's control wheel to the NOSE UP position.

1) Make sure that the stabilizer moves to the nose up position.

(f) Move the captain's control column forward to the maximum travel distance.

1) Make sure that the stabilizer movement stops.

(g) Move the captain's control column to the neutral position.

1) Make sure that the stabilizer moves to the nose up position.

(h) Move and hold the STAB TRIM switches on the captain's control wheel to the NOSE DN position.

1) Make sure that the stabilizer moves to the nose down position.

(i) Move the captain's control column rearward to the maximum travel distance.

1) Make sure that the stabilizer movement stops.

(j) Move the captain's control column to the neutral position.

1) Make sure that the stabilizer moves to the nose down position.

(k) Move the captain's control column rearward slowly until the stabilizer movement stops.

(l) Make sure that the captain's control column position is 3.20 to 4.21 degrees rearward from the neutral position.

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(m) Release the STAB TRIM switches on the captain's control wheel.

(n) Move the captain's control column to the neutral position.

(o) Move and hold the STAB TRIM switches on the captain's control wheel to the NOSE UP position.

1) Make sure that the stabilizer moves to the nose up position.

(p) Move the captain's control column forward slowly until the stabilizer movement stops.

(q) Make sure that the captain's control column position is 2.90 to 3.70 degrees forward from the neutral position.

(r) Release the STAB TRIM switches on the captain's control wheel.

(s) Move the captain's control column to the neutral position.

(t) If installed, remove the protractor, COM-1691, from the captain's control column.

SUBTASK 27-41-00-730-002

(3) Do the test of the first officer's column cutout switch:

(a) Do one of the steps that follow:

1) Install the protractor, COM-1691, on the first officer's control column.

NOTE: The neutral position of the control column is 6.9 degrees forward of the vertical position.

2) Use the MDS to view the control column position throughout this test:

a) Select MAINT.

b) Select MAINT DATA PGS.

c) Select 27 FLIGHT CONTROL - DISPLAY.

d) Make sure that COLUMN POSN shows.

NOTE: The neutral position of the control column is 0 degrees.

(b) Make sure that the STAB TRIM PRI switch is set to the NORMAL position.

(c) Make sure that the STAB TRIM override switch is set to the NORM position.

(d) Set the stabilizer to 8 units of trim.

(e) Move and hold the STAB TRIM switches on the first officer's control wheel to the NOSE UP position.

1) Make sure that the stabilizer moves to the nose up position.

(f) Move the first officer's control column forward to the maximum travel distance.

1) Make sure that the stabilizer movement stops.

(g) Move the first officer's control column to the neutral position.

1) Make sure that the stabilizer moves to the nose up position.

(h) Move and hold the STAB TRIM switches on the first officer's control wheel to the NOSE DN position.

1) Make sure that the stabilizer moves to the nose down position.

(i) Move the first officer's control column rearward to the maximum travel distance.

1) Make sure that the stabilizer movement stops.

(j) Move the first officer's control column to the neutral position.

1) Make sure that the stabilizer moves to the nose down position.

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(k) Move and hold the STAB TRIM switches on the first officer's control wheel to the NOSE UP position.

1) Make sure that the stabilizer moves to the nose up position.

(l) Move the first officer's control column forward slowly until the stabilizer movement stops.

(m) Make sure that the first officer's control column position is 2.90 to 3.70 degrees forward from the neutral position.

(n) Release the STAB TRIM switches on the first officer's control wheel.

(o) Move the first officer's control column to the neutral position.

(p) Move and hold the STAB TRIM switches on the first officer's control wheel to the NOSE DN position.

1) Make sure that the stabilizer moves to the nose down position.

(q) Move the first officer's control column rearward slowly until the stabilizer movement stops.

(r) Make sure that the first officer's control column position is 3.20 to 4.21 degrees rearward from the neutral position.

(s) Release the STAB TRIM switches on the first officer's control wheel.

(t) Move the first officer's control column to the neutral position.

(u) If installed, remove the protractor, COM-1691, from the first officer's control column.

(v) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

——— END OF TASK ———

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HORIZONTAL STABILIZER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of a horizontal stabilizer

(2) An installation of a horizontal stabilizer.

B. You can remove and install the horizontal stabilizer with or without the elevator and elevator tab.

C. This procedure is applicable for the left and right horizontal stabilizers.

TASK 27-41-11-000-801

2. Horizontal Stabilizer Removal

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405)

A. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-2030	Pin Removal Equipment - Horizontal Stabilizer Part #: C55005-19 Supplier: 81205
SPL-2035	Sling Equipment - Horizontal Stabilizer Part #: C55014-44 Supplier: 81205
STD-12395	Load Cell Equipment

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
330	Subzone - Left Horizontal Stabilizer and Elevator
340	Subzone - Right Horizontal Stabilizer and Elevator

D. Access Panels

Number	Name/Location
317BL	Tailcone Access Door

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(Continued)

Number	Name/Location
318BR	Tailcone Access Door
331A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
331AZ	Horizontal Stabilizer, Access Panel, Inboard L.E. Closure Rib
332AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AZ	Horizontal Stabilizer, Access Panel, Inboard T.E. Closure Rib
341A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
341AZ	Horizontal Stabilizer, Access Panel - Inbd L.E. Closure Rib
342AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AZ	Horizontal Stabilizer, Access Panel - Inbd T.E. Closure Rib

E. Prepare for the Removal

NOTE: If the horizontal stabilizer [1] is being removed and the same stabilizer, elevator, and elevator tab is going to be re-installed (without elevator or elevator tab rework or adjustment), then the following procedure will determine if a flight test is necessary.

NOTE: If the tailcone is removed or it's position is adjusted during the elevator or stabilizer removal procedure, the left and right elevator index plate [4] positions will be in a different position when the tailcone is re-installed. The elevator position cannot be properly measured with the elevator index plate [4] in a different location, therefore, you must do a power-off flight test.

SUBTASK 27-41-11-700-001

(1) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-41-11-860-001

(2) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-41-11-860-002

CAUTION MAKE SURE THAT YOU ARE CAREFUL WITH THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE SHARP EDGES ON THE TRAMMEL BAR, SPL-1677 CAN CAUSE CHAFING OR DAMAGE TO THE WIRES.

(3) Use the trammel trammel bar, SPL-1677, to set the B dimension of the horizontal stabilizer [1] at 39.17 ±0.01 in. (994.92 ±0.26 mm) (Figure 401).

NOTE: This is the neutral stabilizer setting of zero degrees or 4 units of trim.

(a) Make sure that you protect wire bundles when using the trammel trammel bar, SPL-1677.

SUBTASK 27-41-11-860-003

(4) Move the control column forward and aft several times and then let it go at the neutral position.

SUBTASK 27-41-11-860-004

(5) Measure the elevator position.

(a) Measure the left elevator position on the elevator index plate [4] (Figure 403).

1) Make sure to note if the elevator is above or below the neutral mark.

(b) Measure the right elevator position on the elevator index plate [4] (Figure 403).

1) Make sure to note if the elevator is above or below the neutral mark.

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SUBTASK 27-41-11-860-005

(6) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-41-11-860-006

(7) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the CUTOUT position.

SUBTASK 27-41-11-860-007

(8) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-11-860-008

(9) Open this circuit breaker (if installed) and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 27-41-11-860-009

(10) If it is necessary, use the stabilizer trim wheel on the control stand to set the horizontal stabilizer [1] at zero degrees (4 units of trim).

SUBTASK 27-41-11-480-001

(11) Install the lock assembly, SPL-1672, on the stabilizer trim wheel at the control stand (Figure 402).

(a) Turn the trim wheel to put the handle at the top of the wheel.

(b) Adjust the height of the trim lock to put the trim wheel handle correctly on the yoke.

(c) Install the pin through the yoke.

(d) Install the safety pin.

SUBTASK 27-41-11-410-001

(12) Open this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-41-11-010-001

(13) To access the left horizontal stabilizer, do this step:

Open these access panels:

Number	Name/Location
331A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
331AZ	Horizontal Stabilizer, Access Panel, Inboard L.E. Closure Rib
332AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body

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(Continued)

Number	Name/Location
333AZ	Horizontal Stabilizer, Access Panel, Inboard T.E. Closure Rib

SUBTASK 27-41-11-010-002

(14) To access the right horizontal stabilizer, do this step:

Open these access panels:

Number	Name/Location
341A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
341AZ	Horizontal Stabilizer, Access Panel - Inbd L.E. Closure Rib
342AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AZ	Horizontal Stabilizer, Access Panel - Inbd T.E. Closure Rib

F. Horizontal Stabilizer Removal

SUBTASK 27-41-11-480-002

(1) Install the sling equipment, SPL-2035 (Figure 404).

(a) Remove the bolts [2] and bolts [3] from the horizontal stabilizer [1] top surface.

CAUTION

MAKE SURE YOU INSTALL THE HOISTING SLING CORRECTLY WHEN YOU REMOVE THE HORIZONTAL STABILIZER WITH OR WITHOUT THE ELEVATOR INSTALLED. IF YOU DO NOT INSTALL THE HOISTING SLING CORRECTLY FOR THE REMOVAL OF THE HORIZONTAL STABILIZER, YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(b) Attach the sling equipment, SPL-2035, to the horizontal stabilizer [1] with fasteners.

NOTE: Low clearance sling is used only with elevator removed.

(c) Attach the load cell, STD-12395, to the sling equipment, SPL-2035.

(d) Attach a hoist to the load cell, STD-12395.

CAUTION

YOU MUST CAREFULLY LIFT THE HOISTING SLING. A SUDDEN MOVEMENT OF THE HOISTING SLING CAN CAUSE DAMAGE TO THE HORIZONTAL STABILIZER.

(e) Lift the weight of the horizontal stabilizer [1].

1) Use the load cell, STD-12395, on the sling to make sure that the force used when lifting horizontal stabilizer [1] is not more than:

- 888 lb (403 kg) (830 lb (376 kg) airplane part + 58 lb (26 kg) tool) with elevator attached
- 688 lb (312 kg) (630 lb (286 kg) airplane part + 58 lb (26 kg) tool) without elevator attached.

SUBTASK 27-41-11-020-001

WARNING

DECREASE THE PRESSURE BEFORE YOU DISCONNECT THE HYDRAULIC LINE. PRESSURIZED FLUID CAN CAUSE INJURIES TO PERSONNEL.

(2) Disconnect the hydraulic line [25] of the elevator tab installed aft of the stabilizer rear spar.

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SUBTASK 27-41-11-020-002

(3) Disconnect these electrical connectors:

(a) Flight recorder position sensor

(b) Exterior logo light (if installed).

SUBTASK 27-41-11-020-003

(4) If the elevator is attached to the horizontal stabilizer, do this step:

(a) Open this access panel:

Number	Name/Location
318BR	Tailcone Access Door

or

Open this access panel:

Number	Name/Location
317BL	Tailcone Access Door

(b) Remove the nut [23], nut [24], and bolt [21] to disconnect the elevator control pushrod [5] from the elevator.

SUBTASK 27-41-11-020-004

(5) Remove the horizontal stabilizer [1] (Figure 405).

NOTE: If the bolts are frozen, use pin removal equipment, SPL-2030, to assist in removal.

(a) At the horizontal stabilizer rear spar, do these steps:

1) Remove the nut [32] and washer [34] from the bolt [28].

2) Remove the nuts [35] and washers [34] from the bolts [38].

3) Remove the bolt [28], washer [29], and washer [27].

a) If it is necessary, remove the seal support structure before you remove this bolt.

b) Keep a record of the bolt length and location for the installation.

4) Remove the bolts [38], washers [37], and washers [27].

a) If it is necessary, remove the seal support structure before you remove these bolts.

b) Keep a record of each bolt length and location for the installation.

5) Remove the washer [30], washers [39], pin [33], pins [36], and antirotation lock [31].

a) If it is necessary, adjust the tension of the sling to help remove the pins.

6) Remove and inspect each bolt or pin.

NOTE: If an existing bolt or pin has missing plating, cracked, or flaking, prior to further flight replace with new parts or repair in accordance with a method approved by an Authorized Representative for the Boeing Commercial Airplanes (BCA) Organization Designation Authorization who has been authorized to make those findings.

a) If the bolts or pins show signs of damage: pitting, corrosion, no plating (pins only), galling (bolts only) or wear beyond drawing tolerances, replace them with the improved bolts or pins.

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b) If the pins or bolts show no sign of damage, they can be reinstalled or replaced with an improved front spar bolts or rear spar pins designed to resist wear, galling, or corrosion.

(b) At the horizontal stabilizer front spar, do these steps:

1) Remove the pins [44], nuts [43], washers [42], washers [45], washers [46], washers [40], and bolts [41].

(c) Lift the horizontal stabilizer [1] out horizontally, then lower to the support stand.

(d) Remove and inspect each bolt or pin.

1) If the bolts or pins show signs of damage: pitting, corrosion, no plating (pins only), galling (bolts only) or wear beyond drawing tolerances, replace them with the improved bolts or pins.

2) If the pins or bolts show no sign of damage, they can be reinstalled or replaced with an improved front spar bolts or rear spar pins designed to resist wear, galling, or corrosion.

(e) If there is an indication of hydraulic fluid contamination on the lugs or clevises, examine the components in the vertical fin for external leakage:

- Main and standby rudder Power Control Unit (PCU).
- Hydraulic hoses and tubing.

NOTE: An example of hydraulic fluid contamination is cracked, peeled, or blistered paint.

SUBTASK 27-41-11-080-003

(6) If it is necessary, remove the sling equipment, SPL-2035.

(a) Release tension applied to the sling equipment, SPL-2035.

(b) Remove the hoist from the load cell, STD-12395.

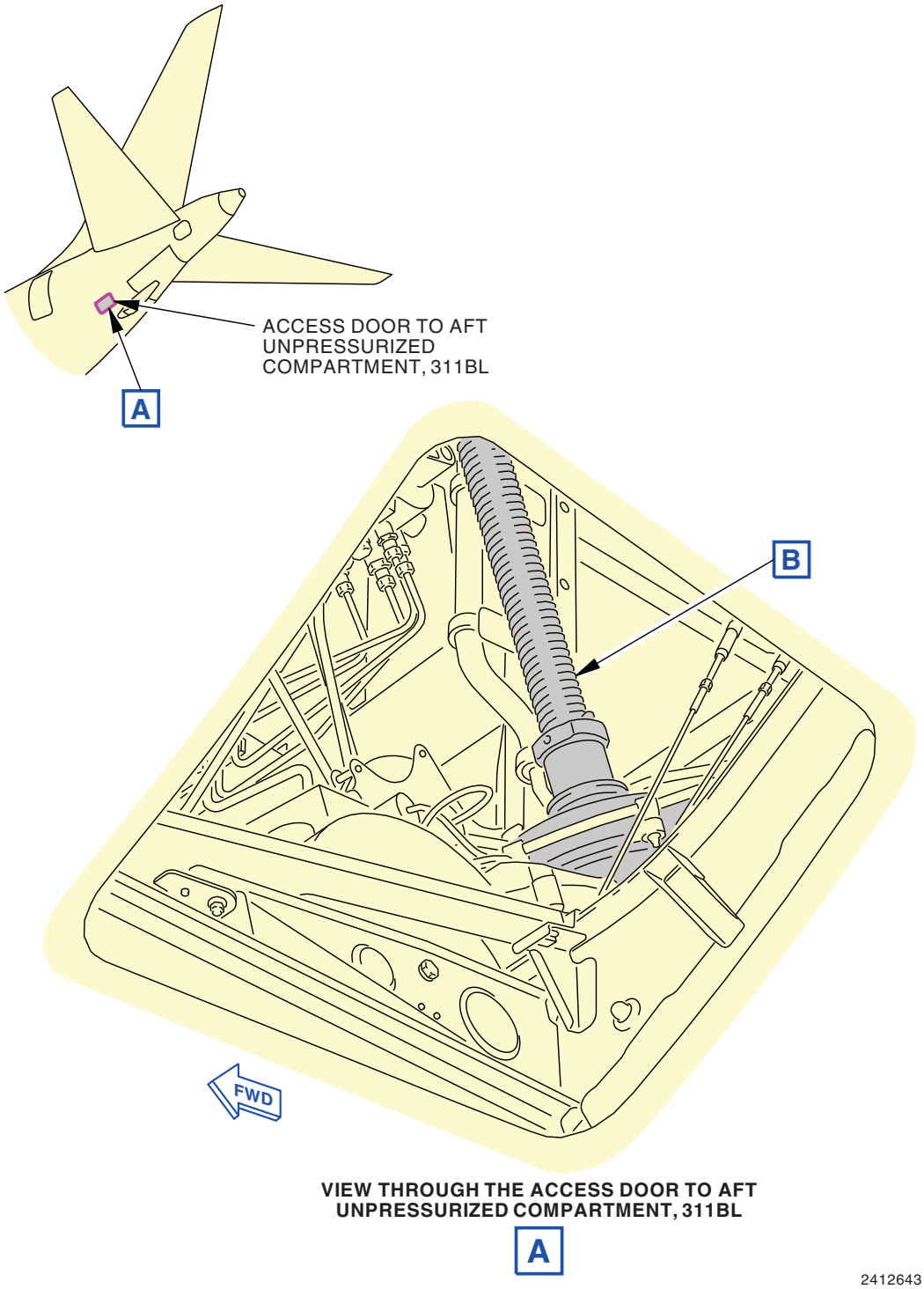
(c) Remove the load cell, STD-12395, from the sling equipment, SPL-2035.

(d) Remove the fasteners that attach the sling equipment, SPL-2035.

(e) Install the bolts [2] and bolts [3] at the horizontal stabilizer [1] top surface.

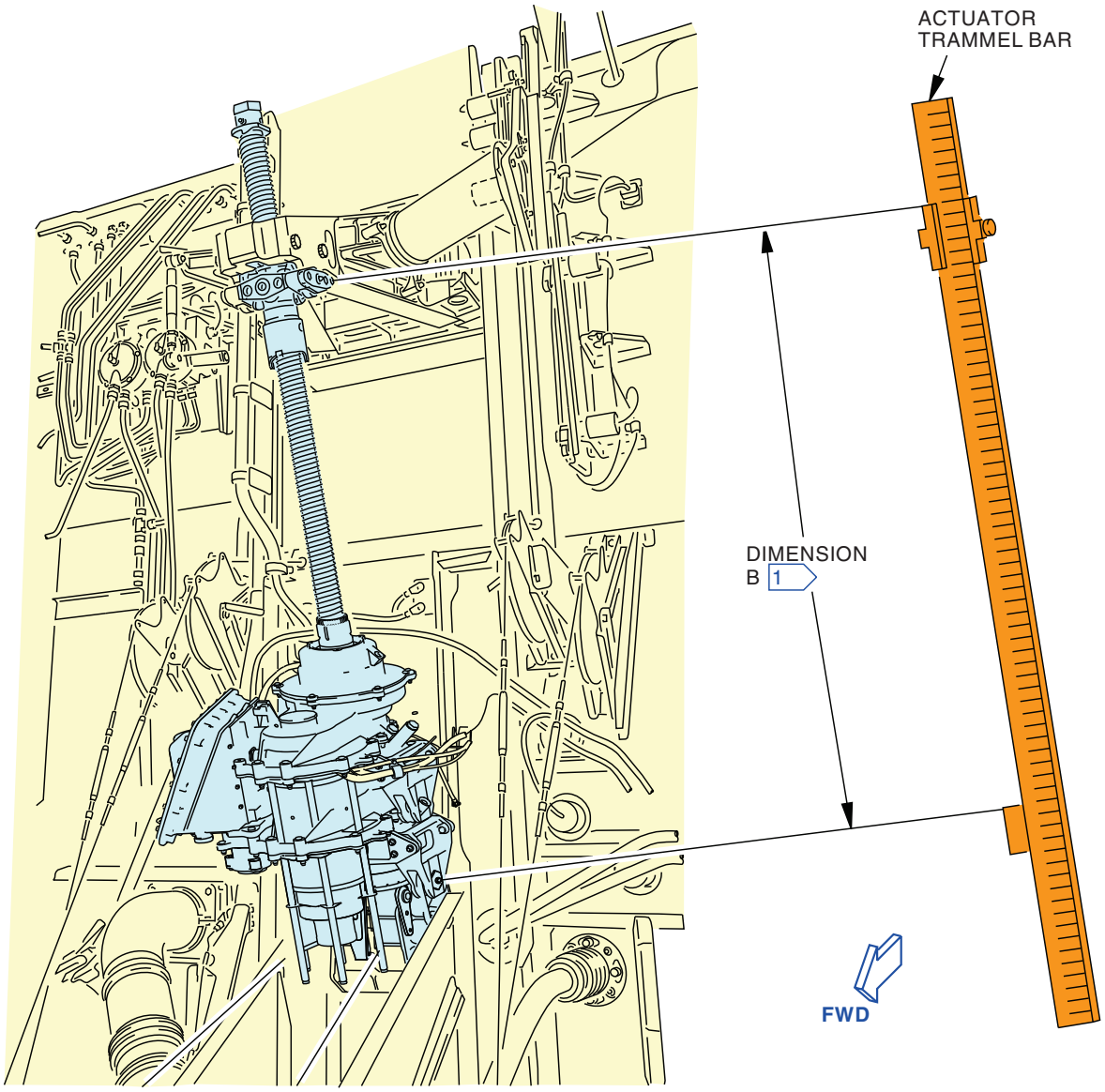
——— END OF TASK ———

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Stabilizer B Dimension Setting
Figure 401/27-41-11-990-801 (Sheet 1 of 2)

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STABILIZER TRIM ACTUATOR

B

NOTE:

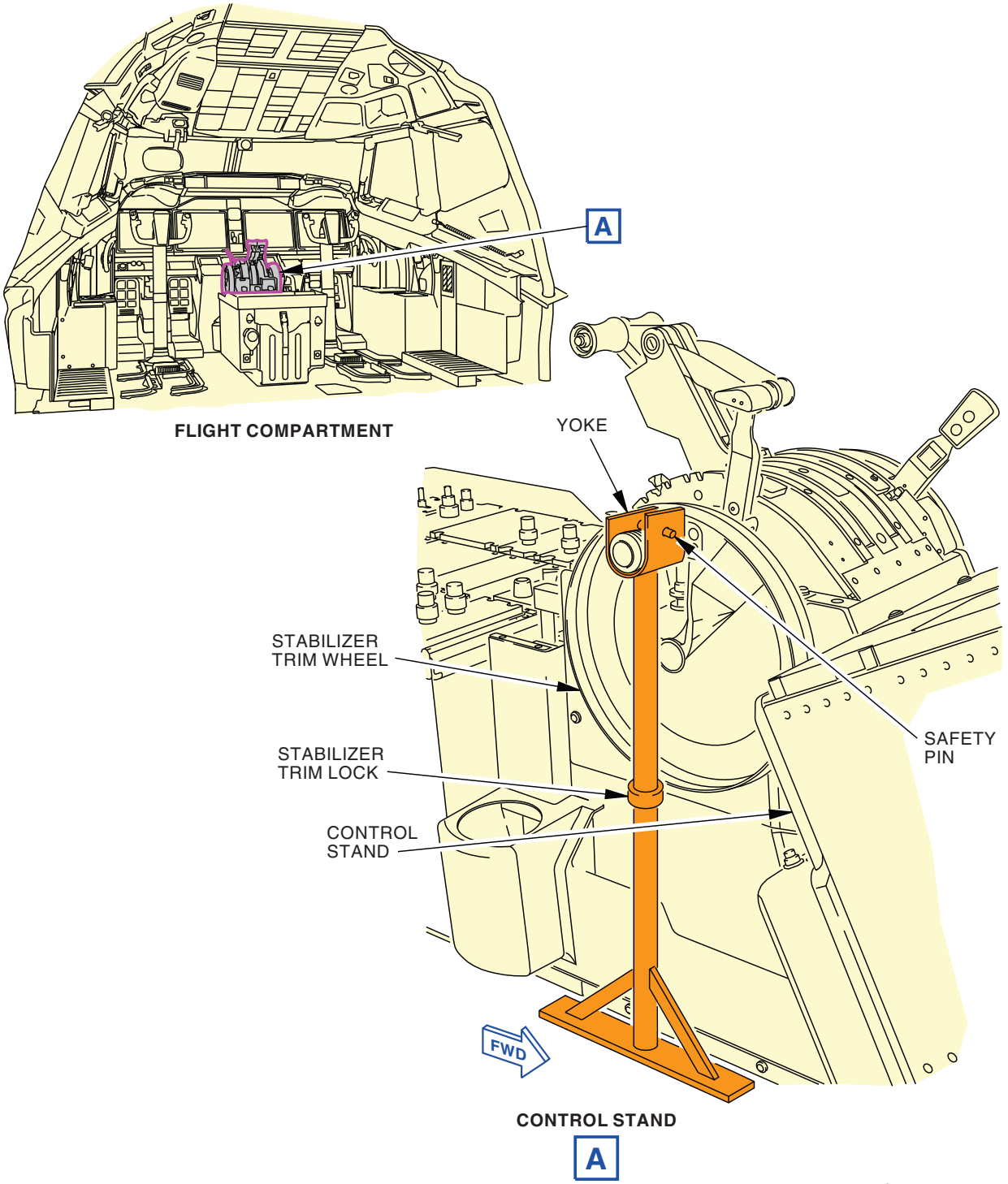
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412644 S00061531073_V2

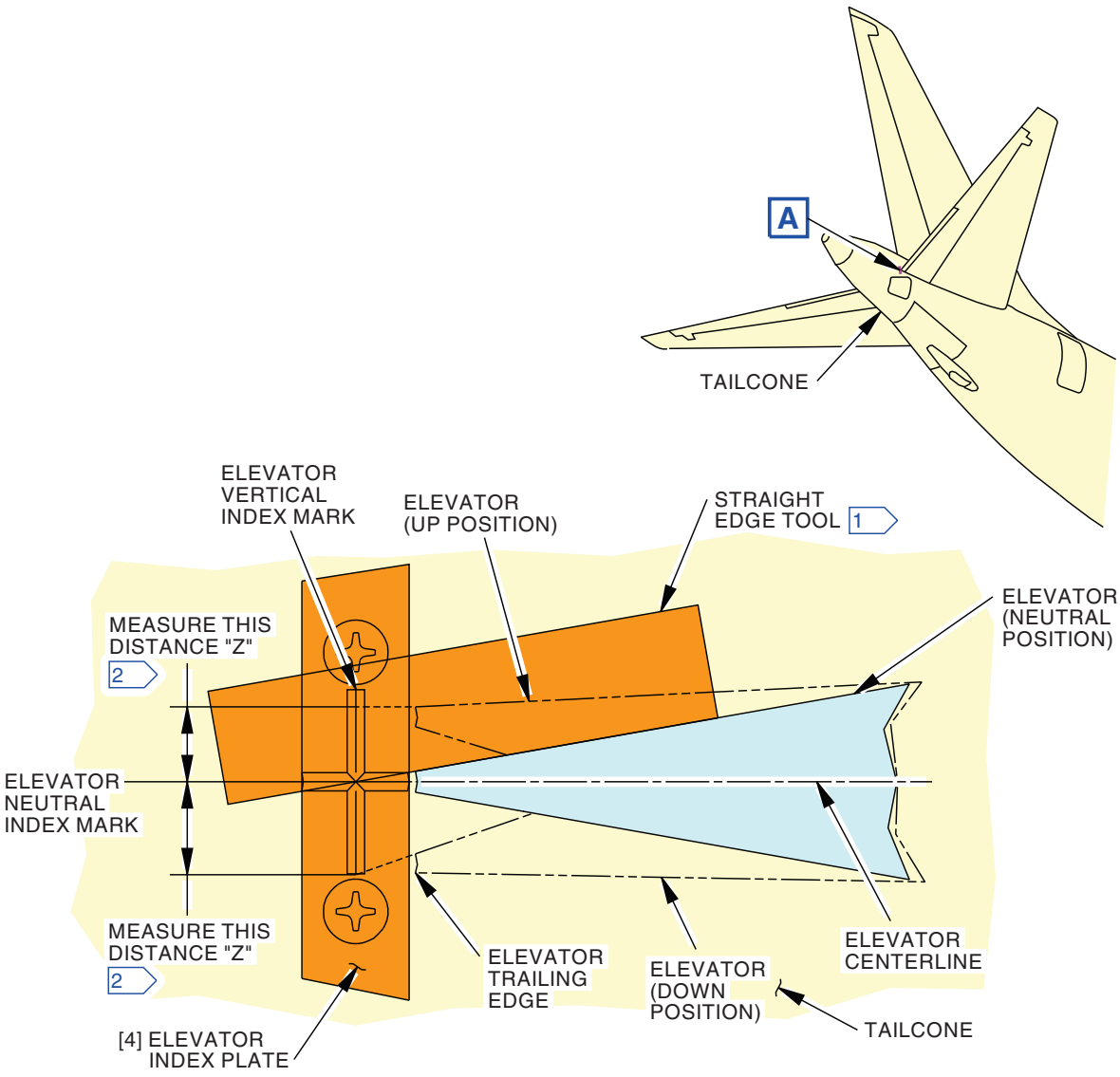
Stabilizer B Dimension Setting
Figure 401/27-41-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Lock Installation
Figure 402/27-41-11-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR POSITION ADJUSTMENT MEASUREMENTS (RIGHT SIDE)

A

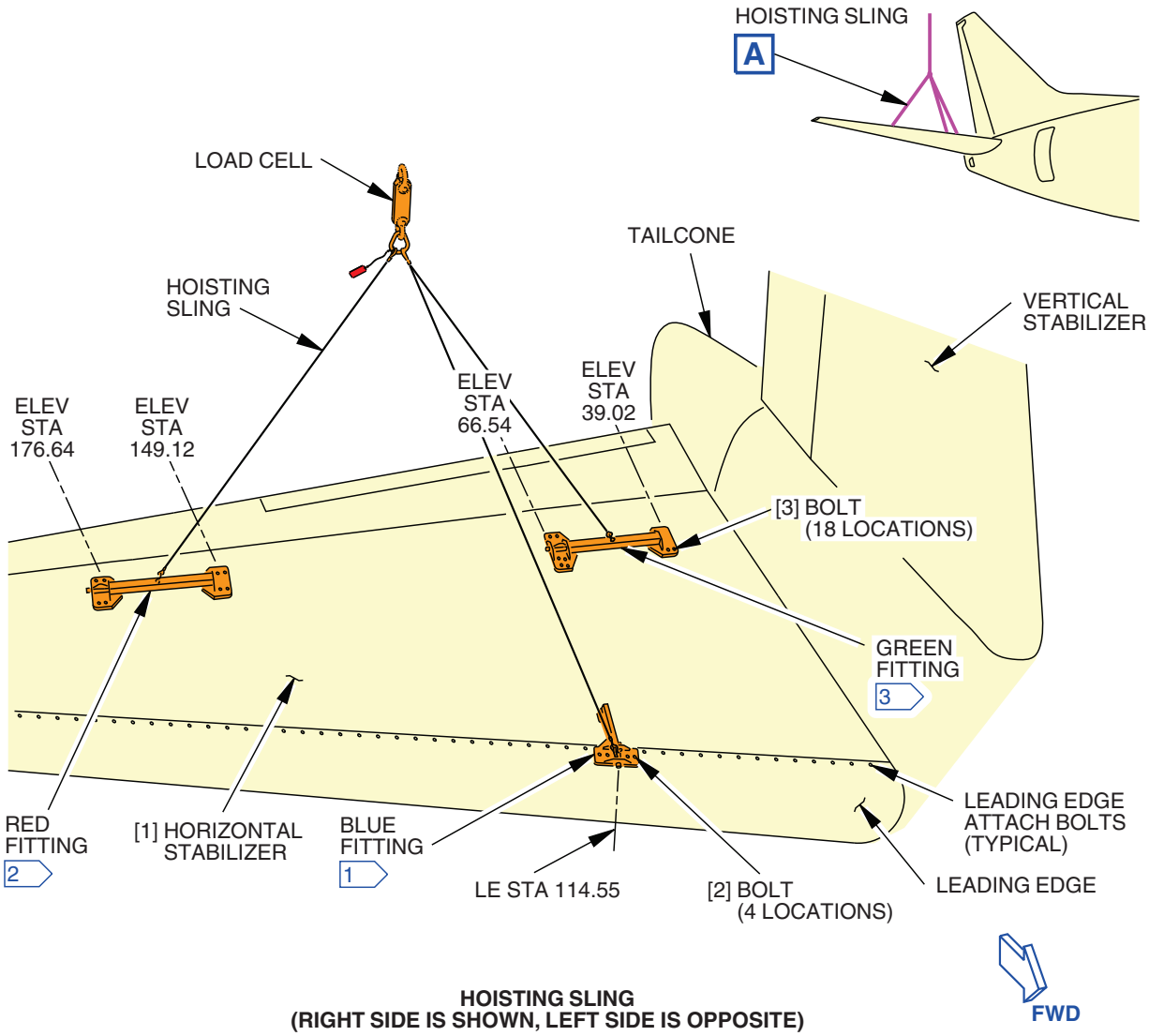
1 ELEVATOR IS AT NEUTRAL WHEN THE BOTTOM EDGE OF A STRAIGHT EDGE TOOL PLACED ON THE UPPER ELEVATOR SURFACE PASSES THROUGH THE MIDDLE OF THE ELEVATOR INDEX PLATE, AS SHOWN.

2 THE DISTANCE "Z" IS MEASURED USING A STRAIGHT EDGE TOOL ON THE UPPER SURFACE OF THE ELEVATOR TO INTERSECT WITH THE ELEVATOR VERTICAL INDEX MARK AT THE ELEVATOR NEUTRAL INDEX MARK.

2412645 S00061531074_V1

Elevator Position Measurement
Figure 403/27-41-11-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HOISTING SLING
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

A

1 THE FORWARD ATTACH (BLUE) FITTING IS ATTACHED TO THE STABILIZER FRONT SPAR (LEADING EDGE ATTACH BOLTS).

TWO BOLTS ARE USED ON THE INBOARD SIDE OF THE LEADING EDGE RIB (LE STA 114.55) AND TWO BOLTS ARE USED ON THE OUTBOARD SIDE OF THE LEADING EDGE RIB (LE STA 114.55).

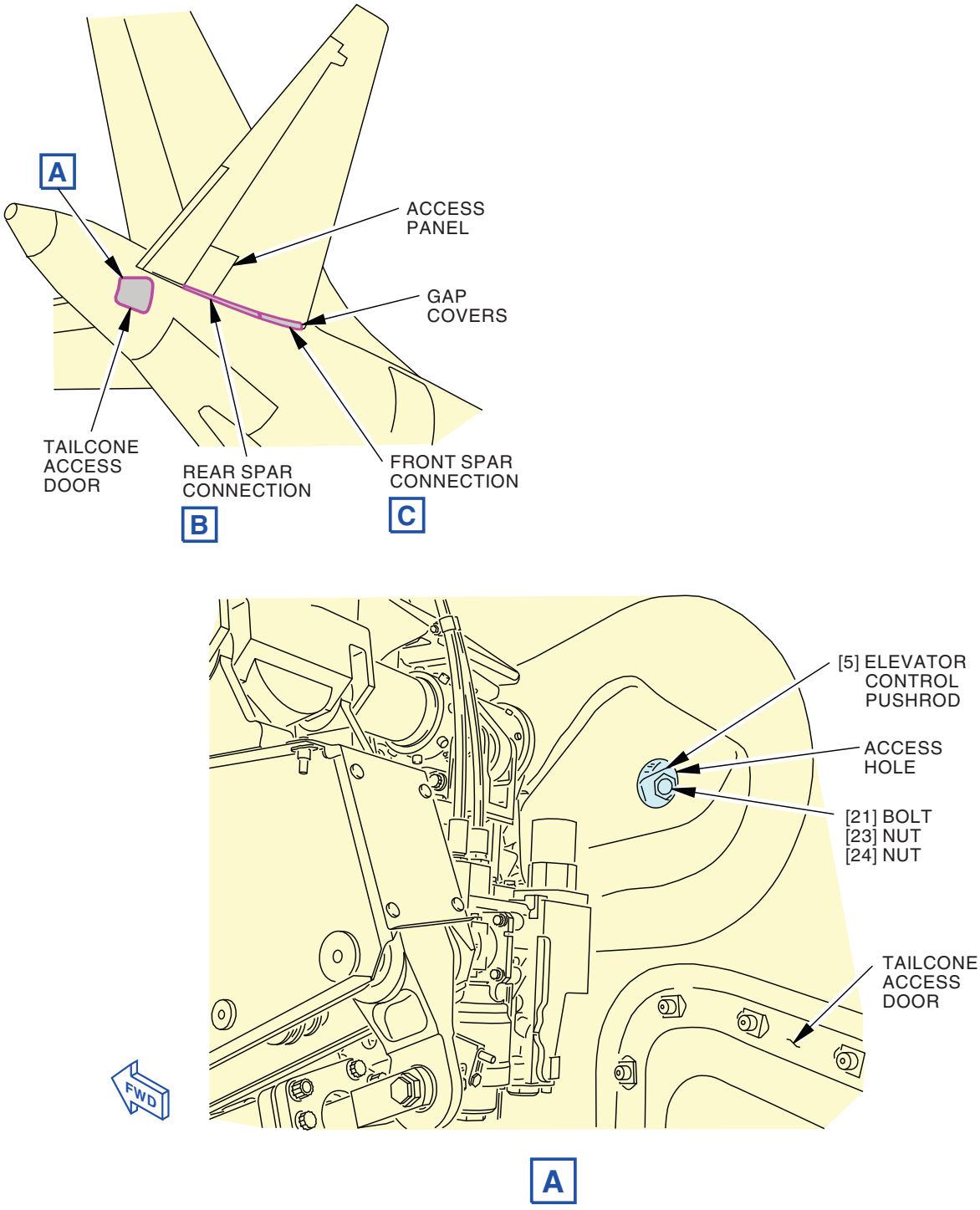
2 THE AFT OUTBOARD ATTACH (RED) FITTING IS ATTACHED TO THE ELEVATOR HINGE SUPPORT RIBS, FOUR BOLTS ARE USED AT ELEV STA 176.64 AND FOUR BOLTS ARE USED AT ELEV STA 149.12.

3 THE AFT INBOARD ATTACH (GREEN) FITTING IS ATTACHED TO THE ELEVATOR HINGE SUPPORT RIBS, SIX BOLTS ARE USED AT ELEV STA 66.54 AND FOUR BOLTS ARE USED AT ELEV STA 39.02.

2412646 S00061531075_V4

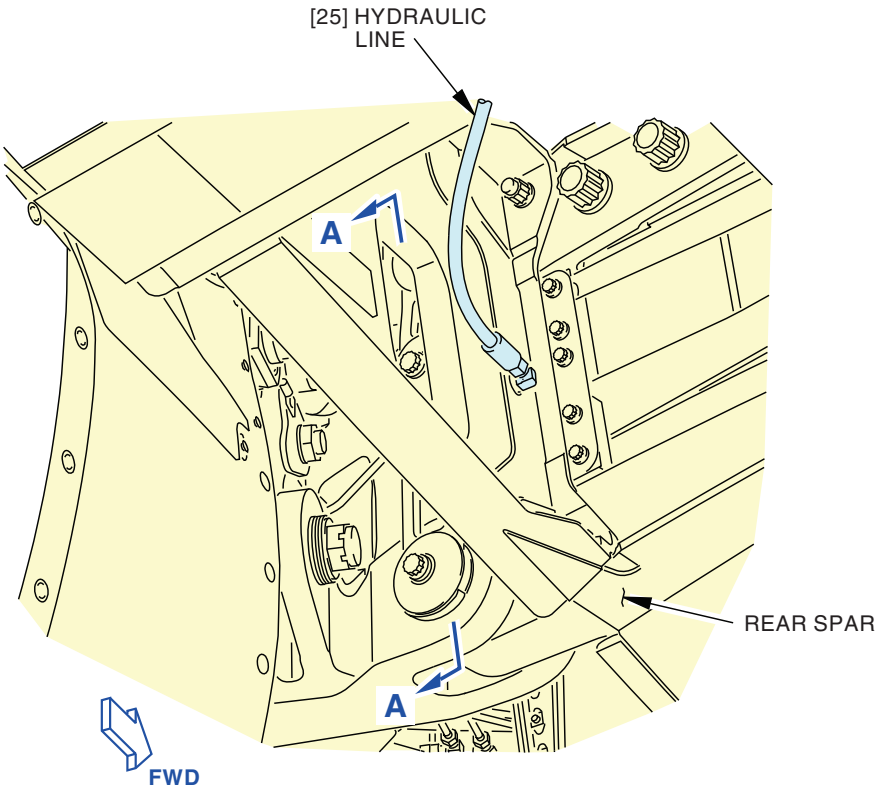
Horizontal Stabilizer Hoisting Sling Installation
Figure 404/27-41-11-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Horizontal Stabilizer Installation
Figure 405/27-41-11-990-805 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



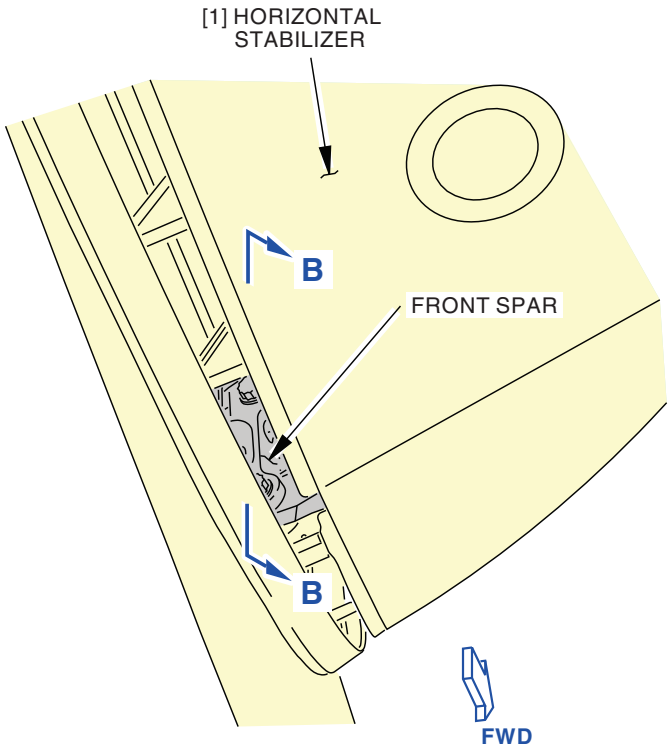
REAR SPAR CONNECTION
(ACCESS PANELS AND GAP COVERS IS REMOVED)
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

B

2412648 S00061531077_V1

Horizontal Stabilizer Installation
Figure 405/27-41-11-990-805 (Sheet 2 of 6)

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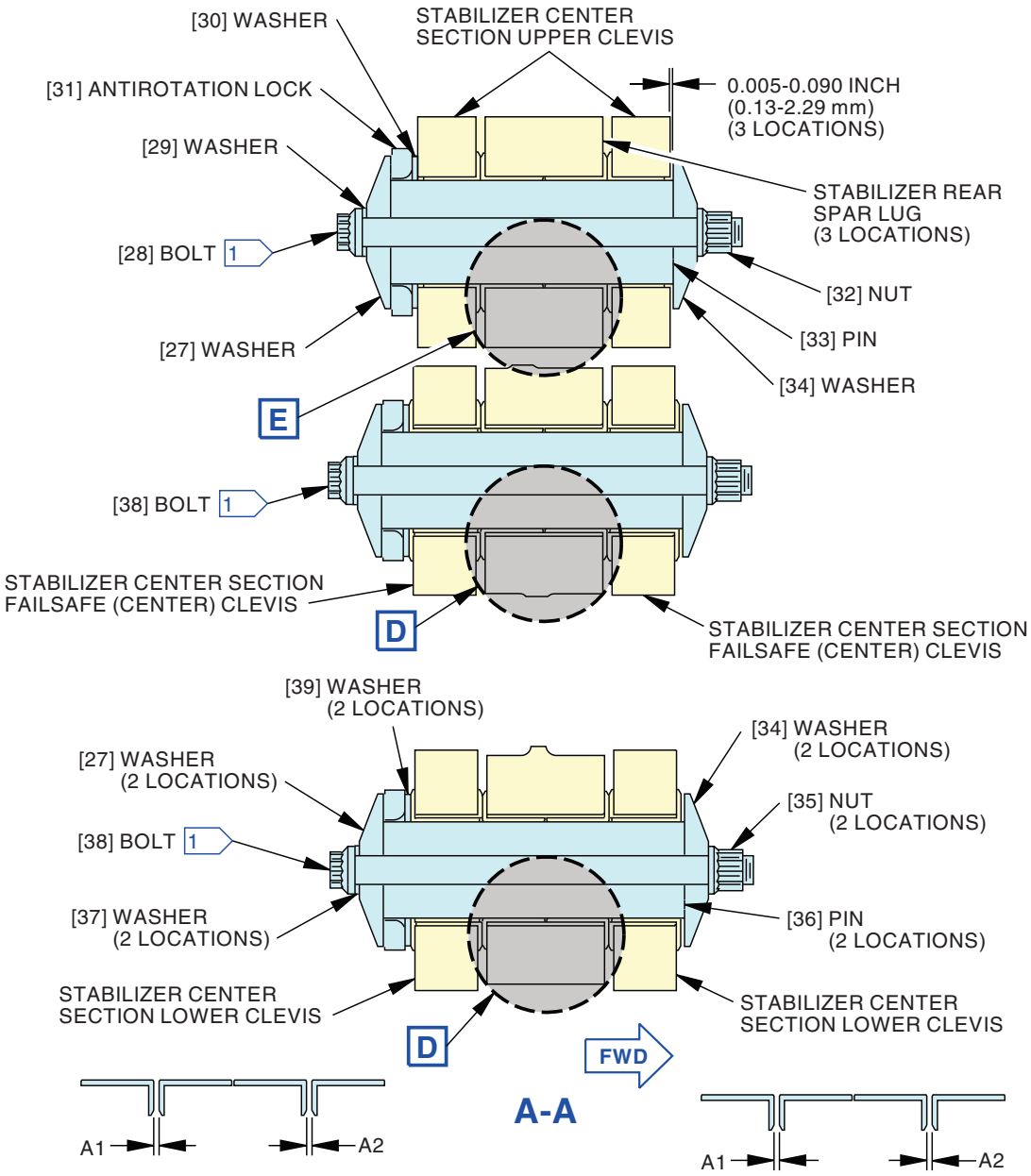
**FRONT SPAR CONNECTION
(GAP COVERS IS REMOVED)
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)**

C

2412649 S00061531078_V1

**Horizontal Stabilizer Installation
Figure 405/27-41-11-990-805 (Sheet 3 of 6)**

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NOTE:

COMBINED GAPS BETWEEN BUSHINGS AT A1 AND A2 SHOULD BE 0.032 INCH (0.81 mm), OR LESS.

D

NOTE:

COMBINED GAPS BETWEEN BUSHINGS AT A1 AND A2 SHOULD BE 0.046 INCH (1.17 mm), OR LESS.

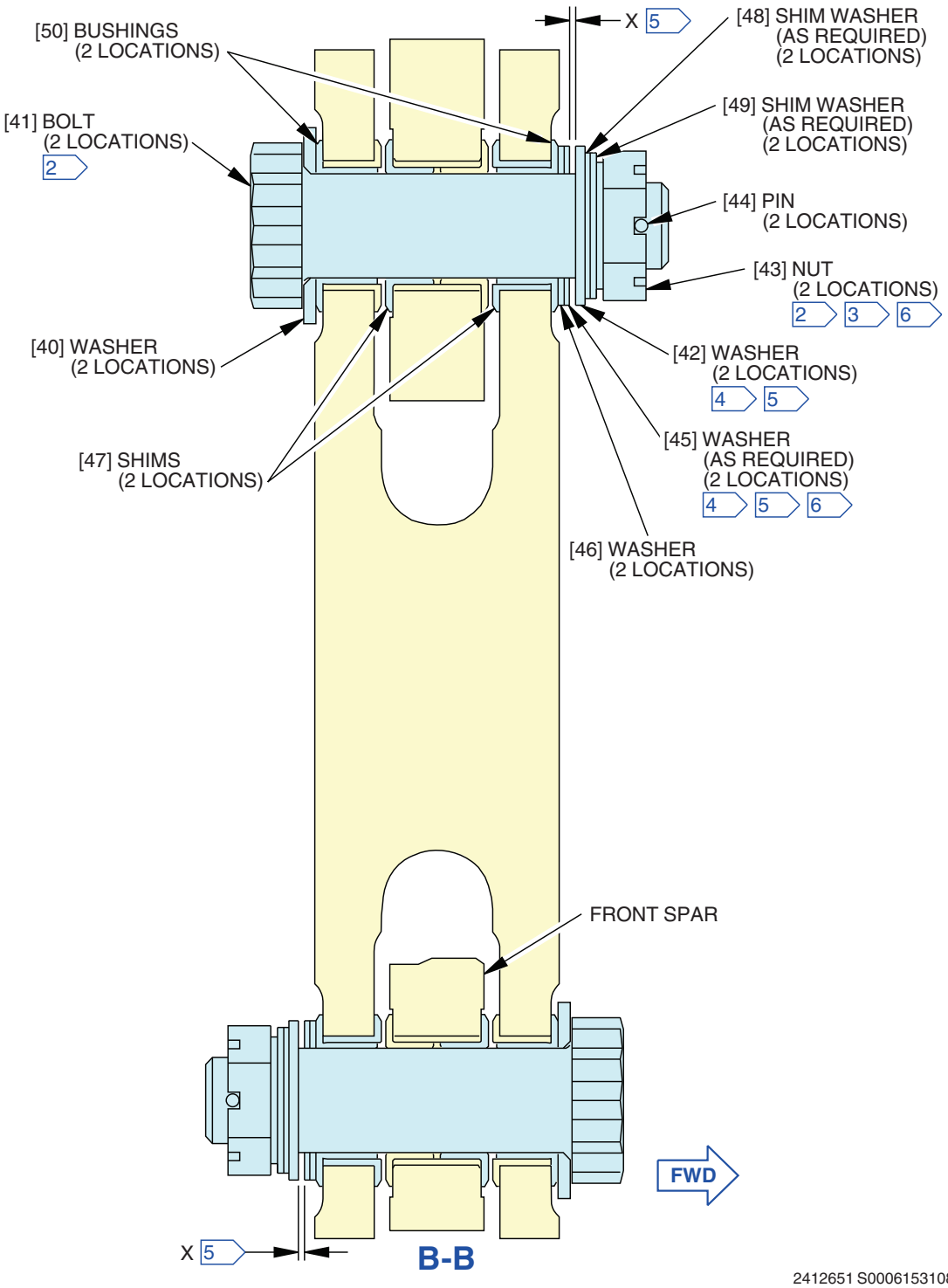
E

1 BOLT HEAD DIRECTION OPTIONAL (WASHER NEEDS TO BE INSTALLED AT BOLT HEAD LOCATION).

2412650 S00061531079_V1

Horizontal Stabilizer Installation
Figure 405/27-41-11-990-805 (Sheet 4 of 6)

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Horizontal Stabilizer Installation
Figure 405/27-41-11-990-805 (Sheet 5 of 6)

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2 BEFORE INSTALLING NUT, COAT THREADS OF BOLT AND NUT WITH BMS 3-28 ANTI-SEIZE COMPOUND.

3 TORQUE NUT TO AN INITIAL TORQUE OF 50 POUND-INCHES (5.6 NEWTON-METERS).

4 AFTER INITIAL TORQUE, WASHER [42] MUST TOUCH STEPPED PART OF BOLT [41]. IF THE WASHER [42] DOES NOT CONTACT STEPPED PART OF BOLT [41], REMOVE WASHERS [45] UNTIL IT DOES.

5 IF THE GAP "X" BETWEEN THE WASHERS [45] AND WASHER [42] IS MORE THAN 0.002 INCH (0.05 mm), ADD WASHERS [45] UNTIL THE GAP "X" IS BETWEEN 0.000-0.002 INCH (0.00-0.05 mm).

6 WHEN WASHERS [45] ADJUSTMENTS ARE COMPLETE, TORQUE NUT TO 950-1400 POUND-INCHES (107.3-158.2 NEWTON-METERS).

2412652 S00061531081_V2

Horizontal Stabilizer Installation
Figure 405/27-41-11-990-805 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-11-400-801

3. Horizontal Stabilizer Installation

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, and Figure 406)

A. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-700-802	Elevator Tab Control System - Operational Test (P/B 501)
27-31-00-700-805	Elevator Travel Limit - Test (P/B 501)
27-31-00-710-803	Elevator Power-Off Flight Test (P/B 501)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-820-802	Elevator Control Pushrods - Adjustment (P/B 501)
27-31-00-820-803	Input Rods of the Elevator Power Control Unit - Adjustment (P/B 501)
27-31-11-400-801	Elevator - Installation (P/B 401)
31-31-51-700-801	Elevator Position Transmitter Installation Test (P/B 401)
33-49-00-960-802	Logo Light - Light Assembly Replacement (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-2035	Sling Equipment - Horizontal Stabilizer Part #: C55014-44 Supplier: 81205
SPL-13581	Alignment Eqpt - Horizontal Stab Installation Part #: C55008-18 Supplier: 81205
STD-12395	Load Cell Equipment

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
D50004	Compound - Antiseize	BMS3-28
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

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(Continued)

Zone	Area
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge

E. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door
331A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
331AZ	Horizontal Stabilizer, Access Panel, Inboard L.E. Closure Rib
332AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AZ	Horizontal Stabilizer, Access Panel, Inboard T.E. Closure Rib
341A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
341AZ	Horizontal Stabilizer, Access Panel - Inbd L.E. Closure Rib
342AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AZ	Horizontal Stabilizer, Access Panel - Inbd T.E. Closure Rib

F. Horizontal Stabilizer Installation

NOTE: Do an inspection of the area near the horizontal stabilizer identification plate for a stenciled note or placard which restricts the horizontal stabilizer to an individual airplane because of a matched set approval.

SUBTASK 27-41-11-480-003

(1) Install the sling equipment, SPL-2035 (Figure 404):

(a) Remove the bolts [2] and bolts [3] from the horizontal stabilizer [1] top surface.

(b) Attach the sling equipment, SPL-2035, to the horizontal stabilizer [1] with fasteners.

(c) Attach the load cell, STD-12395, to the sling equipment, SPL-2035.

NOTE: Low clearance sling is used only with elevator removed.

(d) Attach a hoist to the load cell, STD-12395.

SUBTASK 27-41-11-640-001

(2) Lubricate the bolts, washers, bushings in the front and rear spar fittings, and pins with corrosion inhibiting material, G50136.

(a) Do not put the corrosion inhibiting material, G50136, on the threaded portion of the bolts.

SUBTASK 27-41-11-640-002

(3) Lubricate the bolt threads and nut threads with compound, D50004.

NOTE: This anti-sieze compound prevents galling of the threads.

SUBTASK 27-41-11-420-001

(4) Install the horizontal stabilizer [1] (Figure 405).

CAUTION YOU MUST CAREFULLY LIFT THE HOISTING SLING. A SUDDEN MOVEMENT OF THE HOISTING SLING CAN CAUSE DAMAGE TO THE HORIZONTAL STABILIZER.

(a) Lift and put the horizontal stabilizer [1] in its position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Use the load cell, STD-12395, on the sling to make sure that the force used when lifting horizontal stabilizer [1] is not more than:
   - 888 lb (403 kg) (830 lb (376 kg) airplane part + 58 lb (26 kg) tool) with elevator attached
   - 688 lb (312 kg) (630 lb (286 kg) airplane part + 58 lb (26 kg) tool) without elevator attached.

(b) Align the horizontal stabilizer [1] with the rear and front fittings on the airplane.

(c) Temporarily install the bolts [41] on the front spar of the horizontal stabilizer [1].

NOTE: You can use the stabilizer alignment tool, alignment equipment, SPL-13581, to align the stabilizer in place of using the bolts [41].

(d) At the rear spar of the horizontal stabilizer [1], do these steps:

1) Install the antirotation lock [31], pin [33], washer [30], washer [27], washer [29], bolt [28] with the bolt heads either forward or aft, washer [34], and nut [32].

2) Install the pins [36], washers [39], washers [27], washers [37], bolts [38] with the bolt heads either forward or aft, washers [34], and nuts [35].

3) Measure the clearance between the bushing faces of the external and internal lugs (Figure 405).

NOTE: This clearance measurement is for the sum of the two clearances (A1 and A2) at each of the three rear spar clevises.

a) Make sure that the maximum allowable clearance for each clevis at the center (failsafe) and lower lugs (sum of the two clearances, A1 and A2, at each lug) is 0.032 in. (0.81 mm).

b) Make sure that the maximum allowable clearance for the upper clevis (sum of the two clearances, A1 and A2) is 0.046 in. (1.17 mm).

4) Tighten the nut [32] to 575 ±75 in-lb (65 ±9 N·m) and nuts [35] to 575 ±75 in-lb (65 ±9 N·m) but keep a clearance of 0.005 in. (0.13 mm) – 0.090 in. (2.29 mm) after the nut is tightened.

NOTE: The clearance of 0.005 in. (0.13 mm) – 0.090 in. (2.29 mm) after the nut [32], and nut [35] are tightened, is taken between the Stabilizer Center Section Clevis and washer [34].

(e) At the front spar of the horizontal stabilizer [1], do these steps:

1) Remove the bolts [41].

2) Measure the clearance between the bushing faces of the lugs and clevises.

3) Install the shim [47] to get a maximum clearance of 0.005 in. (0.13 mm) for each side of the lugs (maximum of 0.010 in. (0.25 mm) total allowable clearance for each clevis).

4) Install the washers [40], bolts [41], washers [46], washers [42], washers [45], and nuts [43].

a) If it is necessary, add the shim washers [48] and/or shim washers [49] between the nut [43] and washer [42] to align the pin hole of the bolt [41] and nut [43].

b) Tighten the nuts [43] to an initial torque of 50 in-lb (5.6 N·m).

c) Make sure that the stepped portion of the bolts [41] touch the washers [42].

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<1> If it is necessary, remove the washers [45] until the stepped portion of the bolts [41] touch the washers [42]

d) Make sure that the gap "X" between the washers [45] and washers [42] is 0.002 in. (0.05 mm) or less.

<1> If it is necessary, add washers [45] until the gap "X" is between 0.000 in. (0.00 mm) - 0.002 in. (0.05 mm).

e) Tighten the nuts [43] to 950 in-lb (107.3 N·m) -1400 in-lb (158.2 N·m).

f) Install the new pins [44].

<1> If it is necessary, tighten or loosen the nuts [43], to get the pin hole alignment.

SUBTASK 27-41-11-080-001

(5) Remove the sling equipment, SPL-2035, from the horizontal stabilizer [1].

(a) Release tension applied to the sling equipment, SPL-2035.

(b) Remove the hoist from the load cell, STD-12395.

(c) Remove the load cell, STD-12395 from the sling equipment, SPL-2035.

(d) Remove the fasteners that attach the sling equipment, SPL-2035.

(e) Install the bolts [2] and bolts [3] at the horizontal stabilizer [1] top surface.

SUBTASK 27-41-11-420-002

(6) If the elevator is not attached to the horizontal stabilizer [1], do this task: Elevator - Installation, TASK 27-31-11-400-801.

SUBTASK 27-41-11-420-003

(7) If the elevator is attached to the horizontal stabilizer [1], connect the elevator control pushrod [5] to the elevator:

(a) Move the elevator by hand to the neutral position.

(b) Install the bolt [21] with the bolt head inboard and nuts [23], nuts [24] to connect the elevator control pushrod [5] to the elevator.

(c) Close this access door:

Number	Name/Location
318BR	Tailcone Access Door

or

Close this access door:

Number	Name/Location
317BL	Tailcone Access Door

SUBTASK 27-41-11-420-004

(8) Connect the hydraulic line [25] (that is installed just aft of the stabilizer rear spar) of the elevator tab mechanism actuator.

(a) Tighten the B-nut on hydraulic line [25] to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 27-41-11-420-005

(9) Connect these electrical connectors:

(a) Flight recorder position sensor

(b) Exterior logo light (if it is installed).

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SUBTASK 27-41-11-410-002

(10) For the left horizontal stabilizer, do this step:

(a) Close these access panels:

Number	Name/Location
331A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
331AZ	Horizontal Stabilizer, Access Panel, Inboard L.E. Closure Rib
332AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AZ	Horizontal Stabilizer, Access Panel, Inboard T.E. Closure Rib

SUBTASK 27-41-11-410-003

(11) For the right horizontal stabilizer, do this step:

(a) Close these access panels:

Number	Name/Location
341A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
341AZ	Horizontal Stabilizer, Access Panel - Inbd L.E. Closure Rib
342AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AZ	Horizontal Stabilizer, Access Panel - Inbd T.E. Closure Rib

SUBTASK 27-41-11-390-001

(12) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

SUBTASK 27-41-11-080-002

(13) Remove the lock assembly, SPL-1672, from the stabilizer trim wheel at the control stand.

SUBTASK 27-41-11-820-001

(14) Adjust the stabilizer-to-body seals (Figure 406):

(a) Set the horizontal stabilizer [1] to a zero degree trim (four units of trim).

(b) Make sure that the clearance between the stabilizer-to-body blade seals and the airplane body surface along the upper and lower horizontal edge is 0.00 in. (0.00 mm) - 0.04 in. (1.02 mm).

1) If it is necessary to adjust the stabilizer-to-body blade seals, you must first set the horizontal stabilizer [1] to 5 units of trim.

2) Adjust the stabilizer-to-body blade seals in or out, if it is necessary.

(c) Make sure that the stabilizer-to-body bulb seals touch and push against the airplane body surface.

(d) Move the stabilizer leading edge to the full down position (the ball nut is against the downstop).

(e) Make sure that the clearance is 0.00 in. (0.00 mm) - 0.04 in. (1.02 mm) between the airplane body surface and the stabilizer-to-body blade seals at these locations:

1) The blade seals adjacent to the gap cover of the lower spar.

a) If it is necessary to adjust the stabilizer-to-body blade seals, you must first set the horizontal stabilizer [1] to 5 units of trim.

b) Adjust the stabilizer-to-body blade seals in or out, as necessary.

2) The blade seal adjacent to the gap cover of the upper spar.

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a) If it is necessary to adjust the stabilizer-to-body blade seals, you must first set the horizontal stabilizer [1] to 5 units of trim.

b) Adjust the stabilizer-to-body blade seals in or out, if it is necessary.

SUBTASK 27-41-11-860-010

(15) Set the STAB TRIM PRI switch, S272, to the NORMAL position.

SUBTASK 27-41-11-820-002

(16) If the same stabilizer, elevator, and elevator tab that was removed was re-installed (without elevator or elevator tab rework or adjustment - and the tailcone was not removed), do this check to find if a flight test is necessary:

NOTE: If the tailcone was removed, or its position adjusted during the stabilizer or elevator removal procedure, the left and right elevator index plate [4] positions will be in a different position when the tailcone is re-installed. The elevator cannot be properly measured with the elevator index plate [4] in a different location, therefore, you must do a power-off flight test.

(a) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

(b) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

CAUTION

MAKE SURE THAT YOU ARE CAREFUL WITH THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE SHARP EDGES ON THE TRAMMEL BAR, SPL-1677 CAN CAUSE CHAFING OR DAMAGE TO THE WIRES.

(c) Use the trammel trammel bar, SPL-1677, to set the B dimension of the horizontal stabilizer [1] to 39.17 ±0.01 in. (994.92 ±0.26 mm) Figure 401.

NOTE: This is the neutral stabilizer setting of zero degrees or 4 units of trim.

1) Make sure that you protect wire bundles when you use the trammel trammel bar, SPL-1677.

(d) Move the control column forward and aft several times and then let it go at the neutral position.

(e) Make sure that the right elevator position, “Z”, is within ± 0.06 in. (1.5 mm) of the neutral mark on the right elevator index plate [4] (Figure 403).

1) If it is necessary, adjust the PCU input rods to achieve the ± 0.06 in. (1.5 mm) measurement on the right elevator index plate [4] (TASK 27-31-00-820-803).

(f) Make sure that the stabilizer geometry is correct:

1) Make sure that the left elevator is within 0.12 in. (3.0 mm) of the initial left elevator position.

NOTE: The initial left elevator position is the position, “Z”, measured before the removal of the stabilizer.

2) Make sure that the right elevator is within 0.12 in. (3.0 mm) of the initial right elevator position.

NOTE: The initial right elevator position is the position, “Z”, measured before the removal of the stabilizer.

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3) Make sure that the left elevator is within 0.12 in. (3.0 mm) of the right elevator position.

NOTE: This measurement compares the left and right stabilizer half positions after the installation of the two sides of the stabilizer. It makes sure that the left and right elevators are within 0.12 in. (3.0 mm) of their original positions. The 0.12 in. (3.0 mm) difference is calculated to have a negligible effect on manual reversion.

(g) If the measurements in the above step are not within limits, the elevator system should be adjusted and test flown for manual reversion.

NOTE: You cannot adjust the elevator control pushrod [5] to make sure that the elevator is within the limits. Adjustment of the elevator control pushrod [5] is too coarse to get the minimum tolerance of 0.12 in. (3.0 mm). One half turn of the elevator control pushrod [5] moves the trailing edge of the elevator up or down 0.20 in. (5.1 mm).

SUBTASK 27-41-11-860-011

(17) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-11-860-012

(18) Remove the safety tag and close this circuit breaker (if installed):

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 27-41-11-820-003

(19) Do these system tests (if it is necessary):

(a) Do this task: Elevator Travel Limit - Test, TASK 27-31-00-700-805.

(b) Make sure that the elevator pushrods are correctly adjusted (TASK 27-31-00-820-802).

NOTE: If the elevator pushrods are adjusted, a flight test is necessary.

(c) Do this task: Elevator Tab Control System - Operational Test, TASK 27-31-00-700-802.

(d) Use the stabilizer trim wheel to make sure that the horizontal stabilizer [1] operates through full travel between the stops.

(e) Do this task: Elevator Position Transmitter Installation Test, TASK 31-31-51-700-801.

(f) To test the logo lights, do the test portion of this task: Logo Light - Light Assembly Replacement, TASK 33-49-00-960-802.

(g) If it is necessary, do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

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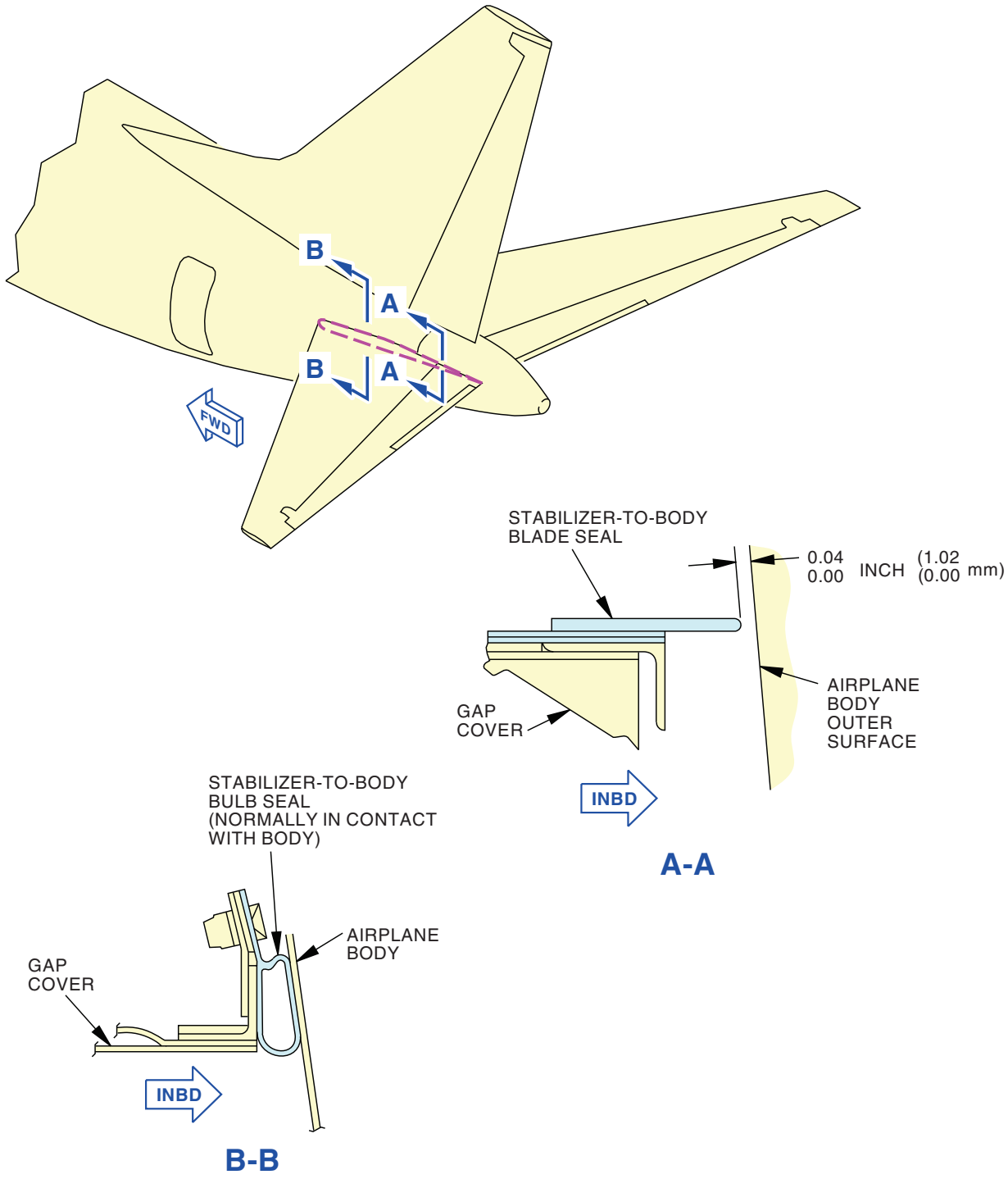
SUBTASK 27-41-11-410-004

(20) Close this access door:

Number	Name/Location
318BR	Tailcone Access Door

——— END OF TASK ———

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Stabilizer-to-Body Seal Adjustment
Figure 406/27-41-11-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HORIZONTAL STABILIZER LEADING EDGE - REMOVAL/INSTALLATION

1. General

A. This procedure contains these tasks:

(1) Horizontal stabilizer leading edge removal

(2) Horizontal stabilizer leading edge installation.

B. The horizontal stabilizer leading edge is referred to as the "leading edge" in this procedure.

C. This procedure is applicable for the left and right leading edges.

TASK 27-41-21-000-801

2. Horizontal Stabilizer Leading Edge Removal

(Figure 401)

A. References

Reference	Title
55-10-00-000-801	Horizontal Stabilizer Tip - Removal (P/B 401)

B. Location Zones

Zone	Area
331	Left Horizontal Stabilizer - Removable Leading Edge
341	Right Horizontal Stabilizer - Removable Leading Edge

C. Horizontal Stabilizer Leading Edge Removal

SUBTASK 27-41-21-020-001

(1) Remove the leading edge [3] of the horizontal stabilizer [1]:

(a) Use non-metallic tools to remove sealant from leading edge [3] and tip cap assembly [4].

(b) Remove the tip cap assembly [4], do this task: Horizontal Stabilizer Tip - Removal, TASK 55-10-00-000-801.

NOTE: The tip cap assembly can be slightly moved out until the leading edge can be removed.

(c) Remove the bolts [2] that attach the lower surface and the inboard of the leading edge [3] to the front spar.

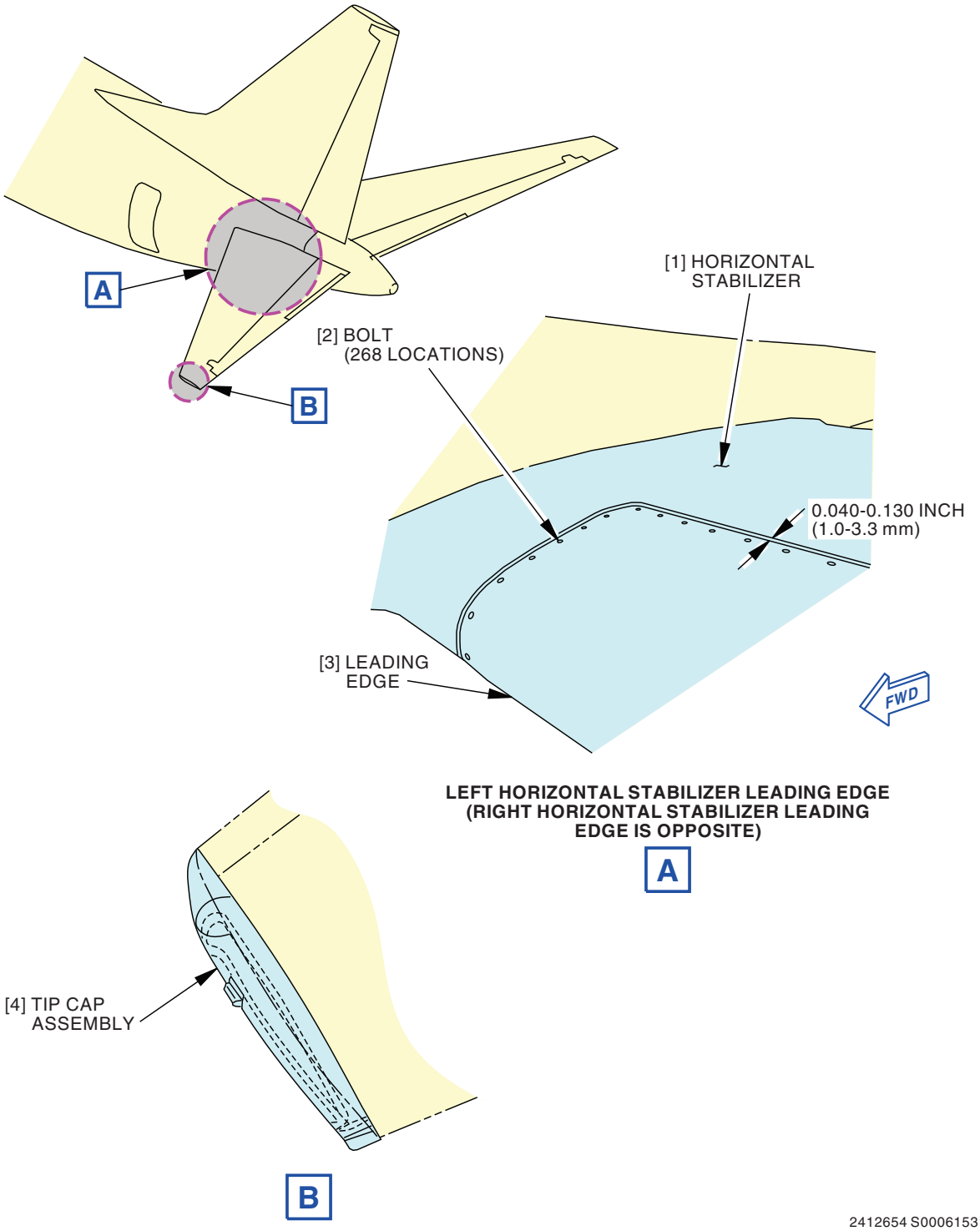
(d) Hold the leading edge [3] and remove the bolts [2] that attach the top surface of the leading edge [3] to the front spar.

(e) Hold each end of the leading edge [3].

(f) Remove the leading edge [3] from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Horizontal Stabilizer Leading Edge Installation
Figure 401/27-41-21-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-21-400-801

3. Horizontal Stabilizer Leading Edge Installation

(Figure 401)

A. References

Reference	Title
55-10-00-400-801	Horizontal Stabilizer Tip - Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A00900	Sealant - Silicone, RTV - Dow Corning 93-006-1RF (Formerly 93-006-1. Use until Stock depleted)	BMS5-175 Type I Class B-1/2

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Leading edge	55-10-52-04A-190	PTF ALL
		55-10-52-04A-195	PTF ALL
		55-10-52-04A-200	PTF ALL
		55-10-52-04A-205	PTF ALL
		55-10-52-04A-210	PTF ALL
		55-10-52-04A-215	PTF ALL
		55-10-52-04A-220	PTF ALL
		55-10-52-04A-225	PTF ALL

D. Location Zones

Zone	Area
331	Left Horizontal Stabilizer - Removable Leading Edge
341	Right Horizontal Stabilizer - Removable Leading Edge

E. Horizontal Stabilizer Leading Edge Installation

SUBTASK 27-41-21-420-001

(1) Install the leading edge [3]:

(a) Lift and put the leading edge [3] in its position.

(b) Make sure that the clearance between the leading edge [3] and horizontal stabilizer [1] is 0.040 in. (1.02 mm) - 0.130 in. (3.30 mm).

CAUTION

ALIGN THE LEADING EDGE ASSEMBLY ALONG THE FULL LENGTH OF THE EDGE BEFORE YOU INSTALL THE FASTENERS. IF YOU PUSH THE ASSEMBLY INTO ITS POSITION AND INSTALL THE FASTENERS, DAMAGE CAN OCCUR TO THE RIB CHORDS.

1) Make sure that the leading edge assembly is aligned along the full length before the fasteners are installed.

NOTE: If the leading edge assembly is not aligned or forced into position during installation, then the induced strains can cause tension cracks in the rib chords.

(c) Install the bolts [2] that attach the top surface of the leading edge [3] to the front spar.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Tighten the bolts [2] to 20 in-lb (2.3 N·m).

(d) Install the bolts [2] that attach the lower surface and inboard of the leading edge [3] to the front spar.

1) Tighten the bolts [2] to 20 in-lb (2.3 N·m).

(e) Install the tip cap assembly [4], do this task: Horizontal Stabilizer Tip - Installation, TASK 55-10-00-400-801.

(f) Fill the clearance between the leading edge [3] and horizontal stabilizer [1] with Dow Corning 93-006-1 RF sealant, A00900, or sealant, A00247.

——— END OF TASK ———

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HORIZONTAL STABILIZER CENTER SECTION HINGE FITTING - REMOVAL/INSTALLATION

1. General

A. This procedure contains these tasks:

(1) A removal of the horizontal stabilizer center section hinge fitting

(2) An installation of the horizontal stabilizer center section hinge fitting.

B. The horizontal stabilizer center section hinge fitting is referred to as the "hinge fitting".

C. This procedure is applicable for the left and right hinge fittings.

TASK 27-41-31-000-801

2. Horizontal Stabilizer Center Section Hinge Fitting Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-41-11-000-801	Horizontal Stabilizer Removal (P/B 401)
53-31-31-000-801	Stabilizer-to-Body Upper Front Spar Sliding Seal Removal (P/B 401)
53-31-31-000-802	Stabilizer-to-Body Lower Front Spar Sliding Seal Removal (P/B 401)
55-10-10-000-801	Thrust Brace Links Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205
SPL-2026	Set - Punch, Horizontal Stabilizer Center Section Part #: C55002-1 Supplier: 81205
SPL-24249	Offset Wrench Kit - Hinge Fitting, Horizontal Stabilizer Part #: C27109-1 Supplier: 81205
STD-9463	Beam - Wood, 4 Inch, by 4 Inch, by 60 inch

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge

D. Access Panels

Number	Name/Location
318BR	Tailcone Access Door
331A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
331AZ	Horizontal Stabilizer, Access Panel, Inboard L.E. Closure Rib
332AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body

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(Continued)

Number	Name/Location
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AZ	Horizontal Stabilizer, Access Panel, Inboard T.E. Closure Rib
341A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
341AZ	Horizontal Stabilizer, Access Panel - Inbd L.E. Closure Rib
342AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AZ	Horizontal Stabilizer, Access Panel - Inbd T.E. Closure Rib

E. Prepare for the Removal

SUBTASK 27-41-31-020-001

WARNING

MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Remove the horizontal stabilizers (TASK 27-41-11-000-801).

SUBTASK 27-41-31-020-002

(2) Remove the stabilizer to body front spar sliding seals and pedestals (TASK 53-31-31-000-801, TASK 53-31-31-000-802).

NOTE: This will give additional space for the stabilizer center section to move forward.

SUBTASK 27-41-31-860-001

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-31-860-002

WARNING

MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Make sure that the horizontal stabilizer is in the neutral position, if not, do these steps:

(a) If it is installed, remove the lock assembly, SPL-1672 (Figure 401).

(b) On the control stand, set the stabilizer trim wheel to put the horizontal stabilizer at zero degrees (4 units of trim).

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SUBTASK 27-41-31-480-001

(5) Install the lock assembly, SPL-1672, on the stabilizer trim wheel at the control stand (Figure 401):

(a) Turn the trim wheel to put the handle at the top of the wheel.

(b) Adjust the height of the trim lock to put the trim wheel handle correctly on the yoke.

(c) Install the pin through the yoke.

(d) Install the safety pin.

SUBTASK 27-41-31-010-001

(6) Open this access door:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-41-31-010-002

(7) For the left hinge fitting, do this step:

(a) Open these access panels:

Number	Name/Location
331A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
331AZ	Horizontal Stabilizer, Access Panel, Inboard L.E. Closure Rib
332AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AZ	Horizontal Stabilizer, Access Panel, Inboard T.E. Closure Rib

SUBTASK 27-41-31-010-003

(8) For the right hinge fitting, do this step:

(a) Open these access panels:

Number	Name/Location
341A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
341AZ	Horizontal Stabilizer, Access Panel - Inbd L.E. Closure Rib
342AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AZ	Horizontal Stabilizer, Access Panel - Inbd T.E. Closure Rib

F. Hinge Fitting Removal

(Figure 402)

SUBTASK 27-41-31-010-004

(1) Remove bracket [57], bracket [58], bracket [59], bracket [61], bracket [62], and wire bundle [60].

SUBTASK 27-41-31-480-002

(2) Use wood beam, STD-9463, or wood beams that are approximately 4 in. (10 cm) by 4 in. (10 cm) cross-section to hold the horizontal stabilizer center section (Figure 402).

(a) Use as many wood beams, STD-9463, or wood beams as necessary to make sure that the horizontal stabilizer center section is level and supported before the hinge pin [26] is removed.

SUBTASK 27-41-31-020-003

(3) Remove the hinge fittings [24] from the bulkhead support structure:

NOTE: Both hinge fittings are to be removed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the pin [20], nut [19], and washer [21].

1) Discard the pin [20].

(b) Remove the pin [22], nut [18], and bushing spacer [23].

1) Discard the pin [22].

(c) Remove the bolts [2] and washers [3] from the bushing retainer [1].

(d) Remove the bushing retainer [1].

(e) Use the set, SPL-2026, and push out these parts:

1) The hinge insert pin [27]

2) The bushing retainer [25]

3) The hinge pin [26].

(f) To remove the adjustable hinge bushing [28], bushing spacer [16], and hinge bushing [17], do these steps:

1) Turn the adjustable hinge bushing [28] to remove it.

NOTE: The adjustable hinge bushing has threads and must be turned for removal or installation.

2) Remove the bushing spacer [16] and hinge bushing [17].

NOTE: The bushing spacer will come off when you remove the adjustable hinge bushing.

SUBTASK 27-41-31-020-004

(4) Remove the upper thrust brace [38] and lower thrust brace [39] (TASK 55-10-10-000-801).

SUBTASK 27-41-31-020-005

(5) Remove the thrust brace fitting [51] from the hinge fitting [24]:

(a) Remove the nuts [50], washers [49], washers [48], and bolts [47] that attach the thrust brace fitting [51] to the hinge fitting [24].

1) Use a marker or pen to make marks on the thrust brace fitting [51] to help make the installation easier.

a) Do not use sharp objects to make marks.

(b) Remove the thrust brace fitting [51] and shim [46].

SUBTASK 27-41-31-020-007

(6) Remove the hinge fitting [24]:

(a) Remove the bolts [52] from the bracket [53].

(b) Remove the bolts [54] from the bracket [55].

(c) Remove the nut [7], washer [5], washers [6], bolt [9], and bonding jumper [8] from the hinge fitting [24].

(d) Remove the pin [56], nut [15], washer [14], washer [13], and bolt [12] from the hinge fitting [24].

1) If it is necessary, use the offset wrench kit, SPL-24249.

2) Discard the pin [56].

(e) Remove the pin [44], nut [43], washer [42], washer [41], and bolt [40] from the hinge fitting [24].

1) If it is necessary, use the offset wrench kit, SPL-24249.

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2) Discard the pin [44].

(f) Remove the pin [37], nut [36], washer [35], washer [34], and bolt [33] from the hinge fitting [24].

1) If it is necessary, use the offset wrench kit, SPL-24249.

2) Discard the pin [37].

(g) With the boards in place, carefully move the horizontal stabilizer center section forward as much as the cutout in the fuselage structure will let you.

NOTE: Use as many maintenance personnel as necessary on the left and right sides of the rear spar connection to lift and move forward.

NOTE: With the boards in position, the Center Section will be able to move approximately 1.5 in. (3.8 cm) forward.

1) Make sure that the horizontal stabilizer center section does not hit the neutral shift input rods.

NOTE: The neutral shift input rods can be damaged if they are hit by the horizontal stabilizer center section.

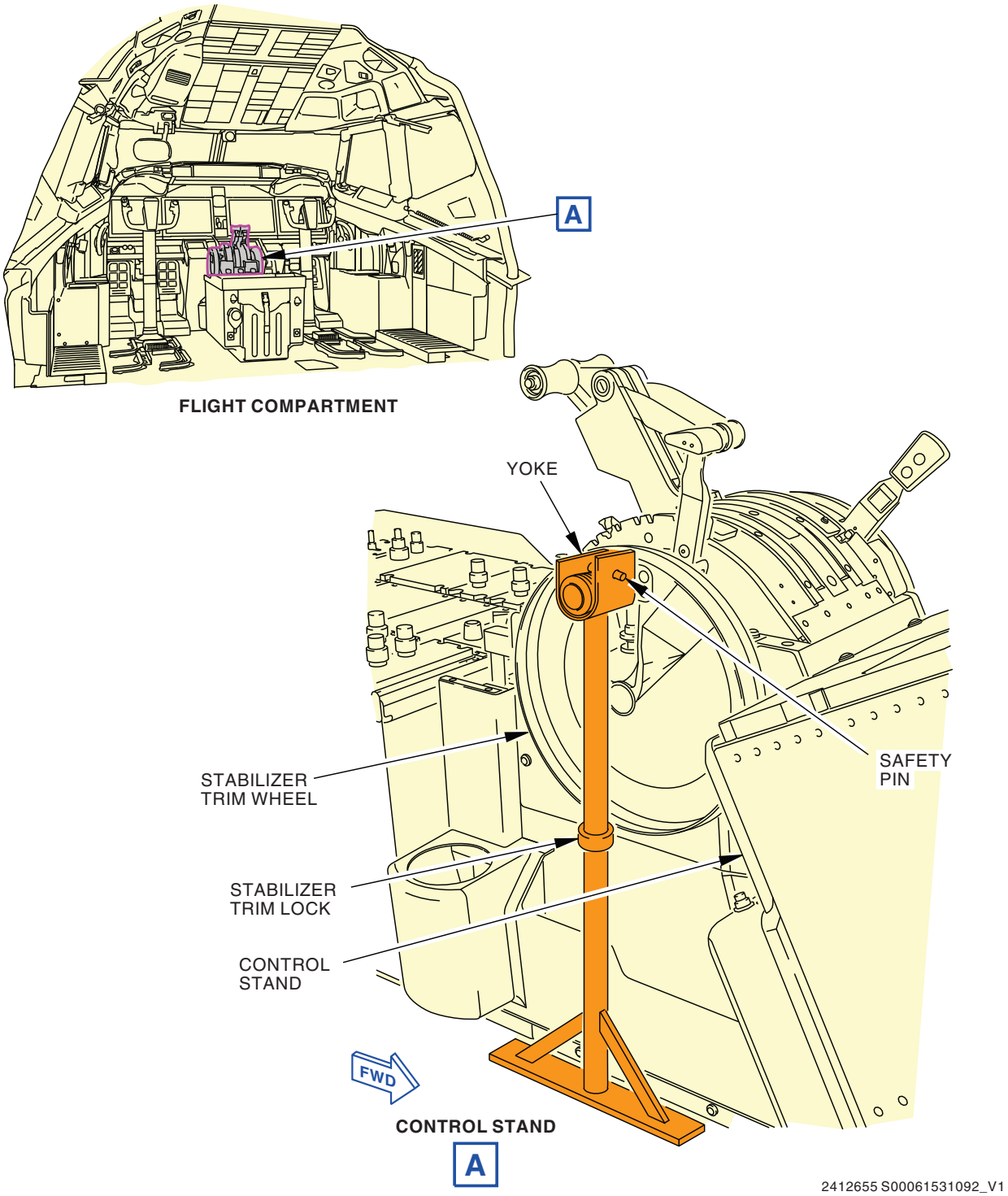
(h) Move the hinge fitting [24] in a clockwise or counterclockwise direction while you move it back and forth until it is removed.

1) If moved in either the clockwise or counterclockwise direction, alternate back and forth until the hinge fitting comes out.

(i) Remove the hinge fitting [24] from the airplane.

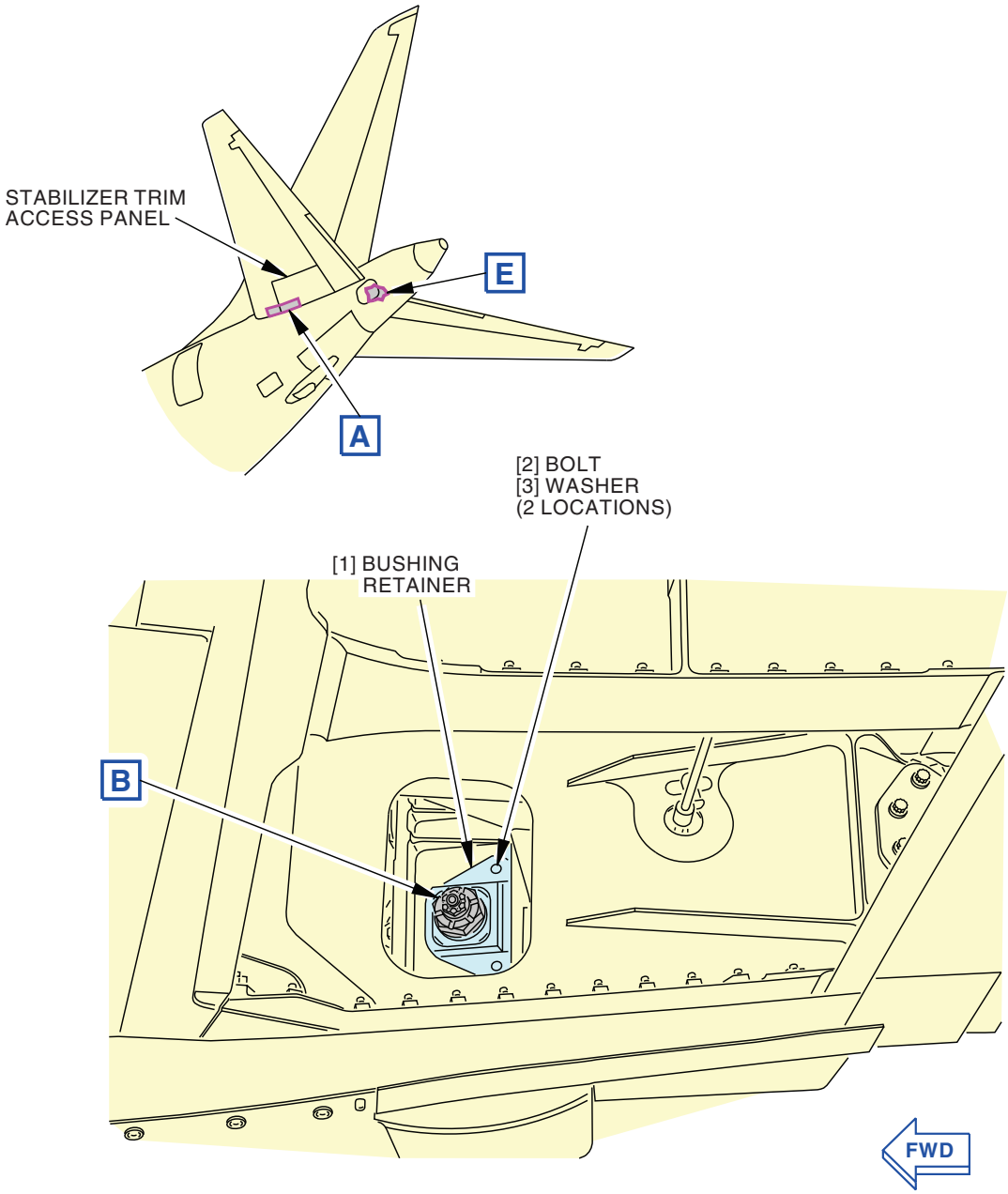
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Lock Installation
Figure 401/27-41-31-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



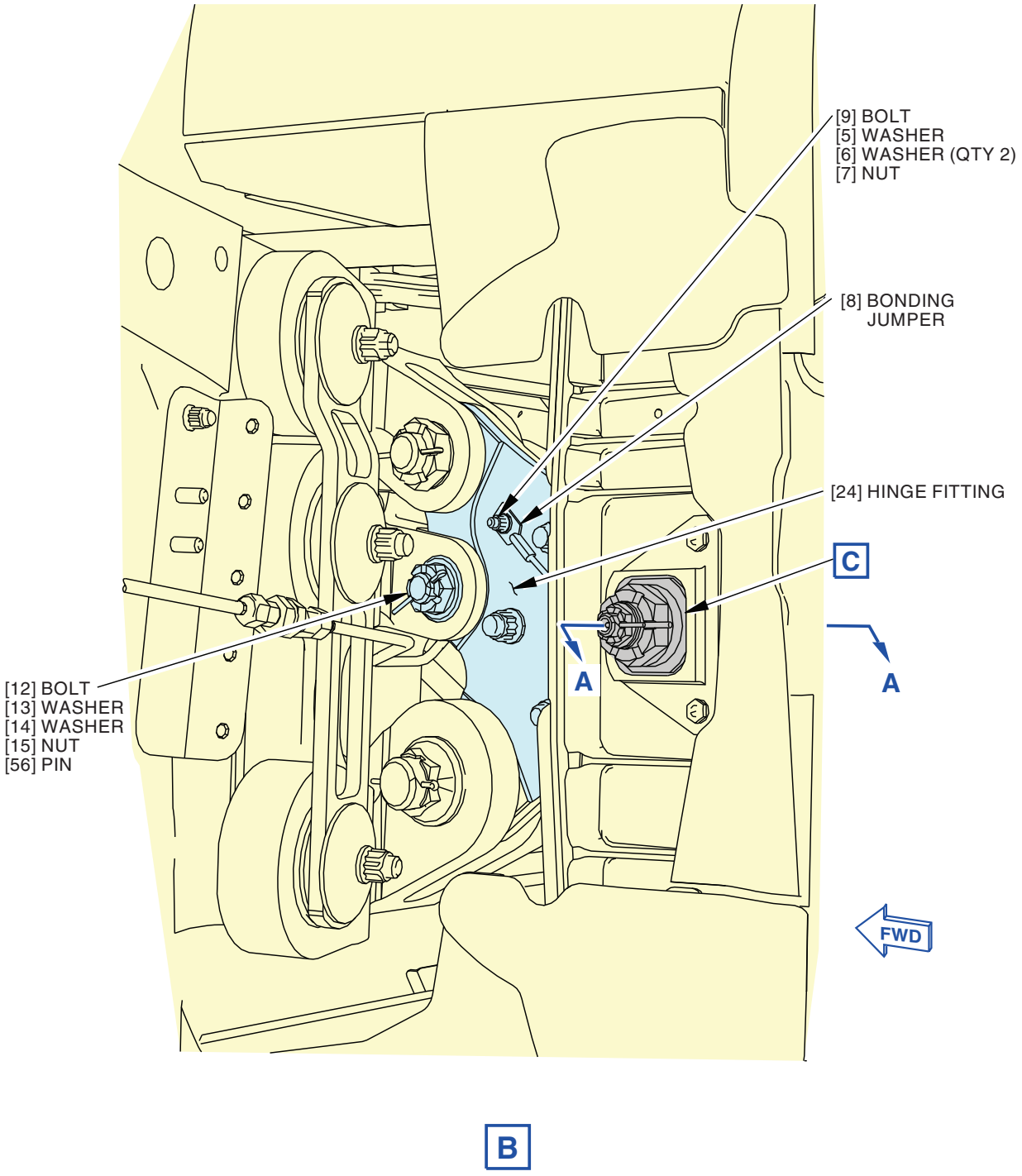
**VIEW THROUGH ACCESS PANEL
(PANEL SHOWN REMOVED)**

A

2412656 S00061531093_V1

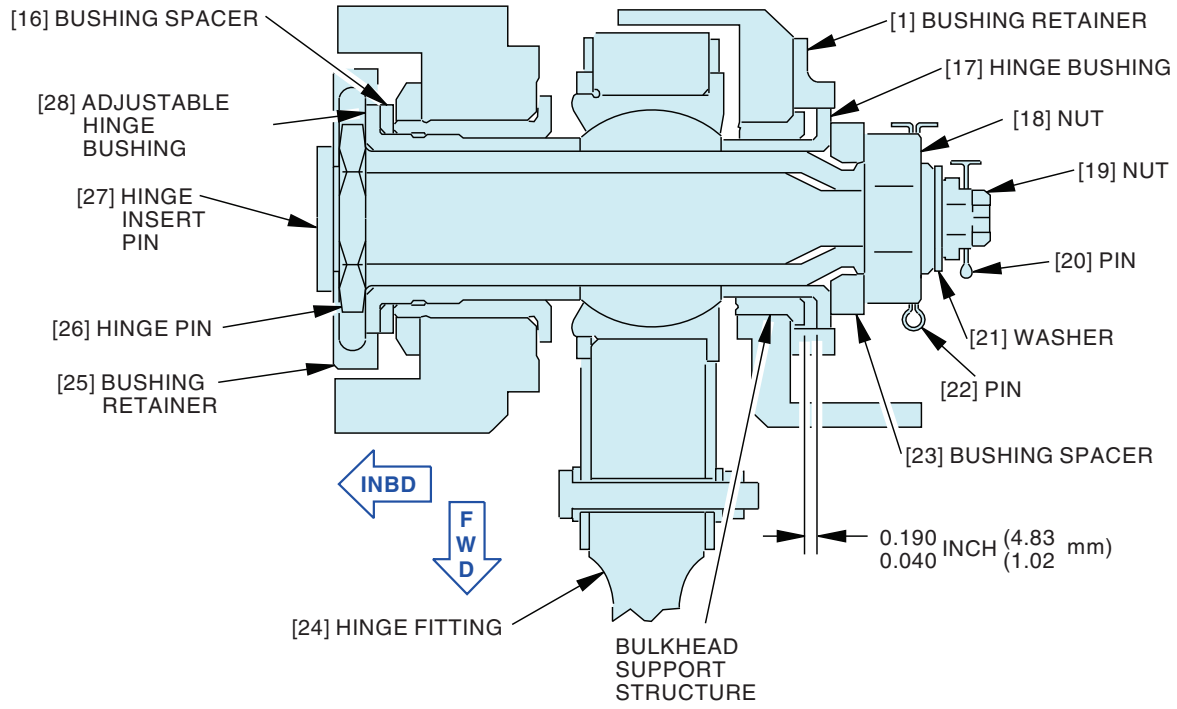
**Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 1 of 12)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



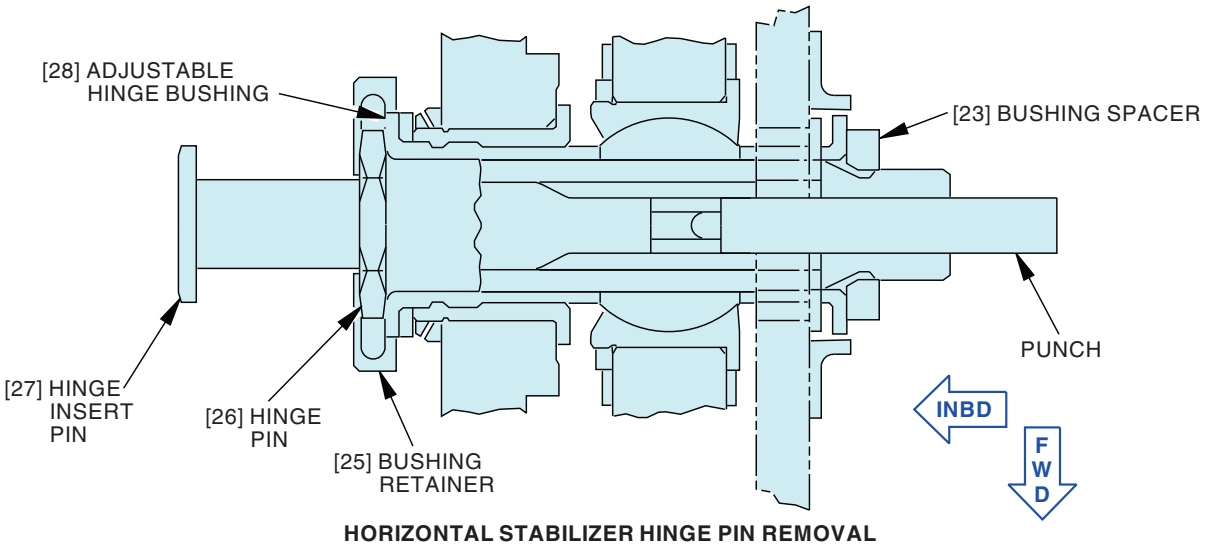
Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 2 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HORIZONTAL STABILIZER HINGE PIN REMOVAL (STEP 1)

A-A



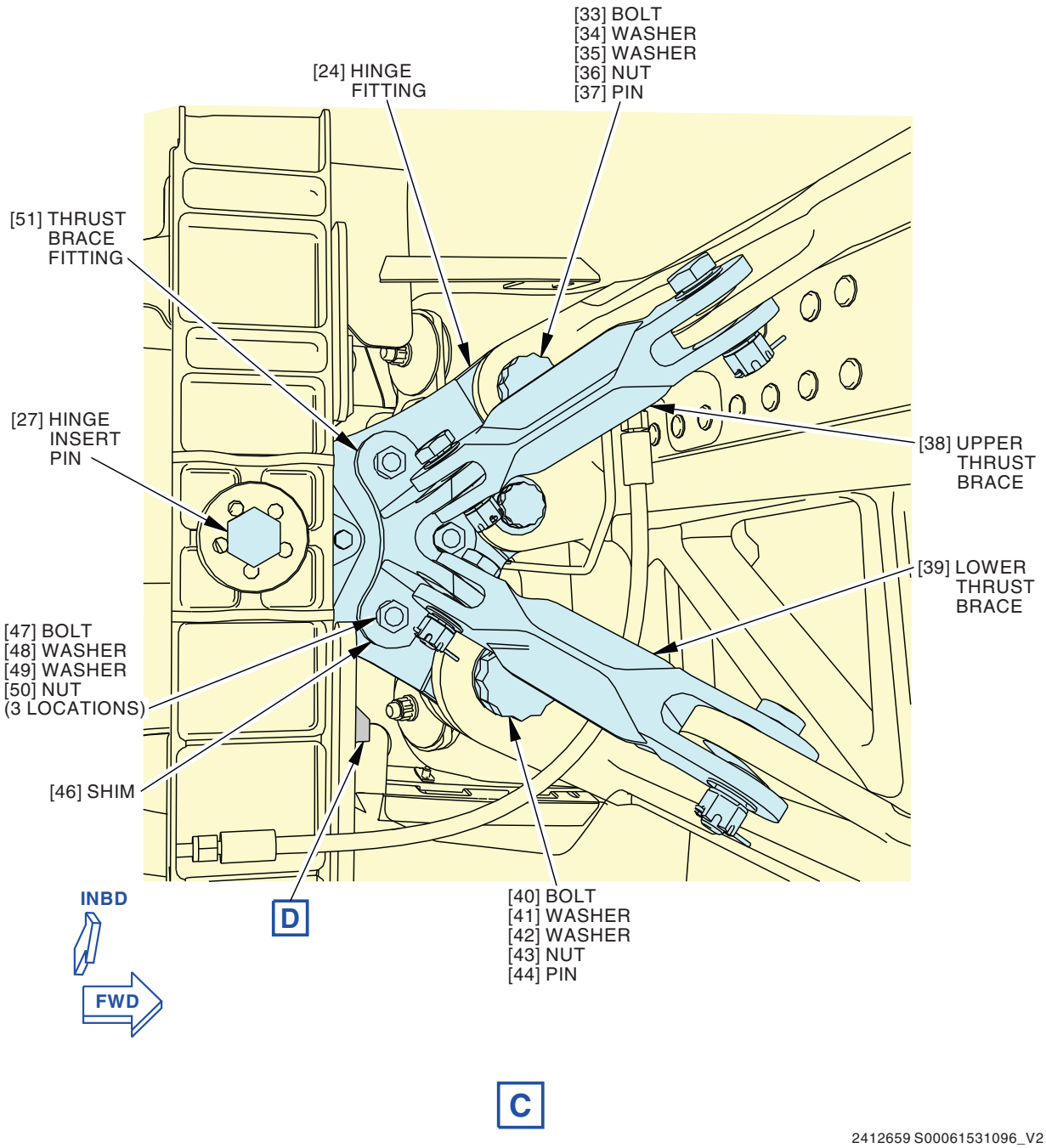
HORIZONTAL STABILIZER HINGE PIN REMOVAL (STEP 2)

A-A

2412658 S00061531095_V1

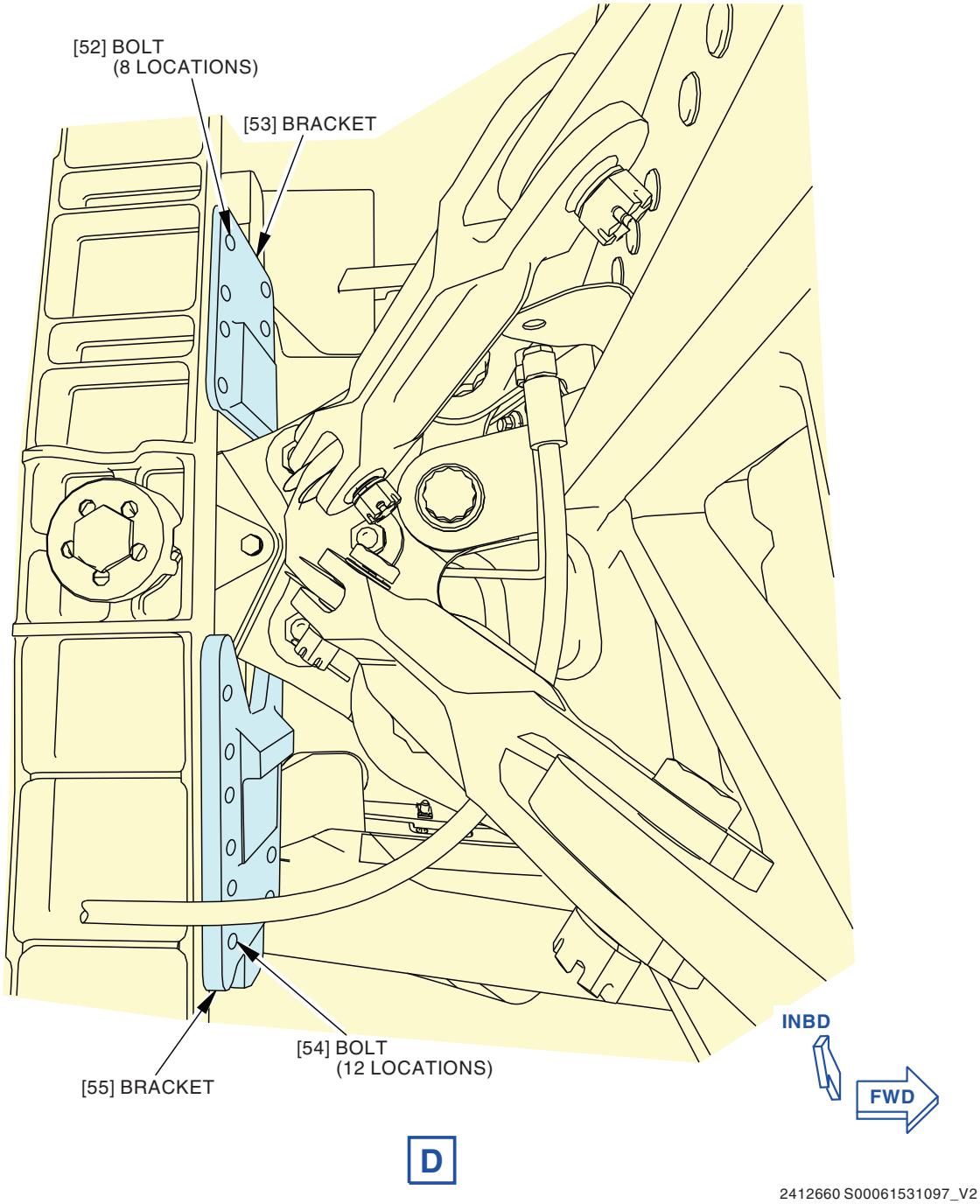
Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 3 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



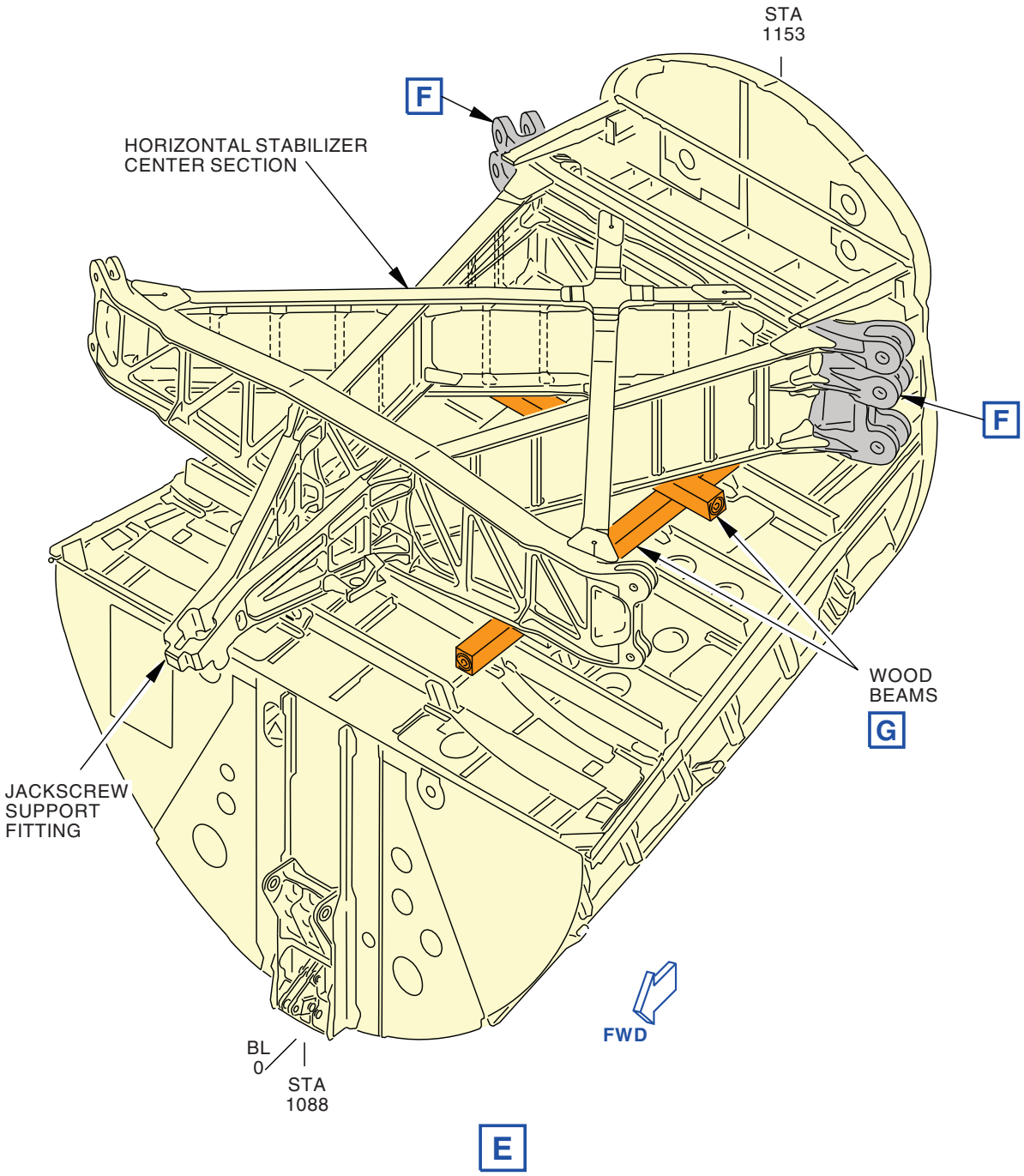
Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 4 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 5 of 12)

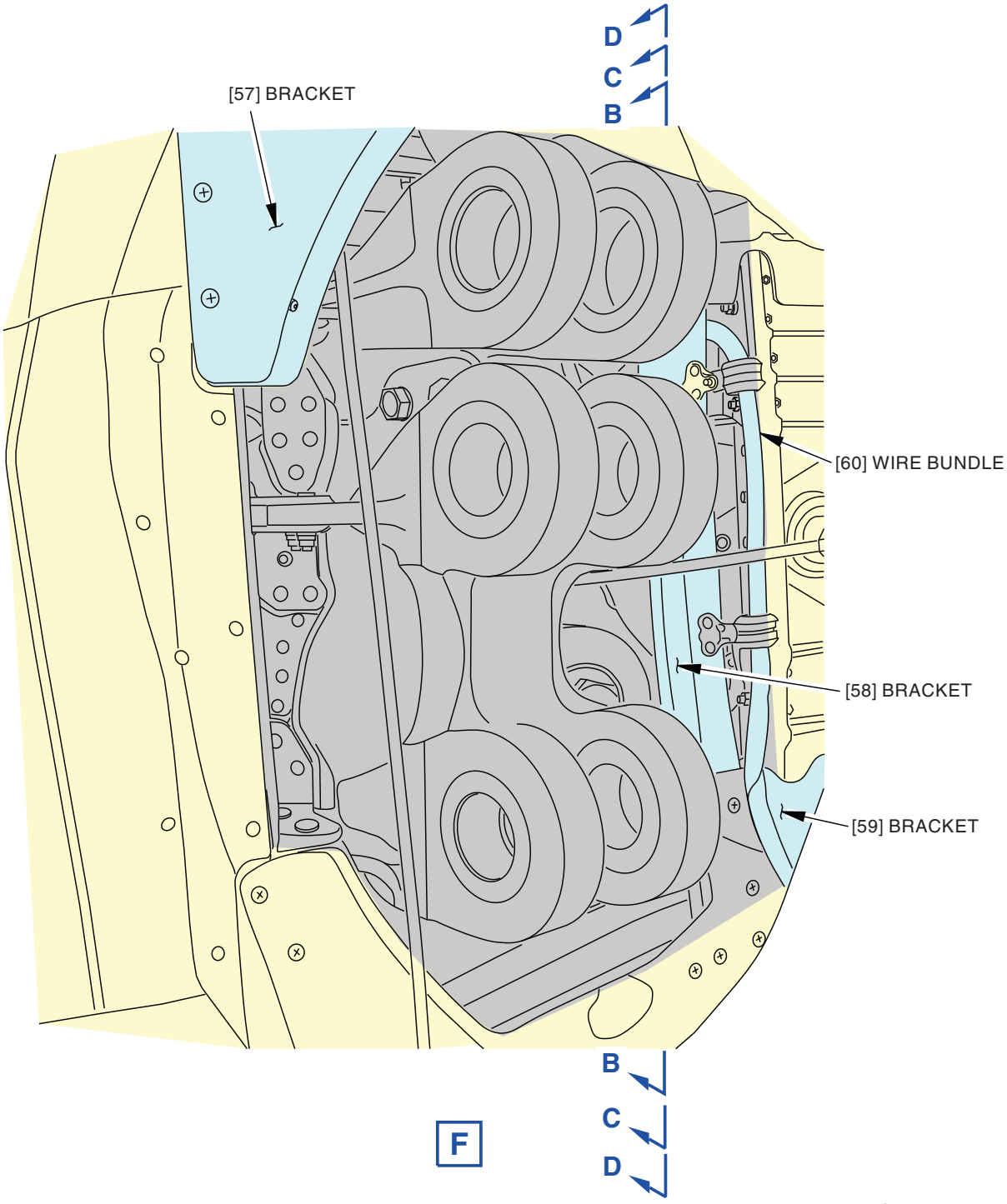
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412661 S00061531098_V1

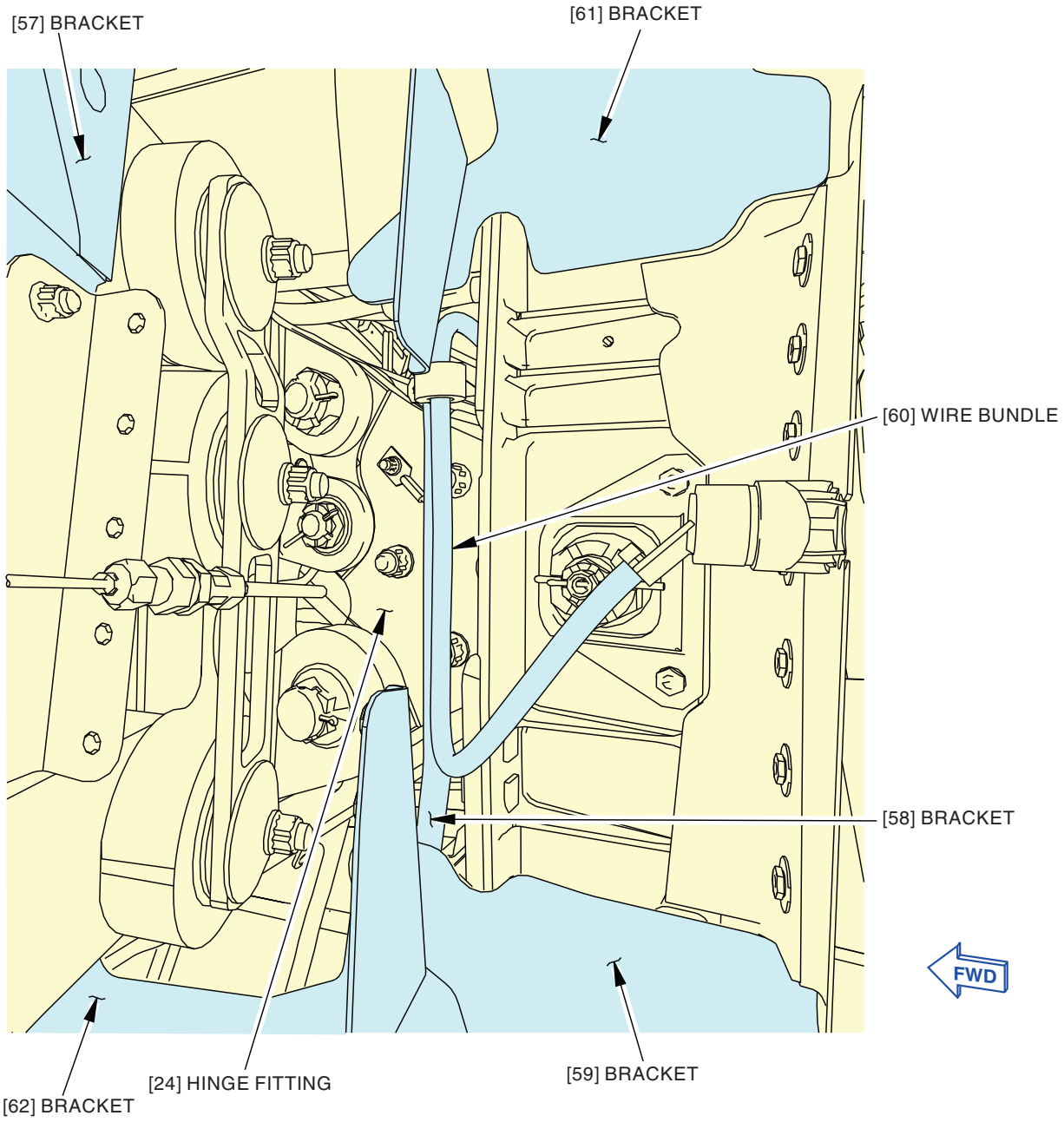
Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 6 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



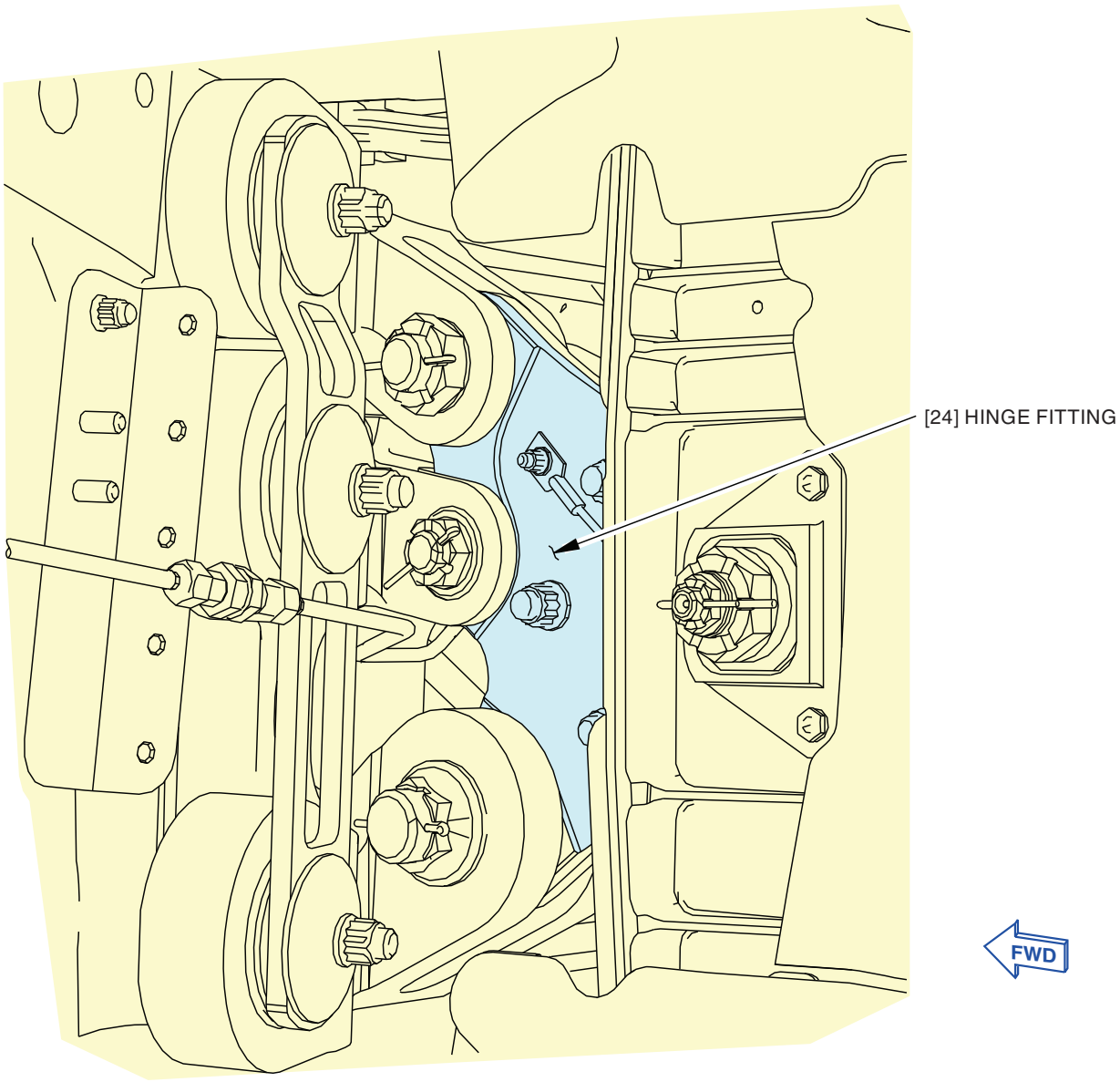
Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 7 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 8 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

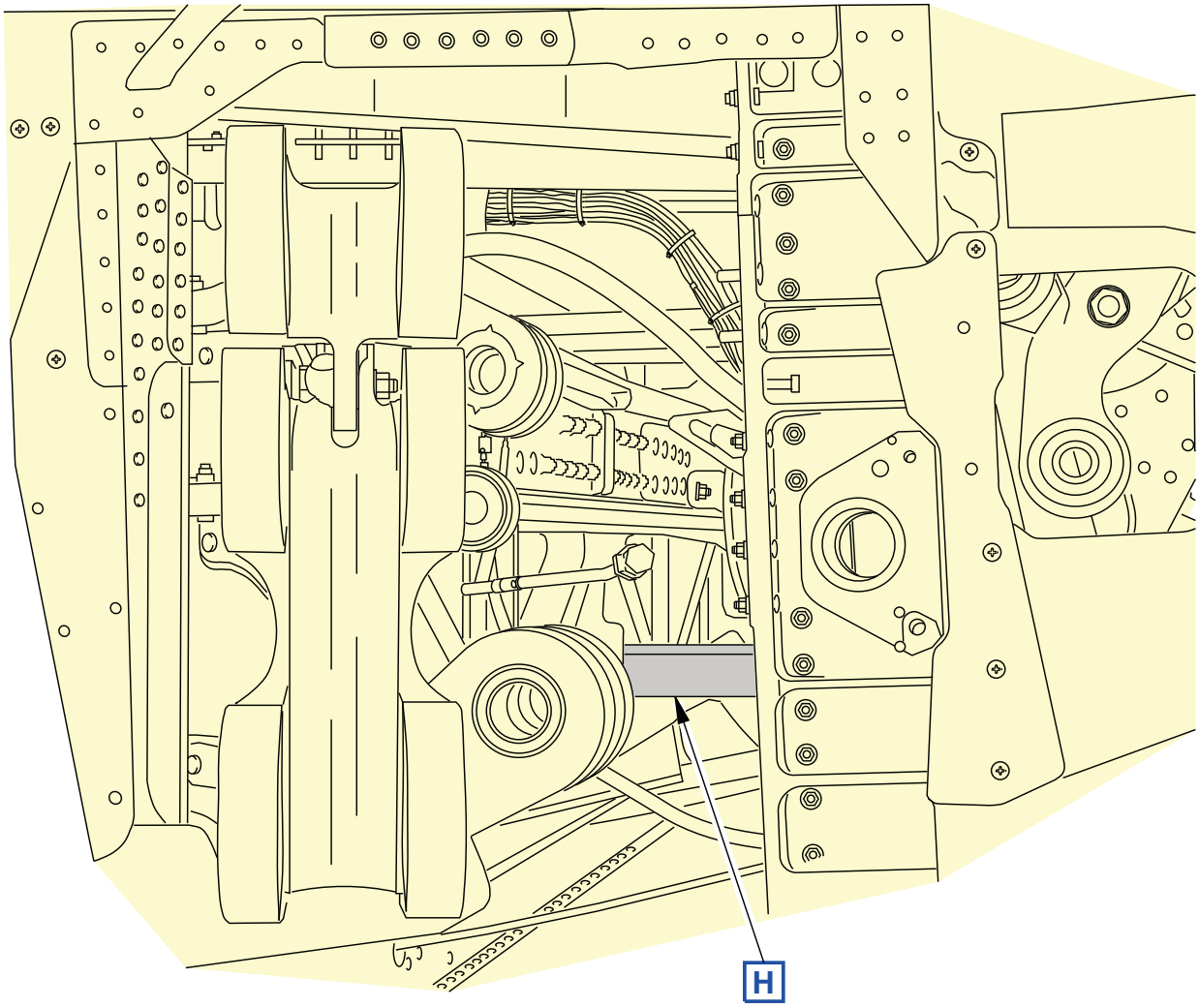


1 THIS VIEW SHOWS THE BRACKETS AND WIRE BUNDLE REMOVED.

2412664 S00061531101_V2

Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 9 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



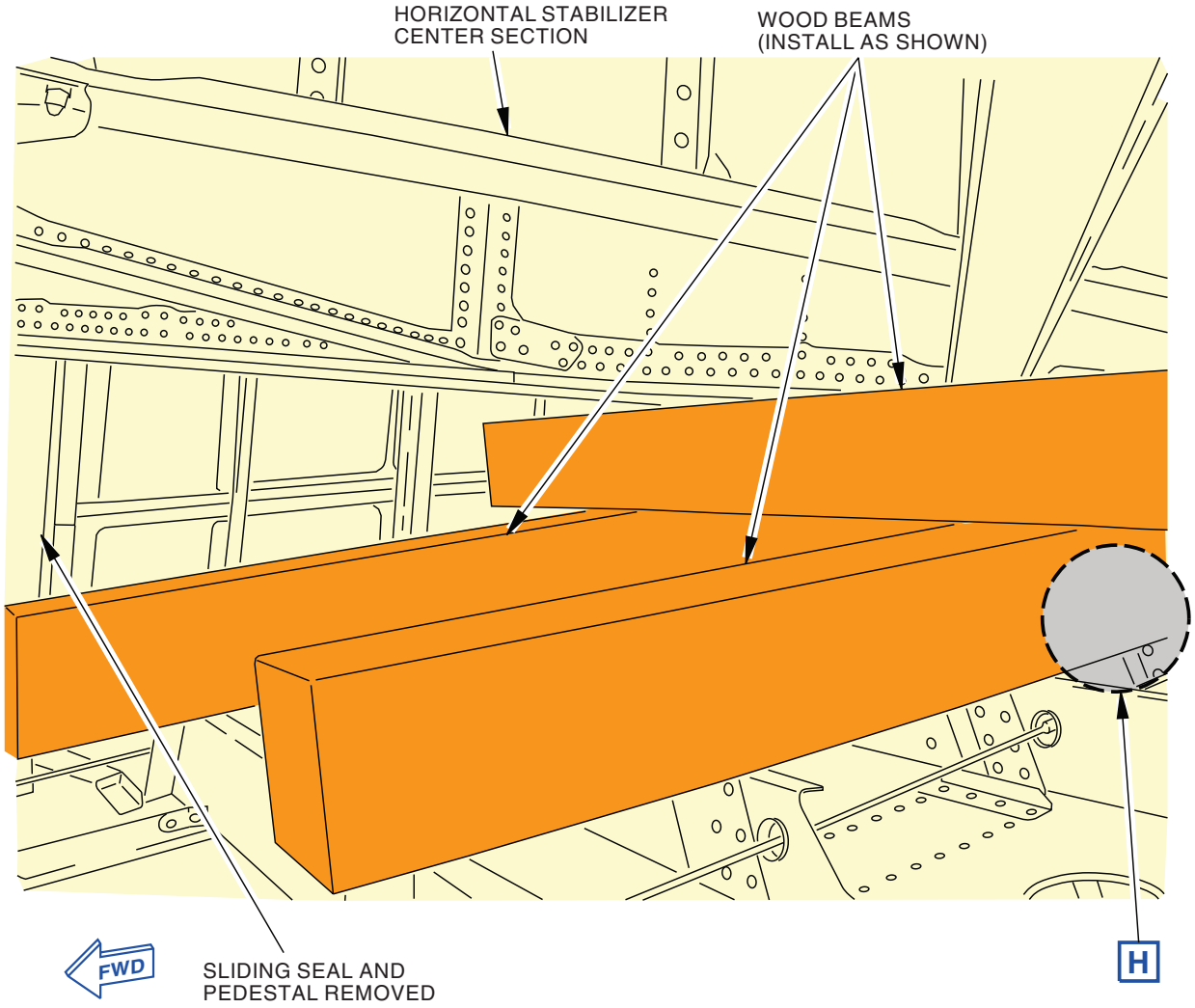
D-D 2

2 THIS VIEW SHOWS THE BRACKETS, WIRE BUNDLE AND HINGE FITTING REMOVED.

2412665 S00061531102_V1

Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 10 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

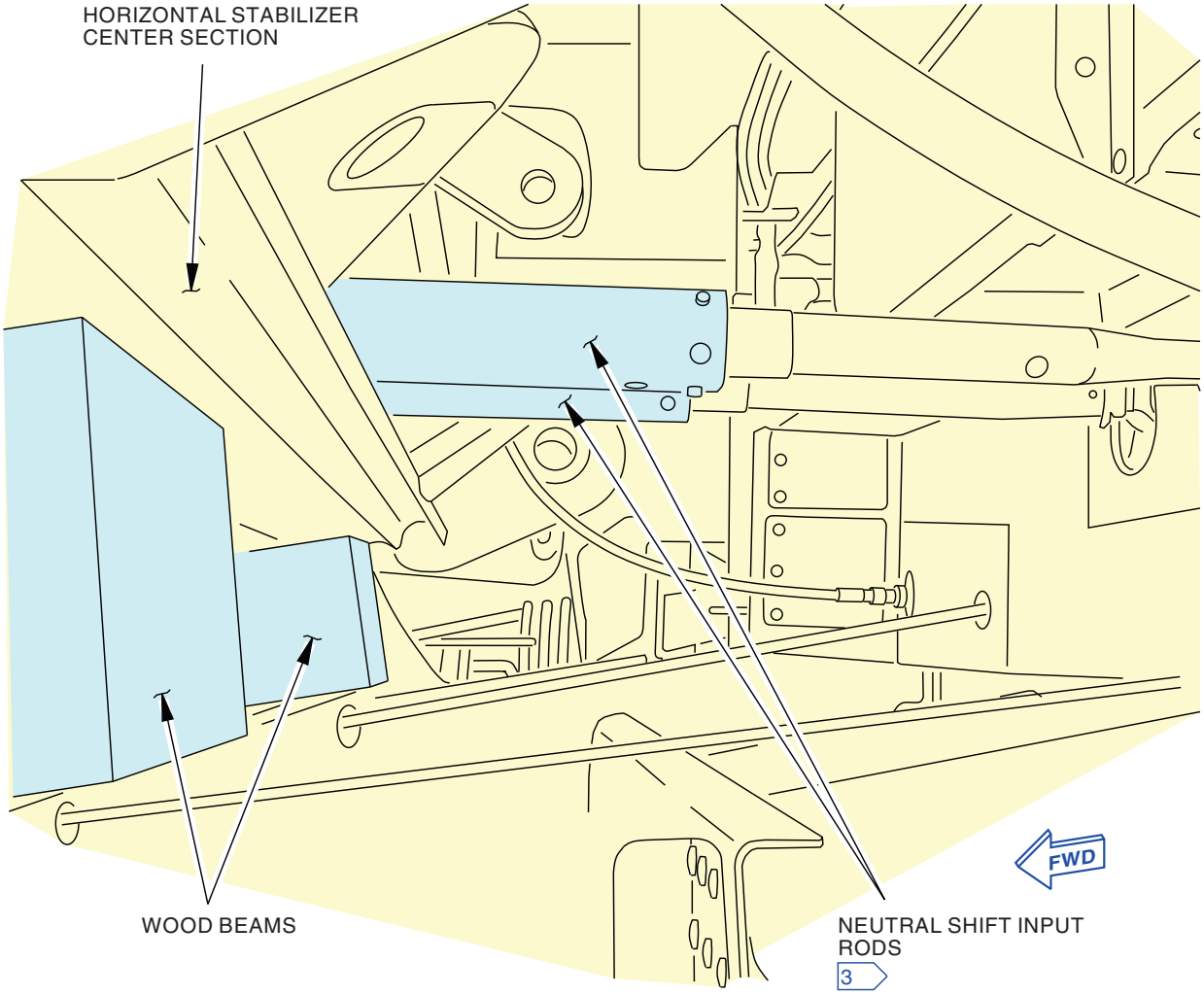


G

2412666 S00061531103_V1

Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 11 of 12)

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3 BE CAREFUL NOT TO DAMAGE THE NEUTRAL SHIFT INPUT RODS WHEN MOVING THE HORIZONTAL STABILIZER CENTER SECTION.

2412667 S00061531104_V1

Horizontal Stabilizer Center Section Hinge Fitting Installation
Figure 402/27-41-31-990-802 (Sheet 12 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-31-400-801

3. Horizontal Stabilizer Center Section Hinge Fitting Installation

(Figure 401, Figure 402)

A. References

Reference	Title
27-41-11-400-801	Horizontal Stabilizer Installation (P/B 401)
53-31-31-400-801	Stabilizer-to-Body Upper Front Spar Sliding Seal Installation (P/B 401)
53-31-31-400-802	Stabilizer-to-Body Lower Front Spar Sliding Seal Installation (P/B 401)
55-10-10-400-801	Thrust Brace Link Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205
SPL-2036	Protector - Horizontal Stabilizer Hinge Pin Thread Part #: C27054-1 Supplier: 81205
SPL-24249	Offset Wrench Kit - Hinge Fitting, Horizontal Stabilizer Part #: C27109-1 Supplier: 81205
STD-1238	Indicator - Dial
STD-9463	Beam - Wood, 4 Inch, by 4 Inch, by 60 inch

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
D00633	Grease - Aircraft General Purpose	BMS3-33
D50004	Compound - Antiseize	BMS3-28

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
20	Pin	55-10-51-14-005	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
22	Pin	55-10-51-14-010	PTF ALL
24	Hinge fitting	55-10-52-01Q-170	PTF ALL
		55-10-52-01Q-175	PTF ALL
37	Pin	55-10-52-01Q-012	PTF ALL
44	Pin	55-10-52-01Q-018	PTF ALL
56	Pin	55-10-52-01Q-005	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge

F. Access Panels

Number	Name/Location
318BR	Tailcone Access Door
331A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
331AZ	Horizontal Stabilizer, Access Panel, Inboard L.E. Closure Rib
332AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AZ	Horizontal Stabilizer, Access Panel, Inboard T.E. Closure Rib
341A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
341AZ	Horizontal Stabilizer, Access Panel - Inbd L.E. Closure Rib
342AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AZ	Horizontal Stabilizer, Access Panel - Inbd T.E. Closure Rib

G. Hinge Fitting Installation

(Figure 402)

SUBTASK 27-41-31-420-001

(1) Install each hinge fitting [24]:

(a) Lubricate the bolt threads of bolt [12], bolt [33], and bolt [40] with grease, D00633.

(b) Put the hinge fittings [24] in their positions.

(c) Install the bolt [33], washer [34], washer [35], and nut [36].

1) Tighten the nut [36] to 2750 ±550 in-lb (310.7 ±62.1 N·m).

a) If it is necessary, use the offset wrench kit, SPL-24249.

2) Carefully move the horizontal stabilizer center section into its position.

3) Install a new pin [37].

a) Tighten nut [36] if it is necessary to get the pin hole alignment but do not exceed 5300 in-lb (598.8 N·m).

(d) Install the bolt [40], washer [41], washer [42], and nut [43].

1) Tighten the nut [43] to 5925 ±525 in-lb (669 ±60 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) If it is necessary, use the offset wrench kit, SPL-24249.

2) Install a new pin [44].

a) Tighten the nut [43] if it is necessary to get the pin hole alignment but do not exceed 8600 in-lb (971.7 N·m).

(e) Install the bolt [12], washer [13], washer [14], and nut [15].

1) Tighten the nut [15] to 1415 ±115 in-lb (159.9 ±13.0 N·m).

a) If it is necessary, use the offset wrench kit, SPL-24249.

2) Install a new pin [56].

a) Tighten the nut [15] if it is necessary to get the pin hole alignment but do not exceed 2000 in-lb (226.0 N·m).

SUBTASK 27-41-31-420-002

(2) Install the bonding jumper [8] for each hinge fitting [24]:

(a) Install the bolt [9], washers [5], washers [6], bonding jumper [8] (between washers [6]), and nut [7].

1) Tighten the nut [7] to 85 ±3 in-lb (9.6 ±0.3 N·m).

2) Measure the electrical bonding resistance between the free end of the bonding jumper [8] and hinge fitting [24].

a) Use an intrinsically safe approved bonding meter, COM-1550.

b) Make sure that the resistance is 0.00216 ohm (2.16 milliohms) or less.

3) Apply sealant, A50359, at the end of the bonding jumper [8].

SUBTASK 27-41-31-420-003

(3) Install the thrust brace fitting [51] on each hinge fitting [24]:

(a) Lubricate the bolts [47] with sealant, A00247.

(b) Put the thrust brace fitting [51] and shim [46] on the hinge fitting [24].

1) Remove or add shim [46] until you can install the bolts [47] and nuts [50] easily.

(c) Install the bolts [47], washers [48], washers [49], and nuts [50].

1) Tighten the nuts [50] to 850 ±220 in-lb (96.0 ±24.9 N·m).

SUBTASK 27-41-31-420-004

(4) Connect the upper thrust brace [38] and lower thrust brace [39] to the thrust brace fitting (TASK 55-10-10-400-801).

SUBTASK 27-41-31-220-001

(5) Do a check of the dimension between the hinge fittings [24] and bulkhead support structure:

(a) Attach an L-angle shape to the forward side of the bulkhead flange to use as a reference plane.

(b) Install the dial indicator, STD-1238.

(c) Measure the dimension between the hinge fitting [24] and the bulkhead support structure.

SUBTASK 27-41-31-420-005

(6) Install the hinge fittings [24] to the bulkhead support structure:

(a) Install each adjustable hinge bushing [28]:

1) Install the adjustable hinge bushing [28] by hand until it touches the hinge fitting bearing.

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2) Use the dial indicator, STD-1238, to measure the dimension between the hinge fitting [24] and the bulkhead support structure.

3) Make sure that this dimension is the same as you measured before.

4) If it is necessary, loosen or tighten the adjustable hinge bushing [28] until you get the correct dimension.

(b) Install the hinge bushing [17]:

1) Lubricate the hinge bushing [17] with compound, D50004.

2) Install the hinge bushing [17].

(c) Install the hinge pin [26]:

1) Lubricate hinge pin [26] and protector, SPL-2036, with grease, D00633.

NOTE: The protector, SPL-2036, is used to protect the head of the hinge pin [26].

2) Use the protector, SPL-2036, to install the hinge pin [26].

CAUTION

DO NOT USE TOO MUCH FORCE WHEN YOU INSTALL THE HINGE PIN. DAMAGE TO THE HINGE PIN CAN OCCUR.

a) Do not apply too much force to install the hinge pin [26].

(d) Install the bushing spacer [23] and nut [18].

1) Tighten the nut [18] by hand.

(e) Install the bolts [2] and washers [3] to attach the bushing retainer [1] to the hinge fitting [24].

(f) Tighten the bolts [2] to 62.5 ±12.5 in-lb (7.1 ±1.5 N·m).

(g) Install the bushing retainer [25] on the adjustable hinge bushing [28].

CAUTION

DO NOT APPLY TORQUE TO THE WRENCH FLATS ON THE HINGE PINS. THIS CAN CAUSE DAMAGE TO THE EQUIPMENT.

(h) Temporarily remove the bushing retainer [25].

(i) Tighten the nut [18] to 750 ±100 in-lb (84.7 ±11.3 N·m).

1) If the pin holes do not align, tighten the nut [18] a maximum 30 degrees as necessary to align the nearest pin hole.

2) Install a new pins [22].

(j) Use the dial indicator, STD-1238, to measure the dimension between the hinge fitting [24] and bulkhead support structure.

1) Make sure that the dimension is the same as you measured before and after you installed the adjustable hinge bushing [28] in ± 0.005 in. (0.127 mm).

(k) Do a check of the clearance between the bulkhead support structure and hinge bushing [17].

1) Remove the bolts [2] and washers [3] to remove the bushing retainer [1].

2) Do a check of the clearance between the bulkhead support structure and hinge bushing [17].

3) Make sure that the clearance is 0.12 +0.07 / -0.08 in. (3.05 +1.78 / -2.03 mm).

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4) Install the bolts [2] and washers [3] to attach the bushing retainer [1] to the hinge fitting [24].

(l) Install the bushing retainer [25] with the hinge insert pin [27].

1) Turn the hinge insert pin [27] until you align the pin holes.

CAUTION

DO NOT APPLY TORQUE TO THE WRENCH FLATS ON THE HINGE PINS. THIS CAN CAUSE DAMAGE TO THE EQUIPMENT.

(m) Install the washer [21] and nut [19].

1) Tighten the nut [19] to a maximum of 50 in-lb (5.6 N·m).

2) If the pin hole of the nut [19] does not align, loosen the nut [19] a maximum of 30 degrees as necessary to align the nearest pin hole of the nut [19].

3) Install a new pin [20].

(n) Install the bolts [52] to attach the bracket [53] to the bulkhead support structure.

(o) Install the bolts [54] to attach the bracket [55] to the bulkhead support structure.

(p) Remove the wood beams, STD-9463, that held the horizontal stabilizer center section (Figure 402).

SUBTASK 27-41-31-410-001

(7) Install the bracket [57], bracket [58], bracket [59], bracket [61], bracket [62], and wire bundle [60].

SUBTASK 27-41-31-080-001

(8) Remove the L-shape and the dial indicator, STD-1238.

SUBTASK 27-41-31-420-006

(9) Install the stabilizer to body front spar sliding seals and pedestal (TASK 53-31-31-400-801, TASK 53-31-31-400-802)..

SUBTASK 27-41-31-420-007

(10) Install the horizontal stabilizers (TASK 27-41-11-400-801).

SUBTASK 27-41-31-420-008

(11) Install the stabilizer to body front spar sliding seals and pedestals (TASK 53-31-31-400-801, TASK 53-31-31-400-802).

SUBTASK 27-41-31-080-002

(12) Make sure that the lock assembly, SPL-1672, has been removed from the stabilizer trim wheel at the control stand (Figure 401).

SUBTASK 27-41-31-710-001

(13) Move the trim control wheel to make sure that the stabilizer operates smoothly through the full travel between the jackscrew stops.

SUBTASK 27-41-31-220-002

(14) Make sure that the clearance between the jackscrew support fitting and the fuselage structure is 0.030 ±0.020 in. (0.76 ±0.51 mm) on the two sides of the stabilizer.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-31-410-002

(1) For the left hinge fitting, do this step:

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(a) Close these access panels:

Number	Name/Location
331A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
331AZ	Horizontal Stabilizer, Access Panel, Inboard L.E. Closure Rib
332AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
333AZ	Horizontal Stabilizer, Access Panel, Inboard T.E. Closure Rib

SUBTASK 27-41-31-410-003

(2) For the right hinge fitting, do this step:

(a) Close these access panels:

Number	Name/Location
341A	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer To Body
341AZ	Horizontal Stabilizer, Access Panel - Inbd L.E. Closure Rib
342AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
343AZ	Horizontal Stabilizer, Access Panel - Inbd T.E. Closure Rib

SUBTASK 27-41-31-390-001

(3) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

SUBTASK 27-41-31-410-004

(4) Close this access door:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-41-31-860-003

(5) Make sure that the stabilizer trim cutout switches are set to the NORMAL position.

SUBTASK 27-41-31-860-004

(6) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

——— END OF TASK ———

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STABILIZER FORWARD CONTROL MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the stabilizer forward control mechanism

(2) An installation of the stabilizer forward control mechanism.

B. The stabilizer forward control mechanism is referred to as the "forward mechanism" in this procedure.

TASK 27-41-41-000-801

2. Stabilizer Forward Control Mechanism Removal

(Figure 401, Figure 402)

A. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Access Panels

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door

C. Prepare for the Removal

SUBTASK 27-41-41-860-003

(1) Move the stabilizer to the full APL NOSE UP position.

SUBTASK 27-41-41-860-004

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-41-010-001

(3) To get access to the forward stabilizer forward control mechanism [43], do this step:

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-41-41-010-002

(4) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Forward Mechanism Removal

SUBTASK 27-41-41-820-001

CAUTION MAKE SURE THAT THE CHAIN IS LOOSE BEFORE YOU RELEASE THE CABLE TENSION. TOO MUCH CABLE TENSION CAN CAUSE DAMAGE TO THE CHAIN.

(1) Adjust the upper adjusting nuts [24] of the aft support link [22] to loosen the chain [8].

SUBTASK 27-41-41-020-014

(2) Remove the lockwire from the upper turnbuckle [35].

SUBTASK 27-41-41-020-013

(3) Remove the nut [34], washer [33], bushing [32], washer [31], washer [51], washer [50], and bolt [30] to disconnect the upper turnbuckle [35] from the bracket.

SUBTASK 27-41-41-020-002

(4) Loosen the lower turnbuckle [1] to release the cable tension.

SUBTASK 27-41-41-020-003

(5) Remove the nuts [42] and washers [41] to remove the cable guard [40] from the cable drum [39].

SUBTASK 27-41-41-020-004

(6) Remove the STA cable from the bottom of the cable drum [39].

(a) Put a 50 lbf (222.4 N) load on the trim control wheel to extend the STB cable and make the removal of the STA cable easier.

(b) Install a tag to the STA cable for easy installation.

SUBTASK 27-41-41-020-005

(7) Disconnect the flexible shaft [21] from the forward stabilizer forward control mechanism [43].

(a) Remove the lockwire from the coupling nut of the flexible shaft [21]

SUBTASK 27-41-41-020-006

(8) Turn the trim control wheel until the connector link [11] of the chain [8] is between the idler sprocket [9] and the forward mechanism sprocket [10].

SUBTASK 27-41-41-020-007

(9) Remove the pins [12] from the connector link [11].

SUBTASK 27-41-41-020-008

(10) Disconnect the chain [8] at the connector link [11].

(a) Make sure that the loose ends of the chain [8] are clear of the forward stabilizer forward control mechanism [43].

SUBTASK 27-41-41-020-009

(11) Remove the STB cable from the top of the cable drum [39].

(a) Install a tag to the STB cable for easy installation.

SUBTASK 27-41-41-020-010

(12) Remove the forward stabilizer forward control mechanism [43]:

(a) Remove the nut [7], washer [6], washers [5], bushing [4], washer [3], and bolt [2] to disconnect the lower turnbuckle [1] from the brackets.

(b) Hold the forward stabilizer forward control mechanism [43].

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(c) Remove nut [29], washer [28], bushing [27], washer [26], and bolt [25] from the forward support link [37].

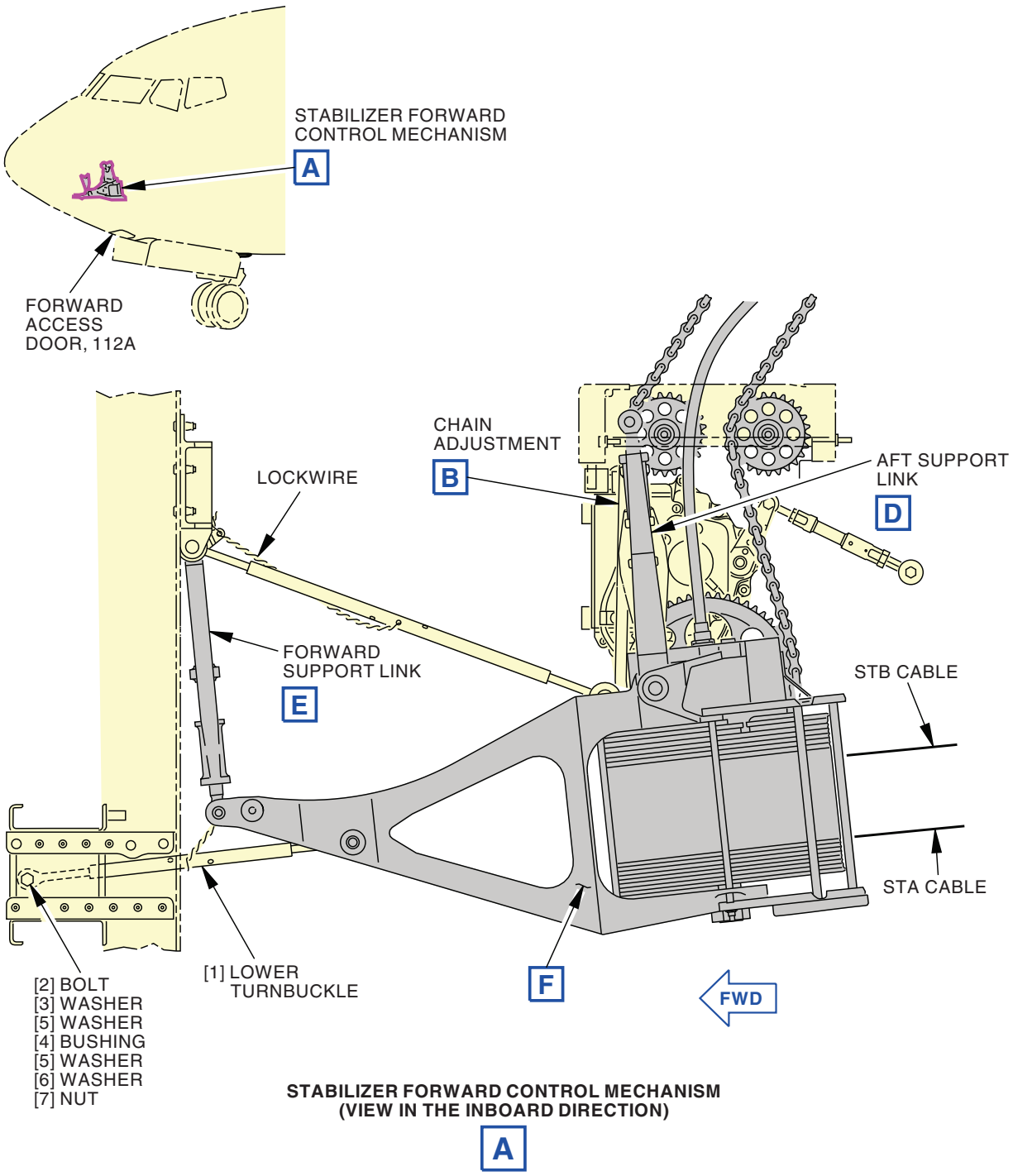
(d) Remove the nut [16], washer [15], bushing [14], washer [52], and bolt [13] from the aft support link [22].

(e) Remove the nut [20], washer [19], bushing [18], washer [53], and bolt [17] from the aft support link [22].

(f) Remove the forward stabilizer forward control mechanism [43] from the airplane.

——— END OF TASK ———

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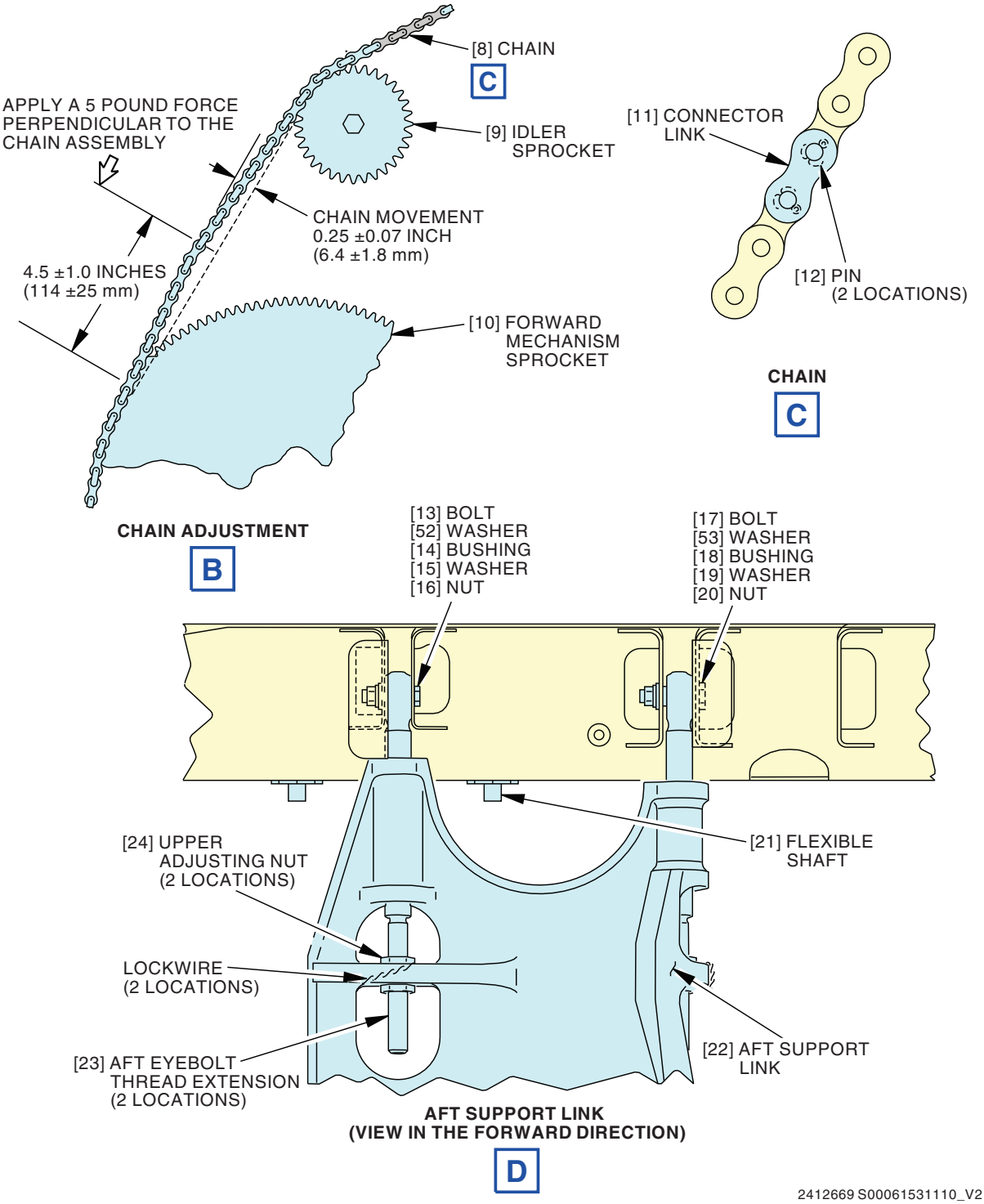
STABILIZER FORWARD CONTROL MECHANISM
(VIEW IN THE INBOARD DIRECTION)

A

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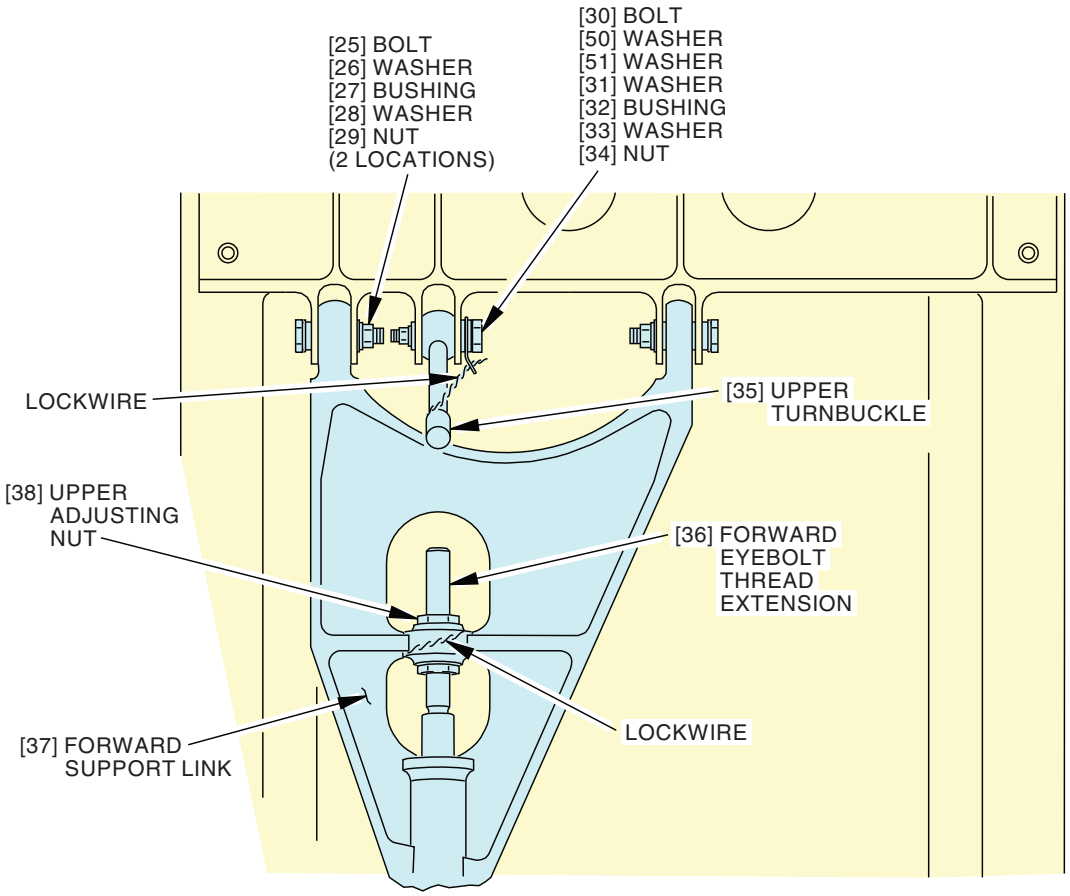
Stabilizer Forward Control Mechanism Installation
Figure 401/27-41-41-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Forward Control Mechanism Installation
Figure 401/27-41-41-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



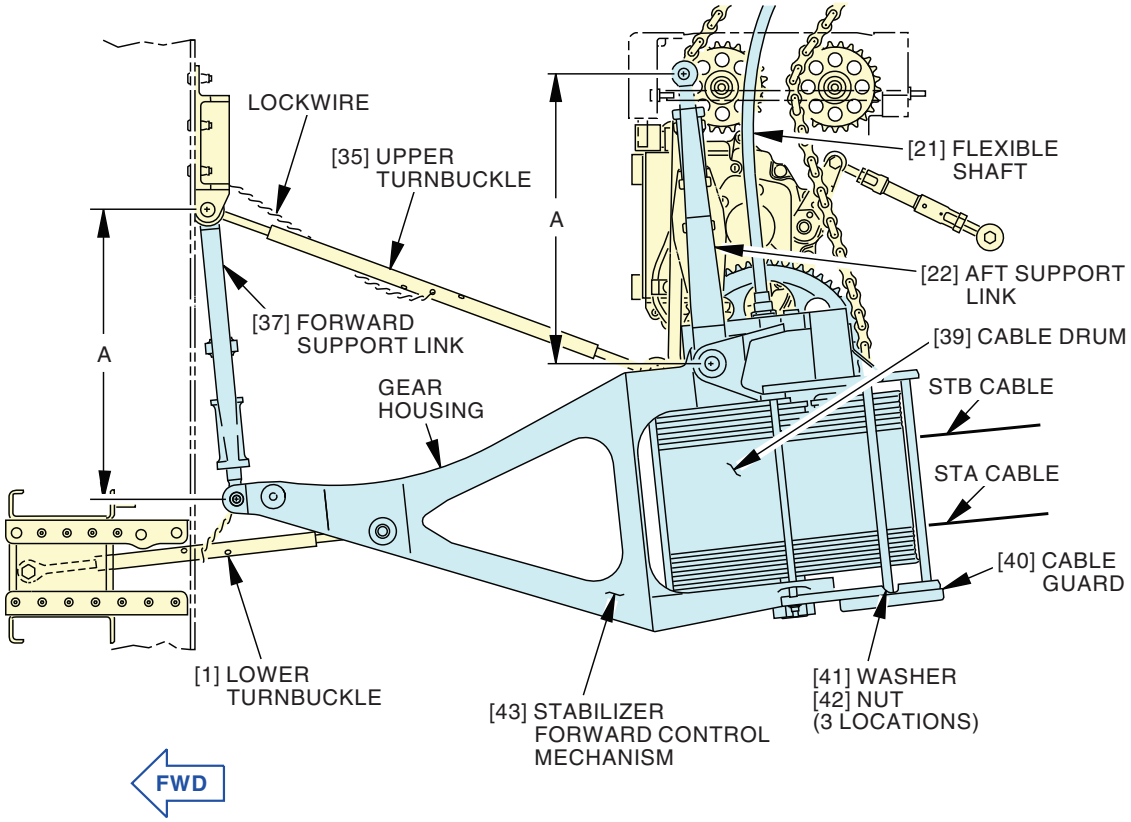
FORWARD SUPPORT LINK
(VIEW IN FORWARD DIRECTION)

E

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Stabilizer Forward Control Mechanism Installation
Figure 401/27-41-41-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



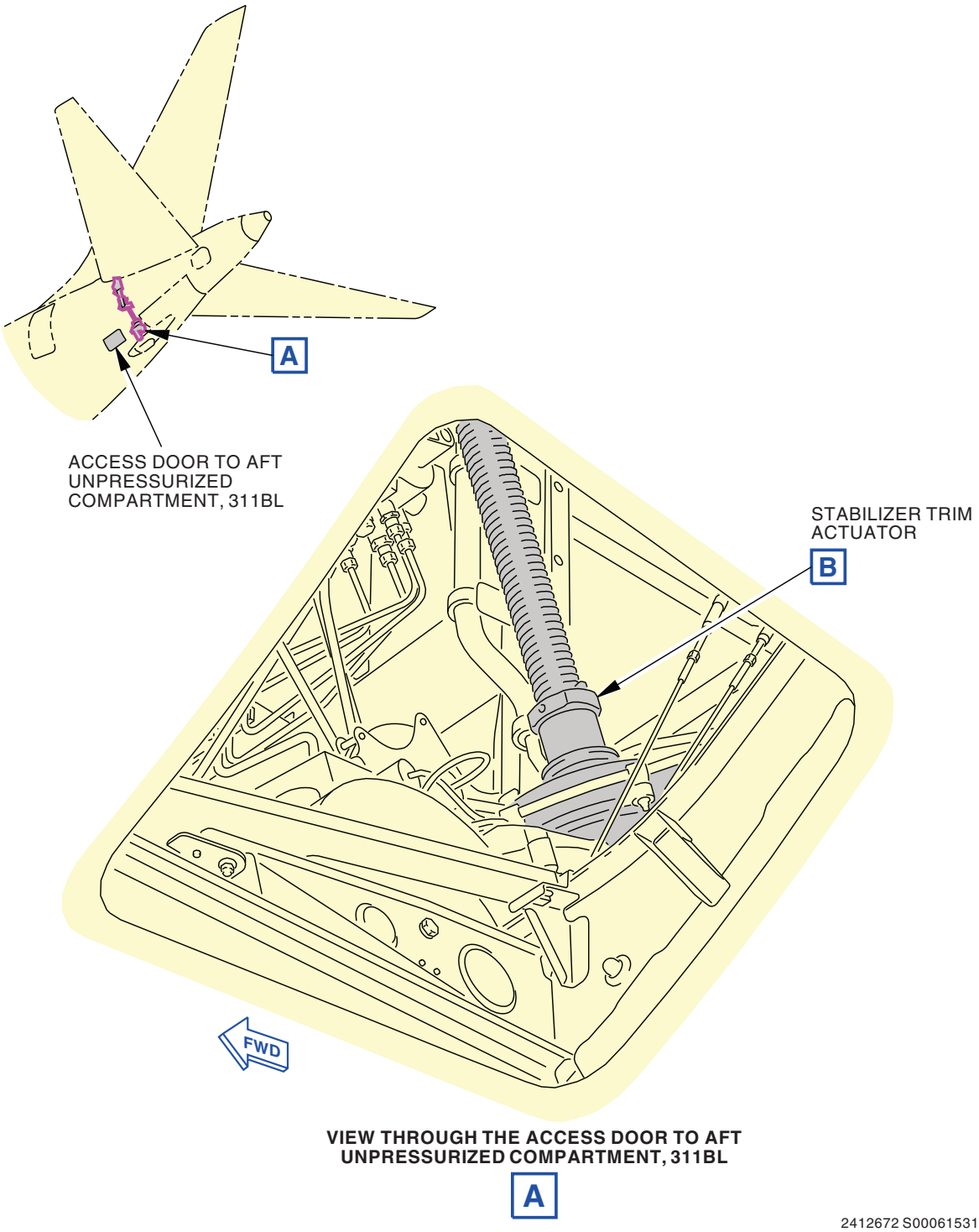
STABILIZER FORWARD CONTROL MECHANISM

F

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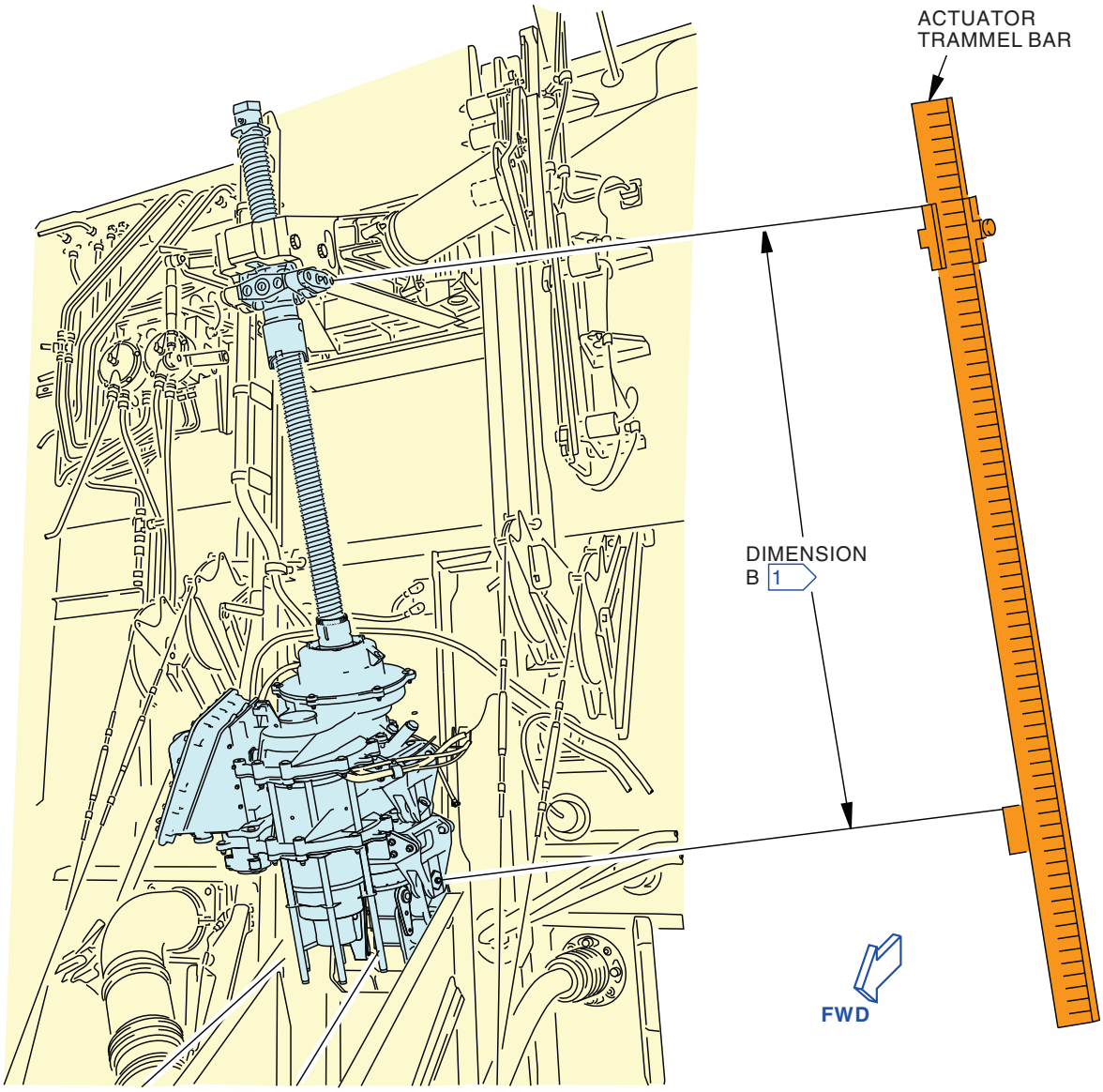
Stabilizer Forward Control Mechanism Installation
Figure 401/27-41-41-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Jackscrew Setting
Figure 402/27-41-41-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

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Stabilizer Trim Jackscrew Setting
Figure 402/27-41-41-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-41-400-801

3. Stabilizer Forward Control Mechanism - Installation

(Figure 401, Figure 402)

A. References

Reference	Title
27-41-00-700-801	Stabilizer Manual Trim and Trim Indicator - Test (P/B 501)
27-41-00-820-801	Stabilizer Control Cable and Chain - Adjustment (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
COM-25414	Repair Tool - Chain, Standard No. 35 Part #: G605C Supplier: 18202

C. Consumable Materials

Reference	Description	Specification
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
43	Stabilizer forward control mechanism	27-41-41-01-095	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

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F. Access Panels

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door

G. Stabilizer Forward Control Mechanism Installation

SUBTASK 27-41-41-420-001

(1) Install the forward stabilizer forward control mechanism [43].

(a) Put the forward stabilizer forward control mechanism [43] in its position.

(b) Install the bolt [17], washer [53], bushing [18], washer [19], and nut [20] to attach the aft support link [22] to the structure.

1) Apply compound, C00528, to the washer [53].

(c) Install the bolt [13], washer [52], bushing [14], washer [15], and nut [16] to attach the aft support link [22] to the structure.

1) Apply compound, C00528, to the washer [52].

(d) Install the bolts [25], washers [26], bushings [27], washers [28], and nuts [29] to attach the forward support link [37] to the structure.

1) Apply compound, C00528, to the bushings [27].

(e) Install the bolt [2], washer [3], bushing [4], washers [5], washer [6], and nut [7] to connect the lower turnbuckle [1] to the brackets.

SUBTASK 27-41-41-420-002

(2) Adjust the lower turnbuckle [1] to get approximately 12.43 in. (315.7 mm) between the connections.

(a) Make sure that there is approximately equal extension of the screws on each side of the sleeve.

SUBTASK 27-41-41-020-011

(3) Connect the chain [8].

(a) Put the chain [8] on the idler sprocket [9] and the forward mechanism sprocket [10].

(b) If it is necessary, add or remove the links from the chain [8], to correct the length of the chain [8].

1) Use the chain repair tool, COM-25414, to install or remove the links from the chain [8].

a) Grind the pin head before pin removal from the chain [8].

b) Do not grind the chain [8].

(c) Connect the chain [8] at the connector link [11].

(d) Install the pins [12] on the outboard side of the chain [8].

SUBTASK 27-41-41-820-002

(4) Adjust the chain [8] tension.

(a) Apply a 5 lbf (22.2 N) force at approximately 4.5 ±1.0 in. (114 ±26 mm) from where the chain [8] touches the forward mechanism sprocket [10].

(b) Make sure that the movement of the chain [8] is 0.25 ±0.07 in. (6.4 ±1.8 mm).

(c) If it is necessary, do these steps to get the correct chain movement:

1) Adjust the upper adjusting nuts [24] and upper adjusting nut [38].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure that all three A dimensions equal ± 0.06 in. (1.52 mm) from each other:
   a) Measure the A dimensions of left and right aft support link [22].
      NOTE: The A dimension is the distance of the aft support link between the center of the two connection bolts.
   b) Measure the A dimension of the forward support link [37].
      NOTE: The A dimension is the distance of the forward support link between the center of the two connection bolts.
   c) Make sure that all three A dimensions are ± 0.06 in. (1.52 mm) from each other.
   d) Install MS20995NC32 lockwire, G01912.

SUBTASK 27-41-41-420-003

(5) Connect the STB cable.
   (a) Attach the eye of the STB cable in the slot in the top of the cable drum [39].
   (b) Turn the trim control wheel to wind the STB cable 28-1/2 turns on the front of the cable drum [39].

SUBTASK 27-41-41-420-004

(6) Connect the STA cable.
   (a) Attach the STA cable to the cable drum [39].
      1) Use a force gauge, COM-1557, to put a 50 lbf (222 N) load on the trim control wheel to pull on the STB cable and make installation of the cable STA easier.
   (b) Wind the STA cable approximately 4-1/2 turns on the front of the cable drum [39].

SUBTASK 27-41-41-420-005

(7) Install the washers [41] and nuts [42] to attach the cable guard [40] to the forward stabilizer forward control mechanism [43].

SUBTASK 27-41-41-860-005

(8) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-41-820-003

(9) Adjust the cables STA and STB (TASK 27-41-00-820-801).

SUBTASK 27-41-41-210-003

(10) Make sure that the flexible shaft [21] is connected to the stabilizer forward control mechanism [43].
   (a) Make sure that the lockwire is installed on the coupling nut of the flexible shaft [21].

SUBTASK 27-41-41-420-010

(11) Install the bolt [30], washer [50], washer [51], washer [31], bushing [32], washer [33], and nut [34] to connect the upper turnbuckle [35] to the brackets.
   (a) Install MS20995NC32 lockwire, G01912.
      NOTE: The lockwire must be attached to the tab of the washer of the upper turnbuckle.

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SUBTASK 27-41-41-710-001

(12) Do this task: Stabilizer Manual Trim and Trim Indicator - Test, TASK 27-41-00-700-801.

SUBTASK 27-41-41-410-001

(13) Close this access door:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-41-41-410-002

(14) Close this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER FORWARD CONTROL MECHANISM - INSPECTION/CHECK

1. General

A. This procedure has the following tasks:

(1) Stabilizer forward control mechanism inspection.

(2) Stabilizer Trim Forward Mechanism Retention Turnbuckles and Turnbuckle Attachment Points Inspection.

(3) Stabilizer Forward Trim Control Mechanism General Visual Inspection.

TASK 27-41-41-210-801

2. Stabilizer Control Mechanism Detail Visual Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

B. Access Panels

Number	Name/Location
112A	Forward Access Door

C. Stabilizer Control Mechanism Detail Visual Inspection

SUBTASK 27-41-41-010-003

(1) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-41-41-210-001

(2) Do a detail visual inspection of the stabilizer forward control mechanism:

(a) Examine all attachment points for security and damage.

(b) Examine upper and lower turnbuckles for damage or binding.

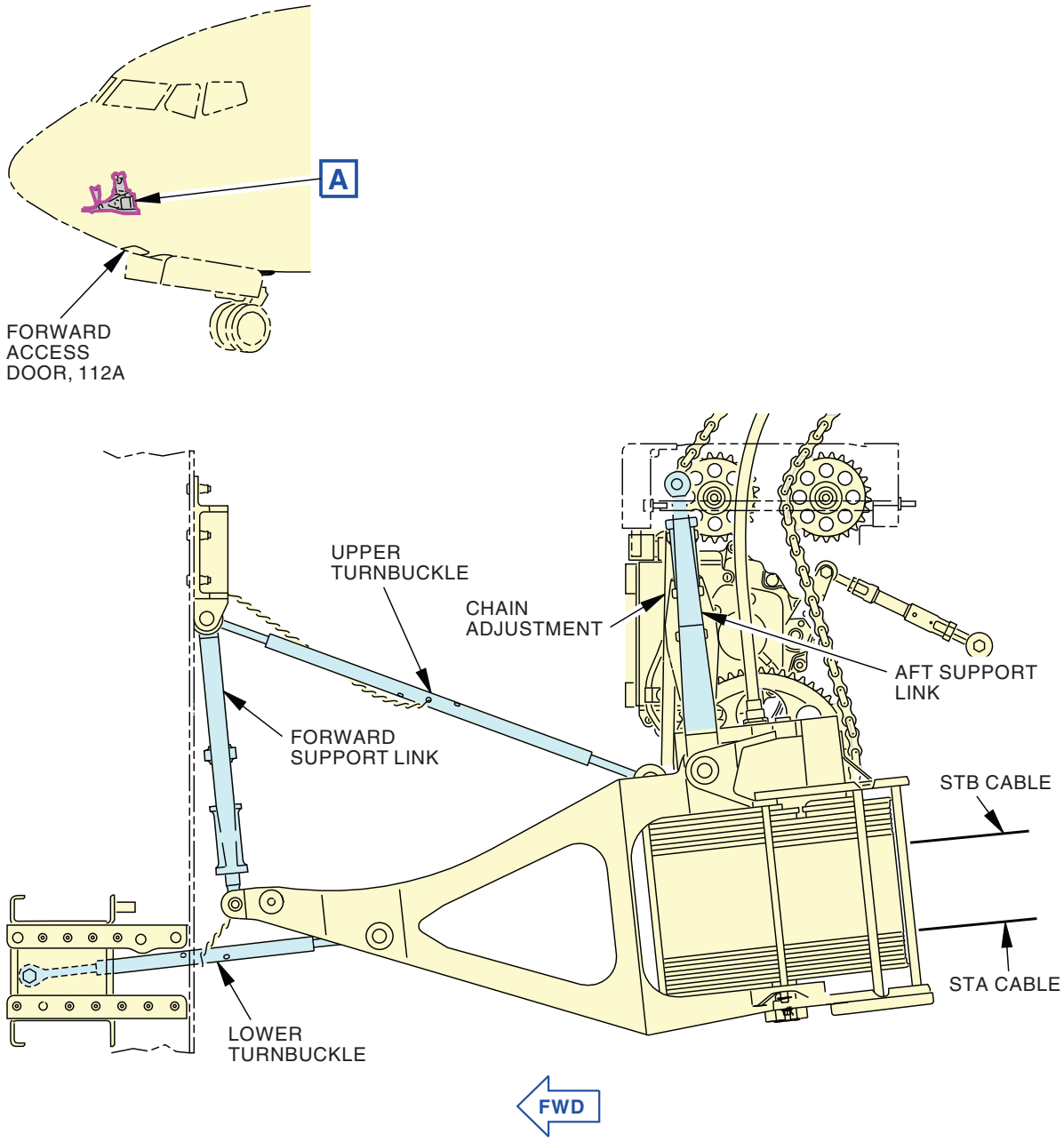
SUBTASK 27-41-41-410-003

(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER FORWARD CONTROL MECHANISM
(VIEW IN THE INBOARD DIRECTION)

A

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Stabilizer Forward Control Mechanism
Figure 601/27-41-41-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-41-210-802

3. Upper and Lower Stabilizer Trim Forward Mechanism Retention Turnbuckles and Turnbuckle Attachment Points Detail Visual Inspection

(Figure 601)

A. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

B. Access Panels

Number	Name/Location
112A	Forward Access Door

C. Procedure

SUBTASK 27-41-41-020-012

(1) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-41-41-211-001

(2) Do a detail visual inspection of the upper and lower stabilizer trim forward mechanism retention turnbuckles and turnbuckle attachment points: (Figure 601)

(a) Examine upper and lower turnbuckles and turnbuckle attachment points for security, damage or binding.

SUBTASK 27-41-41-410-004

(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

TASK 27-41-41-210-803

4. Stabilizer Forward Trim Control Mechanism Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

B. Access Panels

Number	Name/Location
112A	Forward Access Door

C. Stabilizer Forward Trim Control Mechanism Inspection

SUBTASK 27-41-41-010-004

(1) To get access to the stabilizer forward trim control mechanism,

Open this access panel:

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-41-210-002

(2) Do a general visual inspection of the stabilizer forward trim control mechanism.

SUBTASK 27-41-41-410-005

(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COLUMN CUTOUT SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Column cutout switch removal.

(2) Column cutout switch installation.

B. This procedure is applicable for the captain's and first officer's column cutout switches.

(1) Electrical connectors D12156 and D817 are located on the captain's column cutout switch, M1983.

(2) Electrical connectors D10553, D2257 and D3124 are located on the first officer's column cutout switch, M1201.

TASK 27-41-51-000-801

2. Column Cutout Switch Removal

(Figure 401)

A. References

Reference	Title
31-31-52-000-801	Control Column Position Sensor Removal (P/B 401)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

D. Prepare for the Removal

SUBTASK 27-41-51-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-51-010-001

(2) Open this access door:

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-51-010-002

(3) Remove this access panel for the captain's cutout switch (TASK 53-14-01-020-801).

(a) Remove this access panel:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-41-51-010-003

(4) Remove this access panel for the first officer's cutout switch (TASK 53-14-01-020-801).

(a) Remove this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

E. Column Cutout Switch Removal

SUBTASK 27-41-51-020-001

(1) Disconnect the electrical connectors for the column cutout switches.

SUBTASK 27-41-51-020-002

(2) Do these steps before the removal of the captain's column cutout switch [9]:

(a) Remove the nut [6], washer [5], and bolt [4] to disconnect the rod assembly [8] from the column cutout switch arm [3].

(b) Remove the screw [13] from the standoffs on the column cutout switches.

SUBTASK 27-41-51-020-003

(3) Do these steps to remove the captain's column cutout switch [9]:

(a) Remove the bolts [1] and washers [2] that attach the captain's column cutout switch [9] to the beam assembly.

(b) Remove the captain's column cutout switch [9].

SUBTASK 27-41-51-020-006

(4) Do these steps before the removal of the first officer's column cutout switch [12]:

(a) Remove the nut [6], washer [5], and bolt [4] to disconnect the rod assembly [8] from the column cutout switch arm [3].

(b) Remove the screw [13] from the standoffs on the column cutout switches.

SUBTASK 27-41-51-020-004

(5) Do these steps to remove the first officer's column cutout switch [12]:

(a) Remove the bolts [10] and washers [11] that attach the first officer's column cutout switch [12] to the beam assembly.

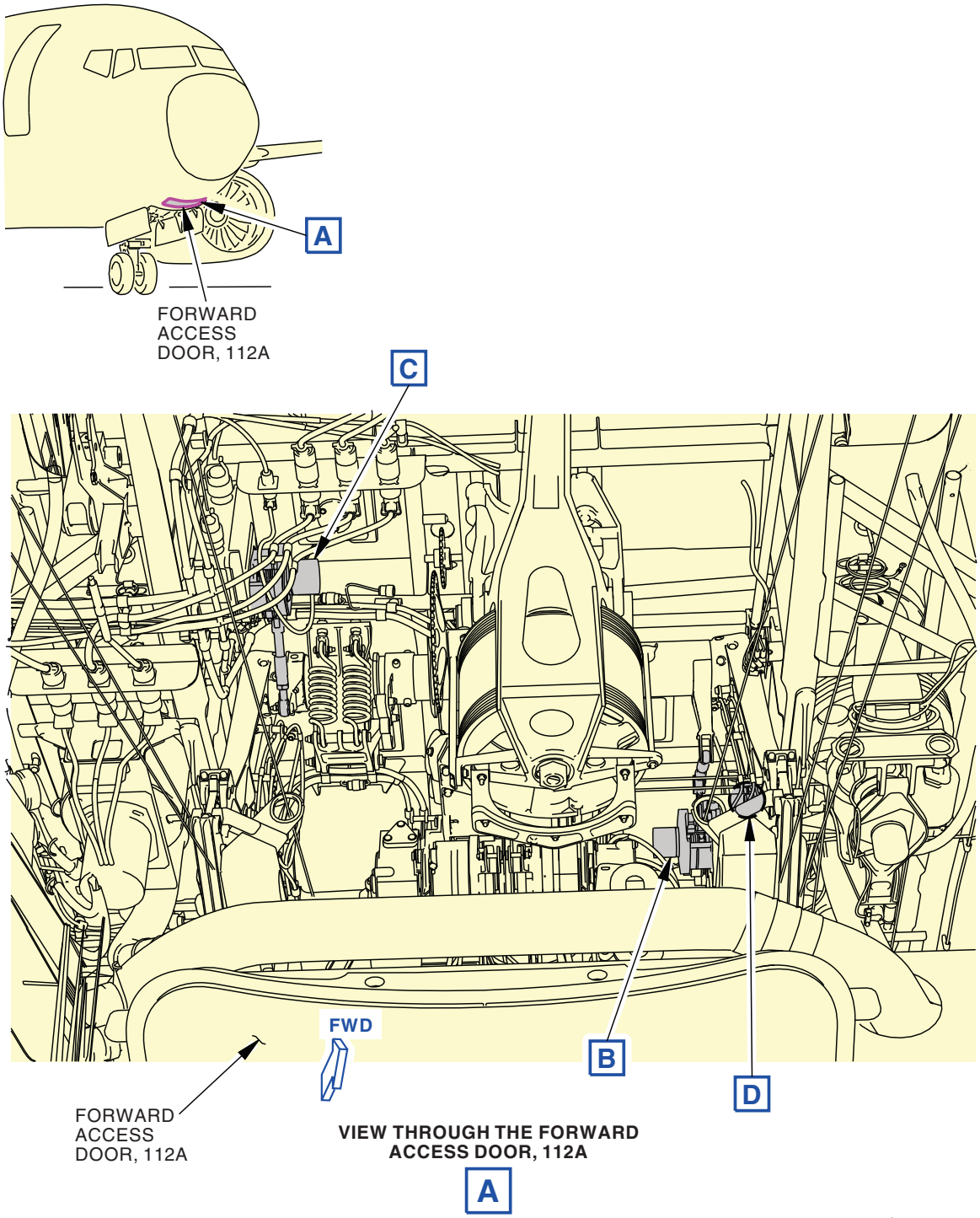
(b) Remove the first officer's column cutout switch [12].

SUBTASK 27-41-51-020-005

(6) If it is necessary, remove the control column position sensor from the column cutout switch. To remove it, do this task: Control Column Position Sensor Removal, TASK 31-31-52-000-801.

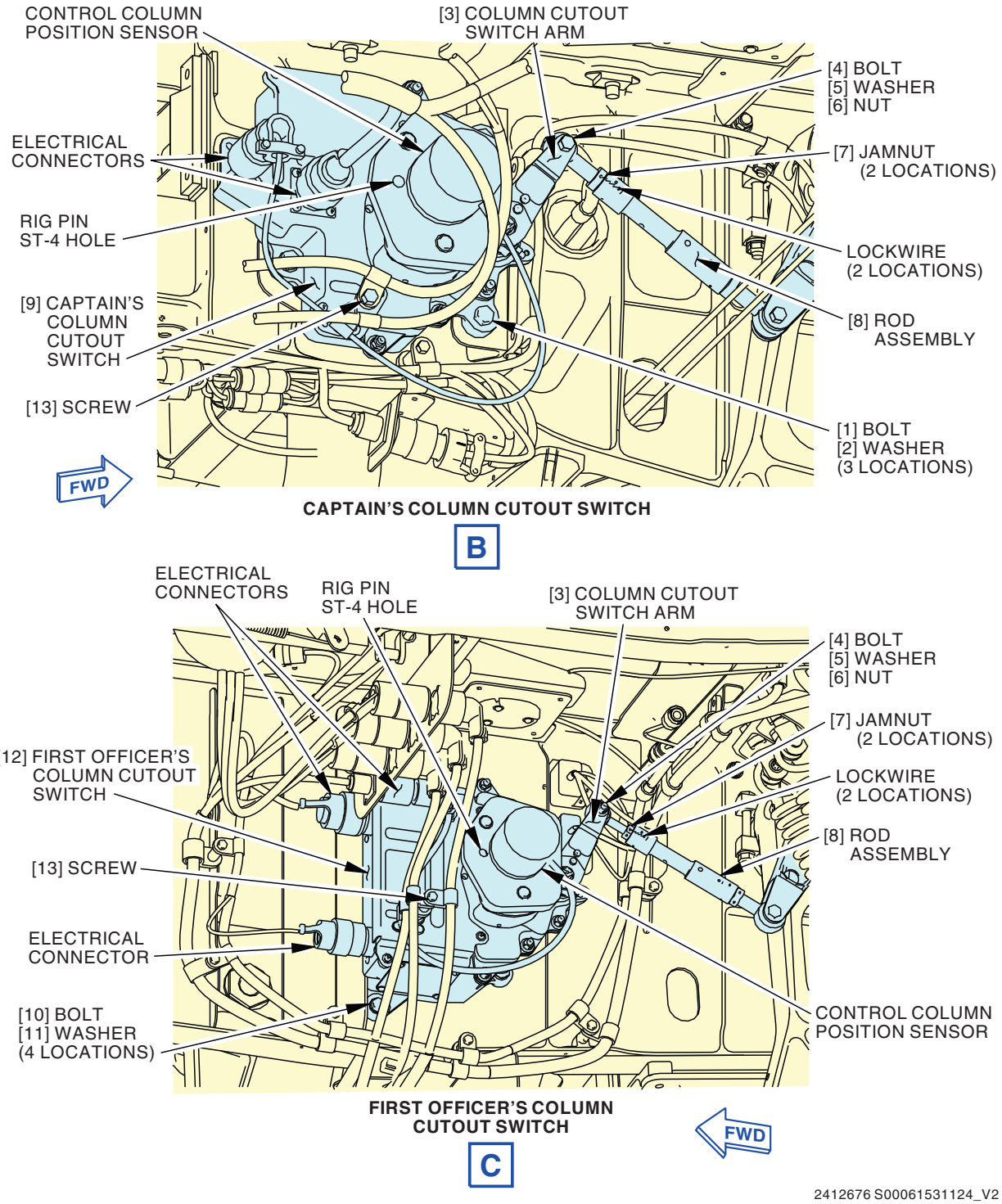
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



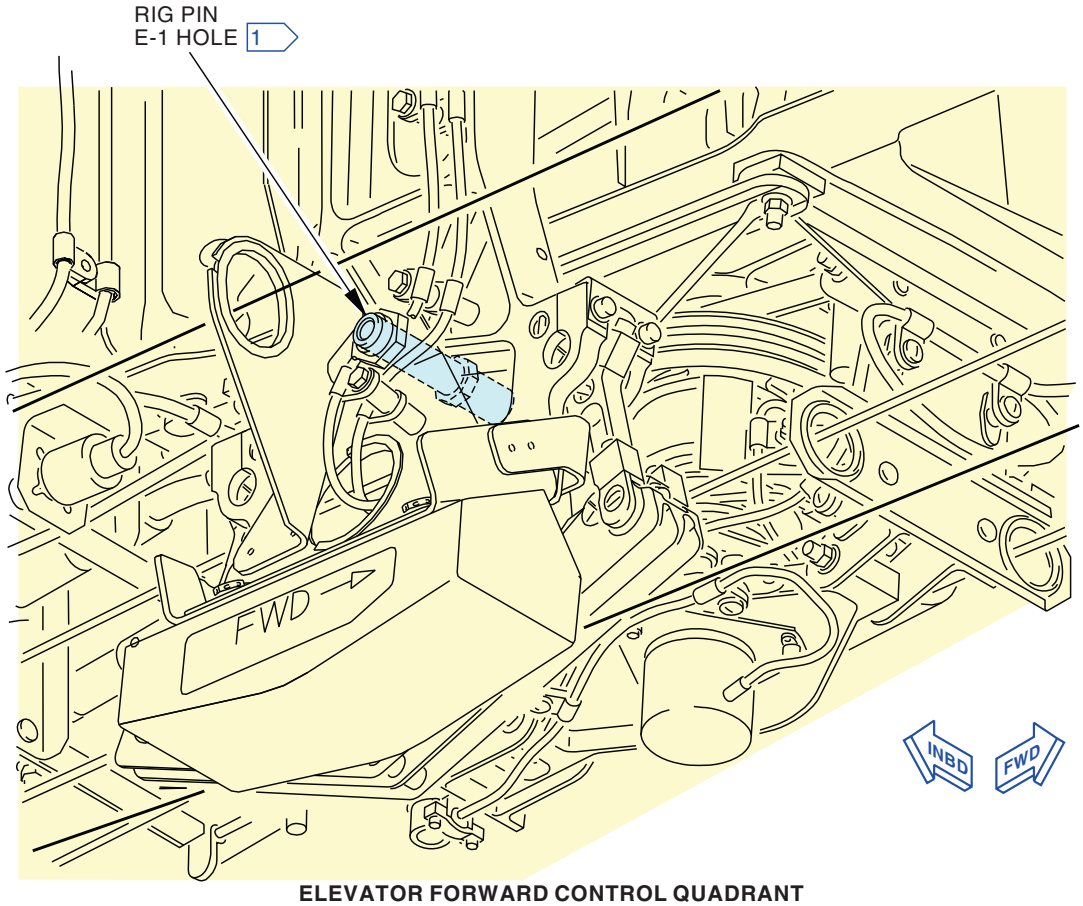
Column Cutout Switch Installation
Figure 401/27-41-51-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Column Cutout Switch Installation
Figure 401/27-41-51-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR FORWARD CONTROL QUADRANT

D

1 CAPTAIN'S SIDE ONLY

2412677 S00061531125_V1

Column Cutout Switch Installation
Figure 401/27-41-51-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-51-400-801

3. Column Cutout Switch Installation

(Figure 401)

A. References

Reference	Title
27-41-00-710-801	Column Cutout Switches Test (P/B 501)
31-31-52-400-801	Control Column Position Sensor Installation (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Captain's column cutout switch	27-41-51-01-055	PTF ALL
12	First officer's column cutout switch	27-41-51-02-055	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

F. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

G. Column Cutout Switch Installation

SUBTASK 27-41-51-420-001

(1) If it is necessary, install the control column position sensor to the column cutout switch. To install it, do this task: Control Column Position Sensor Installation, TASK 31-31-52-400-801.

SUBTASK 27-41-51-420-002

(2) Do these steps to install the captain's column cutout switch [9]:

(a) Put the captain's column cutout switch [9] in its position.

(b) Install the bolts [1] and washers [2] to attach the captain's column cutout switch [9] to the beam assembly.

(c) Connect the electrical connectors.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-51-420-003

(3) Do these steps to install the first officer's column cutout switch [12]:

(a) Put the first officer's column cutout switch [12] in its position.

(b) Install the bolts [10] and washers [11] to attach the first officer's column cutout switch [12] to the beam assembly.

(c) Connect the electrical connectors.

SUBTASK 27-41-51-420-004

(4) Install the screw [13] to the standoffs on the column cutout switches.

SUBTASK 27-41-51-420-005

(5) Connect the rod assembly [8] to the column cutout switch arm [3]:

(a) Install the rig pin E-1 from the rigging pin kit, SPL-1585 through the left elevator forward control quadrant.

(b) Install the rig pin ST-4 from the rigging pin kit, SPL-1585 in the column cutout switch.

(c) Remove the lockwires from the rod assembly [8].

(d) Turn the jamnuts [7] until you can easily install the bolt [4] that connect the rod assembly [8] to the column cutout switch arm [3].

(e) Install the bolt [4], washer [5], and nut [6] to connect the rod assembly [8] to the column cutout switch arm [3].

(f) Tighten the jamnuts [7].

(g) Install the MS20995C32 lockwires, G01048.

(h) Remove the rig pin E-1 from the left elevator forward control quadrant.

(i) Remove the rig pin ST-4 from the control column switch.

SUBTASK 27-41-51-710-001

(6) Do this task: Column Cutout Switches Test, TASK 27-41-00-710-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-51-860-002

(1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-51-410-002

(2) Install this access panel for the captain's column cutout switch (TASK 53-14-01-420-801).

(a) Install this access panel:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-41-51-410-004

(3) Install this access panel for the first officer's column cutout switch (TASK 53-14-01-420-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-41-51-410-005

(4) Close this access door:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HORIZONTAL STABILIZER TRIM CONTROL WHEEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Stabilizer Trim Control Wheel Removal.

(2) Stabilizer Trim Control Wheel Installation.

TASK 27-41-61-010-801

2. Stabilizer Trim Control Wheel Removal

(Figure 401)

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-41-61-860-001

(1) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the CUTOUT position.

SUBTASK 27-41-61-860-002

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

C. Stabilizer Trim Control Wheel Removal

SUBTASK 27-41-61-010-001

(1) Remove the stabilizer trim wheel [1] from the control stand [7]:

(a) Remove the bolt [9] and spacer [8] that attaches the stabilizer trim wheel [1] to the shaft.

(b) Remove the stabilizer trim wheel [1] from the shaft.

SUBTASK 27-41-61-010-002

(2) As necessary, remove the upper side cover [3] and lower side cover [5] from the right side of the control stand [7]:

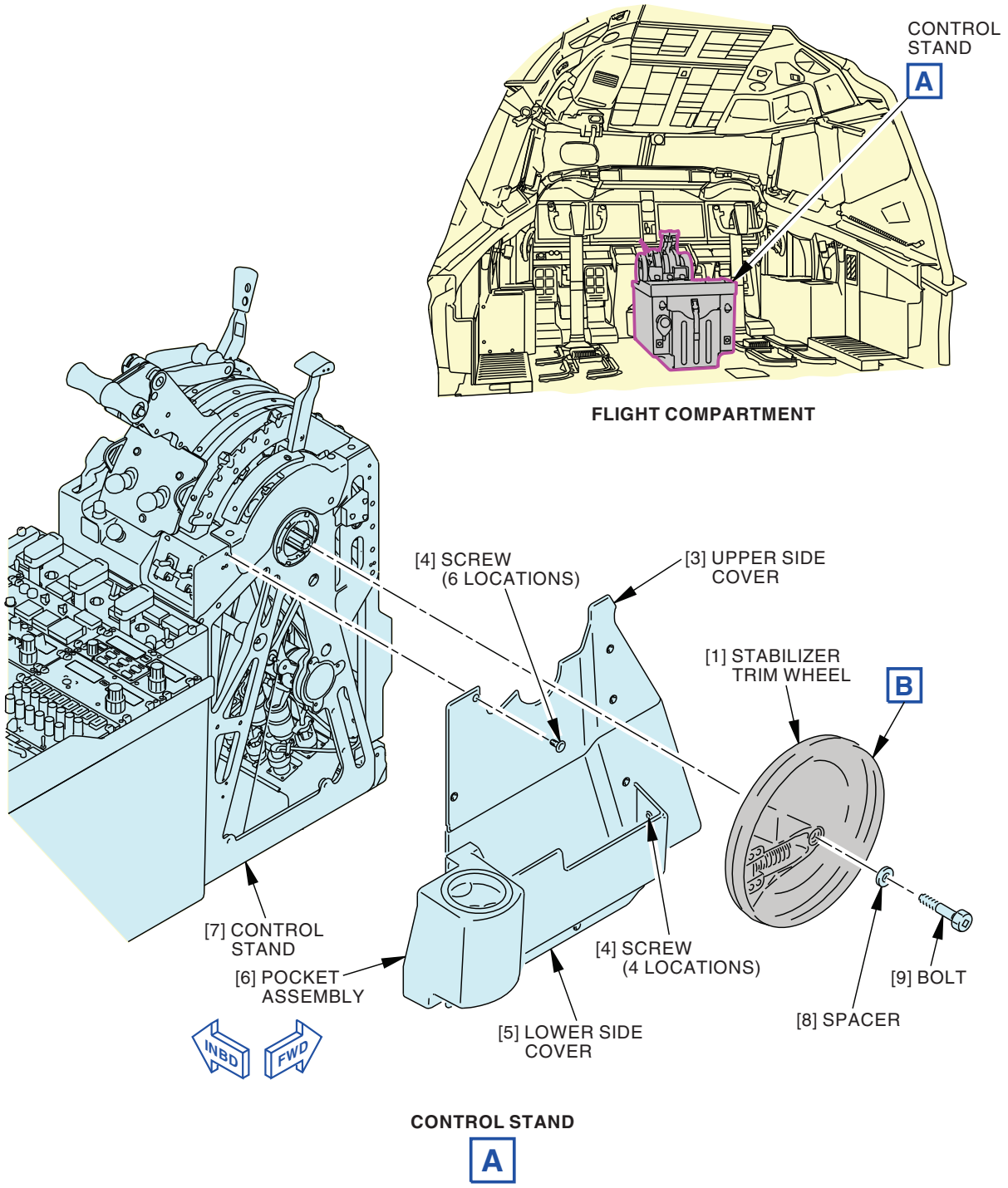
(a) Remove the screws [4] and the pocket assembly [6] from the lower side cover [5].

(b) Remove the screws [4] that attach the upper side cover [3] and the lower side cover [5] to the control stand [7].

(c) Remove the upper side cover [3] and the lower side cover [5] from the control stand [7].

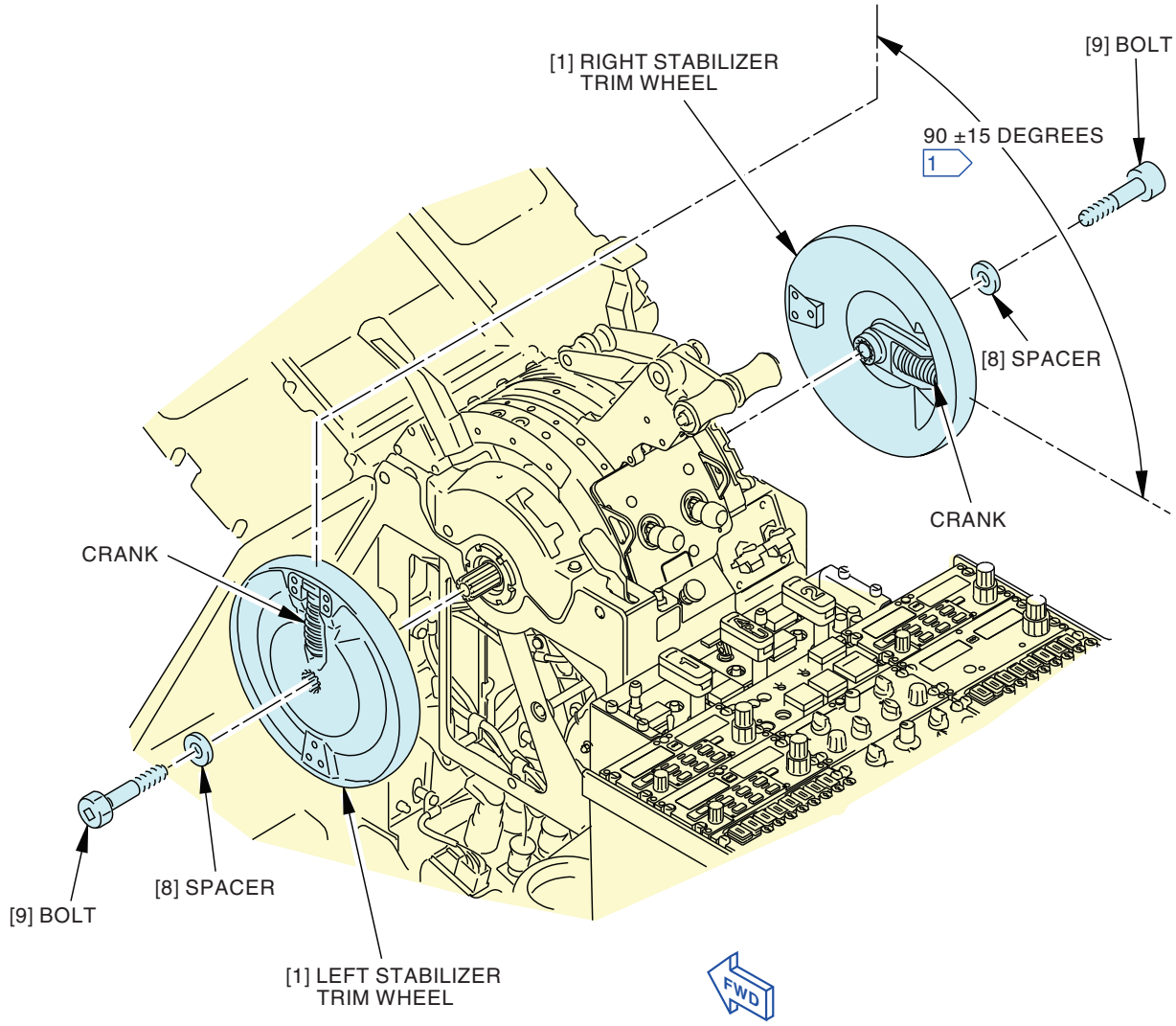
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Control Wheel Installation
Figure 401/27-41-61-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 INSTALL THE LEFT AND RIGHT TRIM ON THE SHAFT WITH THE CRANKS 90°±15 DEGREES APART.

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Stabilizer Trim Control Wheel Installation
Figure 401/27-41-61-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-61-410-801

3. Stabilizer Trim Control Wheel Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Stabilizer trim wheel	27-41-00-19-095	PTF ALL

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Stabilizer Trim Control Wheel Installation

SUBTASK 27-41-61-410-001

(1) As necessary, install the upper side cover [3] and lower side cover [5] on the control stand [7]:

(a) Install the upper side cover [3] and the lower side cover [5] on the control stand [7].

(b) Install the screws [4] that attach the upper side cover [3] and the lower side cover [5] to the control stand [7].

(c) Install the screws [4] in the pocket assembly [6] and the lower side cover [5].

SUBTASK 27-41-61-410-002

(2) Install the stabilizer trim wheel [1] on the control stand [7]:

(a) Install the stabilizer trim wheel [1] on the shaft with the handle 90 +/- 15 degrees apart from the handle on the other stabilizer trim wheel [1].

NOTE: Make sure an equal amount of the shaft is visible on both sides of the pedestal.

(b) Install the bolt [9] and spacer [8] that attaches the stabilizer trim wheel [1] to the shaft.

(c) Tighten the bolt to 150 in-lb (17 N·m) - 160 in-lb (18 N·m) plus the run-on torque.

SUBTASK 27-41-61-860-004

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-61-860-006

(4) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the NORMAL position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER TRIM MOTOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the stabilizer trim motor

(2) An installation of the stabilizer trim motor.

TASK 27-41-71-000-801

2. Stabilizer Trim Motor Removal

(Figure 401)

A. Consumable Materials

Reference	Description	Specification
B50124	Cleaner - Thermoplastic	

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

C. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

D. Prepare for the Removal

SUBTASK 27-41-71-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-71-010-001

(2) To get access to the stabilizer trim motor [1], do this step:

Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Stabilizer Trim Motor Removal

SUBTASK 27-41-71-020-001

(1) Remove the stabilizer trim motor [1]:

(a) Clean the electrical connectors [2] using thermoplastic cleaner, B50124, (isopropyl alcohol) (SWPM 20-60-01).

(b) Disconnect the electrical connectors [2] from the stabilizer trim motor [1].

NOTE: A blue filter seal can be found in the connector. The filter is for Electro-Magnetic Interference (EMI) protection and must stay in the connector.

(c) Install protective caps on the electrical connectors [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Remove the nuts [6], washers [5], washers [11] and bolts [4] that attaches the stabilizer trim motor [1] to the stabilizer trim actuator [3].

(e) Remove the sealant from bolt [7].

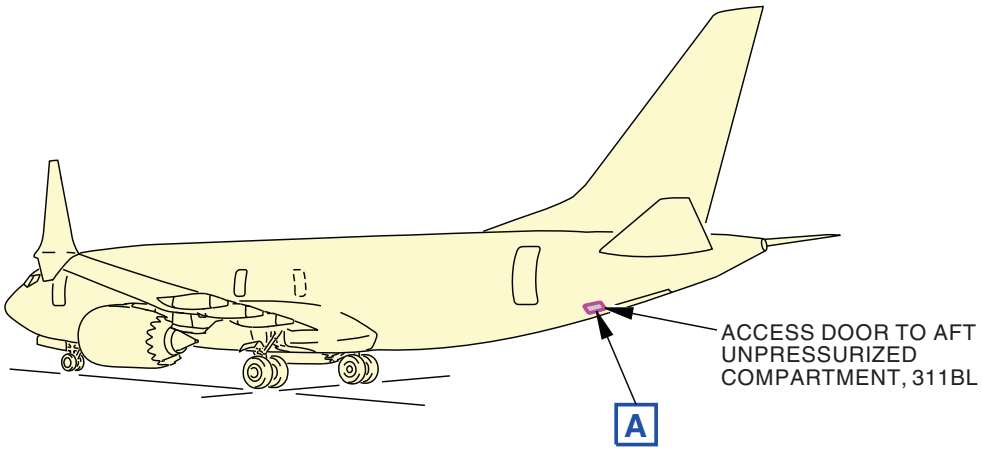
(f) Remove the nut [10], washer [9], washer [8] and bolt [7] that attaches the stabilizer trim motor [1] to the stabilizer trim actuator [3].

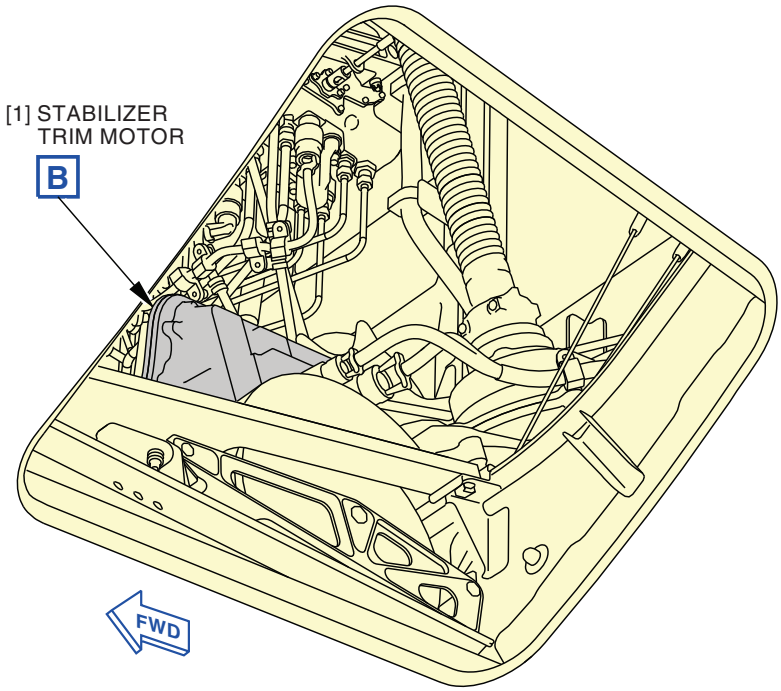
(g) Remove the stabilizer trim motor [1].

(h) Put a cover on the mounting pad to keep unwanted materials out of the stabilizer trim actuator [3].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





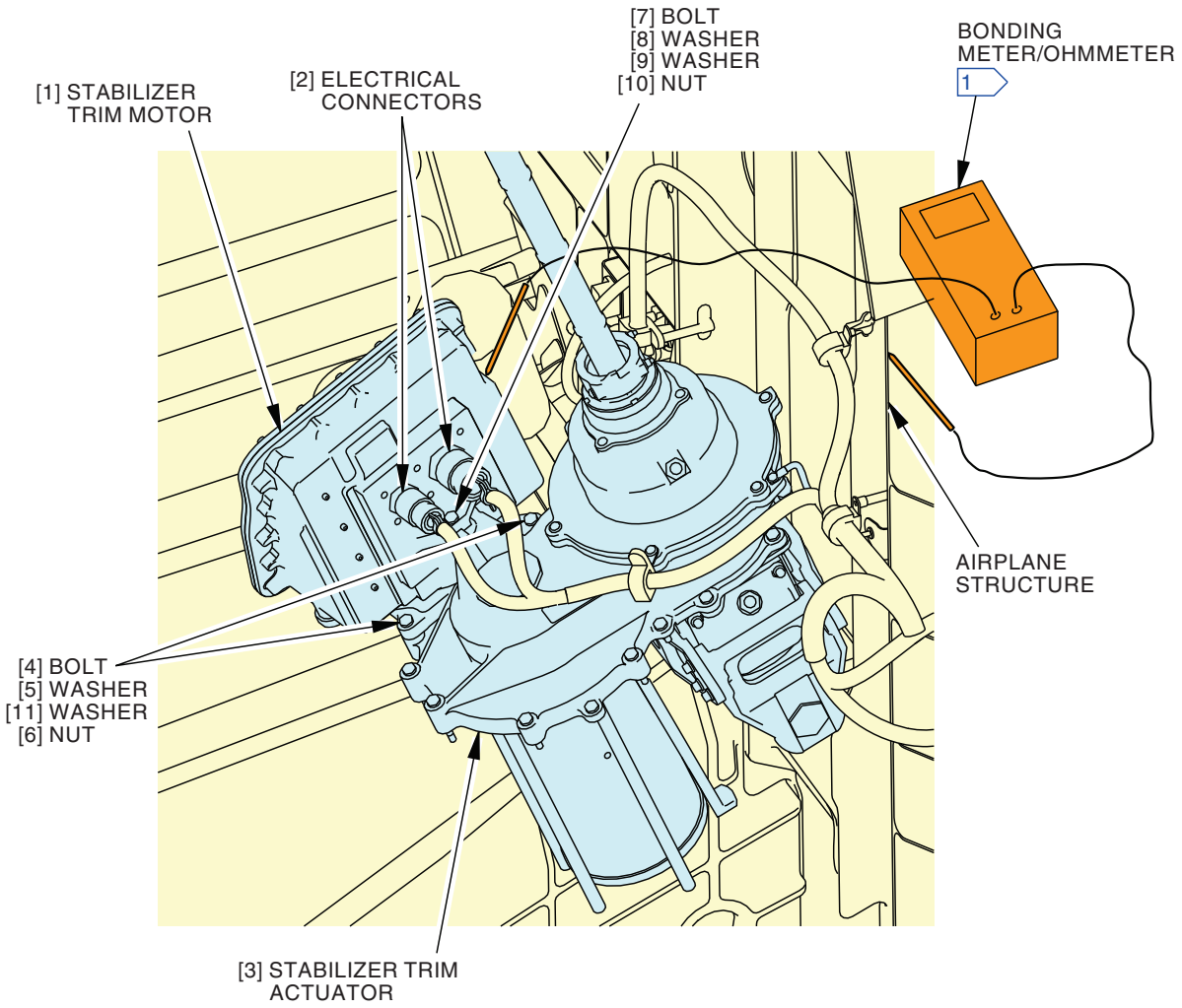
VIEW THROUGH THE ACCESS DOOR TO AFT UNPRESSURIZED COMPARTMENT, 311BL

A

2412682 S00061531138_V2

Stabilizer Trim Motor Installation
Figure 401/27-41-71-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 PLACE THE PROBES FROM THE BONDING METER/OHMMETER AS SHOWN TO MEASURE THE RESISTANCE BETWEEN THE STABILIZER TRIM MOTOR AND THE AIRPLANE STRUCTURE. THE RESISTANCE SHOULD BE 0.0025 OHM OR LESS.

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Stabilizer Trim Motor Installation
Figure 401/27-41-71-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-71-400-801

3. Stabilizer Trim Motor Installation

(Figure 401)

A. References

Reference	Title
27-41-71-700-801	Stabilizer Trim Motor Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
A50337	Sealant - Fuel Tank	BMS5-45 Class B
D00633	Grease - Aircraft General Purpose	BMS3-33

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Stabilizer trim motor	27-41-71-01-180	PTF ALL

E. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Stabilizer Trim Motor Installation

SUBTASK 27-41-71-860-007

(1) Make sure that the DO NOT OPERATE tag, STD-858, is installed on the stabilizer trim wheel at the control stand.

SUBTASK 27-41-71-020-002

(2) Remove the cover from the mounting pad.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-71-160-001

(3) Make sure that the mating surfaces of the stabilizer trim motor [1] and stabilizer trim actuator [3] are clean and free of paint.

SUBTASK 27-41-71-640-001

(4) Fill the internal spline and motor mount area of the stabilizer trim actuator [3] with grease, D00633.

SUBTASK 27-41-71-420-001

(5) Install the stabilizer trim motor [1].

(a) Engage the splined shaft of the stabilizer trim motor [1] with the internal spline of the stabilizer trim actuator [3].

(b) Install the bolts [4], washer [5], washers [11], and nuts [6].

(c) Install the bolt [7], washer [8], washer [9], and nut [10].

(d) Connect the electrical connectors [2].

NOTE: A blue filter seal can be found in the connector. The filter is for Electro-Magnetic Interference (EMI) protection and must stay in the connector.

SUBTASK 27-41-71-220-001

(6) Measure the electrical bonding resistance of the stabilizer trim motor [1] (SWPM 20-20-00) (Figure 401).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical bonding resistance between the stabilizer trim motor [1] and airplane structure is 0.00275 ohms (2.75 milliohms) or less.

(c) Make sure that the electrical bonding resistance between the stabilizer trim motor [1] and gearbox housing is 0.0005 ohms (0.5 milliohms) or less.

SUBTASK 27-41-71-390-001

(7) Apply sealant, A50337, to fully cover the bolt [7], washer [8], washer [9], and nut [10].

SUBTASK 27-41-71-860-002

(8) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-71-860-008

(9) Remove the DO NOT OPERATE tag, from the stabilizer trim wheel at the control stand.

SUBTASK 27-41-71-820-001

(10) Do this task: Stabilizer Trim Motor Test, TASK 27-41-71-700-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-71-410-001

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER TRIM MOTOR - ADJUSTMENT/TEST

1. General

A. This procedure has one task. The task is the test of the stabilizer trim motor.

TASK 27-41-71-700-801

2. Stabilizer Trim Motor Test

A. References

Reference	Title
22-11-81-710-801	Stabilizer Trim Motor Autopilot Test (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
330	Subzone - Left Horizontal Stabilizer and Elevator
340	Subzone - Right Horizontal Stabilizer and Elevator

C. Prepare for the Test

SUBTASK 27-41-71-860-004

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-71-860-005

(2) Make sure that the STAB TRIM PRI (S272) and B/U (S149) switches on the aft area of the control stand are in the NORMAL position.

SUBTASK 27-41-71-860-006

(3) Make sure that the STAB TRIM override switch, on the P8 panel, is in the NORM position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

D. Stabilizer Trim Motor Test

SUBTASK 27-41-71-710-001

(1) Do the test of the stabilizer trim motor:

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Supply system B hydraulic power (TASK 29-11-00-860-801).

(b) Set the flaps to a position from 1-40 units.

1) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE DOWN position.

a) Make sure that the stabilizer leading edge moves up.

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b) Release the stabilizer trim control switches.

2) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE UP position.
   a) Make sure that the stabilizer leading edge moves down.
   b) Release the stabilizer trim control switches.
3) Push and hold the stabilizer trim control switches on the first officer's control wheel to the NOSE DOWN position.
   a) Make sure that the stabilizer leading edge moves up.
   b) Release the stabilizer trim control switches.
4) Push and hold the stabilizer trim control switches on the first officer's control wheel to the NOSE UP position.
   a) Make sure that the stabilizer leading edge moves down.
   b) Release the stabilizer trim control switches.
5) Check the travel time for the stabilizer leading edge to move from the NOSE UP position to the NOSE DOWN position:
   a) Push the stabilizer trim control switches on the captain's control wheel to the NOSE UP position and hold until the stabilizer reaches it's lower limit of travel.
   b) Push the stabilizer trim control switches on the captain's control wheel to the NOSE DOWN position and hold until the stabilizer reaches it's upper limit of travel.
      <1> Make sure that the time for the stabilizer leading edge to move from its lower limit of travel to its upper limit of travel is between 31-39 seconds.
      <2> Make sure that the operation is smooth and there is no unusual noise.
6) Use the stabilizer trim control switches on the captain's control wheel to move the stabilizer many times in the two directions, one after the other for 3 seconds each.
   a) Make sure that the operation is smooth and there is no unusual noise.
7) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE UP position until the system comes to the full trim speed.
   a) Move the stabilizer trim control switches momentarily to OFF and then to the NOSE DOWN position.
   b) Make sure that the stabilizer leading edge moves up immediately after you move the stabilizer trim control switches to the NOSE DOWN position.
   c) Release the stabilizer trim control switches after confirming that the leading edge moves up.
8) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE DOWN position until the system comes to the full trim speed.
   a) Move the stabilizer trim control switches momentarily to OFF and then to the NOSE UP position.
   b) Make sure that the stabilizer leading edge moves down immediately after you move the stabilizer trim control switches to the NOSE UP position.
   c) Release the stabilizer trim control switches after confirming that the leading edge moves down.
9) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE UP position until the system comes to full trim speed.

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a) Release the stabilizer trim control switches.

b) Make sure that the stabilizer trim wheel moves no more than 2.5 turns (900 degrees).

NOTE: Reverse coasting (backdriving) of the trim wheel is allowed if it does not exceed 27 degrees NOSE DOWN after trimming NOSE UP.

10) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE DOWN position until the system comes to full trim speed.

a) Release the stabilizer trim control switches.

b) Make sure that the stabilizer trim wheel moves no more than 2.5 turns (900 degrees).

NOTE: Reverse coasting (backdriving) of the trim wheel is allowed if it does not exceed 100 degrees NOSE UP after trimming NOSE DOWN.

(c) Set the flaps to the full up position.

1) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE DOWN position.

a) Make sure that the stabilizer leading edge moves up.

b) Release the stabilizer trim control switches.

2) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE UP position.

a) Make sure that the stabilizer leading edge moves down.

b) Release the stabilizer trim control switches.

3) Check the travel time for the stabilizer leading edge to move from the NOSE UP position to the NOSE DOWN position:

a) Push the stabilizer trim control switches on the captain's control wheel to the NOSE UP position and hold until the stabilizer reaches it's lower limit of travel.

b) Push the stabilizer trim control switches on the captain's control wheel to the NOSE DOWN position and hold until the stabilizer reaches it's upper limit of travel.

<1> Make sure that the time for the stabilizer leading edge to move from its lower limit of travel to its upper limit of travel is between 47–58 seconds.

<2> Make sure that the operation is smooth and there is no unusual noise.

4) Use the stabilizer trim control switches on the captain's control wheel to move the stabilizer many time in the two directions one after the other for 3 seconds each.

a) Make sure that the operation is smooth and there is no unusual noise.

5) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE UP position until the system comes to the full trim speed.

a) Move the stabilizer trim control switches momentarily to OFF and then to the NOSE DOWN position.

b) Make sure that the stabilizer leading edge moves up immediately after you move the stabilizer trim control switches to the NOSE DOWN position.

c) Release the stabilizer trim control switches.

6) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE DOWN position until the system comes to the full trim speed.

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a) Move the stabilizer trim control switches momentarily to OFF and then to the NOSE UP position.

b) Make sure that the stabilizer leading edge moves down immediately after you move the stabilizer trim control switches to the NOSE UP position.

c) Release the stabilizer trim control switches.

7) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE UP position until the system comes to full trim speed.

a) Release the stabilizer trim control switches.

b) Make sure that the stabilizer trim wheel moves no more than 0.5 turns (180 degrees).

NOTE: Reverse coasting (backdriving) of the trim wheel is allowed if it does not exceed 65 degrees.

8) Push and hold the stabilizer trim control switches on the captain's control wheel to the NOSE DOWN position until the system comes to full trim speed.

a) Release the stabilizer trim control switches.

b) Make sure that the stabilizer trim wheel moves no more than 0.5 turns (180 degrees).

NOTE: Reverse coasting (backdriving) of the trim wheel is allowed if it does not exceed 65 degrees.

(d) Remove system B hydraulic power (TASK 29-11-00-860-805).

(e) Do this task: Stabilizer Trim Motor Autopilot Test, TASK 22-11-81-710-801.

——— END OF TASK ———

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STABILIZER BALL NUT AND JACKSCREW GEARBOX - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the primary stabilizer ball nut and jackscrew gearbox

(2) An installation of the stabilizer ball nut and jackscrew gearbox.

B. The stabilizer ball nut and jackscrew gearbox is referred to as the stabilizer trim actuator in this procedure.

C. The stabilizer trim actuator includes the ballnut, jackscrew, and jackscrew gearbox.

27-CMR-08

TASK 27-41-81-000-801

2. Stabilizer Ball Nut and Jackscrew Gearbox Removal

(Figure 401, Figure 402, Figure 403)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-41-71-000-801	Stabilizer Trim Motor Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9

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(Continued)

Reference	Description
COM-1563	Hoist - Fishpole, 500 lb Safe Working Limit (SWL) Part #: 10/4180 Supplier: K1425 Part #: IA5101-1 Supplier: 053H3 Part #: IA5101-501 Supplier: 053H3 Part #: IA5119-1 Supplier: 053H3 Part #: IA5119-501 Supplier: 053H3 Part #: PF51-003-1 Supplier: 1YRX6 Part #: PF51-009-1 Supplier: 1YRX6 Opt Part #: 10/3641 Supplier: K1425 Opt Part #: 10/3641C1 Supplier: K1425 Opt Part #: MINILIFT Supplier: K1425
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205
SPL-1678	Lock Equipment - Horizontal Stabilizer Part #: F80027-44 Supplier: 81205 Opt Part #: F80027-41 Supplier: 81205
SPL-1728	Equipment - Handling, Stabilizer Jackscrew (Dome Pressure Bulkhead) Part #: C27050-82 Supplier: 81205
SPL-10319	Wrench - Pinion Gear (Rotate) Part #: F71252 Supplier: 81205 Part #: F71267 Supplier: 81205
STD-858	Tag - DO NOT OPERATE
STD-22106	Cable Spool (Minium Spool Diameter 5", Capacity 200 ft, Cable Diameter 0.125")

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door

E. Prepare for the Removal

SUBTASK 27-41-81-860-001

(1) Set the stabilizer leading edge to the full down (airplane nose up) position.

SUBTASK 27-41-81-860-002

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

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SUBTASK 27-41-81-010-001

(3) To get access to the stabilizer forward control mechanism, do this step:

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-41-81-010-002

(4) To get access to the stabilizer trim actuator, do this step:

(a) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-41-81-020-015

(5) Remove the lockwire from the aft support link [6] and the upper turnbuckle [4].

SUBTASK 27-41-81-820-001

CAUTION

MAKE SURE THAT THE CHAIN IS LOOSE BEFORE YOU RELEASE THE CABLE TENSION. TOO MUCH CABLE TENSION CAN CAUSE DAMAGE TO THE CHAIN.

(6) Adjust the upper adjusting nuts [5] of the aft support link [6] to loosen the chain (Figure 401).

SUBTASK 27-41-81-020-002

(7) Disconnect the upper turnbuckle [4].

SUBTASK 27-41-81-020-003

(8) Loosen the lower turnbuckle [3] to release the tension of the STA cable [66] and STB cable [65] (Figure 401).

SUBTASK 27-41-81-020-004

(9) Do this task: Stabilizer Trim Motor Removal, TASK 27-41-71-000-801.

F. Stabilizer Trim Actuator Removal

SUBTASK 27-41-81-020-005

(1) Remove the clamps [44] for the electrical connectors [47] from the jackscrew gearbox [24].

(a) Remove the nuts [46], washers [45], and bolts [43] to remove the clamps [44] from the jackscrew gearbox [24].

(b) Remove the electrical connectors [47] from the jackscrew gearbox [24].

(c) Install the bolt [43], washers [45], and nuts [46] back to the jackscrew gearbox [24].

SUBTASK 27-41-81-020-006

(2) Remove the cable drum guard [67] and spacers [26] from the cable drum [27].

(a) Remove the pin [83] and castellated nut [82] from the support plate [62] that is attached to the cable drum [27].

1) Discard the pin [83].

(b) Remove the nut [60], washer [59], bolt [57], and washer [58], that attach the cable drum guard [67] to the support plate [62].

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(c) Remove the bolts [63], washers [64], support plate [62], spacers [26], and washer [84] from the cable drum [27].

NOTE: Because work space is small, keep the bolts [63] in the support plate [62] and in spacers [26] to aid with the removal procedure.

(d) Remove the bolts [48] and washers [49] that attach the cable drum guard [67] to the cable drum [27].

(e) Remove the cable drum guard [67] and spacers [26].

SUBTASK 27-41-81-020-007

CAUTION

KEEP TENSION ON THE CABLE AT ALL TIMES. WHEN THE CABLE IS DISCONNECTED FROM THE CABLE DRUM, TENSION NEEDS TO BE KEPT ON THE CABLE SO THAT THE CABLE DOES NOT BECOME CAUGHT IN THE CABLE GUARDS ON THE OTHER CABLE DRUM. THIS CAN CAUSE DAMAGE TO THE CABLE AND THE CABLE MAY NEED TO BE REPLACED.

(3) Disconnect the STB cable [65] from the cable drum [27].

(a) Use the force gauge, COM-1557, on the stabilizer trim wheel handle to put a 50 lb (22.7 kg) load on the STA cable [66].

NOTE: This will keep the STA cable on the drum, which will make the removal of the STB cable easier.

SUBTASK 27-41-81-480-001

(4) Install the pinion gear wrench, SPL-10319, on the internal spline of the jackscrew gearbox [24].

SUBTASK 27-41-81-020-008

(5) Move the jackscrew [23] to put the stabilizer leading edge to the full up position.

CAUTION

DO NOT DAMAGE OR KINK THE CABLE WHILE YOU UNWIND IT FROM THE FORWARD CABLE DRUM. THE CABLE CAN KINK BETWEEN THE CABLE GUARD AND THE FORWARD CABLE DRUM WHEN YOU REMOVE THE LAST WRAP OF CABLE. IF YOU REMOVE THE CABLE TOO QUICKLY, YOU CAN CAUSE DAMAGE THE END OF THE CABLE.

CAUTION

KEEP TENSION ON THE CABLE AT ALL TIMES. WHEN THE CABLE IS DISCONNECTED FROM THE CABLE DRUM, TENSION NEEDS TO BE KEPT ON THE CABLE SO THAT THE CABLE DOES NOT BECOME CAUGHT IN THE CABLE GUARDS ON THE OTHER CABLE DRUM. THIS CAN CAUSE DAMAGE TO THE CABLE AND THE CABLE MAY NEED TO BE REPLACED.

(a) When the stabilizer starts to move, pull the STB cable [65] into the tail compartment and wind it on a cable spool, STD-22106.

NOTE: It is necessary to use the cable spool, STD-22106, to prevent damage and maintain tension on the cable.

NOTE: It is recommended that a mechanic watch the cable unwind from the forward cable drum to make sure that no damage occurs to the cable as the last three wraps of cable are slowly unwound from the cable drum.

(b) When the ballnut [28] touches the stops, keep a record of the number of turns of the STA cable [66] on the cable drum [27].

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CAUTION

KEEP TENSION ON THE CABLE AT ALL TIMES. WHEN THE CABLE IS DISCONNECTED FROM THE CABLE DRUM, TENSION NEEDS TO BE KEPT ON THE CABLE SO THAT THE CABLE DOES NOT BECOME CAUGHT IN THE CABLE GUARDS ON THE OTHER CABLE DRUM. THIS CAN CAUSE DAMAGE TO THE CABLE AND THE CABLE MAY NEED TO BE REPLACED.

(c) Disconnect the STA cable [66] from the cable drum [27].

SUBTASK 27-41-81-480-002

(6) Install the control cable clamp, SPL-1569, to the cables.

(a) Attach the DO NOT OPERATE tags, STD-858, to the stabilizer trim wheels.

SUBTASK 27-41-81-480-003

WARNING

INSTALL THE HORIZONTAL STABILIZER LOCKS. THE STABILIZER LEADING EDGE CAN MOVE UP QUICKLY WHEN YOU DISCONNECT THE JACKSCREW. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(7) Install the lock equipment, SPL-1678, on the jackscrew fitting [21] (Figure 403).

(a) Install the lower horizontal stabilizer lock.

NOTE: The maximum torque for the lower lock assembly bolts is 12 in-lb (1 N·m).

CAUTION

DO NOT MOVE THE JACKSCREW FITTING INTO THE HORIZONTAL STABILIZER LOCKS. THIS CAN CAUSE DAMAGE TO THE EQUIPMENT.

(b) Turn the motor shaft or cable drum by hand to move the stabilizer leading edge down until the stabilizer truss touches the horizontal stabilizer lock.

(c) Install the upper horizontal stabilizer lock.

NOTE: The maximum torque for the upper lock assembly bolts is 12 in-lb (1 N·m).

1) Make sure that the adjustable leveling feet are adjusted so that they contact the jackscrew fittings [21].

SUBTASK 27-41-81-480-004

(8) Install the equipment, SPL-1728 and fishpole hoist, COM-1563, to the jackscrew gearbox [24] (Figure 403).

(a) Install the hoist adapter to the sleeve of the jackscrew gearbox [24].

(b) Install the hoist bar between the frames at the top of the stabilizer trim compartment.

(c) Attach the hoist to the hoist bar and connect it to the hoist adapter.

SUBTASK 27-41-81-020-009

(9) Remove the stabilizer trim actuator [53] from the airplane (Figure 402).

(a) Remove the nut [42], washers [40], and bolt [39] to disconnect the bonding jumper [41] from the jackscrew gearbox [24].

1) Install the bolt [39], washers [40], and nut [42] back in the jackscrew gearbox [24].

(b) Remove the bolt [72], washer [73], washer [74], and nut [75] from the safety strap [56].

(c) Remove the bolt [68], washer [69], washer [70], and nut [71] from the safety strap [56].

(d) If necessary, remove the bolt [54] and washer [55] from the safety strap [56].

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(e) If it is necessary to remove the second safety strap [56], remove the bolt [72], washer [73], washer [74], nut [75], bolt [68], washer [69], washer [70], nut [71] from the safety strap [56].

NOTE: The second safety strap can be removed to have additional clearance for the stabilizer trim actuator removal.

(f) Remove the pins [38], nuts [37], washers [36], spacers [35], washers [34], and bolts [33] that attaches the ballnut [28] to the jackscrew fitting [21].

1) Discard the pins [38].

(g) Remove the bolts [29] and washers [30] to remove the upper gimbal pins [31] from the upper gimbal fittings [22].

(h) Move the ballnut [28] down and adjust the jackscrew [23] until it is clear from the jackscrew fitting [21].

(i) Install the washers [30] and bolts [29] to attach the upper gimbal pins [31] back in the upper gimbal fittings [22].

(j) Remove the pins [52], nuts [51], washers [50], and lower gimbal pins [25] from the stabilizer trim actuator [53].

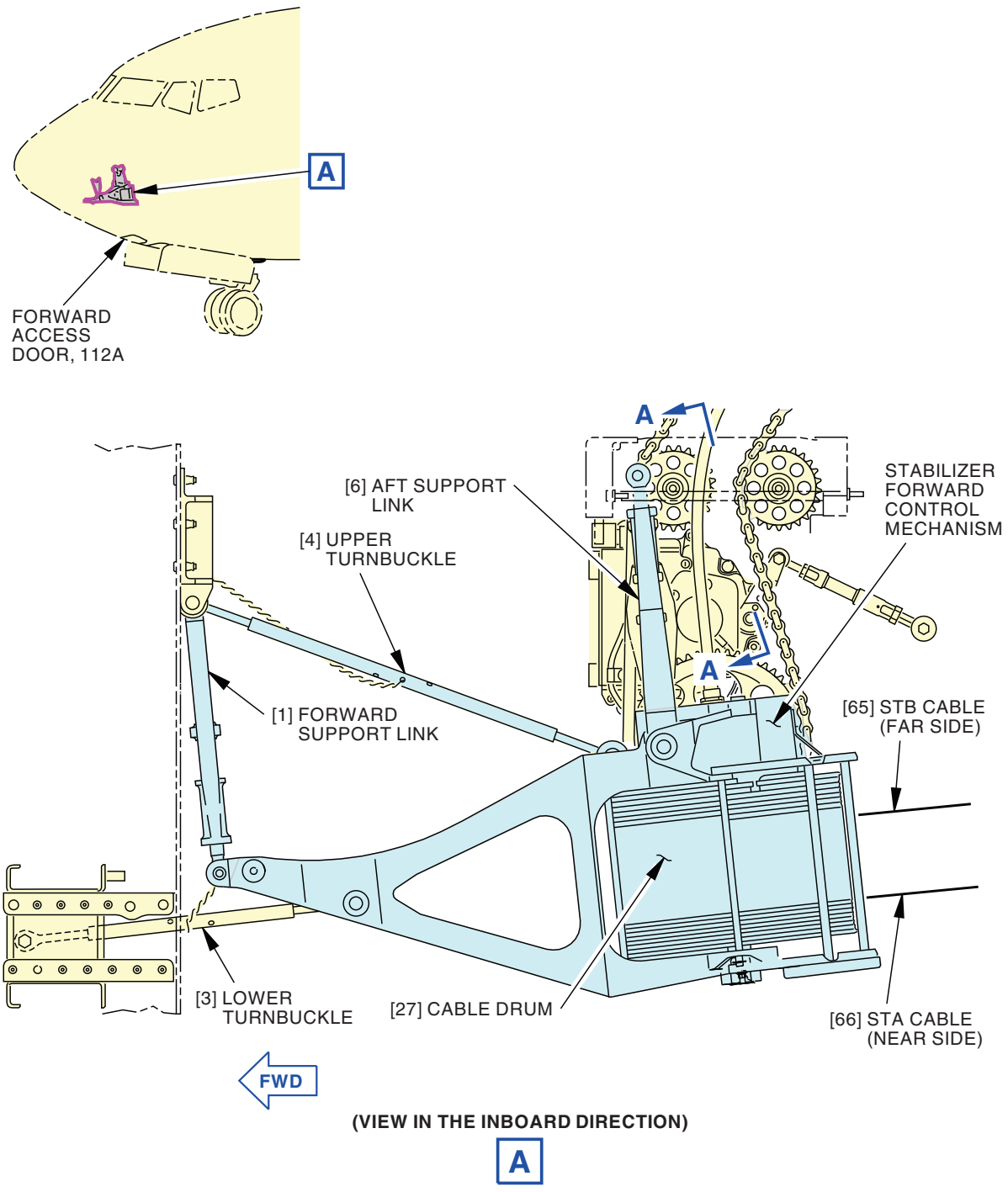
1) Discard the pins [52].

(k) Lower the jackscrew [23] until it is clear of the stabilizer truss.

(l) Lower the stabilizer trim actuator [53] from the airplane.

——— END OF TASK ———

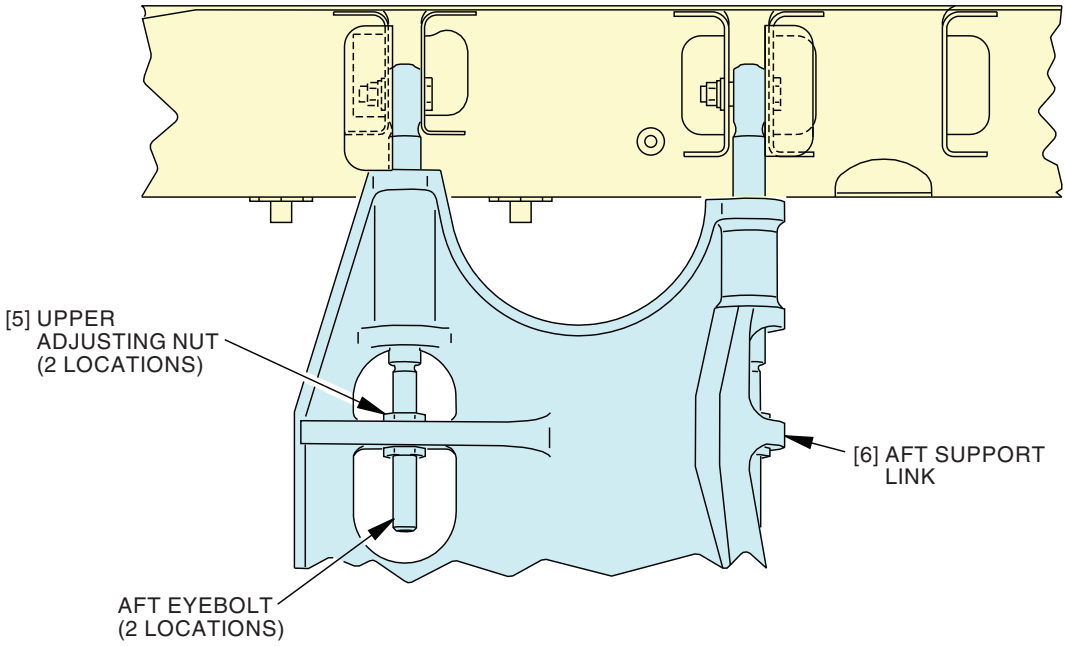
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412684 S00061531146_V3

Stabilizer Forward Control Mechanism Adjustment
Figure 401/27-41-81-990-801 (Sheet 1 of 2)

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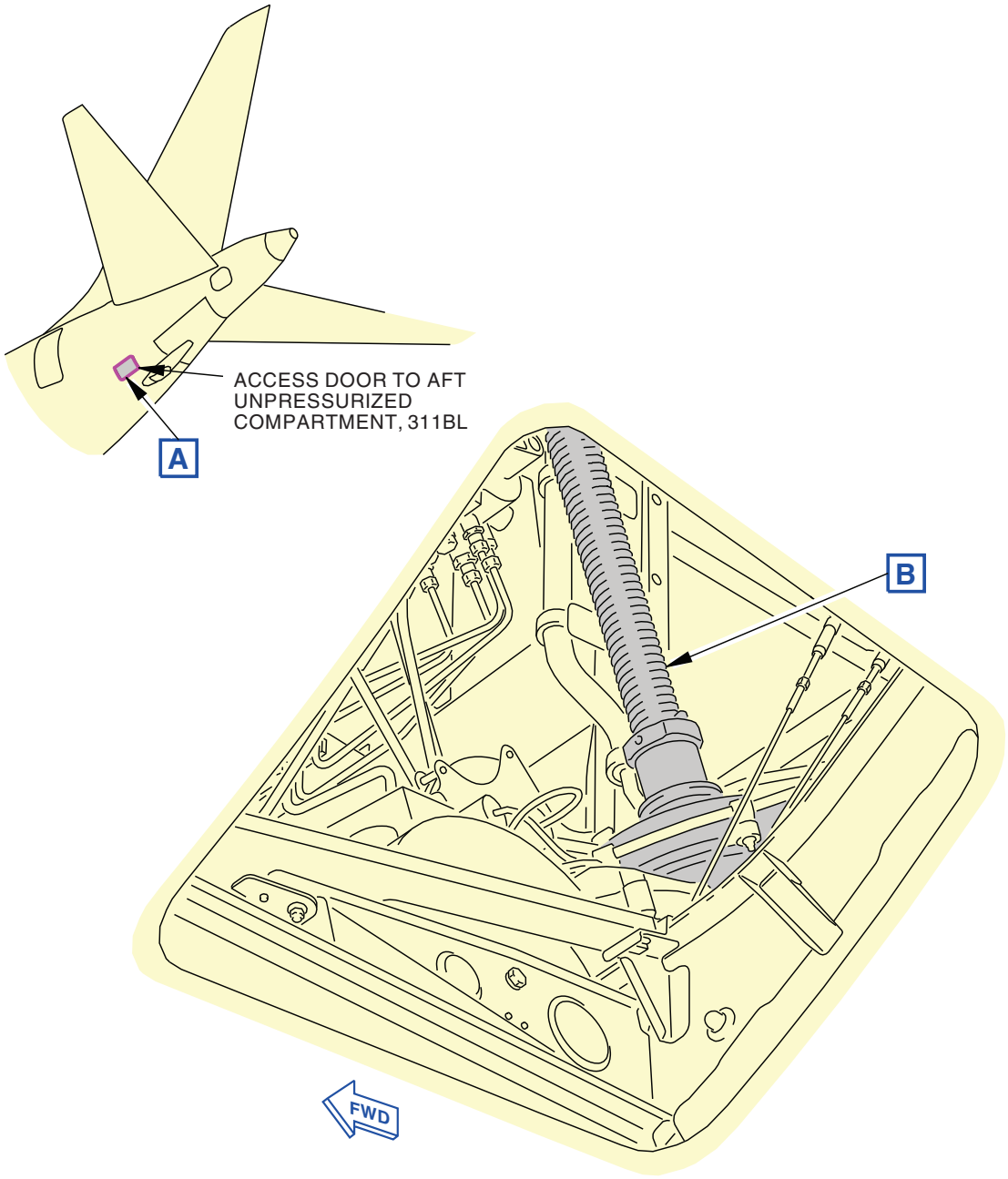
(VIEW IN THE FORWARD DIRECTION)

A-A

2412685 S00061531147_V1

Stabilizer Forward Control Mechanism Adjustment
Figure 401/27-41-81-990-801 (Sheet 2 of 2)

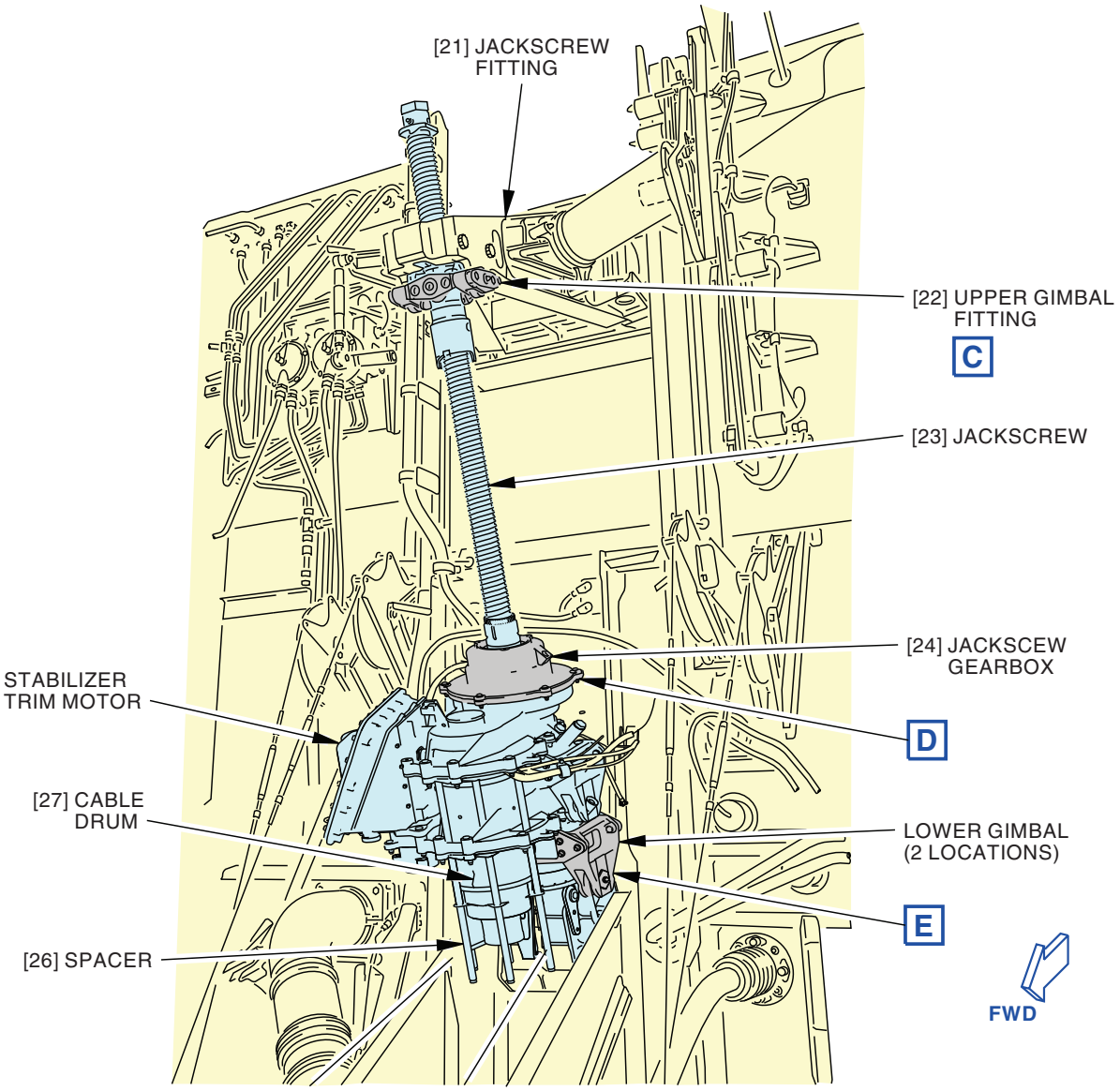
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412687 S00061531149_V1

Stabilizer Ball Nut and Jackscrew Gearbox Installation
Figure 402/27-41-81-990-802 (Sheet 1 of 7)

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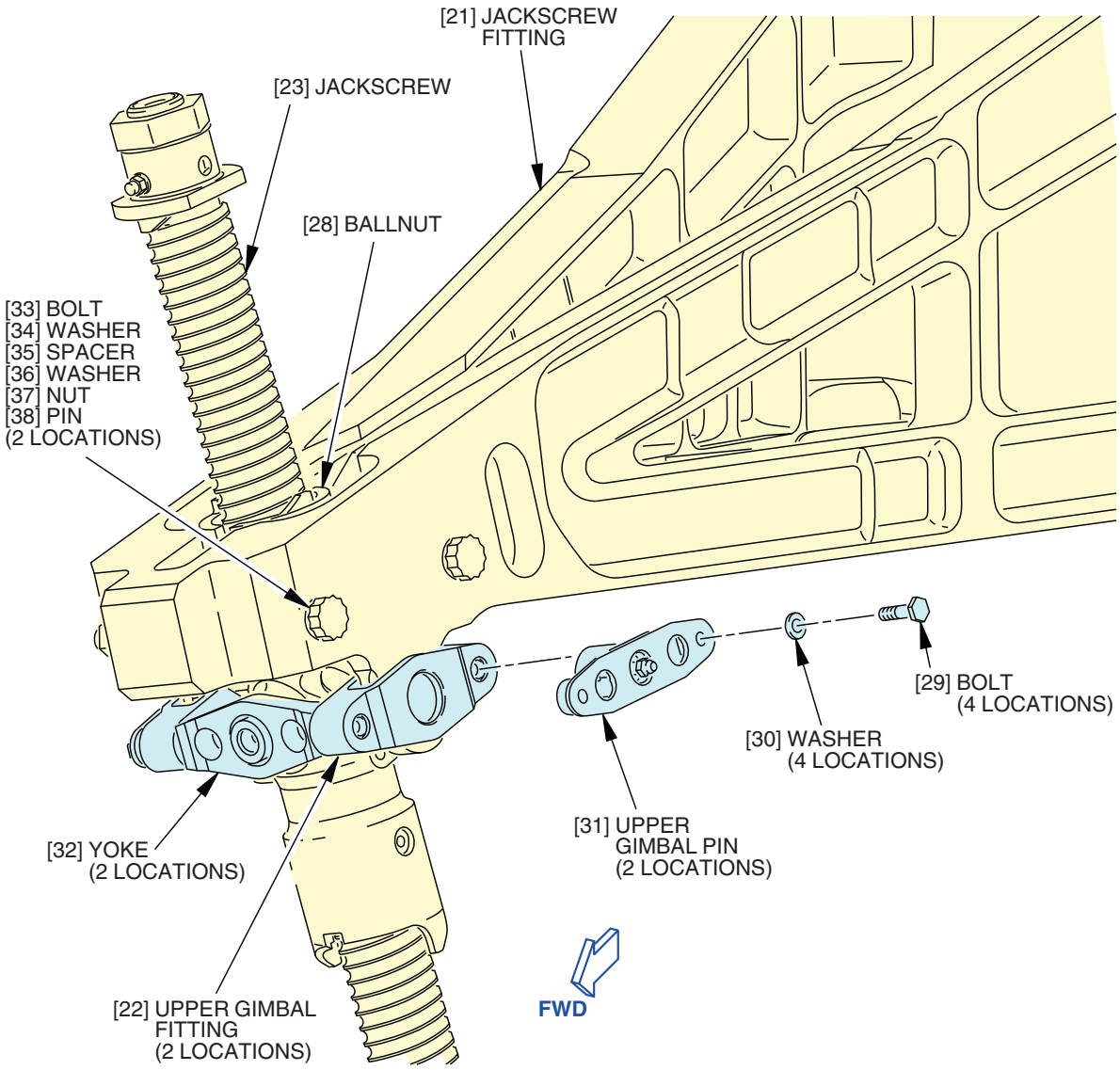
STABILIZER TRIM ACTUATOR

B

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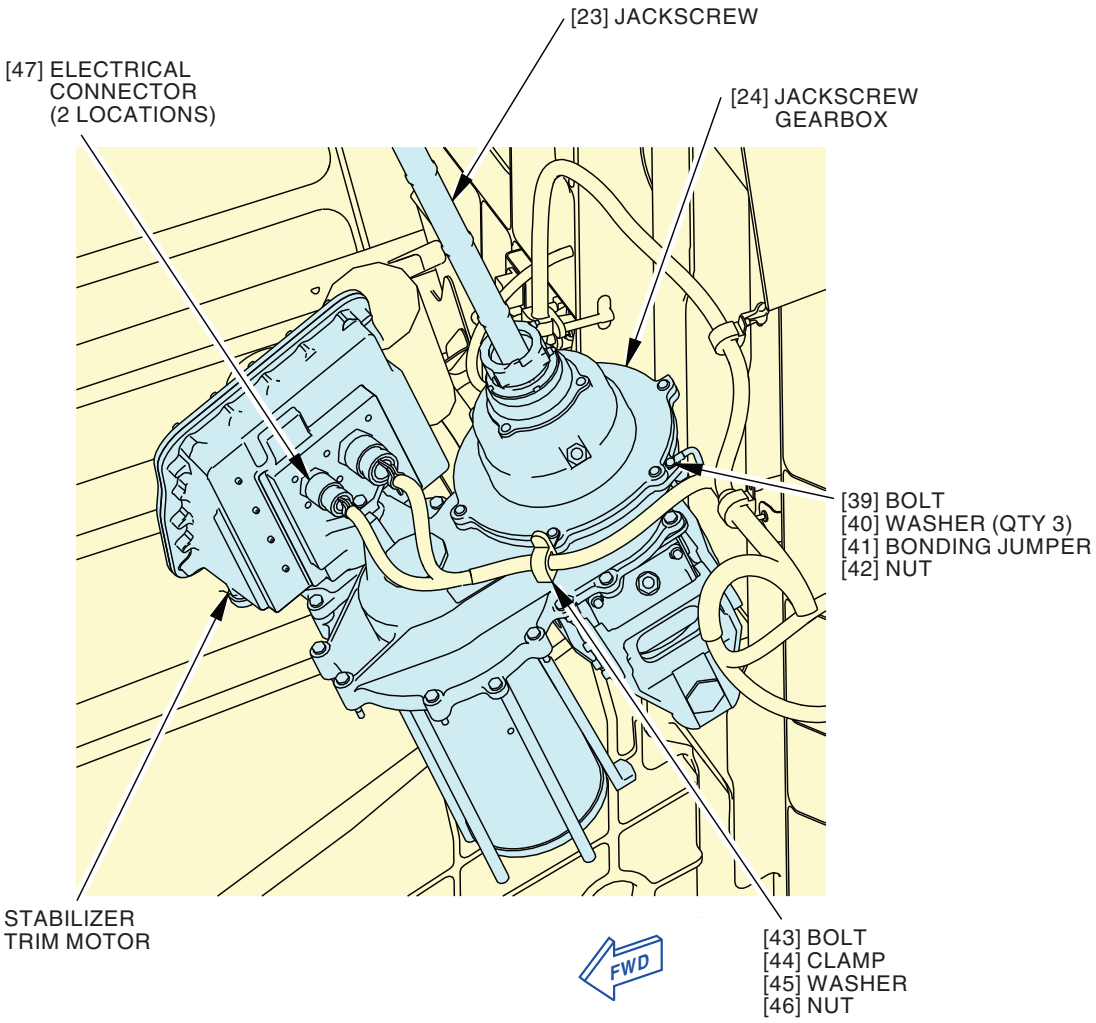
Stabilizer Ball Nut and Jackscrew Gearbox Installation
Figure 402/27-41-81-990-802 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Ball Nut and Jackscrew Gearbox Installation
Figure 402/27-41-81-990-802 (Sheet 3 of 7)

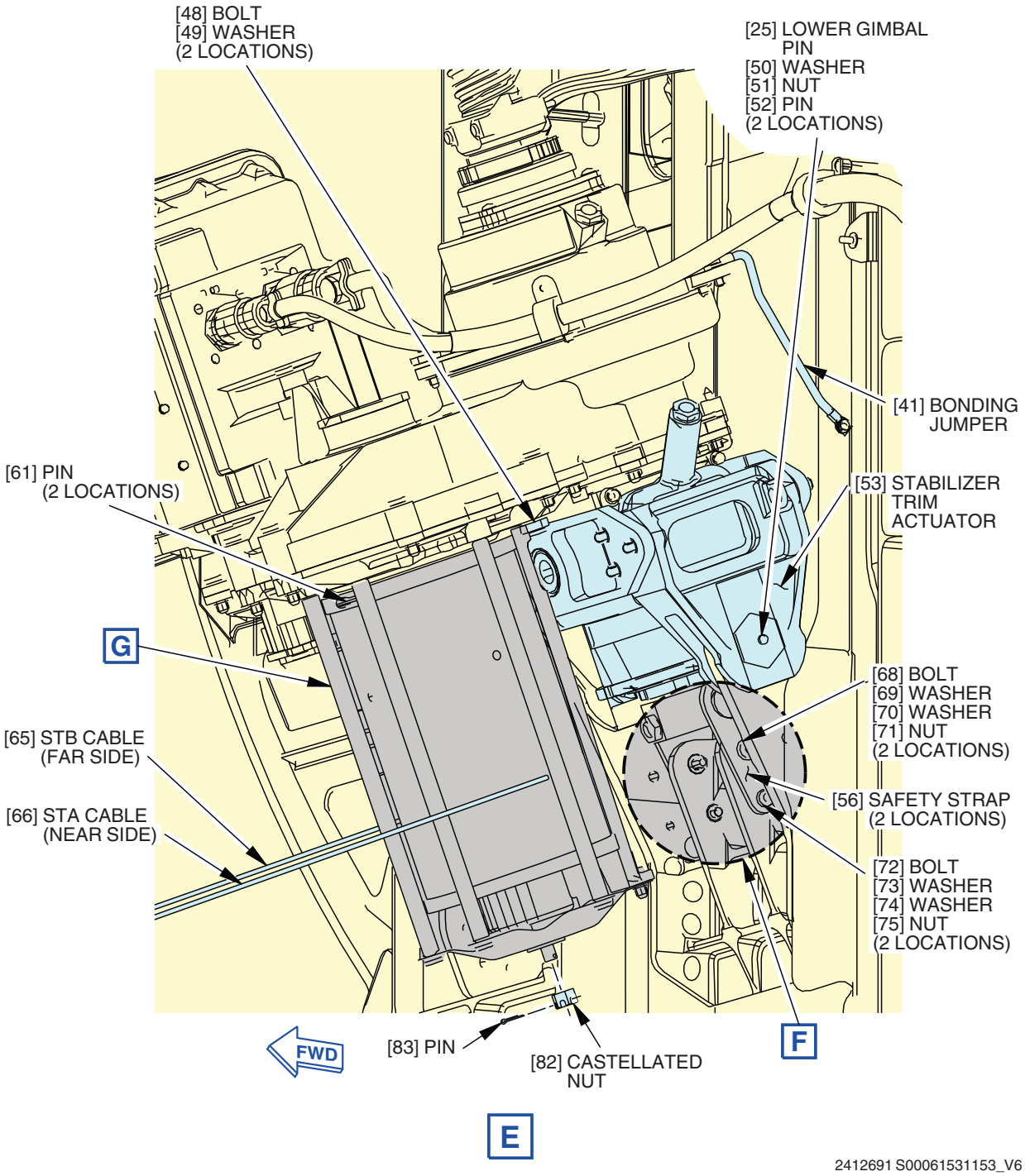
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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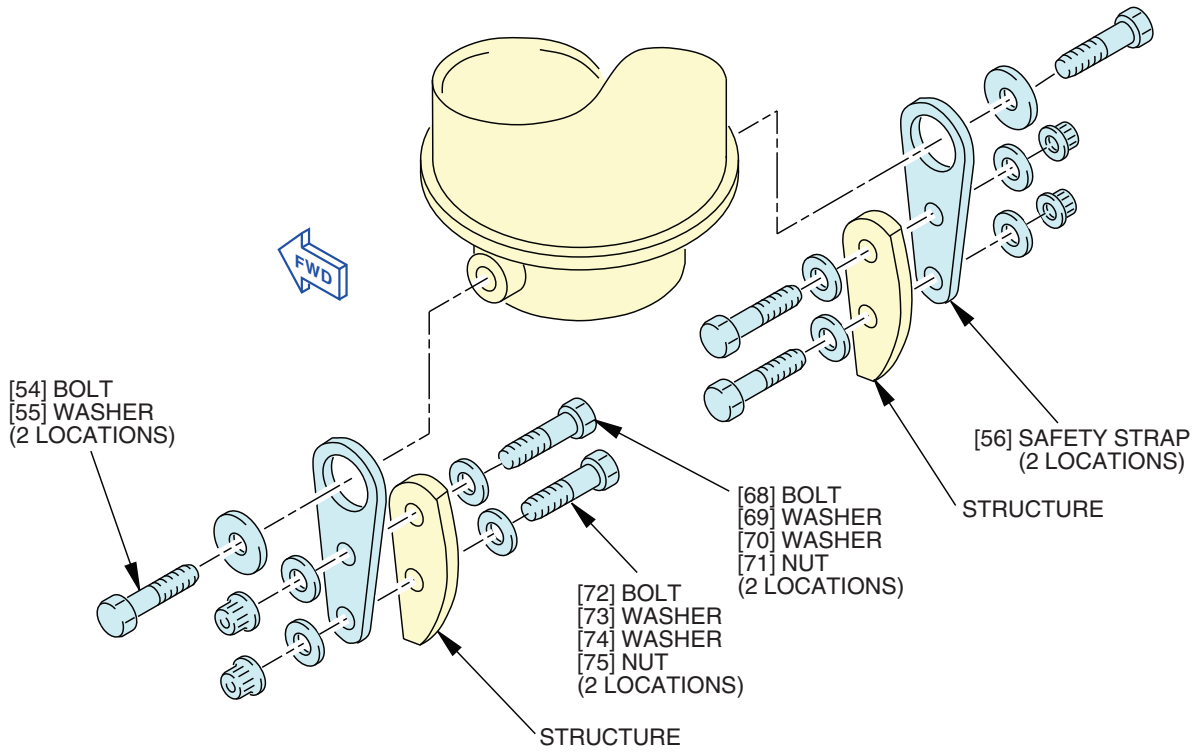
Stabilizer Ball Nut and Jackscrew Gearbox Installation
Figure 402/27-41-81-990-802 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Ball Nut and Jackscrew Gearbox Installation
Figure 402/27-41-81-990-802 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



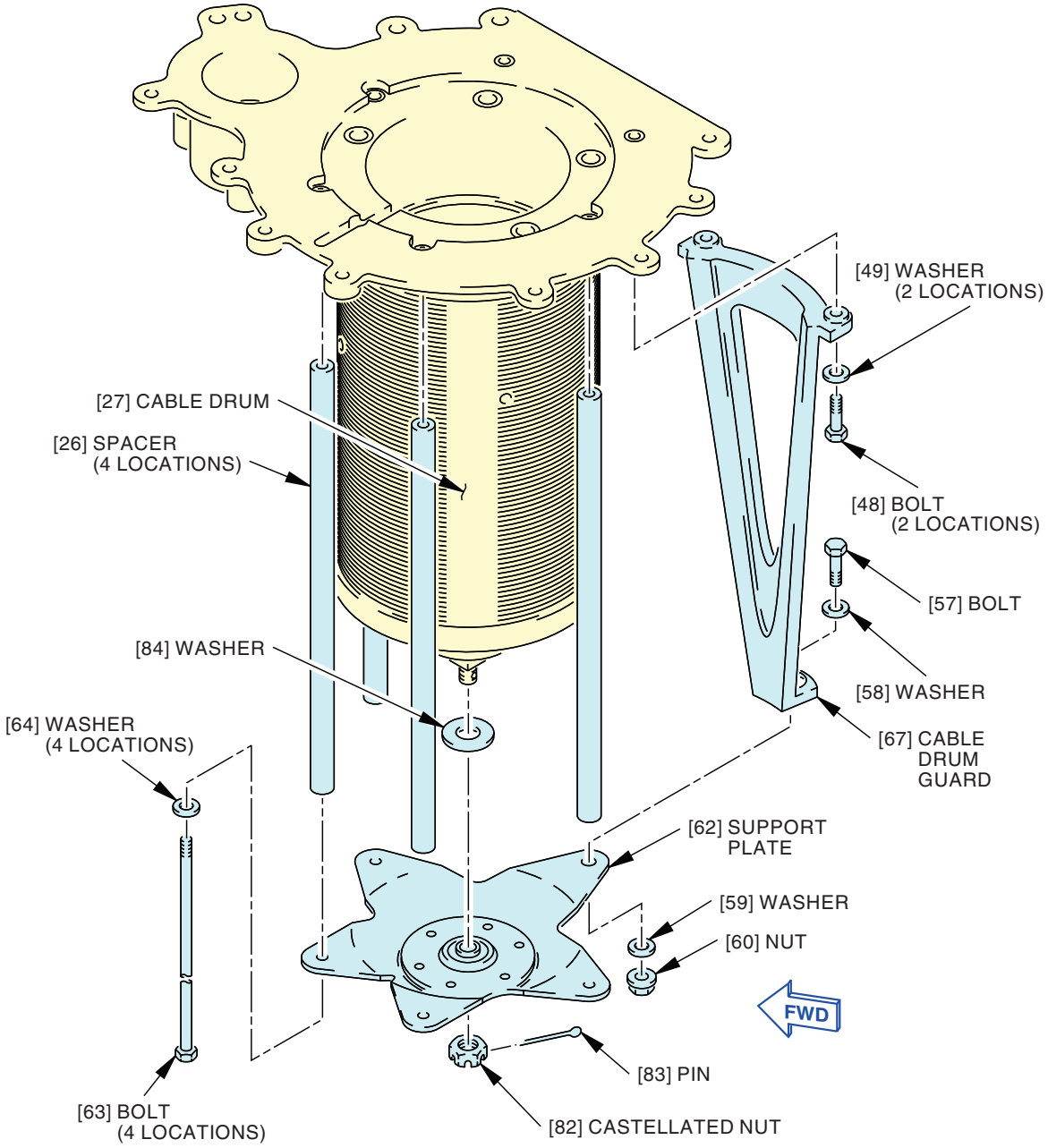
(EXAMPLE)

F

2412692 S00061531154_V3

Stabilizer Ball Nut and Jackscrew Gearbox Installation
Figure 402/27-41-81-990-802 (Sheet 6 of 7)

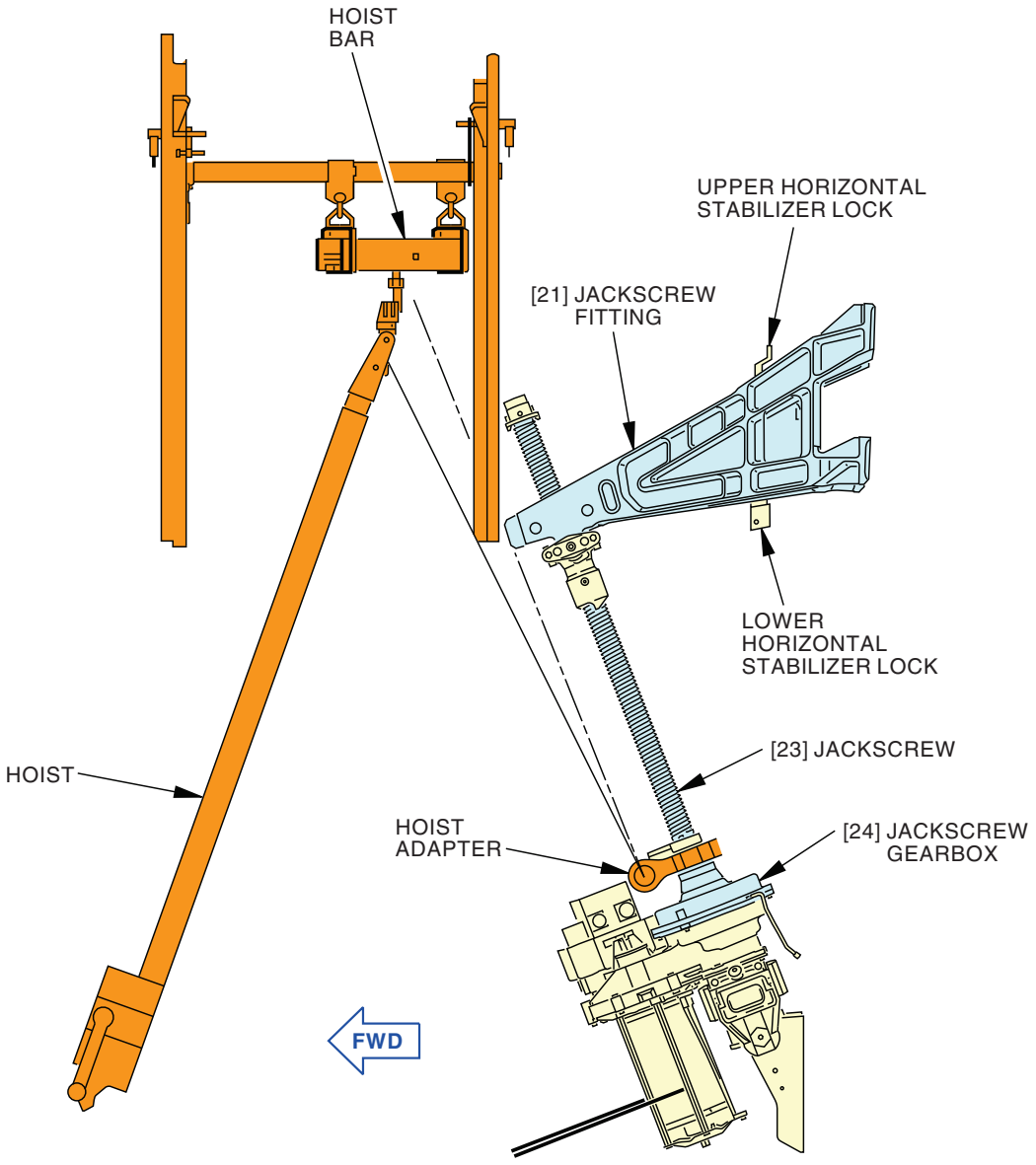
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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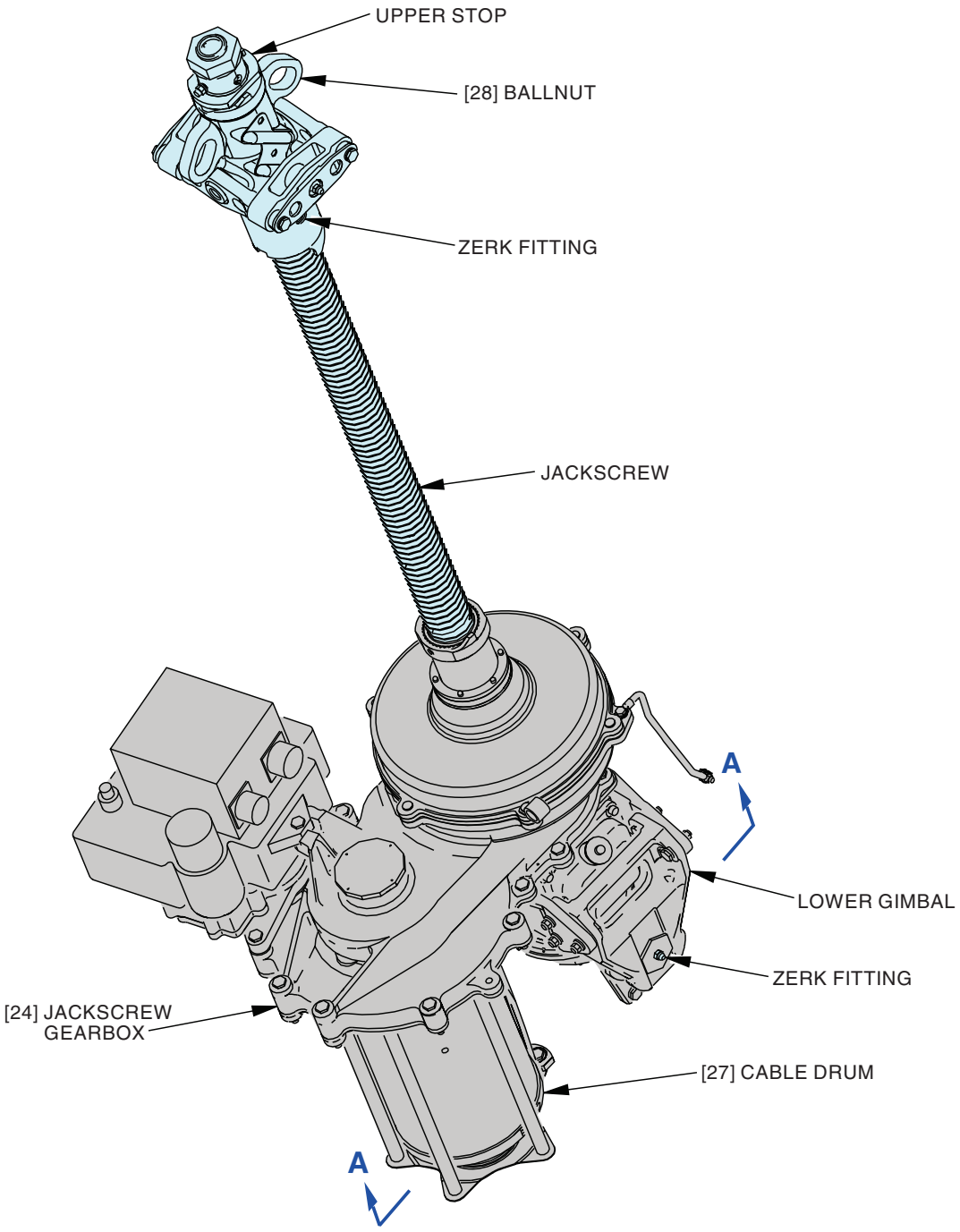
Stabilizer Ball Nut and Jackscrew Gearbox Installation
Figure 402/27-41-81-990-802 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Lock and Jackscrew Handling Equipment Installation
Figure 403/27-41-81-990-803

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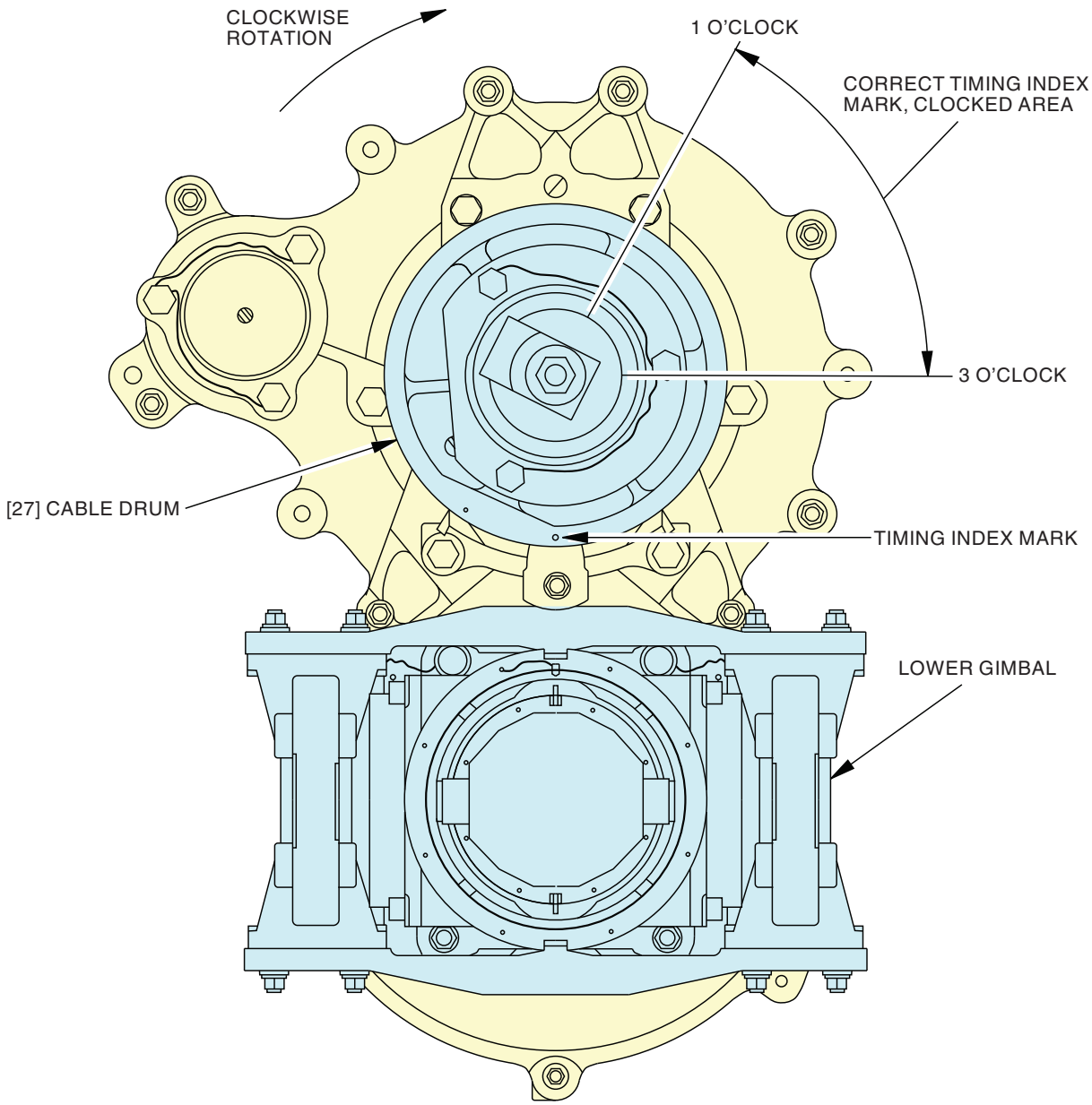


STABILIZER TRIM ACTUATOR

2796375 S0000637844_V1

Jackscrew Gearbox Clocking
Figure 404/27-41-81-990-808 (Sheet 1 of 2)

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(BOTTOM VIEW)

A-A

2794125 S0000636291_V1

Jackscrew Gearbox Clocking
Figure 404/27-41-81-990-808 (Sheet 2 of 2)

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27-CMR-08

TASK 27-41-81-400-801

3. Stabilizer Ball Nut and Jackscrew Gearbox Installation

(Figure 401, Figure 402, Figure 403, Figure 404)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
12-22-41-600-801	Stabilizer Jackscrew, Ballnut and Gimbal - Lubrication (P/B 301)
12-22-41-610-801	Horizontal Stabilizer Actuator Brake - Servicing (P/B 301)
27-41-00-700-801	Stabilizer Manual Trim and Trim Indicator - Test (P/B 501)
27-41-00-700-802	Horizontal Stabilizer Trim Control System Friction Test (P/B 501)
27-41-00-700-803	Stabilizer Electric Trim System Test (P/B 501)
27-41-00-700-804	Stabilizer Trim Wheel Free Movement Test (P/B 501)
27-41-00-700-806	Stabilizer Clutch Friction System Test (P/B 501)
27-41-00-700-808	Stabilizer Trim Secondary Brake System Test (P/B 501)
27-41-00-820-801	Stabilizer Control Cable and Chain - Adjustment (P/B 501)
27-41-71-400-801	Stabilizer Trim Motor Installation (P/B 401)
27-41-81-200-801	Upper and Lower Gimbal Wear Limits Inspection (P/B 601)
27-41-81-210-801	Stabilizer Trim Ball Nut and Jackscrew Gearbox Detail Visual Inspection (P/B 601)
27-41-81-210-802	Stabilizer Jackscrew Gearbox Backlash Inspection (P/B 601)
51-22-00-390-806	Fillet Seal Application (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

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(Continued)

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
COM-1563	Hoist - Fishpole, 500 lb Safe Working Limit (SWL) Part #: 10/4180 Supplier: K1425 Part #: IA5101-1 Supplier: 053H3 Part #: IA5101-501 Supplier: 053H3 Part #: IA5119-1 Supplier: 053H3 Part #: IA5119-501 Supplier: 053H3 Part #: PF51-003-1 Supplier: 1YRX6 Part #: PF51-009-1 Supplier: 1YRX6 Opt Part #: 10/3641 Supplier: K1425 Opt Part #: 10/3641C1 Supplier: K1425 Opt Part #: MINILIFT Supplier: K1425
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205
SPL-1678	Lock Equipment - Horizontal Stabilizer Part #: F80027-44 Supplier: 81205 Opt Part #: F80027-41 Supplier: 81205
SPL-1728	Equipment - Handling, Stabilizer Jackscrew (Dome Pressure Bulkhead) Part #: C27050-82 Supplier: 81205
SPL-10319	Wrench - Pinion Gear (Rotate) Part #: F71252 Supplier: 81205 Part #: F71267 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
D00633	Grease - Aircraft General Purpose	BMS3-33
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

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D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
38	Pin	27-41-71-01-005	PTF ALL
52	Pin	27-41-71-01-090	PTF ALL
53	Stabilizer trim actuator	27-41-71-01-175	PTF ALL
61	Pin	27-41-82-01-007	PTF ALL
83	Pin	27-41-81-01-601	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

F. Access Panels

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door

G. Prepare for the Installation

SUBTASK 27-41-81-220-002

(1) Do this task: Upper and Lower Gimbal Wear Limits Inspection, TASK 27-41-81-200-801.

SUBTASK 27-41-81-820-004

(2) Do the jackscrew gearbox clocking for a new stabilizer trim actuator [53] as follows (Figure 404):

(a) Put the new stabilizer trim actuator [53] on a table with the lower gimbal on the table.

1) Make sure that the cable drum [27] points up.

CAUTION

HOLD THE JACKSCREW. THIS WILL PREVENT THE MOVEMENT OF THE JACKSCREW AND DAMAGE TO THE EQUIPMENT.

(b) Put a support on the jackscrew [23] with foam or equivalent.

(c) Move the ballnut [28] to the upper stop.

(d) Look through the cable drum [27] bottom and turn the cable drum [27] clockwise until the zerk fitting points to the 3 o'clock position.

1) Make sure that the timing index mark is between the 1 o'clock and 3 o'clock position.

(e) If the zerk fitting does not face the 3 o'clock position and timing index mark is not between the 1 o'clock and 3 o'clock position, do this step:

1) Turn the cable drum [27] clockwise until the timing index mark and zerk fitting is in these positions at the same time.

(f) When the timing index mark and zerk fitting are in the correct position, the actuator is clocked correctly.

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SUBTASK 27-41-81-020-014

(3) If the cable drum guard [67] and spacers [26] are installed on the cable drum [27], do these steps (Figure 402):

(a) Remove the castellated nut [82] and pin [83].

1) Discard the pin [83].

(b) Remove the bolts [63] and washers [64] from the support plate [62].

(c) Remove the bolts [48] and washers [49] that attach the cable drum guard [67] to the cable drum [27].

(d) Remove the nut [60], washer [59], washer [58], and bolt [57] that attach the cable drum guard [67] to the support plate [62].

(e) Remove the cable drum guard [67] and spacers [26].

(f) Remove the washer [84].

SUBTASK 27-41-81-480-011

(4) Install the hoist adapter and hoist to the jackscrew gearbox [24] (Figure 403).

SUBTASK 27-41-81-480-006

(5) Lift the stabilizer trim actuator [53] into the airplane (Figure 402).

SUBTASK 27-41-81-020-016

(6) Remove the bolts [29] and washers [30] from the upper gimbal pins [31]

SUBTASK 27-41-81-020-017

(7) Remove the upper gimbal fittings [22] and yokes [32] from the ballnut [28].

H. Stabilizer Trim Actuator Installation

SUBTASK 27-41-81-420-001

(1) Install the Horizontal Stabilizer Trim Actuator (HSTA).

(a) Install the jackscrew [23] through the jackscrew fitting [21].

(b) Align the stabilizer trim actuator [53] with structure.

(c) Apply grease, D00633, to the lower gimbal pins [25].

(d) Install the lower gimbal pins [25] with the grease fitting outboard, washers [50], nuts [51] to connect the stabilizer trim actuator [53] to the structure.

1) Tighten the nuts [51] by hand and then tighten them to nearest pin hole.

(e) Install the new pins [52].

(f) If necessary, install the bolt [54] and washer [55] to connect the stabilizer trim actuator [53] to the safety strap [56].

(g) Install the bolt [68], washer [69], washer [70], and nut [71] with bolt head inboard to connect the stabilizer trim actuator [53] to the safety strap [56].

(h) Install the bolt [72], washer [73], washer [74], and nut [75] with bolt head inboard to connect the stabilizer trim actuator [53] to the safety strap [56].

(i) If the second safety strap [56] is removed, do the following steps:

1) If removed, install the washer [55] and bolt [54] to connect the stabilizer trim actuator [53] to the safety strap [56].

2) Install the bolt [68], washer [69], washer [70], and nut [71] with bolt head inboard to connect the stabilizer trim actuator [53] to the safety strap [56].

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3) Install the bolt [72], washer [73], washer [74], and nut [75] with bolt head inboard to connect the stabilizer trim actuator [53] to the safety strap [56].

(j) Connect the ballnut [28] to the jackscrew fitting [21].

1) Make sure that the ballnut [28] is installed with the grease fitting toward the access door.

a) If it is necessary, rotate the ballnut [28] half a turn.

2) Lubricate the upper gimbal fittings [22] and yokes [32] with grease, D00633.

a) Put the two spacers [35] in each side of the ballnut [28].

3) Install the yokes [32] on the ballnut [28].

4) Engage the upper gimbal fittings [22] on the yokes [32].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

5) Apply Cor-Ban 27L Compound, G50237, to all the bolt holes on the jackscrew fitting [21].

6) Turn the jackscrew [23] to engage the upper gimbal fittings [22] with the jackscrew fitting [21] and to align the bolt holes on the jackscrew fitting [21] with the spacer [35].

7) Loosely install bolts [33], washers [34], spacers [35], washers [36], and nuts [37].

8) Install the upper gimbal pins [31] to connect the ballnut [28] to the jackscrew fitting [21].

9) Apply grease, D00633 to the shank and threads of the bolts [29].

10) Install the bolts [29] and washers [30] to the upper gimbal pins [31].

11) Install MS20995NC32 lockwire, G01912, to attach the two bolt [29] heads together on each upper gimbal pin [31].

12) Tighten the nuts [37] by hand and then loosen the nuts [37] to the nearest pin hole.

13) Install the new pins [38].

SUBTASK 27-41-81-420-002

(2) Install the bonding jumper [41] to the jackscrew gearbox [24].

(a) Remove the nut [42], washers [40], and bolt [39] from the jackscrew gearbox [24].

(b) Clean the area around the bolt for electrical bonding.

(c) Install the hardware as follows: bolt [39], washer [40], bonding jumper [41], washer [40], jackscrew gearbox [24] structure, washer [40], and nut [42].

SUBTASK 27-41-81-760-001

(3) Measure the electrical bonding resistance between the bonding jumper [41] terminal and the jackscrew gearbox [24] (SWPM 20-20-10).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

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SUBTASK 27-41-81-760-003

(4) Measure the electrical bonding resistance between the jackscrew gearbox [24] and aluminum structure (SWPM 20-20-10).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical bonding resistance is 0.00225 ohm (2.25 milliohm) or less.

SUBTASK 27-41-81-390-001

(5) Apply a fillet seal of sealant, A50337, or sealant, A50231, around the bolt [39] and nut [42] (TASK 51-22-00-390-806).

SUBTASK 27-41-81-080-001

(6) Remove equipment, SPL-1728 and fishpole hoist, COM-1563, from the jackscrew gearbox [24] (Figure 403).

(a) Remove the hoist adapter from the jackscrew [23].

(b) Remove the hoist bar and hoist from the airplane.

SUBTASK 27-41-81-080-002

(7) Remove the lock equipment, SPL-1678.

(a) If it is necessary, install the wire bundle clamps.

SUBTASK 27-41-81-420-003

(8) Connect the STA cable [66] to the cable drum [27] (Figure 402).

(a) With the ballnut [28] against the upper stop, wind the STA cable [66] on the cable drum [27] approximately 1.8 turns.

(b) Install the cable terminal in the lower terminal slot of the cable drum [27].

(c) Install the new pin [61] to connect the cable terminal to the cable drum [27].

1) Install the ends of the pin [61] in the terminal slot.

(d) Remove the control cable clamp, SPL-1569.

CAUTION

MAKE SURE THAT THE CABLE WINDS ONTO THE DRUM GROOVES CORRECTLY WITH EACH TURN OF THE DRUM. IF YOU JAM THE CABLE BETWEEN THE CABLE AND THE CABLE DRUM GUARD, DAMAGE TO THE CABLE WILL OCCUR.

(e) Operate the jackscrew to move the stabilizer leading edge to full down position to wind the cable on the cable drum approximately 50.3 turns.

NOTE: Operate the jackscrew until it stops.

NOTE: It is necessary to use the pinion gear wrench, SPL-10319, or equivalent when you operate the jackscrew to move the stabilizer leading edge.

SUBTASK 27-41-81-420-004

CAUTION

KEEP TENSION ON THE CABLE AT ALL TIMES. WHEN THE CABLE IS DISCONNECTED FROM THE CABLE DRUM, TENSION NEEDS TO BE KEPT ON THE CABLE SO THAT THE CABLE DOES NOT BECOME CAUGHT IN THE CABLE GUARDS ON THE OTHER CABLE DRUM. THIS CAN CAUSE DAMAGE TO THE CABLE AND THE CABLE MAY NEED TO BE REPLACED.

(9) Connect the STB cable [65] to the cable drum [27].

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(a) Use the force gauge, COM-1557, on the stabilizer trim wheel handle to put a 50 lb (22.7 kg) load on the STA cable [66].

NOTE: This will keep the STA cable on the drum, which will make the installation of the STB cable easier.

NOTE: All the grooves of the cable drum will be filled with cable except for 1/2 turn between the STA and STB cable.

(b) Install the cable terminal in the top terminal slot of the cable drum [27].

(c) Install the new pin [61] to connect the cable terminal to the cable drum [27].

1) Install the ends of the pin [61] in the terminal slot.

SUBTASK 27-41-81-420-005

(10) Install the spacers [26] and cable drum guard [67] to the cable drum [27].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply Cor-Ban 27L Compound, G50237, to the bolts [48], bolt [57], nut [60], and bolts [63].

(b) Install the bolts [48] and washers [49] to attach the cable drum guard [67] to the cable drum [27].

1) Tighten the bolts [48].

CAUTION

MAKE SURE THAT YOU INSTALL THE WASHER BETWEEN THE SUPPORT PLATE AND THE CABLE DRUM. IF YOU DO NOT OBEY, THE CABLE DRUM WILL NOT TURN CORRECTLY DURING OPERATION.

(c) Install the washer [84] to the cable drum [27].

CAUTION

MAKE SURE THAT YOU INSTALL THE SUPPORT PLATE CORRECTLY. IF YOU DO NOT INSTALL THE SUPPORT PLATE, THE AMOUNT OF TORQUE NECESSARY TO DISENGAGE THE MANUAL CLUTCH CAN BE MORE THAN THE LIMITS.

(d) Install the bolts [63], washers [64], support plate [62], and spacers [26] to the cable drum [27].

1) Install the bolts [63] to the washers [64], support plate [62], and spacers [26].

NOTE: The bearing assembly of the support plate [62] must be on the same side as the cable drum [27].

2) Install the bolts [63] to the cable drum [27].

3) Tighten the bolts [63].

(e) Install the bolt [57], washer [58], nut [60], and washer [59] to attach the cable drum guard [67] to the support plate [62].

1) Tighten the bolt [57].

(f) Install the castellated nut [82] to the cable drum [27].

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1) Install the castellated nut [82] to the thread of the cable drum [27].
2) Hand tighten the castellated nut [82] against the support plate [62].
3) Align the castellated nut [82] to the nearest pin hole.

(g) Install the new pin [83] to the castellated nut [82].

SUBTASK 27-41-81-730-001

(11) Do this task: Stabilizer Clutch Friction System Test, TASK 27-41-00-700-806.

SUBTASK 27-41-81-080-003

(12) Remove the pinion gear wrench, SPL-10319, from the jackscrew gearbox [24].

SUBTASK 27-41-81-820-002

(13) Adjust the STA cable [66], STB cable [65] and then reconnect the upper turnbuckle [4] (TASK 27-41-00-820-801).

(a) Make sure that lockwire is reinstalled to the upper turnbuckle [4].

SUBTASK 27-41-81-420-008

(14) Install the electrical connectors [47] to the jackscrew gearbox [24].

(a) Remove the nuts [46], washers [45], and bolts [43] from the jackscrew gearbox [24].

(b) Put the clamps [44] for the electrical connectors [47] on the jackscrew gearbox [24].

(c) Install the bolts [43], washers [45], and nuts [46] to install the clamps [44] to the jackscrew gearbox [24].

SUBTASK 27-41-81-420-006

(15) Do this task: Stabilizer Trim Motor Installation, TASK 27-41-71-400-801.

SUBTASK 27-41-81-860-003

(16) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-81-600-001

(17) Do the subsequent inspections only if you replaced a HSTA with a serviceable stabilizer trim actuator that is not new or not overhauled:

(a) Do the Detailed Visual Inspection of the HSTA for metal particles, corrosion, cracks, damage, and worn areas (TASK 27-41-81-210-801).

(b) Do this task: Stabilizer Jackscrew Gearbox Backlash Inspection, TASK 27-41-81-210-802.

SUBTASK 27-41-81-640-001

(18) Lubricate the upper gimbal and lower gimbal at the grease fittings and jackscrew (TASK 12-22-41-600-801).

SUBTASK 27-41-81-730-002

(19) Do this task: Stabilizer Manual Trim and Trim Indicator - Test, TASK 27-41-00-700-801.

SUBTASK 27-41-81-730-003

(20) Do this task: Horizontal Stabilizer Trim Control System Friction Test, TASK 27-41-00-700-802.

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SUBTASK 27-41-81-610-001

(21) Do this task: Horizontal Stabilizer Actuator Brake - Servicing, TASK 12-22-41-610-801.

NOTE: This lubrication procedure must be performed after the installation of any new stabilizer trim actuator.

SUBTASK 27-41-81-730-004

(22) Do this task: Stabilizer Trim Secondary Brake System Test, TASK 27-41-00-700-808.

SUBTASK 27-41-81-730-005

(23) Do this task: Stabilizer Trim Wheel Free Movement Test, TASK 27-41-00-700-804.

SUBTASK 27-41-81-730-006

(24) Do this task: Stabilizer Electric Trim System Test, TASK 27-41-00-700-803.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-81-410-001

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-41-81-410-002

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-41-81-480-007

(3) Remove the DO NOT OPERATE tags from the stabilizer trim control wheels.

——— END OF TASK ———

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STABILIZER BALL NUT AND JACKSCREW GEARBOX - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) A detail visual inspection of the stabilizer trim ball nut and jackscrew gearbox

(2) An inspection of the stabilizer jackscrew gearbox backlash

(3) An inspection of the upper and lower gimbal wear limits.

27-CMR-07

TASK 27-41-81-210-801

2. Stabilizer Trim Ball Nut and Jackscrew Gearbox Detail Visual Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the visual inspection of the stabilizer trim ball nut, jackscrew, gearbox, and load paths in detail.

B. References

Reference	Title
12-22-41-600-801	Stabilizer Jackscrew, Ballnut and Gimbal - Lubrication (P/B 301)
27-41-81-000-801	Stabilizer Ball Nut and Jackscrew Gearbox Removal (P/B 401)
27-41-81-400-801	Stabilizer Ball Nut and Jackscrew Gearbox Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-600	Mirror - Inspection
STD-858	Tag - DO NOT OPERATE
STD-7028	Equipment - Borescope Inspection
STD-14360	Flashlight

D. Consumable Materials

Reference	Description	Specification
G50316	Cloth - Clean, Dry, Lint-free, White, Cotton	

E. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Prepare for the Inspection

SUBTASK 27-41-81-040-001

(1) Set the STAB TRIM PRI switch, S272, located on the aft area of the control stand, to the CUTOUT position.

SUBTASK 27-41-81-480-009

(2) Attach a DO NOT OPERATE tag, STD-858, to the STAB TRIM PRI switch.

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SUBTASK 27-41-81-010-003

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

H. Stabilizer Trim Ball Nut and Jackscrew Gearbox Detail Visual Inspection

SUBTASK 27-41-81-210-001

(1) Do these Horizontal Stabilizer Trim Actuator (HSTA) inspections:

(a) Do a detailed inspection of all attachment points to make sure that they are secure and do not have damage as follows:

1) To make sure that the HSTA is correctly attached to the airplane structure, do an inspection of the components shown on Table 601, Table 602, and Table 603 (Views C and D, Figure 601).

Table 601/27-41-81-993-802 HSTA Upper Gimbal Assembly

Component	QTY
bolts [11] washers [12]	4
yokes [7] upper gimbal fittings [8] upper gimbal pins [10]	2
lockwires [9]	2
bolts [1]*[1] spacers [3]*[1] nuts [5]*[1] pins [6]*[1]	2
washers [2]*[1] washers [4]*[1]	1 or 2

*[1] Rotation of these components are satisfactory because they are loose installed.

Table 602/27-41-81-993-804 HSTA Upper Stop

Component	QTY
pin [32] nut [33]*[1] washer [34] upper stop [35]	1
bolt [36] washer [37] washer [38] nut [39]*[2]	1

*[1] If found loose, tighten to 500 in-lb (56.5 N·m) - 700 in-lb (79.1 N·m).

*[2] If found loose, tighten to 10 in-lb (1.1 N·m) - 20 in-lb (2.3 N·m) beyond run on torque.

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Table 603/27-41-81-993-803 HSTA Lower Gimbal Assembly

Component	QTY
lower gimbal pins [17] washers [18] nuts [19] pins [20]	2
Lower Yoke and Fork: bolts [13] washers [14] washers [15] nuts [16]	12
lower gimbal yoke [40] lower gimbal fork [41]	2
bolts [21] washers [22]	2
safety straps [27] bolts [23] washers [24] washers [25] nuts [26]	2
bolts [28] washers [29] washers [30] nuts [31]	2

a) Examine the components for damage, deformation, corrosion, or loose hardware, unless otherwise specified.

<1> If it is necessary, use a flashlight, STD-14360, and inspection mirror, STD-600, or borescope, STD-7028.

<2> If one or more conditions are found, replace the component.

(b) Do a general visual inspection of the area below, on, and around the HSTA if grease is released from the ballnut.

NOTE: Large quantities of grease around the HSTA can be that the leakage is from the ballnut. The leakage can be because of a defective seal or a return tube was lifted out of position.

1) If a defective seal or a return tube is the cause of the leakage, do a detailed inspection.

2) Examine the ballnut tube retainers for a deformation, corrosion, or a return tube that is lifted.

3) If the ballnut tube retainers have a deformation, corrosion, or the leakage is from the return tube, remove and replace the actuator (TASK 27-41-81-000-801 and TASK 27-41-81-400-801).

(c) Examine cable drum and cables for damage or binding.

(d) Examine the electrical connectors of the stabilizer trim motor for damage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Examine the grease on and around the jackscrew and ballnut for metallic particles, other solid particles, and corrosion products in the grease.

NOTE: Metallic particles in the grease can be an indication of degradation of the balls in the ball nut. If metallic particles are found in the grease, remove for repair and replace the HSTA.

1) If metallic particles, or other solid particles, or corrosion products are found in the grease, do this step:

a) Remove for repair and replace the HSTA (TASK 27-41-81-000-801 and TASK 27-41-81-400-801).

(f) Check if any of the ballnut bearings are found in the vicinity of the ballnut, on the jackscrew, or directly below the HSTA.

NOTE: The ball bearing dimension is 0.21875 in. (5.55625 mm) diameter.

1) If the ball bearings are found, remove for repair and replace the HSTA (TASK 27-41-81-000-801 and TASK 27-41-81-400-801).

(g) Wipe the grease and dirt from the jackscrew threads with a clean, dry, and non-abrasive cotton cloth, G50316.

(h) Do a detailed visual inspection of the upper and lower gimbals for cracks or damage (View E, Figure 601).

1) Use a borescope, STD-7028, to examine the areas around the holes and bushings (Views A-A and B-B, Figure 601).

2) If parts with damage are found, replace the HSTA (TASK 27-41-81-000-801 and TASK 27-41-81-400-801).

3) Examine the upper jackscrew fitting and lower jackscrew fitting (View E, Figure 601).

(i) Do a detailed inspection of the jackscrew as follows:

1) Examine the jackscrew threads for cross-threading, distortion, or stripping.

2) Do a check for metallic particles, pitting, gouging, corrosion, spalling, or brinelling in the jackscrew threads.

NOTE: If metallic particles are observed in the grease, there can be damage in the jackscrew/ball nut assembly. Normal wear of the jackscrew actuator can occur but this will not release visible metallic particles in the grease.

NOTE: Phosphate coating on the horizontal stabilizer trim actuator ballscrew can be normal.

3) Do check the jackscrew for obvious differences in thread shape between thread grooves in the lower, middle, and upper portions of the jackscrew.

4) Do a check the jackscrew for damage or cracking.

5) Remove the DO NOT OPERATE tag, STD-858, from the STAB TRIM PRI switch.

6) Make sure that the STAB TRIM switch, on the stab trim and cabin door panel, P8-47 module, is in the NORM position.

7) Set the STAB TRIM PRI switch, S272, located on the aft area of the control stand, to the NORMAL position.

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WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

8) Push and hold the stabilizer trim control switches on the Captain's Control Wheel (NOSE UP or NOSE DOWN) to move the jackscrew to a different position.

9) Set the STAB TRIM PRI switch, S272, to the CUTOUT position.

10) Attach the DO NOT OPERATE tag, STD-858, to the STAB TRIM PRI switch.

11) Wipe the grease and dirt from the jackscrew threads with a clean, dry, non-abrasive cotton cloth, G50316.

12) Examine the jackscrew as follows:

a) Examine the jackscrew threads for cross-threading, distortion, or stripping.

b) Examine the jackscrew threads for metallic particles, pitting, gouging, corrosion, spalling, or brinelling.

NOTE: If metallic particles are observed in the grease, there can be damage in the jackscrew/ball nut assembly. Normal wear of the jackscrew actuator can happen but this will not release visible metallic particles in the grease.

c) Examine the jackscrew for obvious differences in thread shape between thread grooves in the lower, middle, and upper portions.

13) If damage is found, replace the stabilizer jackscrew actuator (TASK 27-41-81-000-801 and TASK 27-41-81-400-801).

SUBTASK 27-41-81-640-002

(2) Lubricate the stabilizer trim jackscrew (TASK 12-22-41-600-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-81-410-003

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-41-81-080-005

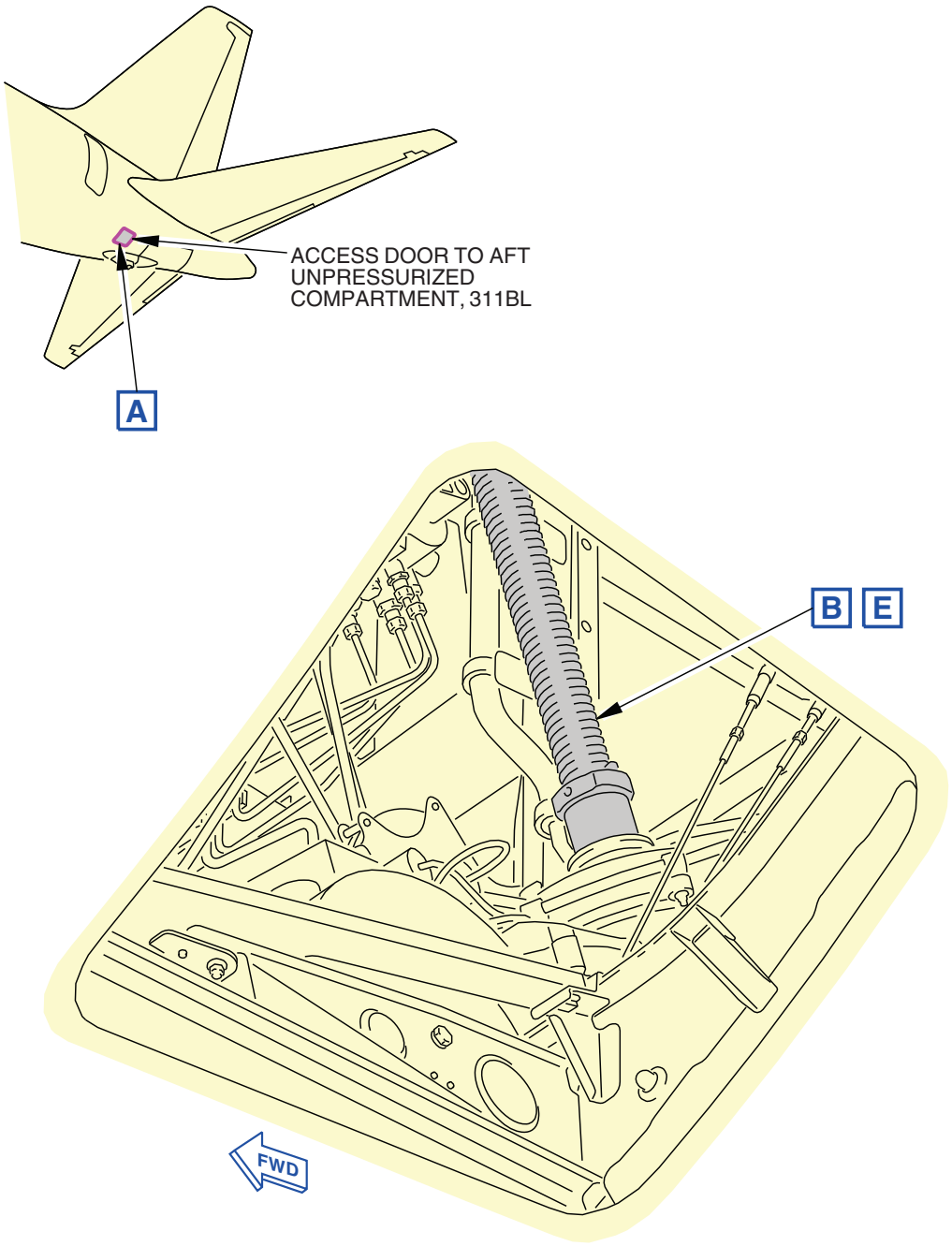
(2) Remove the DO NOT OPERATE tag, STD-858, from the STAB TRIM PRI switch, S272.

SUBTASK 27-41-81-860-010

(3) Set the STAB TRIM PRI switch, S272, to the NORMAL position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



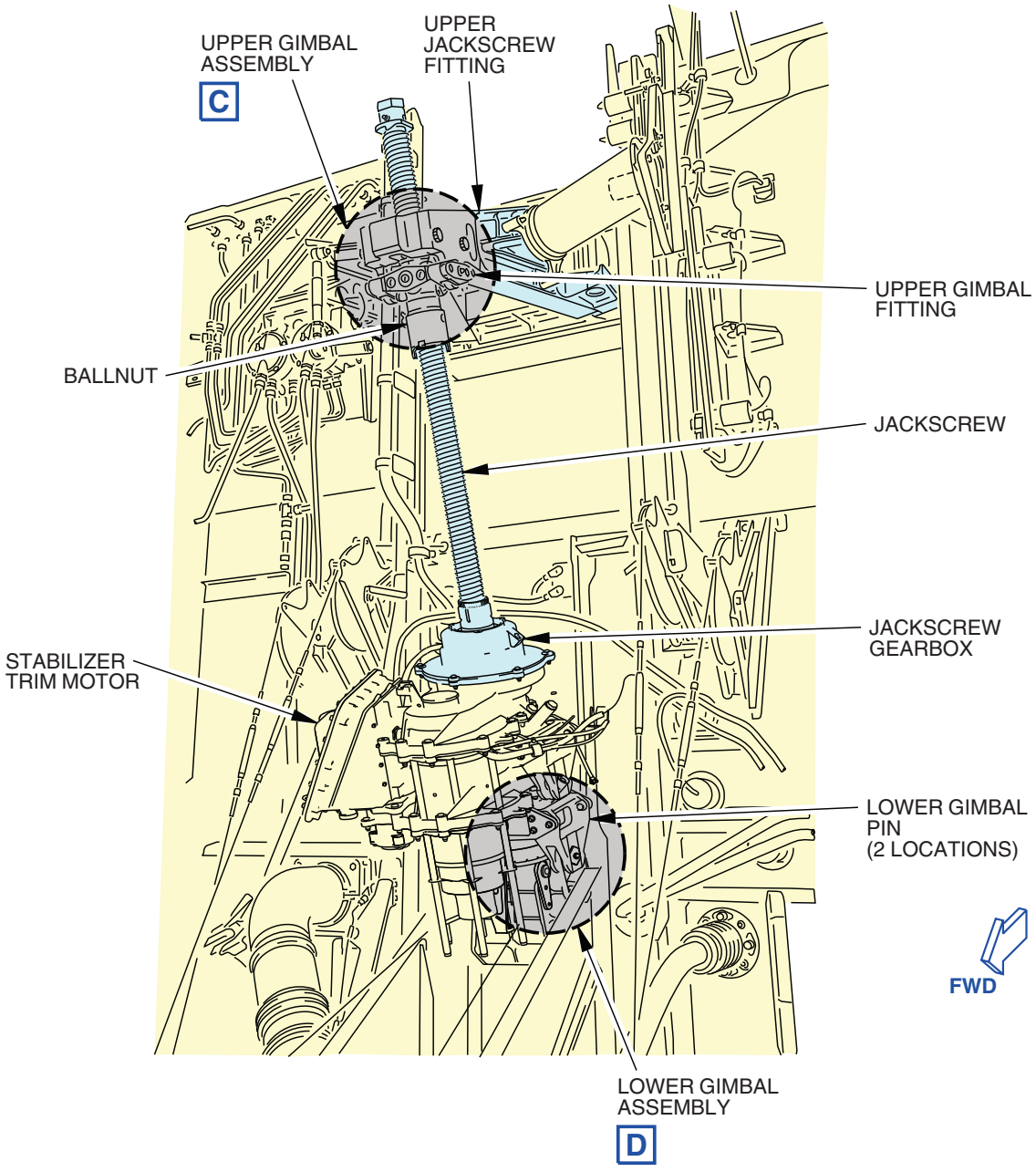
VIEW THROUGH THE ACCESS DOOR TO AFT UNPRESSURIZED COMPARTMENT, 311BL

A

2412694 S00061531159_V2

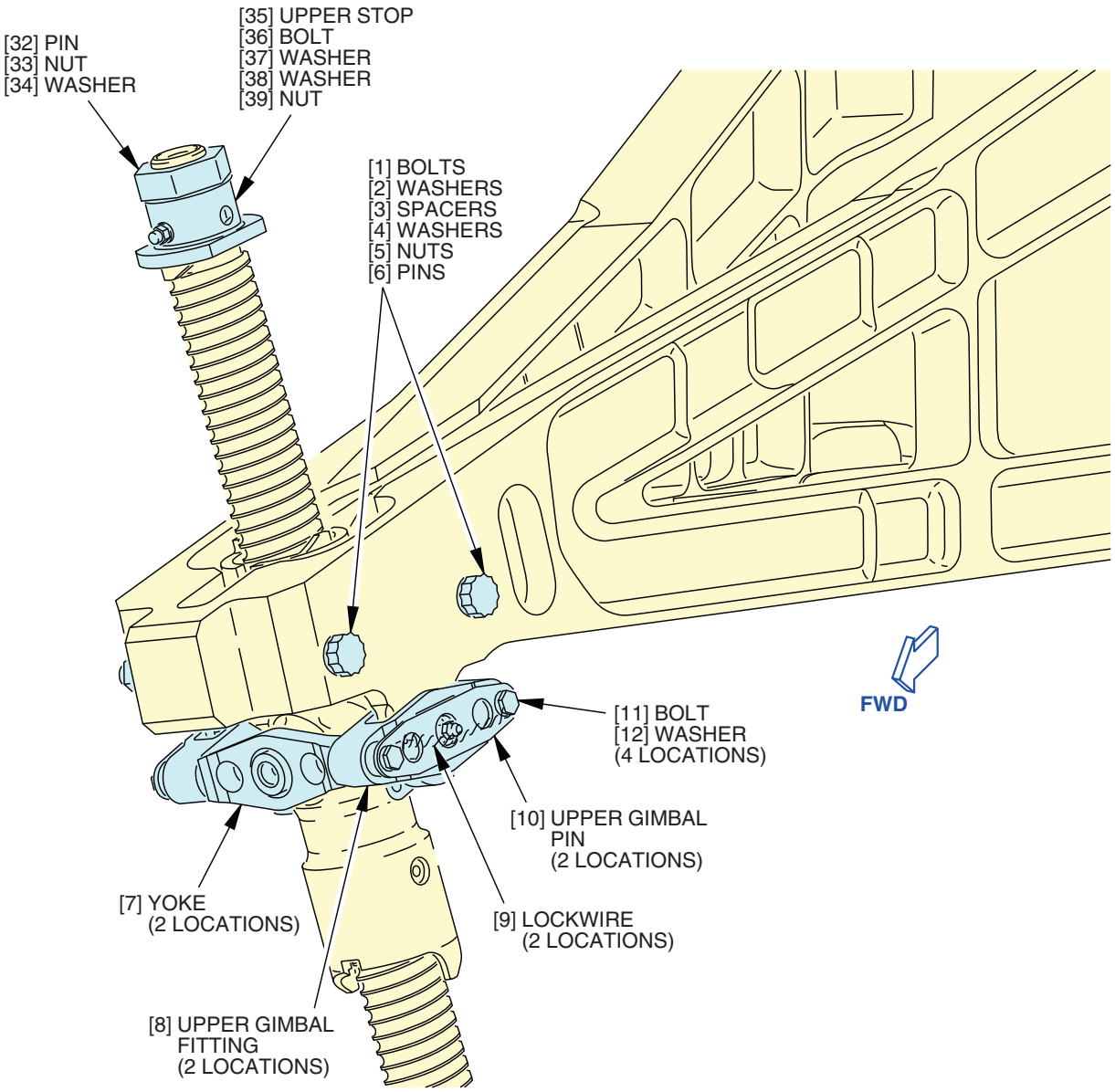
Stabilizer Ball Nut and Jackscrew Gearbox Visual Inspection
Figure 601/27-41-81-990-804 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Ball Nut and Jackscrew Gearbox Visual Inspection
Figure 601/27-41-81-990-804 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



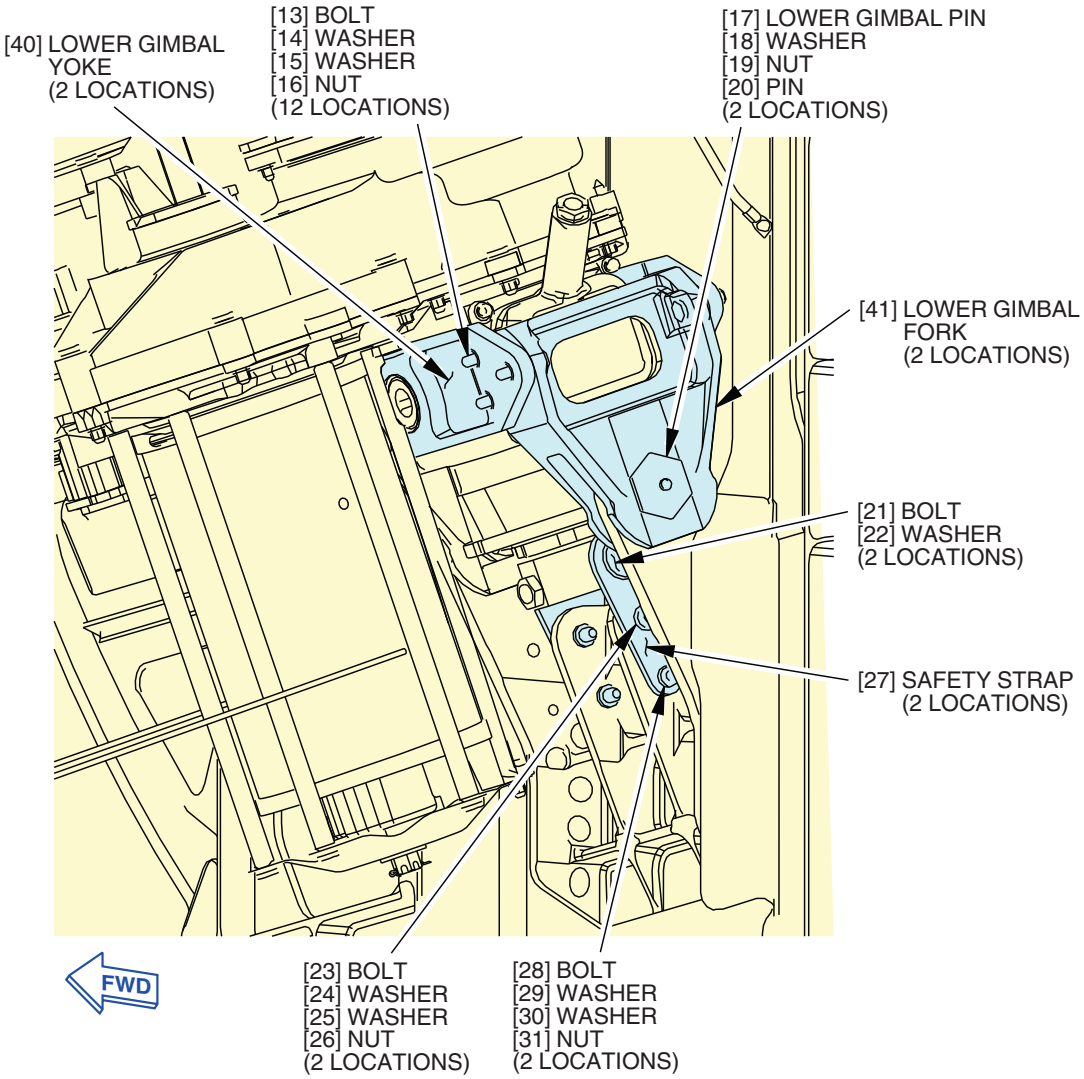
UPPER GIMBAL ASSEMBLY

C

3111780 S0000850673_V1

Stabilizer Ball Nut and Jackscrew Gearbox Visual Inspection
Figure 601/27-41-81-990-804 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



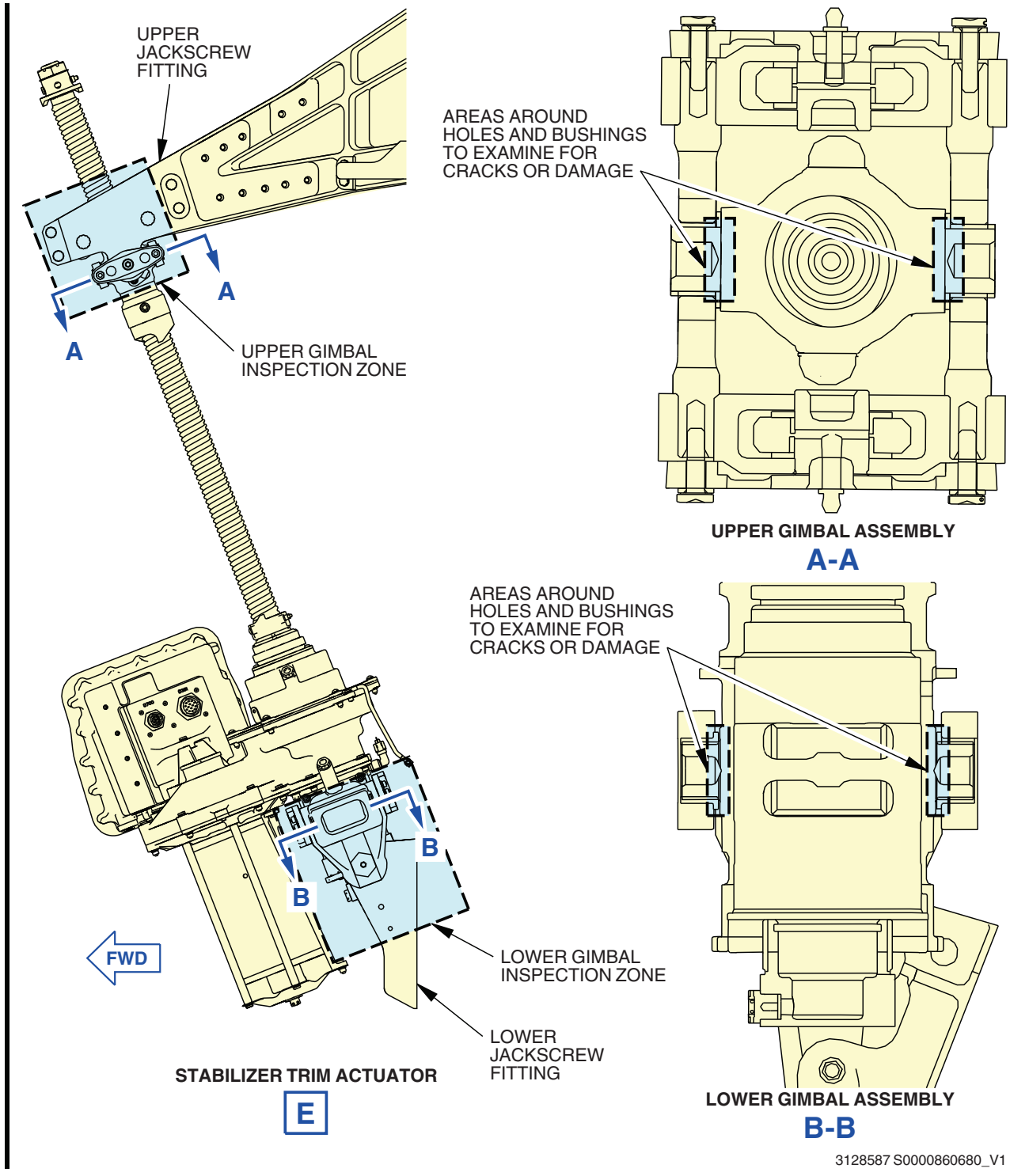
LOWER GIMBAL ASSEMBLY

D

3111781 S0000850674_V1

Stabilizer Ball Nut and Jackscrew Gearbox Visual Inspection
Figure 601/27-41-81-990-804 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Ball Nut and Jackscrew Gearbox Visual Inspection
Figure 601/27-41-81-990-804 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-81-210-802

3. Stabilizer Jackscrew Gearbox Backlash Inspection

(Figure 602)

A. General

(1) This task contains the instructions for the backlash inspection of stabilizer jackscrew gearbox.

B. References

Reference	Title
27-41-81-000-801	Stabilizer Ball Nut and Jackscrew Gearbox Removal (P/B 401)
27-41-81-400-801	Stabilizer Ball Nut and Jackscrew Gearbox Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1752	Check Equipment - Jackscrew Backlash, Horizontal Stabilizer Part #: C27065-20 Supplier: 81205 Opt Part #: C27065-1 Supplier: 81205
STD-1238	Indicator - Dial

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Inspection

SUBTASK 27-41-81-860-006

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT

SUBTASK 27-41-81-860-011

(2) Set the STAB TRIM PRI switch, S272, on the control stand, to the CUTOUT position.

SUBTASK 27-41-81-010-004

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Stabilizer Jackscrew Gearbox Backlash Inspection

SUBTASK 27-41-81-860-004

(1) Move the stabilizer to 8.5 units of trim.

SUBTASK 27-41-81-480-008

(2) Install the check equipment, SPL-1752, and the dial indicator, STD-1238, on the jackscrew as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the plate assembly of the check equipment, SPL-1752, on the jackscrew fitting with the screws.

(b) Install the left and right bulkhead plate assembly of the check equipment, SPL-1752, to the bulkhead stiffeners with the screws.

(c) Install the C-channel and hydraulic cylinder assembly of the check equipment, SPL-1752, to the left and right bulkhead plate assembly of the check equipment, SPL-1752, with the pins.

NOTE: The flat side of the C-channel assembly of the check equipment, SPL-1752, should point forward.

(d) Hold the upper end of the jackscrew with both hands and turn clockwise and counter-clockwise to ensure the brake ratchet pawls are engaged.

NOTE: This will lock the jackscrew in place so that the jackscrew will not turn when the load is applied.

(e) Connect the clevis eye of the hydraulic cylinder of the check equipment, SPL-1752, to the clevis on the plate assembly of the check equipment, SPL-1752, with the pin and lock pin.

(f) Connect the hydraulic assembly hose of the check equipment, SPL-1752, to the hydraulic cylinder coupling of the check equipment, SPL-1752.

(g) Install the magnetic holder of the check equipment, SPL-1752, and dial indicator, STD-1238, on the jackscrew.

SUBTASK 27-41-81-700-001

(3) Pressurize the check equipment, SPL-1752, to 900 psi (6205 kPa) – 950 psi (6550 kPa) with the hand hydraulic pump of the check equipment, SPL-1752.

SUBTASK 27-41-81-700-002

(4) If the dial indicator, STD-1238, shows more than 0.025 in. (0.64 mm), replace the stabilizer ball nut and jackscrew gearbox (TASK 27-41-81-000-801 and TASK 27-41-81-400-801).

SUBTASK 27-41-81-080-004

(5) Remove the check equipment, SPL-1752, and dial indicator, STD-1238, from the jackscrew:

(a) Remove the magnetic holder of the check equipment, SPL-1752, and dial indicator, STD-1238, from the jackscrew.

(b) Disconnect the clevis eye of the hydraulic assembly of the check equipment, SPL-1752, from the hydraulic cylinder coupling, of the check equipment, SPL-1752, and remove the hydraulic assembly.

(c) Remove the pin and lock pin to disconnect the hydraulic cylinder of the check equipment, SPL-1752, from the clevis on the plate assembly of the check equipment, SPL-1752.

(d) Unlock the jackscrew.

(e) Remove the pins that attach the C-channel and hydraulic cylinder assembly of the check equipment, SPL-1752, to the left and right bulkhead plate assembly of the check equipment, SPL-1752.

(f) Remove the C-channel and hydraulic cylinder assembly of the check equipment, SPL-1752, from the left and right bulkhead plate assembly of the check equipment, SPL-1752.

(g) Remove the screws that attach the left and right bulkhead plate assembly of the check equipment, SPL-1752, to the bulkhead stiffeners.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Remove the left and right bulkhead plate assembly of the check equipment, SPL-1752, from the bulkhead stiffeners.

(i) Remove the screws that attach the plate assembly of the check equipment, SPL-1752, to the jackscrew fitting.

(j) Remove the plate assembly of the check equipment, SPL-1752, from the jackscrew fitting.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-81-410-004

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-41-81-860-007

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

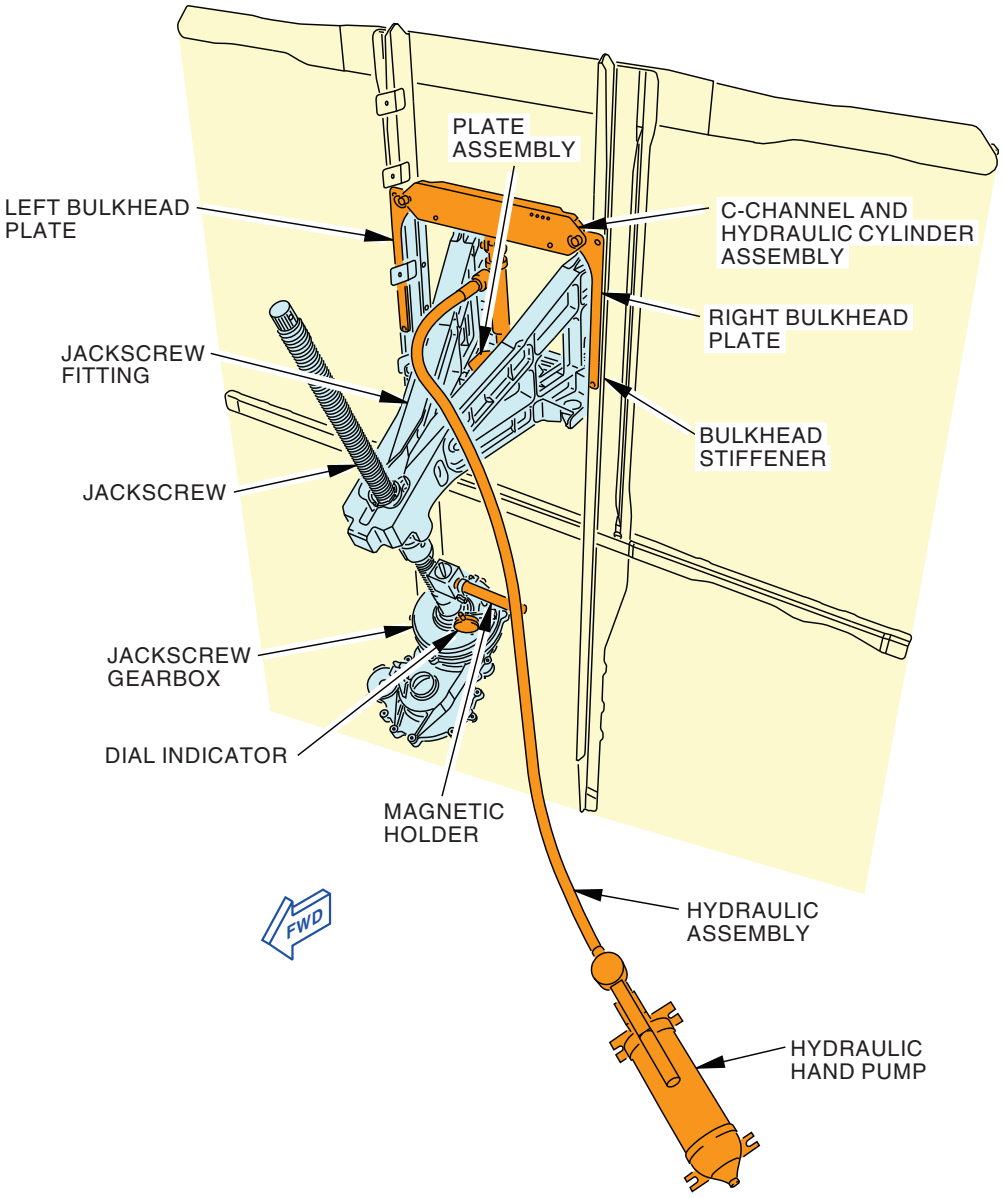
Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT

SUBTASK 27-41-81-860-009

(3) Set the STAB TRIM PRI switch, S272, on the control stand, to the NORMAL position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412696 S00061531162_V2

Stabilizer Jackscrew Gearbox Backlash Inspection
Figure 602/27-41-81-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-81-200-801

4. Upper and Lower Gimbal Wear Limits Inspection

(Figure 603)

A. General

(1) This task gives the wear limits for specific areas of the Horizontal Stabilizer Trim Actuator (HSTA).

(2) This task supplies the data for a dimensional inspection for specific areas of the HSTA.

B. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left

D. Upper and Lower Gimbal Wear Limits Inspection

SUBTASK 27-41-81-020-012

(1) Examine the parts of the lower and upper gimbals for worn areas:

(a) Measure the parts as necessary with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(b) Compare the dimensions you measured with the permitted wear limit dimensions and the maximum service diametric clearance from (Table 604).

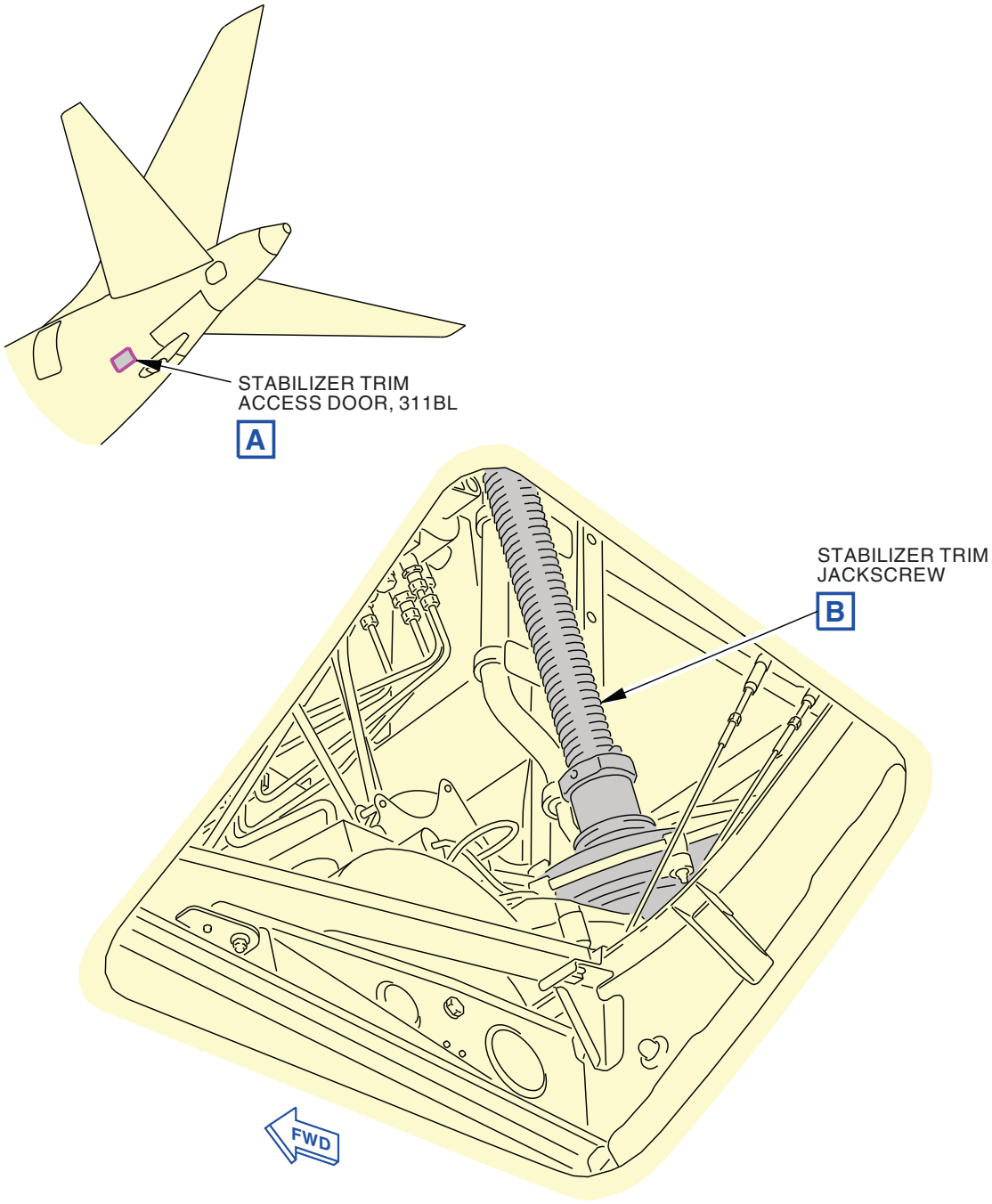
(c) If the parts are not in the tolerance, replace or repair the parts as required.

Table 604/27-41-81-993-801

NAME OF THE PARTS THAT ARE IN CONTACT	QTY	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
			MINIMUM inches / mm	MAXIMUM inches / mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 JACKSCREW FITTING BUSHING	4	I.D.	1.0000 in. (25.40 mm)	1.0010 in. (25.43 mm)	1.0030 in. (25.48 mm)	0.0050 in. (0.127 mm)
#1 UPPER GIMBAL PIN	2	O.D.	0.9980 in. (25.35 mm)	0.9990 in. (25.37 mm)	0.9970 in. (25.32 mm)	0.0050 in. (0.127 mm)
#2 STRUCTURE FITTING BUSHING	4	I.D.	1.0000 in. (25.40 mm)	1.0010 in. (25.43 mm)	1.0030 in. (25.48 mm)	0.0060 in. (0.152 mm)
#2 LOWER GIMBAL PIN	2	O.D.	0.9980 in. (25.35 mm)	0.9990 in. (25.37 mm)	0.9940 in. (25.25 mm)	0.0060 in. (0.152 mm)

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

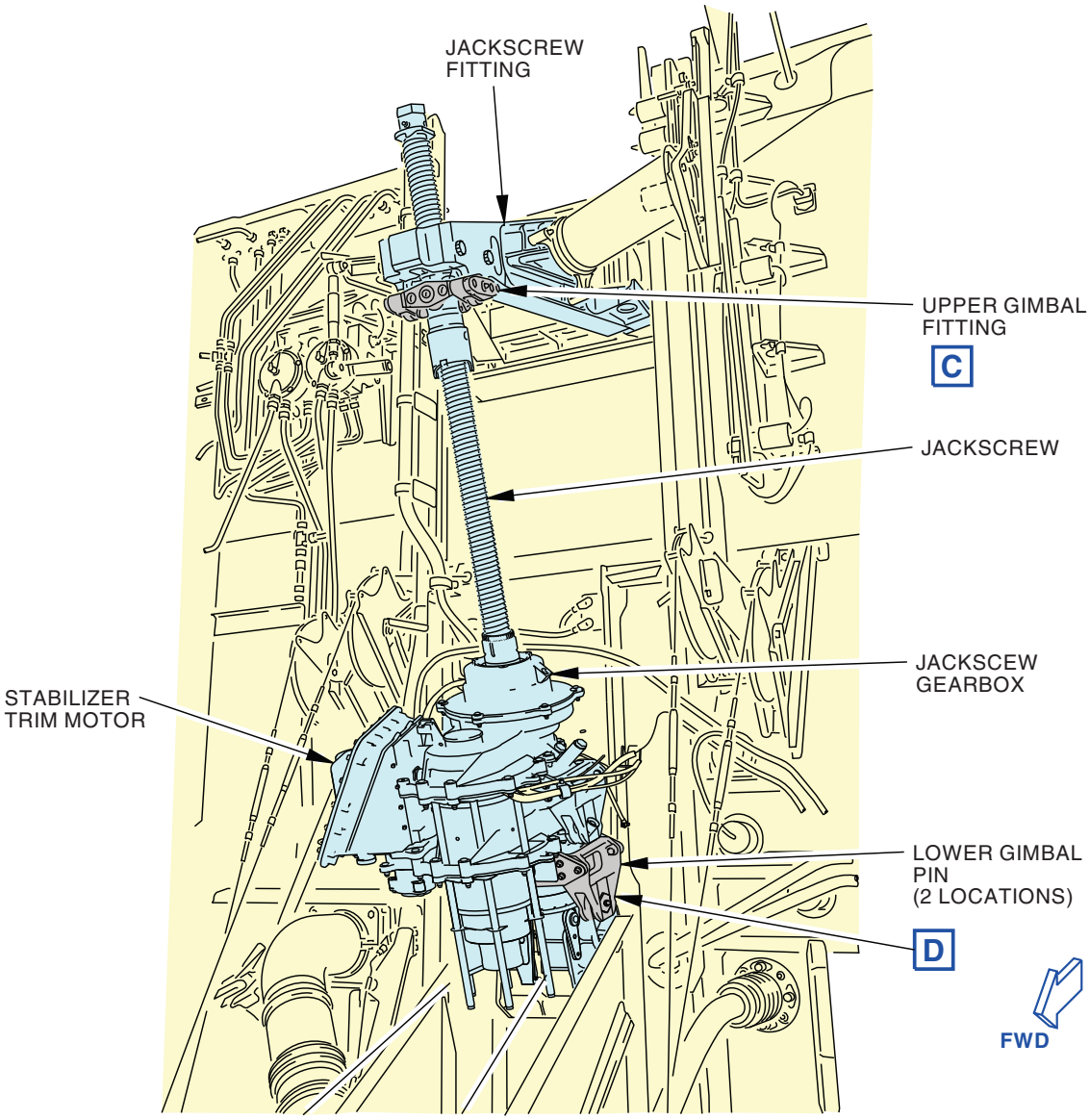


VIEW THROUGH THE STABILIZER TRIM ACCESS DOOR, 311BL
A

2412697 S00061531164_V1

Stabilizer Ball Nut and Jackscrew Gearbox Wear Limits
Figure 603/27-41-81-990-806 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



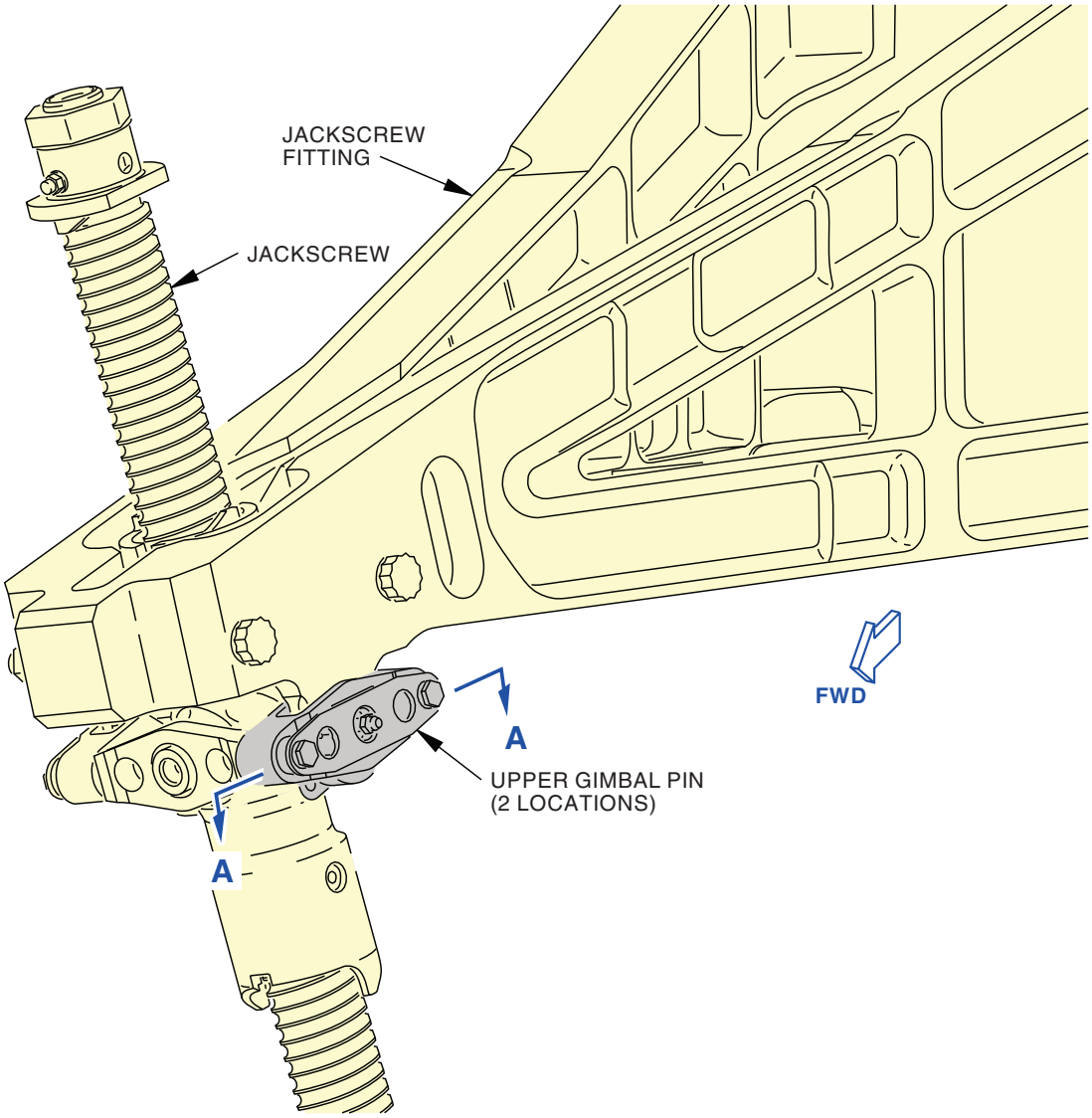
STABILIZER TRIM ACTUATOR

B

2412698 S00061531165_V2

Stabilizer Ball Nut and Jackscrew Gearbox Wear Limits
Figure 603/27-41-81-990-806 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



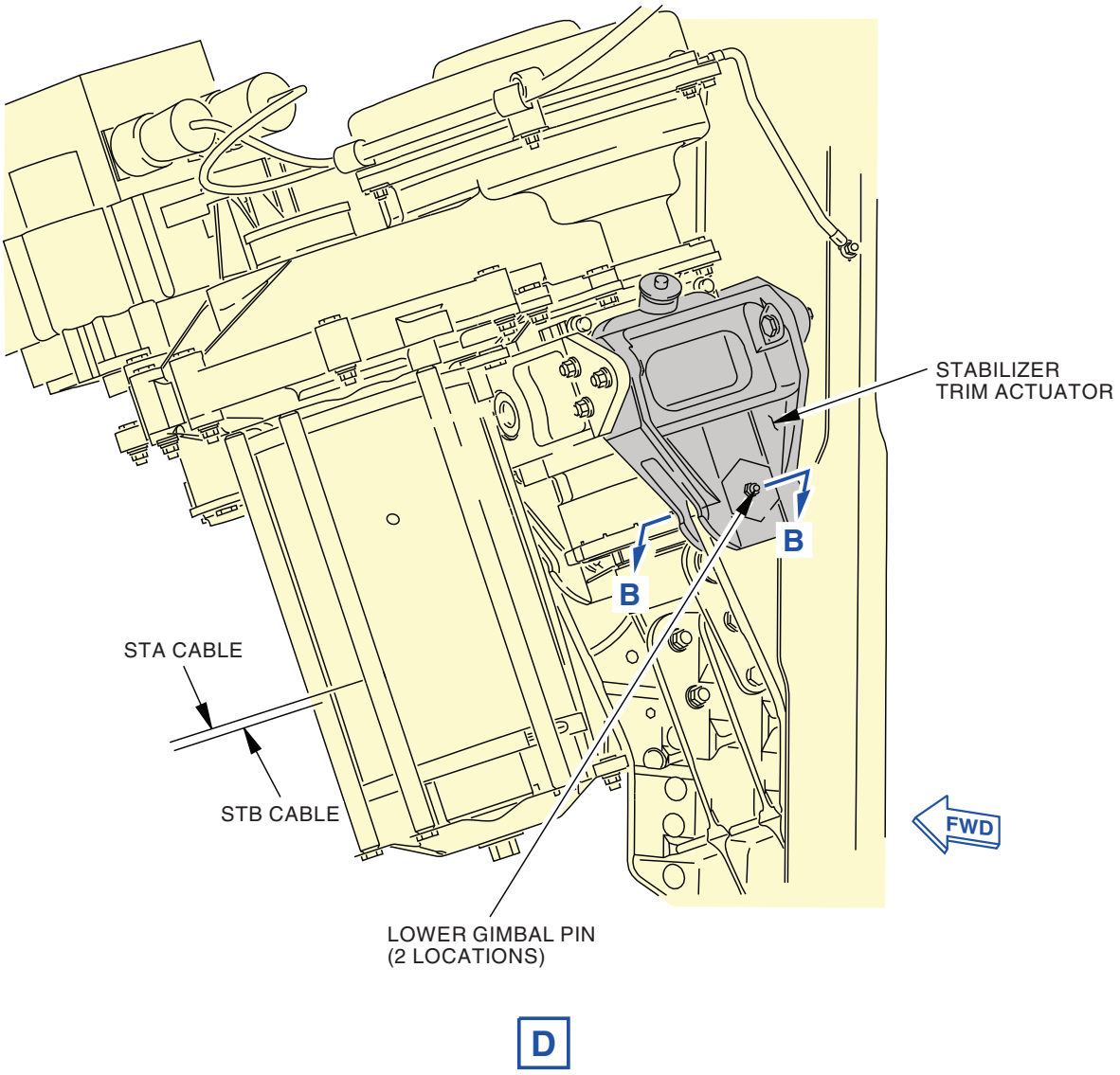
UPPER GIMBAL FITTING

C

2412699 S00061531166_V3

Stabilizer Ball Nut and Jackscrew Gearbox Wear Limits
Figure 603/27-41-81-990-806 (Sheet 3 of 5)

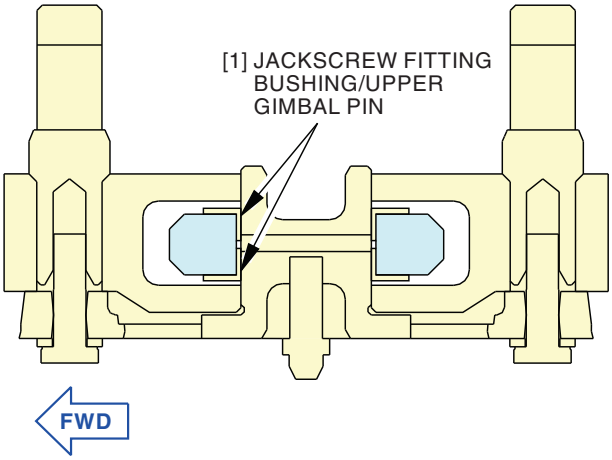
ECCN 9E991 BOEING PROPRIETARY - See title page for details



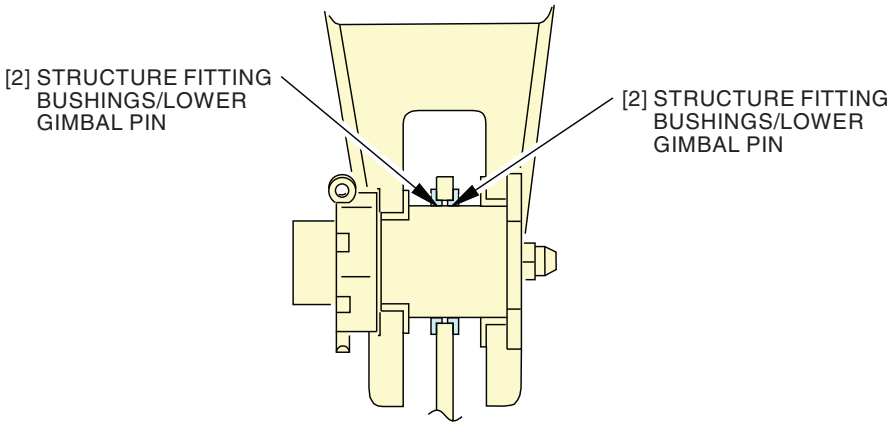
2412701 S00061531168_V5

Stabilizer Ball Nut and Jackscrew Gearbox Wear Limits
Figure 603/27-41-81-990-806 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

3058820 S0000822120_V1

Stabilizer Ball Nut and Jackscrew Gearbox Wear Limits
Figure 603/27-41-81-990-806 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER CONTROL CABLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the stabilizer trim control cables

(2) An installation of the stabilizer trim control cables.

B. The stabilizer trim control cables are referred to as the STA and STB cables in this procedure. The STA cable is located to the left of the STB cable.

C. The stabilizer trim control cable includes the cable with a swaged end connected to the aft cable drum and an end with an eyelet that connects it to the forward cable drum.

TASK 27-41-82-000-801

2. Stabilizer Control Cable Removal

(Figure 401, Figure 402)

A. References

Reference	Title
20-10-04-000-802	Control Cable Half Ball Pressure Seal Removal (P/B 401)
27-41-71-000-801	Stabilizer Trim Motor Removal (P/B 401)
50-10-03-000-801	Cargo Compartment Ceiling Liner Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205
SPL-10319	Wrench - Pinion Gear (Rotate) Part #: F71252 Supplier: 81205 Part #: F71267 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door
311BL	Stabilizer Trim Access Door

E. Prepare for the Removal

SUBTASK 27-41-82-860-001

(1) Set the stabilizer leading edge to the full down (airplane nose up) position.

SUBTASK 27-41-82-860-002

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-82-010-001

(3) To get access to the stabilizer forward control mechanism and the forward pulleys, do this step:

(a) Open these access panels:

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

SUBTASK 27-41-82-010-002

(4) To get access to the stabilizer trim actuator and the aft pulley, do this step:

(a) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-41-82-010-003

(5) Remove the cargo liners and floor panels (Figure 402).

(a) Do this task: Cargo Compartment Ceiling Liner Removal, TASK 50-10-03-000-801.

(b) Do this task: Passenger Cabin Floor Panel Removal, TASK 53-21-00-000-801.

SUBTASK 27-41-82-020-001

(6) Disconnect the upper turnbuckle [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-82-820-001

CAUTION MAKE SURE THAT THE CHAIN IS LOOSE BEFORE YOU RELEASE THE CABLE TENSION. TOO MUCH CABLE TENSION CAN CAUSE DAMAGE TO THE CHAIN.

(7) Adjust the upper adjusting nuts [7] of the aft support link [8] to loosen the chain (Figure 401).

SUBTASK 27-41-82-020-010

WARNING DO NOT OPERATE THE STABILIZER ELECTRICALLY UNTIL YOU ADJUST THE STABILIZER TRIM LIMIT SWITCHES, CABLES, AND CHAIN. THE HORIZONTAL STABILIZER CAN MOVE QUICKLY WHEN YOU OPERATE THE STABILIZER TRIM LIMIT SWITCHES. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Disconnect the flexible shaft [36] from the stabilizer forward control mechanism.

(a) Remove the lockwire from the coupling nut of the flexible shaft [36].

SUBTASK 27-41-82-020-002

(9) Loosen the lower turnbuckle [3] to release the tension of the STA cable [6] and STB cable [5].

SUBTASK 27-41-82-020-009

(10) Remove the half ball pressure seals, do this task: Control Cable Half Ball Pressure Seal Removal, TASK 20-10-04-000-802.

SUBTASK 27-41-82-020-003

(11) Do this task: Stabilizer Trim Motor Removal, TASK 27-41-71-000-801.

F. Stabilizer Control Cable Removal

SUBTASK 27-41-82-020-004

(1) Remove the cable drum guard [28] and spacers [14] from the cable drum [15]:

(a) Remove the pin [83] and castellated nut [82] from the support plate [25] that is attached to the cable drum [15].

1) Discard the pin [83].

(b) Remove the nut [23], washer [22], washer [21], and bolt [20] that attach the cable drum guard [28] to the support plate [25].

(c) Remove the bolts [26], washers [27], support plate [25], spacers [14], and washer [84] from the cable drum [15].

NOTE: Because work space is small, you must keep the bolts in the support plate and in the spacers to help the removal procedure.

(d) Remove the bolts [16] and washers [17] that attach the cable drum guard [28] to the cable drum [15].

(e) Remove the cable drum guard [28] from the cable drum [15].

SUBTASK 27-41-82-020-005

(2) Do these steps to disconnect the STB cable [5] from the cable drum [15]:

(a) Install the pinion gear wrench, SPL-10319, on the internal spline of the jackscrew gearbox [12] (Figure 401).

NOTE: Insert the arm of the pinion gear wrench, SPL-10319, into the motor mount hole to keep the internal spline of the jackscrew gearbox from turning.

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(b) Use the force gauge, COM-1557, on the captain's or first officer's stabilizer trim wheel handle to put a 50 lb (22.7 kg) load on the handle to stretch the STA cable [6].

NOTE: This will keep the STA cable on the drum and will make the removal of the STB cable easier.

(c) Disconnect the STB cable [5] from the cable drum [15].

SUBTASK 27-41-82-020-006

(3) Remove the STB cable [5] from the cable run.

(a) Remove the arm of the pinion gear wrench, SPL-10319, from the stabilizer trim motor mount hole.

(b) Remove the DO NOT OPERATE tag, from the stabilizer trim wheel at the control stand.

(c) Using the captain's or first officer's stabilizer trim control wheel, move the jackscrew [11] to put the stabilizer leading edge to the full up (airplane nose down) position.

CAUTION MAKE SURE THAT YOU DO NOT DAMAGE OR KINK THE CABLE WHILE YOU WIND IT ONTO THE SPOOL OR REEL. DAMAGE TO THE CABLE CAN OCCUR.

CAUTION IF YOU ARE TO USE THE CABLE AGAIN, DO NOT LET IT TOUCH THE DIRTY SURFACES. DIRT IN THE GREASE ON THE CABLES CAN CAUSE FAILURE OF THE CABLE OR CAN CAUSE PROBLEMS WHEN RE-RIGGING THE CABLE.

(d) When the stabilizer starts to move, pull the STB cable [5] into the tail compartment and wind it on a spool or a reel.

NOTE: Tension must always be kept on the STB cable while pulling it into the tail compartment. The best way to keep steady tension on the STB cable is to have a mechanic pulling the cable into the tail compartment while another winds it onto a spool or reel.

NOTE: To ensure the STB cable is not damaged or kinked as it is unwound from the forward drum, it is best to have a mechanic in the forward compartment to watch the cable as it unwinds from the forward drum. This will ensure that the mechanic operating the stabilizer trim control wheel does not turn the forward drum too far and kink the cable on the right cable guard spacer (Figure 401).

(e) When the ballnut touches the upper stop, keep a record of the number of turns of the STA cable [6] on the cable drum [15].

(f) Disconnect the STA cable [6] from the cable drum [15].

NOTE: Use a control cable clamp, SPL-1569, to keep tension on the STA cable to keep it from unwinding on the forward cable drum.

(g) Disconnect the STB cable [5] from the forward cable drum [15].

1) Remove the nut and washer from the clevis bolt.

2) Remove the clevis bolt from the STB cable [5] eyelet.

SUBTASK 27-41-82-480-001

(4) Remove the STB cable [5] from the aircraft.

(a) Attach lockstitch or some similar material to the aft end of the STB cable [5].

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CAUTION MAKE SURE THAT YOU DO NOT DAMAGE OR KINK THE CABLE WHILE YOU WIND IT ONTO THE SPOOL OR REEL. DAMAGE TO THE CABLE CAN OCCUR.

CAUTION IF YOU ARE TO USE THE CABLE AGAIN, DO NOT LET IT TOUCH THE DIRTY SURFACES. DIRT IN THE GREASE ON THE CABLES CAN CAUSE FAILURE OF THE CABLE OR CAN CAUSE PROBLEMS WHEN RE-RIGGING THE CABLE.

(b) Using a spool or reel, wind the STB cable [5] up in the forward mechanism compartment.

1) Remove the idler pulleys to let the aft end of the cable through the pulley stations (Figure 402).

2) Remove the aft pulley as required to let the end of the cable through the pulley stations (Figure 402).

NOTE: The aft pulley must only be removed while the cable end passes through the bracket to decrease the cable rubbing.

SUBTASK 27-41-82-020-007

(5) Do the following steps to remove the STA cable [6] from the aircraft:

(a) Remove the cable clamp from the STA cable [6] at the aft cabin pressure bulkhead while maintaining force on the STA cable [6].

(b) Using the captain's or first officer's stabilizer trim control wheel, unwind the STA cable [6] from the forward drum.

CAUTION MAKE SURE THAT YOU DO NOT DAMAGE OR KINK THE CABLE WHILE YOU WIND IT ONTO THE SPOOL OR REEL. DAMAGE TO THE CABLE CAN OCCUR.

CAUTION IF YOU ARE TO USE THE CABLE AGAIN, DO NOT LET IT TOUCH THE DIRTY SURFACES. DIRT IN THE GREASE ON THE CABLES CAN CAUSE FAILURE OF THE CABLE OR CAN CAUSE PROBLEMS WHEN RE-RIGGING THE CABLE.

(c) Pull the STA cable [6] into the tail compartment and wind it on a spool or a reel.

NOTE: Tension must always be kept on the STA cable while pulling it into the tail compartment. The best way to keep steady tension on the STA cable is to have a mechanic pulling the cable into the tail compartment while another mechanic winds the cable onto a spool or reel.

NOTE: To ensure the STA cable is not damaged or kinked as it is unwound from the forward drum, it is recommended to have a mechanic in the forward compartment to watch the cable as it slowly unwinds from the forward drum. This will ensure that the mechanic pulling the cable into the aft unpressurized compartment does not pull the cable too far and kink the cable on the left cable guard spacer (Figure 401).

(d) Disconnect the STA cable [6] from the forward cable drum [15].

1) Remove the cotter pin [18], the nut, the washer, and the washers from the clevis bolt.

a) Discard the cotter pin [18].

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2) Remove the clevis bolt from the STA cable [6] eyelet.

SUBTASK 27-41-82-480-002

(6) Remove the STA cable [6] from the aircraft.

(a) Attach lockstitch or some similar material to the aft end of the STA cable [6].

CAUTION

MAKE SURE THAT YOU DO NOT DAMAGE OR KINK THE CABLE WHILE YOU WIND IT ONTO THE SPOOL OR REEL. DAMAGE TO THE CABLE CAN OCCUR.

CAUTION

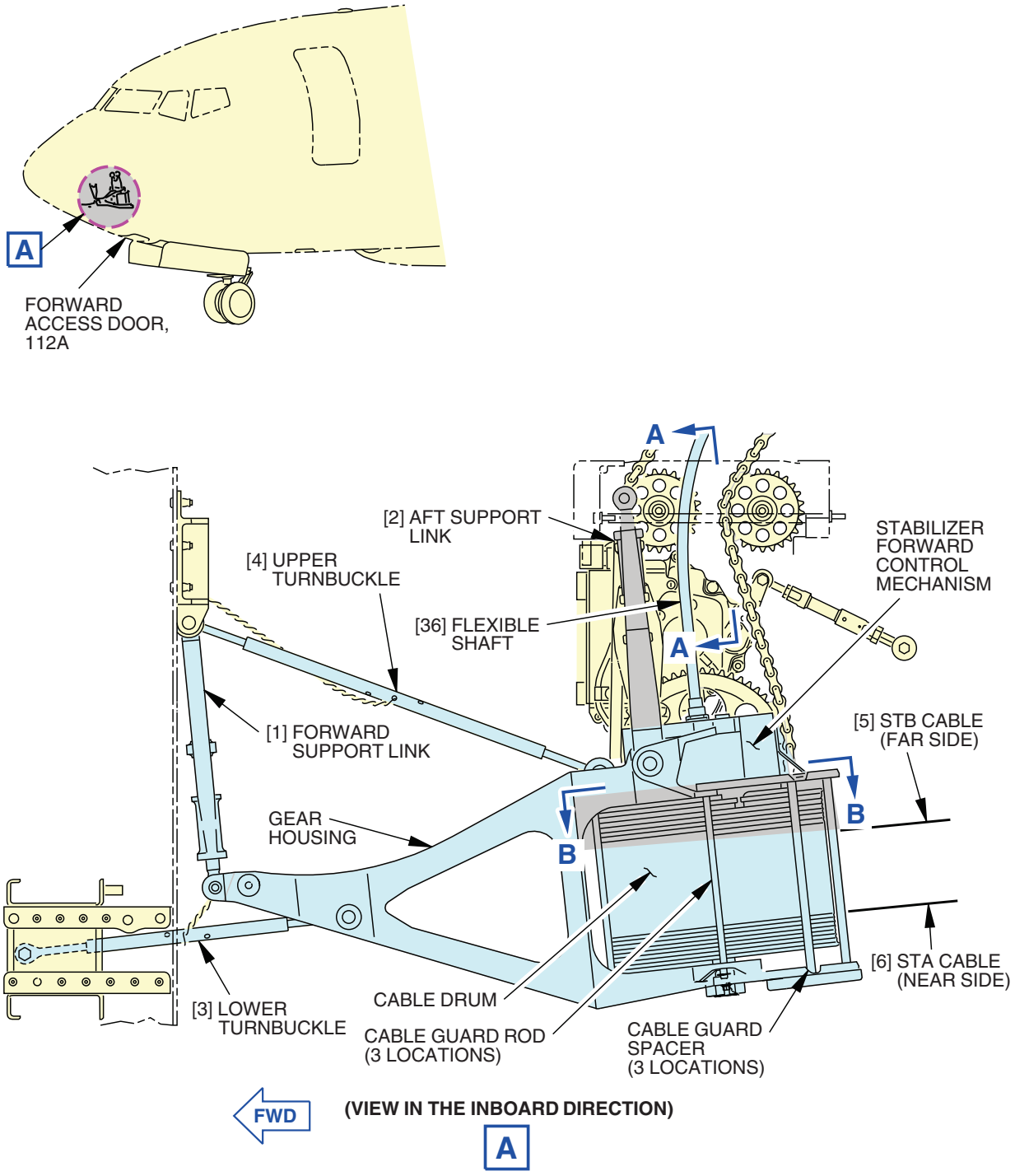
IF YOU ARE TO USE THE CABLE AGAIN, DO NOT LET IT TOUCH THE DIRTY SURFACES. DIRT IN THE GREASE ON THE CABLES CAN CAUSE FAILURE OF THE CABLE OR CAN CAUSE PROBLEMS WHEN RE-RIGGING THE CABLE.

(b) Using a spool or reel, wind the STA cable [6] up in the forward mechanism compartment.

1) Remove the idler pulleys to let the aft end of the cable through the pulley stations (Figure 402).

END OF TASK

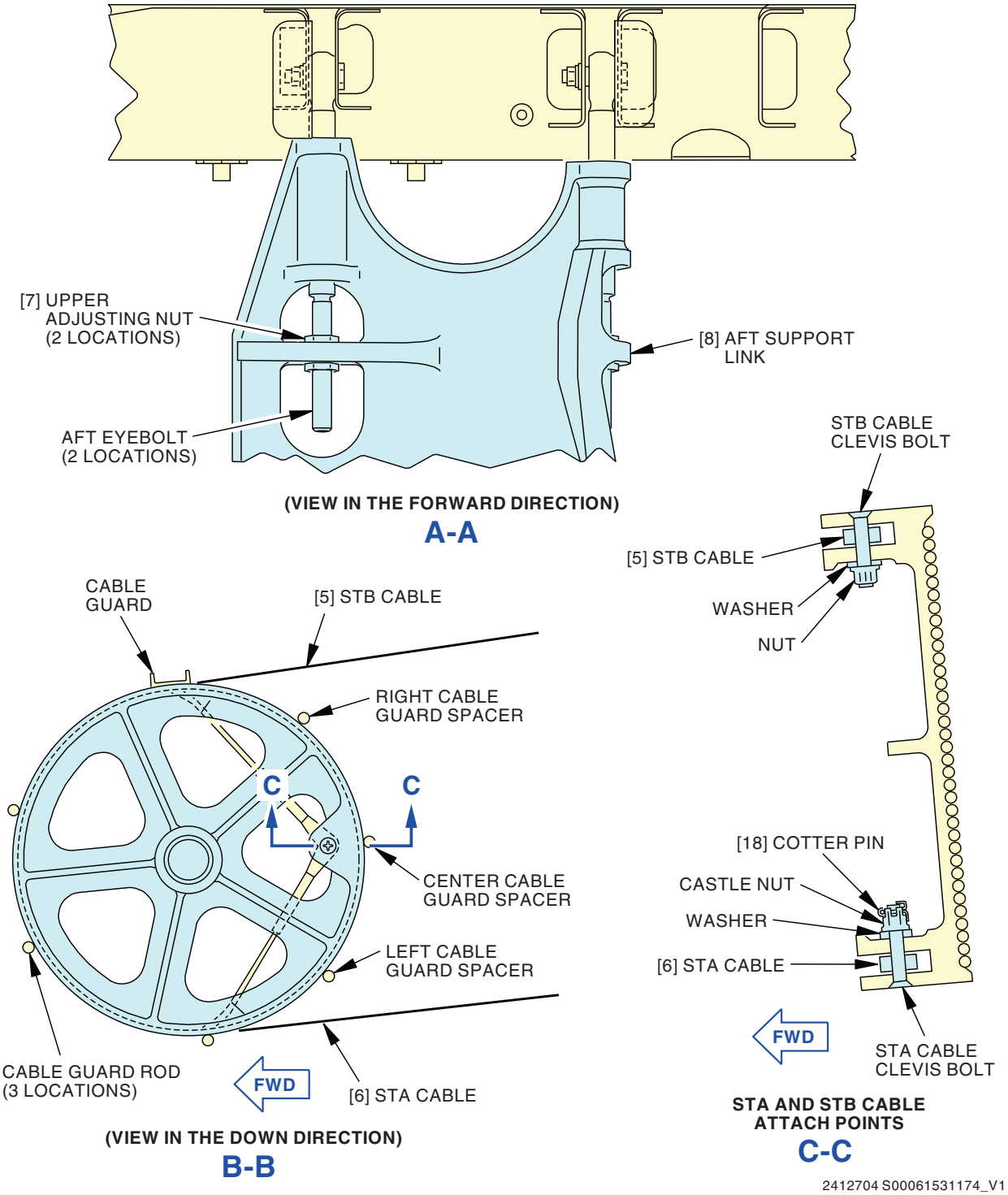
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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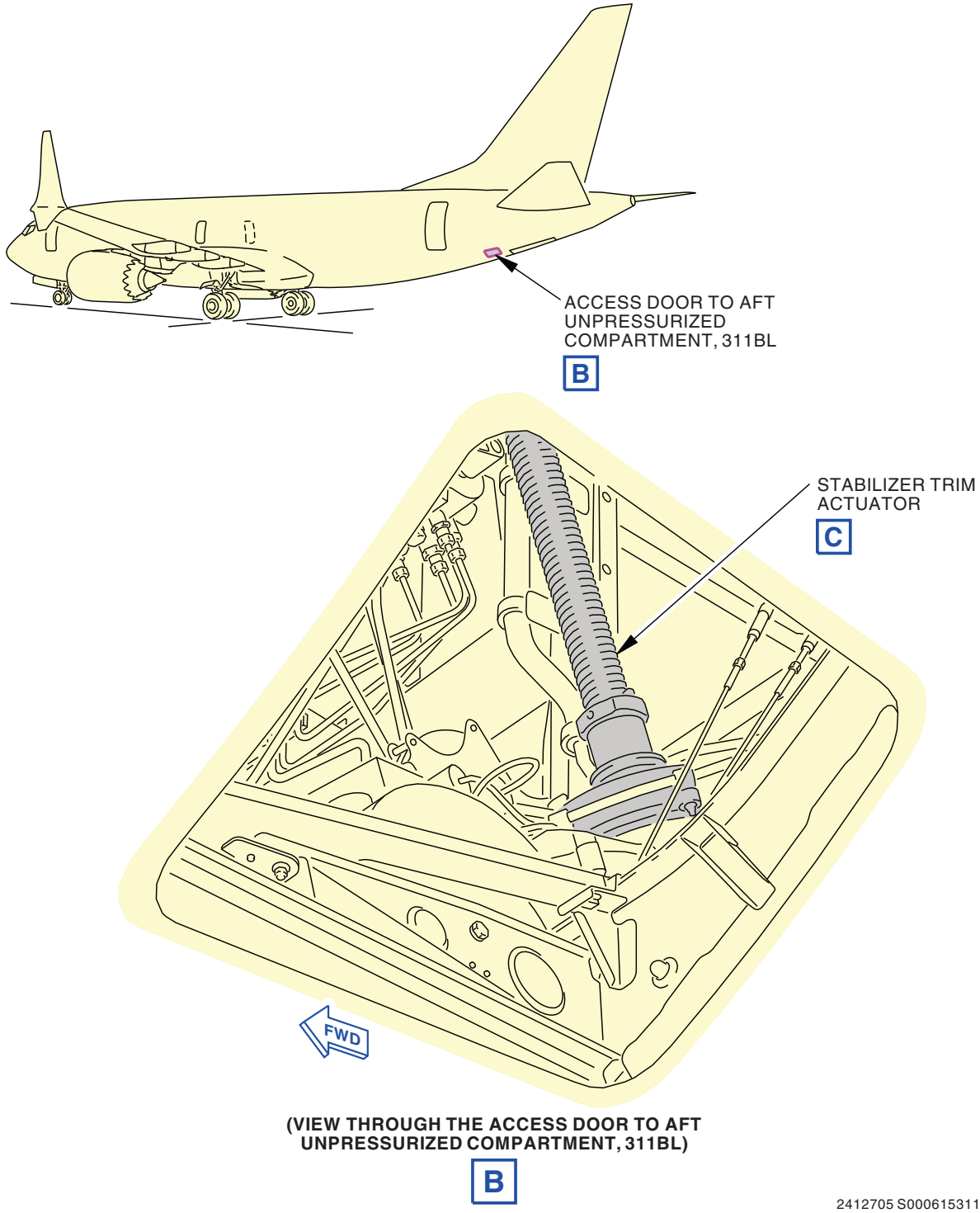
Stabilizer Control Cable Replacement
Figure 401/27-41-82-990-801 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



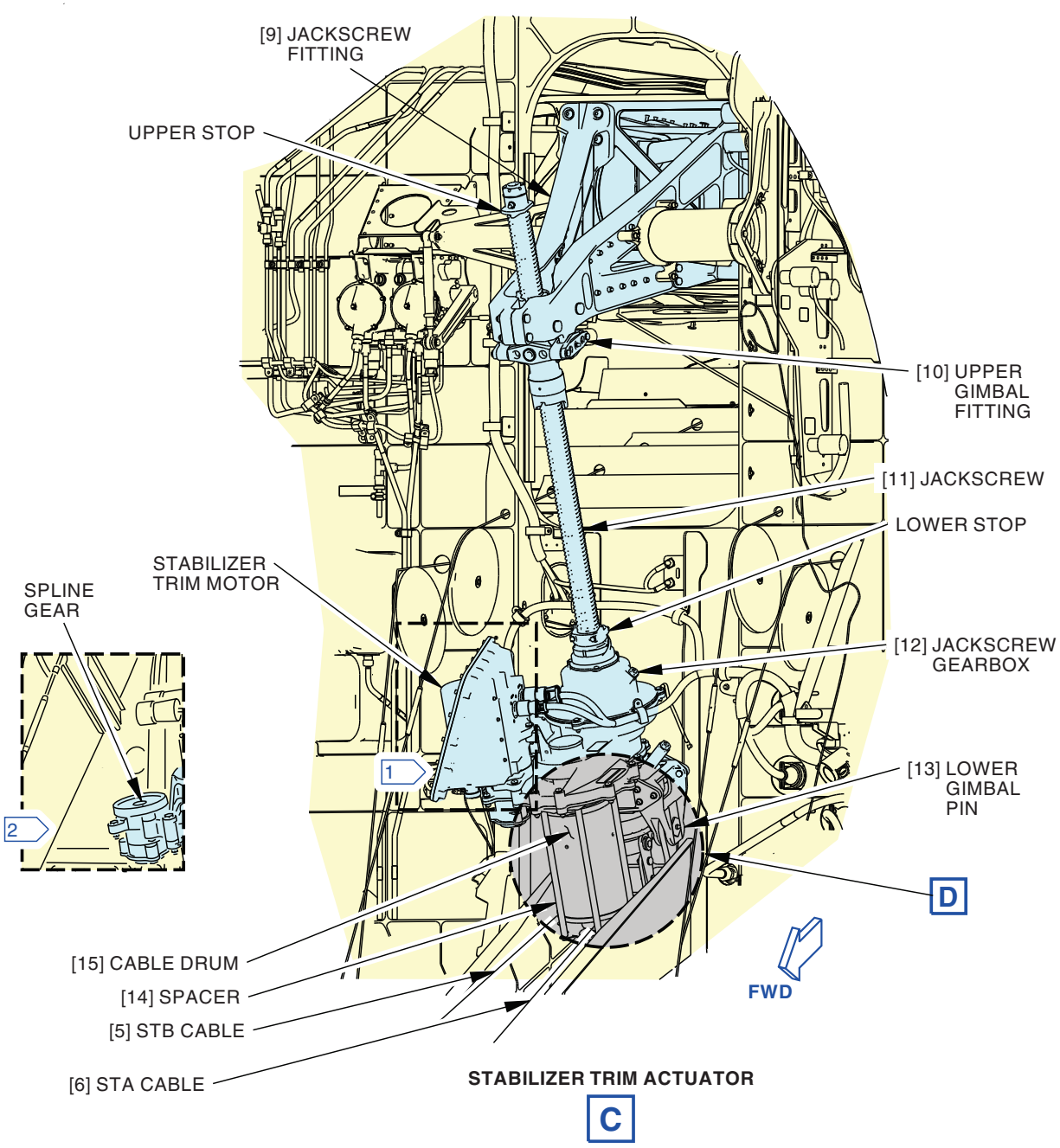
Stabilizer Control Cable Replacement
Figure 401/27-41-82-990-801 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Control Cable Replacement
Figure 401/27-41-82-990-801 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

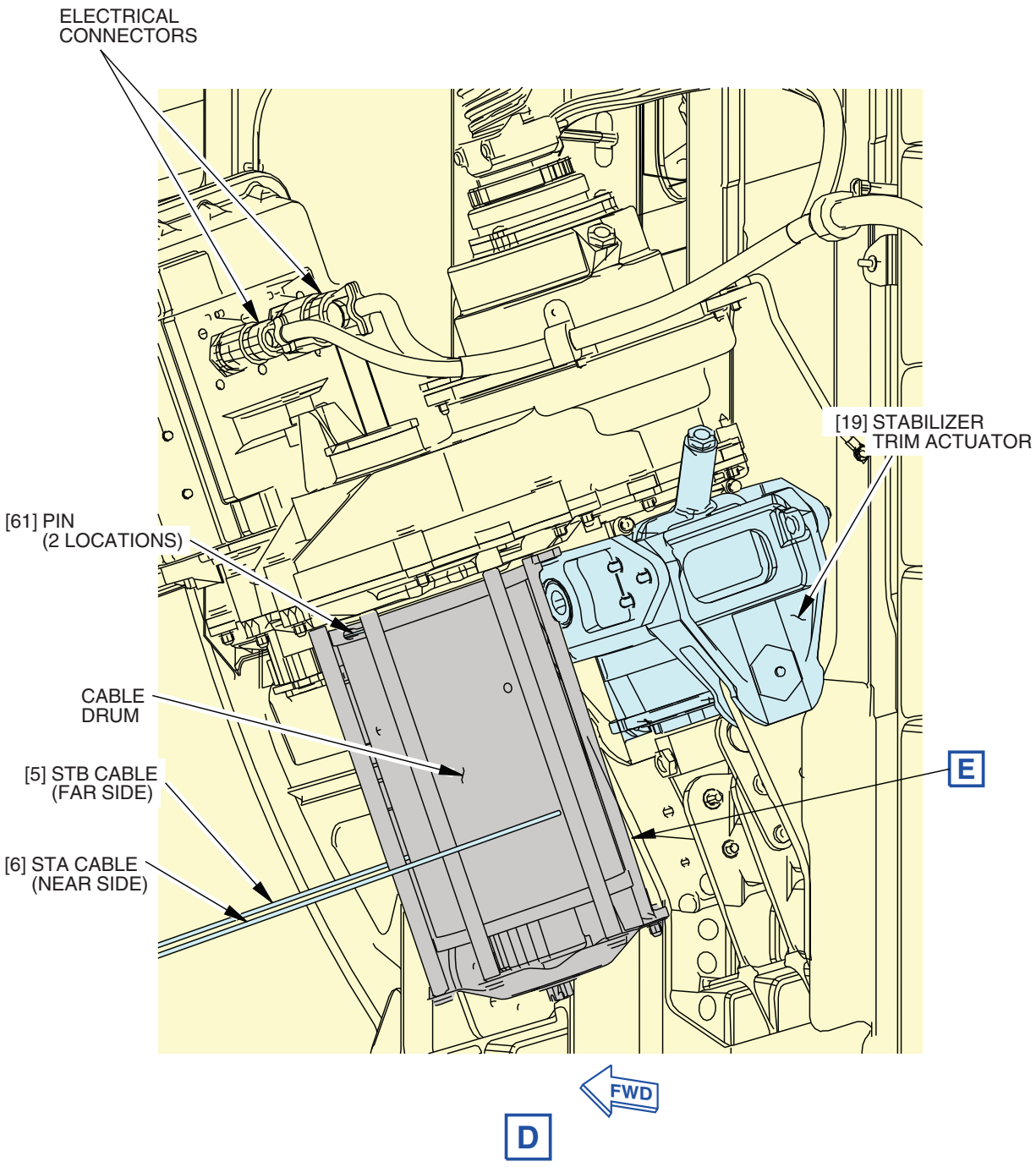


1 STABILIZER ACTUATOR ASSEMBLY WITH STABILIZER TRIM MOTOR (STM).
2 STABILIZER ACTUATOR ASSEMBLY WITHOUT STABILIZER TRIM MOTOR (STM).

2412706 S00061531176_V4

Stabilizer Control Cable Replacement
Figure 401/27-41-82-990-801 (Sheet 4 of 6)

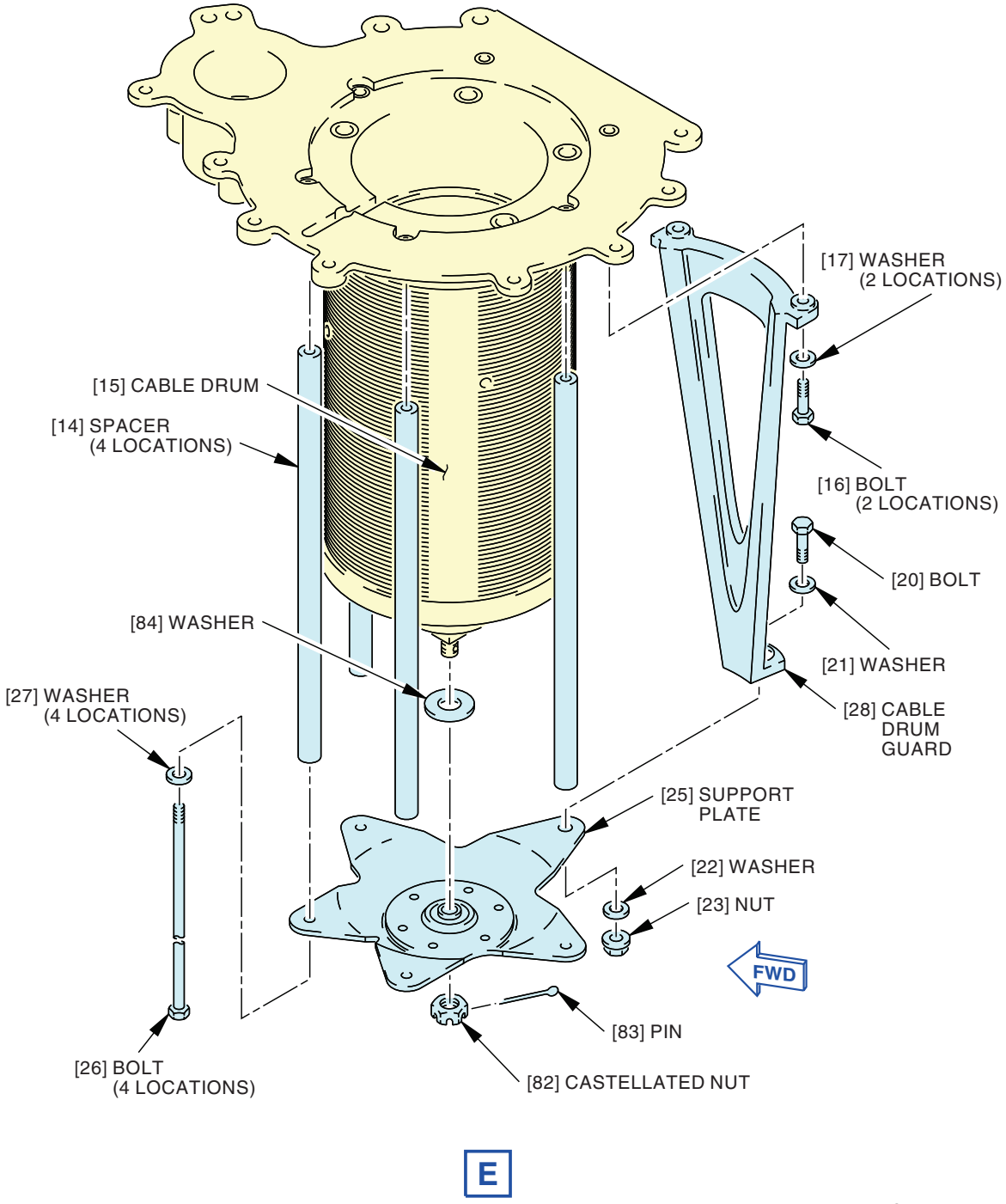
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2412707 S00061531177_V4

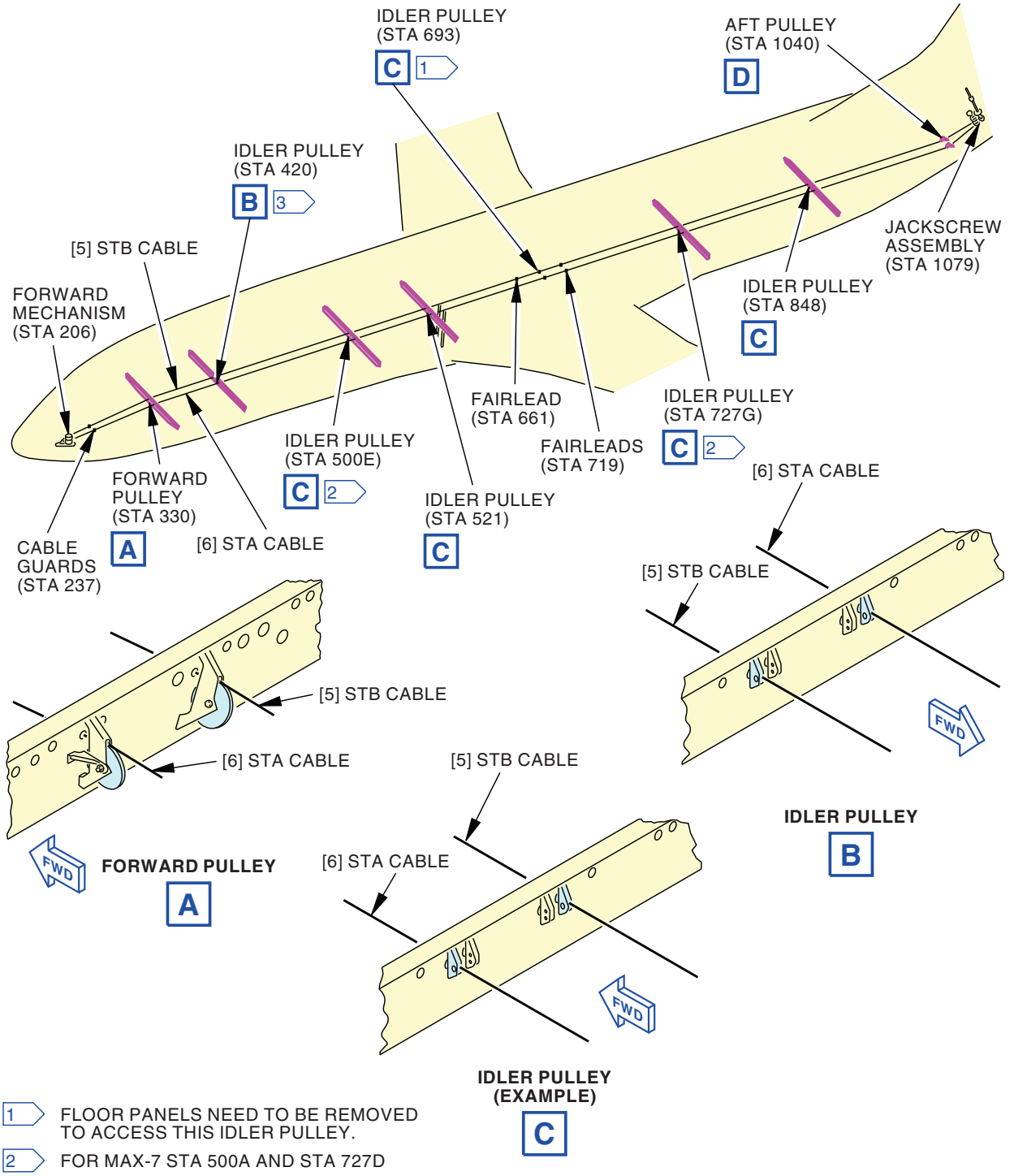
Stabilizer Control Cable Replacement
Figure 401/27-41-82-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Control Cable Replacement
Figure 401/27-41-82-990-801 (Sheet 6 of 6)

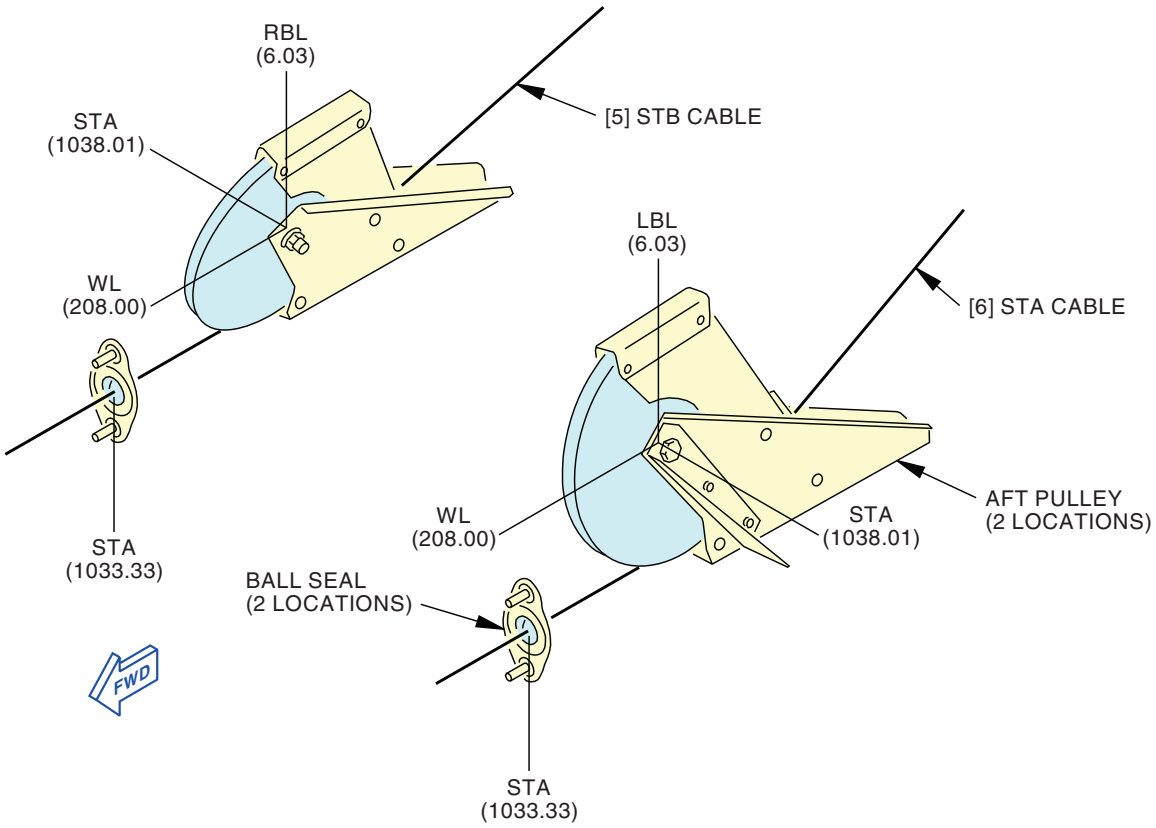
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412708 S00061531178_V2

Stabilizer Control Cable Installation
Figure 402/27-41-82-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AFT PULLEY AND BALL SEAL

D

2905860 S0000695857_V1

Stabilizer Control Cable Installation
Figure 402/27-41-82-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-82-400-801

3. Stabilizer Control Cable Installation

(Figure 401, Figure 402)

A. General

(1) It is recommended that cable assemblies on each side of a cable loop be replaced at the same time using the same grade material in order to minimize asymmetric relaxation.

(2) It is recommended that airplanes supplied with the cables that contain higher zinc coating (Grade D or Grade K) be re-tensioned 3 to 4 months into service. This will bring the tension of the control cables within the desired range. After the cables are re-tensioned, the high zinc cables (Grade D or Grade K) will continue to relax at a rate that is equivalent to the Grade B cables. To identify the grade of the cable, refer to the table below:

Table 401/27-41-82-993-801 Control Cable Grades and Diameters

CONTROL CABLE DIAMETER (TERMINAL SIZE REF)	CONSTRUCTION FIRST DASH NUMBER *[1]									
	GRADE A					GRADE B		GRADE D		GRADE K
	ZINC			TIN OVER ZINC		ZINC	TIN OVER ZINC	ZINC	TIN OVER ZINC	ZINC
	7 X 7	7 X 7 (RT)	7 X 19	7 X 7	7 X 7 (RT)	7 X 19	7 X 19	7 X 19	7 X 19	7 X 19
.094 (3/32)	3B	3G*[2]	3A	3D	3E *[2]	- — —	- — —	- — —	- — —	- — —
.125 (1/8)	- — —	- — —	4A	- — —	- — —	4F	4C	4J*[3]	4H*[3]	4K*[3]
.188 (3/16)	- — —	- — —	- — —	- — —	- — —	- — —	- — —	6J*[4]	- — —	- — —

*[1] First dash number in the part number of the cable. Example: BACC2C-3D-01085-CE is Grade A, Tin Over Zinc, 7 x 7.

*[2] Code 3E and 3G are specified only where there is a specific requirement for REDUCED TOLERANCE (RT) percent elongation wire rope, 7 X 7 RT, per BMS7-265.

*[3] Only Grade D and Grade K .125 (1/8) cables (4J, 4H and 4K) are high zinc cables.

*[4] It is not necessary to adjust the Grade D 0.188 (3/16) cables (6J) 3 to 4 months after installation.

B. References

Reference	Title
12-22-41-610-801	Horizontal Stabilizer Actuator Brake - Servicing (P/B 301)
20-10-04-400-802	Control Cable Half Ball Pressure Seal Installation (P/B 401)
27-41-00-700-801	Stabilizer Manual Trim and Trim Indicator - Test (P/B 501)
27-41-00-700-802	Horizontal Stabilizer Trim Control System Friction Test (P/B 501)
27-41-00-700-803	Stabilizer Electric Trim System Test (P/B 501)
27-41-00-700-804	Stabilizer Trim Wheel Free Movement Test (P/B 501)
27-41-00-700-806	Stabilizer Clutch Friction System Test (P/B 501)
27-41-00-820-801	Stabilizer Control Cable and Chain - Adjustment (P/B 501)
27-41-71-400-801	Stabilizer Trim Motor Installation (P/B 401)
50-10-03-400-801	Cargo Compartment Ceiling Liner Installation (P/B 401)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

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C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205
SPL-10319	Wrench - Pinion Gear (Rotate) Part #: F71252 Supplier: 81205 Part #: F71267 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	STB cable	Not Specified	
6	STA cable	Not Specified	
18	Pin	Not Specified	
61	Pin	Not Specified	

F. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

G. Access Panels

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door
311BL	Stabilizer Trim Access Door

H. Prepare for the Installation

SUBTASK 27-41-82-860-003

(1) Use the pinion gear wrench, SPL-10319, to set the stabilizer leading edge to the full down (airplane nose up) position.

SUBTASK 27-41-82-100-001

(2) Use a clean cloth to clean the grease and dirt from the forward and aft cable drums.

SUBTASK 27-41-82-420-001

(3) Install the STA cable [6] and STB cable [5] into the cable runs (Figure 402).

(a) Put the new, or used, cables in the forward mechanism compartment.

(b) Attach the lockstitch to the aft end of STA cable [6] and STB cable [5].

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(c) Pull the lockstitch from the aft unpressurized compartment and guide the stabilizer control cables through the cable runs of the airplane.

1) Make sure that the idler pulleys are removed.

2) Remove the aft pulley as required to let the end of the cable through the pulley stations (Figure 402).

NOTE: The aft pulley must only be removed while the cable end passes through the bracket to decrease the cable rubbing.

SUBTASK 27-41-82-420-002

(4) Wind the STA cable [6] and STB cable [5] onto a spool or reel in the aft unpressurized compartment.

SUBTASK 27-41-82-420-003

(5) Install the idler pulleys in the idler pulley brackets at the locations shown in (Figure 402).

(a) Make sure that the STA cable [6] and STB cable [5] are in the groove above each idler pulley.

I. Stabilizer Control Cable Installation

SUBTASK 27-41-82-420-004

(1) Connect the STA cable [6] to the bottom slot of the forward cable drum (Figure 401):

(a) Using the captain's or first officer's stabilizer control wheel, rotate the forward drum so that the lower cable connection point is on the left side of the airplane.

1) Install the eyelet bolt through the STA cable [6] eyelet, the washers on the eyelet bolt, the castle nut on the eyelet bolt, and a new cotter pin [18] through the eyelet bolt (Figure 401).

CAUTION

MAKE SURE THAT THE CABLE WINDS ONTO THE DRUM GROOVES CORRECTLY WITH EACH TURN OF THE DRUM. IF YOU JAM THE CABLE BETWEEN THE CABLE AND THE CABLE DRUM GUARD, DAMAGE TO THE CABLE WILL OCCUR.

(b) Use the captain's or first officer's stabilizer trim control wheel to wind the STA cable [6] "completely" onto the forward cable drum (Approximately 33 1/2 turns).

NOTE: Tension must always be kept on the STA cable while winding it onto the forward cable drum. The best way to keep steady tension on the STA cable is to have a mechanic applying tension to the cable in the aft unpressurized compartment while another mechanic keeps the cable in position in the forward mechanism compartment.

NOTE: To ensure the STA cable is not damaged or kinked as it is wound onto the forward drum, it is recommended to have a mechanic in the forward mechanism compartment to watch the cable as it slowly winds onto the forward cable drum. This will ensure that the mechanic that is turning the stabilizer trim control wheel, does not turn the forward drum too much and damage or kink the cable.

SUBTASK 27-41-82-420-005

(2) Connect the STB cable [5] to the top slot of the forward cable drum (Figure 401):

(a) While keeping tension on the STA cable [6], use the captain's or first officer's stabilizer control wheel to rotate the forward drum so that the upper cable connection point is on the right side of the airplane.

1) Install the STB cable [5] eyelet in the location on the upper right side of the forward cable drum.

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a) Install the eyelet bolt through the STB cable [5] eyelet and the washer and nut on the eyelet bolt.

CAUTION

MAKE SURE THAT THE CABLE WINDS ONTO THE DRUM GROOVES CORRECTLY WITH EACH TURN OF THE DRUM. IF YOU JAM THE CABLE BETWEEN THE CABLE AND THE CABLE DRUM GUARD, DAMAGE TO THE CABLE WILL OCCUR.

(b) Use the captain's or first officer's stabilizer trim control wheel to wind the STB cable [5] onto the forward cable drum approximately 2 turns.

NOTE: Tension must always be kept on the STA cable and STB cable while winding them onto the forward cable drum. The best way to keep steady tension on the cables is to have a mechanic applying tension to the cable in the aft unpressurized compartment while another mechanic keeps the cable in position at the forward cable drum.

(c) Install non-metallic wedges between the STA cable [6], STB cable [5], and cable guards on the left and right sides of the forward cable drum.

NOTE: The wedges are used to keep tension on the STA cable and STB cable while the cables are rigged to the rear cable drum.

SUBTASK 27-41-82-420-006

(3) Install the STB cable [5] on the cable drum [15]:

(a) Install the cable terminal in the top terminal slot of the cable drum [15].

(b) Install a new pin [61] to connect the cable terminal to the cable drum [15].

1) Install the ends of the pin [61] in the terminal slot.

(c) Place approximately 1.7 wraps of the STB cable [5] on the cable drum [15].

CAUTION

MAKE SURE THAT THE STB CABLE IS IN THE GROOVE OF THE FORWARD PULLEY AND AFT PULLEY BEFORE YOU TIGHTEN THE CABLE. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(d) Using the pinion gear wrench, SPL-10319, wind the STB cable [5] onto the cable drum [15] until the ballscrew reaches the upper stops.

1) Use the pinion gear wrench, SPL-10319, to keep tension on the STB cable [5].

2) Make sure that the STB cable [5] is in the groove of the forward (STA 330) and aft (STA 1038) pulley (Figure 402).

SUBTASK 27-41-82-420-007

(4) Install the STA cable [6] on the cable drum [15]:

(a) Place approximately 1.8 wraps of cable around the aft cable drum and install the cable terminal in the bottom terminal slot of the cable drum [15].

(b) Install the new pin [61] to connect the cable terminal to the cable drum [15].

1) Install the ends of the pin [61] in the terminal slot.

(c) Remove the non-metallic wedges from the forward cable drum while keeping tension on the STA cable [6] at the aft cable drum.

(d) Attach a control cable clamp, SPL-1569, to the STA cable [6], forward of the aft cable drum.

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WARNING MAKE SURE THAT YOU DO NOT GET FINGERS OR HANDS CAUGHT BETWEEN THE CABLE AND THE DRUM OR INTO A LOOP OF CABLE. IF THE CABLE MOVES SUDDENLY, DANGEROUS INJURY TO PERSONNEL CAN OCCUR.

CAUTION MAKE SURE THAT THE STA CABLE IS IN THE GROOVE OF THE FORWARD PULLEY AND AFT PULLEY BEFORE YOU TIGHTEN THE CABLE. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(e) Apply tension to the STA cable [6] with the control cable clamp, SPL-1569, until there is sufficient cable to complete the last wrapping around the aft cable drum [15].

1) If necessary, loosen the lower turnbuckle [3], on the forward mechanism, to get sufficient play in the cable.

2) Make sure that the STA cable [6] is in the groove of the forward (STA 330) and aft (STA 1038) pulley (Figure 402).

SUBTASK 27-41-82-080-001

(5) Remove the control cable clamp, SPL-1569.

SUBTASK 27-41-82-420-008

(6) Install the spacers [14] and cable drum guard [28] to the cable drum [15]:

(a) Apply grease, D00633, to the bolts [16], bolt [20], and bolts [26].

(b) Install the bolts [16] and washers [17] to attach the cable drum guard [28] to the cable drum [15].

1) Tighten the bolts [16].

CAUTION MAKE SURE THAT YOU INSTALL THE WASHER BETWEEN THE SUPPORT PLATE AND THE CABLE DRUM. IF YOU DO NOT OBEY, THE CABLE DRUM WILL NOT TURN CORRECTLY DURING OPERATION.

(c) Install the washer [84] to the cable drum [15].

CAUTION MAKE SURE THAT YOU INSTALL THE SUPPORT PLATE CORRECTLY. THE SUM OF TORQUE TO DISENGAGE THE MANUAL CLUTCH CAN BE MORE THAN THE LIMITS. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(d) Install the bolts [26], washers [27], support plate [25], and spacers [14] to the cable drum [15].

1) Install the bolts [26] to the washers [27], support plate [25], and spacers [14].

NOTE: The bearing assembly of the support plate must be on the same side as the cable drum.

2) Install the bolts [26] to the cable drum [15].

3) Tighten the bolts [26].

(e) Install the bolt [20], washer [21], washer [22], and nut [23] to attach the cable drum guard [28] to the support plate [25].

1) Tighten the bolt [20].

(f) Install the castellated nut [82] to the cable drum [15].

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1) Install the castellated nut [82] to the thread of the cable drum [15].
2) Hand tighten the castellated nut [82] against the support plate [25].
3) Align the castellated nut [82] back to the nearest pin hole.

(g) Install the new pin [83] to the castellated nut [82].

SUBTASK 27-41-82-820-002

CAUTION

MAKE SURE THAT THE ROUTING OF STABILIZER CABLES STA AND STB ARE CORRECT ON ALL PULLEYS AND FLOOR BEAM FAIRLEADS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE CAN OCCUR.

(7) Adjust the STA cable [6] and STB cable [5], do this task: Stabilizer Control Cable and Chain - Adjustment, TASK 27-41-00-820-801.

SUBTASK 27-41-82-420-011

(8) Make sure that the flexible shaft [36] is connected to the stabilizer forward control mechanism.

(a) Make sure that MS20995NC32 lockwire, G50347, is installed on the coupling nut of the flexible shaft [36].

SUBTASK 27-41-82-020-008

(9) Reconnect the upper turnbuckle [4] after the STA and STB cables are completely rigged.

SUBTASK 27-41-82-080-002

(10) Remove the pinion gear wrench, SPL-10319, from the jackscrew gearbox [12].

SUBTASK 27-41-82-730-001

(11) Do this task: Stabilizer Clutch Friction System Test, TASK 27-41-00-700-806.

SUBTASK 27-41-82-860-005

(12) Attach DO NOT OPERATE tag, STD-858, on the stabilizer trim wheel at the control stand.

SUBTASK 27-41-82-420-009

(13) Do this task: Stabilizer Trim Motor Installation, TASK 27-41-71-400-801.

SUBTASK 27-41-82-860-004

(14) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-82-730-002

(15) Do this task: Stabilizer Manual Trim and Trim Indicator - Test, TASK 27-41-00-700-801.

SUBTASK 27-41-82-730-003

(16) Do this task: Horizontal Stabilizer Trim Control System Friction Test, TASK 27-41-00-700-802.

SUBTASK 27-41-82-610-001

(17) Do this task: Horizontal Stabilizer Actuator Brake - Servicing, TASK 12-22-41-610-801.

NOTE: This lubrication procedure must be performed after the installation of any new stabilizer trim actuator.

SUBTASK 27-41-82-730-004

(18) Do this task: Stabilizer Trim Wheel Free Movement Test, TASK 27-41-00-700-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-82-730-005

(19) Do this task: Stabilizer Electric Trim System Test, TASK 27-41-00-700-803.

SUBTASK 27-41-82-420-010

(20) Install the half ball pressure seals, do this task: Control Cable Half Ball Pressure Seal Installation, TASK 20-10-04-400-802.

SUBTASK 27-41-82-700-001

(21) Do a cycle check for the flight control system along the cable runs STA cable [6] and STB cable [5]:

(a) If necessary, use a video camera on areas that are not visible.

(b) Make sure that the cable routing is correct.

(c) Operate the system along the full range of travel.

(d) Make sure that there is free movement in all directions of travel.

(e) Make sure that the flight control surfaces align correctly.

(f) Make sure that the flight control surfaces have the correct routing and clearances.

(g) Make sure that the bellcranks and control rods move correctly.

(h) Make sure that the detents align correctly for the full range of the cable control mechanism.

(i) Make sure that the controls have the correct cushion for the full range of the cable control mechanism.

(j) Make a note of usual sounds from closed areas while the system is in operation.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-82-410-001

(1) Install the cargo liners and floor panels that were removed for access to the idler pulleys (TASK 50-10-03-400-801, TASK 53-21-00-400-801).

SUBTASK 27-41-82-410-002

(2) Close these access panels:

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

SUBTASK 27-41-82-410-003

(3) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER TRIM CUTOUT SWITCHES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Stabilizer trim cutout switch removal

(2) Stabilizer trim cutout switch installation.

B. This procedure is applicable to the primary cutout switch (S272) and back up cutout switch (S149).

TASK 27-41-91-000-801

2. Stabilizer Trim Cutout Switch Removal

(Figure 401)

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-41-91-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

C. Cutout Switch Removal

SUBTASK 27-41-91-010-001

(1) Remove the screws [6] and pull the panel out from the control stand.

SUBTASK 27-41-91-020-001

(2) Disconnect the electrical terminal [7] from the back of the cutout switch [1].

SUBTASK 27-41-91-020-002

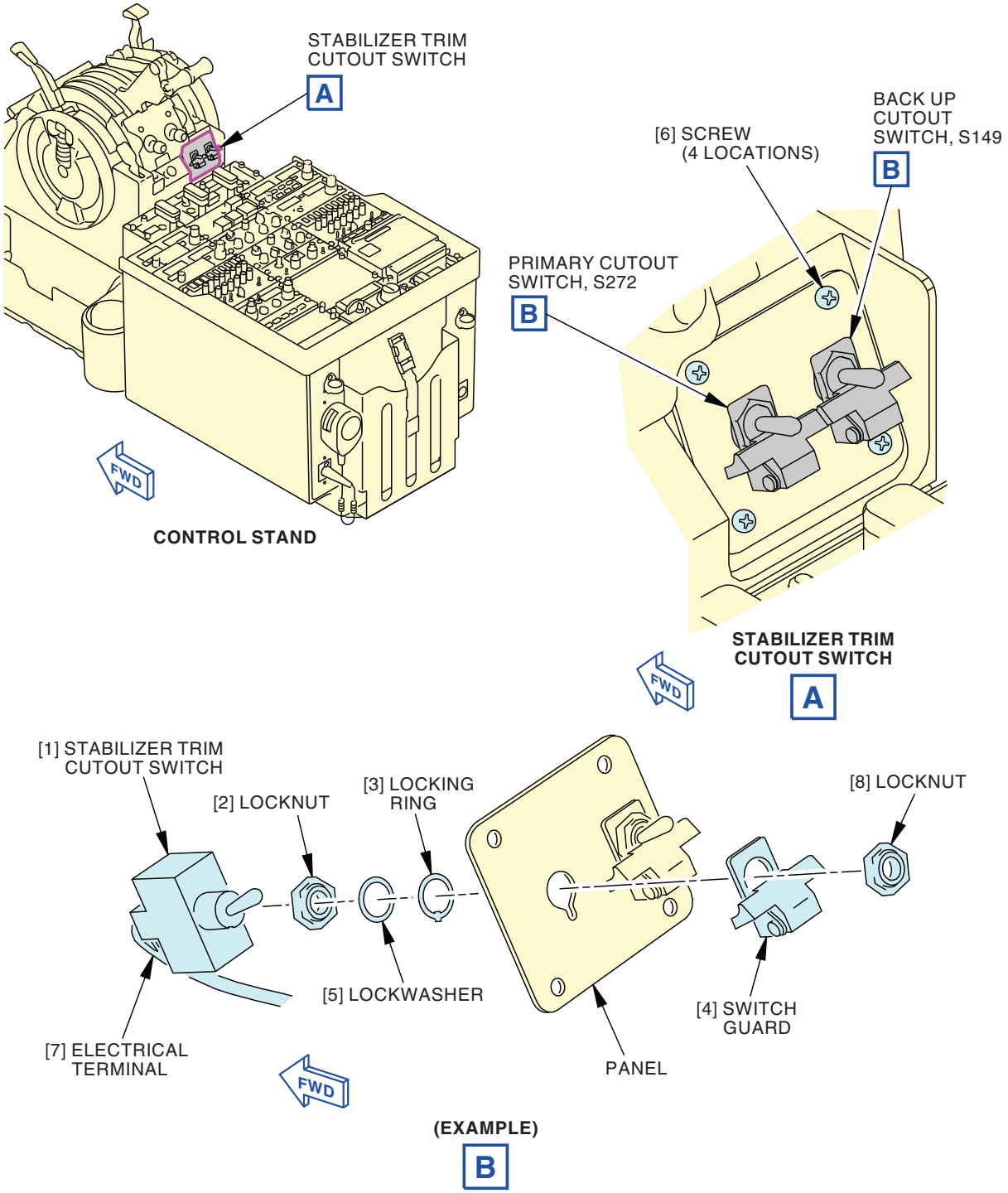
(3) Remove the cutout switch [1] from the panel:

(a) Remove the locknut [8], switch guard [4], locking ring [3], lockwasher [5], and locknut [2] from the cutout switch [1].

(b) Remove the cutout switch [1] from the back side of the panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Cutout Switch Installation
Figure 401/27-41-91-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-91-400-801

3. Stabilizer Trim Cutout Switch Installation

(Figure 401)

A. References

Reference	Title
27-41-91-700-801	Primary Stabilizer Trim Cutout Switch Test (P/B 501)
27-41-91-700-802	Back Up Stabilizer Trim Cutout Switch Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cutout switch	27-41-91-01-120	PTF ALL

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Procedure

SUBTASK 27-41-91-420-001

(1) Install the cutout switch [1] on the panel:

(a) Install the cutout switch [1], locknut [2], lockwasher [5], and locking ring [3] on the back side of the panel.

(b) Install the switch guard [4] and locknut [8] on the front side of the panel.

NOTE: The cutout switch [1], locking ring [3], and switch guard [4] have keys to ensure correct installation.

SUBTASK 27-41-91-420-002

(2) Connect the electrical terminal [7] to the back of the cutout switch [1].

SUBTASK 27-41-91-420-003

(3) Install the panel on the control stand:

(a) Put the panel in its position on the control stand.

(b) Install the screws [6].

SUBTASK 27-41-91-860-002

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-91-820-001

(5) Do this task: Primary Stabilizer Trim Cutout Switch Test, TASK 27-41-91-700-801.

SUBTASK 27-41-91-820-002

(6) Do this task: Back Up Stabilizer Trim Cutout Switch Test, TASK 27-41-91-700-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER TRIM CUTOUT SWITCHES - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Primary stabilizer trim cutout switch test

(2) Back up stabilizer trim cutout switch test.

27-CMR-09

TASK 27-41-91-700-801

2. Primary Stabilizer Trim Cutout Switch Test

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Prepare for the Test

SUBTASK 27-41-91-860-004

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-91-860-005

(2) Make sure that the STAB TRIM override switch on the P8 panel is in the NORM position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

C. Primary Stabilizer Trim Cutout Switch Test

SUBTASK 27-41-91-710-001

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do the test of the STAB TRIM PRI switch, S272, on the aft area of the control stand as follows:

(a) Make sure that the STAB TRIM PRI switch is in the NORMAL position.

(b) Push and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

(c) Make sure that the stabilizer leading edge start to move down.

(d) Set the STAB TRIM PRI switch to the CUTOUT position.

(e) Make sure that the movement of the stabilizer stops immediately.

(f) Release the stabilizer trim switches.

(g) Set the STAB TRIM PRI switch to the NORMAL position.

(h) Push and hold the STAB TRIM switches on the captain's control wheel to the NOSE DOWN position.

(i) Make sure that the stabilizer leading edge moves up.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Set the STAB TRIM PRI switch to the CUTOUT position.

(k) Make sure that the movement of the stabilizer stops immediately.

(l) Release the stabilizer trim switches.

(m) Set the STAB TRIM PRI switch back to the NORMAL position.

——— END OF TASK ———

27-CMR-09

TASK 27-41-91-700-802

3. Back Up Stabilizer Trim Cutout Switch Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-41-91-860-007

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-91-860-008

(2) Make sure that the STAB TRIM override switch, on P8 panel is in the NORM position.

SUBTASK 27-41-91-860-009

(3) Make sure that the VHF NAV and IRS switches, on the P5 Forward Overhead Panel, are in the NORMAL positions.

SUBTASK 27-41-91-860-010

(4) Do this task: Air Data Inertial Reference System - Alignment from the ISDU, TASK 34-21-00-820-802 or Air Data Inertial Reference System - Alignment from the FMC CDU, TASK 34-21-00-820-801 or Air Data Inertial Reference System - Automatic Alignment from the IRS MSU, TASK 34-21-00-820-803.

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SUBTASK 27-41-91-860-011

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Supply hydraulic power to hydraulic systems A and B (Hydraulic System A or B Pressurization, TASK 29-11-00-860-801).

SUBTASK 27-41-91-860-012

(6) Make sure that the trailing edge flaps are in the UP position.

SUBTASK 27-41-91-860-015

(7) Make sure that the PRI cutout switch is in the NORMAL position.

SUBTASK 27-41-91-860-016

(8) Make sure that the B/U cutout switch is in the NORMAL position.

SUBTASK 27-41-91-860-017

(9) Make sure that both flight director switches are OFF.

SUBTASK 27-41-91-860-018

(10) Make sure that the autopilot is not engaged.

SUBTASK 27-41-91-860-019

(11) Set the altitude on the Mode Control Panel (MCP) to a minimum of 10,000 ft (3048 m) above the current field elevation.

D. Back Up Stabilizer Trim Cutout Switch Test

SUBTASK 27-41-91-710-002

(1) Do the test of the B/U (back up) stabilizer trim cutout switch, S149, on the aft area of the control stand as follows:

(a) Push the CMD A or CMD B autopilot engage switch, on the P7 panel, to engage the autopilot.

(b) Make sure that the CMD A or CMD B autopilot engage switch light comes on.

NOTE: A flashing yellow box will momentarily appear around the CWS P and/or CWS R on the Primary Flight Display (PFD) flight mode annunciator.

(c) Move the control column forward approximately 2 in. (5 cm), then release the control column.

NOTE: If still present, the flashing yellow box around the CWS P and/or CWS R will disappear when the control column is released.

1) Do not apply forces to the control column or stabilizer trim wheel.

NOTE: This is to prevent an accidental autopilot disengagement and failure of the test.

(d) Make sure that the stabilizer trim wheel is rotating in the nose-down direction.

(e) While the stabilizer trim wheel is rotating, move the B/U cutout switch to the CUTOUT position.

(f) Make sure that the autopilot disengages and that the CMD A or CMD B autopilot engage switch light turns off.

(g) Make sure that the stabilizer trim wheel stops moving.

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(h) Push the captain's or first officer's autopilot disengage switch on the control wheel to reset the autopilot disengage warnings.

(i) Set the B/U cutout switch to the NORMAL position.

(j) Push the CMD A or CMD B autopilot engage switch, on the P7 panel, to engage the autopilot.

(k) Make sure that the CMD A or CMD B autopilot engage switch light comes on.

NOTE: A flashing yellow box will momentarily appear around the CWS P and/or CWS R on the PFD flight mode annunciator.

(l) Move the control column aft approximately 2 in. (5 cm), then release the control column.

NOTE: If still present, the flashing yellow box around the CWS P and/or CWS R will disappear when the control column is released.

1) Do not apply forces to the control column or stabilizer trim wheel.

NOTE: This is to prevent an accidental autopilot disengagement and failure of the test.

(m) Make sure that the stabilizer trim wheel is rotating in the nose up direction.

(n) While the stabilizer trim wheel is rotating, move the B/U cutout switch to the CUTOUT position.

(o) Make sure that the autopilot disengages and that the CMD A or CMD B autopilot engage switch light turns off.

(p) Make sure that the stabilizer trim wheel stops moving.

(q) Push the captain's or first officer's autopilot disengage switch on the control wheel to reset the autopilot disengage warnings.

(r) Set the B/U cutout switch to the NORMAL position.

E. Put the Airplane Back to its Usual Condition

SUBTASK 27-41-91-860-013

(1) Remove hydraulic power from Hydraulic Systems A and B (Hydraulic System A or B Power Removal, TASK 29-11-00-860-805).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER TRIM LIMIT SWITCHES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of a stabilizer trim limit switch

(2) An installation of a stabilizer trim limit switch.

B. The stabilizer trim limit switch is referred to as the "limit switch" in this procedure.

C. There are three limit switches (S144, S145, and S844) and four stabilizer takeoff warning switches (S132, S1184, S546 and S1183) on the brackets along the stabilizer jackscrew.

D. This procedure is applicable to all three limit switches.

TASK 27-41-94-000-801

2. Stabilizer Trim Limit Switch Removal

(Figure 401)

A. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

B. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

C. Prepare for the Removal

SUBTASK 27-41-94-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT

SUBTASK 27-41-94-010-001

(2) To get access to the limit switch, do this step:

Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-41-94-860-002

(3) Move the stabilizer until the cam is not in contact with the limit switch.

D. Limit Switch Removal

SUBTASK 27-41-94-020-001

(1) Remove the limit switch [1]:

(a) Disconnect the limit switch [1] wire from the wire connection.

(b) Remove the locknut [4] and washer [3] which attach the limit switch arm [5] to the limit switch [1].

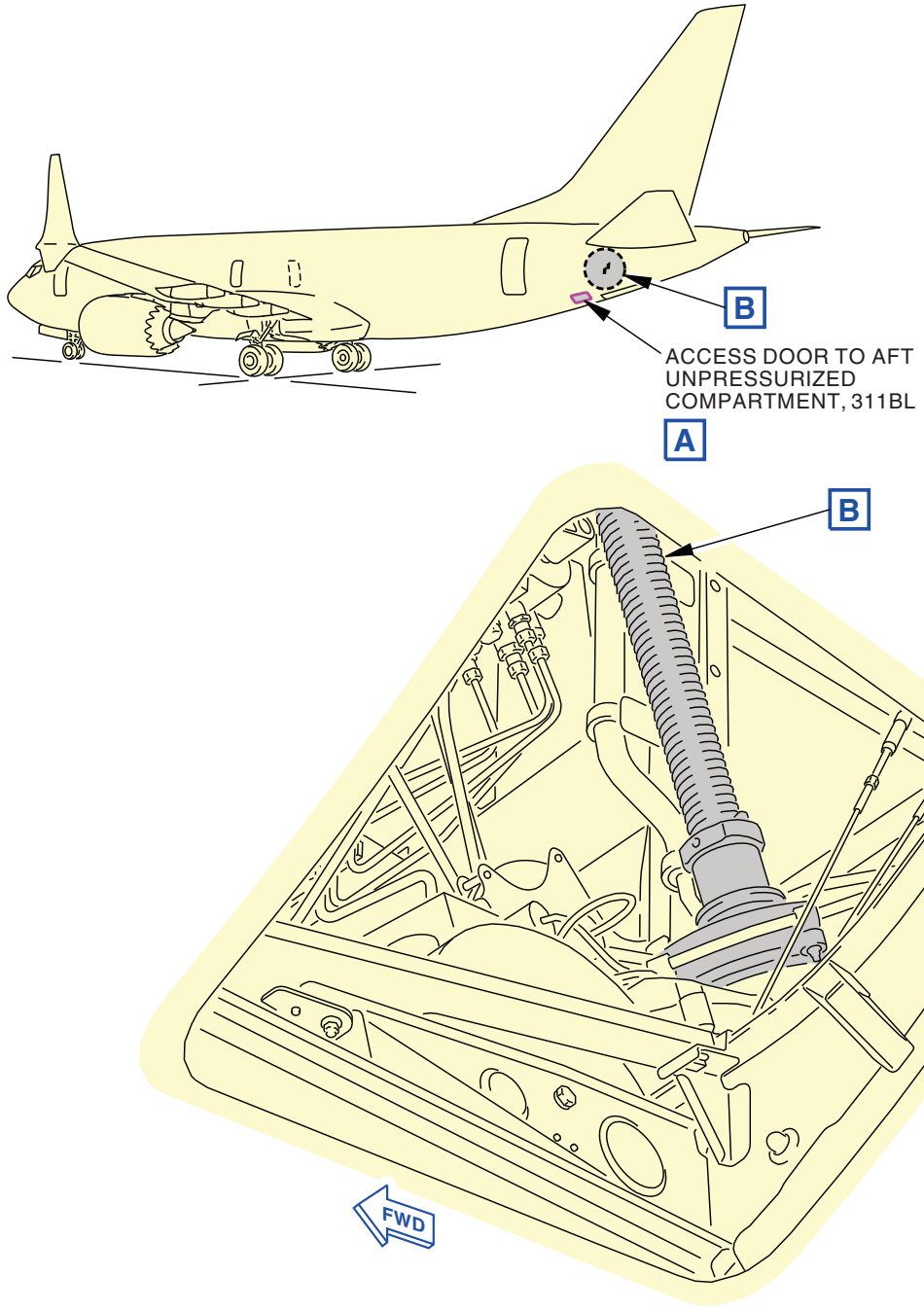
(c) Remove the lockwasher [6] and retainer nut [7] which attach the limit switch [1] to the bracket assembly [2].

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(d) Remove the limit switch [1] from the bracket assembly [2].

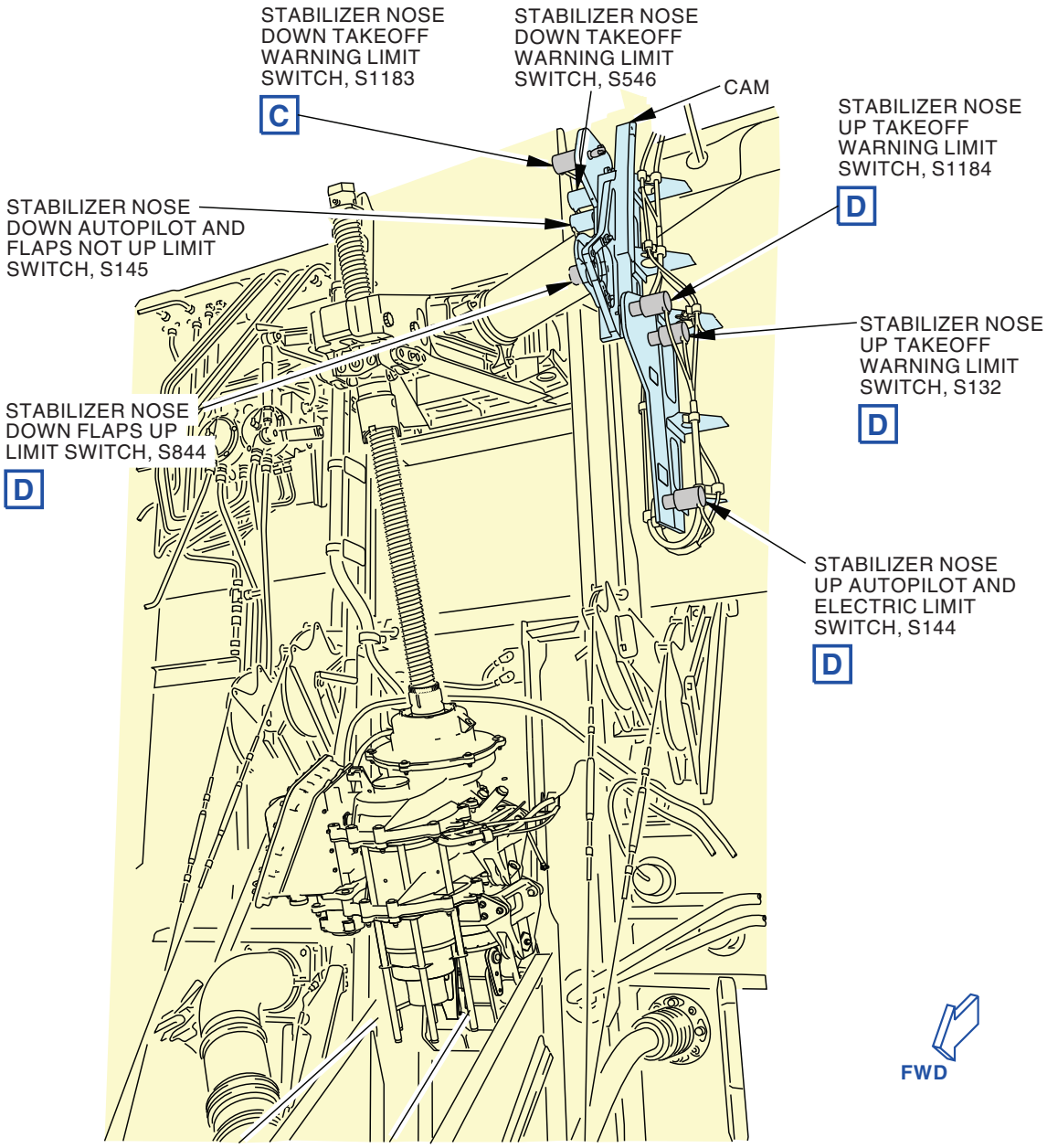
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Limit Switches Installation
Figure 401/27-41-94-990-801 (Sheet 1 of 3)

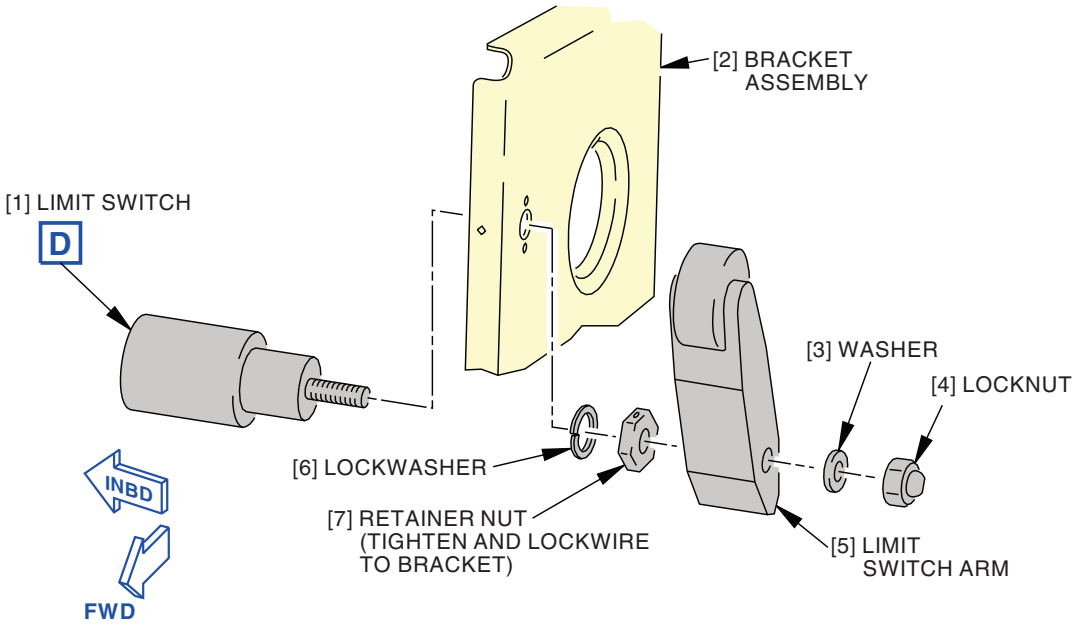
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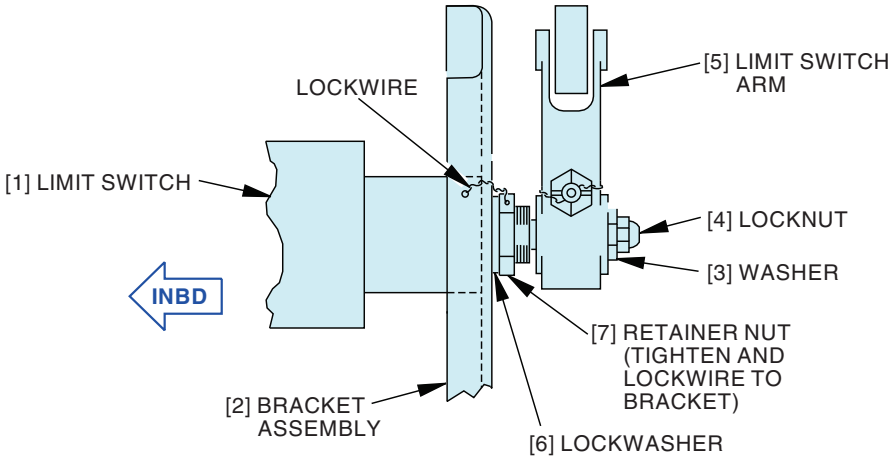
Stabilizer Trim Limit Switches Installation
Figure 401/27-41-94-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER AUTOPILOT AND ELECTRIC ACTUATOR LIMIT SWITCHES (EXAMPLE)

C



STABILIZER AUTOPILOT AND ELECTRIC ACTUATOR LIMIT SWITCHES (EXAMPLE)

D

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Stabilizer Trim Limit Switches Installation
Figure 401/27-41-94-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-94-400-801

3. Stabilizer Trim Limit Switch Installation

(Figure 401)

A. References

Reference	Title
27-41-94-700-801	Stabilizer Trim Limit Switches Test (P/B 501)
27-41-94-820-801	Stabilizer Trim Limit Switches Adjustment (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Limit switch	27-41-94-01-010	PTF ALL
		31-51-02-01-010	PTF ALL
		31-51-02-01-015	PTF ALL

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Limit Switch Installation

SUBTASK 27-41-94-420-001

(1) Install the limit switch [1] as follows:

(a) Install the limit switch [1] through the bracket assembly [2].

(b) Install the retainer nut [7] and lockwasher [6] to attach the limit switch [1] to the bracket assembly [2].

(c) Tighten the retainer nut [7] to 15 in-lb (2 N·m) - 25 in-lb (3 N·m).

(d) Install lockwire on retainer nut [7].

(e) Turn the limit switch arm [5] to the approximate position for installation but make sure that it does not touch the cam.

(f) Put the limit switch arm [5] on the limit switch [1] to engage the serrations.

(g) Install the washer [3] and locknut [4].

1) Tighten the locknut [4] by hand.

NOTE: You will tighten the locknut [4] after you adjust the limit switch [1].

(h) Connect the limit switch [1] wire at the wire connection.

SUBTASK 27-41-94-860-003

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT

SUBTASK 27-41-94-820-001

(3) Do this task: Stabilizer Trim Limit Switches Adjustment, TASK 27-41-94-820-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-94-710-001

(4) Do this task: Stabilizer Trim Limit Switches Test, TASK 27-41-94-700-801.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-94-410-001

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER TRIM LIMIT SWITCHES - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An adjustment of the stabilizer trim limit switches

(2) A test of the stabilizer trim limit switches.

B. The stabilizer trim limit switch is referred to as the "limit switch" in this procedure.

C. There are three limit switches (S144, S145, and S844) and two stabilizer takeoff warning switches (S132, S1184, S546 and S1183) on the brackets along the stabilizer jackscrew.

D. This procedure is applicable for all three limit switches.

TASK 27-41-94-820-801

2. Stabilizer Trim Limit Switches Adjustment

(Figure 501)

A. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1587	Wrench - Torque, 30 in-lbs, +/-5% accuracy Part #: 5718A42 Supplier: 3A054 Opt Part #: TE3FUA Supplier: 55719
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G02166	Lockwire - MS20995NC20, Monel - 0.020 Inch (0.508 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-41-94-010-002

(1) To get access to the limit switches, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-41-94-860-004

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(2) Use the trammel bar, SPL-1677, to set the applicable "B" dimensions as follows:

(a) With the stabilizer autopilot and electric actuator nose down (flaps not up), make sure that the switch S145 "B" dimension at -0.45 units of trim is 45.48 ±0.05 in. (1155.2 ±1.3 mm).

(b) With the stabilizer electric actuator nose down (flaps up), make sure that the switch S844 "B" dimension at 3.85 units of trim is 39.39 ±0.05 in. (1000.5 ±1.3 mm).

(c) With the stabilizer autopilot and electric actuator nose up, make sure that the switch S144 "B" dimension at 14.00 units of trim is 24.80 ±0.05 in. (629.9 ±1.3 mm).

G. Stabilizer Trim Limit Switches Adjustment

SUBTASK 27-41-94-820-002

(1) Adjust the Type 2 style stabilizer trim limit switch (Figure 501 (Sheet 4)):

(a) Remove the lockwire from the lockscrews [3].

(b) Loosen the locknut [4].

(c) Loosen the lockscrews [3].

(d) Use a 3/32 inch (2.4 millimeters) HEX torque wrench (0-30 lbs), COM-1587, to turn the worm gear with no more than 6 in-lb (0.7 N·m) until the limit switch [1] starts to operate (when you hear a click).

(e) If you cannot turn the worm gear with 6 in-lb (0.7 N·m) of torque, do these steps:

1) Move the stabilizer until the limit switch [1] does not touch the cam.

2) Adjust the limit switch arm [2].

CAUTION

DO NOT MOVE THE LIMIT SWITCH ARM MORE THAN ITS RANGE OF MOVEMENT. IT CAN CAUSE DAMAGE TO THE LIMIT SWITCH

3) Move the stabilizer by hand until the limit switch [1] clicks.

4) Make sure that the "B" dimension is the same as follows:

a) With the stabilizer autopilot and electric actuator nose down (flaps not up), make sure that the switch S145 "B" dimension at -0.45 units of trim is 45.48 ±0.05 in. (1155.2 ±1.3 mm).

b) With the stabilizer electric actuator nose down (flaps up), make sure that the switch S844 "B" dimension at 3.85 units of trim is 39.39 ±0.05 in. (1000.5 ±1.3 mm).

c) With the stabilizer autopilot and electric actuator nose up, make sure that the switch S144 "B" dimension at 14.00 units of trim is 24.80 ±0.05 in. (629.9 ±1.3 mm).

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(f) Tighten the lockscrews [3] with the fingers.

1) Apply an additional 1/6 of a turn with the fingers.

(g) Tighten the locknut [4] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

(h) Install MS20995NC20 lockwire, G02166, on the lockscrews [3] (TASK 20-10-44-400-801).

SUBTASK 27-41-94-820-003

(2) Adjust the Type 1 style stabilizer trim limit switch (Figure 501 (Sheet 5)):

(a) Remove the lockwire from the lockscrew [3].

(b) Loosen the locknut [4].

(c) Loosen the lockscrew [3].

(d) Using a slot-head screwdriver, turn the lockscrew [3] worm gear until the switch is in the desired position.

(e) If you can not turn the worm gear with 6 in-lb (0.7 N·m) of torque, do these steps:

1) Move the stabilizer until the limit switch [1] does not touch the cam.

2) Adjust the limit switch arm [2].

CAUTION

DO NOT MOVE THE LIMIT SWITCH ARM MORE THAN ITS RANGE OF MOVEMENT. IT CAN CAUSE DAMAGE TO THE LIMIT SWITCH

3) Move the stabilizer by hand until the limit switch [1] clicks.

4) Make sure that the "B" dimension is the same as follows:

a) With the stabilizer autopilot and electric actuator nose down (flaps not up), make sure that he switch S145 "B" dimension at -0.45 units of trim is 45.48 ±0.05 in. (1155.2 ±1.3 mm).

b) With the stabilizer electric actuator nose down (flaps up), make sure that the switch S844 "B" dimension at 3.85 units of trim is 39.39 ±0.05 in. (1000.5 ±1.3 mm).

c) With the stabilizer autopilot and electric actuator nose up, make sure that the switch S144 "B" dimension at 14.00 units of trim is 24.80 ±0.05 in. (629.9 ±1.3 mm).

(f) Tighten the lockscrew [3] with the fingers.

1) Apply an additional 1/6 of a turn with the fingers.

(g) Tighten the locknut [4] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

(h) Install MS20995NC20 lockwire, G02166, on the lockscrew [3] (TASK 20-10-44-400-801).

SUBTASK 27-41-94-820-004

(3) Make sure that the limit switch arm [2] overtravel is sufficient:

(a) With the stabilizer leading edge in the full down position (airplane nose up), make sure that the switch S144 can be rotated away from the cam surface by a minimum of 0.05 in. (1.27 mm) with finger pressure only.

(b) With the stabilizer leading edge in the full up position (airplane nose down), make sure that the switches S145 and S844 can be rotated away from the cam surface by a minimum of 0.05 in. (1.27 mm) with finger pressure only.

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SUBTASK 27-41-94-710-002

(4) Do this task: Stabilizer Trim Limit Switches Test, TASK 27-41-94-700-801.

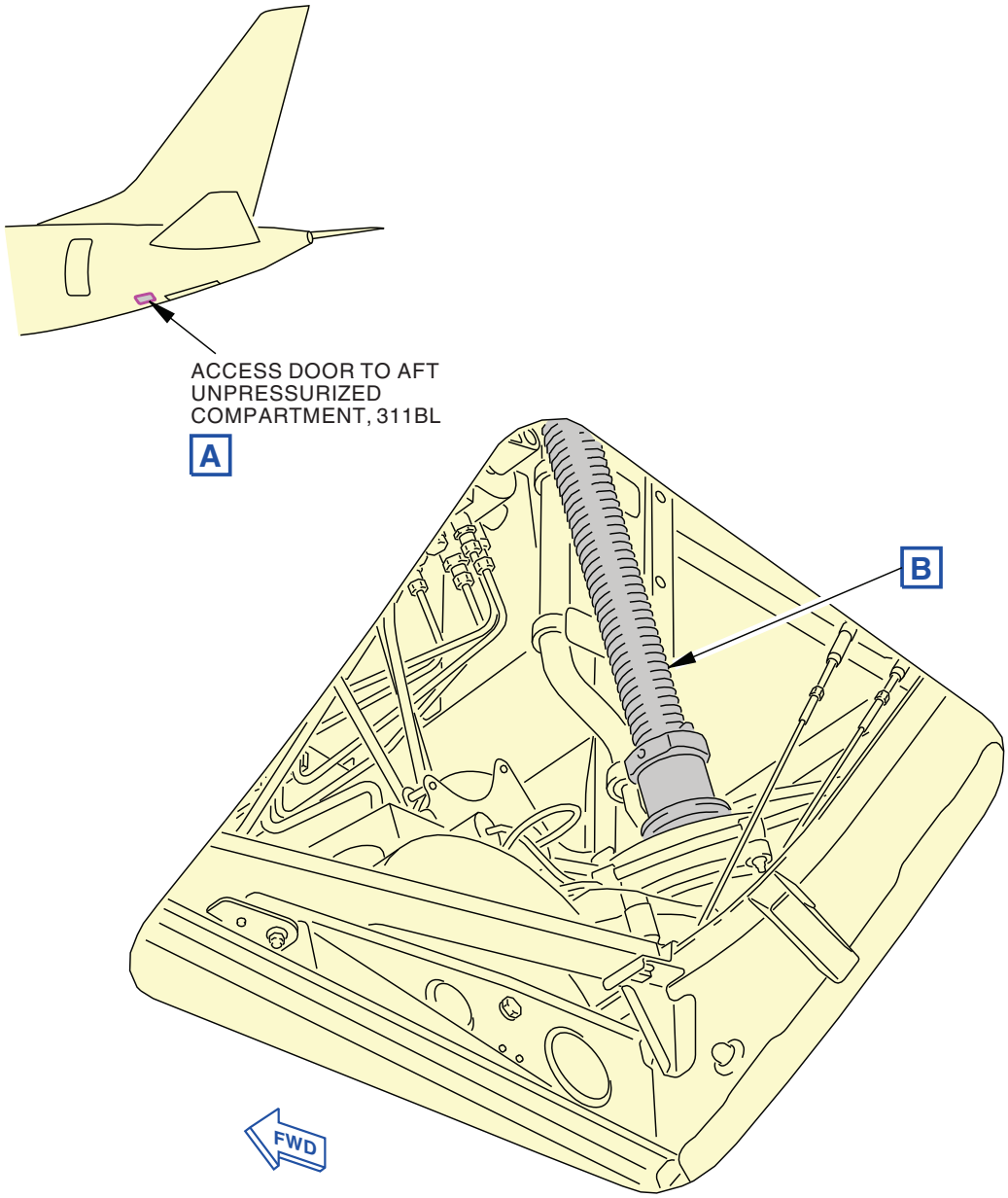
SUBTASK 27-41-94-410-002

(5) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

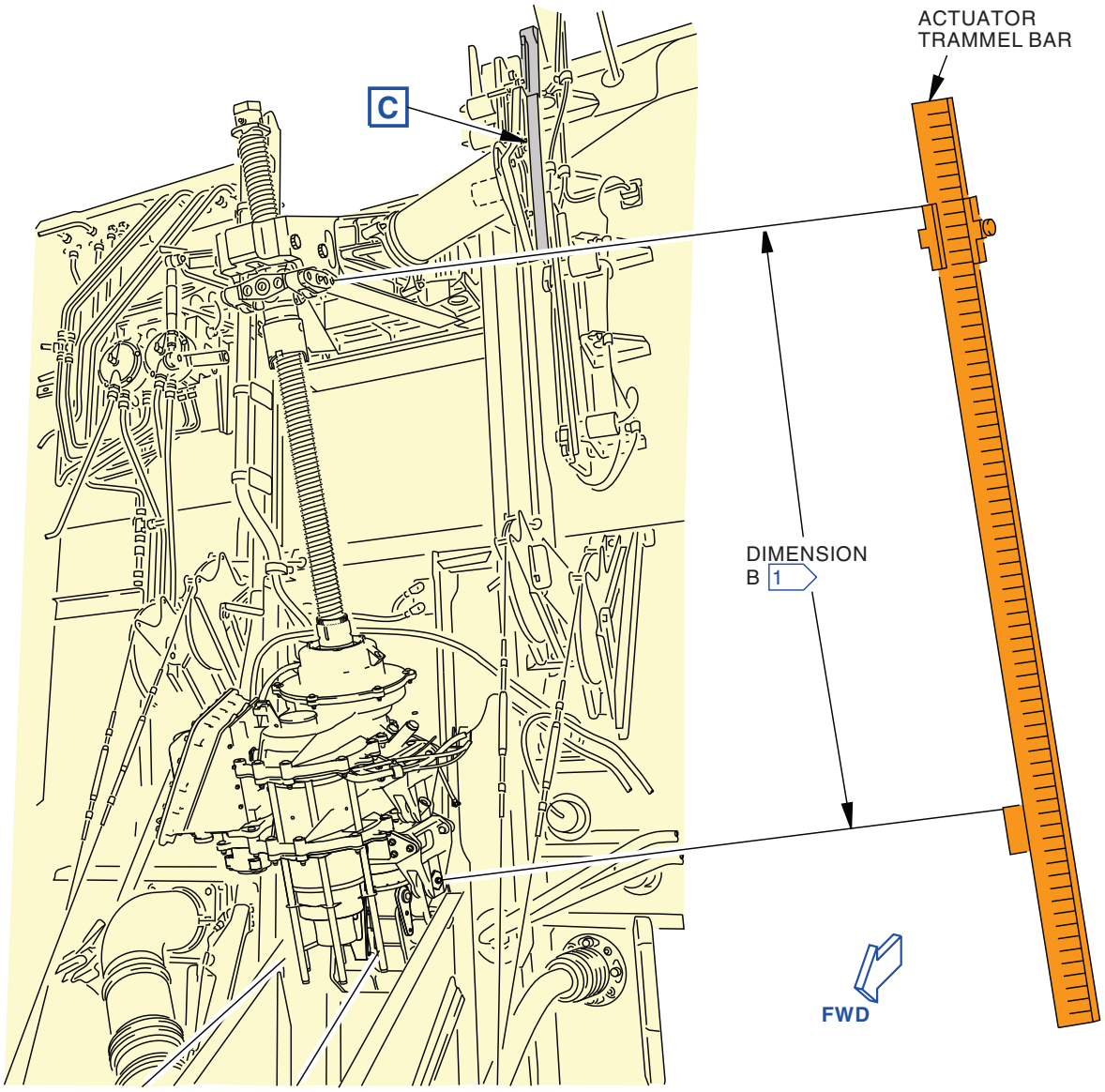


VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL
A

2412713 S00061531197_V1

Stabilizer Trim Limit Switches Adjustment
Figure 501/27-41-94-990-802 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

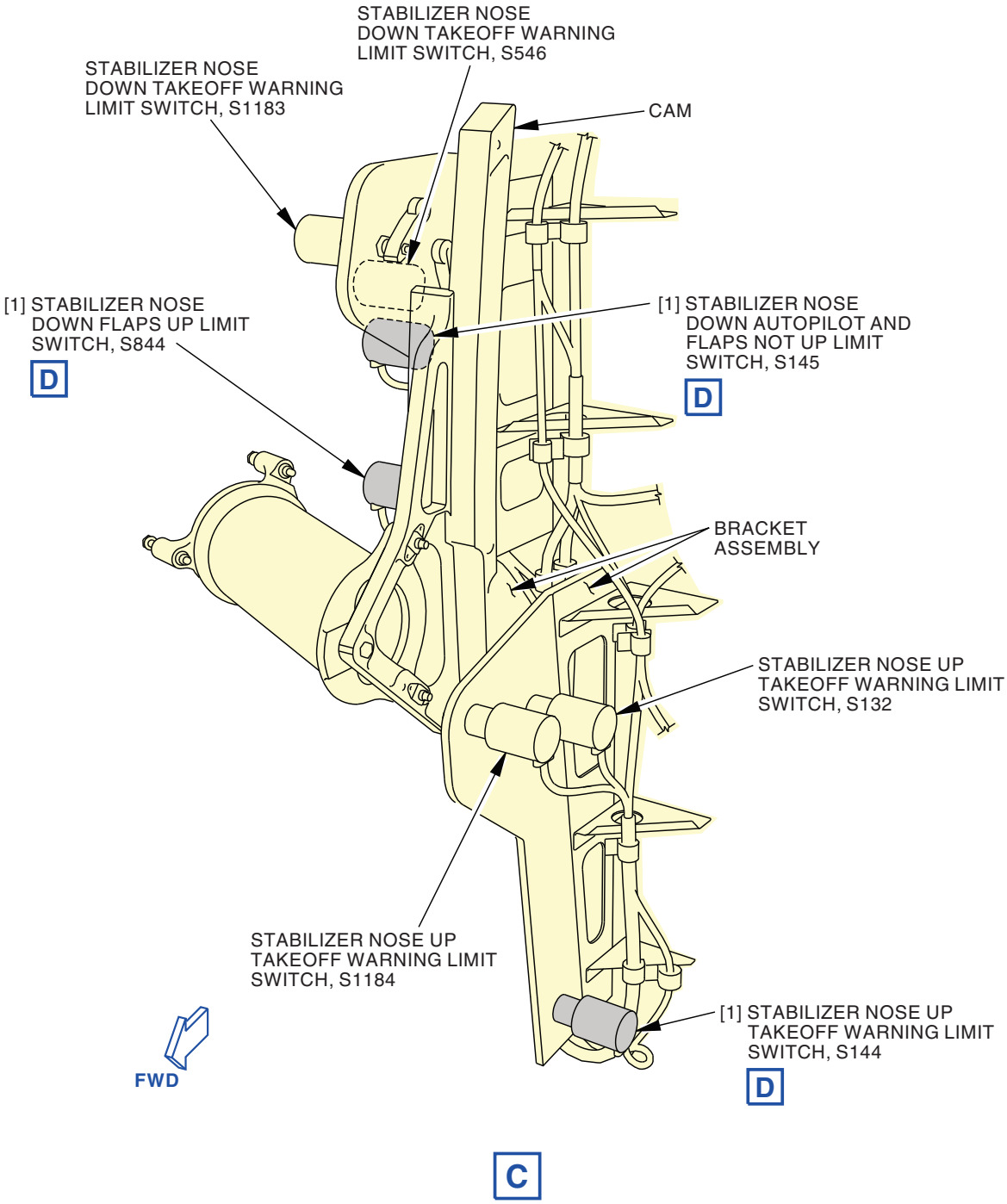
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412714 S00061531198_V2

Stabilizer Trim Limit Switches Adjustment
Figure 501/27-41-94-990-802 (Sheet 2 of 5)

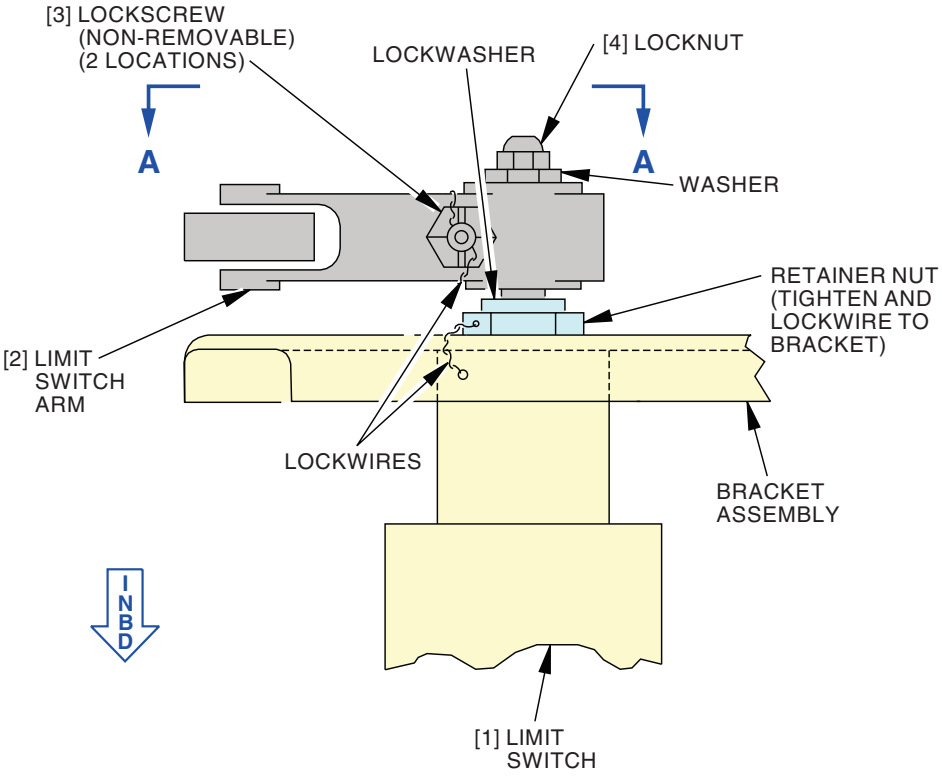
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412715 S00061531199_V2

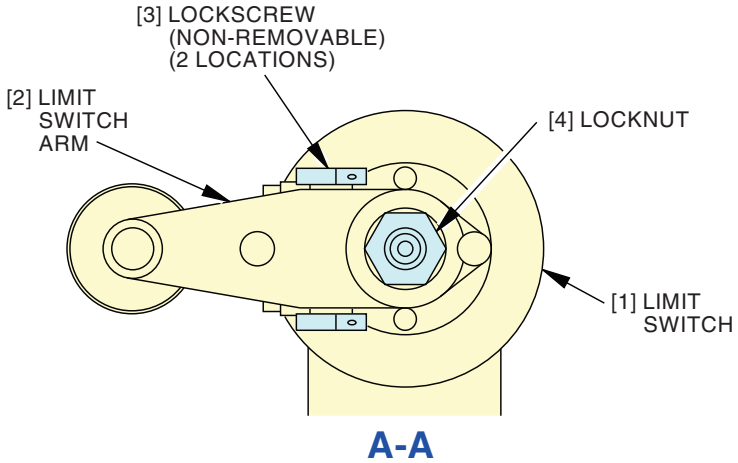
Stabilizer Trim Limit Switches Adjustment
Figure 501/27-41-94-990-802 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER AUTOPILOT AND ELECTRIC ACTUATOR LIMIT SWITCH (TYPE 2)

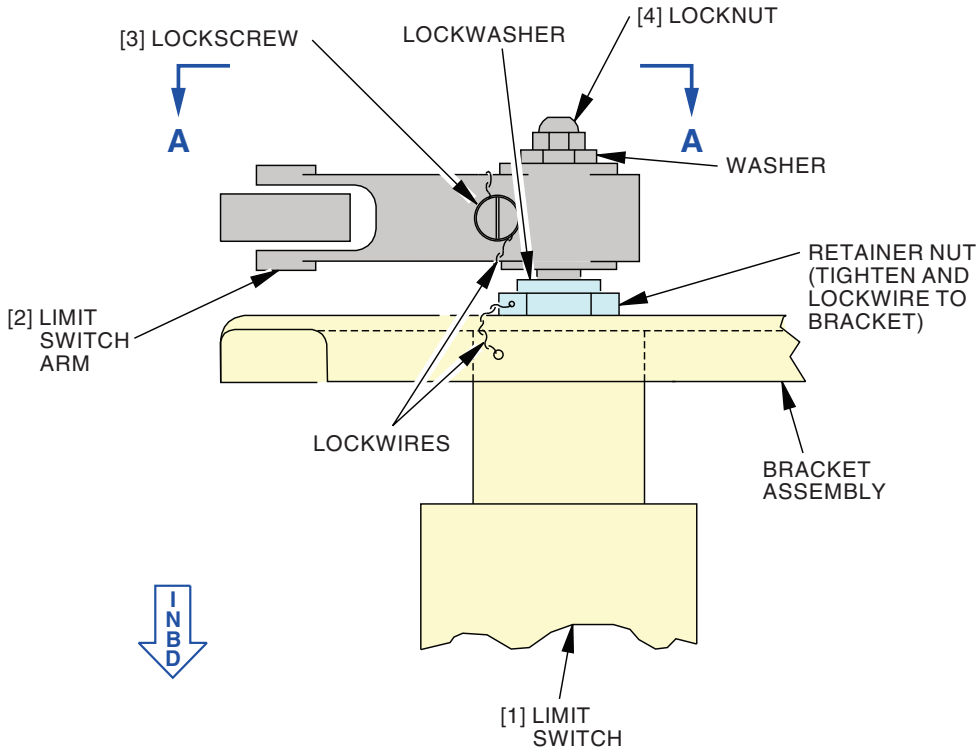
D



2412716 S00061531200_V2

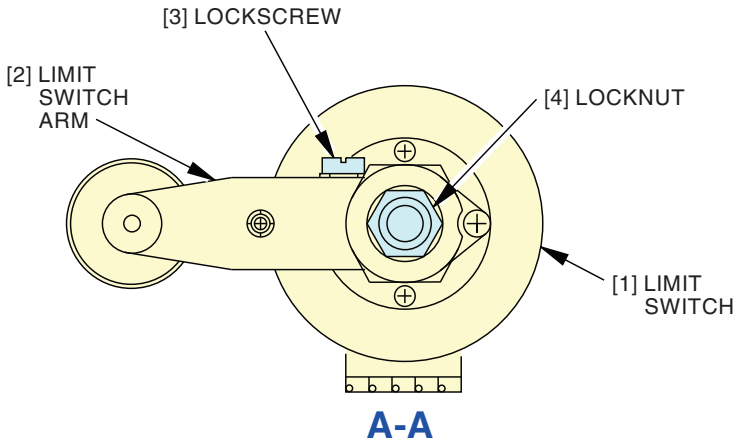
Stabilizer Trim Limit Switches Adjustment
Figure 501/27-41-94-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER AUTOPILOT AND ELECTRIC ACTUATOR LIMIT SWITCH (TYPE 1)

D



2412717 S00061531201_V2

Stabilizer Trim Limit Switches Adjustment
Figure 501/27-41-94-990-802 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-94-700-801

3. Stabilizer Trim Limit Switches Test

A. General

(1) The trim system can move the stabilizer to a position more than the limits that were set during the adjustment of the limit switches because of the coasting. This occurs when power has been removed from the stab trim actuator but the ballscrew continues to move as it comes to a stop. This condition is normal. This makes the "B" dimension tolerances for the test are larger than the tolerance for the adjustment.

(2) This test is applicable only for limit switches S145, S844, and S144.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-41-94-860-006

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-94-860-007

(2) Make sure that the STAB TRIM PRI switch, S272, on the aft area of the control stand is in the NORMAL position.

SUBTASK 27-41-94-860-008

(3) Make sure that the STAB TRIM override switch on the P8 panel is in the NORM position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-94-860-009

(4) Make sure that the flaps are in the full up position.

SUBTASK 27-41-94-010-003

(5) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Stabilizer Trim Limit Switches Test

SUBTASK 27-41-94-710-003

(1) Do the test of the limit switch S844:

(a) Set the STAB TRIM PRI switch, S272, in the CUTOUT position.

(b) Use the stabilizer trim wheel to move the stabilizer to the maximum leading edge up (APL NOSE DOWN) position.

(c) Set the STAB TRIM PRI switch in the NORMAL position.

(d) Turn the captain's stabilizer trim wheel clockwise approximately 3 turns.

(e) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

(f) Make sure that the stabilizer leading edge does not move.

(g) Release the stabilizer trim switches.

(h) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

(i) Make sure that the stabilizer leading edge moves down (APL NOSE UP).

(j) Release the stabilizer trim switches.

(k) Move the stabilizer to approximately 3.85 units of trim

(l) Move the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until the stabilizer stops automatically.

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(m) Use the trammel bar, SPL-1677, to measure the "B" dimension.

(n) Make sure that the "B" dimension is 39.39 ±0.10 in. (1000.5 ±2.5 mm).

SUBTASK 27-41-94-710-004

(2) Do the test of the limit switch, S145, as follows:

(a) If you will use the standby hydraulic pump motor to move the flaps, then do these steps:

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT OPERATE THE TRAILING EDGE FLAPS WITH ELECTRICAL POWER MORE THAN ONE FULL EXTEND/RETRACT CYCLE IN 25 MINUTES. YOU CAN CAUSE DAMAGE TO THE FLAP ELECTRIC MOTORS IF YOU OPERATE THEM TOO FREQUENTLY.

1) Put the ALTERNATE FLAP arm switch in the ARM position.

   NOTE: The standby hydraulic pump motor will operate when the ALTERNATE FLAP arm switch is in the ARM position.

2) Move the alternate flap drive control switch to extend the flaps to the 5-unit position.

(b) If you use the system B hydraulic power to move the flap, then do these steps:

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

1) Pressurize hydraulic system B (TASK 29-11-00-860-801).

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

2) Move the flap control lever to extend the flaps to the 5-unit position.

(c) Set the STAB TRIM PRI switch in the CUTOUT position.

(d) Use the stabilizer trim wheel to move the stabilizer to the maximum leading edge up (APL NOSE DOWN) position.

(e) Set the STAB TRIM PRI switch in the NORMAL position.

(f) Turn the captain's stabilizer trim wheel clockwise approximately 2 +/- 0.25 turns.

(g) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

(h) Make sure that the stabilizer leading edge does not move.

(i) Release the stabilizer trim switches.

(j) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

(k) Make sure that stabilizer leading edge moves down (APL NOSE UP).

(l) Release the stabilizer trim switches.

(m) Move the stabilizer to approximately 1 units of trim.

(n) Move the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position until the stabilizer stops automatically.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(o) Use the trammel bar, SPL-1677, to measure the "B" dimension.

(p) Make sure that the "B" dimension is 45.48 ±0.10 in. (1155.2 ±2.5 mm).

(q) Move the flap control lever or the alternate flap drive control switch to retract the flaps.

(r) Remove the pressure from hydraulic system B (TASK 29-11-00-860-805).

SUBTASK 27-41-94-710-005

(3) Do the test of the limit switch S144 as follows:

(a) Move the stabilizer to the maximum leading edge down (APL NOSE UP) position.

(b) Turn the captain's stabilizer trim wheel counterclockwise approximately 3 turns.

(c) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE UP position.

(d) Make sure that the stabilizer leading edge does not move.

(e) Release the stabilizer trim switches.

(f) Move and hold the stabilizer trim switches on the captain's control wheel to the NOSE DOWN position.

(g) Make sure that stabilizer leading edge moves up (APL NOSE DOWN).

(h) Release the stabilizer trim switches.

(i) Move the stabilizer to approximately 12 units of trim.

(j) Move the stabilizer trim switches on the captain's control wheel to the NOSE UP position until the stabilizer stops automatically.

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(k) Use the trammel bar, SPL-1677, to measure the "B" dimension.

(l) Make sure that the "B" dimension is 24.80 ±0.10 in. (629.9 ±2.5 mm).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-41-94-410-003

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER TRIM CONTROL SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Stabilizer trim control switch removal

(2) Stabilizer trim control switch installation.

B. The stabilizer trim control switch is referred to as the "switch assembly" in this procedure.

C. The switch assembly has an electrical cable which connects to the terminal block in the control wheel. If a switch becomes defective, the switch and electrical cable are replaced as an assembly.

D. This procedure is applicable for the captain's and first officer's switch assembly.

TASK 27-41-97-000-801

2. Stabilizer Trim Control Switch Removal

(Figure 401)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-41-97-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT

C. Switch Assembly Removal

SUBTASK 27-41-97-020-001

(1) Remove the medallion [2] from the control wheel.

(a) Remove the screws [3].

(b) Remove the pad [4].

SUBTASK 27-41-97-020-002

(2) Remove the screws [6] from the terminal block cover [5].

(a) Remove the terminal block cover [5].

SUBTASK 27-41-97-020-003

(3) Remove the stabilizer trim control switch [1]:

(a) Remove the screws [11], washers [10], and washers [12] to disconnect the switch wires from the terminal block [9].

(b) Remove the terminals from the switch wires to help the removal.

(c) Attach a threading wire to the switch electrical cable.

NOTE: This threading wire that is attached to the electrical cable is to help with the installation of the switch assembly.

(d) Remove the screws [7] which attach the stabilizer trim control switch [1] to the control wheel handle.

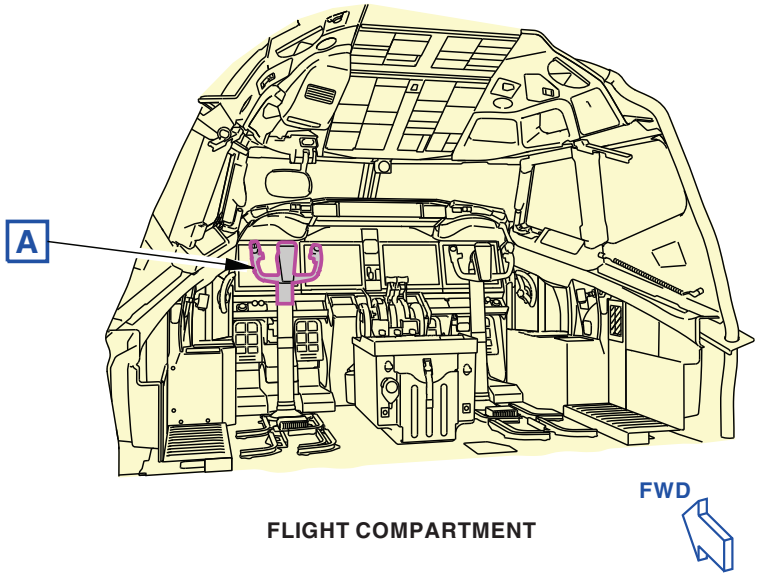
ECCN 9E991 BOEING PROPRIETARY - See title page for details

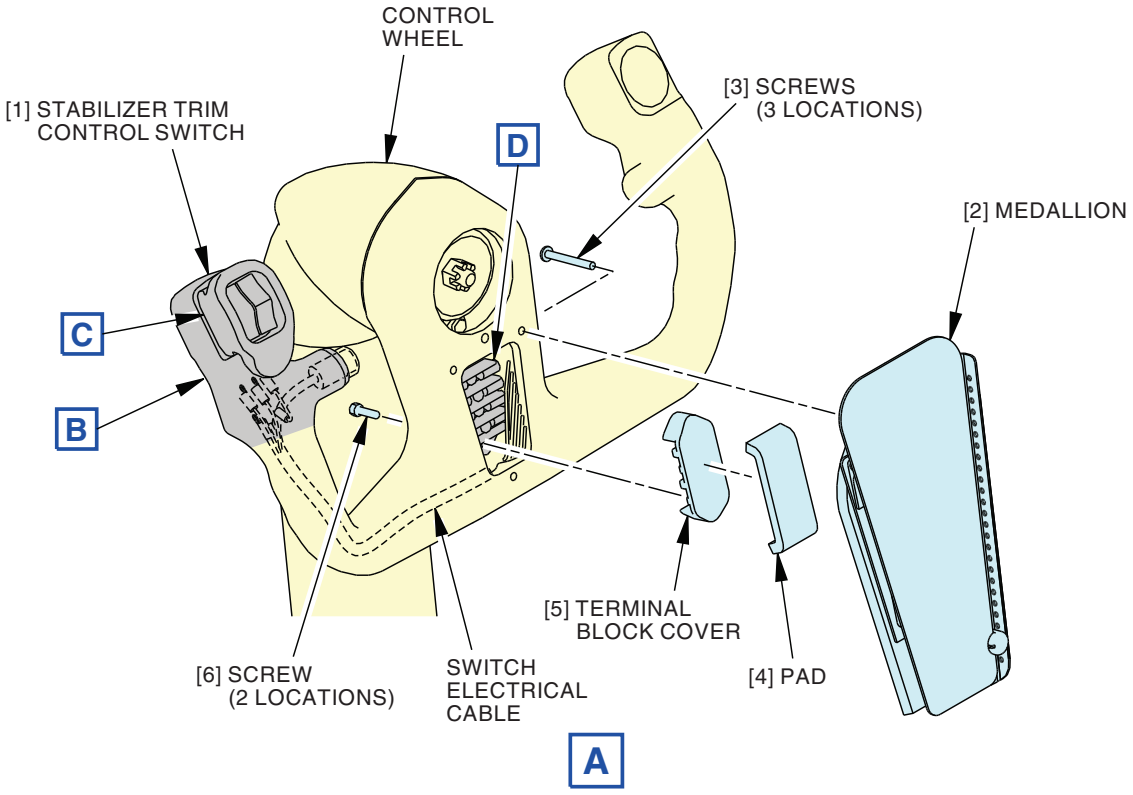
(e) Slowly pull the stabilizer trim control switch [1] from the horn of the control wheel until the switch wires show.

NOTE: Pull until approximately 3 in. (76 mm) of the threaded wire comes out of the control wheel handle. Do not remove the threaded wire from the control wheel. It is necessary to use this threaded wire for installation.

——— END OF TASK ———

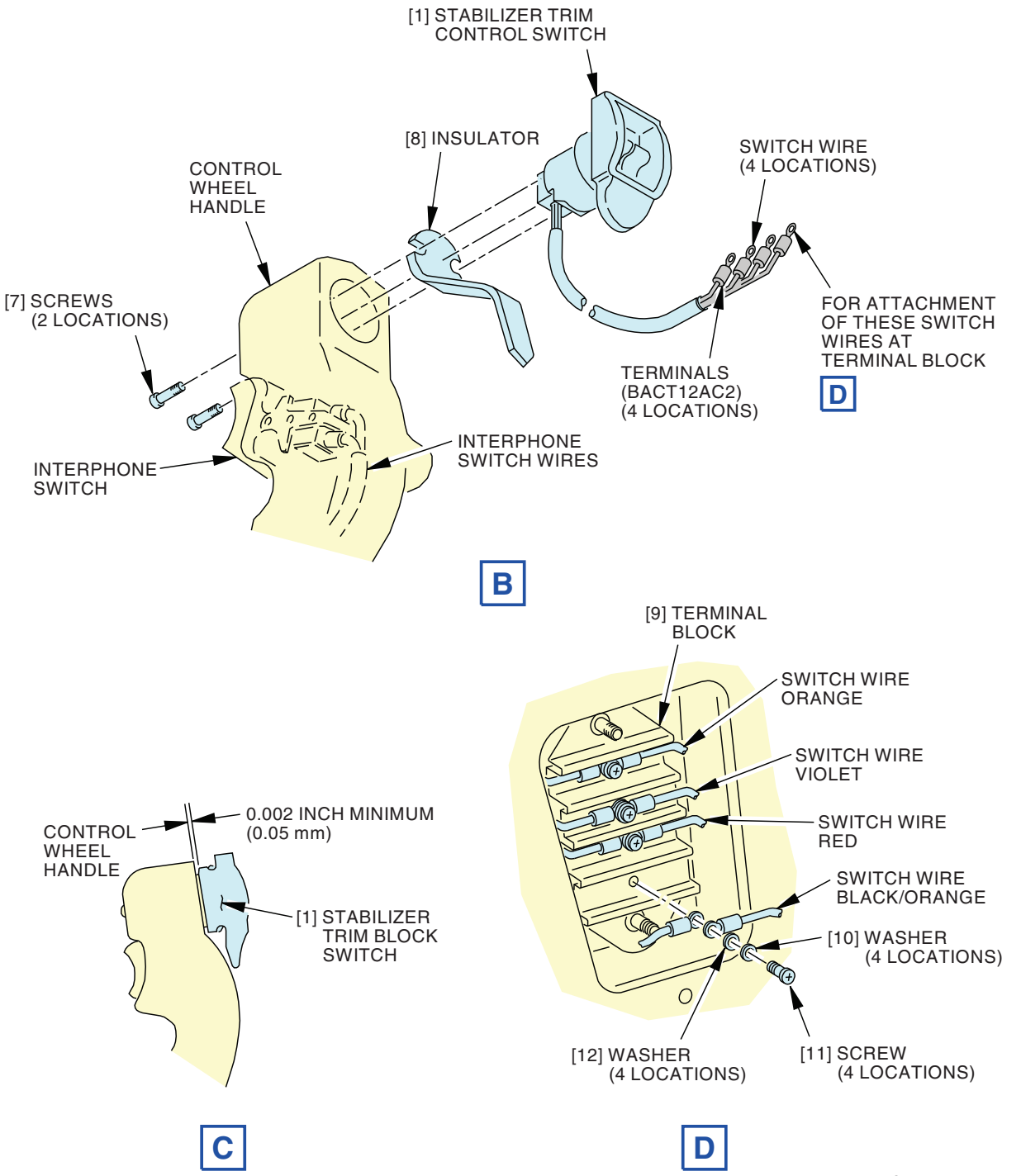
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Stabilizer Trim Control Switch Installation
Figure 401/27-41-97-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Control Switch Installation
Figure 401/27-41-97-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-41-97-400-801

3. Stabilizer Trim Control Switch Installation

(Figure 401)

A. References

Reference	Title
27-41-97-700-801	Stabilizer Trim Control Switch Test (P/B 501)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Switch Assembly Installation

SUBTASK 27-41-97-420-001

(1) Install the stabilizer trim control switch [1]:

NOTE: A threaded wire is attached to the switch electrical cable.

(a) Slowly pull the switch electrical cable through the control wheel handle with the other end of the threaded wire.

(b) Install the insulator [8] between the stabilizer trim control switch [1] and the interphone switch wires.

(c) Put the stabilizer trim control switch [1] in its position.

(d) Install the screws [7] to hold the stabilizer trim control switch [1] to the control wheel.

1) Tighten the screw [7] to 4.0 ±0.5 in-lb (0.45 ±0.06 N·m).

2) Make sure there is 0.002 in. (0.05 mm) clearance between the control wheel handle and stabilizer trim control switch [1].

(e) Remove the threaded wire from the switch electrical cable.

(f) Install the terminals on the switch wires.

(g) Install the screws [11], washers [10], and washers [12] to connect the terminals of the switch wire to the terminal block [9].

1) Make sure that the terminals are installed correctly according to the color code identification (Figure 401).

SUBTASK 27-41-97-420-002

(2) Install the screws [6] to attach the terminal block cover [5] to the control wheel.

SUBTASK 27-41-97-420-003

(3) Install the medallion [2] on the control wheel.

(a) Install the pad [4].

(b) Install the screws [3].

SUBTASK 27-41-97-860-002

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-41-97-710-001

(5) Do this task: Stabilizer Trim Control Switch Test, TASK 27-41-97-700-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STABILIZER TRIM CONTROL SWITCH - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Stabilizer Trim Control Switch Test - Use this test to make sure that the stabilizer trim control switches operate correctly.

B. The stabilizer trim control switches are installed on the control wheels. Each control wheel has two stabilizer trim control switches.

C. This procedure is applicable for the stabilizer trim control switches on the captain's and first officer's control wheels.

TASK 27-41-97-700-801

2. Stabilizer Trim Control Switch Test

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Test

SUBTASK 27-41-97-860-004

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-41-97-860-005

(2) Make sure that the STAB TRIM PRI switch, S272 on the control stand is in the NORMAL position.

SUBTASK 27-41-97-860-006

(3) Make sure that the STAB TRIM override switch on P8 panel is in the NORM position.

NOTE: The override switch is on the same control panel for the FLT DK DOOR.

C. Stabilizer Trim Control Switch Test

SUBTASK 27-41-97-710-002

(1) Do a test of the stabilizer trim control switches on the applicable control wheel:

(a) Push and hold the two stabilizer trim control switches on the applicable control wheel to the NOSE DOWN position.

(b) Make sure that the stabilizer leading edge moves up.

(c) Make sure that the trim wheels turn in the nose down position.

(d) Make sure that the STAB TRIM indicator for the applicable control wheel moves in the NOSE DOWN position.

(e) Release the two stabilizer trim control switches.

(f) Push and hold the two stabilizer trim control switches on the applicable control wheel to the NOSE UP position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Make sure that the stabilizer leading edge moves down.

(h) Make sure that the trim wheels turn in the nose up position.

(i) Make sure that the STAB TRIM indicator for the applicable control wheel moves in the NOSE UP position.

(j) Release the two stabilizer trim control switches.

(k) Using either the captain's or first officer's trim control switch, move the stabilizer to 4 units of trim as indicated by the stabilizer position indicator.

(l) Push and hold the left stabilizer trim control switch on the applicable control wheel to the NOSE UP position.

(m) Make sure that the stabilizer leading edge does not move.

(n) Release the left stabilizer trim control switch.

(o) Push and hold the left stabilizer trim control switch on the applicable control wheel to the NOSE DOWN position.

(p) Make sure that the stabilizer leading edge does not move.

(q) Release the left stabilizer trim control switch.

(r) Push and hold the right stabilizer trim control switch on the applicable control wheel to the NOSE UP position.

(s) Make sure that the stabilizer leading edge does not move.

(t) Release the right stabilizer trim control switch.

(u) Push and hold the right stabilizer trim control switch on the applicable control wheel to the NOSE DOWN position.

(v) Make sure that the stabilizer leading edge does not move.

(w) Release the right stabilizer trim control switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRAILING EDGE FLAP SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks for the trailing edge flap system:

(1) A task to deactivate the trailing edge flap system

(2) A task to reactivate the trailing edge flap system

(3) A task to operate the trailing edge flaps with primary control

(4) A task to operate the trailing edge flaps with alternate control

(5) A task to manually operate the trailing edge flaps without airplane power

(6) A task to extend the trailing edge flaps

(7) A task to retract the trailing edge flaps

(8) A task to operate the trailing edge flaps with the leading edge flaps and slats retracted and disabled

(9) A task to install the flap control handle lockout assembly

(10) A task to remove the flap control handle lockout assembly.

TASK 27-51-00-040-801

2. Trailing Edge Flap System Deactivation

(Figure 201, Figure 202)

A. General

(1) Use this task to deactivate the trailing edge flap system.

NOTE: This task will also deactivate the leading edge system.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1676	Lock Assembly - Flight Controls (used on valve p/n 10-60710-2 and 10-60710-6) Part #: F80049-66 Supplier: 81205 Opt Part #: F80049-17 Supplier: 81205 Opt Part #: F80049-31 Supplier: 81205 Opt Part #: F80049-46 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Flap Deactivation

SUBTASK 27-51-00-210-001

(1) Make sure that the ALTERNATE FLAPS ARM switch on the P5 overhead panel is in the OFF position (Figure 201).

SUBTASK 27-51-00-860-001

(2) Open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

SUBTASK 27-51-00-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-00-040-001

(4) Deactivate the trailing edge flap bypass valve (Figure 202):

CAUTION MAKE SURE THAT YOU PREVENT BENDS OR ABRASION OF THE WIRES WHEN THE WIRING HARNESS FOR THE BYPASS VALVE IS DISCONNECTED. IF YOU DO NOT OBEY, DAMAGE TO THE WIRING HARNESS COULD OCCUR.

(a) Disconnect the electrical connector from the bypass valve.

NOTE: If desired, inspect the connector for corrosion now. If there is corrosion, refer to the SWPM 20-60-03 to correct the problem. This step is also done in the installation task.

(b) Move the manual override lever on the bypass valve to position 2.

(c) Install the lock, SPL-1676, on the bypass valve.

1) Make sure that the detent pin is installed.

SUBTASK 27-51-00-210-002

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Trailing Edge Flap System - Tryout

NOTE: This tryout is to make sure that the trailing edge flap system is in zero energy state.

(a) Make sure that the trailing edge flaps do not move using primary flight controls:

1) Move the flap control lever to a minimum of one lever detent position away from the current position.

2) Make sure that the trailing edge flaps do not move.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Return the flap control lever to the detent position corresponding to the current flap position.

(b) Make sure that the trailing edge flaps do not move using the alternate flight controls:

1) Set the alternate flaps arm switch to the ARM position.

a) If flaps are in retracted position, set the alternate flaps control switch to DOWN position.

b) If flaps are in the extended position, set the alternate flaps control switch to UP position.

2) Make sure that the trailing edge flaps do not move.

3) Return the alternate flaps control switch to OFF, and the alternate flaps arm switch to OFF.

SUBTASK 27-51-00-420-006

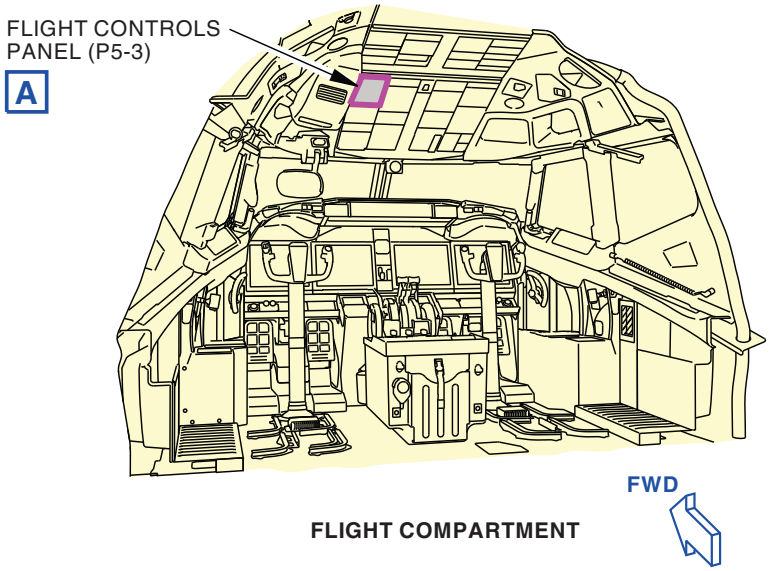
(6) If the flaps are in the retracted position, do this task to install the flap control handle lockout assembly: Flap Control Handle Lockout Assembly Installation, TASK 27-51-00-400-801.

SUBTASK 27-51-00-480-001

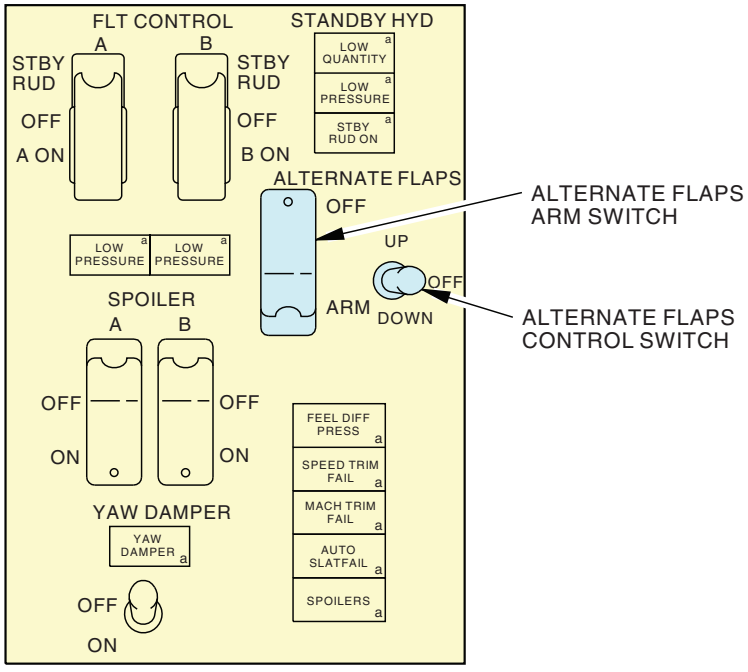
(7) Put DO NOT OPERATE tags, STD-858, on the ALTERNATE FLAPS switches.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



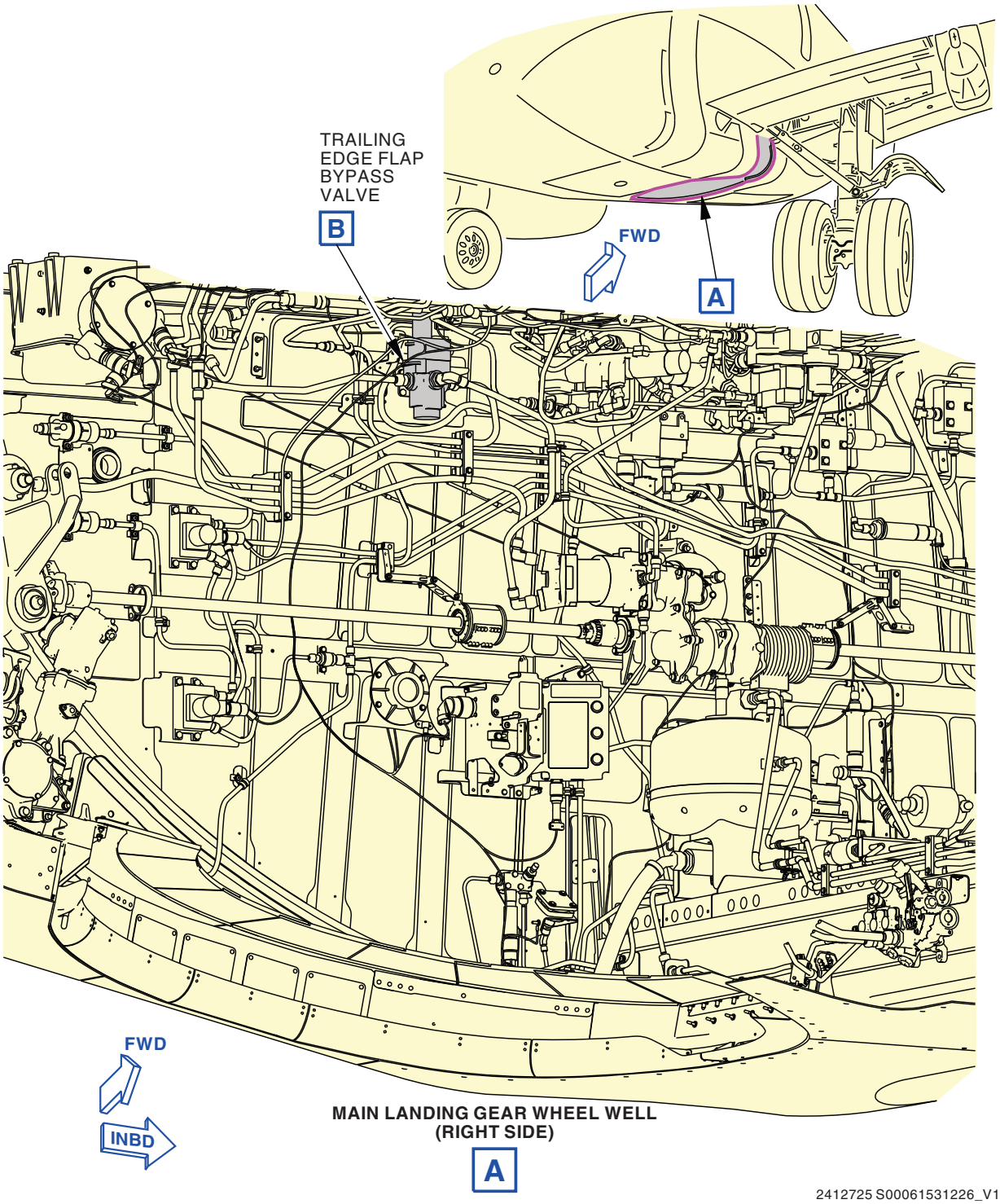
FLIGHT CONTROLS PANEL (P5-3)

A

2412724 S00061531225_V1

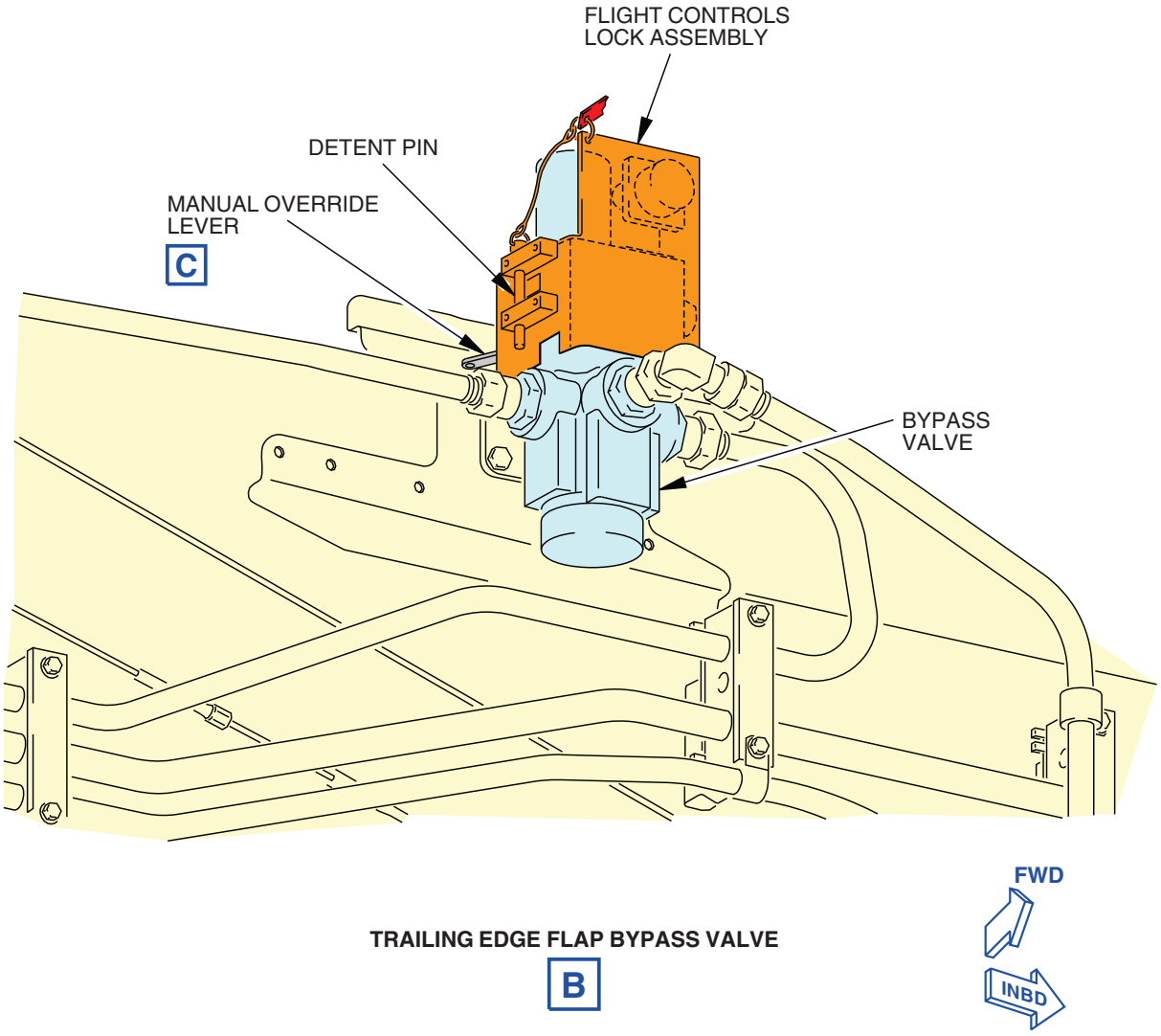
Trailing Edge Flap Controls
Figure 201/27-51-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



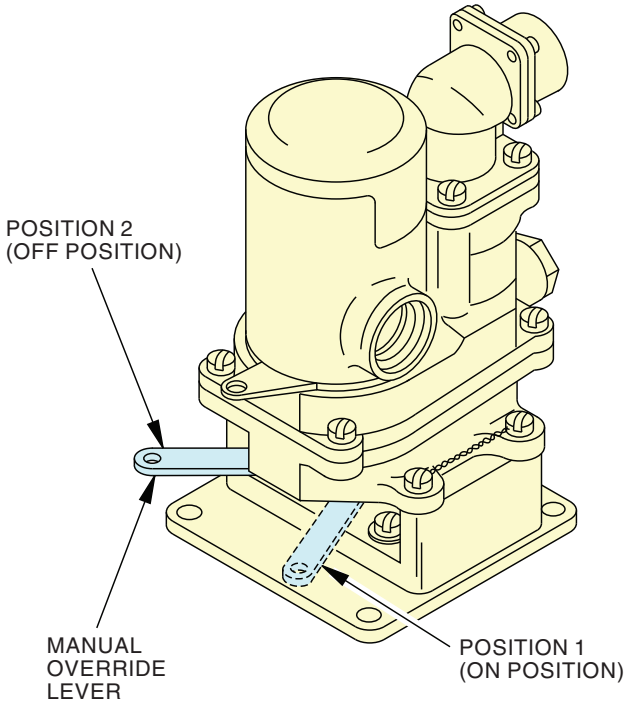
Trailing Edge Flap Bypass Valve
Figure 202/27-51-00-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trailing Edge Flap Bypass Valve
Figure 202/27-51-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**MANUAL OVERRIDE LEVER POSITION
(EXAMPLE)**

C

2849983 S0000666270_V1

**Trailing Edge Flap Bypass Valve
Figure 202/27-51-00-990-802 (Sheet 3 of 3)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-440-801

3. Trailing Edge Flap System Reactivation

(Figure 201, Figure 202)

A. General

(1) Use this procedure to reactivate the trailing edge flap system.

NOTE: This task will also reactivate the leading edge system.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1676	Lock Assembly - Flight Controls (used on valve p/n 10-60710-2 and 10-60710-6) Part #: F80049-66 Supplier: 81205 Opt Part #: F80049-17 Supplier: 81205 Opt Part #: F80049-31 Supplier: 81205 Opt Part #: F80049-46 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

E. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Procedure

SUBTASK 27-51-00-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-00-440-001

(2) Reactivate the trailing edge flap bypass valve (Figure 202):

(a) Remove the lock, SPL-1676 from the bypass valve.

(b) Move the manual override lever on the bypass valve to position 1.

(c) Before you connect the electrical connector, examine the connector for corrosion.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO THE STEPS BELOW IF THE AIRPLANE OPERATES AT AIRPORTS THAT USE DEICING FLUID WHICH CONTAINS POTASSIUM FORMATE. ALSO DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTOR IS PART OF A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(d) If there was corrosion, refer to SWPM 20-60-03 to correct the problem.

(e) Apply the D5026NS or ZC-026 compound, G50171 to the connector (SWPM 20-60-03).

(f) Connect the electrical connector to the bypass valve.

SUBTASK 27-51-00-860-002

(3) Remove the safety locks and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

SUBTASK 27-51-00-080-001

(4) Remove the DO-NOT-OPERATE tags from the ALTERNATE FLAPS switches (Figure 201).

SUBTASK 27-51-00-020-001

(5) Do this task to remove the lockout assembly if it is installed: Flap Control Handle Lockout Assembly Removal, TASK 27-51-00-000-801.

END OF TASK

TASK 27-51-00-860-801

4. Trailing Edge Flap System Operation With Primary Control

(Figure 203)

A. General

(1) The primary control system uses the hydraulic system B to move the flaps.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Trailing Edge Flap System Operation With Primary Control Procedure

SUBTASK 27-51-00-860-003

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-00-860-004

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-51-00-860-005

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Move the flap control lever to the applicable detent position.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-006

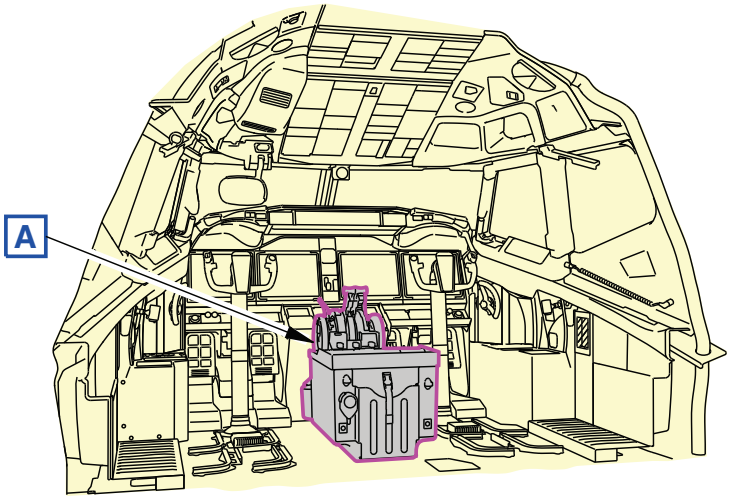
(1) Move the flap control lever to the UP position to retract the flaps and slats.

SUBTASK 27-51-00-860-007

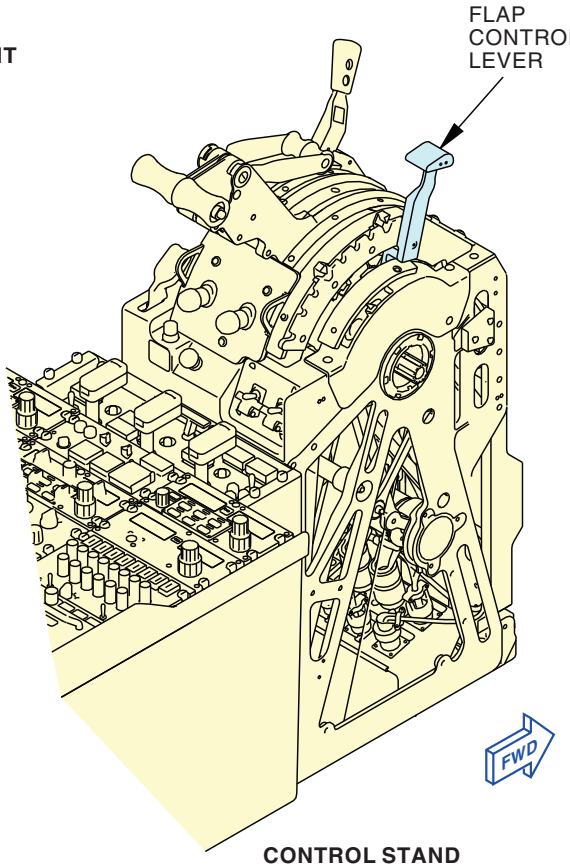
(2) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



CONTROL STAND

A

2482650 S0000583545_V1

Trailing Edge Flap System Operation With Primary Control
Figure 203/27-51-00-990-813

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-860-802

5. Trailing Edge Flap System Operation With Alternate Control

(Figure 201)

A. General

(1) The alternate control system uses electrical power to move the flaps.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Trailing Edge Flap System Operation with Alternate Control

SUBTASK 27-51-00-860-009

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

DO NOT OPERATE THE TRAILING EDGE FLAPS WITH ELECTRICAL POWER MORE THAN ONE FULL EXTEND/RETRACT CYCLE IN 25 MINUTES. YOU CAN CAUSE DAMAGE TO THE FLAP ELECTRIC MOTORS IF YOU OPERATE THEM TOO FREQUENTLY.

(1) Move the ALTERNATE FLAPS ARM switch, on the P5 panel, to the ARM position to arm the alternate control system.

NOTE: The Standby Hydraulic System activates when the ALTERNATE FLAPS ARM switch is set to the ARM position. The Standby Hydraulic System also operates the rudder, thrust reversers, leading edge flaps and slats.

SUBTASK 27-51-00-860-010

CAUTION

DO NOT OPERATE THE TRAILING EDGE FLAPS WITH ELECTRICAL POWER MORE THAN ONE FULL EXTEND/RETRACT CYCLE IN 25 MINUTES. YOU CAN CAUSE DAMAGE TO THE FLAP ELECTRIC MOTORS IF YOU OPERATE THEM TOO FREQUENTLY.

(2) Move the flap control lever to the 40-unit position.

NOTE: This will decrease the load on the flap electric (alternate) motor. You can move the lever past where you will move the flaps to.

SUBTASK 27-51-00-860-011

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION

WAIT A MINIMUM OF 15 SECONDS AFTER THE FLAPS STOP BEFORE YOU USE THE ALTERNATE SYSTEM TO MOVE THE FLAPS AGAIN. YOU CAN CAUSE DAMAGE TO THE FLAP ALTERNATE MOTOR.

(3) Move the ALTERNATE FLAPS control switch to the UP position or to the DOWN position to extend or retract the flaps as it is necessary.

NOTE: The leading edge flaps and slats will extend with alternate control.

(a) To deactivate the leading edge flaps and slats, do this task: Trailing Edge Flap System Operation With Leading Edge Flaps and Slats Disabled, TASK 27-51-00-860-805.

SUBTASK 27-51-00-860-012

(4) Move the ALTERNATE FLAPS control switch to the OFF position.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-013

WARNING

MAKE SURE THE POSITION OF THE FLAPS AGREES WITH THE POSITION OF THE FLAP CONTROL LEVER. WHEN YOU SUPPLY HYDRAULIC POWER, THE FLAPS AND SLATS WILL MOVE AUTOMATICALLY TO THE POSITION OF THE FLAP CONTROL LEVER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Move the ALTERNATE FLAPS control switch until the flaps are in the same position as the flap control lever or move the flap control lever to the position of the flaps.

SUBTASK 27-51-00-860-014

(2) Move the ALTERNATE FLAPS control switch to the OFF position.

SUBTASK 27-51-00-860-015

(3) Move the ALTERNATE FLAPS ARM switch to the OFF position to disarm the alternate control system.

END OF TASK

TASK 27-51-00-980-801

6. Trailing Edge Flap System Manual Operation

(Figure 204)

A. General

(1) Use this task to manually operate the trailing edge flap system without airplane power.

B. References

Reference	Title
27-51-51-960-801	Flap Power Drive Unit Manual Drive Shearout Replacement (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Trailing Edge Flap System Manual Operation

SUBTASK 27-51-00-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-00-040-002

WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-00-010-001

WARNING MAKE SURE THAT THERE IS NO RIG PIN ON THE FLAP CONTROL UNIT BEFORE MANUAL OPERATION OF THE TRAILING EDGE FLAPS. IF YOU DO NOT OBEY, IT CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Remove the cover [4] from the flap power drive unit [1]:

(a) Remove the bolts [3] and washers [2] that attach the cover [4] to the flap power drive unit [1].

(b) Remove the cover [4] from the flap power drive unit [1]

SUBTASK 27-51-00-980-001

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION DO NOT TURN THE MANUAL DRIVE WITH MORE THAN 350 POUND-INCHES (39.5 NEWTON-METERS) TORQUE OR FASTER THAN 1000 RPM. YOU CAN CAUSE DAMAGE TO THE POSITION SENSOR GEARBOX AND THE TORQUE TUBES.

(4) Use a 3/8 inch square drive to turn the manual drive on the PDU to extend or retract the trailing edge flaps to the applicable position.

NOTE: Refer to the cover for the correct direction to turn the manual drive. Look at the number 4 or 5 ballscrew to find the position of the flaps.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If the pin is sheared, do this task: Flap Power Drive Unit Manual Drive Shearout Replacement, TASK 27-51-51-960-801.

NOTE: The manual drive on the PDU has a shearout installed to prevent damage to the system from too much drive torque.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-980-002

WARNING

MAKE SURE THE POSITION OF THE FLAPS AGREES WITH THE POSITION OF THE FLAP CONTROL LEVER. WHEN YOU SUPPLY HYDRAULIC POWER, THE FLAPS AND SLATS WILL MOVE AUTOMATICALLY TO THE POSITION OF THE FLAP CONTROL LEVER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Turn the manual drive until the flaps are in a position which agrees with the flap control lever.

SUBTASK 27-51-00-410-001

(2) Install the cover [4] on the flap power drive unit [1]:

(a) Apply fluid, D00467, to the packing [5] on the cover [4].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin layer of Cor-Ban 27L Compound, G50237, to the mating surface of the cover [4] and the flap power drive unit [1].

(c) Put the cover [4] in its position on the flap power drive unit [1].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(d) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the bolts [3].

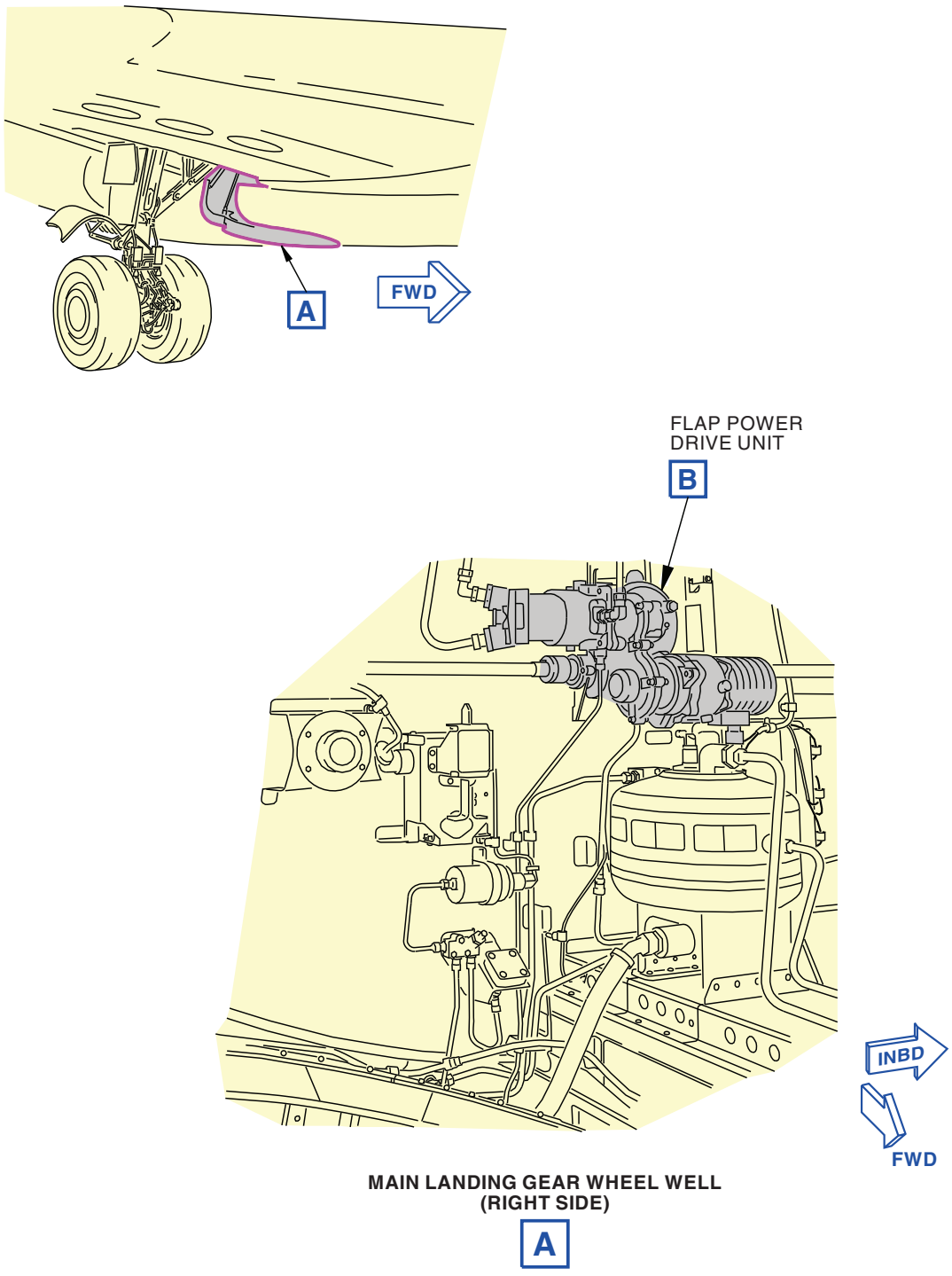
(e) Install the bolts [3] and washers [2] to attach the cover [4].

SUBTASK 27-51-00-440-002

(3) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

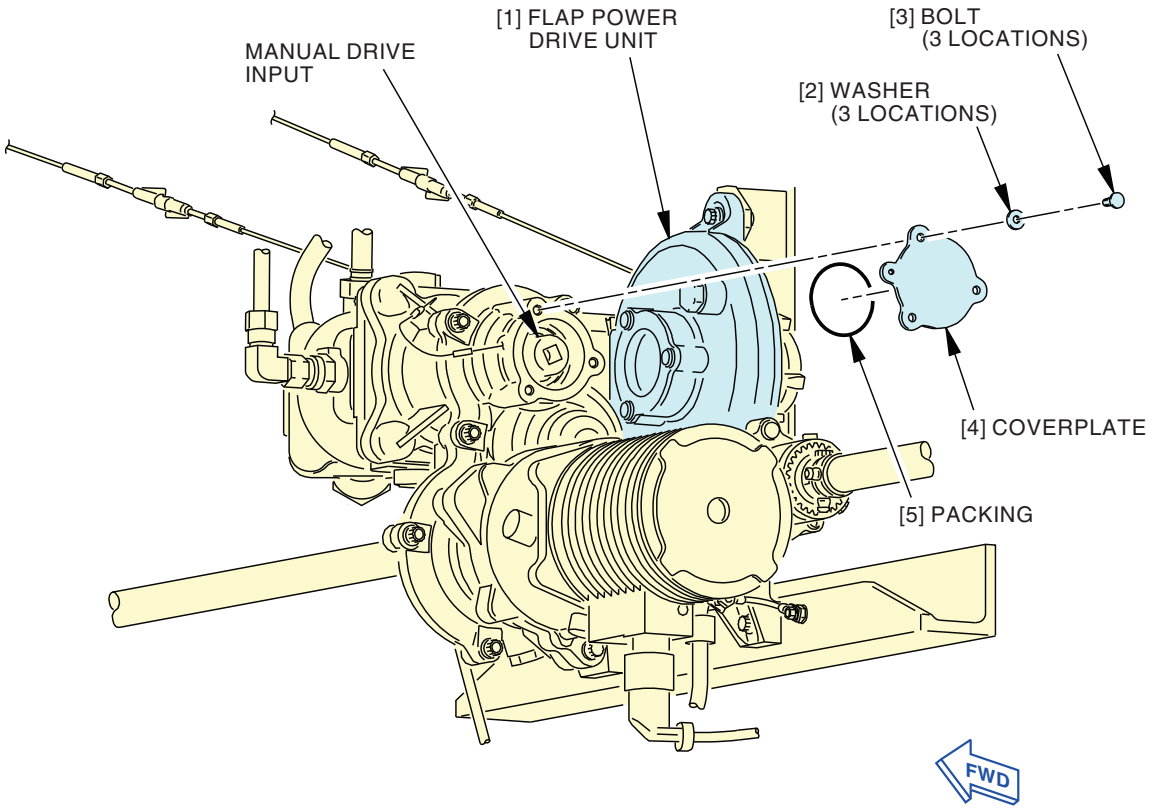
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trailing Edge Flap Manual Operation
Figure 204/27-51-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLAP POWER DRIVE UNIT

B

2412728 S00061531233_V1

Trailing Edge Flap Manual Operation
Figure 204/27-51-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-860-803

7. Extend the Trailing Edge Flaps

A. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Procedure

SUBTASK 27-51-00-860-016

(1) Do one of these tasks to extend the trailing edge flaps:

(a) To extend the flaps with hydraulic power, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(b) To extend the flaps with electrical power in alternate mode, do this task: Trailing Edge Flap System Operation With Alternate Control, TASK 27-51-00-860-802.

(c) To manually extend the flaps, do this task: Trailing Edge Flap System Manual Operation, TASK 27-51-00-980-801.

——— END OF TASK ———

TASK 27-51-00-860-804

8. Retract the Trailing Edge Flaps

A. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Procedure

SUBTASK 27-51-00-860-017

(1) Do one of these tasks to retract the trailing edge flaps:

(a) To retract the flaps with hydraulic power, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(b) To retract the flaps with electrical power in alternate mode, do this task: Trailing Edge Flap System Operation With Alternate Control, TASK 27-51-00-860-802.

(c) To manually retract the flaps, do this task: Trailing Edge Flap System Manual Operation, TASK 27-51-00-980-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-860-805

9. Trailing Edge Flap System Operation With Leading Edge Flaps and Slats Disabled

WARNING

MAKE SURE THAT FOR AIRPLANES WITH THE -1001 CRUISE DEPRESSURIZATION VALVE, YOU REVIEW THIS WARNING. MAKE SURE THAT YOU CLOSE THE FAN COWLS AND THRUST REVERSERS IF YOU MOVE THE TRAILING EDGE FLAPS WHILE THE LEADING EDGE FLAPS AND SLATS ARE DISCONNECTED. THE -1001 CRUISE DEPRESSURIZATION CREEP CAN CAUSE MOVEMENT OF THE LEADING EDGE FLAPS AND SLATS WHEN YOU MOVE THE FLAP CONTROL LEVER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

A. General

(1) This task gives instructions to operate the Trailing Edge (TE) flap system with the Leading Edge (LE) flaps and slats retracted and disabled.

(2) You can operate the TE flaps with primary or alternate control or manually after you do the Disable the LE Slats part of this task.

(3) An LE device creep condition has been associated with the -1001 cruise depressurization valve. It can cause hydraulic pressure leakage through the valve when manually closed.

(a) This cruise depressurization creep condition can permit movement of the deactivated LE flaps and slats when the TE flaps are moved.

(b) The redesigned -1003 cruise depressurization valve is not susceptible to this creep condition.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00467	Fluid - Landing Gear Shock Strut	BMS3-32 Type II
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

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E. Prepare to Operate the Trailing Edge Flaps

SUBTASK 27-51-00-860-114

(1) Make sure that the TE flaps and the LE flaps and slats are in the retracted position.

(a) If they are not in the retracted position, do these tasks:

- Retract the Trailing Edge Flaps, TASK 27-51-00-860-804
- Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804

SUBTASK 27-51-00-040-010

(2) Deactivate the LE flaps and slats in the retracted position (Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801).

SUBTASK 27-51-00-860-108

WARNING

MAKE SURE THAT FOR AIRPLANES WITH THE -1001 CRUISE DEPRESSURIZATION VALVE, YOU REVIEW THIS WARNING. MAKE SURE THAT YOU CLOSE THE FAN COWLS AND THRUST REVERSERS IF YOU MOVE THE TRAILING EDGE FLAPS WHILE THE LEADING EDGE FLAPS AND SLATS ARE DISCONNECTED. THE -1001 CRUISE DEPRESSURIZATION CREEP CAN CAUSE MOVEMENT OF THE LEADING EDGE FLAPS AND SLATS WHEN YOU MOVE THE FLAP CONTROL LEVER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL OR DAMAGE TO THE EQUIPMENT CAN OCCUR.

(3) Make sure that the fan cowls and thrust reversers are closed.

(a) If necessary, do these tasks:

1) Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00
2) Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 27-51-00-860-109

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-51-00-860-106

(5) Do the LE Flaps and Slats Tryout test (TASK 27-81-00-040-801).

F. Trailing Edge Flap Operation With Primary Control

SUBTASK 27-51-00-860-023

(1) Make sure that you deactivate the LE Flaps and Slats before you operate the TE flaps.

SUBTASK 27-51-00-860-024

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

NOTE: The LE flaps and slats will move to the retracted position when you supply hydraulic power if they are not in the retracted position.

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SUBTASK 27-51-00-860-025

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE TRAILING EDGE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Move the flap control lever to the applicable detent position.

SUBTASK 27-51-00-860-026

(4) To retract the TE flaps, move the flap control lever to the UP position.

SUBTASK 27-51-00-860-027

(5) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

G. Trailing Edge Flap Operation With Alternate Control

SUBTASK 27-51-00-860-028

(1) Make sure that you deactivate the LE Flaps and Slats before you operate the TE flaps.

SUBTASK 27-51-00-860-029

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE TRAILING EDGE FLAPS. THE FLAPS WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Move the ALTERNATE FLAPS ARM switch on the P5 panel to arm the alternate control system.

SUBTASK 27-51-00-860-030

CAUTION WAIT A MINIMUM OF 15 SECONDS AFTER THE FLAPS STOP BEFORE YOU USE THE ALTERNATE SYSTEM TO MOVE THE FLAPS AGAIN. YOU CAN CAUSE DAMAGE TO THE FLAP ALTERNATE MOTOR.

(3) Move the ALTERNATE FLAPS control switch to DOWN to extend the TE flaps as it is necessary.

SUBTASK 27-51-00-860-031

(4) Move the ALTERNATE FLAPS control switch to the OFF position.

SUBTASK 27-51-00-860-032

WARNING MAKE SURE THE POSITION OF THE FLAPS AGREES WITH THE POSITION OF THE FLAP CONTROL LEVER. WHEN YOU SUPPLY HYDRAULIC POWER, THE FLAPS AND SLATS WILL MOVE AUTOMATICALLY TO THE POSITION OF THE FLAP CONTROL LEVER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) To retract the TE flaps, move the ALTERNATE FLAPS control switch to UP until the flaps are in the same position as the flap control lever.

SUBTASK 27-51-00-860-033

(6) Move the ALTERNATE FLAPS control switch to the OFF position.

SUBTASK 27-51-00-860-034

(7) Move the ALTERNATE FLAPS ARM switch to the OFF position to disarm the alternate control system.

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H. Trailing Edge Flap Manual Operation

(Figure 204)

SUBTASK 27-51-00-860-035

(1) Make sure that you deactivate the LE Flaps and Slats before you operate the TE flaps.

SUBTASK 27-51-00-480-007

(2) Install DO-NOT-OPERATE tags on the flap control lever, the ALTERNATE FLAPS switches and the hydraulic control switches.

SUBTASK 27-51-00-040-003

WARNING DO THE TRAILING EDGE FLAP DEACTIVATION PROCEDURE BEFORE YOU DO WORK ON THE FLAP SYSTEM. WITH THE FLAPS ACTIVATED, THE FLAPS CAN MOVE AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-00-010-002

(4) Remove the cover [4] from the flap power drive unit [1]:

(a) Remove the bolts [3] and washers [2] that attach the cover [4] to the flap power drive unit [1].

(b) Remove the cover [4] from the flap power drive unit [1].

SUBTASK 27-51-00-980-004

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT TURN THE MANUAL DRIVE WITH MORE THAN 350 POUND-INCHES (39.5 NEWTON-METERS) TORQUE OR FASTER THAN 1000 RPM. YOU CAN CAUSE DAMAGE TO THE POSITION SENSOR GEARBOX AND THE TORQUE TUBES.

(5) Use a 3/8 inch square drive to turn the manual drive on the Power Drive Unit (PDU) to extend the TE flaps to the applicable position.

(a) Refer to the cover [4] for the correct direction to turn the manual drive.

(b) Look at the number 4 or 5 ballscrew to find the position of the flaps.

SUBTASK 27-51-00-980-005

WARNING MAKE SURE THE POSITION OF THE FLAPS AGREES WITH THE POSITION OF THE FLAP CONTROL LEVER. WHEN YOU SUPPLY HYDRAULIC POWER, THE FLAPS AND SLATS WILL MOVE AUTOMATICALLY TO THE POSITION OF THE FLAP CONTROL LEVER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) To retract the TE flaps, turn the manual drive in the retract direction until the flaps are in a position which agrees with the flap control lever.

SUBTASK 27-51-00-410-002

(7) Install the cover [4] on the flap power drive unit [1]:

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(a) Apply fluid, D00467, to the packing [5] on the cover [4].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin layer of Cor-Ban 27L Compound, G50237, to the mating surface of the cover [4] and the flap power drive unit [1].

(c) Put the cover [4] in its position on the flap power drive unit [1].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(d) Apply a thin layer of Cor-Ban 27L Compound, G50237, to the shank and threads of the bolts [3].

(e) Install the bolts [3] and washers [2] to attach the cover [4].

SUBTASK 27-51-00-440-003

(8) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-111

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-51-00-440-007

(2) Reactivate the LE flaps and slats (Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801).

SUBTASK 27-51-00-860-113

(3) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

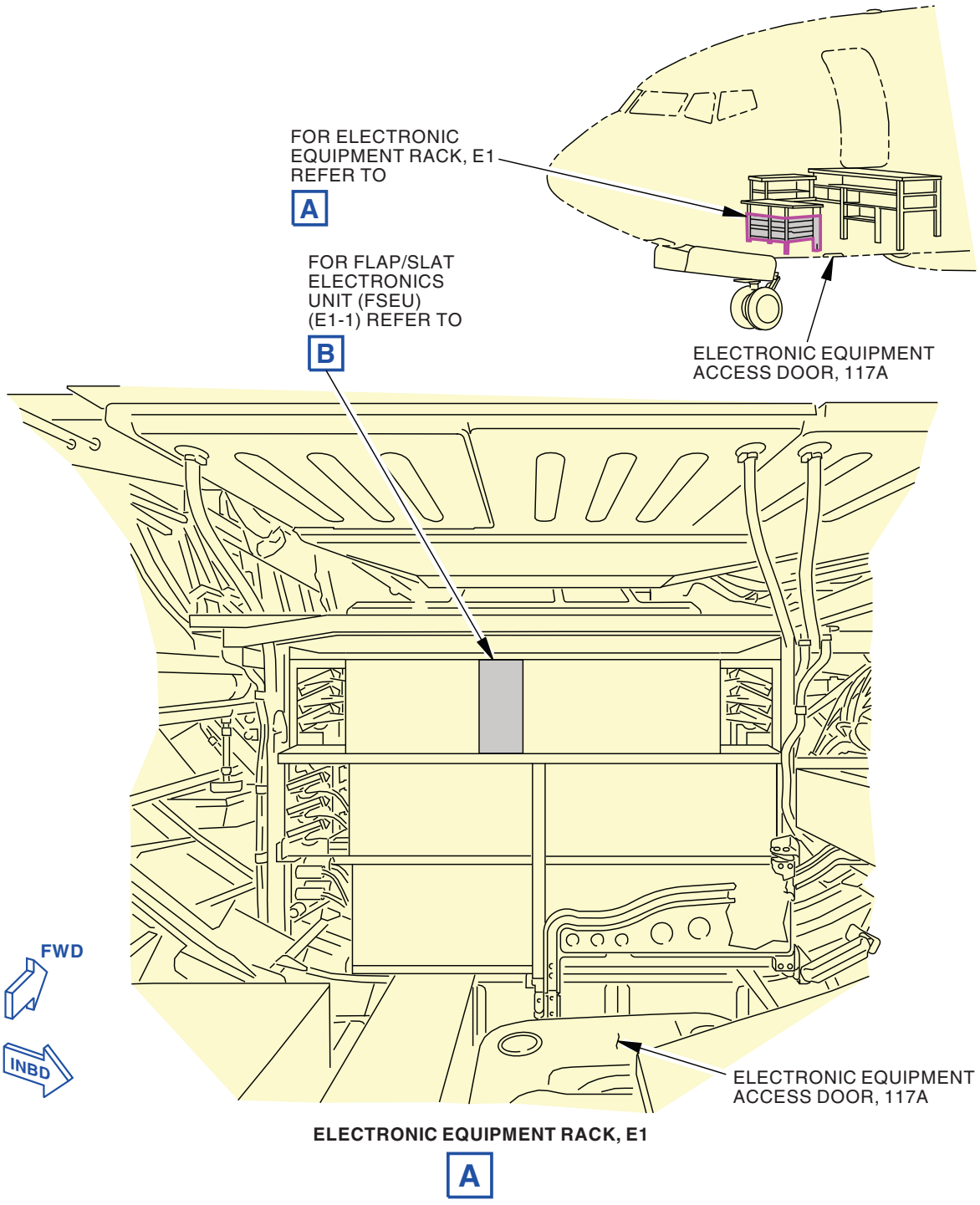
Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-51-00-860-110

(4) Remove the hydraulic power from the airplane. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

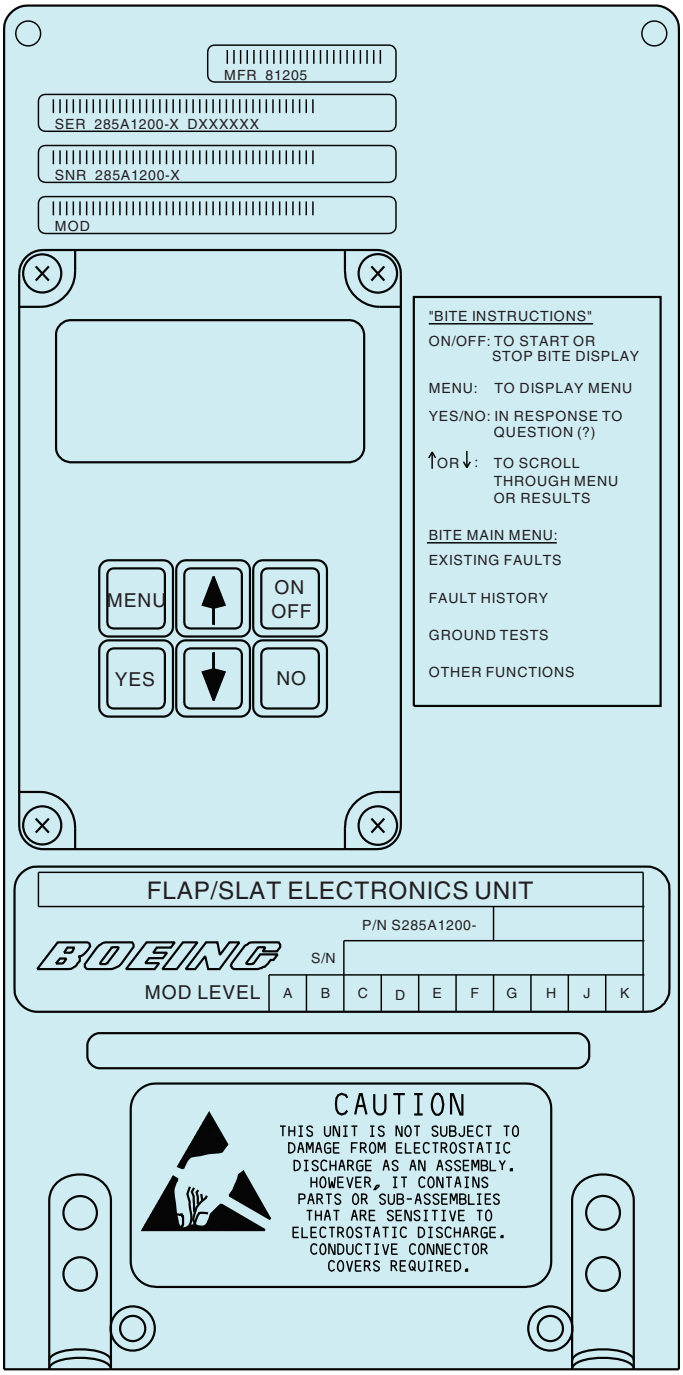
END OF TASK

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Flap/Slat Electronics Unit
Figure 205/27-51-00-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



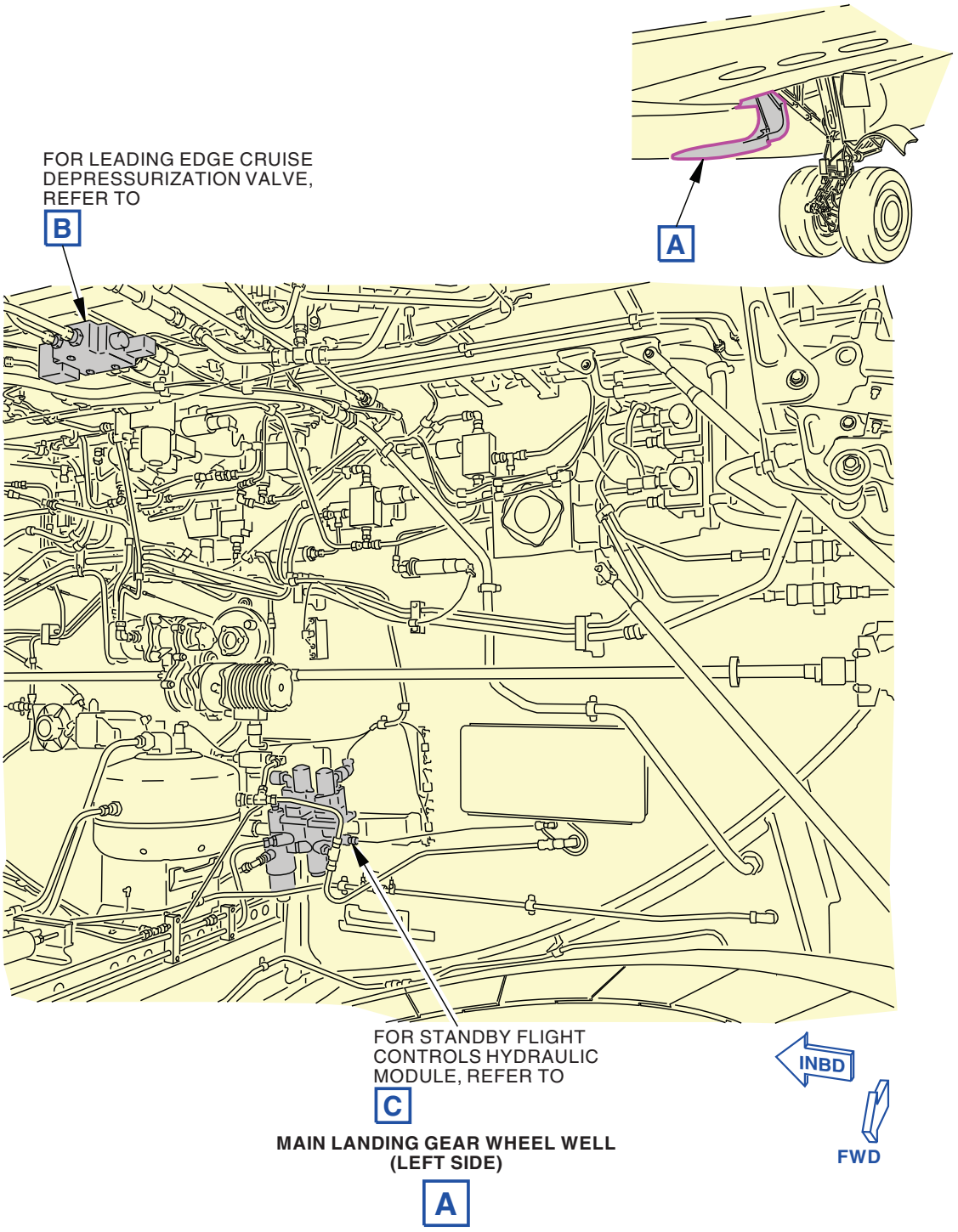
FLAP/SLAT ELECTRONICS UNIT (FSEU)

B

2412730 S00061531238_V1

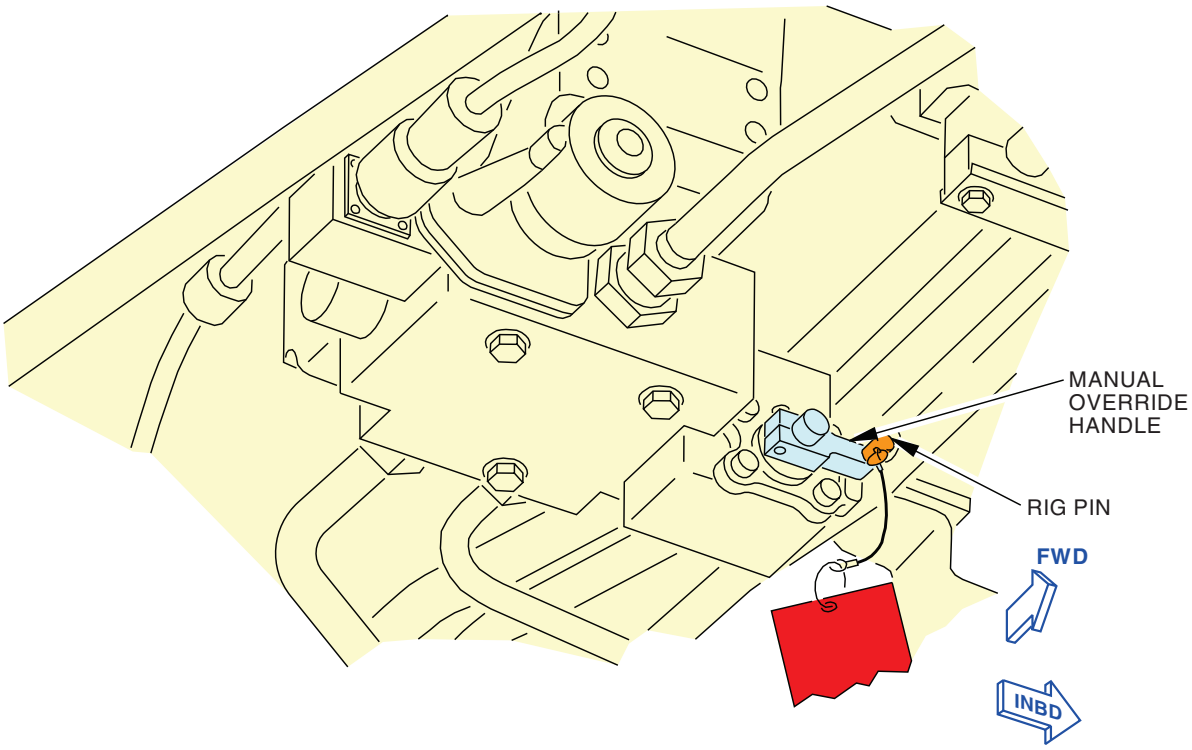
Flap/Slat Electronics Unit
Figure 205/27-51-00-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Flap and Slat Deactivation
Figure 206/27-51-00-990-805 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



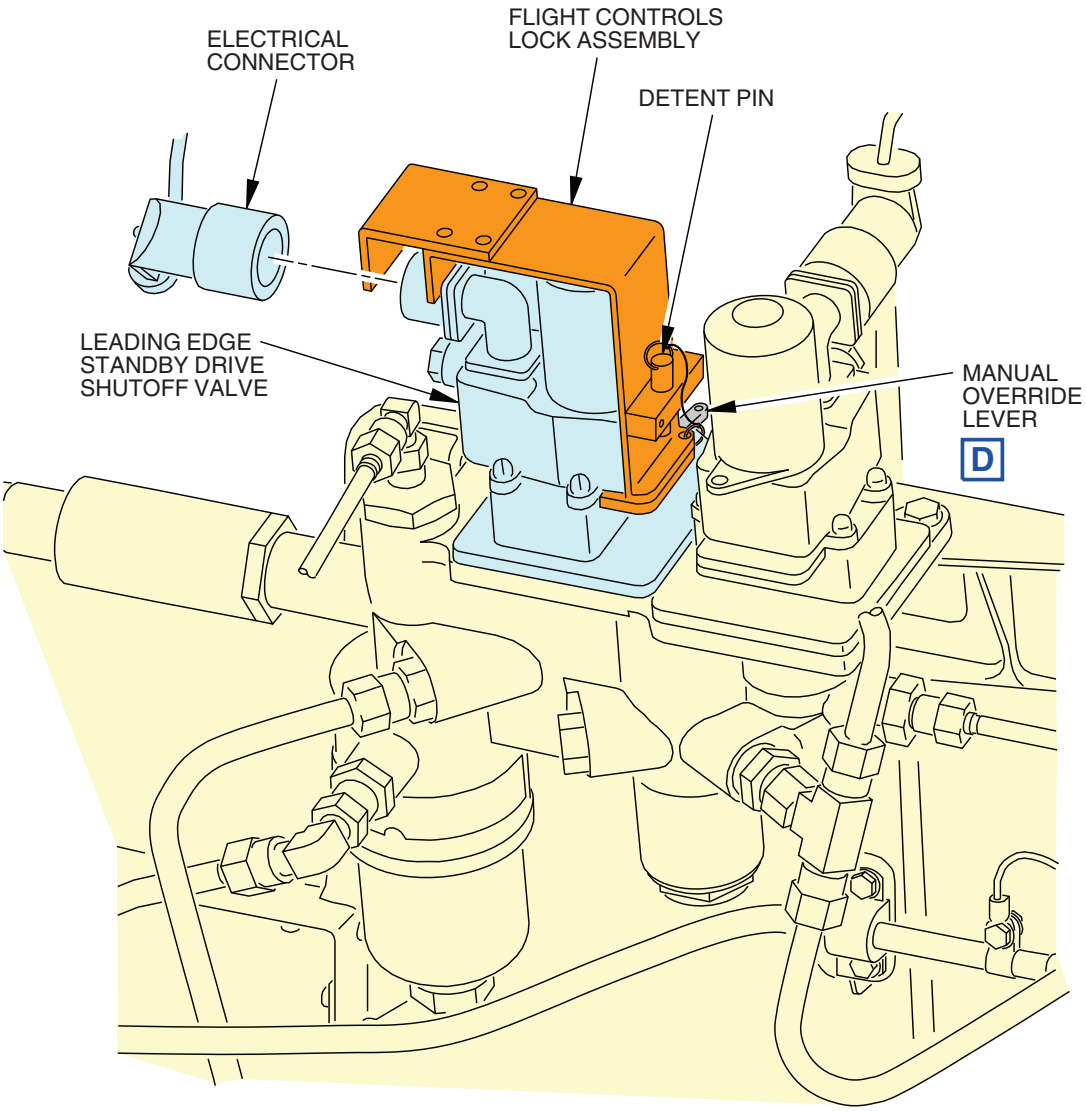
**LEADING EDGE CRUISE
DEPRESSURIZATION VALVE**

B

2412732 S00061531240_V1

**Leading Edge Flap and Slat Deactivation
Figure 206/27-51-00-990-805 (Sheet 2 of 4)**

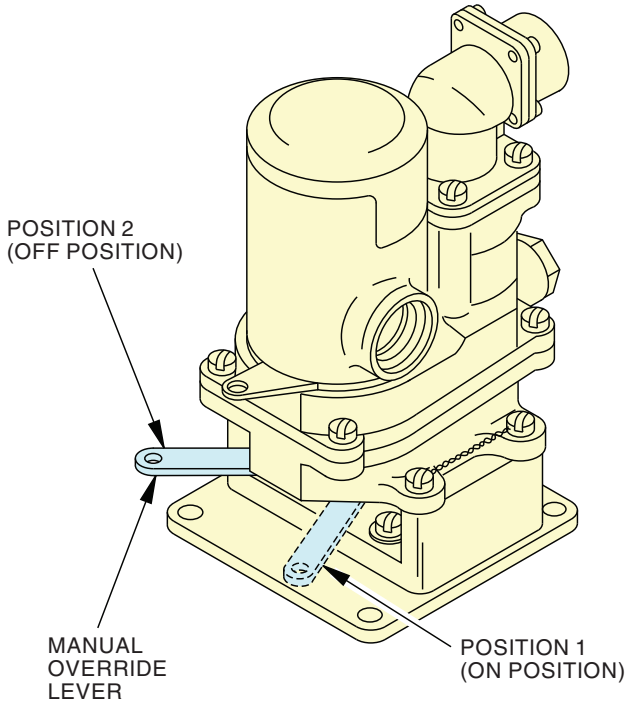
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412733 S00061531241_V4

Leading Edge Flap and Slat Deactivation
Figure 206/27-51-00-990-805 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**MANUAL OVERRIDE LEVER POSITION
(EXAMPLE)**

D

2844536 S0000662328_V3

**Leading Edge Flap and Slat Deactivation
Figure 206/27-51-00-990-805 (Sheet 4 of 4)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-000-801

10. Flap Control Handle Lockout Assembly Removal

(Figure 207)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-13747	Control Handle Lockout Assy, C27008-14 (included in C27008 Lock Equipment) Part #: C27008-22 Supplier: 81205

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Flap Control Handle Lockout Assembly Removal

SUBTASK 27-51-00-080-003

(1) Remove the control handle lockout assembly, C27008-14, SPL-13747:

(a) Remove the t-pin from the control handle lockout assembly.

(b) Remove the control handle lockout assembly from the handle of the flap control lever.

(c) Stow the t-pin in the control handle lockout assembly.

——— END OF TASK ———

TASK 27-51-00-400-801

11. Flap Control Handle Lockout Assembly Installation

(Figure 207)

A. General

(1) The flap control handle lockout assembly prevents the accidental movement of the flap control lever. With electrical power on, the flaps will operate if you move the flap control lever.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-13747	Control Handle Lockout Assy, C27008-14 (included in C27008 Lock Equipment) Part #: C27008-22 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Flap Control Handle Lockout Assembly Installation

SUBTASK 27-51-00-480-009

(1) Install the control handle lockout assembly, C27008-14, SPL-13747:

(a) Remove the t-pin from the control handle lockout assembly.

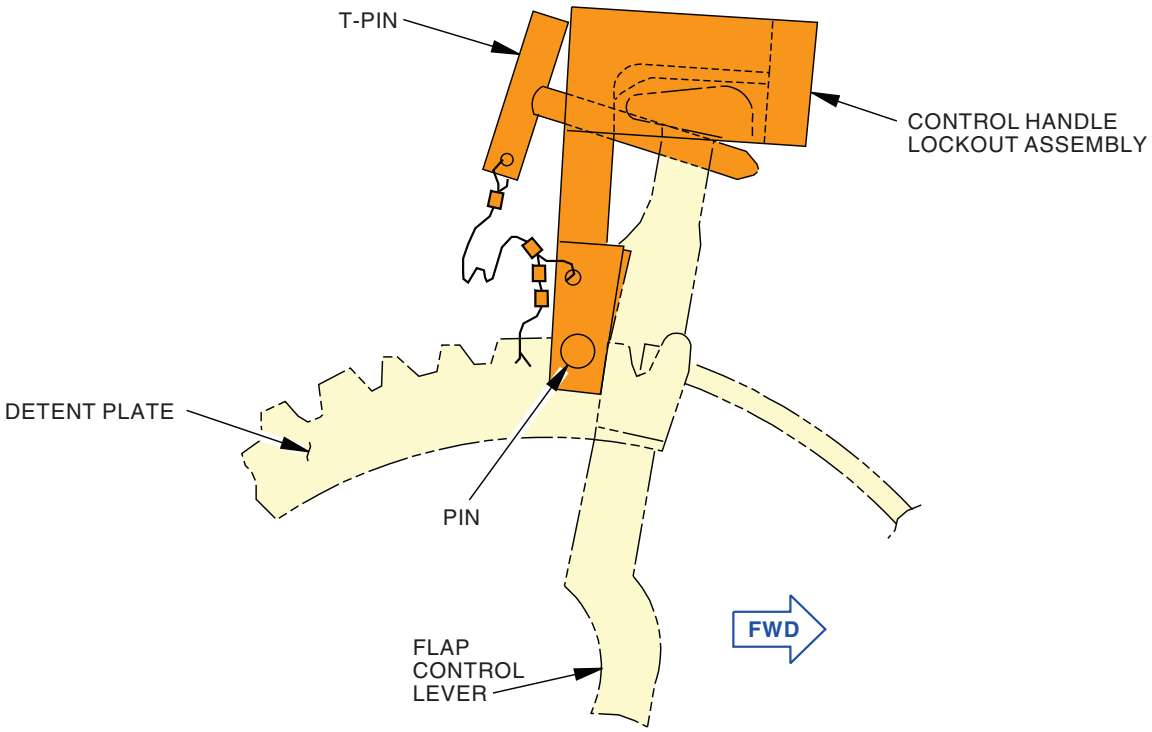
(b) Put the control handle lockout assembly on the handle of the flap control lever.

1) Make sure that the pin at the lower end of the control handle lockout assembly is in the detent on the detent plate.

(c) Install the t-pin to lock the control handle lockout assembly on the flap control lever handle.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONTROL STAND

2412734 S00061531243_V1

Flap Control Handle Lockout Assembly Installation/Removal
Figure 207/27-51-00-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRAILING EDGE FLAP SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An operational test of the trailing edge flap system

(2) An adjustment of the trailing edge flap control system

(3) An adjustment of the trailing edge flap drive system

(4) A functional test of the trailing edge flap system

(5) A Built-In-Test Equipment (BITE) test of the Flap/Slat Electronics Unit (FSEU) display and existing faults

(6) A BITE test of the trailing edge flap asymmetry and skew systems

(7) A BITE test of the trailing edge flap load relief system

(8) A test of flap priority valve

(9) A test of flap control lever force

(10) An functional test of the alternate control system for the trailing edge flaps

(11) An operational test of the no-back brakes inside the transmissions

(12) A functional test of the trailing edge flap load relief system

TASK 27-51-00-710-801

2. Trailing Edge Flap System Operational Test

(Figure 501)

A. General

(1) During the operational test, you make sure the operation of the Trailing Edge (TE) flaps is satisfactory. This test does not make sure the flaps are adjusted correctly. You must not use this test as an alternate for the system test of the TE flaps.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Operational Test

SUBTASK 27-51-00-860-038

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

NOTE: When operating the high lift devices on the ground with only the system B Electric Motor-Driven Pump (EMDP) supplying power the pump low pressure light may momentarily illuminate. This illumination and associated pressure drop should be considered normal as long as the pressure recovers within 10 seconds of initiating movement of the high lift devices.

E. Primary Method for the Trailing Edge Flap System Operation Test

SUBTASK 27-51-00-710-001

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Do these steps for the primary method of the TE flap operation test:

NOTE: Use the flap position indicator to find the position of the flaps.

(a) Do one of these methods to show the flap position sensor and skew sensor inputs to the Flap/Slat Electronics Unit (FSEU):

1) On the FSEU Built-In-Test Equipment (BITE) display (Method 1):

a) Push the ON/OFF button on the FSEU to display “EXISTING FAULTS?”.

b) Push the NO switch three times on the FSEU to display “OTHER FUNCTNS?”.

c) Push the YES switch on the FSEU to display “SET OUTPUTS?”.

d) Push the NO switch on the FSEU to display “I/O MONITOR?”.

e) Push the YES switch on the FSEU to display “PROX SENSORS?”.

f) Push the NO switch on the FSEU to display 'SYNCHRO INPUTS?”.

g) Push the YES switch on the FSEU to display 'LVR POS”.

2) On the MAX Display System (MDS) (Method 2):

a) Push the SYS button on the Multi Function Panel (MFP).

b) Select the Maint button on the MENU bar at the top right of the System Page.

c) Push the SEL button on the SELECTOR knob.

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d) Turn the SELECTOR knob to select MAINT DATA PGS on the Maintenance menu.

e) Move the cursor down and then select the ATA 27 FLIGHT CONTROL DISPLAY button below the REAL column on the MAINT DATA PGS menu.

(b) Move the flap control lever to the UP position and wait for the flaps to reach the UP position.

1) Make sure that the flap position indicator pointers are within the detent Up white band.

(c) Make sure that the position and skew sensor values displayed on the FSEU fall within the following range:

1) For Method 1, push the arrow keys on the FSEU to step through the position and skew sensor values at the UP position.

2) For Method 2, observe the skew sensor values on the MDS.

DISPLAY	FSEU SENSOR VALUE RANGE	MDS SENSOR VALUE RANGE
LVR POS	-2.8 to 2.8	-2.8 to 2.8
L FL POS	-3.7 to 1.8	-3.7 to 1.8
R FL POS	-3.7 to 1.8	-3.7 to 1.8
F1 SKEW	295.7 to 311.0	-64.3 to -49
F2 SKEW	330.5 to 346.0	-29.5 to -14
F3 SKEW	0.0 to 13.2	0.0 to 13.2
F4 SKEW	332.8 to 346.9	-27.2 to -13.1
F5 SKEW	13.1 to 27.2	-27.2 to -13.1
F6 SKEW	346.8 to 360.0	0.0 to 13.2
F7 SKEW	14.0 to 29.5	-29.5 to -14
F8 SKEW	49.0 to 64.3	-64.3 to -49

(d) Move the flap control lever to the 10-unit position and wait for the flaps to reach the 10 position.

1) Make sure that the pointers on the flap position indicator are in the 10-unit white band.

(e) Make sure that the position and skew sensor values displayed on the FSEU fall within the following range:

1) For Method 1, push the arrow keys on the FSEU to step through the position and skew sensor values at the 10 position.

2) For Method 2, observe the skew sensor values on the MDS at the 10 position.

DISPLAY	FSEU SENSOR VALUE RANGE	MDS SENSOR VALUE RANGE
LVR POS	50.6 to 54.6	50.6 to 54.6
L FL POS	146.2 to 154.9	146.2 to 154.9
R FL POS	146.2 to 154.9	146.2 to 154.9
F1 SKEW	95.9 to 113.6	95.9 to 113.6
F2 SKEW	149.1 to 166.7	149.1 to 166.7

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(Continued)

DISPLAY	FSEU SENSOR VALUE RANGE	MDS SENSOR VALUE RANGE
F3 SKEW	164.7 to 181.5	164.7 to 181.5
F4 SKEW	121.0 to 138.5	121.0 to 138.5
F5 SKEW	221.5 to 239.0	121.0 to 138.5
F6 SKEW	178.5 to 195.3	164.7 to 181.5
F7 SKEW	193.3 to 210.9	149.1 to 166.7
F8 SKEW	246.4 to 264.1	95.9 to 113.6

(f) Move the flap control lever to the 40-unit position and wait for the flaps to reach the 40 position.

1) Make sure that the pointers on the flap position indicator are in the 40-unit white band.

(g) Make sure that the position and skew sensor values displayed on the FSEU fall within the following range:

1) For Method 1, push the arrow keys on the FSEU to step through the position and skew sensor values at the 40 position.

2) For Method 2, observe the skew sensor values on the MDS at the 40 position.

DISPLAY	FSEU SENSOR VALUE RANGE	MDS SENSOR VALUE RANGE
LVR POS	105.4 to 110.9	105.4 to 110.9
L FL POS	267.1 to 273.3	267.1 to 273.3
R FL POS	267.1 to 273.3	267.1 to 273.3
F1 SKEW	224.4 to 242.1	224.4 to 242.1
F2 SKEW	279.2 to 296.9	279.2 to 296.9
F3 SKEW	288.7 to 305.5	288.7 to 305.5
F4 SKEW	265.1 to 282.1	265.1 to 282.1
F5 SKEW	77.9 to 94.9	265.1 to 282.1
F6 SKEW	54.5 to 71.3	288.7 to 305.5
F7 SKEW	63.1 to 80.8	279.2 to 296.9
F8 SKEW	117.9 to 135.6	224.4 to 242.1

(h) For Method 1, push the ON/OFF button on the front panel of the FSEU to turn it off.

(i) For Method 2, push the SYS button on the MFP to put the displays back to normal.

(j) Move the flap control lever to the UP position and wait for the flaps to reach the UP position.

1) Make sure that the pointers on the flap position indicator are in the UP white band.

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F. Alternate Method for the Trailing Edge Flap System Operation Test

SUBTASK 27-51-00-860-041

> **WARNING** MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-51-00-710-002

(2) Do these steps for the alternate method of the TE flap operation test:

(a) Move the flap control lever to the UP detent and wait for the flaps to reach the UP position.

(b) Push the ON/OFF button on the FSEU to display “EXISTING FAULTS?”.

(c) Push the NO switch three times on the FSEU to display “OTHER FUNCTIONS?”.

(d) Push the YES switch on the FSEU to display “SET OUTPUTS?”.

(e) Push the NO switch on the FSEU to display “I/O MONITOR?”.

(f) Push the YES switch on the FSEU to display “PROX SENSORS?”.

(g) Push the NO switch twice on the FSEU to display “DISCRETE INPUTS?”.

(h) Push the YES switch on the FSEU to display “ALTN ARM?”.

(i) Make sure that the FSEU displays “ALTN ARM NOT ARM”.

(j) Put the Alternate Flaps Master Arming Switch, that is on the aft overhead panel, to the “Arm” position.

(k) Make sure that the FSEU displays “ALTN ARM ARMED”.

(l) Put the Alternate Flaps Master Arming Switch, that is on the aft overhead panel, to the “Off” position.

(m) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(n) Put the Alternate Flaps Master Arming Switch, that is on the aft overhead panel, to the “Arm” position.

(o) Move the flap control lever to the 1 detent and wait for the flaps to reach the 1 position.

(p) Make sure that the flap position indicator pointers lie within the detent 1 white band.

(q) Move the Alternate Flap Drive Control Switch to the “Down” position for 5 seconds, then release.

1) Make sure that the TE flaps, Leading Edge (LE) flaps and slats do not move.

2) Make sure that the flap position indicator pointers remains within the detent 1 white band.

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(r) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(s) Move the Alternate Flap Drive Control Switch to the “Up” and hold until the flaps stop moving.

(t) Make sure that the flap position indicator pointers lie within the detent Up white band.

(u) Make sure that the “LE FLAPS TRANSIT” and “LE EXTEND” master indication lights on the center panel are off.

(v) Make sure that all the LE flaps and slats indication lights on the aft overhead panel are off.

(w) Move the flap control lever to the 40 detent.

(x) Move the Alternate Flap Drive Control Switch to the “Down” position for 5 seconds, then release.

(y) Wait for the LE surfaces to fully extend.

(z) Make sure that the LE indication flaps “EXTEND” and slats “FULL EXTEND” lights on the aft overhead panel are now on.

(aa) Move the Alternate Flap Drive Control Switch to the “Down” position and hold until the flaps stop moving.

(ab) Make sure that the flap position indicator pointers lie within the detent 40 white band.

(ac) Make sure that the “LE FLAPS EXT” master indication light is on.

(ad) Put the Alternate Flaps Master Arming Switch, that is on the aft overhead panel, to the “Off” position.

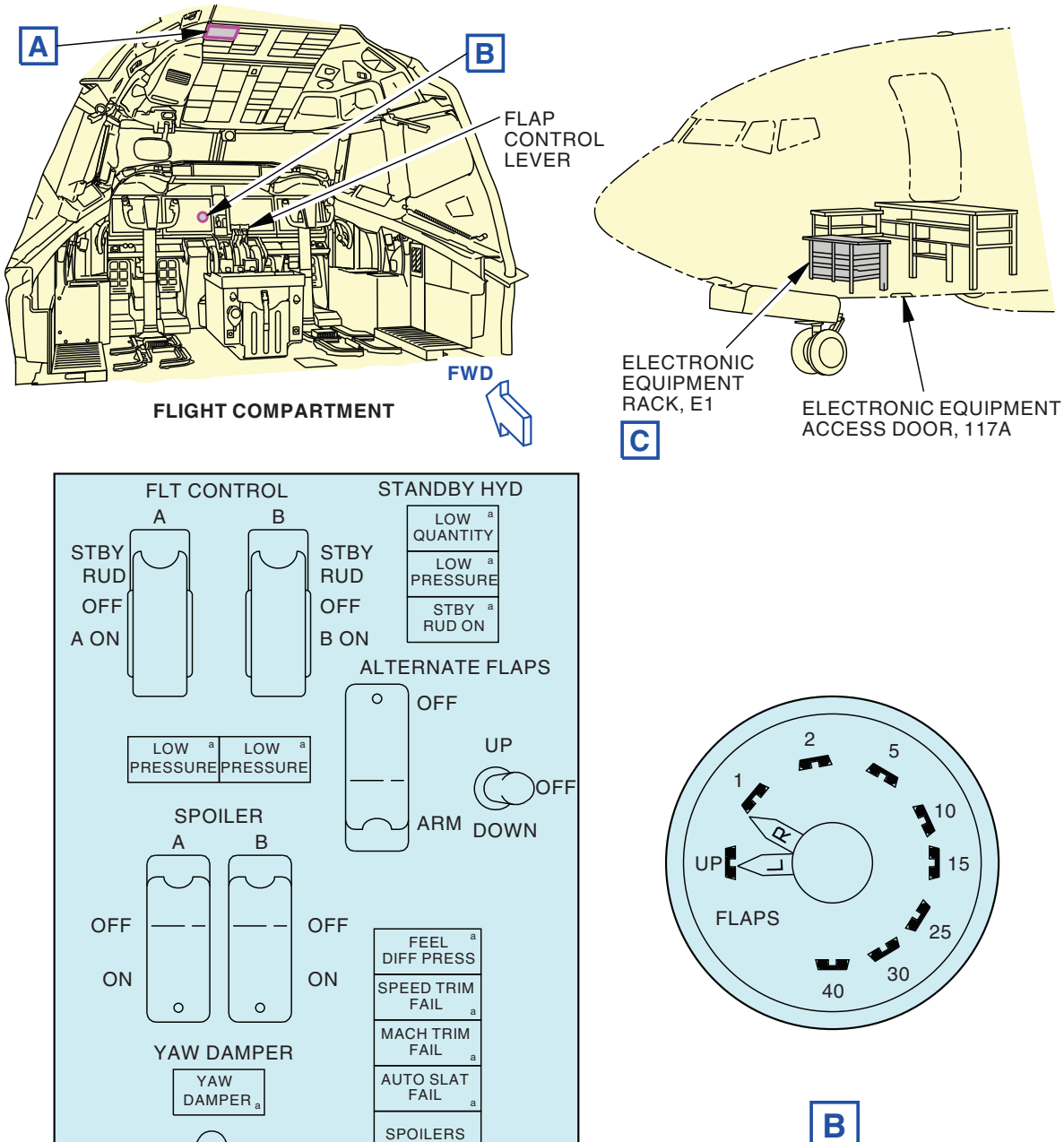
G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-043

(1) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

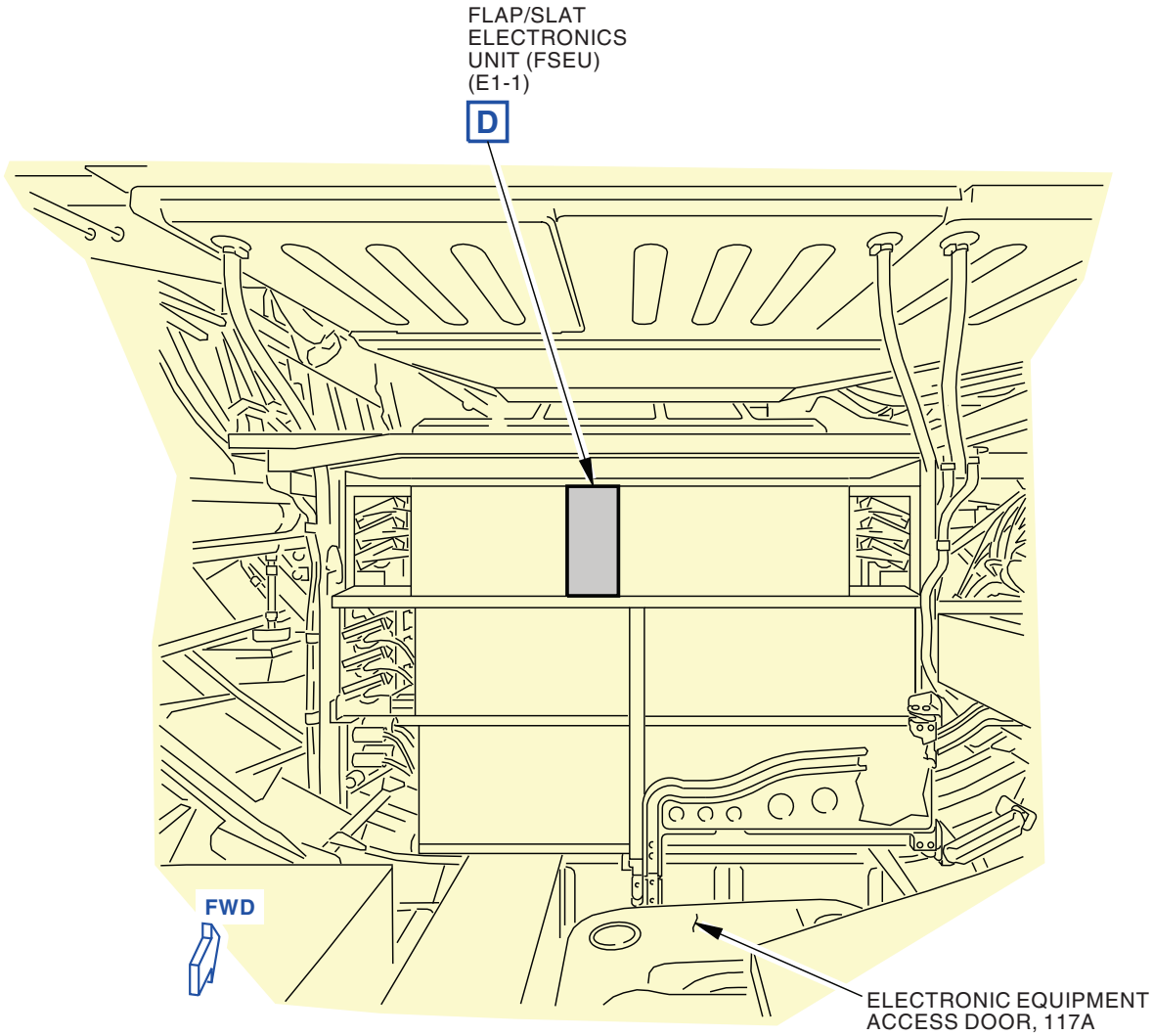
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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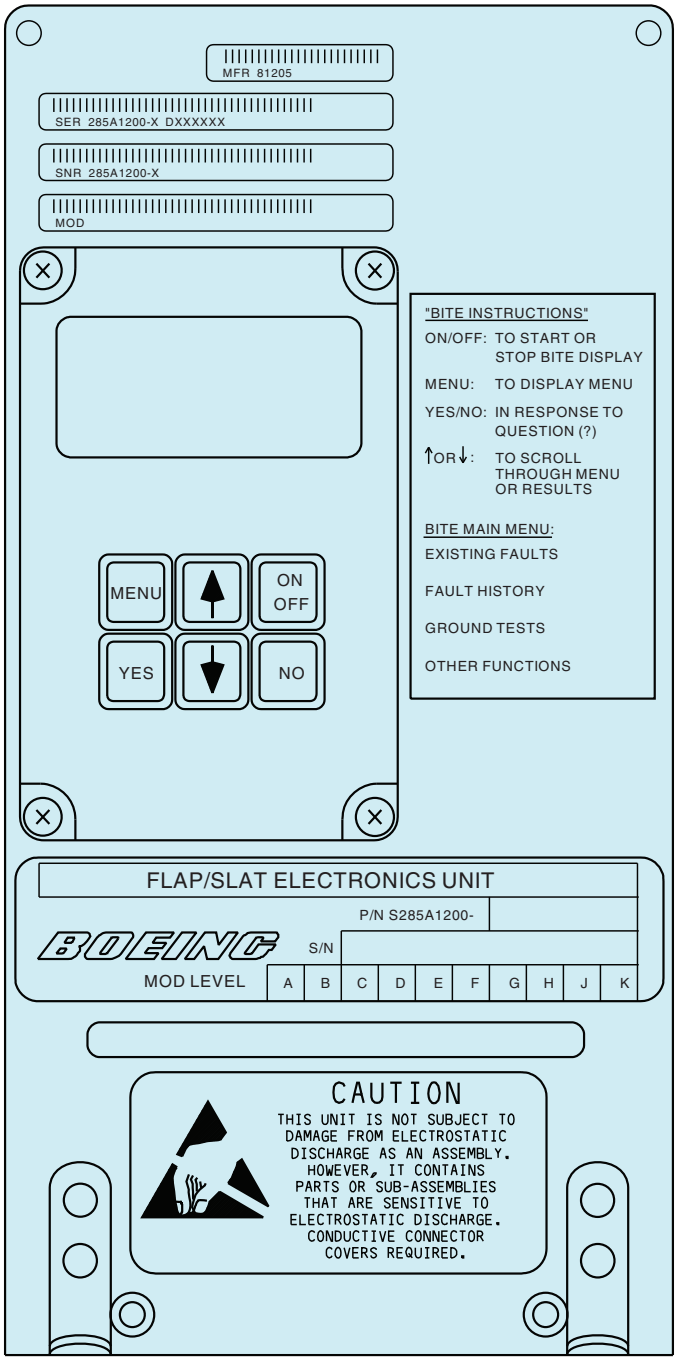
Trailing Edge Flap System Controls
Figure 501/27-51-00-990-807 (Sheet 1 of 3)

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Trailing Edge Flap System Controls
Figure 501/27-51-00-990-807 (Sheet 2 of 3)

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FLAP/SLAT ELECTRONICS UNIT (FSEU)

D

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Trailing Edge Flap System Controls
Figure 501/27-51-00-990-807 (Sheet 3 of 3)

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TASK 27-51-00-820-801

3. Trailing Edge Flap Control System Adjustment and Test

(Figure 502)

A. General

(1) This task does an adjustment of the Trailing Edge (TE) flap control system.

(a) The TE flap control system includes the flap control lever, flap control unit, flap control valve, control cables WFA and WFB, and follow-up cables WFFA and WFFB.

(2) When you retract the TE flaps with hydraulic power, it is necessary to stop for two minutes after you move the flaps. The flaps can move a small distance for 2 minutes after extension or retraction.

(3) The adjustment instructions included in this task are included in the Flap Actuation Rigging Instructions drawing 256A3001.

(4) Before you do cable adjustment, permit a minimum of one hour at a constant ambient temperature of ± 5 degrees F (± 2.8 degrees C) for the airframe temperature to become stable.

(a) It is recommended to do the cable adjustments in a controlled environment such as a hangar.

NOTE: Direct sunlight on the airframe will cause the cable readings to be inaccurate.

(b) All temperature readings will be taken as ambient outside but adjacent to the applicable area of the airframe.

(c) The cable tension values will not be correct when there are temperature differences along the cable run.

(d) It is recommended that airplanes supplied with the cables that contain higher zinc coating (Grade D or Grade K) be re-tensioned 3 to 4 months into service. This will bring the tension of the control cables within the desired range. After the cables are re-tensioned, the high zinc cables (Grade D or Grade K) will continue to relax at a rate that is equivalent to the Grade B cables. To identify the grade of the cable, refer to the table below:

Table 501/27-51-00-993-807 Control Cable Grades and Diameters

CONTROL CABLE DIAMETER (TERMINAL SIZE REF)	CONSTRUCTION FIRST DASH NUMBER *[1]									
	GRADE A					GRADE B		GRADE D		GRADE K
	ZINC			TIN OVER ZINC		ZINC	TIN OVER ZINC	ZINC	TIN OVER ZINC	ZINC
	7 X 7	7 X 7 (RT)	7 X 19	7 X 7	7 X 7 (RT)	7 X 19	7 X 19	7 X 19	7 X 19	7 X 19
.094 (3/32)	3B	3G*[2]	3A	3D	3E *[2]	- — —	- — —	- — —	- — —	- — —
.125 (1/8)	- — —	- — —	4A	- — —	- — —	4F	4C	4J*[3]	4H*[3]	4K*[3]
.188 (3/16)	- — —	- — —	- — —	- — —	- — —	- — —	- — —	6J*[4]	- — —	- — —

*[1] First dash number in the part number of the cable. Example: BACC2C-3D-01085-CE is Grade A, Tin Over Zinc, 7 x 7.

*[2] Code 3E and 3G are specified only where there is a specific requirement for REDUCED TOLERANCE (RT) percent elongation wire rope, 7 X 7 RT, per BMS7-265.

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*[3] Only Grade D and Grade K .125 (1/8) cables (4J, 4H and 4K) are high zinc cables.

*[4] It is not necessary to adjust the Grade D 0.188 (3/16) cables (6J) 3 to 4 months after installation.

(5) Before you do cable adjustment, allow the tensiometer to stabilize to the same temperature as the airplane.

B. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
20-10-44 P/B 401	LOCKING DEVICES - REMOVAL/INSTALLATION
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-801	Trailing Edge Flap System Operation With Primary Control (P/B 201)
27-58-01-000-801	Flap Position Transmitter Removal (P/B 401)
27-58-01-400-801	Flap Position Transmitter Installation (P/B 401)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1726	Clamp Set - Trailing Edge Flap Torque Tube Part #: C27022-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38
G51490	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 18 Inches Long	BACC13AT3K18, AMS 5689

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E. Location Zones

Zone	Area
116	Nose Landing Gear Wheel Well - Right
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

G. Prepare for the Adjustment

SUBTASK 27-51-00-020-003

(1) For the flap position transmitters on the left and right wings, do this task: Flap Position Transmitter Removal, TASK 27-58-01-000-801.

NOTE: If you do not remove the flap position transmitters, they can become incorrectly adjusted.

SUBTASK 27-51-00-480-010

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-00-860-045

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE TE FLAPS AND FLAP DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION CAREFULLY MONITOR THE FLAP MOVEMENT AFTER YOU COMPLETE THE WORK ON THE FLAP CONTROL SYSTEM OR THE FLAP DRIVE SYSTEM. IF YOU DO NOT ADJUST THE COMPONENTS CORRECTLY, YOU CAN CAUSE DAMAGE TO THE TRAILING-EDGE FLAP SYSTEM.

(3) Use the alternate system or the manual drive to retract the TE flaps to the UP position.

SUBTASK 27-51-00-040-005

(4) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

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H. Trailing Edge Flap Control System Adjustment

SUBTASK 27-51-00-020-004

CAUTION DISCONNECT THE TORQUE TUBES FROM THE FLAP-POWER DRIVE UNIT BEFORE YOU ADJUST THE FLAP CONTROL SYSTEM. ADJUST THE FLAP DRIVE SYSTEM CORRECTLY. IF YOU DO NOT, IT CAN CAUSE DAMAGE TO THE FLAP SYSTEM.

(1) If the flap drive system is not adjusted correctly, disconnect the torque tubes [18] from the power drive unit [17].

(a) Install the clamp set, SPL-1726, on the torque tubes [18] on each side of the power drive unit [17].

(b) Remove lockwire or safety cable from the screws [14].

(c) Remove the screws [14] that attach the coupling sleeves [15] to the splined couplings [16].

(d) Move the coupling sleeves [15] off the splined couplings [16].

(e) Move the torque tubes [18] away from the flap power drive unit [17].

SUBTASK 27-51-00-010-003

(2) Remove the floor panel to get access to the top of the flap control unit [4], do this task: Passenger Cabin Floor Panel Removal, TASK 53-21-00-000-801.

NOTE: The floor panel is located between stations 706 and 727B on the right side of the airplane.

SUBTASK 27-51-00-010-016

(3) To get access to the turnbuckles [3] for the cable assemblies [1] WFA and WFB, do this step:

Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

SUBTASK 27-51-00-820-001

(4) Do the adjustment of the flap control lever [7] and cable assemblies [1] WFA and WFB.

(a) Put the flap control lever [7] in the rigged position.

1) Move the flap control lever [7] aft until the detent pin is out of the UP detent.

2) Put a 0.19 in. (4.83 mm) thick rigging block between the detent pin on the flap control lever [7] and the upstop on the detent plate.

a) Make sure that the rigging block does not go into the flaps UP detent.

(b) Install a clamp to hold the quadrant [6] against the aft side of the stop [5] on the flap control unit [4].

(c) Remove the clips [2] from the turnbuckles [3] on the cable assemblies [1] WFA and WFB.

(d) If you installed a new cable assembly [1], do these steps:

1) Use the turnbuckles [3] to adjust the tension in the cable assemblies [1] to 275 ±15 lbf (1.22 ±0.07 kN).

2) Remove the clamp from the quadrant [6].

3) Remove the rigging block from the flap control lever [7].

4) Move the flap control lever [7] through its full travel 25 times.

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5) Put the flap control lever [7] in the rigged position with the rigging block.

6) Install the clamp to hold the quadrant [6] against the aft side of the stop [5] on the flap control unit [4].

(e) Use the turnbuckles [3] to adjust the tension in the cable assemblies [1] WFA and WFB to the limit shown in (Table 502).

1) Adjust the tensions close to the maximum values in the table.

NOTE: The cables will stretch over a period of time.

2) During the adjustment, make sure that the tension in the cable assemblies [1] is never less than 50 lbf (0.22 kN).

NOTE: To make sure that the tension is correct, the temperature of the airframe must be constant in the limit of ±5 degrees for a minimum of one hour.

(f) Remove the clamp from the quadrant [6].

(g) Remove the rigging block from the flap control lever [7].

(h) Move the flap control lever [7] through its full travel a minimum of 5 times. Stop with the flap control lever [7] in the 1-unit position.

(i) Carefully move the flap control lever [7] forward until you feel resistance.

(j) Measure the distance between the detent pin on the flap control lever [7] and the upstop on the detent plate.

(k) If the distance is not in the limit of 0.19 in. (4.83 mm), then adjust the cable assembly [1] and the flap control lever [7] again.

(l) Install the clips [2] on the turnbuckles [3].

Table 502/27-51-00-993-801

Temperature	Cable Tension Pounds (kilograms)			
Degrees F (Degrees C)	WFA/WFB Cables		WFFA/WFFB Cables	
	Minimum	Maximum	Minimum	Maximum
120 (48.9)	159 (72)	189 (86)	73 (33)	83 (37)
118 (47.8)	158 (72)	188 (85)	73 (33)	83 (37)
116 (46.6)	157 (71)	187 (85)	72 (33)	82 (37)
114 (45.5)	154 (70)	184 (83)	72 (33)	82 (37)
112 (44.4)	153 (69)	183 (83)	71 (32)	81 (37)
110 (43.3)	152 (69)	182 (83)	71 (32)	81 (37)
108 (42.2)	150 (68)	180 (82)	70 (32)	80 (36)
106 (41.1)	149 (68)	179 (81)	70 (32)	80 (36)
104 (40.0)	148 (67)	178 (81)	69 (31)	79 (36)
102 (38.9)	147 (67)	177 (80)	68 (31)	78 (35)
100 (37.8)	144 (65)	174 (79)	68 (31)	78 (35)
98 (36.7)	143 (65)	173 (78)	67 (30)	77 (35)
96 (35.5)	142 (64)	172 (78)	67 (30)	77 (35)
94 (34.4)	140 (64)	170 (77)	66 (30)	76 (34)

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Table 502/27-51-00-993-801 (Continued)

Temperature	Cable Tension Pounds (kilograms)			
Degrees F (Degrees C)	WFA/WFB Cables		WFFA/WFFB Cables	
	Minimum	Maximum	Minimum	Maximum
92 (33.3)	139 (63)	169 (77)	66 (30)	76 (34)
90 (32.2)	103 (47)	168 (76)	65 (30)	75 (34)
88 (31.1)	102 (46)	167 (76)	65 (30)	75 (34)
86 (30.5)	100 (45)	165 (75)	64 (29)	74 (34)
84 (28.9)	98 (44)	163 (74)	64 (29)	74 (34)
82 (27.8)	97 (44)	162 (73)	63 (29)	73 (33)
80 (26.7)	95 (43)	160 (72)	63 (29)	73 (33)
78 (25.5)	94 (43)	159 (72)	62 (28)	72 (33)
76 (24.4)	93 (42)	158 (72)	62 (28)	72 (33)
74 (23.3)	92 (42)	157 (71)	61 (28)	71 (32)
72 (22.2)	90 (41)	155 (70)	61 (28)	71 (32)
70 (21.1)	89 (40)	154 (70)	60 (27)	70 (32)
68 (20.0)	87 (39)	152 (69)	59 (27)	69 (31)
66 (18.9)	85 (39)	150 (68)	59 (27)	69 (31)
64 (17.8)	84 (38)	149 (68)	58 (26)	68 (31)
62 (16.7)	83 (37)	148 (67)	58 (26)	68 (31)
60 (15.5)	82 (37)	147 (67)	57 (26)	67 (30)
58 (14.4)	80 (36)	145 (66)	57 (26)	67 (30)
56 (13.3)	79 (36)	144 (65)	56 (25)	66 (30)
54 (12.2)	78 (35)	143 (65)	56 (25)	66 (30)
52 (11.1)	75 (34)	140 (64)	55 (25)	65 (30)
50 (10.0)	74 (34)	139 (63)	55 (25)	65 (30)
48 (8.9)	108 (49)	138 (63)	54 (24)	64 (29)
46 (7.8)	107 (49)	137 (62)	54 (24)	64 (29)
44 (6.7)	105 (48)	135 (61)	53 (24)	63 (29)
42 (5.5)	104 (47)	134 (61)	53 (24)	63 (29)
40 (4.4)	103 (47)	133 (60)	52 (24)	62 (28)
38 (3.3)	102 (46)	132 (60)	52 (24)	62 (28)
36 (2.2)	99 (45)	129 (59)	51 (23)	61 (28)
34 (1.1)	98 (44)	128 (58)	51 (23)	61 (28)
32 (0.0)	97 (44)	127 (58)	50 (23)	60 (27)
30 (-1.1)	95 (43)	125 (57)	50 (23)	60 (27)

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Table 502/27-51-00-993-801 (Continued)

Temperature	Cable Tension Pounds (kilograms)			
Degrees F (Degrees C)	WFA/WFB Cables		WFFA/WFFB Cables	
	Minimum	Maximum	Minimum	Maximum
28 (-2.2)	94 (43)	124 (56)	50 (23)	60 (27)
26 (-3.3)	93 (42)	123 (56)	49 (22)	59 (27)
24 (-4.4)	92 (42)	122 (55)	49 (22)	59 (27)
22 (-5.5)	89 (40)	119 (54)	48 (22)	58 (26)
20 (-6.6)	88 (40)	118 (54)	48 (22)	58 (26)
18 (-7.8)	87 (39)	117 (53)	47 (21)	57 (26)
16 (-8.9)	86 (39)	116 (53)	47 (21)	57 (26)
14 (-10.0)	84 (38)	114 (52)	46 (21)	56 (25)
12 (-11.1)	83 (37)	113 (51)	46 (21)	56 (25)
10 (-12.2)	82 (37)	112 (51)	45 (20)	55 (25)
8 (-13.3)	81 (37)	111 (50)	45 (20)	55 (25)
6 (-14.4)	80 (36)	110 (50)	44 (20)	54 (24)
4 (-15.5)	77 (35)	107 (49)	44 (20)	54 (24)
2 (-16.6)	76 (34)	106 (48)	43 (20)	53 (24)
0 (-17.8)	75 (34)	105 (48)	43 (20)	53 (24)
-2 (-18.9)	74 (34)	104 (47)	42 (19)	52 (24)
-4 (-20.0)	72 (33)	102 (46)	42 (19)	52 (24)
-6 (-21.1)	71 (32)	101 (46)	42 (19)	52 (24)
-8 (-22.2)	70 (32)	100 (45)	41 (19)	51 (23)
-10 (-23.3)	69 (31)	99 (45)	41 (19)	51 (23)
-12 (-24.4)	67 (30)	97 (44)	40 (18)	50 (23)
-14 (-25.5)	66 (30)	96 (44)	40 (18)	50 (23)
-16 (-26.6)	64 (29)	94 (43)	39 (18)	49 (22)
-18 (-27.7)	63 (29)	93 (42)	39 (18)	49 (22)
-20 (-28.8)	62 (28)	92 (42)	38 (17)	48 (22)
-22 (-30.0)	61 (28)	91 (41)	38 (17)	48 (22)
-24 (-31.1)	60 (27)	90 (41)	37 (17)	47 (21)
-26 (-32.2)	58 (26)	88 (40)	37 (17)	47 (21)
-28 (-33.3)	56 (25)	86 (39)	37 (17)	47 (21)
-30 (-34.4)	55 (25)	85 (39)	36 (16)	46 (21)
-32 (-35.5)	54 (24)	84 (38)	36 (16)	46 (21)
-34 (-36.7)	53 (24)	83 (37)	35 (16)	45 (20)

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Table 502/27-51-00-993-801 (Continued)

Temperature	Cable Tension Pounds (kilograms)			
Degrees F (Degrees C)	WFA/WFB Cables		WFFA/WFFB Cables	
	Minimum	Maximum	Minimum	Maximum
-36 (-37.8)	52 (24)	82 (37)	35 (16)	45 (20)
-38 (-38.9)	51 (23)	81 (37)	34 (15)	44 (20)
-40 (-40.0)	50 (23)	80 (36)	34 (15)	44 (20)

NOTE: The minimum (Low) cable tension listed in the table above maintenance action and should not be construed as an in-service operational limit for cable tension.

SUBTASK 27-51-00-820-002

(5) Make sure that the flap power drive unit [17] is in the rigged position.

(a) Make sure that the black rig mark on the cable drum [12] extends past the white cable guard [13] a minimum of 0.03 in. (0.76 mm) on each side.

WARNING DO NOT RETRACT THE POWER DRIVE UNIT PAST THE RIG MARK IN THE UP DIRECTION. YOU CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) If the power drive unit [17] is not in the rigged position, use the manual drive to move the power drive unit [17].

SUBTASK 27-51-00-820-003

(6) Install the rig pin [27] from the rigging pin kit, SPL-1585, in the flap control unit [4].

NOTE: This will make sure the flap control unit [4] is in the adjusted position.

(a) Remove the screws [25], washers [32] and the lower pan [26] from the flap control unit [4].

(b) Install the rig pin [27] through the cam follower arm and the valve cam.

SUBTASK 27-51-00-820-004

(7) Adjust the tension in the follow-up cables WFFA and WFFB.

(a) Remove the clips [8] from the turnbuckles [10] on the cables [9] and cables [11].

(b) If you installed a new cable [9] or cable [11], use the turnbuckles to adjust the tension in the cables to 125 +10 / -0 lbf (0.56 +0.05 / -0.00 kN)).

1) Use the low tension cable tensiometer, COM-1553, to measure the tension of the cable.

(c) Use the turnbuckles [10] to adjust the tension in the cables WFFA and WFFB to the limit shown in (Table 502).

NOTE: If the temperature is 30°F (-1.11°C) or higher the minimum tension is 50 lbf (0.22 kN). If the temperature is less than 30°F (-1.11°C) the minimum tension is 30 +10 / -0 lbf (0.13 +0.05 / -0.00 kN).

NOTE: To make sure that the tension is correct, the temperature of the airframe must be constant in the limit of ±5 degrees for a minimum of one hour.

1) During the adjustment, make sure that the tension in the cables is never less than the minimum.

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2) Use the low tension cable tensiometer, COM-1553, to measure the tension of the cable.

(d) Make sure you can easily remove and install the rig pin [27].

(e) Install the clips [8] on the turnbuckles [10].

SUBTASK 27-51-00-820-005

(8) Make a preliminary adjustment of the trailing edge flap control valve [19].

NOTE: This adjustment will make sure that the ballscrew nuts do not hit the upstops on the ballscrews when you retract the flaps. The final adjustment will be done after the flap drive system is adjusted.

(a) Measure the distance the valve slide [29] extends past the index surface [28] on the trailing edge flap control valve [19].

(b) If the valve slide [29] does not extend past the index surface [28] in the limit of 0.0 in. (0.00 mm) to 0.03 in. (0.76 mm), then adjust the trailing edge flap control valve [19]:

1) Loosen the nut [30] and nut [33] on the adjustable sleeve [31].

2) Turn the adjustable sleeve [31] to change the position of the rod end [24] until the valve slide [29] and the index surface [28] align in the limit.

3) Tighten the nut [30] and nut [33] on the adjustable sleeve [31].

4) Make sure that the distance between the center of the rod end [24] and the nut [33] is less than 0.94 in. (23.88 mm).

5) Make sure that the distance between the nut [30] and the hex on the adjustable sleeve [31] is less than 0.19 in. (4.83 mm).

CAUTION

MAKE SURE THAT YOU REMOVE THE HYDRAULIC POWER BEFORE YOU DISCONNECT THE ROD END FROM THE SUMMING LEVER. IF YOU DISCONNECT THE ROD END WITH HYDRAULIC POWER ON, YOU CAN CAUSE DAMAGE TO THE FLAP CONTROL UNIT.

6) If it is necessary, disconnect the rod end [24] from the summing lever [20].

NOTE: It is necessary to disconnect the rod end [24] if you cannot align the valve slide [29] and index surface [28] while the adjustable sleeve [31] is in the limits.

a) Remove the nut [23], washer [22], and bolt [21] that attach the rod end [24] to the summing lever [20].

b) Change the position of the adjustable sleeve [31].

c) Install the bolt [21], washer [22] and nut [23] to attach the rod end [24] to the summing lever [20].

7) Tighten the nut [30] and nut [33].

NOTE: It is not necessary to install the lockwire on the nuts now. You will install lockwire after the final adjustment is complete.

SUBTASK 27-51-00-820-006

(9) Do this task: Trailing Edge Flap Drive System Adjustment, TASK 27-51-00-820-802.

SUBTASK 27-51-00-420-002

(10) Connect the torque tubes [18] to the flap power drive unit [17].

(a) Remove the clamp set, SPL-1726, from the torque tubes [18].

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(b) Make sure that the ballscrew nuts for transmission No. 4 and No. 5 are 1/2 turn ±10 degrees (±0.25 inch) (±6.3 millimeters) from the upstop on the ballscrew yoke.

(c) Apply grease, D00633, to the coupling sleeve [15].

(d) Put the splined couplings [16] in the coupling sleeve [15].

(e) Install the screws [14] to attach the torque tubes [18] to the power drive unit [17].

(f) Install MS20995C32 lockwire, G01048, or safety cable kit, G51490, on the screws [14] (PAGEBLOCK 20-10-44/401).

SUBTASK 27-51-00-420-003

(11) For the flap position transmitters on the left and right wings, do this task: Flap Position Transmitter Installation, TASK 27-58-01-400-801.

SUBTASK 27-51-00-860-046

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(12) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-51-00-860-047

(13) Use the primary system to extend the TE flaps to the 40-unit position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

SUBTASK 27-51-00-860-048

(14) Use the primary system to retract the TE flaps to the UP position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

NOTE: Wait a minimum of 2 minutes before you remove the hydraulic power. The flaps can move a small distance during this time.

SUBTASK 27-51-00-860-049

(15) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-51-00-820-007

(16) Do these steps to do a check of the adjustment of the trailing edge flap control valve [19].

(a) Make sure that you can install the rig pin [27] in the flap control unit [4].

(b) Make sure that the ballscrew nuts for transmission No. 4 and No. 5 are 1/2 turn ±10 degrees (±0.25 inch) (±6.3 millimeters) from the upstop on the ballscrew yoke.

SUBTASK 27-51-00-820-008

(17) If you cannot install the rig pin [27] or if the ballscrew nuts are not in the correct position, then do a final adjustment of the trailing edge flap control valve [19].

NOTE: It can be necessary to adjust the trailing flap control valve more than once to get it to the correct position.

(a) Loosen the nut [30] and nut [33] on the adjustable sleeve [31].

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CAUTION ADJUST THE FLAP CONTROL VALVE A SMALL DISTANCE AT A TIME. IF YOU ADJUST THE FLAP CONTROL VALVE TOO FAR, THE BALLSCREW NUTS CAN HIT THE UPSTOPS. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(b) Turn the adjustable sleeve [31] to change the position of the valve slide [29].

NOTE: If the flaps did not retract far enough, adjust the valve slide [29] to be further out of the trailing edge flap control valve [19]. If the flaps retracted too far, adjust the valve slide [29] to be further in the trailing edge flap control valve [19].

(c) Tighten the nut [30] and nut [33] on the adjustable sleeve [31].

NOTE: Make sure that the distance between the center of the rod end [24] and the nut [33] is less than 0.94 in. (23.88 mm). Make sure the distance between the nut [30] and the hex on the adjustable sleeve [31] is less than 0.19 in. (4.83 mm).

CAUTION MAKE SURE THAT YOU REMOVE THE HYDRAULIC POWER BEFORE YOU DISCONNECT THE ROD END FROM THE SUMMING LEVER. IF YOU DISCONNECT THE ROD END WITH HYDRAULIC POWER ON, YOU CAN CAUSE DAMAGE TO THE FLAP CONTROL UNIT.

(d) If it is necessary, disconnect the rod end [24] from the summing lever [20].

NOTE: It is necessary to disconnect the rod end [24] if you cannot adjust the valve slide [29] enough to get the flaps in the correct position.

1) Remove the nut [23], washer [22] and bolt [21] that attach the rod end [24] to the summing lever [20].
2) Change the position of the adjustable sleeve [31].
3) Install the bolt [21], washer [22] and nut [23] to attach the rod end [24] to the summing lever [20].

(e) Tighten the nut [30] and nut [33].

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES, CABLES, AND LANDING GEAR AREAS. PRESSURIZATION OF A HYDRAULIC SYSTEM CAN MOVE CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS. MOVEMENT OF CONTROL SURFACES, CABLES, LANDING GEAR, AND LANDING GEAR DOORS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(f) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(g) Use the primary system to fully extend and retract the TE flaps a minimum of five times, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(h) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(i) Make sure that you can install the rig pin [27] in the flap control unit [4].

(j) Make sure that the ballscrew nuts for transmission No. 4 and No. 5 are 1/2 turn ±10 degrees (±0.25 inch) (±6.3 millimeters) from the upstop on the ballscrew yoke.

SUBTASK 27-51-00-420-004

(18) Install lockwire on the nut [30], nut [33] and the adjustable sleeve [31].

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I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-410-014

(1) Close this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

(b) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

SUBTASK 27-51-00-410-005

(2) Install the floor panel above the flap control unit [4], do this task: Passenger Cabin Floor Panel Installation, TASK 53-21-00-400-801.

SUBTASK 27-51-00-410-006

(3) Install the lower pan [26] on the flap control unit [4].

(a) Remove the rig pin [27].

(b) Put the lower pan [26] in its location on the flap control unit [4].

WARNING

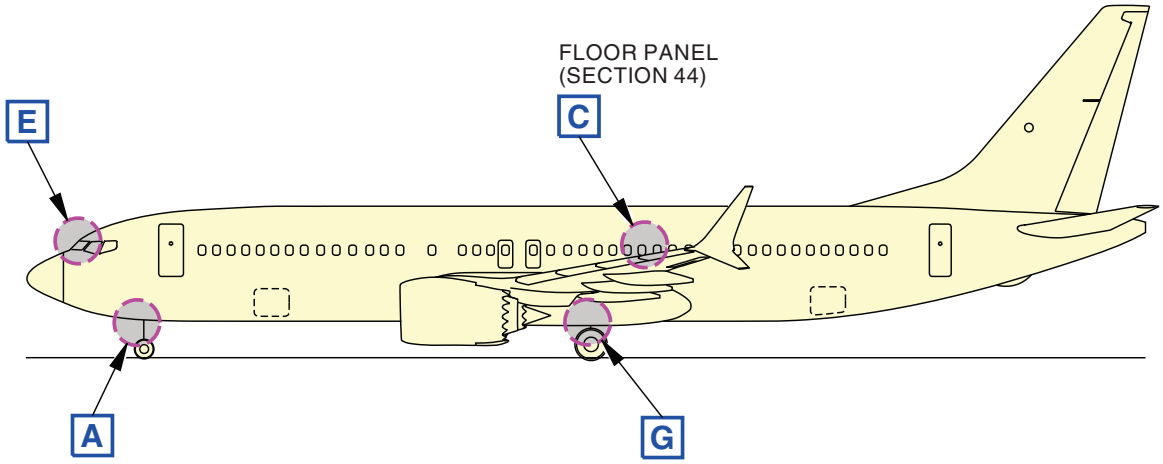
USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

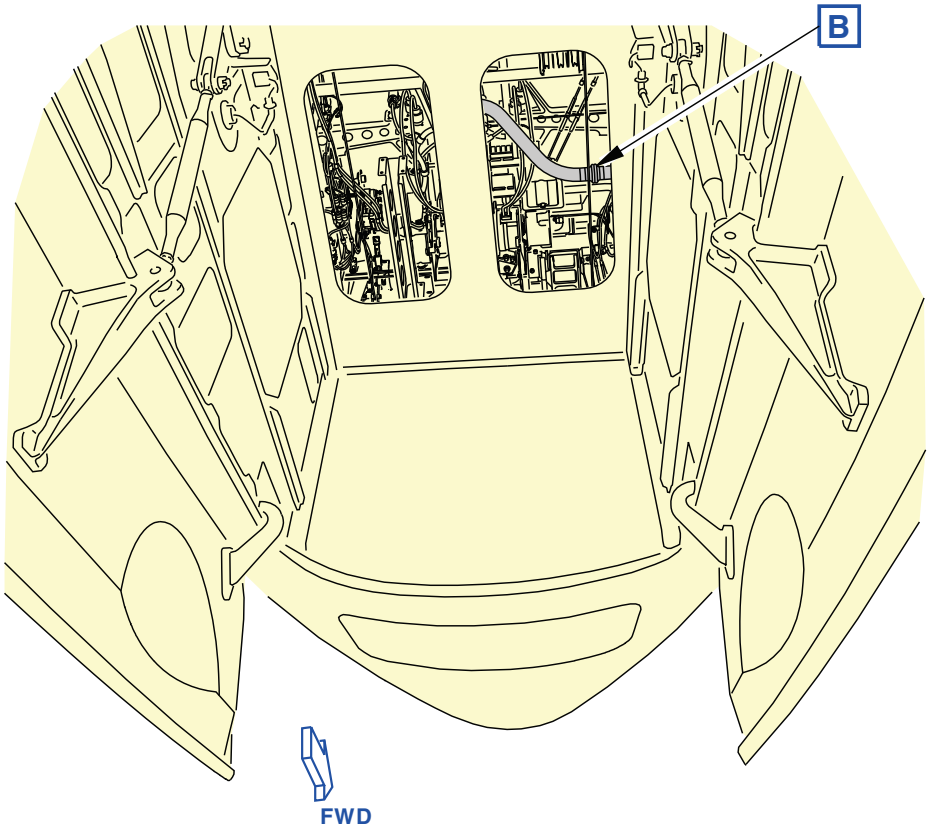
(c) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the screws [25].

(d) Install the washers [32] and screw [25] to attach the lower pan [26].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





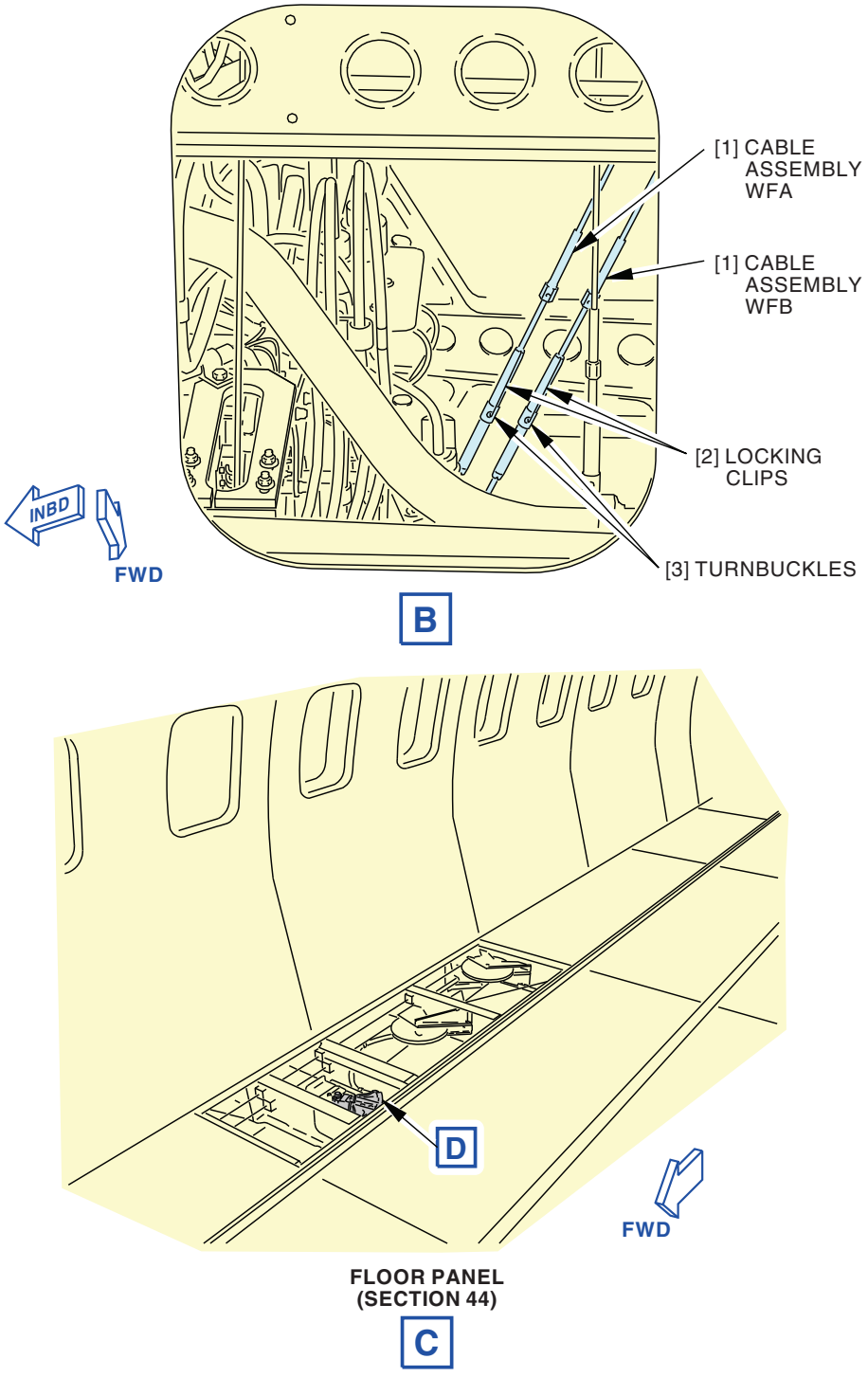
NOSE LANDING GEAR WHEEL WELL

A

2412738 S00061531251_V2

Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 1 of 9)

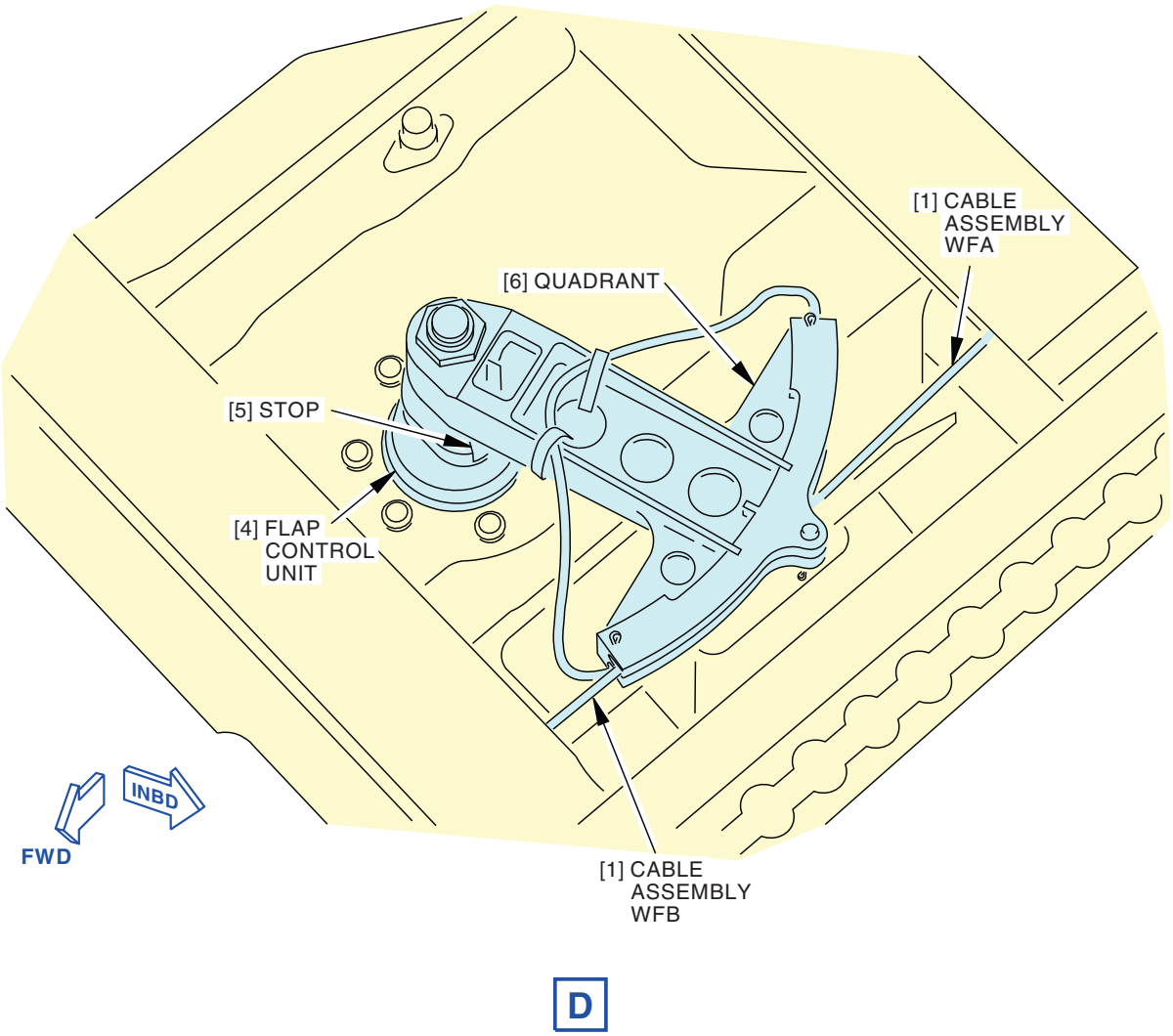
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412739 S00061531252_V1

Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 2 of 9)

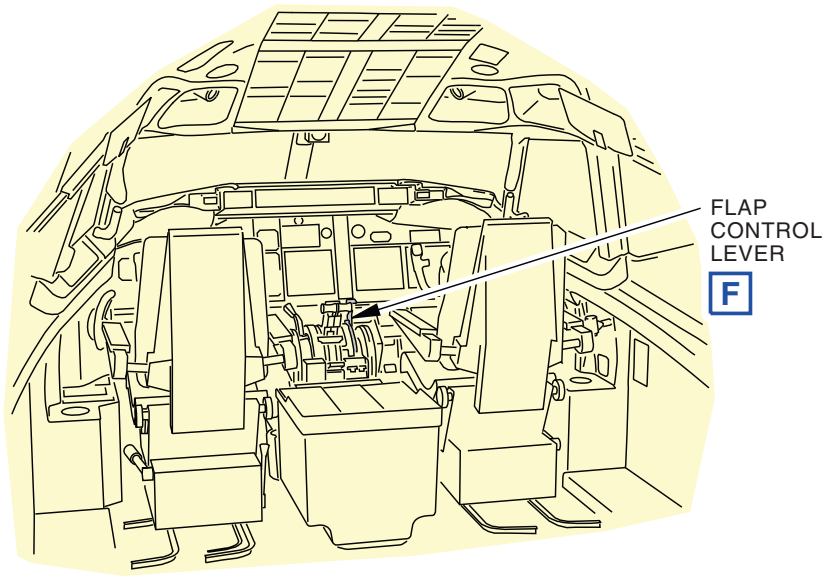
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412740 S00061531253_V1

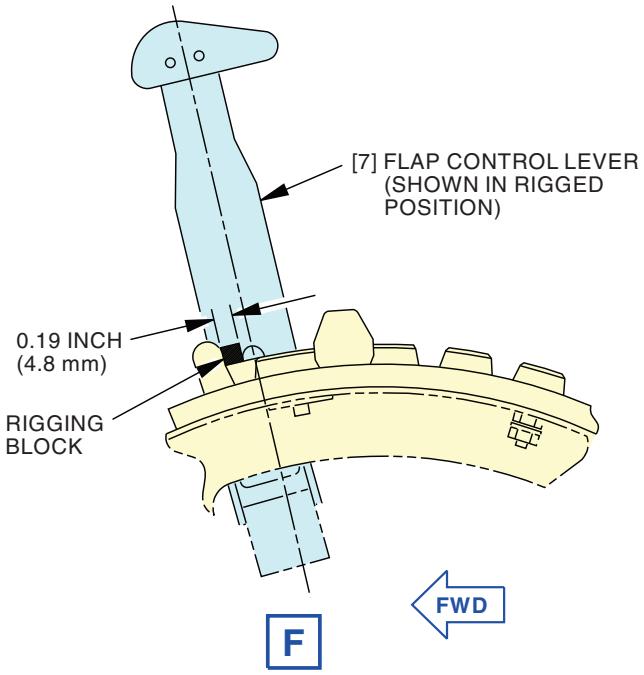
Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT

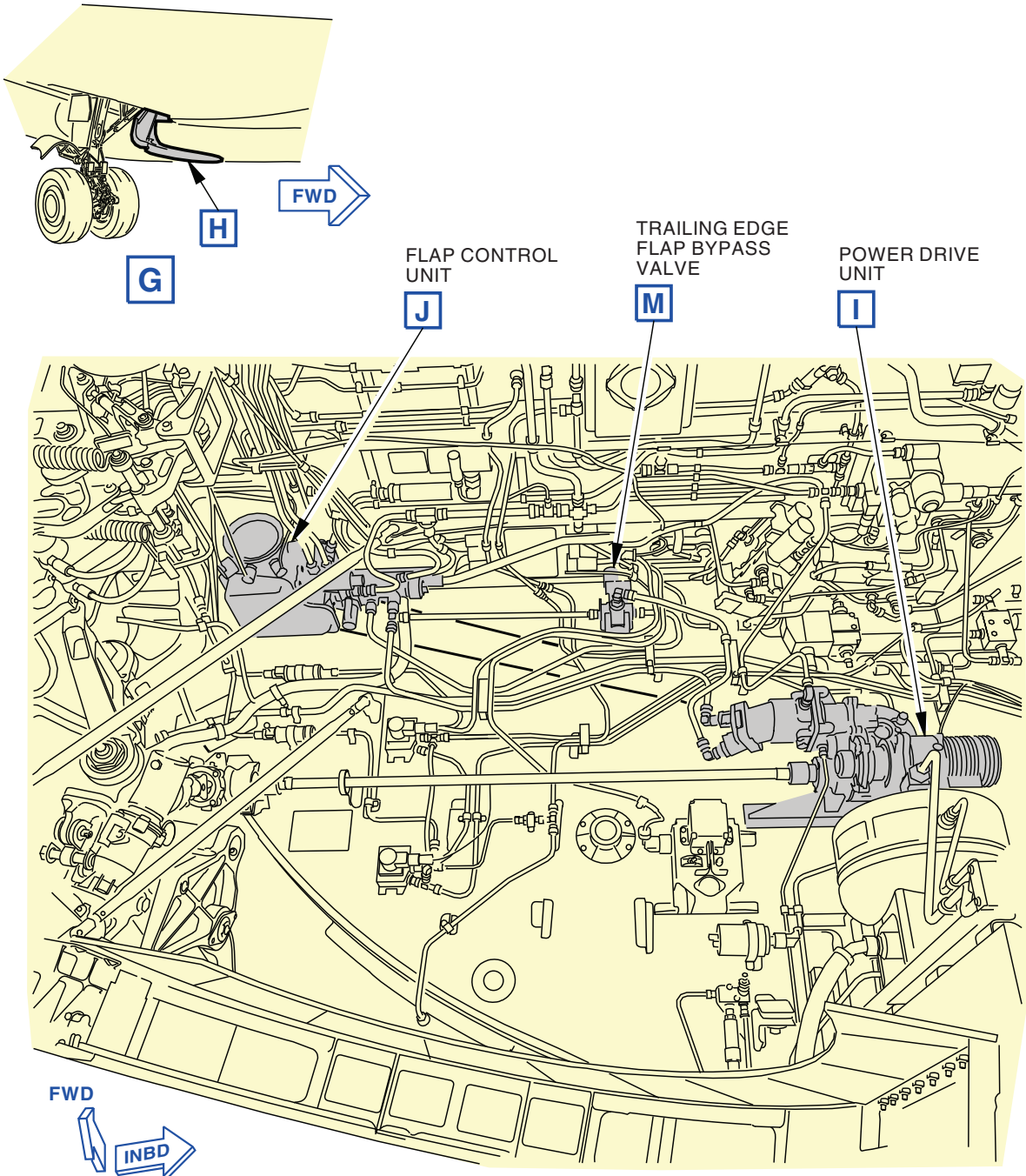
E



2412741 S00061531254_V1

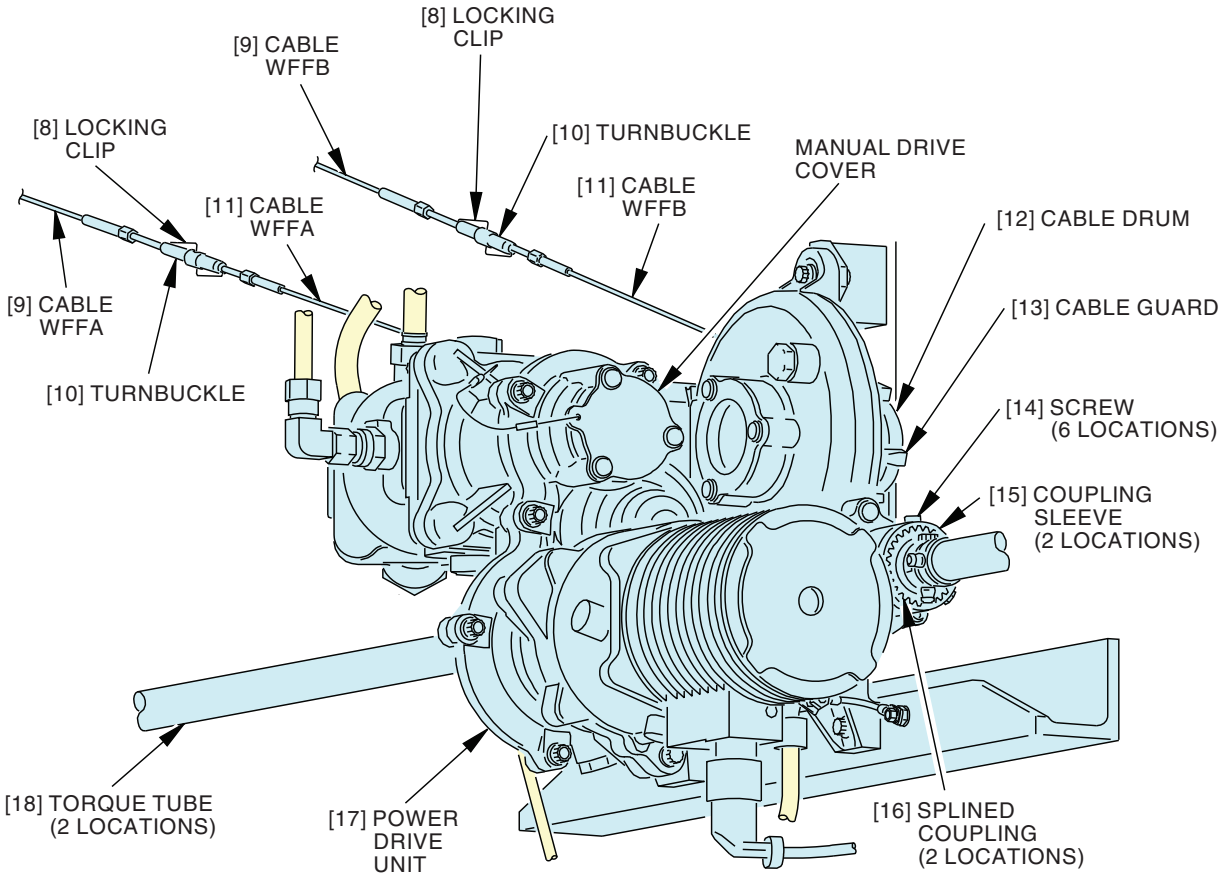
Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



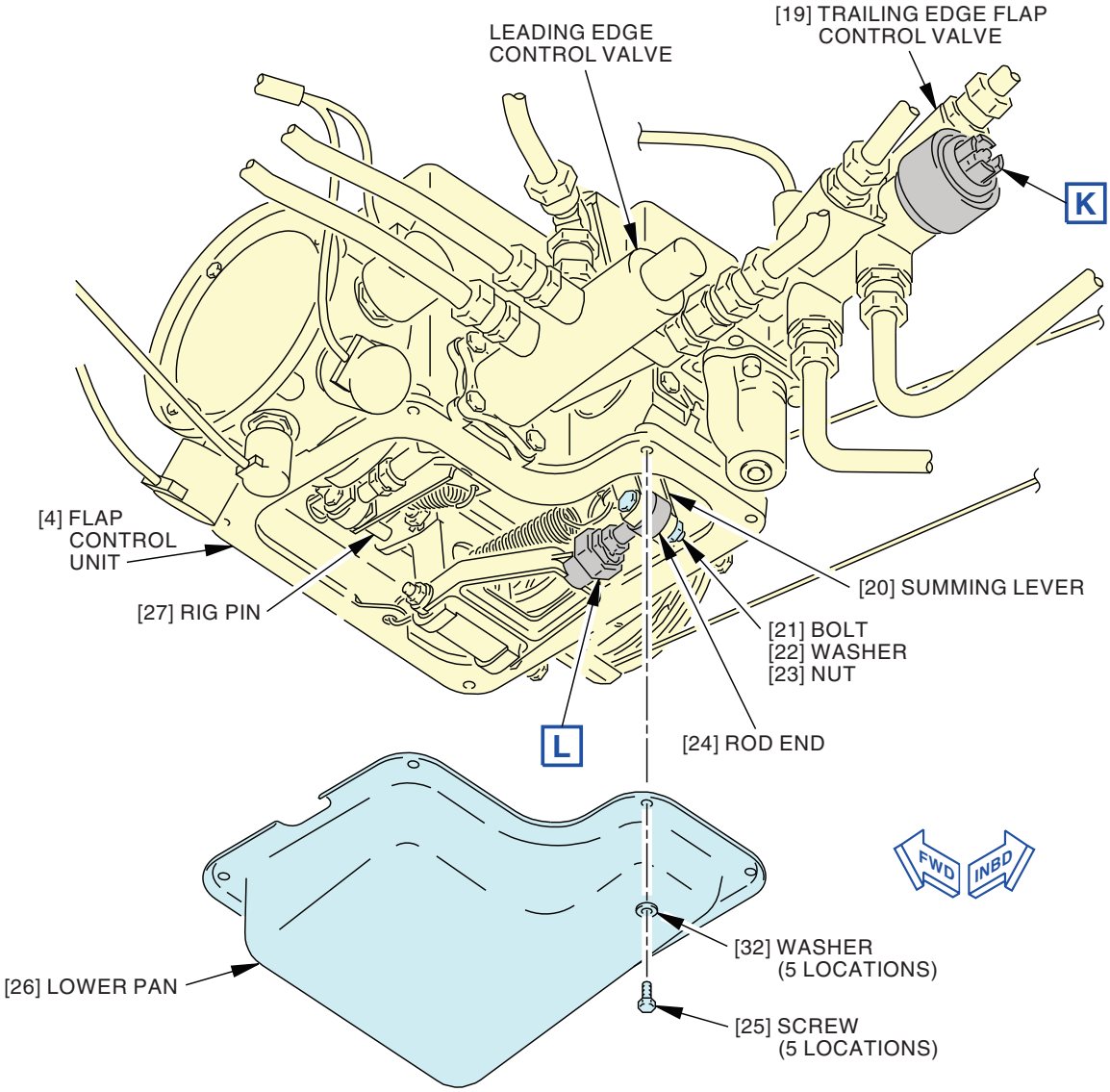
POWER DRIVE UNIT

I

2412743 S00061531256_V1

Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 6 of 9)

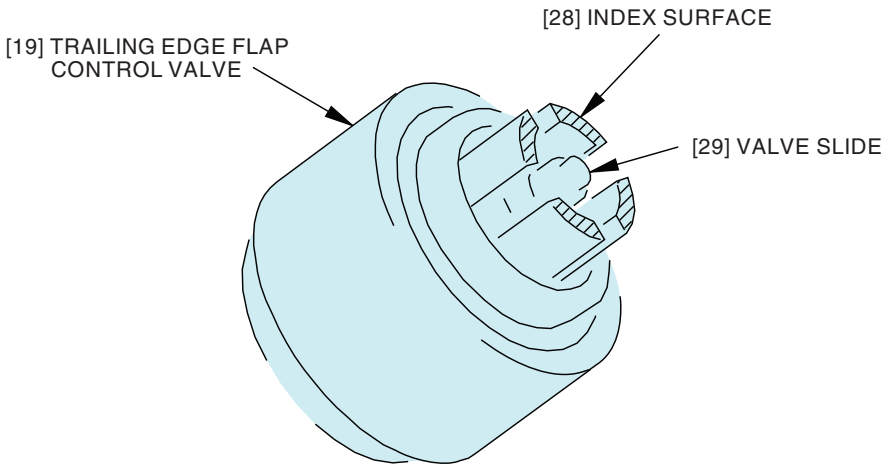
ECCN 9E991 BOEING PROPRIETARY - See title page for details

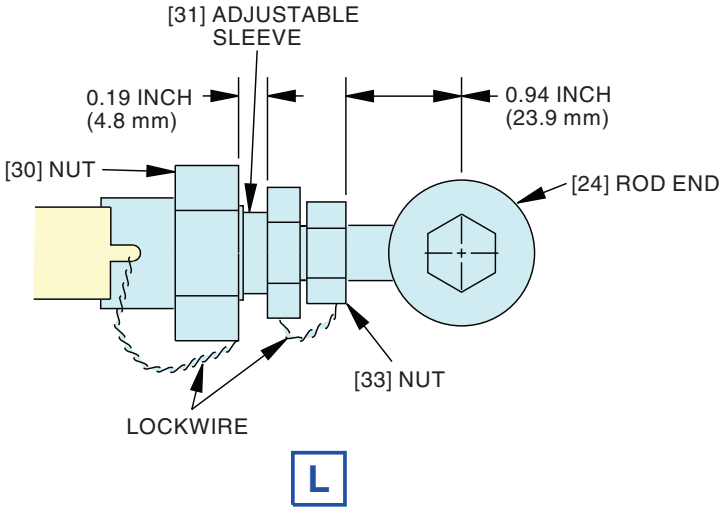


2412744 S00061531257_V1

Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

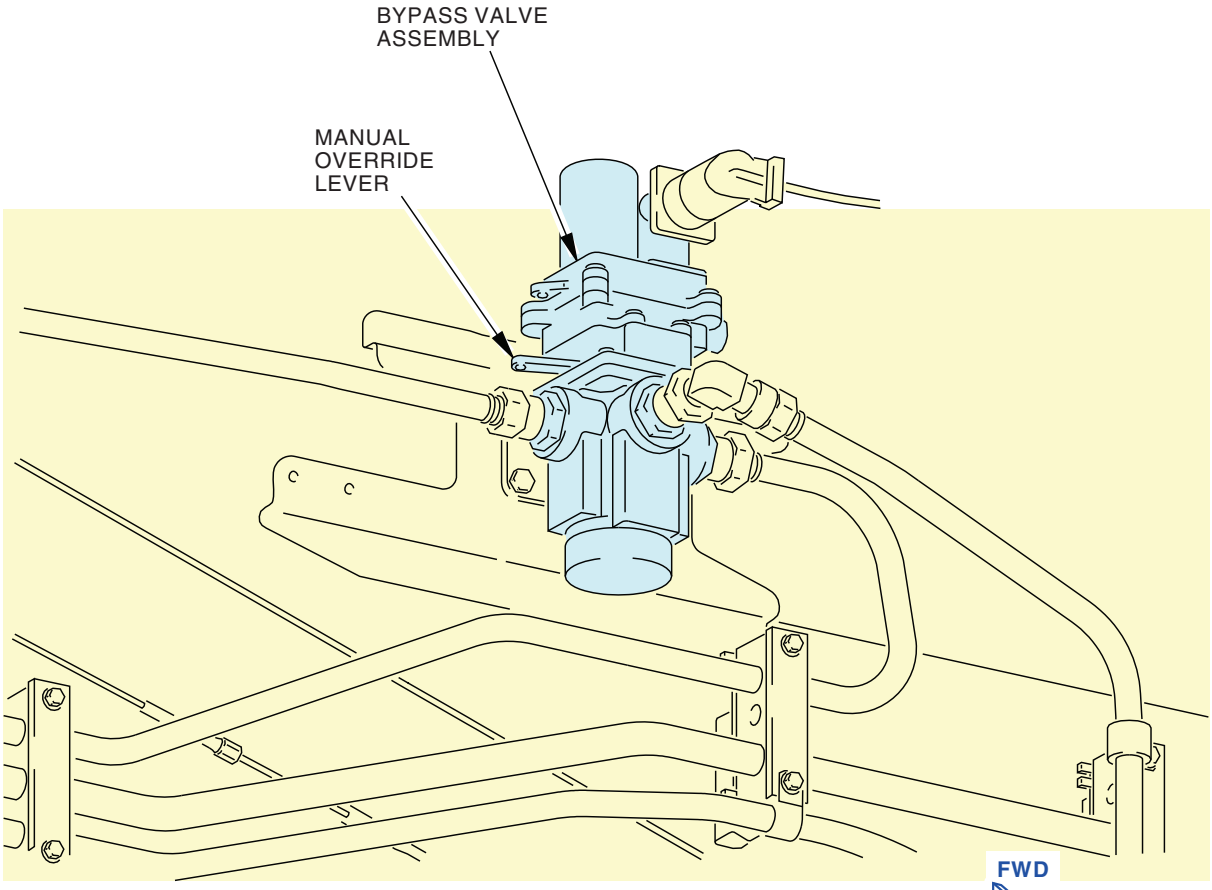




2412745 S00061531258_V1

Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TRAILING EDGE FLAP BYPASS VALVE

M

2804321 S0000643010_V1

Trailing Edge Flap Control System Adjustment
Figure 502/27-51-00-990-808 (Sheet 9 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-820-802

4. Trailing Edge Flap Drive System Adjustment

(Figure 503)

A. General

(1) This task does a check of the adjustment and an adjustment of the trailing edge flap drive system. The procedure to adjust the inboard flaps is in (TASK 27-51-11-820-801). The procedure to adjust the outboard flaps is in (TASK 27-51-21-820-801).

(a) The trailing edge flap drive system includes the flap transmissions, angle gearboxes, torque tubes, and the flap power drive unit.

B. References

Reference	Title
20-10-44 P/B 401	LOCKING DEVICES - REMOVAL/INSTALLATION
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-980-801	Trailing Edge Flap System Manual Operation (P/B 201)
27-51-11-820-801	Inboard Trailing Edge Flap Adjustment (P/B 501)
27-51-21-820-801	Outboard Trailing Edge Flap Adjustment (P/B 501)
27-58-01-000-801	Flap Position Transmitter Removal (P/B 401)
27-58-01-400-801	Flap Position Transmitter Installation (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-5384	Wrench - Coupling Sleeve Flap Actuation Part #: C27041-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51490	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 18 Inches Long	BACC13AT3K18, AMS 5689

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
541	Left Wing - Fairing Flap Support No. 4

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(Continued)

Zone	Area
542	Left Wing - Fairing Flap Support No. 3
543	Left Wing - Fairing Flap Support No. 2
544	Left Wing - Fairing Flap Support No. 1
550	Subzone - Left Wing: Trailing Edge, Aft of Rear Spar, Inbd of Outboard Trailing Edge Flap
560	Subzone - Left Wing: Trailing Edge , Aft of Rear Spar, Outboard of Inbd Trailing Edge Flap, Inbd of Fixed Trailing Edge
641	Right Wing - Fairing Flap Support No. 5
642	Right Wing - Fairing Flap Support No. 6
643	Right Wing - Fairing Flap Support No. 7
644	Right Wing - Fairing Flap Support No. 8
650	Subzone - Right Wing: Trailing Edge, Aft of Rear Spar, Inboard of Outboard Trailing Edge Flap
660	Subzone - Right Wing: Trailing Edge, Aft of Rear Spar, Outboard of Inboard Trailing Edge Flap, Inboard of Fixed Trailing Edge

F. Do a Check of the Trailing Edge Flap Drive System Adjustment

SUBTASK 27-51-00-040-006

(1) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-00-480-011

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-00-980-006

(3) Manually extend the flaps until the ballscrew nuts touch the downstops on the ballscrews:

(a) If the torque tubes are not connected to the flap power drive unit, manually extend the flaps on each wing:

1) If all of the torque tubes on the wing are connected, use a coupling sleeve wrench, SPL-5384, on the torque tube couplings inboard of the tee-angle gearbox.

CAUTION DO NOT MOVE ONE END OF A FLAP MORE THAN 1.4 INCHES (35.5 MILLIMETERS) OF BALLSCREW TRAVEL MORE THAN THE OTHER END. YOU CAN CAUSE THE FLAP TO SKEW TO MUCH. THIS CAN CAUSE DAMAGE TO THE FLAP.

2) If all of the torque tubes on the wing are not connected, use a coupling sleeve wrench, SPL-5384, to drive each flap independently.

(b) If the torque tubes are connected to the flap power drive unit, and all of the torque tubes are connected on each wing, use the manual drive to extend the flaps. To use the manual drive, do this task: Trailing Edge Flap System Manual Operation, TASK 27-51-00-980-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-00-220-001

(4) Do a check of the adjustment of the flap drive system on each wing:

NOTE: If all of the torque tubes on the wing are not connected, it is not necessary to do this check.

(a) At all of the transmissions, measure and record the dimension from the downstop on the ballscrew nut to the downstop on the ballscrew.

NOTE: This dimension is referred to as Dimension Y.

(b) Make sure the Dimension Y for the transmission No. 4 and transmission No. 5 are in these limits:

1) If the torque tubes are connected between the wings, Dimension Y must be 0.0 (the ballscrew nut touches the downstop) for one transmission and in the limit of 0.0 in. (0.00 mm) to 0.1 in. (2.54 mm) for the other transmission.

2) If the torque tubes are not connected between the wings, Dimension Y must be 0.0 (the ballscrew nut touches the downstop) for the two transmission.

(c) On the left wing flap drive system:

1) Make sure Dimension Y for transmissions No. 1 and No. 3 are in the limit of 0.0 in. (0.00 mm) to 0.2 in. (5.08 mm) more than Dimension Y at transmission No. 4.

2) Make sure Dimension Y for transmission No. 2 is in the limit of 0.0 in. (0.00 mm) to 0.22 in. (5.59 mm) more than Dimension Y at transmission No. 1.

(d) On the right wing flap drive system:

1) Make sure Dimension Y for transmissions No. 6 and No. 8 are in the limit of 0.0 in. (0.00 mm) to 0.2 in. (5.08 mm) more than Dimension Y at transmission No. 5.

2) Make sure Dimension Y for transmission No. 7 is in the limit of 0.0 in. (0.00 mm) to 0.22 in. (5.59 mm) more than Dimension Y at transmission No. 8.

SUBTASK 27-51-00-820-009

(5) If the flap drive system is not adjusted correctly on each wing, do the trailing edge flap drive system adjustment.

G. Trailing Edge Flap Drive System Adjustment

SUBTASK 27-51-00-020-005

(1) For the flap position transmitters on the left and right wings, do this task: Flap Position Transmitter Removal, TASK 27-58-01-000-801.

NOTE: If you do not remove the flap position transmitters before you disconnect the torque tubes, they will become incorrectly adjusted.

SUBTASK 27-51-00-020-006

(2) Disconnect the torque tubes [18] from the flap power drive unit [17] (Figure 502):

NOTE: This will disconnect the flap drive system for the left wing from the right wing. It is necessary to adjust the left and the right wings independently, then connect them together.

(a) Remove lockwire or safety cable from the screws [14].

(b) Remove the screws [14] that attach the coupling sleeves [15] to the splined couplings [16].

(c) Move the coupling sleeves [15] off the splined couplings [16].

(d) Move the torque tubes [18] away from the flap power drive unit [17].

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SUBTASK 27-51-00-020-007

(3) Disconnect the torque tube from the inboard side of transmission No. 2 or transmission No. 7 (Figure 503):

NOTE: This will disconnect the flap drive system for the outboard flaps from the inboard flaps.

(a) Remove lockwire or safety cable from the screws [14].

(b) Remove the screws [14] that attach the coupling sleeves [15] to the splined couplings [16].

(c) Move the coupling sleeves [15] off the splined couplings [16].

(d) Move the torque tubes away from the transmission.

SUBTASK 27-51-00-820-010

(4) Adjust the flap drive system for the outboard flap:

NOTE: This procedure is applicable to the drive system for the outboard flap on the left wing and the right wing.

CAUTION

DO NOT MOVE ONE END OF A FLAP MORE THAN 1.4 INCHES (35.5 MILLIMETERS) OF BALLSCREW TRAVEL MORE THAN THE OTHER END. YOU CAN CAUSE THE FLAP TO SKEW TO MUCH. THIS CAN CAUSE DAMAGE TO THE FLAP.

(a) Use a coupling sleeve wrench, SPL-5384, on the coupling on the inboard side of transmission No. 2 to extend the outboard flap until one of the ballscrew nuts touch the downstop on the ballscrew.

(b) Make sure the ballscrew nut at transmission No. 1 touches the downstop (Dimension Y = 0).

(c) Make sure the Dimension Y for transmission No. 2 is in the limit of 0.0 in. (0.00 mm) to 0.2 in. (5.08 mm).

(d) If the ballscrew nuts for transmissions No. 1 and No. 2 are not in the limits, adjust the drive system for the outboard flap:

1) Disconnect the torque tubes between transmissions No. 1 and No. 2:

a) Remove lockwire or safety cable from the screws [14].

b) Remove the screws [14] that attach the coupling sleeves [15] to the splined couplings [16].

c) Move the coupling sleeve [15] off the splined coupling [16].

d) Move the torque tube away from the transmission.

2) Manually extend the transmissions until the ballscrew nuts for transmissions No. 1 and transmission No. 2 touch the downstops.

3) Connect the torque tubes between the transmissions No. 1 and No. 2:

NOTE: To align the torque tube couplings, you can retract the transmission No. 2 a maximum of 1/3 of a torque tube turn.

a) Apply grease, D00633, to the coupling sleeve [15].

b) Put the splined coupling [16] in the coupling sleeve [15].

c) Install the screws [14] to attach the torque tube to the transmission.

d) Install MS20995NC32 lockwire, G50347, or safety cable kit, G51490, on the screws [14] (PAGEBLOCK 20-10-44/401).

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SUBTASK 27-51-00-820-011

(5) Adjust the flap drive system for the inboard flap:

NOTE: This procedure is applicable to the drive system for the inboard flap on the left wing and the right wing.

CAUTION

DO NOT MOVE ONE END OF A FLAP MORE THAN 1.4 INCHES (35.5 MILLIMETERS) OF BALLSCREW TRAVEL MORE THAN THE OTHER END. YOU CAN CAUSE THE FLAP TO SKEW TO MUCH. THIS CAN CAUSE DAMAGE TO THE FLAP.

(a) Use a coupling sleeve wrench, SPL-5384, on the coupling on the inboard side of transmission No. 4 to extend the inboard flap until one of the ballscrew nuts touch the downstop on the ballscrew.

(b) Make sure the ballscrew nut at transmission No. 4 touches the downstop (Dimension Y = 0).

(c) Make sure the Dimension Y for transmission No. 3 is in the limit of 0.0 in. (0.00 mm) to 0.18 in. (4.57 mm).

(d) If the ballscrew nuts for transmissions No. 3 and No. 4 are not in the limits, adjust the drive system for the inboard flap:

1) Disconnect the torque tubes between transmissions No. 3 and No. 4:

a) Remove lockwire or safety cable from the screws [14].

b) Remove the screws [14] that attach the coupling sleeve [15] to the splined coupling [16].

c) Move the coupling sleeve [15] off the splined coupling [16].

d) Move the torque tube away from the transmission.

2) Manually extend the transmissions until the ballscrew nuts for transmissions No. 3 and transmission No. 4 touch the downstops.

3) Connect the torque tubes between the transmissions No. 3 and No. 4.

NOTE: To align the torque tube couplings, you can retract the transmission No. 3 a maximum of 1/3 of a torque tube turn.

a) Apply grease, D00633, to the coupling sleeve [15].

b) Put the splined coupling [16] in the coupling sleeve [15].

c) Install the screws [14] to attach the torque tube to the transmission.

d) Install MS20995NC32 lockwire, G50347, or safety cable kit, G51490, on the screws [14] (PAGEBLOCK 20-10-44/401).

SUBTASK 27-51-00-820-012

(6) Connect the inboard and the outboard flaps:

NOTE: This procedure is applicable to the drive system for the inboard and outboard flaps on the left wing and the right wing.

(a) Make sure the ballscrew nuts at transmission No. 1 and transmission No. 4 touch the downstop (Dimension Y = 0).

(b) Make sure the Dimension Y for transmission No. 2 is in the limit of 0.0 in. (0.00 mm) to 0.2 in. (5.08 mm).

(c) Make sure the Dimension Y for transmission No. 3 is in the limit of 0.0 in. (0.00 mm) to 0.18 in. (4.57 mm).

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(d) Connect the torque tubes between the transmissions No. 2 and No. 3.

NOTE: To align the torque tube couplings, you can retract the transmission No. 2 a maximum of 1/3 of a torque tube turn.

1) Apply grease, D00633, to the coupling sleeve [15].

2) Put the splined coupling [16] in the coupling sleeve [15].

3) Install the screws [14] to attach the torque tube to the transmission.

4) Install MS20995NC32 lockwire, G50347, or safety cable kit, G51490, on the screws [14] (PAGEBLOCK 20-10-44/401).

(e) Make sure that the Dimension Y for transmission No. 2 is in the limit of 0.0 in. (0.00 mm) to 0.4 in. (10.16 mm).

SUBTASK 27-51-00-820-013

(7) Move the flap drive system to the retracted position (Figure 502):

(a) Make sure that the Trailing Edge (TE) Flap Power Bypass Valve is in position 2.

WARNING DO NOT RETRACT THE POWER DRIVE UNIT PAST THE RIG MARK IN THE UP DIRECTION. YOU CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Use the manual drive on the flap power drive unit [17] to retract it to the retracted position.

NOTE: The flap power drive unit is in the retracted position when the black rig mark on the cable drum [12] extends past the white cable guard [13] a minimum of 0.03 in. (0.76 mm) on each side.

(c) Make sure that the flap power drive unit is in the retracted position and install the rig pin [27] in the flap control unit using rig pin F70207-116 from rigging pin kit, SPL-1585.

(d) Use a coupling sleeve wrench, SPL-5384, on the torque tube couplings inboard of the tee-angle gearbox to retract the left and right wings to the retracted position.

NOTE: The flap drive systems for the wings are in the retracted position when the ballscrew nuts are 1/2 turn +/- 10 degrees (+/- 0.25 inch) from the upstop on the yoke.

1) Retract the flap drive systems until the ballscrew nuts for transmission No. 4 and transmission No. 5 are approximately 1/2 turn from the upstop on the yokes.

2) Make sure the distance from the upstop for transmission No. 4 and transmission No. 5 are the same in the limit of 0.25 in. (6.35 mm).

SUBTASK 27-51-00-820-014

(8) Connect the torque tubes [18] to the flap power drive unit [17]:

NOTE: This will connect the flap drive system for the left wing to the right wing.

(a) Do not rotate the torque tubes more than 5 spline teeth, in either direction, to the nearest position where the bolt holes align at the torque tube coupling connection.

1) Do not rotate the Power Drive Unit (PDU) at this step.

(b) Apply grease, D00633, to the coupling sleeve [15].

(c) Put the splined coupling [16] in the coupling sleeve [15].

(d) Install the screws [14] to attach the torque tube [18] to the power drive unit [17].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Install MS20995NC32 lockwire, G50347, or safety cable kit, G51490, on the screws [14] (PAGEBLOCK 20-10-44/401).

SUBTASK 27-51-00-210-003

(9) Make sure the torque trip indicators on the transmissions are set.

SUBTASK 27-51-00-020-008

(10) Remove the rig pin [27] from the flap power control unit.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-420-005

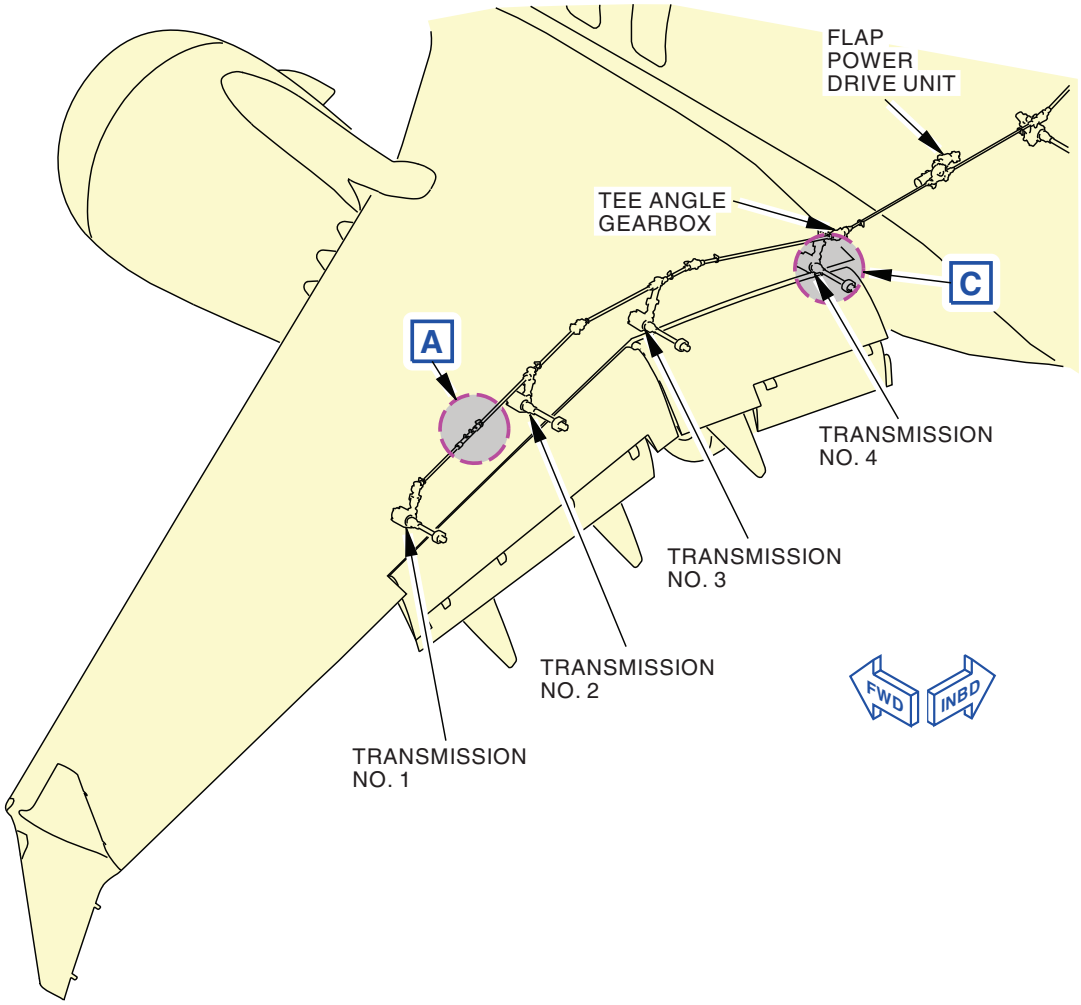
(1) For the flap position transmitters on the left and the right wing, do this task: Flap Position Transmitter Installation, TASK 27-58-01-400-801.

SUBTASK 27-51-00-440-004

(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

——— END OF TASK ———

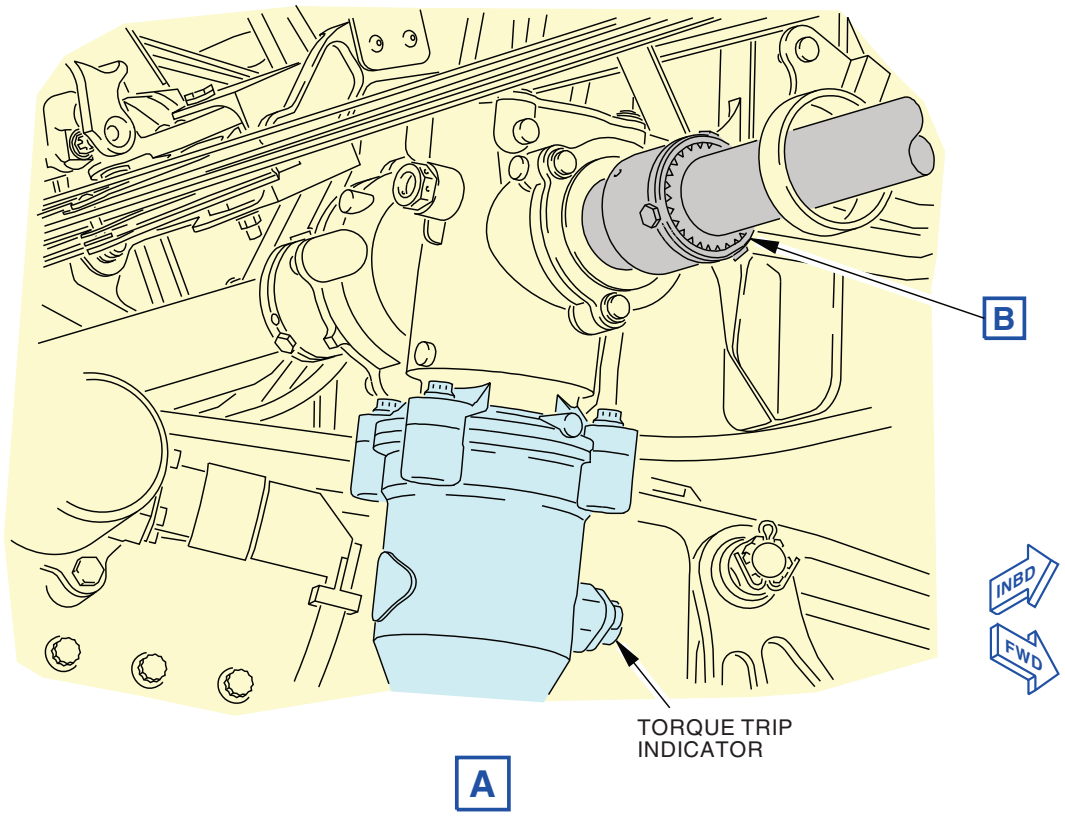
ECCN 9E991 BOEING PROPRIETARY - See title page for details

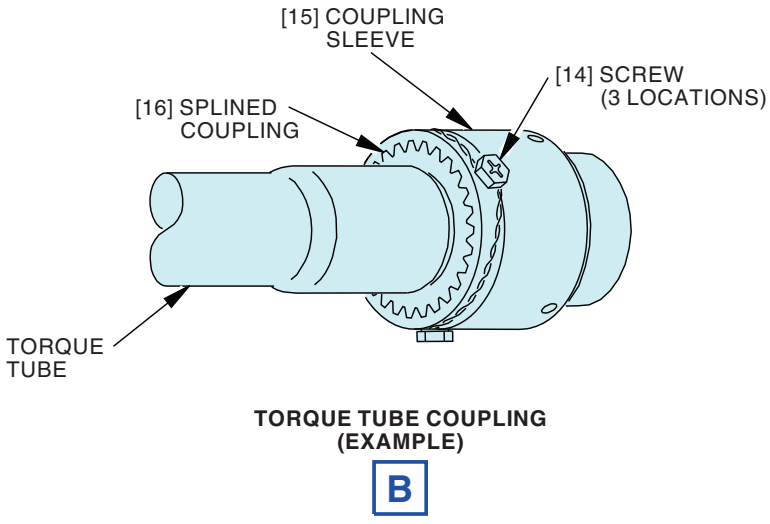


2412746 S00061531260_V1

Trailing Edge Flap Drive System Adjustment
Figure 503/27-51-00-990-809 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





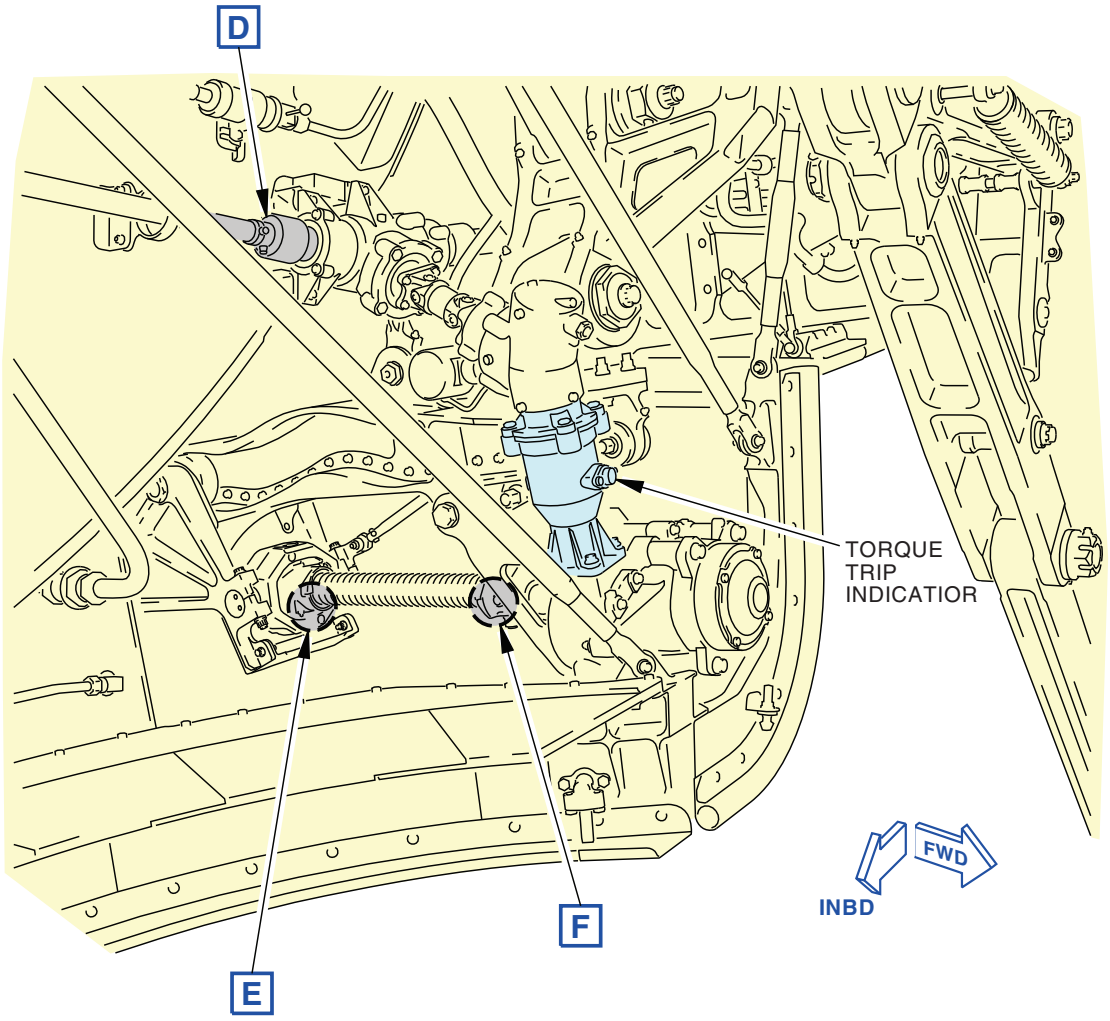
TORQUE TUBE COUPLING
(EXAMPLE)

B

2412748 S00061531262_V1

Trailing Edge Flap Drive System Adjustment
Figure 503/27-51-00-990-809 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



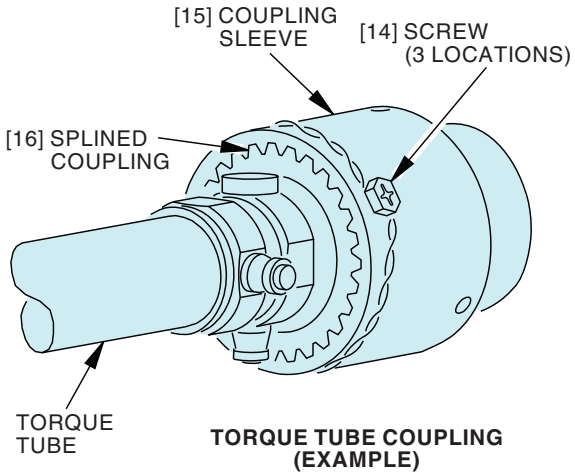
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

C

2412750 S00061531264_V1

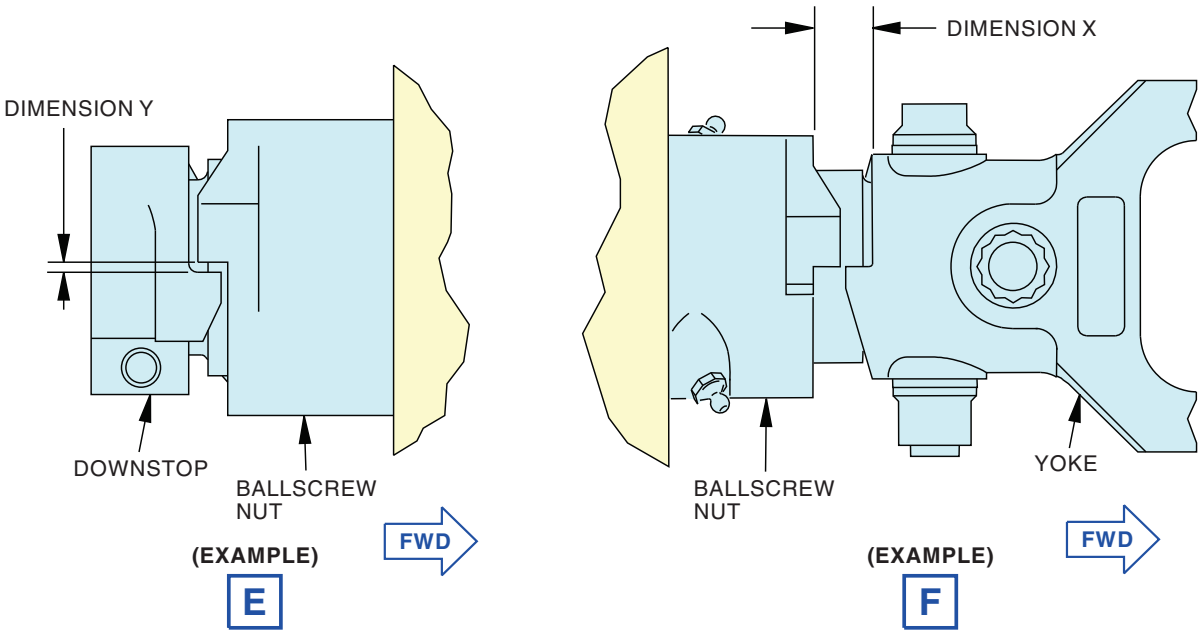
Trailing Edge Flap Drive System Adjustment
Figure 503/27-51-00-990-809 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TORQUE TUBE COUPLING (EXAMPLE)

D



(EXAMPLE) E

(EXAMPLE) F

2412751 S00061531265_V1

Trailing Edge Flap Drive System Adjustment
Figure 503/27-51-00-990-809 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-730-801

5. Trailing Edge Flap System Test

(Figure 501)

A. General

(1) This task does a functional test of the trailing edge flap system. This task includes these parts:

(a) A test of the Flap/Slat Electronics Unit (FSEU) display and a check of existing faults

(b) A test of the asymmetry and skew systems

(c) A test of the load relief system

(d) A test of the primary operation

(e) A test of the power supply for the ALTERNATE FLAPS switches and the bypass valve

(f) A test of the power supply for the flap electric motor

(g) A test of the UP limit switch

(h) A test of the DOWN limit switch

(i) A test of the flap priority valve

(j) A test of the force required to move the flap control lever.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Trailing Edge Flap System Test

SUBTASK 27-51-00-010-005

(1) To get access to the main equipment center, do this step:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Trailing Edge Flap System Test

SUBTASK 27-51-00-730-001

(1) Do this task: FSEU Display and Existing Faults Test, TASK 27-51-00-740-801.

SUBTASK 27-51-00-730-002

(2) Do this task: Trailing Edge Flap Asymmetry Skew System BITE Test, TASK 27-51-00-740-802.

SUBTASK 27-51-00-730-006

(3) Do this task: Trailing Edge Flap Load Relief System BITE Test, TASK 27-51-00-740-803.

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SUBTASK 27-51-00-860-052

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-51-00-730-004

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP-DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Do a test of the primary operation of the trailing edge flaps:

(a) Move the flap control lever to the UP position.

(b) Make sure that the flaps reach the UP position.

(c) Do one of these methods to show the flap lever sensor, position sensors and skew sensor inputs to the FSEU:

1) On the FSEU Built-In-Test Equipment (BITE) display (Method 1):

a) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?

b) Push the NO button until OTHER FUNCTNS? shows on the display.

c) Push the YES button to show SET OUTPUTS? on the display.

d) Push the NO button until I/O MONITOR? shows on the display.

e) Push the YES button to select I/O MONITOR? on the menu.

f) Push the NO button until SYNCHRO INPUTS? shows on the display.

g) Push the YES button to select SYNCHRO INPUTS? from the menu.

2) On the MAX Display System (MDS) (Method 2):

a) Push the SYS button on the Multi Function Display (MFD).

b) Select the MAINT button on the MENU bar at the top right of the System Page.

c) Push the SEL button on the SELECTOR knob.

d) Turn the SELECTOR knob to select MAINT DATA PGS on the Maintenance menu.

e) Move the cursor down and then select the ATA 27 FLIGHT CONTROL DISPLAY button below the REAL column on the MAINT DATA PGS menu.

(d) Make sure the inputs for the sensors are in the limits shown in Table 503 or Table 504 for the UP position.

1) For Method 1, push the arrow keys on the FSEU to step through the sensor values.

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Table 503/27-51-00-993-802

Display (Sensor)	FSEU Sensor Value Range (Flap Position)			
	UP	1-Unit	15-Units	40-Units
LVR POS	-2.8 to 2.8	13.8 to 21.4	61.4 to 68.3	105.4 to 110.9
L FL POS	-3.7 to 1.8	34.5 to 40.8	176.1 to 185.0	267.1 to 273.3
R FL POS	-3.7 to 1.8	34.5 to 40.8	176.1 to 185.0	267.1 to 273.3
F1 SKEW	295.7 to 311.0	12.2 to 29.1	117.8 to 135.6	224.4 to 242.1
F2 SKEW	330.5 to 346.0	60.9 to 78.1	171.0 to 188.6	279.2 to 296.9
F3 SKEW	0.0 to 13.2	82.9 to 99.2	185.2 to 201.8	288.7 to 305.5
F4 SKEW	332.8 to 346.9	24.3 to 40.0	147.3 to 164.6	265.1 to 282.1
F5 SKEW	13.1 to 27.2	320.0 to 335.7	195.4 to 212.7	77.9 to 94.9
F6 SKEW	346.8 to 360.0	260.8 to 277.1	158.2 to 174.8	54.5 to 71.3
F7 SKEW	14.0 to 29.5	281.9 to 299.1	171.4 to 189.0	63.1 to 80.8
F8 SKEW	49.0 to 64.3	330.9 to 347.8	224.4 to 242.2	117.9 to 135.6

2) For Method 2, observe the sensor values on the MDS.

Table 504/27-51-00-993-805

Display (Sensor)	MDS Sensor Value Range (Flap Position)			
	UP	1-Unit	15-Units	40-Units
LVR POS	-2.8 to 2.8	13.8 to 21.4	61.4 to 68.3	105.4 to 110.9
L FL POS	-3.7 to 1.8	34.5 to 40.8	176.1 to 185.0	267.1 to 273.3
R FL POS	-3.7 to 1.8	34.5 to 40.8	176.1 to 185.0	267.1 to 273.3
F1 SKEW	-64.3 to -49	12.2 to 29.1	117.8 to 135.6	224.4 to 242.1
F2 SKEW	-29.5 to -14	60.9 to 78.1	171.0 to 188.6	279.2 to 296.9
F3 SKEW	0.0 to 13.2	82.9 to 99.2	185.2 to 201.8	288.7 to 305.5
F4 SKEW	-27.2 to -13.1	24.3 to 40.0	147.3 to 164.6	265.1 to 282.1
F5 SKEW	-27.2 to -13.1	24.3 to 40.0	147.3 to 164.6	265.1 to 282.1
F6 SKEW	0.0 to 13.2	82.9 to 99.2	185.2 to 201.8	288.7 to 305.5
F7 SKEW	-29.5 to -14	60.9 to 78.1	171.0 to 188.6	279.2 to 296.9
F8 SKEW	-64.3 to -49	12.2 to 29.1	117.8 to 135.6	224.4 to 242.1

(e) Move the flap control lever to the 1-unit detent.

1) Measure the time it takes for the leading edge flaps and slats to extend from the UP to the extend position.

a) If a hydraulic cart is used to supply System B hydraulic power, it should take the leading edge flaps 5.5 to 7.5 seconds and the slats 7.5 to 9.5 seconds to extend.

b) If the Electric Motor-Driven Pump (EMDP) is used to supply System B hydraulic power, it should take the leading edge flaps 7.0 to 9.0 seconds and the slats 9.0 to 11.0 seconds to extend.

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(f) Make sure the inputs for the sensors are in the limits shown in (Table 503 or Table 504) for the 1-unit position.

1) For Method 1, push the arrow keys on the FSEU to step through the sensor values.

2) For Method 2, observe the sensor values on the MDS.

(g) Move the flap control lever to the 40-unit position to extend the flaps.

1) Measure the time it takes for the trailing edge flaps to extend from the 1-unit position to the 40-unit position.

a) If a hydraulic cart or Engine Driven Pump (EDP) was used to supply System B hydraulic power, it should take 22 to 27 seconds for the flaps to extend.

b) If the EMDP was used to supply System B hydraulic power, it should take 55 to 69 seconds for the flaps to extend.

(h) Make sure the inputs for the sensors are in the limits shown in Table 503 or Table 504 for the 40-unit position.

1) For Method 1, push the arrow keys on the FSEU to step through the sensor values.

2) For Method 2, observe the sensor values on the MDS.

(i) Move the flap control lever to the 1-unit position to retract the flaps.

1) Measure the time it takes for the trailing edge flaps to retract from the 40-unit position to the 1-unit position.

a) If a hydraulic cart was used to supply System B hydraulic power, it should take 22 to 27 seconds for the flaps to retract.

b) If the EMDP was used to supply System B hydraulic power, it should take 55 to 69 seconds for the flaps to retract.

(j) Move the flap control lever to the UP detent.

1) Measure the time it takes for the leading edge flaps and slats to retract.

a) If a hydraulic cart is used to supply System B hydraulic power, it should take the leading edge flaps 7.0 to 9.0 seconds and the slats 6.5 to 8.5 seconds to retract.

b) If the EMDP is used to supply System B hydraulic power, it should take the leading edge flaps 8.0 to 10.0 seconds and the slats 7.0 to 9.0 seconds to retract.

(k) Move the flap control lever to the 15-unit position to extend the flaps.

(l) Make sure the inputs for the sensors are in the limits shown in Table 503 or Table 504 for the 15-unit position.

1) For Method 1, push the arrow keys on the FSEU to step through the sensor values.

2) For Method 2, observe the sensor values on the MDS.

(m) For Method 1, push the ON/OFF button on the front panel of the FSEU to turn it off.

(n) For Method 2, push the SYS button on the Multi Function Panel (MFP) to put the displays back to normal.

(o) Move the flap control lever to the UP position to retract the flaps.

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SUBTASK 27-51-00-720-001

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP-DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Do a test of the power supply for the ALTERNATE FLAPS switches and the bypass valve:

(a) Move the flap control lever to the UP position.

(b) Make sure that the Leading Edge (LE) surfaces and Trailing Edge (TE) surfaces to reach their UP position.

(c) Push the ON/OFF button on the FSEU to display “EXISTING FAULTS?”.

(d) Push the NO button three times on the FSEU to display “OTHER FUNCTIONS?”.

(e) Push the YES button on the FSEU to display “SET OUTPUTS?”.

(f) Push the NO button on the FSEU to display “I/O MONITOR?”.

(g) Push the YES button on the FSEU to display “PROX SENSORS?”.

(h) Push the NO button twice on the FSEU to display “DISCRETE INPUTS?”.

(i) Push the YES button on the FSEU to display “ALTN ARM”.

(j) Make sure that the FSEU displays “ALTN ARM NOT ARM”.

(k) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the ARM position.

(l) Make sure that the FSEU displays “ALTN ARM ARMED”.

(m) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the OFF position.

(n) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(o) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the ARM position.

(p) Move the flap control lever to the 1-unit position to extend the flaps.

(q) Wait for the LE and TE surfaces to reach their 1-unit position.

(r) Make sure that the pointers on the flap position indicator are in the 1-unit white band.

(s) Move and hold the ALTERNATE FLAPS control switch in the DOWN position for 5 seconds, then release.

NOTE: With the circuit breaker open, the bypass valve will not operate.

(t) Make sure that the flaps do not extend and that the flap position indicator pointers remain within the detent 1-unit white band.

(u) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

(v) Move the ALTERNATE FLAPS control switch to the UP position and hold until the flaps stop moving.

(w) Make sure that the flap position indicator pointer lies within the detent UP white band.

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(x) Make sure that the “LE FLAPS TRANSIT” and “LE EXTEND” master indication lights on the center panel are off.

(y) Make sure that all the LE flaps and slats indication lights on the overhead panel, P5, are off.

(z) Move the flap control lever to the 40-unit detent.

(aa) Move and hold the ALTERNATE FLAPS control switch to the DOWN position for 5 seconds, then release.

(ab) Wait for the LE surfaces to fully extend.

(ac) Make sure that the LE flaps “EXTEND” and slats “FULL EXTEND” indication lights on the aft overhead panel are now on.

(ad) Move the ALTERNATE FLAPS control switch to the DOWN position and hold until the flaps stop moving.

(ae) Make sure that the flap position indicator pointer lies within the detent 40-unit white band.

(af) Make sure that the “LE FLAPS EXT” master indication light on the center panel is on.

(ag) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the OFF position.

SUBTASK 27-51-00-720-002

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP-DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Do a test of the power supply for the flap electric motor and the time necessary to retract the flaps:

(a) Move the flap control lever to the 1-unit position to extend the flaps.

(b) Make sure that the pointers on the flap position indicator are in the 1-unit white band.

(c) Open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

(d) Move the ALTERNATE FLAPS arm switch to the ARM position.

(e) Move and hold the ALTERNATE FLAPS control switch in the UP position for 5 seconds.

NOTE: With the circuit breaker open, the electric motor will not operate.

(f) Make sure that the flaps do not retract.

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE SLATS DURING THIS PROCEDURE. THE SLATS CAN MOVE AUTOMATICALLY WHEN THE HYDRAULIC SYSTEM IS PRESSURIZED. THIS CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(g) Move and hold the ALTERNATE FLAPS control switch in the DOWN position for 5 seconds.

(h) Make sure that the flaps do not move and that the flap position indicator pointers remain within the 1-unit white band.

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(i) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

(j) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the OFF position.

(k) Wait for the LE and TE surfaces to hydraulically reach their 1-unit position.

(l) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the ARM position.

(m) Measure the time it takes for the TE flaps to go from the 1-unit position to the UP position.

1) Move the ALTERNATE FLAPS control switch to the UP position and hold until the flaps stop.

2) Make sure that the time for the retraction of the TE flaps is between 47 to 58 seconds.

(n) Make sure that the flap position indicator pointer is within the detent UP white band.

(o) Move the flap control lever to the UP detent.

(p) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the OFF position.

SUBTASK 27-51-00-720-003

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP-DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(8) Do a test of the UP limit switch:

(a) Move the flap control lever to the 1-unit detent position.

(b) Wait for the LE surfaces and TE surfaces to reach the 1-unit position.

(c) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the ARM position.

(d) Move and hold the ALTERNATE FLAPS control switch in the UP position until the flaps stop.

(e) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?

(f) Do a check of the position sensor inputs to the FSEU:

1) Push the NO button three times on the FSEU to display “OTHER FUNCTNS?”.

2) Push the YES button on the FSEU to display “SET OUTPUTS?”.

3) Push the NO button on the FSEU to display “I/O MONITOR?”.

4) Push the YES button on the FSEU to display “PROX SENSORS?”.

5) Push the NO button on the FSEU to display “SYNCHRO INPUTS?”.

6) Push the YES button on the FSEU to display “LVR POS”.

7) Push the arrow keys to step through the position sensor values at the UP position.

8) Make sure that the inputs for the sensors are in the limits shown in (Table 505) for the UP position.

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Table 505/27-51-00-993-803

Display (Sensor)	Sensor Input (Flap Position)	
	UP	40-Units
L FL POS	-1.2 to 4.7	259.9 to 270.3
R FL POS	-1.2 to 4.7	259.9 to 270.3

9) Move the flap control lever to the UP detent.

(g) Put the ALTERNATE FLAPS arm switch, on the overhead panel, P5, in the OFF position.

SUBTASK 27-51-00-720-004

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP-DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(9) Do a test of the DOWN limit switch:

(a) Move the ALTERNATE FLAPS arm switch to the OFF position.

(b) Move the flap control lever to the 30-unit position to extend the flaps.

(c) Wait for the LE surfaces and TE surfaces to reach the 30-unit position.

(d) Make sure that the pointers on the flap position indicator are in the 30-unit white band.

(e) Move the ALTERNATE FLAPS arm switch to the ARM position.

(f) Move the flap control lever to the 40-unit position.

(g) Move and hold the ALTERNATE FLAPS control switch in the DOWN position until the flaps stop.

(h) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?

(i) Do a check of the position sensor inputs to the FSEU:

1) Push the NO button three times on the FSEU to display “OTHER FUNCTIONS?”.

2) Push the YES button on the FSEU to display “SET OUTPUTS?”.

3) Push the NO button on the FSEU to display “I/O MONITOR?”.

4) Push the YES button on the FSEU to display “PROX SENSORS?”.

5) Push the NO button on the FSEU to display “SYNCHRO INPUTS?”.

6) Push the YES button on the FSEU to display “LVR POS?”.

7) Push the arrow keys to step through the position sensor values at the 40-unit position.

8) Make sure that the inputs for the sensors are in the limits shown in (Table 505) for the 40-unit position.

(j) Push the ON/OFF button on the front panel of the FSEU to turn it off.

(k) Move the ALTERNATE FLAPS arm switch to the OFF position.

SUBTASK 27-51-00-720-005

(10) Do this task: Flap Priority Valve Test, TASK 27-51-00-720-801.

SUBTASK 27-51-00-720-006

(11) Do this task: Flap Control Lever Force Test, TASK 27-51-00-720-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-054

(1) If the FSEU is not turned off, push the ON/OFF button on the front panel of the FSEU to turn it off.

SUBTASK 27-51-00-410-007

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-51-00-860-055

(3) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-51-00-740-801

6. FSEU Display and Existing Faults Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test does a check of the Built-In-Test Equipment (BITE) display and power supplies for the Flap/Slat Electronics Unit (FSEU). It also does a check for existing faults in the high lift system detected by the FSEU.

B. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Test

SUBTASK 27-51-00-740-012

(1) Make sure that the Trailing Edge (TE) flaps are in the fully retracted position.

SUBTASK 27-51-00-710-004

(2) Make sure that the B system hydraulic power is removed prior to performing the test.

(a) If it is necessary, remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-51-00-010-006

(3) To get access to the main equipment center, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. FSEU Display and Existing Faults Test

SUBTASK 27-51-00-740-001

WARNING IF THERE IS HYDRAULIC PRESSURE DURING THIS TEST, KEEP PERSONNEL AWAY FROM THE FLAPS. THE FLAPS CAN MOVE AND CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(1) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?

SUBTASK 27-51-00-750-001

(2) Do a test of the Direct Current (DC) power supplies for the FSEU:

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	14	C01070	FLIGHT CONTROL AUTOSLAT DC 2
C	14	C01068	FLIGHT CONTROL AUTOSLAT DC 1

Open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1

(b) Make sure that the front panel display on the FSEU is still on.

(c) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1

(d) Open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	9	C01468	FLIGHT CONTROL FSEU DC 2

(e) Make sure that the front panel display on the FSEU is still on.

(f) Open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1

(g) Make sure that the front panel display on the FSEU is off.

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(h) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2

(i) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

(j) Make sure that the front panel display on the FSEU comes on.

SUBTASK 27-51-00-740-002

(3) Do a test of the BITE display on the FSEU:

(a) Push the down arrow until GROUND TESTS? shows on the display.

(b) Push the YES button to select GROUND TESTS? on the FSEU.

(c) Push the down arrow until DISPLAY TEST? shows on the display.

(d) Push the YES button to select DISPLAY TEST? on the FSEU.

(e) Make sure that the display characters on the front panel of the FSEU turn on in groups of four for approximately 2 seconds.

SUBTASK 27-51-00-740-003

(4) Do a check for EXISTING FAULTS with the FSEU:

NOTE: During the following BITE test, all the Leading Edge (LE) indication lights (12 amber transit and 20 green extend/full extend lights) on the aft overhead panel will be on for about 5 seconds.

(a) Push the MENU button twice.

NOTE: The display will show EXISTING FAULTS?

(b) Push the YES button to select the EXISTING FAULTS? test.

NOTE: The display will show TEST IN PROGRESS.

(c) Make sure that the lights come on for 5 seconds while the display shows TEST IN PROGRESS:

1) P5 Aft Overhead Panel

a) LE Devices TRANSIT

b) LE Devices EXT

c) LE Devices FULL EXT

(d) Make sure there are no EXISTING FAULTS.

SUBTASK 27-51-00-740-004

(5) Do a test of the ASYM MODE on the FSEU:

(a) Push the MENU switch on the FSEU to display “EXISTING FAULTS?”.

NOTE: The takeoff and landing warning horn might sound when the following steps are performed. If the horns sound, no action is required.

(b) Push the NO switch three times on the FSEU to display “OTHER FUNCTIONS?”.

(c) Push the YES switch on the FSEU to display “SET OUTPUTS?”.

(d) Push the NO switch on the FSEU to display “I/O MONITOR?”.

(e) Push the YES switch on the FSEU to display “PROX SENSORS?”.

(f) Push the NO switch twice on the FSEU to display “DISCRETE INPUTS?”.

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(g) Push the YES switch on the FSEU to display “ALTN ARM NOT ARM”.

(h) Push the NO switch five times on the FSEU to display “ASY MODE CLOSED”.

(i) Open this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

(j) Make sure that the FSEU displays “ASY MODE OPEN”.

(k) Close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

(l) Make sure that the FSEU displays “ASY MODE CLOSED”.

(m) Push the ON/OFF button on the front panel of the FSEU to turn off the display.

SUBTASK 27-51-00-740-011

(6) Do a check for the autoslat analog discrete input to the FSEU:

(a) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: Select MENU on the display to show EXISTING FAULTS?.

(b) Push the NO button until OTHER FUNCTIONS? shows on the display.

(c) Push the YES button to select OTHER FUNCTIONS? on the FSEU.

(d) Push the NO button until I/O MONITOR? shows on the display.

(e) Push the YES button to select I/O MONITOR? on the FSEU.

(f) Push the NO button until DISCRETE INPUTS? shows on the display.

(g) Push the YES button to select DISCRETE INPUTS? on the FSEU.

(h) Push the NO button until AUTOSLT1 shows on the display.

NOTE: Line 2 on the display will show either COMMAND or NOT CMD.

(i) Push the down arrow to show AUTOSLT2 on the display.

NOTE: Line 2 on the display will show either COMMAND or NOT CMD.

(j) If AUTOSLT1 or AUTOSLT2 shows COMMAND on line 2, the FSEU receives an autoslat command discrete input from the Stall Management Yaw Damper Computer (SMYDC).

1) If AUTOSLT1 or AUTOSLT2 shows COMMAND on line 2, refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

NOTE: The autoslat command discrete from the SMYDC prevents the FSEU LE un-commanded motion monitor from being set. AUTOSLT1 comes from SMYDC 1 and AUTOSLT2 comes from SMYDC 2.

(k) If AUTOSLT1 and AUTOSLT2 shows NOT CMD on line 2, then continue.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-058

(1) Push the ON/OFF button on the front panel of the FSEU to turn it off.

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SUBTASK 27-51-00-410-008

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 27-51-00-740-802

7. Trailing Edge Flap Asymmetry Skew System BITE Test

(Figure 501 and Figure 504)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test does a check of the asymmetry and skew detection functions for the Trailing Edge (TE) flaps.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-51-00-010-007

(1) To get access to the main equipment center, do this step:

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Trailing Edge Flap Asymmetry and Skew BITE Test

SUBTASK 27-51-00-740-005

(1) Push the ON/OFF button, on the front panel of the Flap/Slat Electronics Unit (FSEU), to turn on the display.

NOTE: The display will show EXISTING FAULTS?

SUBTASK 27-51-00-740-006

(2) Do the Built-In-Test Equipment (BITE) test of the asymmetry and skew detection systems:

(a) Push the NO button until GROUND TESTS? shows on the display.

(b) Push the YES button to select TE ASYM-SKW-IND? on the FSEU.

(c) Push the Yes button to show TEST IN PROGRESS on the FSEU.

(d) Make sure that the left indicator needle moves in the clockwise direction.

(e) Push the Yes button in response to IS L IND NDL CW? on the FSEU.

(f) Make sure that the left indicator needle moves in the counterclockwise direction.

(g) Push the Yes button in response to IS L IND NDL CCW? on the FSEU.

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(h) Make sure that the right indicator needle moves in the clockwise direction.

(i) Push the Yes button in response to IS R IND NDL CW? on the FSEU.

(j) Make sure that the right indicator needle moves in the counterclockwise direction.

(k) Push the Yes button in response to IS R IND NDL CCW? on the FSEU.

(l) Make sure that TEST PASSED shows at the end of the test.

SUBTASK 27-51-00-210-004

(3) Make sure that the manual override lever, on the flap bypass valve, is in POSITION 1 (View B, Figure 504).

NOTE: The flap bypass valve is installed in the right main wheel well.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-062

(1) Push the ON/OFF button, on the front panel of the FSEU, to turn it off.

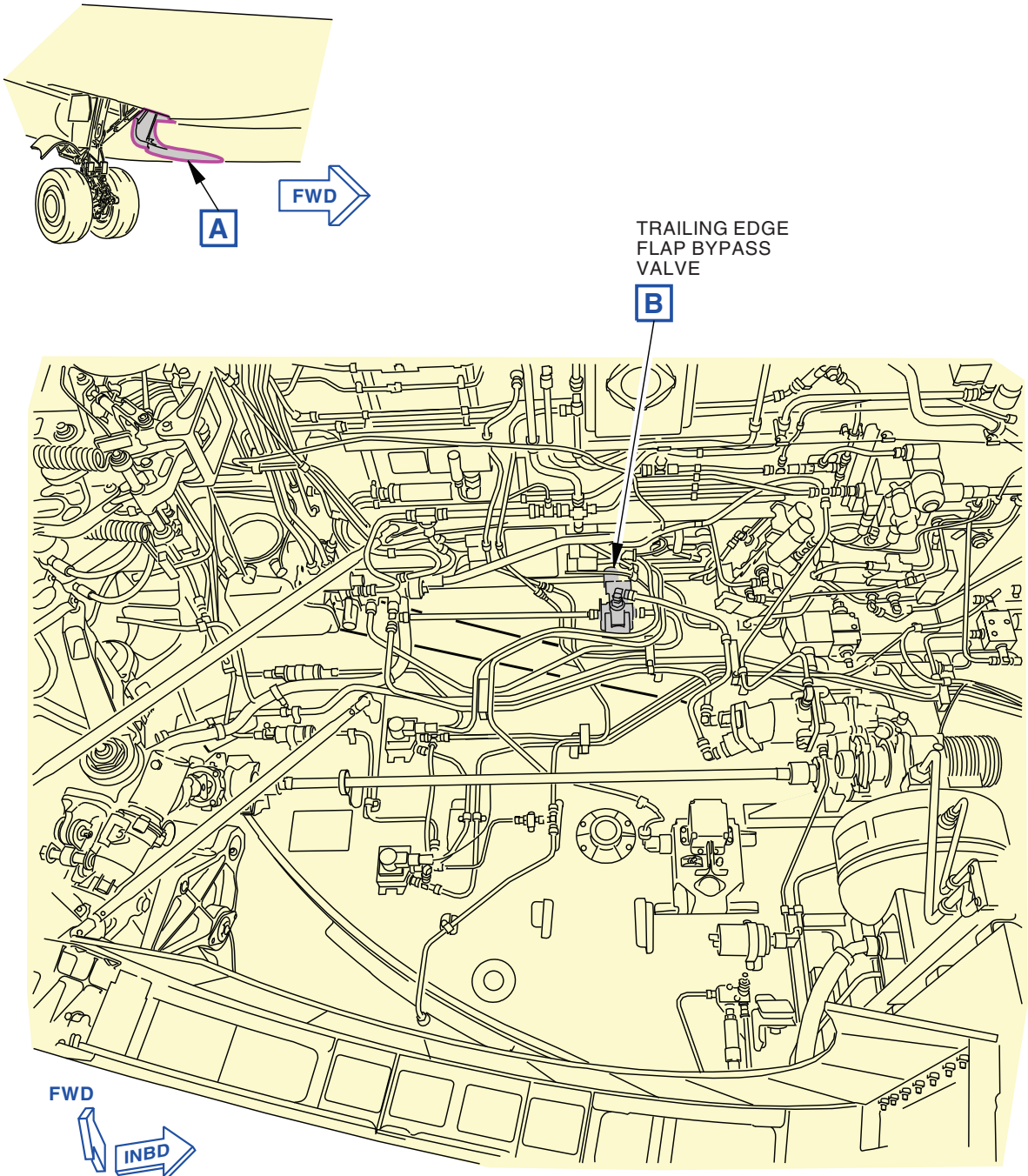
SUBTASK 27-51-00-410-009

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

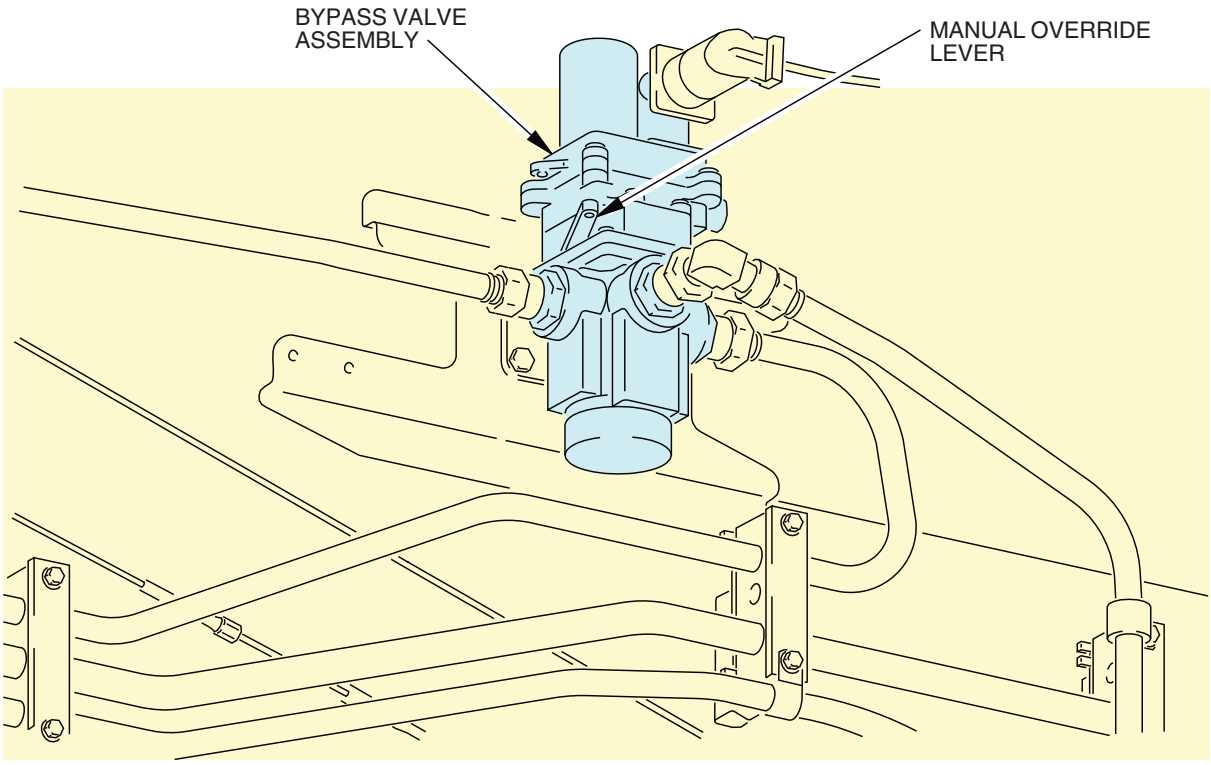
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trailing Edge Flap Bypass Valve
Figure 504/27-51-00-990-817 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trailing Edge Flap Bypass Valve
Figure 504/27-51-00-990-817 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-740-803

8. Trailing Edge Flap Load Relief System BITE Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test does a check of the load relief functions for the Trailing Edge (TE) flaps.

(2) In this procedure it is necessary to move the flap control lever to move the TE flaps.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Test

SUBTASK 27-51-00-860-067

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-51-00-860-100

(2) Set the FLT CONTROL and SPOILER switches on the P5-3 panel to the OFF position.

SUBTASK 27-51-00-010-008

(3) To get access to the main equipment center, do this step:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Trailing Edge Flap Load Relief System BITE Test

SUBTASK 27-51-00-740-007

(1) Push the ON/OFF button on the front panel of the Flap/Slat Electronics Unit (FSEU) to turn on the display.

NOTE: The display will show EXISTING FAULTS?

SUBTASK 27-51-00-740-008

(2) Do the Built-In-Test Equipment (BITE) test of the load relief system:

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(a) Make sure that the TE flaps are at the lever position and then push the YES button during the test.

NOTE: The test is time sensitive.

(b) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF

(c) Push the NO switch twice on the FSEU to display “GROUND TESTS?”.

(d) Push the YES switch on the FSEU to display “TE ASYM-SKW-IND?”.

(e) Push the NO switch on the FSEU to display “TE FLAP LD REL?”.

(f) Push the YES switch on the FSEU to display “WARNING:FLP CLR?”.

(g) Push the YES switch on the FSEU to display “FLP LVR TO 40?”.

(h) Move the flap control lever to the 40-unit detent.

(i) Push the YES switch on the FSEU to display “TEST IN PROGRESS?”.

(j) Make sure the “FLP LOAD RLF 40” is displayed on the FSEU front panel.

(k) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF

NOTE: The flaps will move in the subsequent steps and the flap indicator will move accordingly.

(l) Push the MENU switch on the FSEU to display “TE FLAP LD REL?”.

(m) Push the YES switch on the FSEU to display “WARNING:FLP CLR?”.

(n) Push the YES switch on the FSEU to display “FLP LVR TO 40?”.

NOTE: The flaps will retract to the 30-unit position before extending to the 40-unit position.

(o) Push the YES switch on the FSEU to display “FLP LVR TO 10?”.

(p) Move the flap control lever to the 10-unit detent.

(q) Wait for all the surfaces to reach the flaps 10-unit position.

(r) Push the YES switch on the FSEU to display “FLP LVR TO 5?”.

NOTE: The flaps will retract to the 5-unit position before extending to the 10-unit position.

(s) Move the flap control lever to the 5-unit detent.

(t) Wait for all the surfaces to reach the flaps 5-unit position.

(u) Push the YES switch on the FSEU to display “FLP LVR TO UP?”.

NOTE: The flaps will stay in its position.

(v) Move the flap control lever to the UP detent.

(w) Wait for all the surfaces to reach the flaps UP position.

(x) Push the YES switch on the FSEU to display “TEST PASSED”.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-068

(1) Push the ON/OFF button on the front panel of the FSEU to turn it off.

SUBTASK 27-51-00-410-010

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-51-00-860-101

(3) Set the FLT CONTROL and SPOILER switches on the P5–3 panel to the ON position.

SUBTASK 27-51-00-860-070

(4) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-51-00-720-801

9. Flap Priority Valve Test

(Figure 501)

A. General

(1) This test does a check of the flap priority valve.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-51-00-860-073

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Use a hydraulic service cart to supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

NOTE: It will be necessary to control the pressure output of the hydraulic service cart.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Flap Priority Valve Test

SUBTASK 27-51-00-860-074

> **WARNING** MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP DRIVE MECHANISMS BEFORE YOU MOVE THE FLAPS. THE FLAPS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Move the flap control lever to the 30-unit position to extend the flaps.

NOTE: Wait for the flaps to get to the 30-unit position.

SUBTASK 27-51-00-730-005

(2) Reduce the pressure output from the hydraulic service cart to 2000 psi (13,790 kPa).

SUBTASK 27-51-00-720-007

(3) Do a check of the time necessary to retract the flaps.

NOTE: The flaps will retract slowly with the hydraulic pressure reduced.

(a) Move the flap control lever to the 25-unit position.

NOTE: The flaps can make noise while they retract slowly.

(b) Make sure the time necessary to retract the flaps is a minimum of 59 seconds.

NOTE: It is not necessary to wait more than 1 minute for the flaps to retract.

(c) Increase the pressure output from the hydraulic service cart to 3000 psi (20,684 kPa).

NOTE: The flaps will retract much faster at normal operating pressure.

(d) Move the flap control lever to the 5-unit position.

NOTE: Wait for the flaps to get to the 5-unit position.

SUBTASK 27-51-00-720-008

(4) Do a check of the time necessary to retract the flaps to the UP position.

(a) Move the flap control lever to the UP position.

(b) Make sure the time necessary to retract the flaps is 17 to 22 seconds.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-075

(1) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-51-00-720-802

10. Flap Control Lever Force Test

(Figure 501)

A. General

(1) This test does a check of the force necessary to move the flap control lever.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Test

SUBTASK 27-51-00-860-078

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

F. Flap Control Lever Force Test

SUBTASK 27-51-00-860-079

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP-DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Move the flap control lever to the UP position to retract the flaps.

SUBTASK 27-51-00-720-009

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP-DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Measure the force necessary to move the flap control lever to the 1-unit position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Move the flap control lever out of the UP detent.

(b) Put a force gauge, COM-1557, on the end of the flap control lever.

(c) Use the force gauge, COM-1557, to move the flap control lever to the 1-unit position.

NOTE: Make sure the force gauge, COM-1557, is perpendicular to the flap control lever when you measure the force.

(d) Make sure the force necessary to move the flap control lever is less than 5 lbf (22.24 N).

SUBTASK 27-51-00-720-010

(3) Use the force gauge, COM-1557, to measure the force necessary to move the flap control lever to the 40-unit position with a stop at each detent.

(a) Make sure the force necessary to move the flap control lever is less than 5 lbf (22.24 N).

SUBTASK 27-51-00-720-011

(4) Use the force gauge, COM-1557, to measure the force necessary to move the flap control lever to the UP position with a stop at each detent.

(a) Make sure the force necessary to move the flap control lever is less than 5 lbf (22.24 N).

SUBTASK 27-51-00-080-004

(5) Remove the force gauge, COM-1557, from the end of the flap control lever.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-080

(1) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-51-00-720-805

11. Flap Control Lever Feel Test

A. General

(1) This test does a check on the Flap Control Lever movement in the up and down movement. To make sure that there is a smooth procedure with the flight handle after control cable adjustments.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Test

SUBTASK 27-51-00-860-103

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B with hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

E. Flap Control Level Feel Test

SUBTASK 27-51-00-720-018

(1) Put the Flap Control Lever between UP and 1 detent.

SUBTASK 27-51-00-720-019

(2) Lift up and move the Flap Control Lever forward to the UP detent.

SUBTASK 27-51-00-720-020

(3) Stow the Flap Control Lever from the lifted position into the UP detent.

(a) Make sure that axial movement is in the bottom direction

(b) Make sure that it is put at the bottom of the detent

(c) Make sure that the Lever goes into the stand smoothly.

NOTE: There will be some permitted resistance because of the cable load in the UP detent.

SUBTASK 27-51-00-720-021

(4) Move the Flap Control Lever from the UP detent.

(a) Make sure that axial movement is in the up direction.

SUBTASK 27-51-00-720-022

(5) Make sure that the Flap Control Lever is smooth when moved in the up direction.

NOTE: There will be some permitted resistance because of the cable load in the UP detent.

SUBTASK 27-51-00-720-023

(6) Stow and attach safely the Flap Control Lever back into the UP detent to complete the test.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-104

(1) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

TASK 27-51-00-720-803

12. Alternate Flap Control System Functional Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) During the functional test, you make sure the alternate operation of the Trailing Edge (TE) flaps is satisfactory.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Functional Test

SUBTASK 27-51-00-860-083

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

E. Procedure

SUBTASK 27-51-00-720-012

(1) Do these steps for the Alternate Flap System Functional Test:

CAUTION

DO NOT CONTINUOUSLY OPERATE THE FLAP ELECTRIC MOTOR FOR MORE THAN 4 MINUTES. YOU MUST WAIT A MINIMUM OF 25 MINUTES BETWEEN OPERATIONS OF THE FLAP ELECTRIC MOTOR. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(a) Move the flap control lever to the 1-unit position to extend the flaps.

(b) Make sure the pointers on the flap position indicator are in the 1-unit white band.

(c) Open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

CAUTION

DO NOT OPERATE THE TRAILING EDGE FLAPS WITH ELECTRICAL POWER MORE THAN ONE FULL EXTEND/RETRACT CYCLE IN 25 MINUTES. YOU CAN CAUSE DAMAGE TO THE FLAP ELECTRIC MOTORS IF YOU OPERATE THEM TOO FREQUENTLY.

(d) Move the ALTERNATE FLAPS ARM switch to the ARM position.

(e) Move and hold the ALTERNATE FLAPS CONTROL switch in the UP position for 5 seconds.

NOTE: With the circuit breaker open, the electric motor will not operate.

(f) Make sure the flaps do not retract.

(g) Move and hold the ALTERNATE FLAPS CONTROL switch in the DOWN position for 5 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Make sure that the flaps do not move and that the flap position indicator pointers remain within the 1-unit white band.

(i) Close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

(j) Put the ALTERNATE FLAPS ARM switch, on the aft overhead panel in the “OFF” position.

(k) Wait for the Leading Edge (LE) and TE surfaces to hydraulically reach their 1-unit position.

(l) Put the ALTERNATE FLAPS ARM switch, on the aft overhead panel in the “ARM” position.

(m) Measure the time it takes for the TE flaps to go from the 1-unit position to the UP position.

(n) Move the ALTERNATE FLAPS CONTROL switch to the UP position and hold until the flaps stop.

(o) Make sure that the time for the retraction of the TE flaps is between 47 to 58 seconds.

(p) Make sure that the flap position indicator pointer is within the detent UP white band.

(q) Move the flap control lever to the UP detent.

(r) Put the ALTERNATE FLAPS ARM switch, on the aft overhead panel in the “OFF” position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-084

(1) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-51-00-710-802

13. Transmission No-Back Brake Operational Test

(Figure 505)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) During this operational test, you make sure the no-back brakes in the transmissions operate.

(2) This test involves listening for a ratchet sound from the transmissions while the flaps extend.

(3) A minimum of two people are necessary to do this test.

B. References

Reference	Title
27-51-00-860-801	Trailing Edge Flap System Operation With Primary Control (P/B 201)
27-51-28-000-803	Outboard Flap Outboard Support Aft Fairing Removal (P/B 401)
27-51-28-000-805	Outboard Flap Inboard Support Aft Fairing Removal (P/B 401)
27-51-28-400-803	Outboard Flap Outboard Support Aft Fairing Installation (P/B 401)
27-51-28-400-805	Outboard Flap Inboard Support Aft Fairing Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-51-31 P/B 401	INBOARD FLAP TRANSMISSIONS - REMOVAL/INSTALLATION
27-51-41 P/B 401	OUTBOARD FLAP TRANSMISSIONS - REMOVAL/INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1797	Stethoscope - Mechanics, 12 Inch Probe Part #: GA111D Supplier: 55719 Opt Part #: GA111C Supplier: 55719
SPL-12536	Test Equipment - Outboard T.E. Flap Transmission, No-Back Brake Operational Test Part #: C27084-1 Supplier: 81205

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
541	Left Wing - Fairing Flap Support No. 4
542	Left Wing - Fairing Flap Support No. 3
543	Left Wing - Fairing Flap Support No. 2
544	Left Wing - Fairing Flap Support No. 1
641	Right Wing - Fairing Flap Support No. 5
642	Right Wing - Fairing Flap Support No. 6
643	Right Wing - Fairing Flap Support No. 7
644	Right Wing - Fairing Flap Support No. 8

E. Prepare for the Operational Test

SUBTASK 27-51-00-480-012

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-00-010-009

(2) Remove the aft fairings for the outboard flaps:

NOTE: Removal of the aft fairings for the inboard flaps is not necessary since the ratcheting sound can be heard easily without removal.

(a) Do this task: Outboard Flap Inboard Support Aft Fairing Removal, TASK 27-51-28-000-805.

(b) Do this task: Outboard Flap Outboard Support Aft Fairing Removal, TASK 27-51-28-000-803.

SUBTASK 27-51-00-860-087

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

F. Operational Test of the Transmission No-Back Brake

NOTE: This test is applicable to all of the transmissions for the trailing edge flaps.

SUBTASK 27-51-00-860-088

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Retract the flaps to the UP position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

SUBTASK 27-51-00-710-003

(2) Do a test of the no-back brake in the transmissions:

(a) Put a stethoscope, COM-1797, on the applicable transmission to listen for a ratchet sound.

NOTE: The ratchet sound from the inboard flaps can be easily heard without the need for a stethoscope.

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Extend the flaps to the 15 unit position and listen for a ratchet sound from the transmissions, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

NOTE: The ratchet sound is the no-back brake inside the transmission. It is necessary to do the test again for the additional transmissions, or have additional people listen to the transmissions.

(c) If the ratchet sound cannot be heard on the outboard flaps transmission, install test equipment, SPL-12536, and perform the test again.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) If it is necessary to do this test on another transmission, retract the flaps to the UP position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(e) If the ratcheting sound is not heard replace the transmission.

1) For the inboard transmission replacement do this task: INBOARD FLAP TRANSMISSIONS - REMOVAL/INSTALLATION, PAGEBLOCK 27-51-31/401.

2) For the outboard transmission replacement do this task: OUTBOARD FLAP TRANSMISSIONS - REMOVAL/INSTALLATION, PAGEBLOCK 27-51-41/401.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-089

(1) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-51-00-080-005

(2) Remove test equipment, SPL-12536, if it was installed.

SUBTASK 27-51-00-410-011

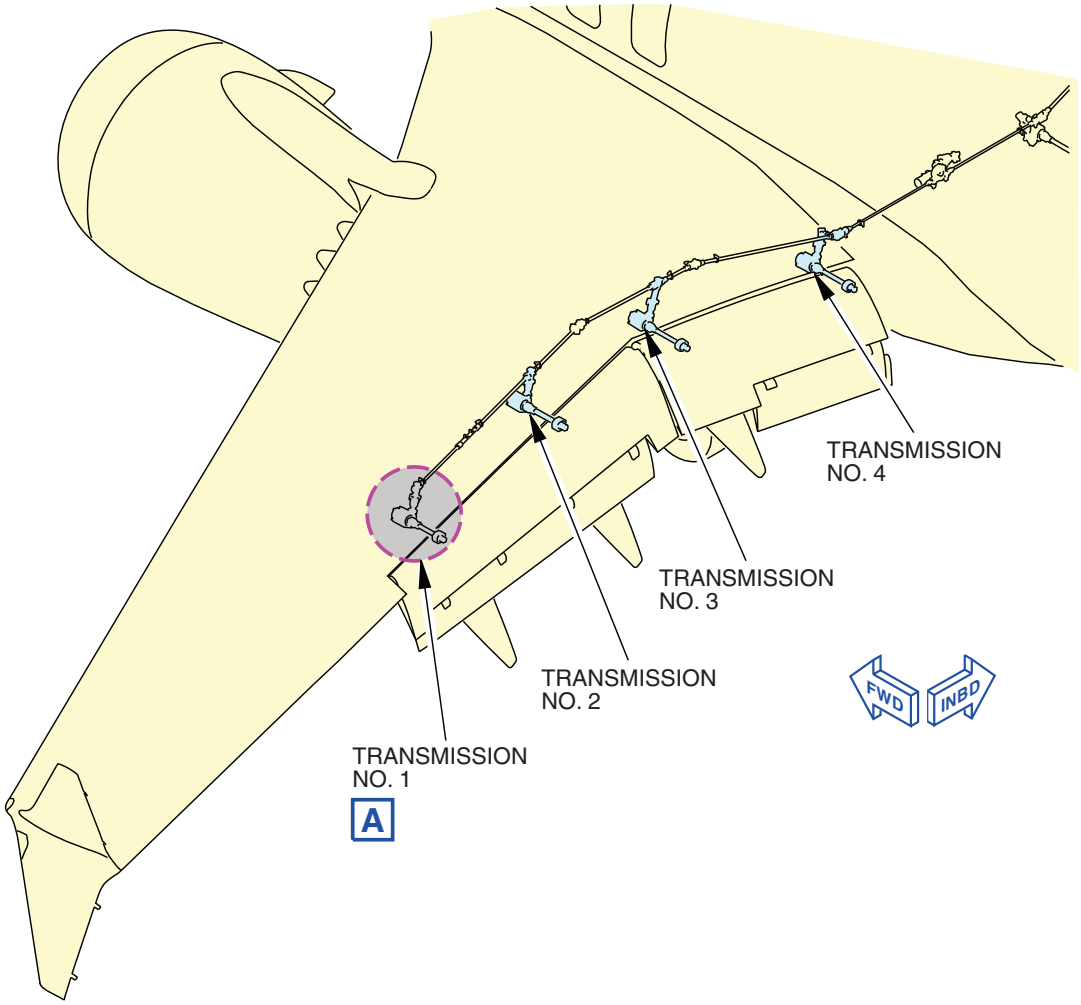
(3) Install the aft fairings for the outboard flaps:

(a) Do this task: Outboard Flap Inboard Support Aft Fairing Installation, TASK 27-51-28-400-805.

(b) Do this task: Outboard Flap Outboard Support Aft Fairing Installation, TASK 27-51-28-400-803.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

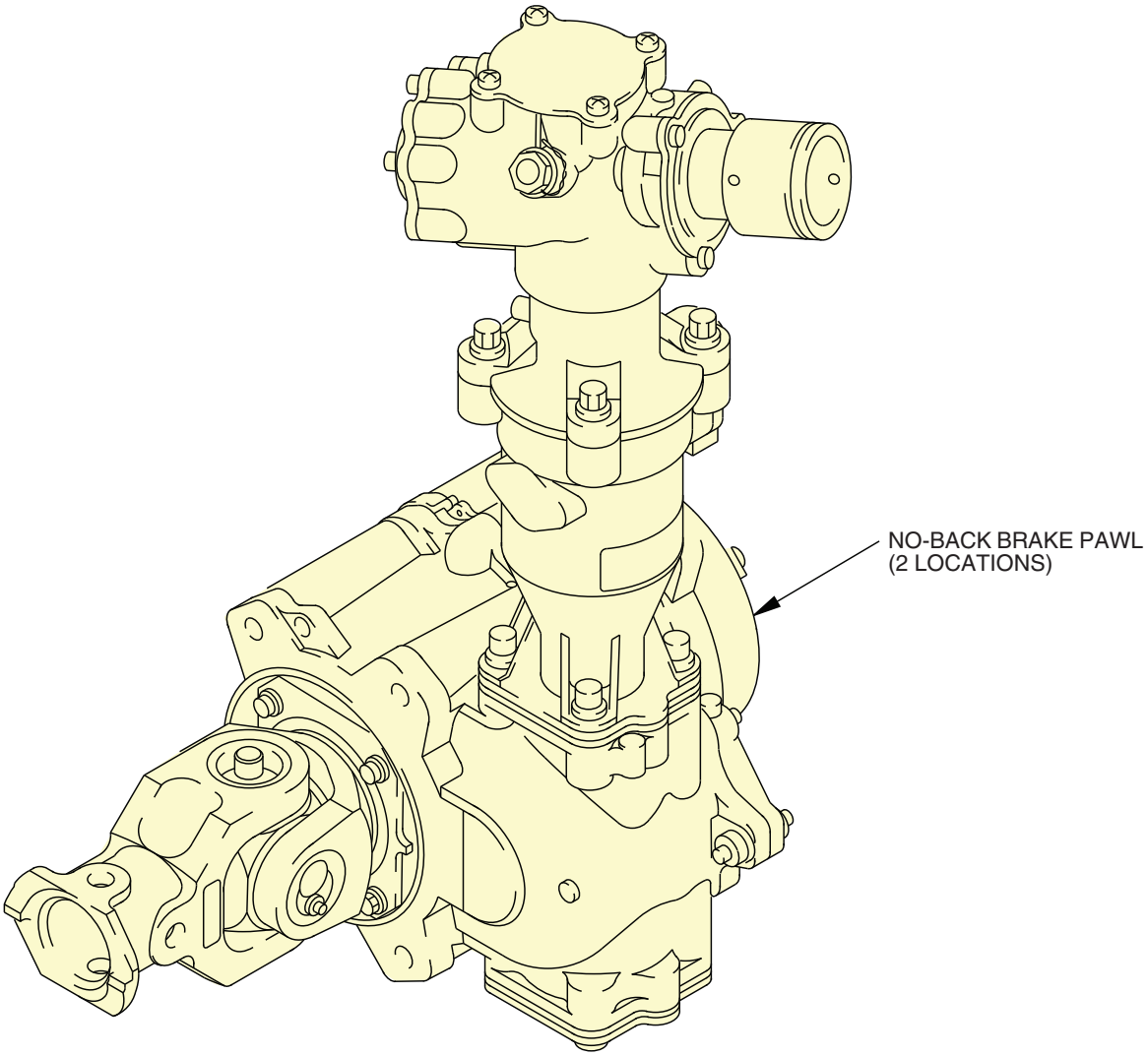


(EXAMPLE)

3089837 S0000841163_V1

Transmission No-Back Brake Operational Test
Figure 505/27-51-00-990-818 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**(FLAP TRANSMISSION 1 AND 8 ARE SHOWN,
FLAP TRANSMISSIONS 2, 3, 4, 5, 6 AND 7 ARE EQUIVALENT)**

A

3089854 S0000841164_V1

**Transmission No-Back Brake Operational Test
Figure 505/27-51-00-990-818 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-720-804

14. Trailing Edge Flap Load Relief System Functional Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task does a functional check of the load relief system for the Trailing Edge (TE) flaps. It makes sure that there is airspeed data from the Air Data Inertial Reference Unit (ADIRU) to the Flap/Slat Electronics Unit (FSEU).

(2) In this procedure it is necessary to use the pitot static system to simulate airspeed.

B. References

Reference	Title
27-51-00-860-801	Trailing Edge Flap System Operation With Primary Control (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
34-11-00-790-801	Captain's Total Air Pressure System - Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Test

SUBTASK 27-51-00-860-092

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-51-00-860-099

(2) Set the FLT CONTROL and SPOILER switches, on the P5-3 panel, to the OFF position.

SUBTASK 27-51-00-010-010

(3) To get access to the main equipment center, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

G. Trailing Edge Flap Load Relief System Functional Test

SUBTASK 27-51-00-740-009

(1) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

NOTE: The display will show EXISTING FAULTS?

SUBTASK 27-51-00-000-001

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF

SUBTASK 27-51-00-740-010

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do the Built-In-Test Equipment (BITE) test of the load relief system:

(a) Push the down arrow until GROUND TESTS? shows on the FSEU.

(b) Push the YES button to display “TE ASYM-SKW-IND?” on the FSEU.

(c) Push the down arrow until TE FLAP LD REL? shows on the display.

(d) Push the YES button until “WARNING: FLP CLR?” shows on the display.

(e) Push the YES button until “FLP LVR TO 40?” shows on the display.

(f) Move the flap control lever to the 40-unit detent.

(g) Push the YES button on the FSEU to display “TEST IN PROGRESS”.

(h) Make sure that “FLP LOAD RLF 40” is displayed on the FSEU front panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF

(j) Push the MENU button on the FSEU to display “TE FLAP LD REL?”.

(k) Push the YES button on the FSEU to display “WARNING: FLP CLR?”.

(l) Push the YES button on the FSEU to display “FLP LVR TO 40?”.

(m) Push the YES button on the FSEU to display “FLP LVR TO 10?”.

(n) Move the flap control lever to the 10-unit detent.

(o) Wait for all surfaces to reach the flaps 10-unit detent position.

(p) Push the YES button on the FSEU to display “FLP LVR TO 5?”.

(q) Move the flap control lever to the 5-unit detent.

(r) Wait for all surfaces to reach the flaps 5-unit detent position.

(s) Push the YES button on the FSEU to display “FLP LVR TO UP?”.

(t) Move the flap control lever to the UP detent.

(u) Wait for all surfaces to reach the flaps UP position.

(v) Push the YES button on the FSEU to display “TEST IN PROGRESS”.

(w) Make sure that the “TEST PASSED” is displayed on the FSEU.

SUBTASK 27-51-00-860-093

(4) Push the ON/OFF button on the front panel of the FSEU to turn it off.

SUBTASK 27-51-00-860-094

(5) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	1	C00523	HEATERS CAPT PITOT
C	2	C00238	HEATERS TEMP PROBE
C	3	C01072	HEATERS ALPHA VANE LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT

SUBTASK 27-51-00-480-013

(6) Use the air data accessory kit, COM-25177, and pitot test adapter, COM-1916, to connect the air data model test set, COM-1914, to the left pitot probe to supply airspeed output from the left ADIRU (TASK 34-11-00-790-801).

NOTE: The air data accessory kit, COM-25177, contains hoses to connect to the air data model test set, COM-1914, and pitot test adapter, COM-1916.

SUBTASK 27-51-00-860-098

(7) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF

SUBTASK 27-51-00-720-013

(8) Do a test of the load relief system with the flaps at 40 units:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Use the primary system to extend the flaps to the 40-unit position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(b) Use the air data model test set, COM-1914, to raise the airspeed to 168 knots (193.33 mph), on the captain's display unit.

NOTE: When you apply or remove pressure to the pitot system, the rate must not be more than 300 knots (345.23 mph) for each minute.

(c) Make sure that the TE flaps retract to the 30-unit position.

(d) Use the air data model test set, COM-1914, to lower the airspeed to 161 knots (185.28 mph).

(e) Make sure that the TE flaps extend to the 40-unit position.

SUBTASK 27-51-00-720-014

(9) Do a test of the load relief system with the flaps at 30 units:

(a) Use the primary system to retract the flaps to the 30-unit position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(b) Use the air data model test set, COM-1914, to raise the airspeed to 177 knots (203.69 mph).

NOTE: When you apply or remove pressure to the pitot system, the rate must not be more than 300 knots (345.23 mph) for each minute.

(c) Make sure that the TE flaps retract to the 25-unit position.

(d) Use the air data model test set, COM-1914, to lower the airspeed to 170 knots (195.63 mph).

(e) Make sure that the TE flaps extend to the 30-unit position.

SUBTASK 27-51-00-720-015

(10) Do a test of the load relief system with the flaps at 25 units:

(a) Use the primary system to retract the flaps to the 25-unit position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(b) Use the air data model test set, COM-1914, to raise the airspeed to 192 knots (220.95 mph).

NOTE: When you apply or remove pressure to the pitot system, the rate must not be more than 300 knots (345.23 mph) for each minute.

(c) Make sure that the TE flaps retract to the 15-unit position.

(d) Use the air data model test set, COM-1914, to lower the airspeed to 185 knots (212.89 mph).

(e) Make sure that the TE flaps extend to the 25-unit position.

SUBTASK 27-51-00-720-016

(11) Do a test of the load relief system with the flaps at 15 units:

(a) Use the primary system to retract the flaps to the 15-unit position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(b) Use the air data model test set, COM-1914, to raise the airspeed to 202 knots (232.46 mph).

NOTE: When you apply or remove pressure to the pitot system, the rate must not be more than 300 knots (345.23 mph) for each minute.

(c) Make sure that the TE flaps retract to the 10-unit position.

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(d) Use the air data model test set, COM-1914, to lower the airspeed to 195 knots (224.40 mph).

(e) Make sure that the TE flaps extend to the 15-unit position.

SUBTASK 27-51-00-720-017

(12) Do a test of the load relief system with the flaps at 10 units:

(a) Use the primary system to retract the flaps to the 10-unit position, do this task: Trailing Edge Flap System Operation With Primary Control, TASK 27-51-00-860-801.

(b) Use the air data model test set, COM-1914, to raise the airspeed to 212 knots (243.97 mph).

NOTE: When you apply or remove pressure to the pitot system, the rate must not be more than 300 knots (345.23 mph) for each minute.

(c) Make sure that the TE flaps retract to the 5-unit position.

(d) Use the air data model test set, COM-1914, to lower the airspeed to 205 knots (235.91 mph).

(e) Make sure that the TE flaps extend to the 10-unit position.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-00-860-095

(1) Move the flap control lever to the UP position to retract the flaps.

SUBTASK 27-51-00-860-096

(2) Put the system back to ambient pressure.

NOTE: When you apply or remove pressure to the pitot system, the rate must not be more than 300 knots (345.23 mph) for each minute.

SUBTASK 27-51-00-080-006

(3) Disconnect the pitot/static system tester from the pitot system:

CAUTION DO NOT DISCONNECT THE PITOT SYSTEM TEST ADAPTER WHEN THE PITOT-STATIC SYSTEM IS NOT AT AMBIENT PRESSURE. IF YOU DO, DAMAGE TO THE ADM CAN OCCUR.

(a) Disconnect the air data model test set, COM-1914, and hose from the pitot test adapter, COM-1916.

(b) Remove the pitot test adapter, COM-1916, from the pitot probe.

(c) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	1	C00523	HEATERS CAPT PITOT
C	2	C00238	HEATERS TEMP PROBE
C	3	C01072	HEATERS ALPHA VANE LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT

SUBTASK 27-51-00-860-115

(4) Set the FLT CONTROL and SPOILER switches, on the P5-3 panel, to the ON position.

SUBTASK 27-51-00-860-097

(5) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 27-51-00-410-012

(6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRAILING EDGE FLAP SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has these inspections:

(1) An inspection of the trailing edge flap drive system components

(2) An inspection of the trailing edged flap actuation system.

TASK 27-51-00-210-801

2. Trailing Edge Flap Drive System Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
541	Left Wing - Fairing Flap Support No. 4
542	Left Wing - Fairing Flap Support No. 3
543	Left Wing - Fairing Flap Support No. 2
544	Left Wing - Fairing Flap Support No. 1
550	Subzone - Left Wing: Trailing Edge, Aft of Rear Spar, Inbd of Outboard Trailing Edge Flap
560	Subzone - Left Wing: Trailing Edge , Aft of Rear Spar, Outboard of Inbd Trailing Edge Flap, Inbd of Fixed Trailing Edge
641	Right Wing - Fairing Flap Support No. 5
642	Right Wing - Fairing Flap Support No. 6
643	Right Wing - Fairing Flap Support No. 7
644	Right Wing - Fairing Flap Support No. 8
650	Subzone - Right Wing: Trailing Edge, Aft of Rear Spar, Inboard of Outboard Trailing Edge Flap
660	Subzone - Right Wing: Trailing Edge, Aft of Rear Spar, Outboard of Inboard Trailing Edge Flap, Inboard of Fixed Trailing Edge

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Trailing Edge Flap Drive System Inspection

SUBTASK 27-51-00-480-014

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-00-010-011

(2) Extend the trailing edge flaps to the 40-unit position, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-51-00-040-008

(3) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-00-210-005

(4) Do a general visual inspection of these items:

(a) Flap transmissions

(b) Torque tube support

(c) Seal rib angle gearbox

(d) Main landing gear beam angle gearbox

(e) Torque tube couplings

(f) Tee angle gearbox and U-joint.

SUBTASK 27-51-00-440-005

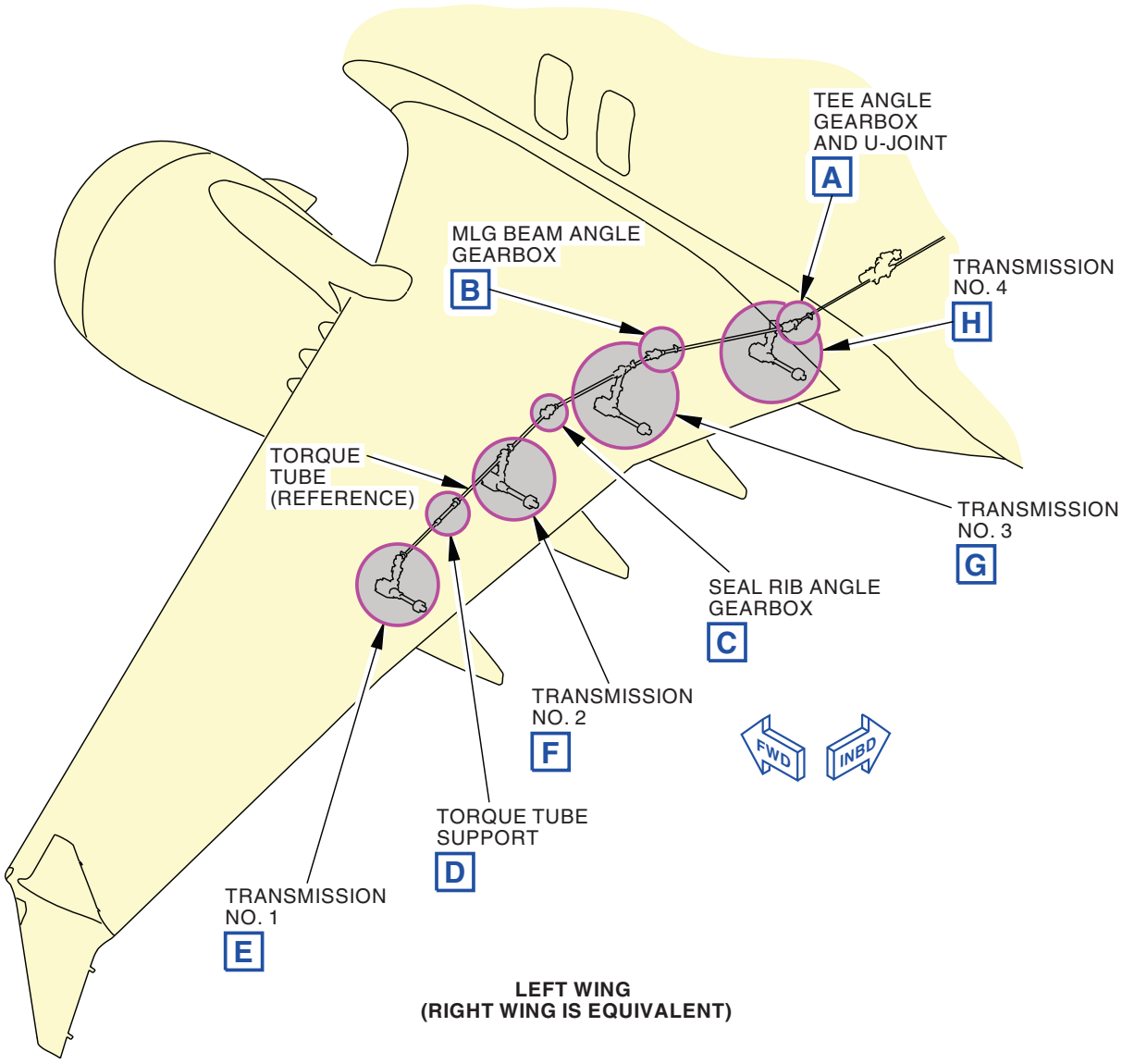
(5) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-00-010-012

(6) Retract the trailing edge flaps to the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

——— END OF TASK ———

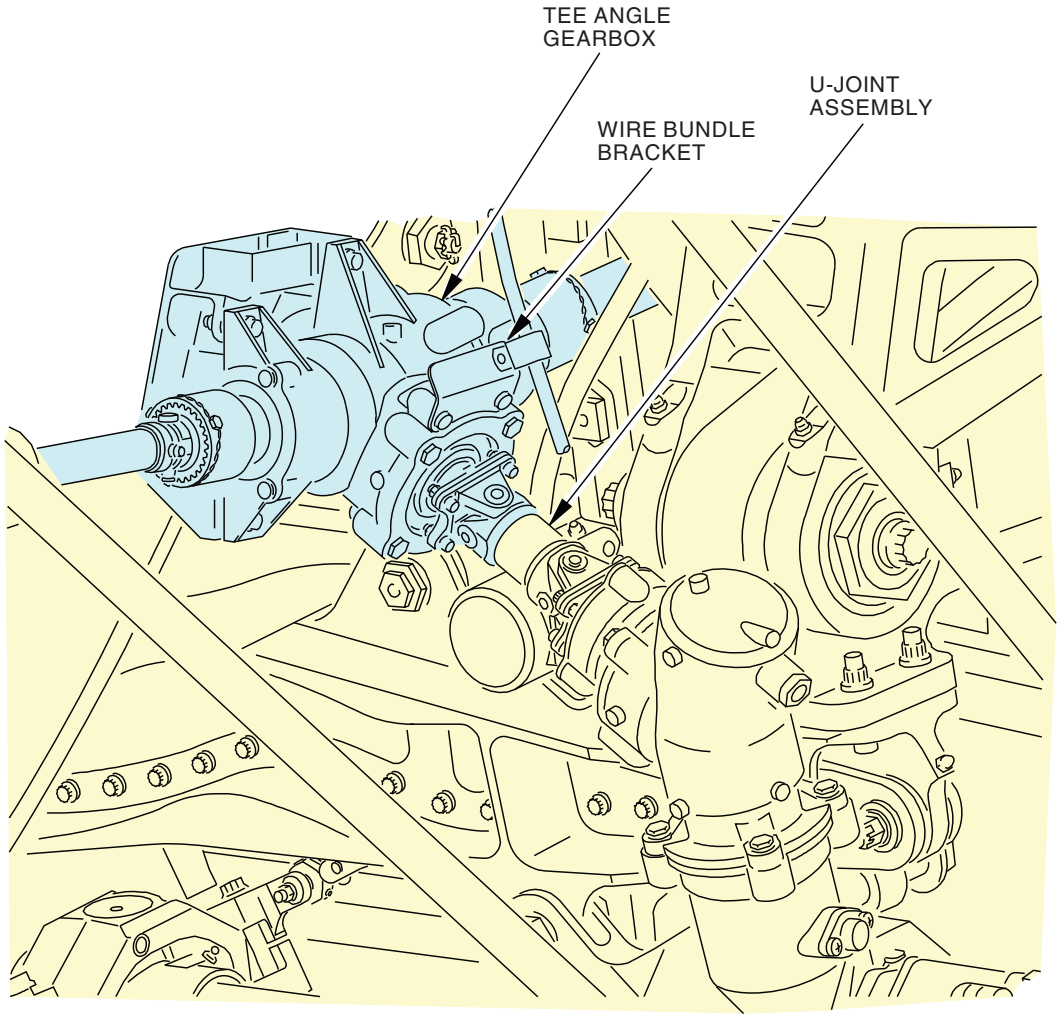
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412752 S00061531277_V2

Trailing Edge Flap Drive System Inspection
Figure 601/27-51-00-990-810 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

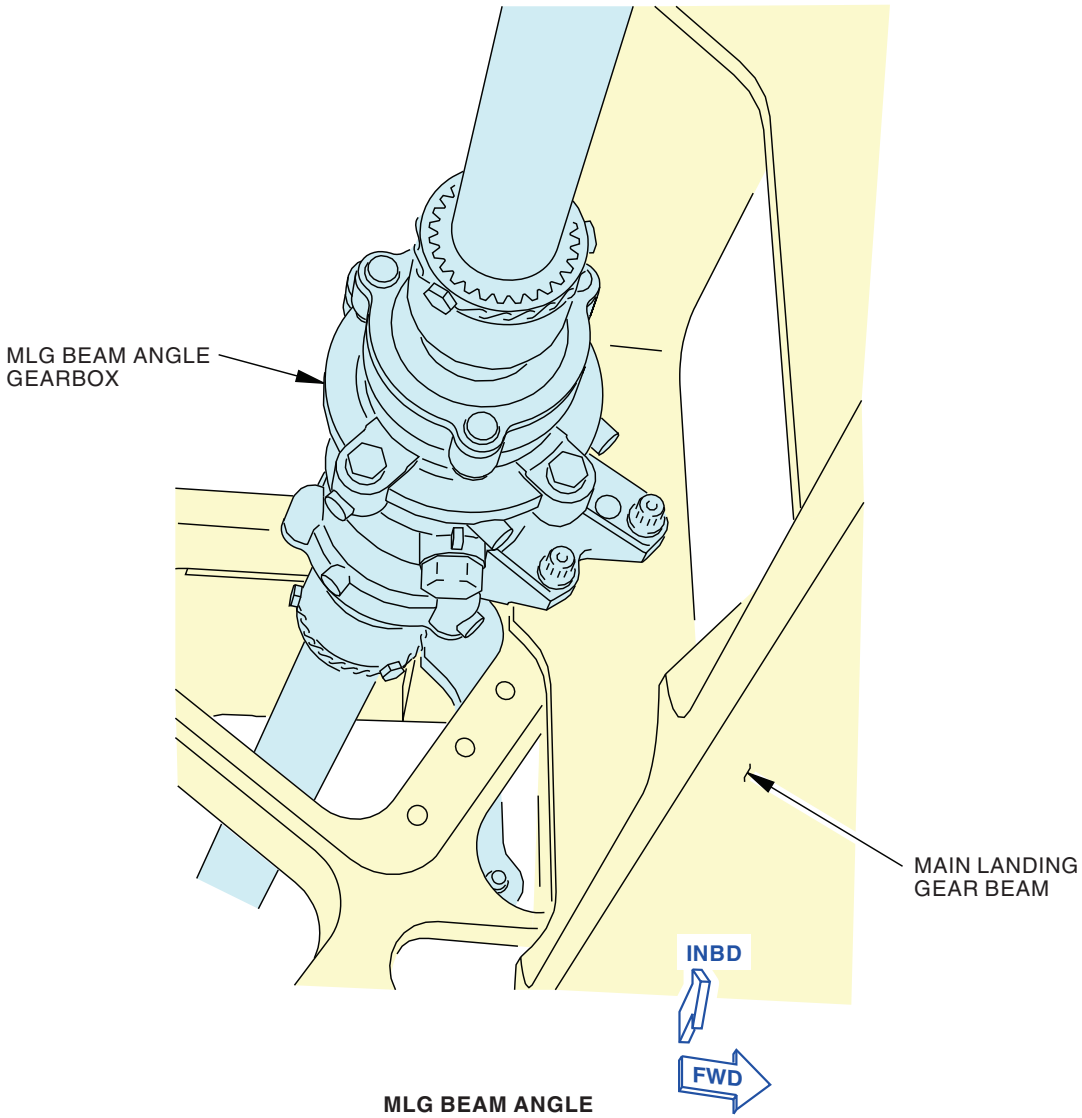


TEE ANGLE GEARBOX AND U-JOINT

3043541 S0000810549_V1

Trailing Edge Flap Drive System Inspection
Figure 601/27-51-00-990-810 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



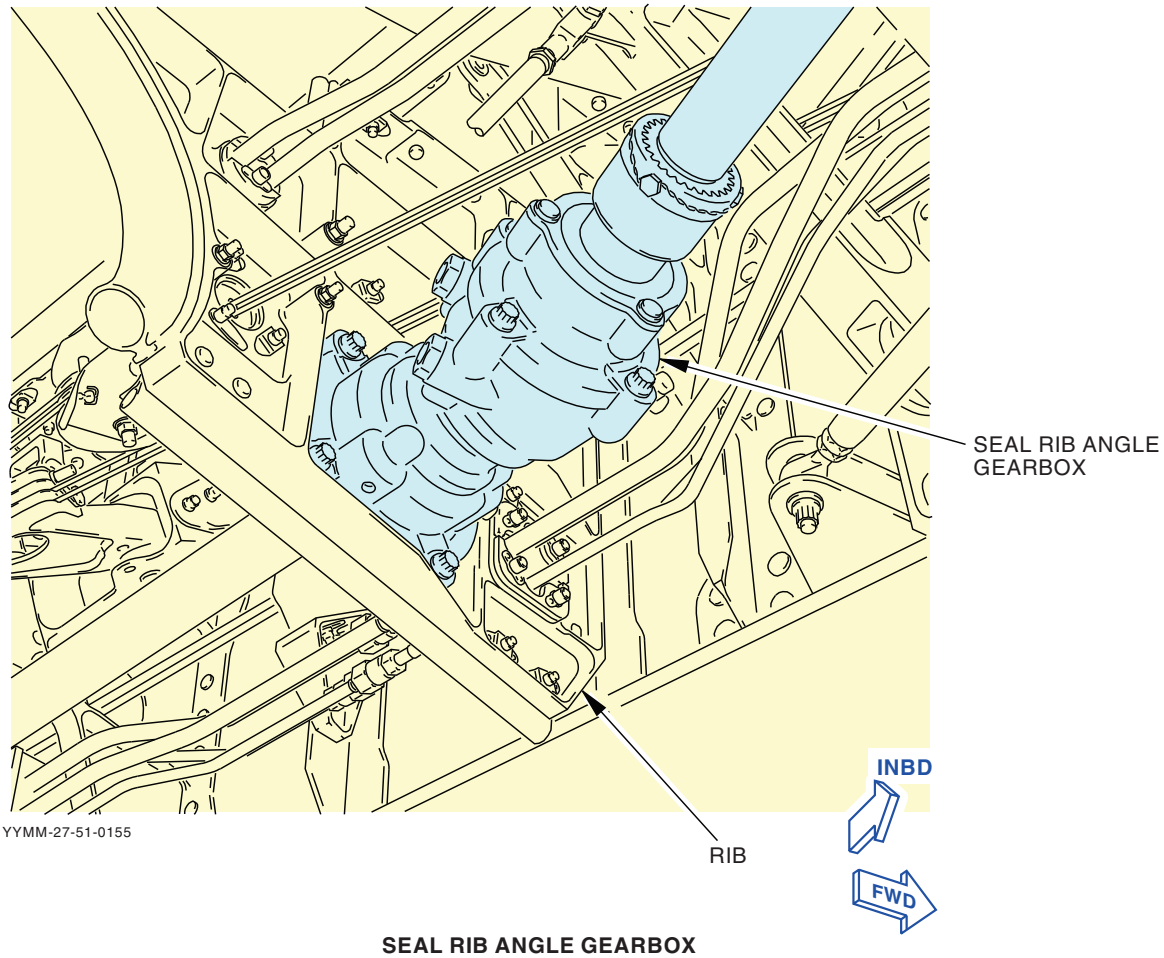
NOTE:

SOME SURROUNDING STRUCTURE REMOVED FOR CLARITY.

3043542 S0000810550_V1

Trailing Edge Flap Drive System Inspection
Figure 601/27-51-00-990-810 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



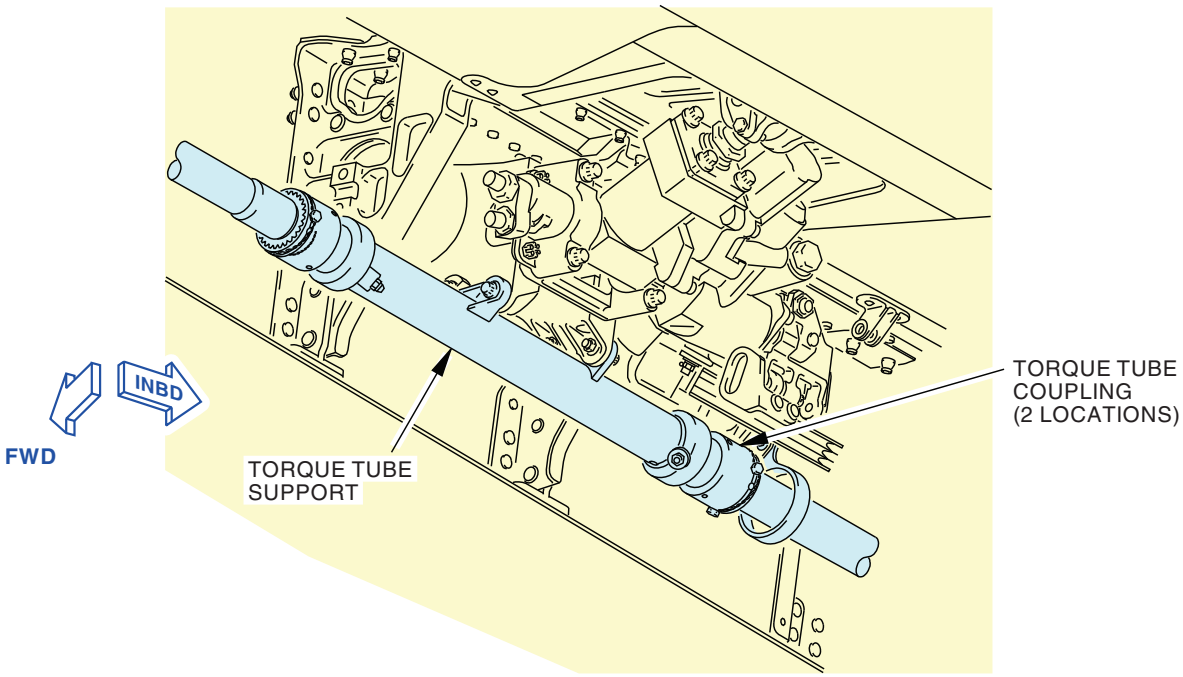
SEAL RIB ANGLE GEARBOX
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

C

3043553 S0000810551_V1

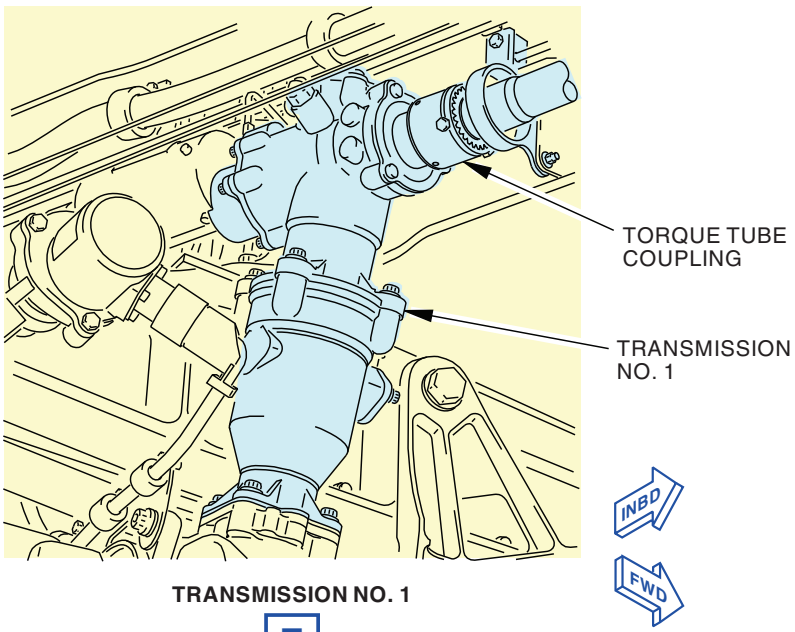
Trailing Edge Flap Drive System Inspection
Figure 601/27-51-00-990-810 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TORQUE TUBE SUPPORT

D



NOTE:

LOWER MIDSPAN PANEL REMOVED FOR CLARITY.

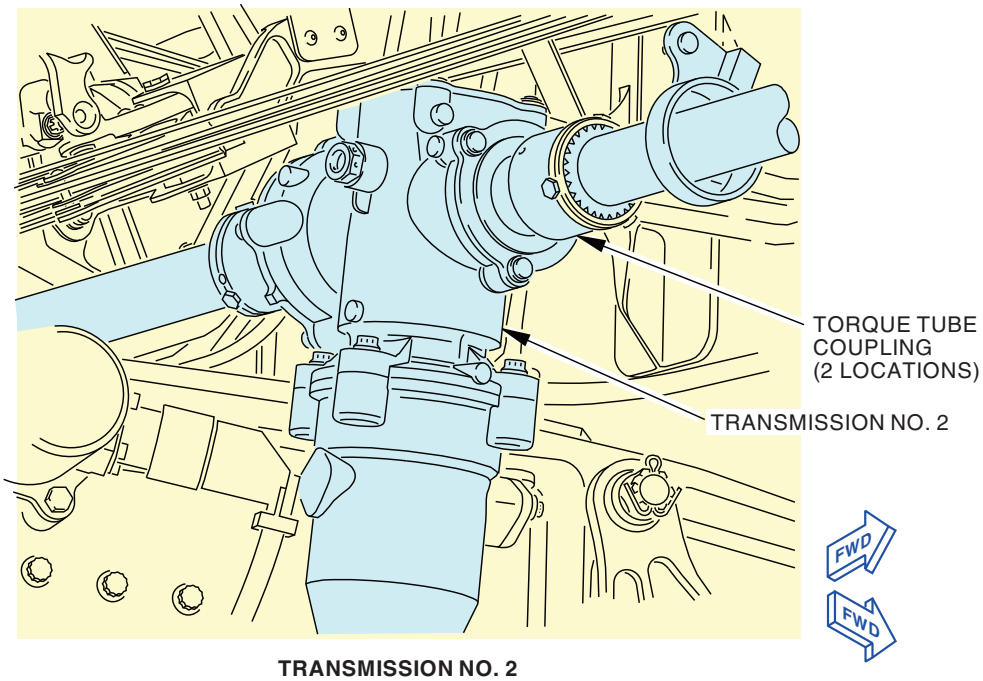
TRANSMISSION NO. 1

E

3043558 S0000810552_V1

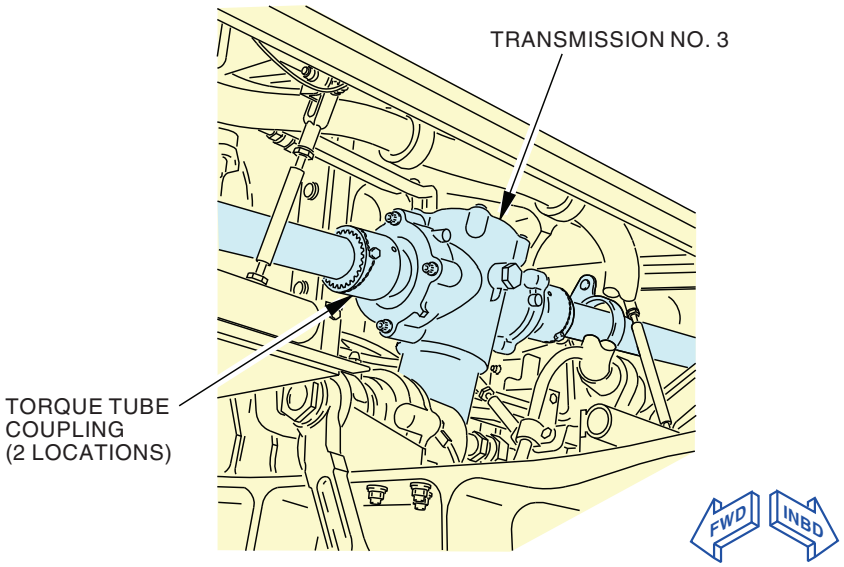
Trailing Edge Flap Drive System Inspection
Figure 601/27-51-00-990-810 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TRANSMISSION NO. 2

F



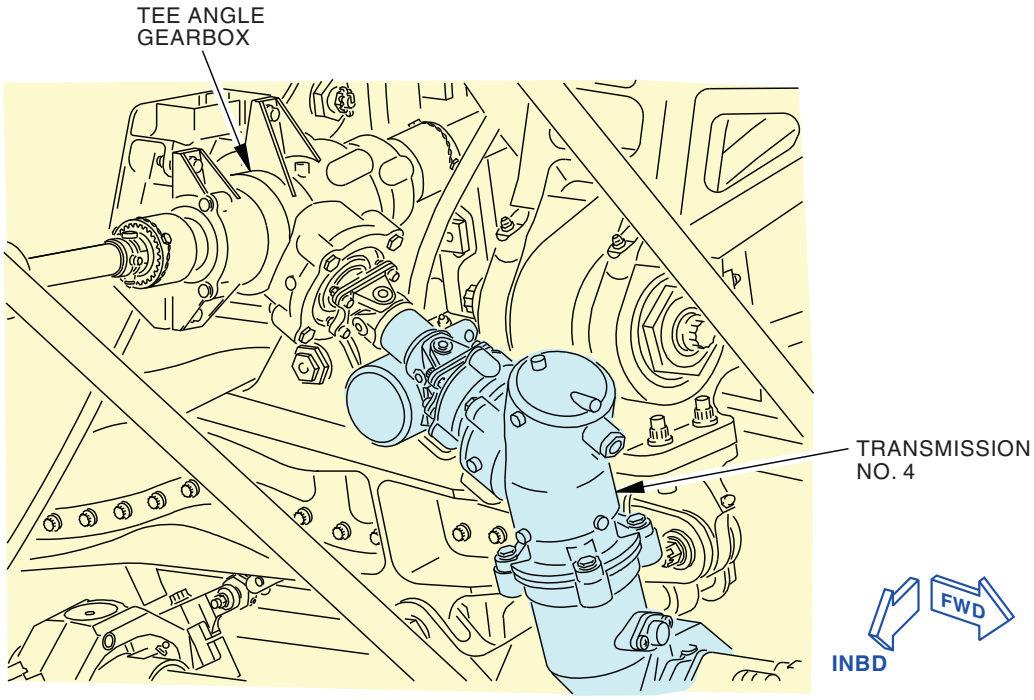
TRANSMISSION NO. 3

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3043581 S0000810553_V1

Trailing Edge Flap Drive System Inspection
Figure 601/27-51-00-990-810 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TRANSMISSION NO. 4

H

3043613 S0000810554_V1

Trailing Edge Flap Drive System Inspection
Figure 601/27-51-00-990-810 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-00-210-802

3. Trailing Edge Flap Actuation System Inspection

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
542	Left Wing - Fairing Flap Support No. 3
543	Left Wing - Fairing Flap Support No. 2
544	Left Wing - Fairing Flap Support No. 1
550	Subzone - Left Wing: Trailing Edge, Aft of Rear Spar, Inbd of Outboard Trailing Edge Flap
560	Subzone - Left Wing: Trailing Edge , Aft of Rear Spar, Outboard of Inbd Trailing Edge Flap, Inbd of Fixed Trailing Edge
642	Right Wing - Fairing Flap Support No. 6
643	Right Wing - Fairing Flap Support No. 7
644	Right Wing - Fairing Flap Support No. 8
650	Subzone - Right Wing: Trailing Edge, Aft of Rear Spar, Inboard of Outboard Trailing Edge Flap
660	Subzone - Right Wing: Trailing Edge, Aft of Rear Spar, Outboard of Inboard Trailing Edge Flap, Inboard of Fixed Trailing Edge

D. Access Panels

Number	Name/Location
567BT	Flap, Forward Carriage Bearing And Fitting
567ET	Flap, Forward Carriage Bearing And Fitting
667BT	Flap, Forward Carriage Bearing And Fitting
667ET	Flap, Forward Carriage Bearing And Fitting

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Inspection

SUBTASK 27-51-00-480-015

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-00-010-013

(2) Extend the trailing edge flaps to the 40-unit position. To extend the flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-51-00-040-009

(3) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

F. Outboard Flap Inspection

(Figure 602)

SUBTASK 27-51-00-010-014

(1) At the flap supports, remove the leading edge skin panel [2] from the main flap [1] forward of the carriage:

(a) Remove the bolts [3] that attach the leading edge skin panel [2].

(b) Remove the leading edge skin panel [2] from the main flap [1].

Open these access panels:

Number	Name/Location
567BT	Flap, Forward Carriage Bearing And Fitting
567ET	Flap, Forward Carriage Bearing And Fitting
667BT	Flap, Forward Carriage Bearing And Fitting
667ET	Flap, Forward Carriage Bearing And Fitting

SUBTASK 27-51-00-210-006

(2) Do a detailed visual inspection of these items at the flap supports:

NOTE: These are flap supports No. 1, 2, 7 and 8.

(a) Aft flap drive rod, push rod and bellcrank

NOTE: Some freeplay in the bellcrank is acceptable.

(b) Carriage rollers

(c) Aft link and fittings

(d) Forward carriage bearing and fitting.

SUBTASK 27-51-00-210-007

(3) Do a detailed visual inspection of the deflection control rollers between the inboard and outboard flaps and at the outboard end of the outboard flap.

SUBTASK 27-51-00-210-008

(4) At the outboard aft flap, do a detailed visual inspection of these items:

(a) Aft flap track attachments.

NOTE: There are four aft flap tracks on the outboard flap.

(b) Inboard and outboard programming rollers.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Inboard and outboard aft flap pushrods.

(d) Rubstrips common to the aft flap support tracks, and common to the inboard and outboard edge program tracks.

(e) Inboard and outboard aft flap track rollers.

SUBTASK 27-51-00-410-013

(5) Install the leading edge skin panel [2] on the main flap [1] forward of the carriage:

(a) Close these access panels:

Number	Name/Location
567BT	Flap, Forward Carriage Bearing And Fitting
567ET	Flap, Forward Carriage Bearing And Fitting
667BT	Flap, Forward Carriage Bearing And Fitting
667ET	Flap, Forward Carriage Bearing And Fitting

(b) Put the leading edge skin panel [2] in its location on the leading edge of the main flap [1].

(c) Apply compound, C00528 to the holes for the bolts [3].

(d) Install the bolts [3] to attach the leading edge skin panel [2].

(e) Apply sealant, A00247 to fill all of the clearances around the leading edge skin panel [2] that are more than 0.05 in. (1.27 mm).

G. Inboard Flap Inspection

(Figure 603)

SUBTASK 27-51-00-210-009

(1) At the outboard flap support for the inboard flap, do a detailed visual inspection of these items:

NOTE: These are flap supports No. 3 and 6.

(a) Aft flap drive rod and crankshaft

(b) Carriage rollers

SUBTASK 27-51-00-210-010

(2) At the inboard flap support for the inboard flap, do a detailed visual inspection of these items:

NOTE: These are flap supports No. 4 and 5.

(a) Carriage rollers

SUBTASK 27-51-00-210-011

(3) At the inboard aft flap, do a detailed visual inspection of these items:

(a) Aft flap track attachments.

NOTE: There are two aft flap tracks on the inboard flap.

(b) Inboard and outboard programming rollers.

(c) Inboard and outboard aft flap pushrods.

(d) Rubstrips common to the aft flap support tracks, and common to the inboard and outboard edge program tracks.

(e) Inboard and outboard aft flap track rollers.

SUBTASK 27-51-00-440-006

(4) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

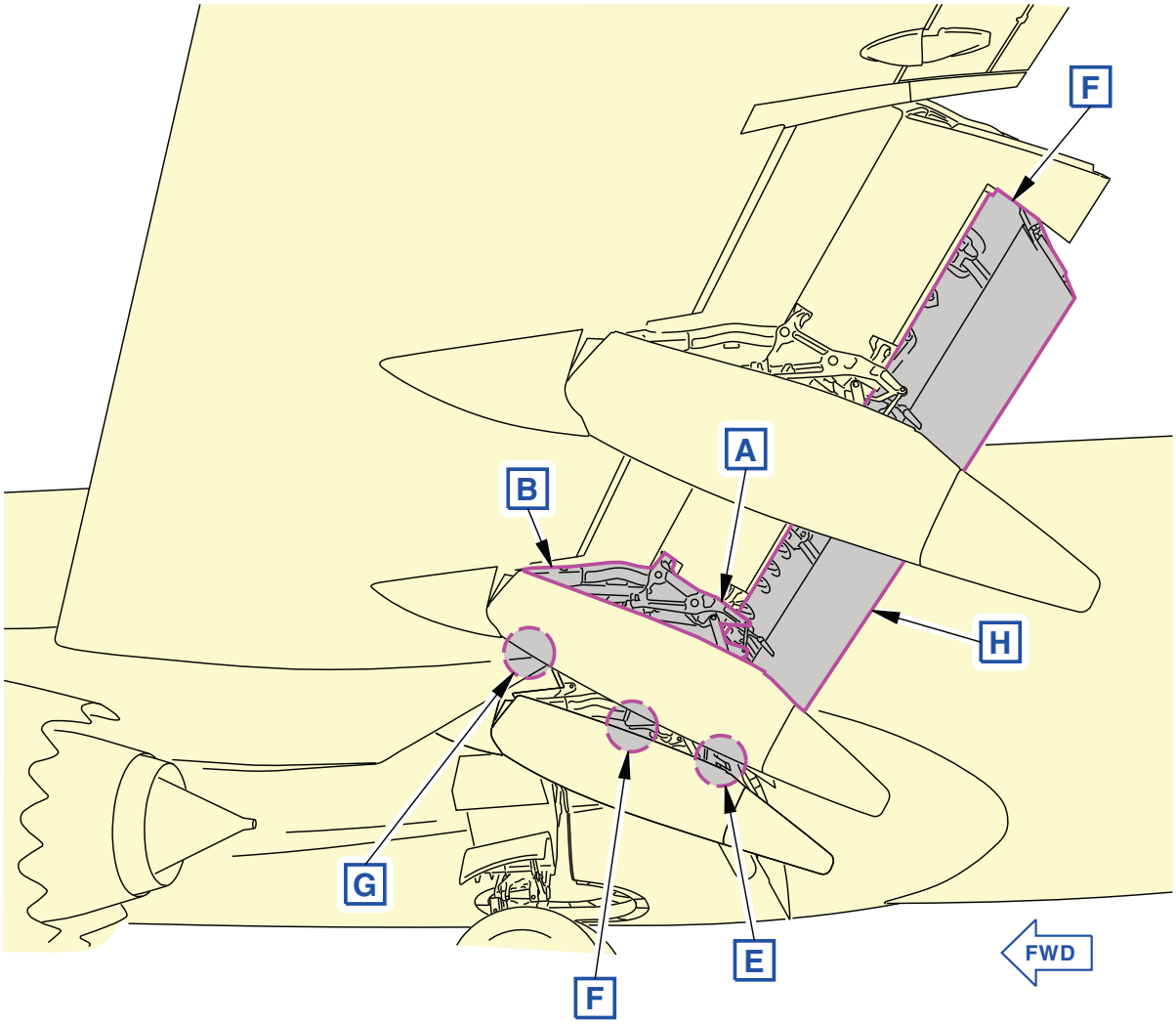
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-00-010-015

(5) Retract the trailing edge flaps to the retracted position. To retract the flaps, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

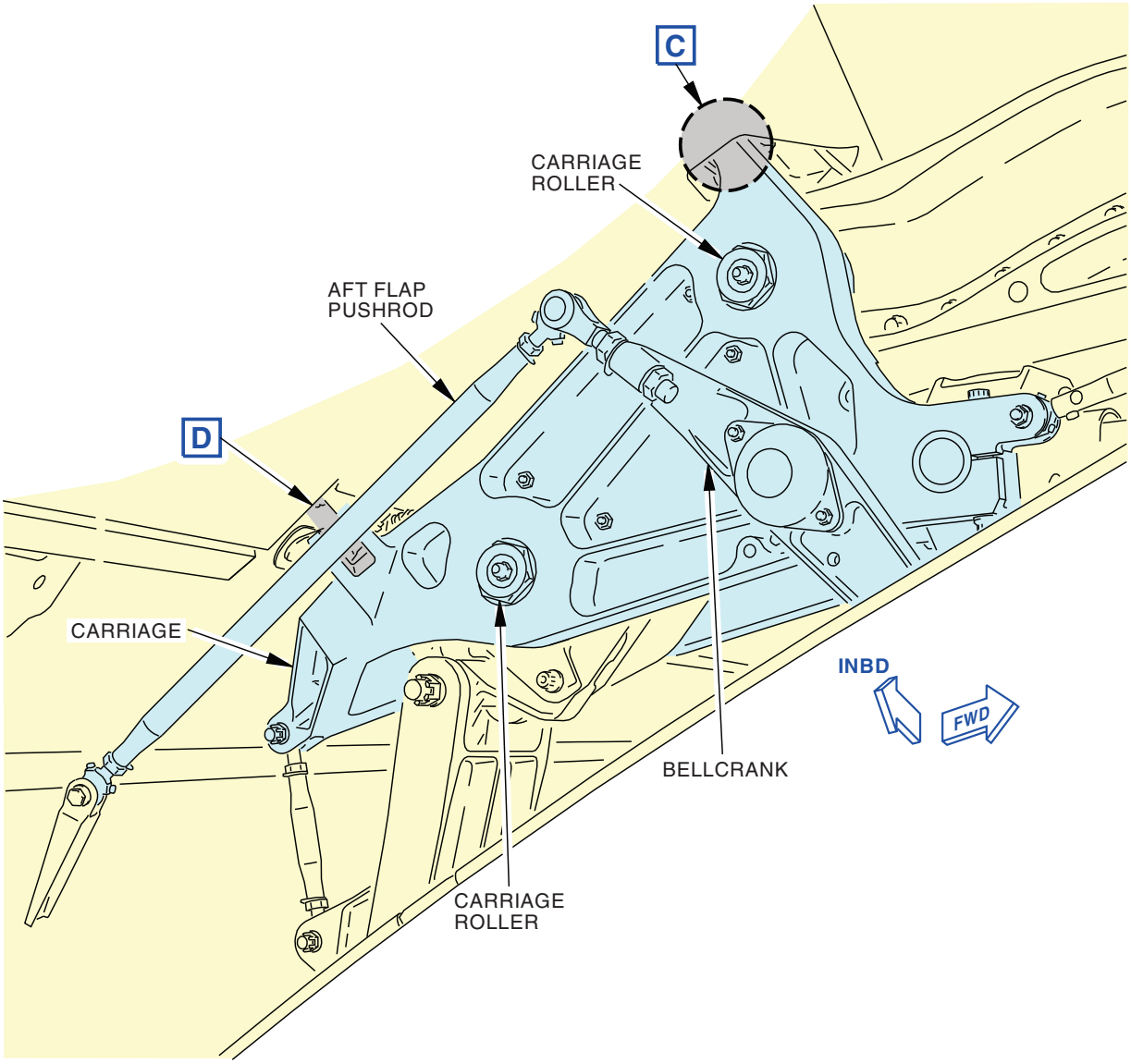


LEFT WING
(RIGHT WING IS EQUIVALENT)

2412753 S00061531279_V1

Outboard Trailing Edge Flap Actuation System Inspection
Figure 602/27-51-00-990-811 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



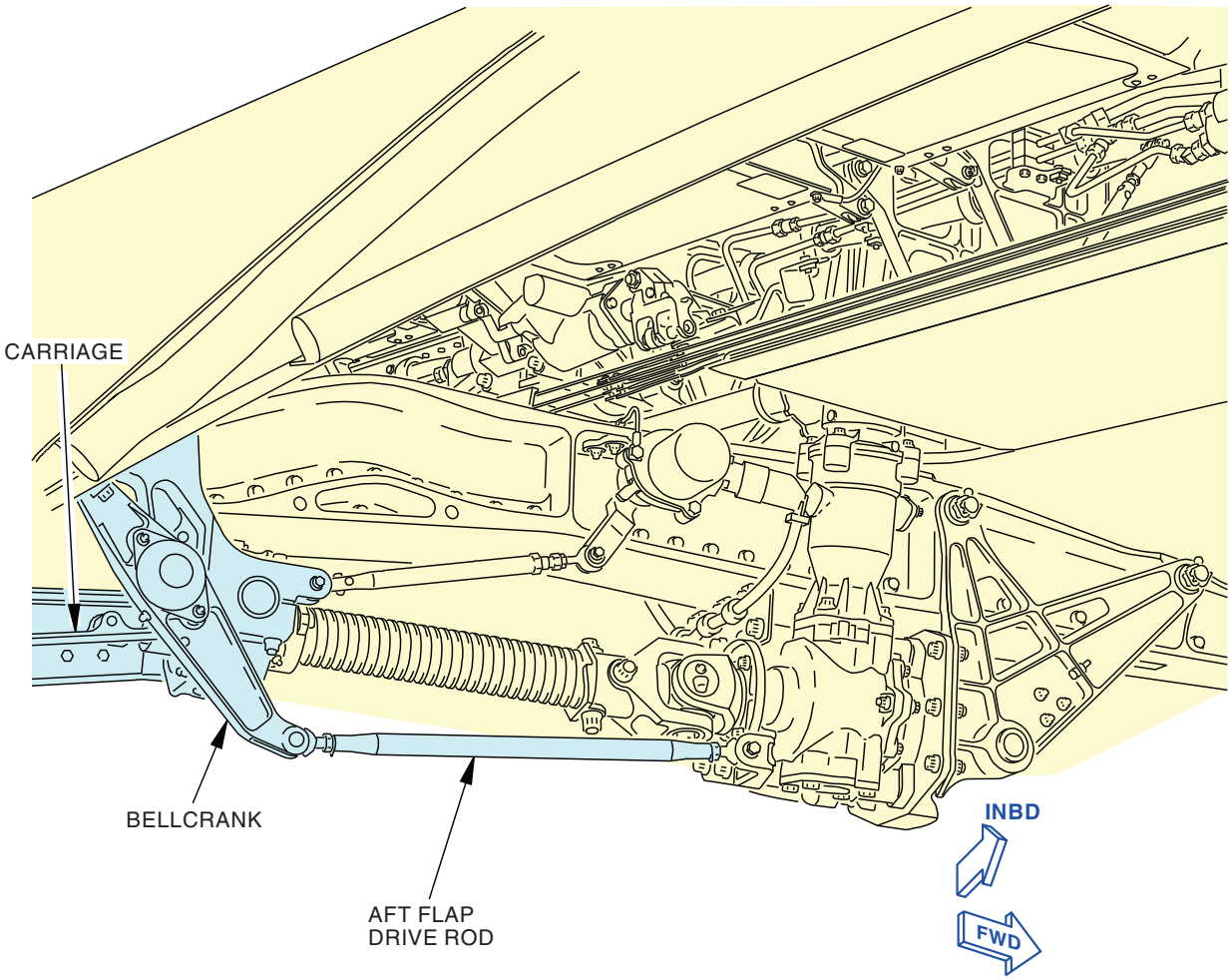
OUTBOARD FLAP INBOARD FLAP SUPPORT
(OUTBOARD FLAP OUTBOARD FLAP SUPPORT IS EQUIVALENT)

A

2412754 S00061531280_V1

Outboard Trailing Edge Flap Actuation System Inspection
Figure 602/27-51-00-990-811 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



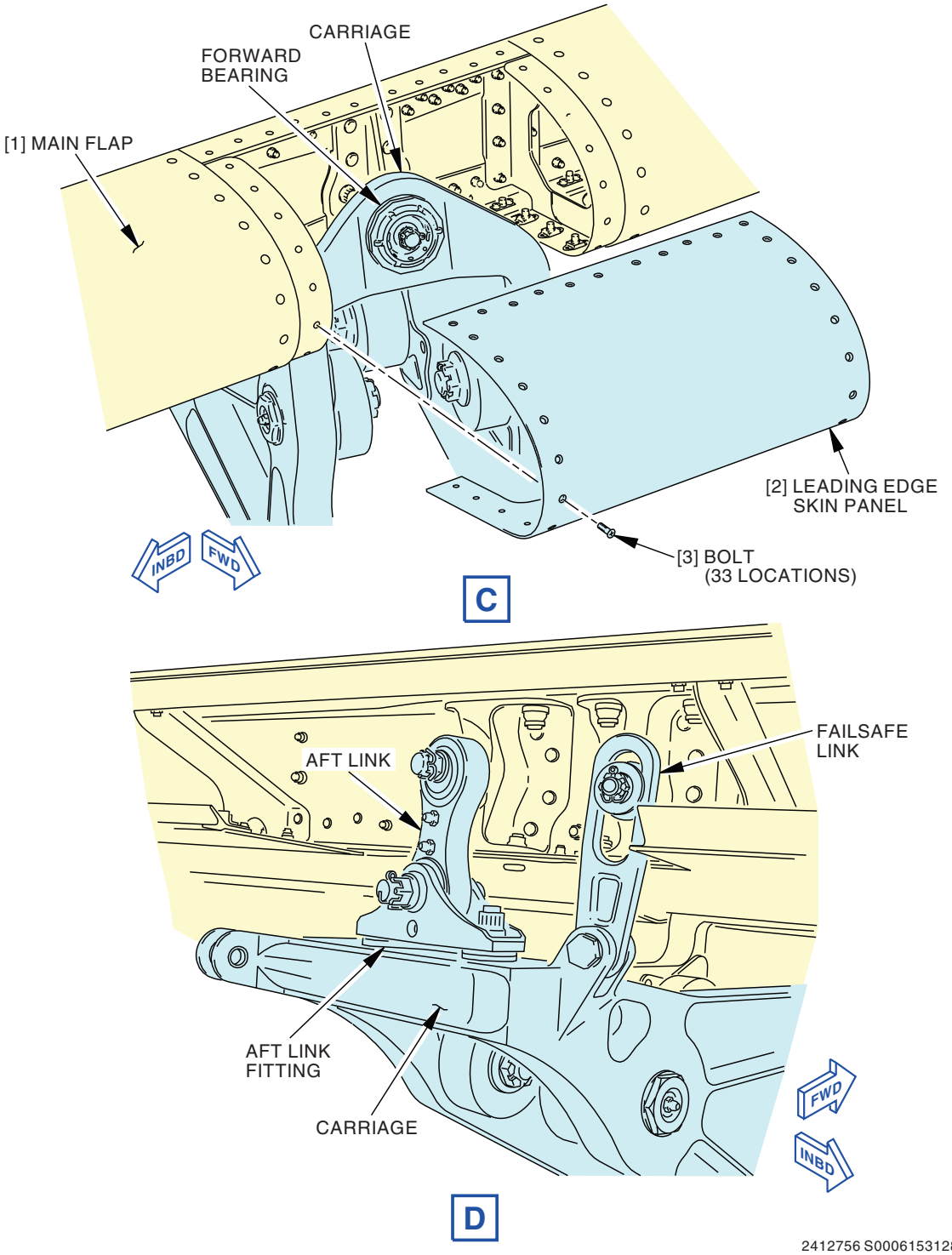
OUTBOARD FLAP INBOARD FLAP SUPPORT
(OUTBOARD FLAP OUTBOARD FLAP SUPPORT IS EQUIVALENT)

B

2412755 S00061531281_V1

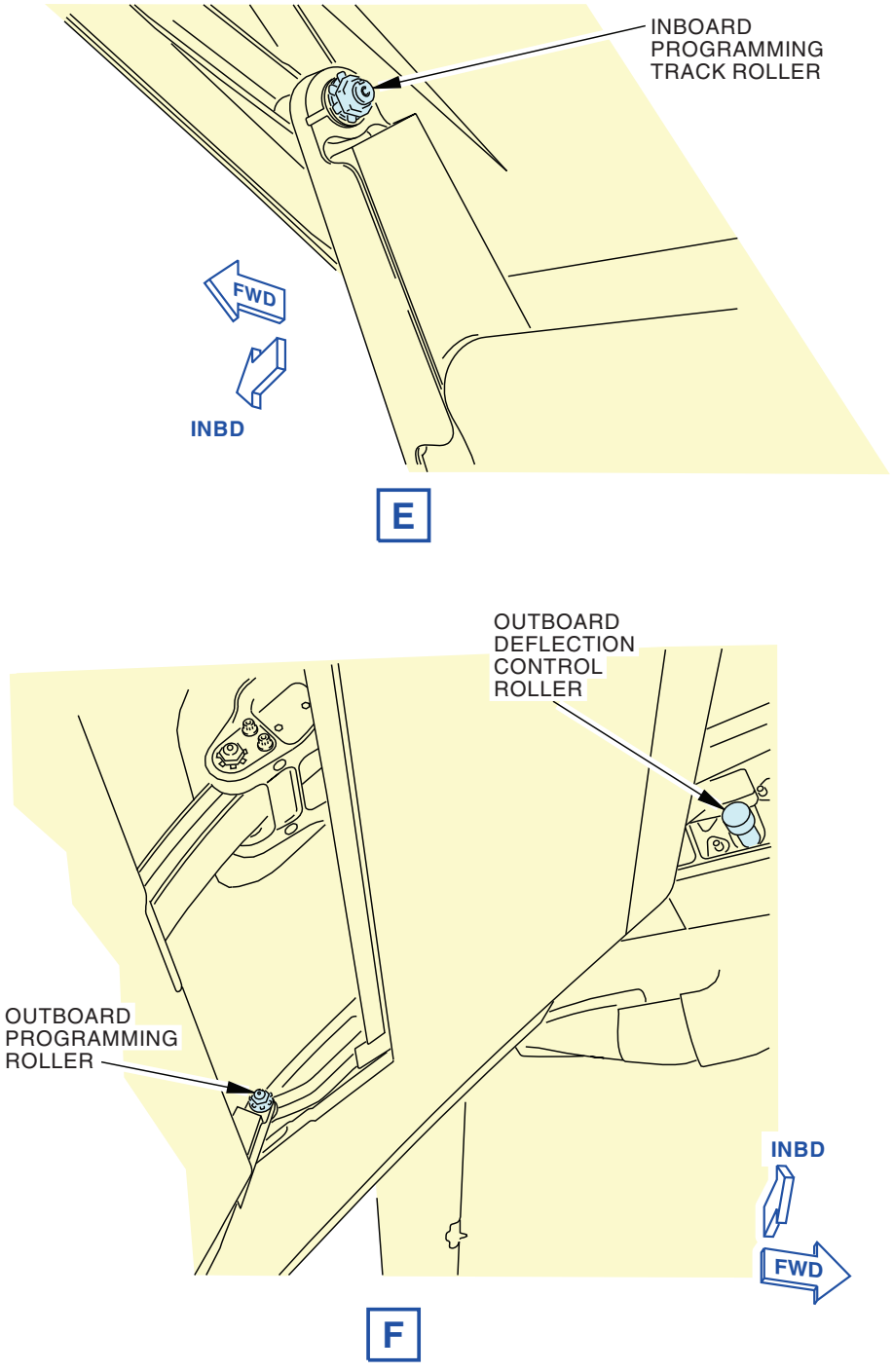
Outboard Trailing Edge Flap Actuation System Inspection
Figure 602/27-51-00-990-811 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outboard Trailing Edge Flap Actuation System Inspection
Figure 602/27-51-00-990-811 (Sheet 4 of 7)

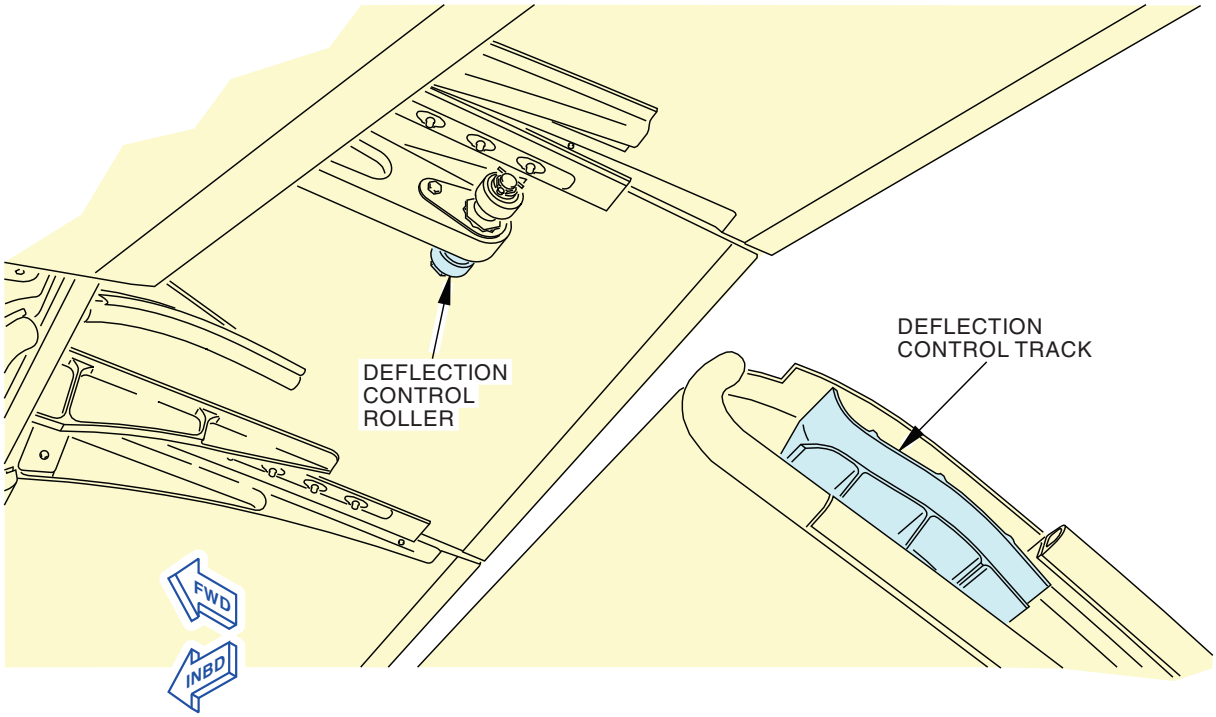
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412757 S00061531283_V1

Outboard Trailing Edge Flap Actuation System Inspection
Figure 602/27-51-00-990-811 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

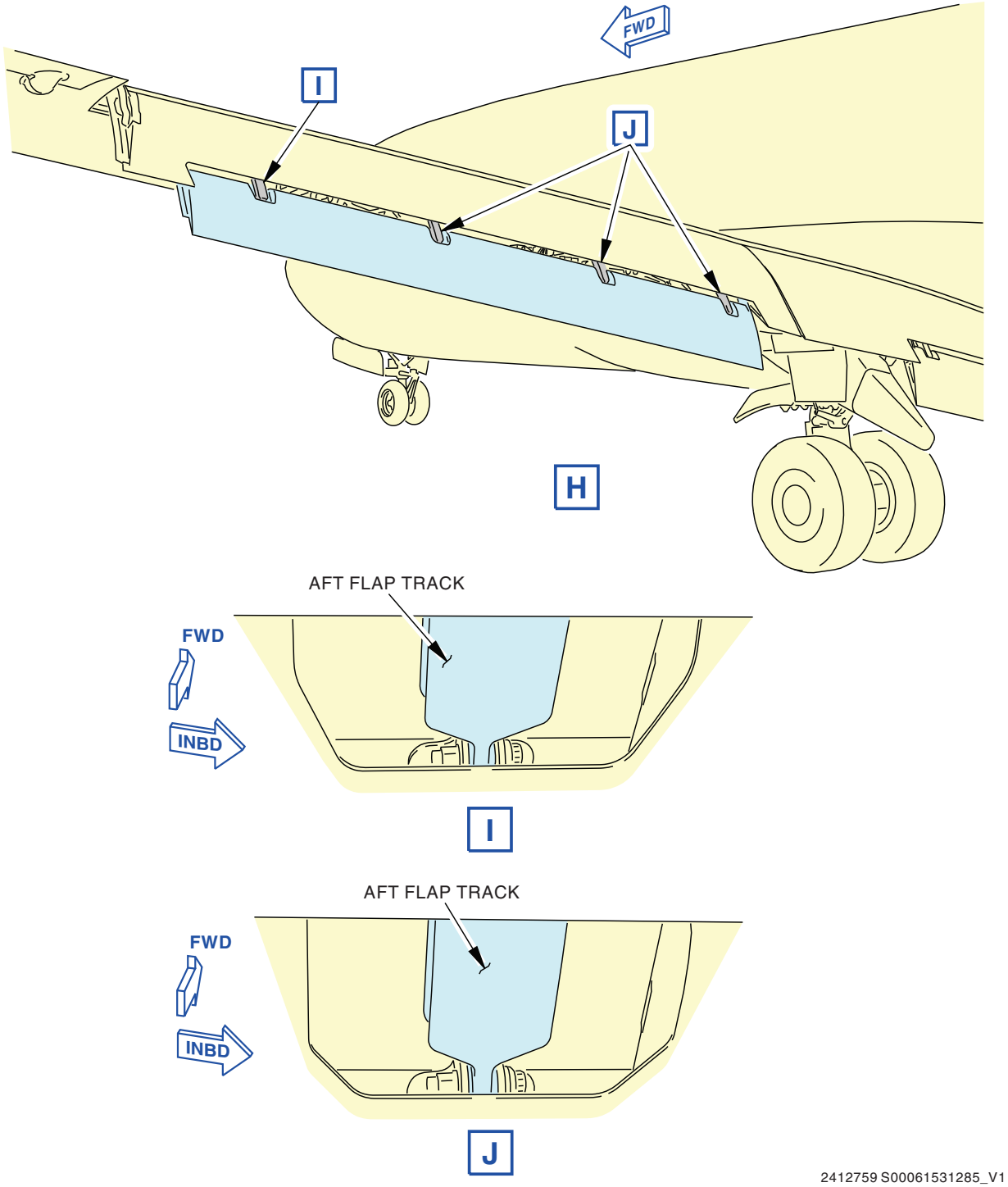


G

2412758 S00061531284_V1

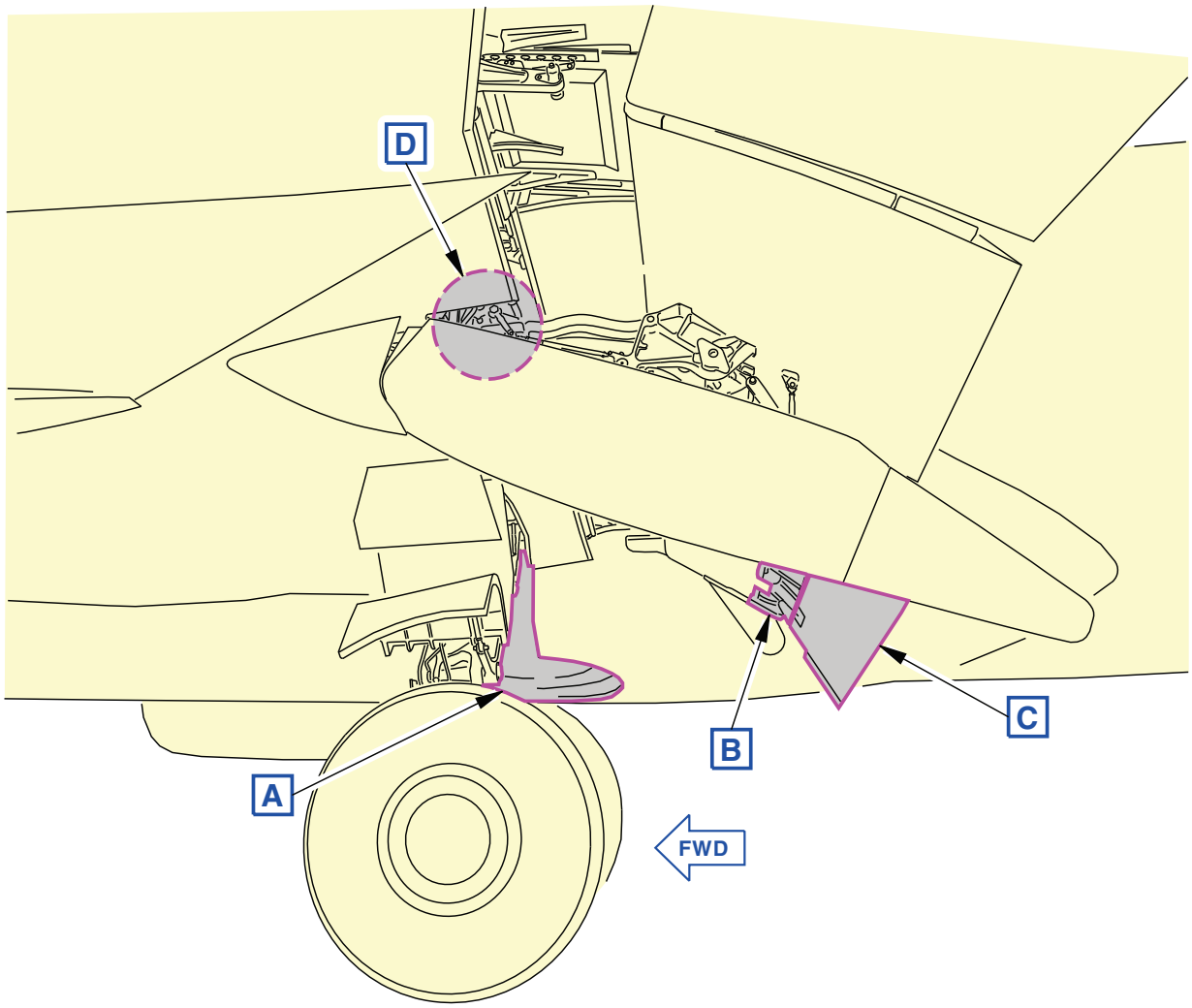
Outboard Trailing Edge Flap Actuation System Inspection
Figure 602/27-51-00-990-811 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outboard Trailing Edge Flap Actuation System Inspection
Figure 602/27-51-00-990-811 (Sheet 7 of 7)

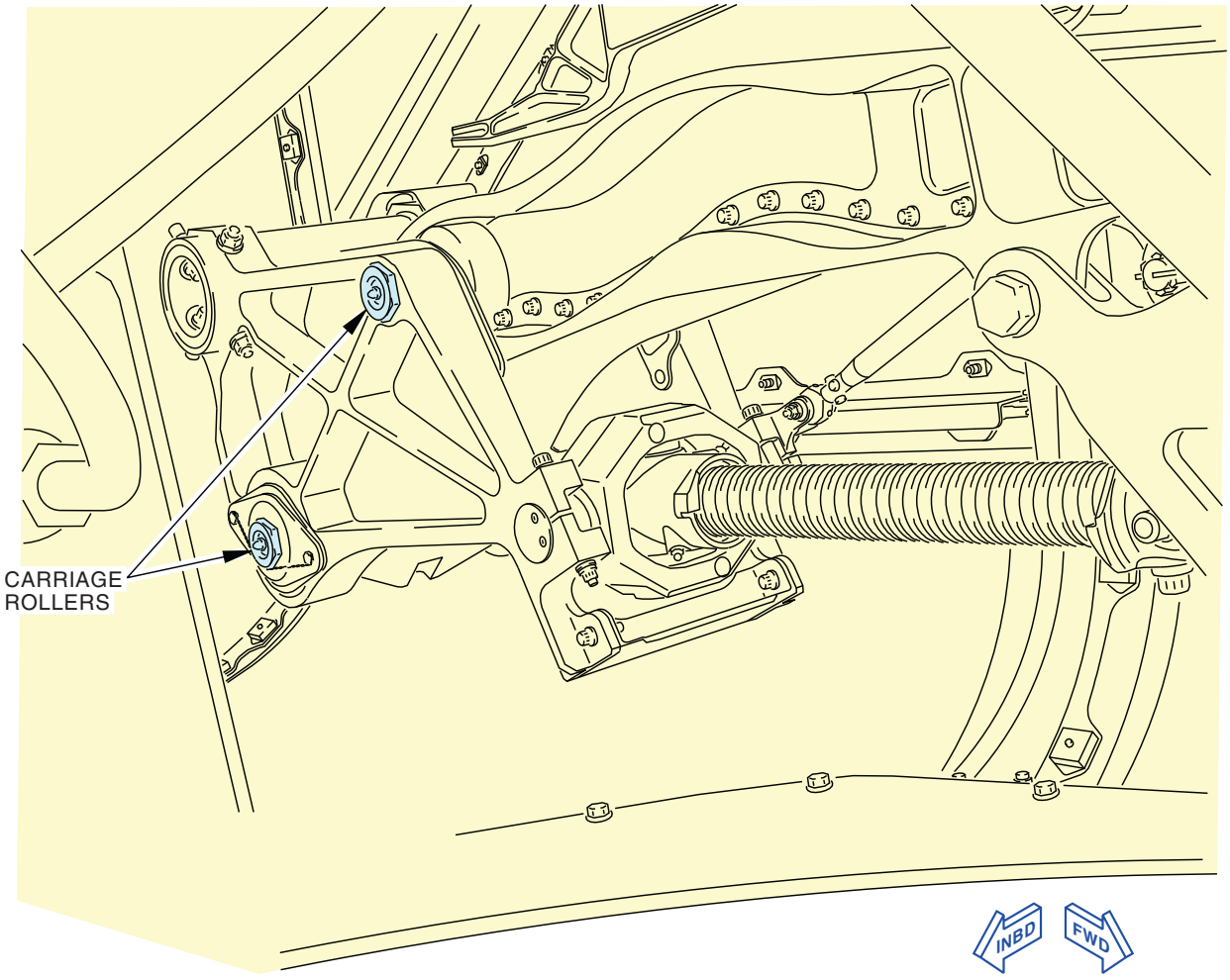
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412760 S00061531286_V1

Inboard Trailing Edge Flap Actuation System Inspection
Figure 603/27-51-00-990-812 (Sheet 1 of 4)

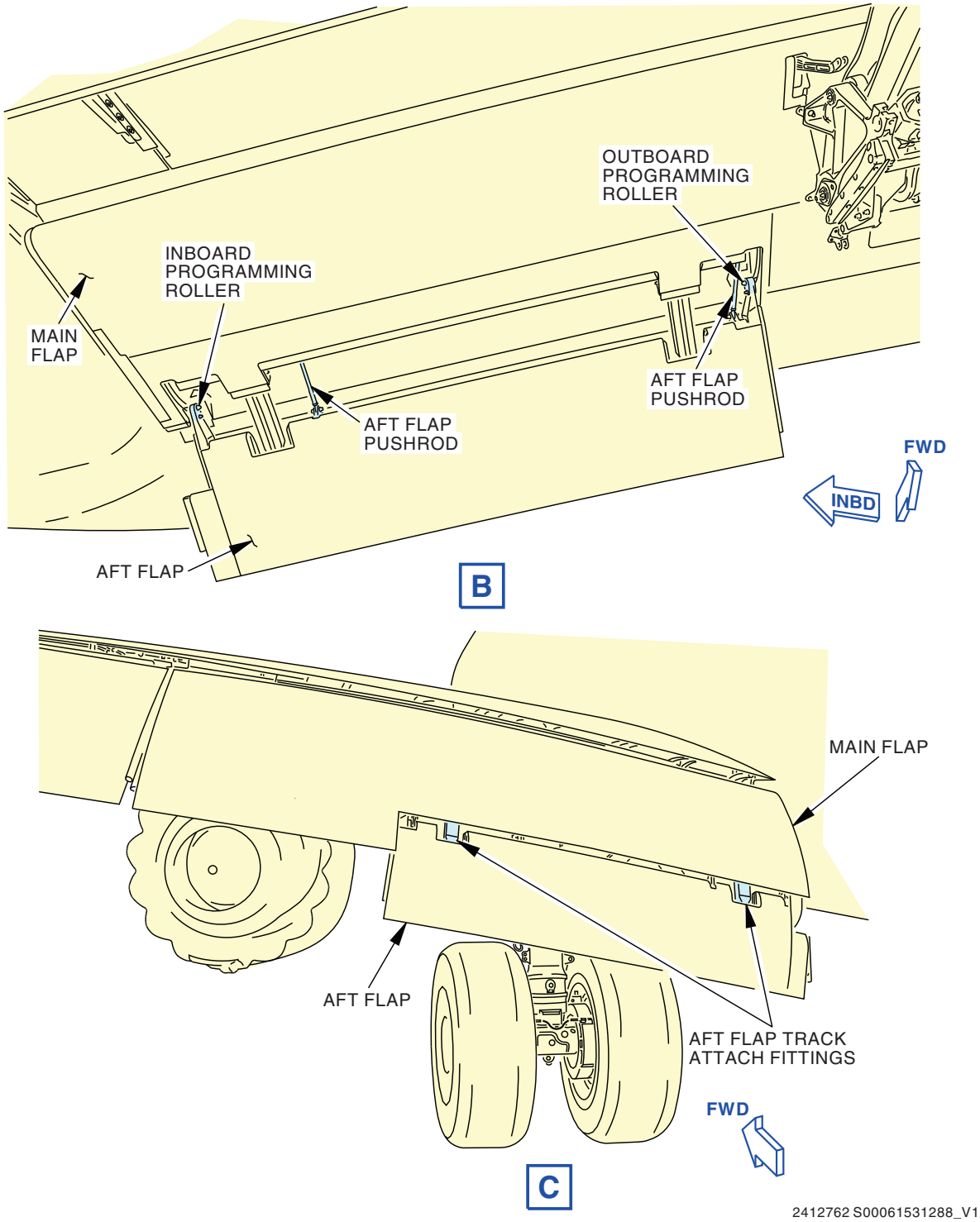
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412761 S00061531287_V1

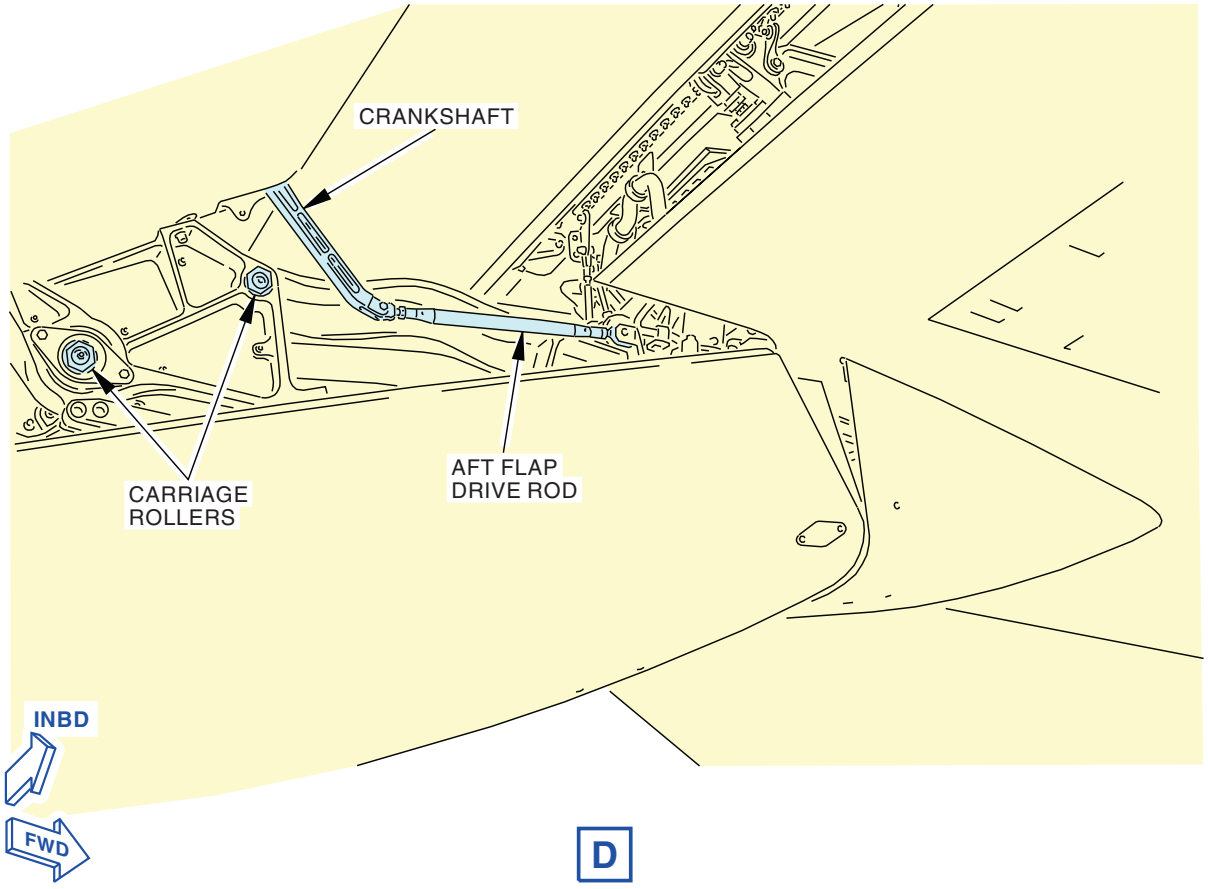
Inboard Trailing Edge Flap Actuation System Inspection
Figure 603/27-51-00-990-812 (Sheet 2 of 4)

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Inboard Trailing Edge Flap Actuation System Inspection
Figure 603/27-51-00-990-812 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Inboard Trailing Edge Flap Actuation System Inspection
Figure 603/27-51-00-990-812 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLAP/SLAT ELECTRONICS UNIT (FSEU) - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Flap/slat electronics unit (FSEU) removal

(2) Flap/slat electronics unit (FSEU) installation.

TASK 27-51-01-000-801

2. Flap/Slat Electronics Unit Removal

(Figure 401)

A. General

(1) The flap/slat electronics unit is installed on the E1-1 shelf in the main equipment center.

B. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Procedure

SUBTASK 27-51-01-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-51-01-010-001

(2) To get access to the main equipment center, do this step:

Open this access panel:

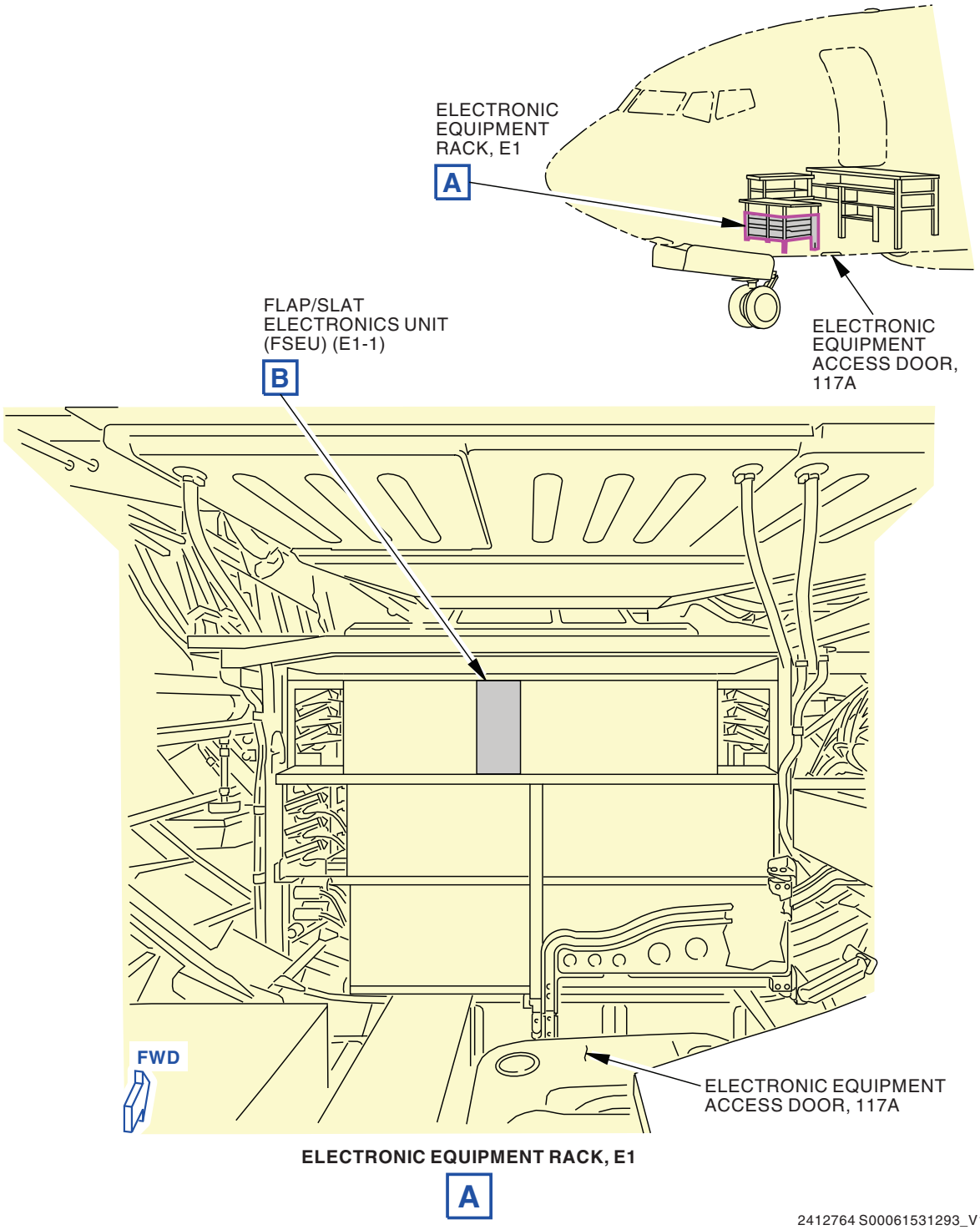
Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-51-01-020-001

(3) Remove the flap/slat electronics unit [1] from the E1-1 shelf, do this task: E/E Box Removal, TASK 20-10-07-000-801.

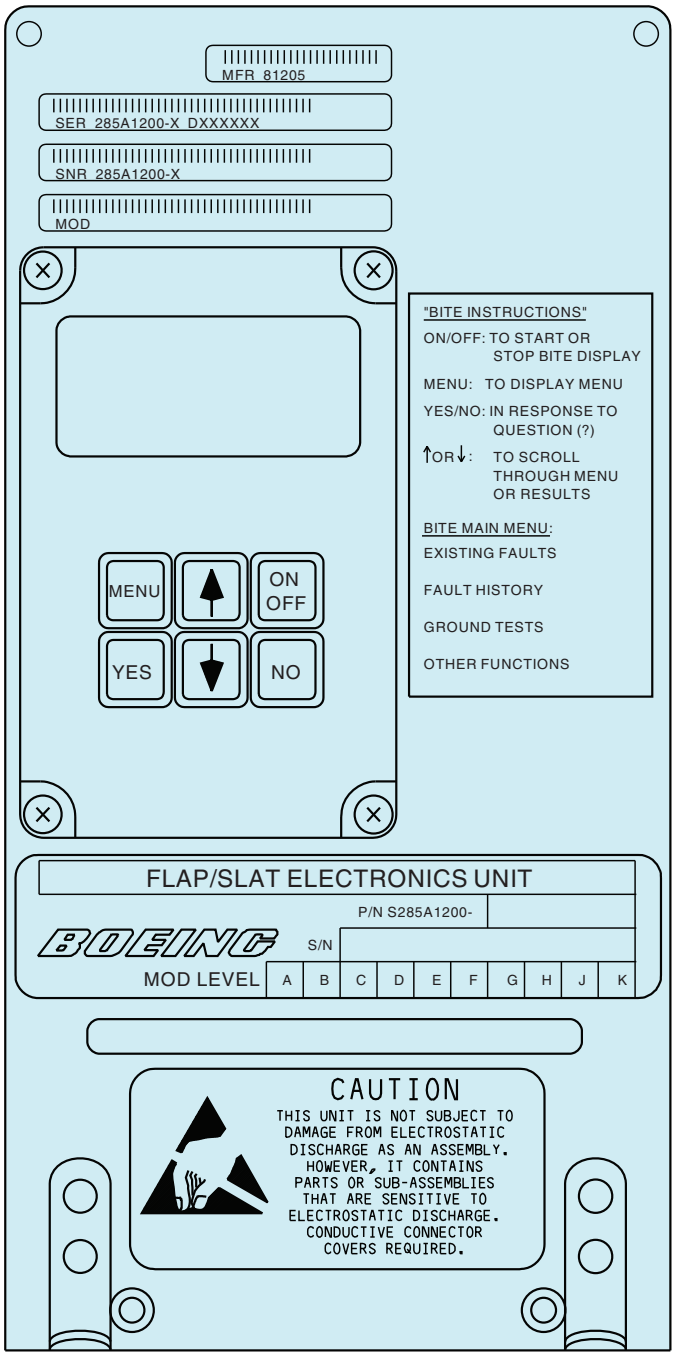
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap/Slat Electronics Unit Installation
Figure 401/27-51-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



[1] FLAP/SLAT ELECTRONICS UNIT (FSEU)

B

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Flap/Slat Electronics Unit Installation
Figure 401/27-51-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-01-400-801

3. Flap/Slat Electronics Unit Installation

(Figure 401)

A. General

(1) The flap/slat electronics unit is installed on the E1-1 shelf in the main equipment center.

B. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
27-51-00-740-801	FSEU Display and Existing Faults Test (P/B 501)
27-51-00-740-802	Trailing Edge Flap Asymmetry Skew System BITE Test (P/B 501)
27-51-00-740-803	Trailing Edge Flap Load Relief System BITE Test (P/B 501)
27-81-00-700-804	Leading Edge Uncommanded Motion Protection Test (P/B 501)
27-81-00-730-801	Leading Edge Cruise Depressurization Test (P/B 501)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Flap/slat electronics unit	27-51-01-01-010	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Flap/Slat Electronic Unit Installation

SUBTASK 27-51-01-420-001

(1) Install the flap/slat electronics unit [1] on the E1-1 shelf, do this task: E/E Box Installation, TASK 20-10-07-400-801.

SUBTASK 27-51-01-860-002

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
A	8	C00211	FLIGHT CONTROL FSEU DC 1
A	9	C01468	FLIGHT CONTROL FSEU DC 2
A	11	C01443	FC TE FLAP POS/SKEW SNSR & IND RIGHT
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

G. Flap/Slat Electronic Unit Installation Test

SUBTASK 27-51-01-740-001

(1) Do the Existing Faults BITE test with the flap/slat electronics unit [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) To do the BITE test, do this task: FSEU Display and Existing Faults Test, TASK 27-51-00-740-801.

SUBTASK 27-51-01-740-002

(2) Do the Leading Edge Cruise Depressurization Test with the flap/slat electronics unit [1].

(a) To do the depressurization test, do this task: Leading Edge Cruise Depressurization Test, TASK 27-81-00-730-801.

SUBTASK 27-51-01-700-001

(3) Do the Leading Edge Uncommanded Motion Protection Test with the flap/slat electronics unit [1].

(a) To do the uncommanded motion protection test, do this task: Leading Edge Uncommanded Motion Protection Test, TASK 27-81-00-700-804.

SUBTASK 27-51-01-740-003

(4) Do the Load Relief BITE test with the flap/slat electronics unit [1].

(a) To do the BITE test, do this task: Trailing Edge Flap Load Relief System BITE Test, TASK 27-51-00-740-803.

SUBTASK 27-51-01-740-004

(5) Do the Asymmetry Skew System BITE test with the flap/slat electronics unit [1].

(a) To do the BITE test, do this task: Trailing Edge Flap Asymmetry Skew System BITE Test, TASK 27-51-00-740-802.

SUBTASK 27-51-01-710-001

(6) Do a test of the air mode inhibit function of the flap/slat electronics unit [1]:

(a) Simulate air mode, do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

NOTE: It is only necessary to simulate air mode with system 1.

(b) Push the ON/OFF button on the front panel of the flap/slat electronics unit [1] to turn it off and then on.

(c) Make sure that INHIBIT: IN AIR flashes on the display window on the flap/slat electronics unit [1] before the display turns off.

(d) Return the airplane to ground mode, do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

(e) Push the ON/OFF button on the front panel of the flap/slat electronics unit [1] to turn it off and then on.

(f) Make sure that INHIBIT: IN AIR does not show on the display window on the flap/slat electronics unit [1].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-01-010-002

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRAILING EDGE FLAP BYPASS VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the trailing edge flap bypass valve assembly

(2) An installation of the trailing edge flap bypass valve assembly

(3) A removal of the trailing edge flap bypass valve motor

(4) An installation of the trailing edge flap bypass valve motor.

B. The trailing edge bypass valve assembly includes the valve housing, the bypass valve and the bypass valve motor.

TASK 27-51-02-000-801

2. Trailing Edge Flap Bypass Valve Assembly Removal

(Figure 401)

A. General

(1) This task includes the removal of the bypass valve assembly as a unit. The bypass valve assembly includes the housing, the valve and the valve motor.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 27-51-02-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-02-040-001

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-02-860-001

(3) Remove the pressure from the system B hydraulic reservoir (TASK 29-09-00-860-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-02-860-002

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

E. Trailing Edge Flap Bypass Valve Assembly Removal

SUBTASK 27-51-02-020-001

(1) Disconnect the electrical connector [5] from the bypass valve assembly [4].

(2) If desired, inspect the connector for corrosion now, do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

NOTE: This step is also done in the installation task.

SUBTASK 27-51-02-020-002

(3) Disconnect the hydraulic lines [1] from the bypass valve assembly [4]:

(a) Put a container below the bypass valve assembly [4] to hold the hydraulic fluid.

(b) Disconnect the hydraulic lines [1] from the bypass valve assembly [4].

(c) Install caps on the hydraulic lines [1] and plugs in the hydraulic ports.

SUBTASK 27-51-02-020-003

(4) Remove the flap bypass valve assembly [4]:

(a) Remove the bolts [6] and washers [7] that attach the bypass valve assembly [4] to the bracket [2].

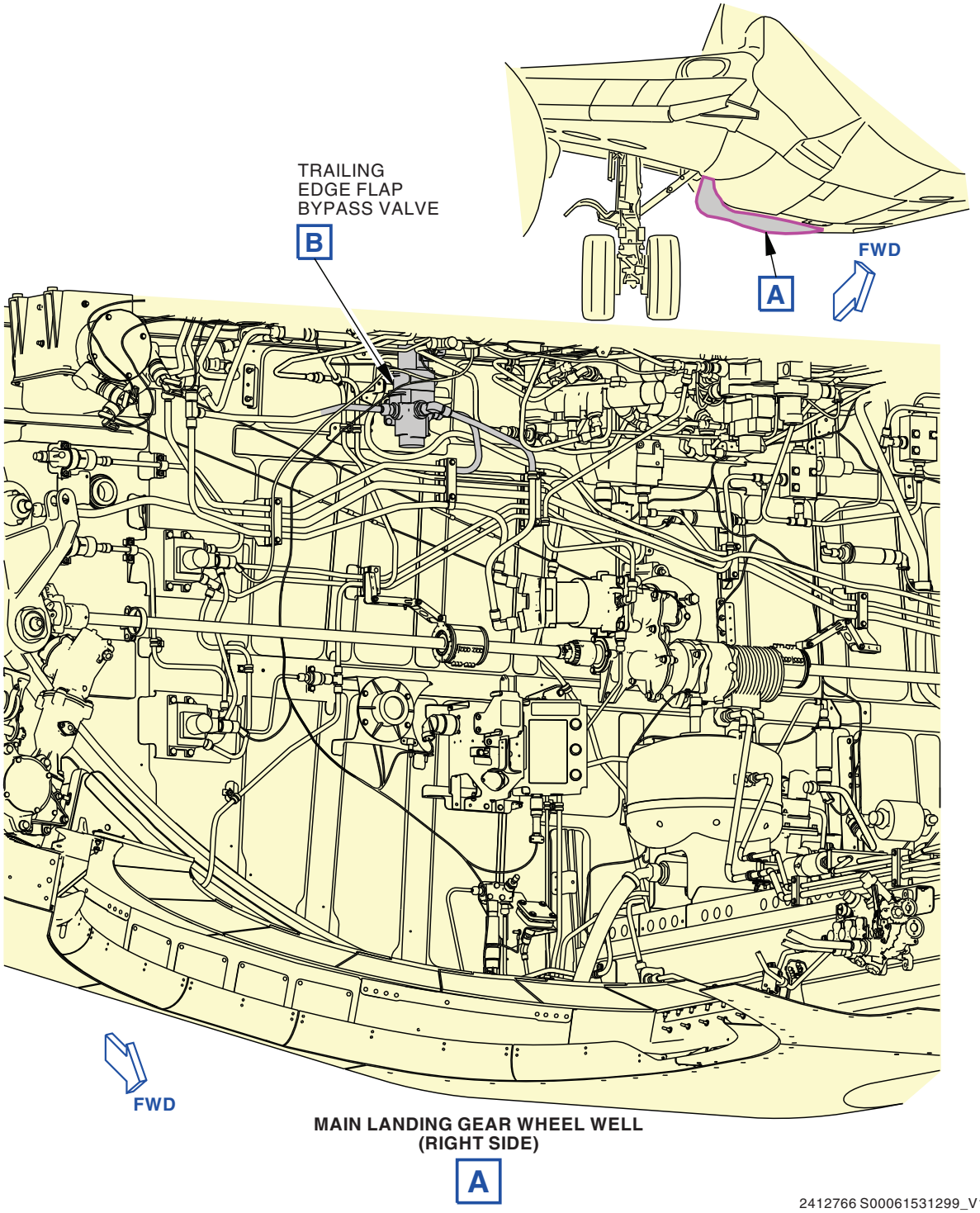
(b) Remove the bypass valve assembly [4] from the bracket [2].

SUBTASK 27-51-02-020-004

(5) If you will install a new bypass valve assembly [4], remove the unions [13] and packings [14] from the bypass valve assembly [4].

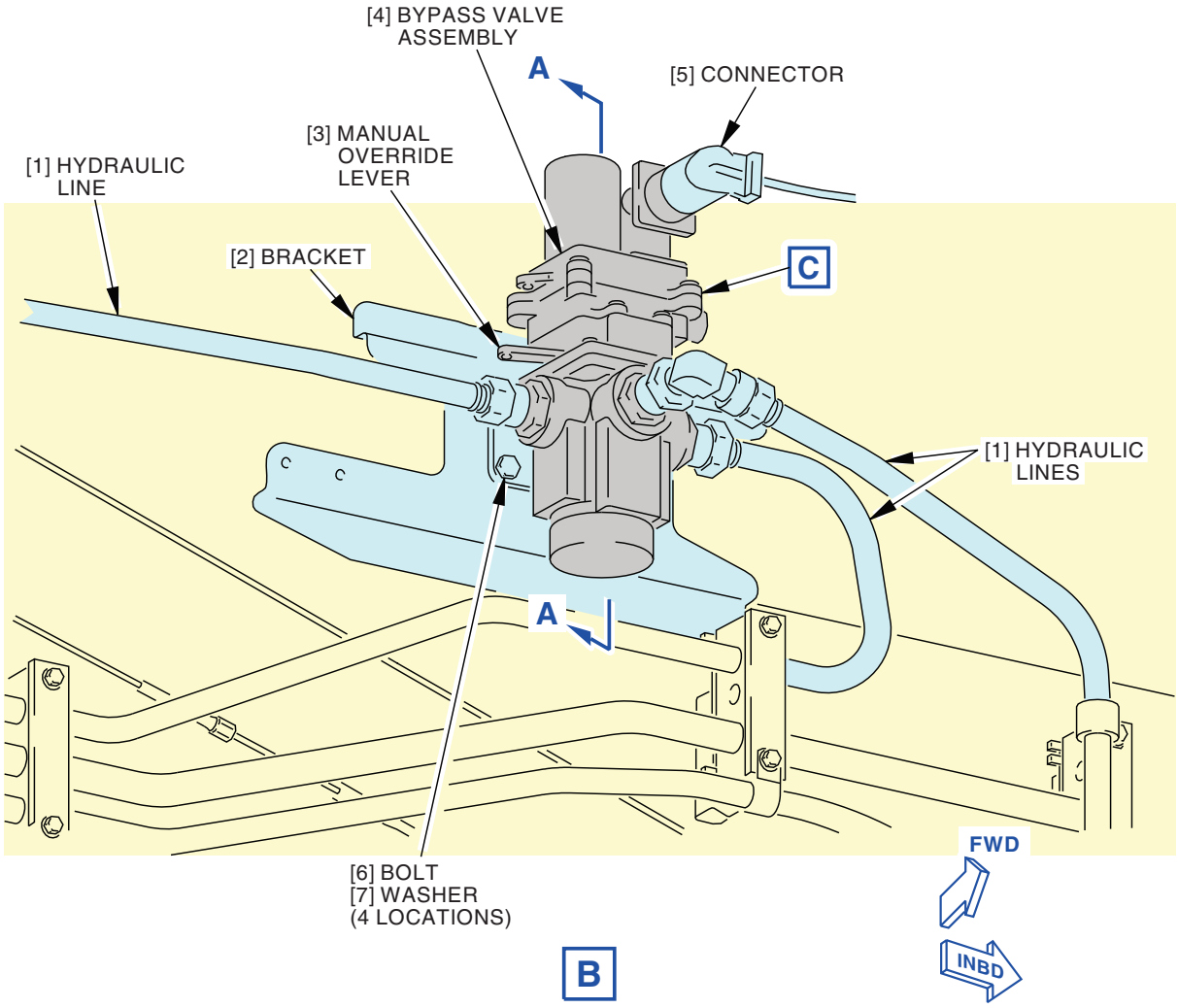
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trailing Edge Flap Bypass Valve Assembly Installation
Figure 401/27-51-02-990-801 (Sheet 1 of 4)

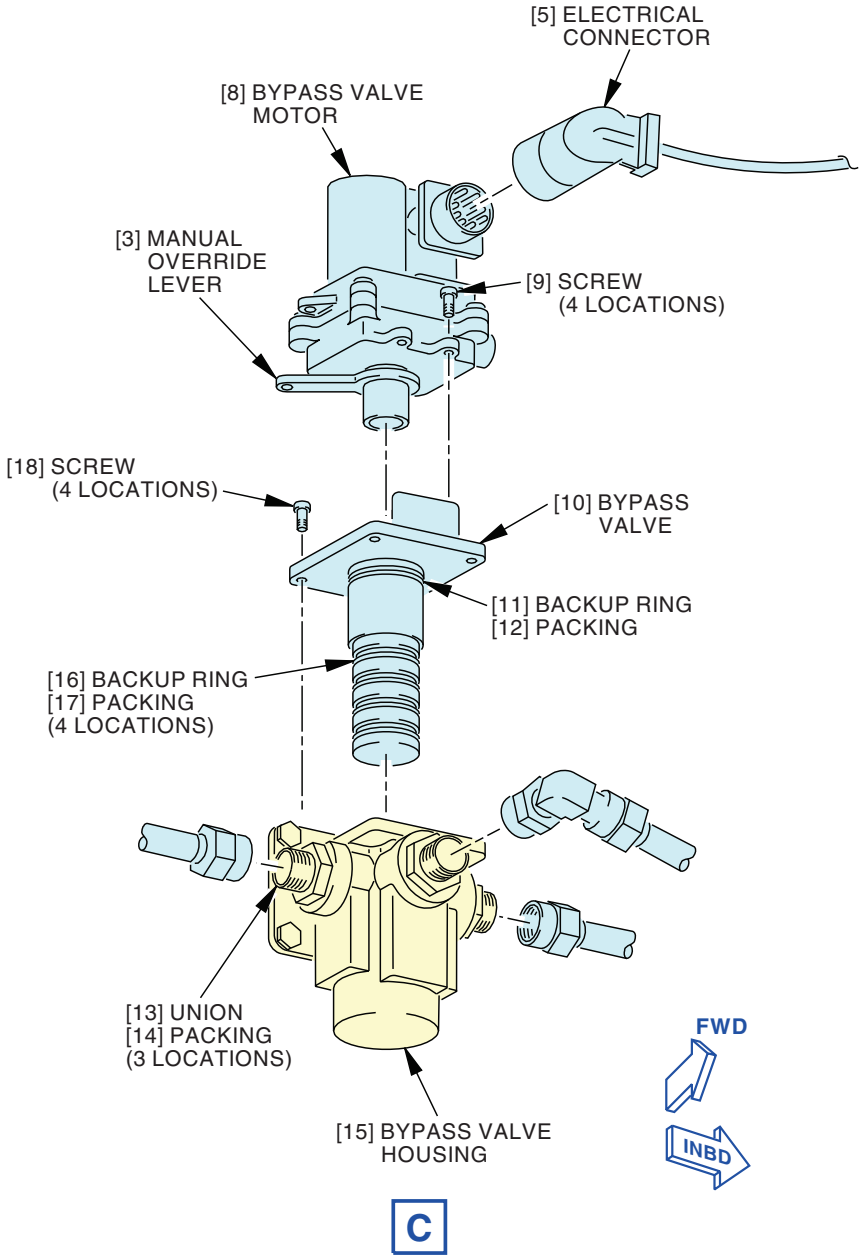
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Trailing Edge Flap Bypass Valve Assembly Installation
Figure 401/27-51-02-990-801 (Sheet 2 of 4)

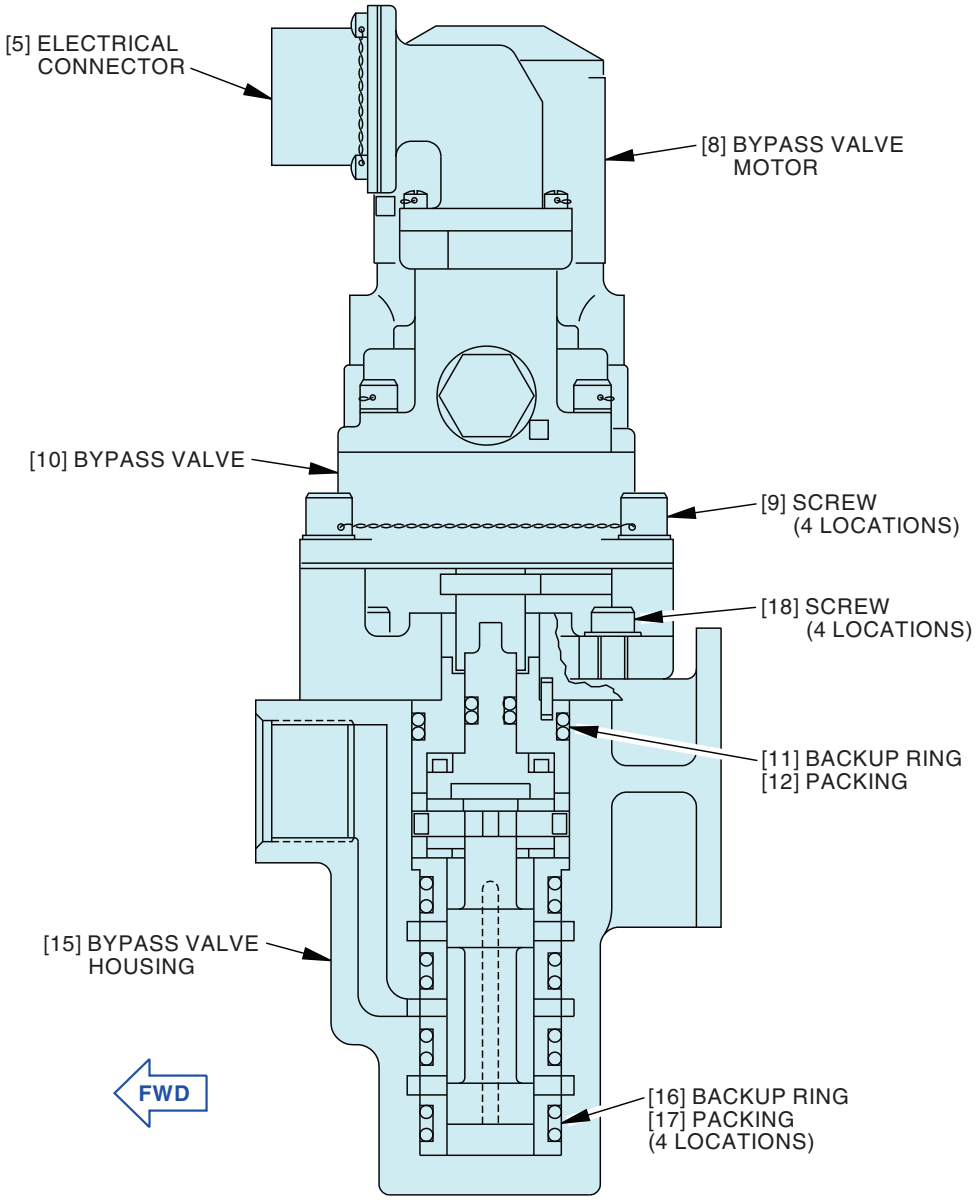
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412768 S00061531301_V1

Trailing Edge Flap Bypass Valve Assembly Installation
Figure 401/27-51-02-990-801 (Sheet 3 of 4)

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2412769 S00061531302_V1

Trailing Edge Flap Bypass Valve Assembly Installation
Figure 401/27-51-02-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-02-400-801

3. Trailing Edge Flap Bypass Valve Assembly Installation

(Figure 401)

A. General

(1) This task includes the installation of the bypass valve assembly as a unit. The bypass valve assembly includes the housing, the valve and the valve motor.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-02-720-801	Trailing Edge Flap Bypass Valve Functional Test (P/B 501)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00070	Fluid - Hydraulic, Petroleum Base	MIL-PRF-5606 (Replaces MIL-H-5606)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Bypass valve assembly	27-51-02-01-010	PTF ALL
14	Packing	27-51-02-01-015	PTF ALL

F. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

G. Trailing Edge Flap Bypass Valve Assembly Installation

SUBTASK 27-51-02-420-001

(1) If they are not installed, install the unions [13] in the bypass valve assembly [4]:

(a) Apply MCS 352B fluid, D00054 or fluid, D00070 to the new packings [14].

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(b) Put the packings [14] and the unions [13] in their location on the bypass valve assembly [4].

SUBTASK 27-51-02-420-002

(2) Install the bypass valve assembly [4]:

(a) Make sure that the mating surface of the bypass valve assembly [4] and the bracket [2] are clean for electrical bonding.

(b) Put the bypass valve assembly [4] in its position on the bracket [2].

(c) Install the bolts [6] and washers [7] to attach the bypass valve assembly [4] to the bracket [2].

(d) With the electrical bonding intrinsically safe approved bonding meter, COM-1550, make sure that the bonding resistance between the bypass valve assembly [4] and the bracket [2] is less than 0.0025 ohm.

SUBTASK 27-51-02-420-003

(3) Connect the hydraulic lines [1] to the bypass valve assembly [4]:

(a) Remove the caps from the hydraulic lines [1] and the plugs from the ports.

(b) Connect the hydraulic lines [1] to the bypass valve assembly [4].

(c) Tighten the B-nuts on the hydraulic lines [1] to 500 ±25 in-lb (56 ±3 N·m).

SUBTASK 27-51-02-210-001

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES AT AIRPORTS WHERE DEICING FLUID THAT CONTAIN POTASSIUM FORMATE IS USED. ALSO DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTOR IS PART OF A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(4) Before you connect the electrical connector [5], examine the connector for corrosion (TASK 20-60-07-910-802).

SUBTASK 27-51-02-420-004

(5) Connect the electrical connector [5] to the bypass valve assembly [4].

SUBTASK 27-51-02-860-004

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-51-02-860-016

(7) Pressurize the system B hydraulic reservoir (TASK 29-09-00-860-801).

SUBTASK 27-51-02-440-001

(8) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-02-720-001

(9) Do this task: Trailing Edge Flap Bypass Valve Functional Test, TASK 27-51-02-720-801.

SUBTASK 27-51-02-210-002

(10) Make sure that there are no hydraulic leaks from the bypass valve assembly [4].

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SUBTASK 27-51-02-860-017

(11) If the landing gear downlock pins are not needed, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

——— END OF TASK ———

TASK 27-51-02-000-803

4. Trailing Edge Flap Bypass Valve Motor Removal

(Figure 401)

A. General

(1) This task includes the removal of the bypass valve motor from the bypass valve.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 27-51-02-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-02-040-003

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-02-860-009

(3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

E. Trailing Edge Flap Bypass Valve Motor Removal

SUBTASK 27-51-02-020-007

(1) Disconnect the electrical connector [5] from the bypass valve motor [8].

(2) If desired, inspect the connector for corrosion now (TASK 20-60-07-910-802).

NOTE: This step is also done in the installation task.

SUBTASK 27-51-02-980-001

(3) Move the manual override lever [3] to POSITION 2.

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SUBTASK 27-51-02-020-008

(4) Remove the bypass valve motor [8] from the bypass valve [10]:

(a) Remove the screws [9] that attach the bypass valve motor [8] to the bypass valve [10].

(b) Carefully lift the bypass valve motor [8] off bypass valve [10].

——— END OF TASK ———

TASK 27-51-02-400-803

5. Trailing Edge Flap Bypass Valve Motor Installation

(Figure 401)

A. General

(1) This task includes the installation of the bypass valve motor on the bypass valve.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-02-720-801	Trailing Edge Flap Bypass Valve Functional Test (P/B 501)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G02421	Lockwire - MS20995C40, Corrosion Resistant Steel - 0.040 Inch (1.016 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Trailing Edge Flap Bypass Valve Motor Installation

SUBTASK 27-51-02-420-007

(1) Install the bypass valve motor [8] on the flap bypass valve [10]:

(a) Carefully put the bypass valve motor [8] in its location on the flap bypass valve [10].

(b) Install the screws [9] to attach the bypass valve motor [8] to the flap bypass valve [10].

(c) Tighten the screws [9] to 12 in-lb (1.4 N·m) - 15 in-lb (1.7 N·m).

(d) Install MS20995C40 lockwire, G02421 on the screws [9].

SUBTASK 27-51-02-210-005

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES AT AIRPORTS WHERE DEICING FLUID THAT CONTAIN POTASSIUM FORMATE IS USED. ALSO DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTOR IS PART OF A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(2) Before you connect the electrical connector [5], examine the connector for corrosion (TASK 20-60-07-910-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-02-420-008

(3) Connect the electrical connector [5] to the bypass valve motor [8].

SUBTASK 27-51-02-860-011

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-51-02-860-020

(5) Pressurize the system B hydraulic reservoir (TASK 29-09-00-860-801).

SUBTASK 27-51-02-440-003

(6) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-02-720-003

(7) Do this task: Trailing Edge Flap Bypass Valve Functional Test, TASK 27-51-02-720-801.

SUBTASK 27-51-02-860-021

(8) If the landing gear downlock pins are not needed, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRAILING EDGE FLAP BYPASS VALVE - ADJUSTMENT/TEST

1. General

A. This procedure has a functional test of the trailing edge flap bypass valve.

TASK 27-51-02-720-801

2. Trailing Edge Flap Bypass Valve Functional Test

(Figure 501)

A. References

Reference	Title
27-51-00-740-802	Trailing Edge Flap Asymmetry Skew System BITE Test (P/B 501)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-51-02-860-013

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system B hydraulic power, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-51-02-860-014

(2) If the trailing edge flaps are not in the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

D. Trailing Edge Flap Bypass Valve Functional Test

SUBTASK 27-51-02-740-001

(1) Do this task: Trailing Edge Flap Asymmetry Skew System BITE Test, TASK 27-51-00-740-802.

SUBTASK 27-51-02-720-004

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LE AND TE FLAPS AND FLAP-DRIVE MECHANISMS. THE FLAPS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Do a test of the operational speed of the trailing edge flaps:

(a) Move the flap control lever to the UP position and wait for the flaps to reach the UP position.

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(b) Measure the time it takes for the trailing edge flaps to extend from the UP position to the 1-unit position:

1) Move the flap control lever to the 1-unit position to extend the flaps.

2) Make sure the time it takes for the flaps to extend to the 1-unit position is 11 to 14 seconds.

NOTE: If you will use the system B Electric Motor-Driven Pump (EMDP) only, the extension time will be 34 to 39 seconds.

(c) Measure the time it takes for the trailing edge flaps to retract from the 1-unit position to the UP position:

1) Move the flap control lever to the UP position to retract the flaps.

2) Make sure the time it takes for the flaps to retract to the UP position is 11 to 14 seconds.

NOTE: If you will use the system B EMDP only, the retraction time will be 34 to 39 seconds.

SUBTASK 27-51-02-720-005

(3) Do a test of the operation of the bypass valve while the trailing edge flaps are moving. Do the next two steps within approximately 10 seconds:

NOTE: It is necessary to do the following two steps within approximately 10 seconds. The flaps must still be extending when you do the second step.

(a) Move the flap control lever to the 1-unit position.

(b) While the trailing edge flaps are extending, move the ALTERNATE FLAPS ARM switch to the ARM position.

(c) Make sure the flaps stop extending when you move the ARM switch.

SUBTASK 27-51-02-720-006

(4) Move the flap control lever to the UP position.

(a) Make sure the flaps do not move.

SUBTASK 27-51-02-720-007

(5) Put the ALTERNATE FLAPS ARM switch in the OFF position.

(a) Make sure the flaps retract to the retracted position.

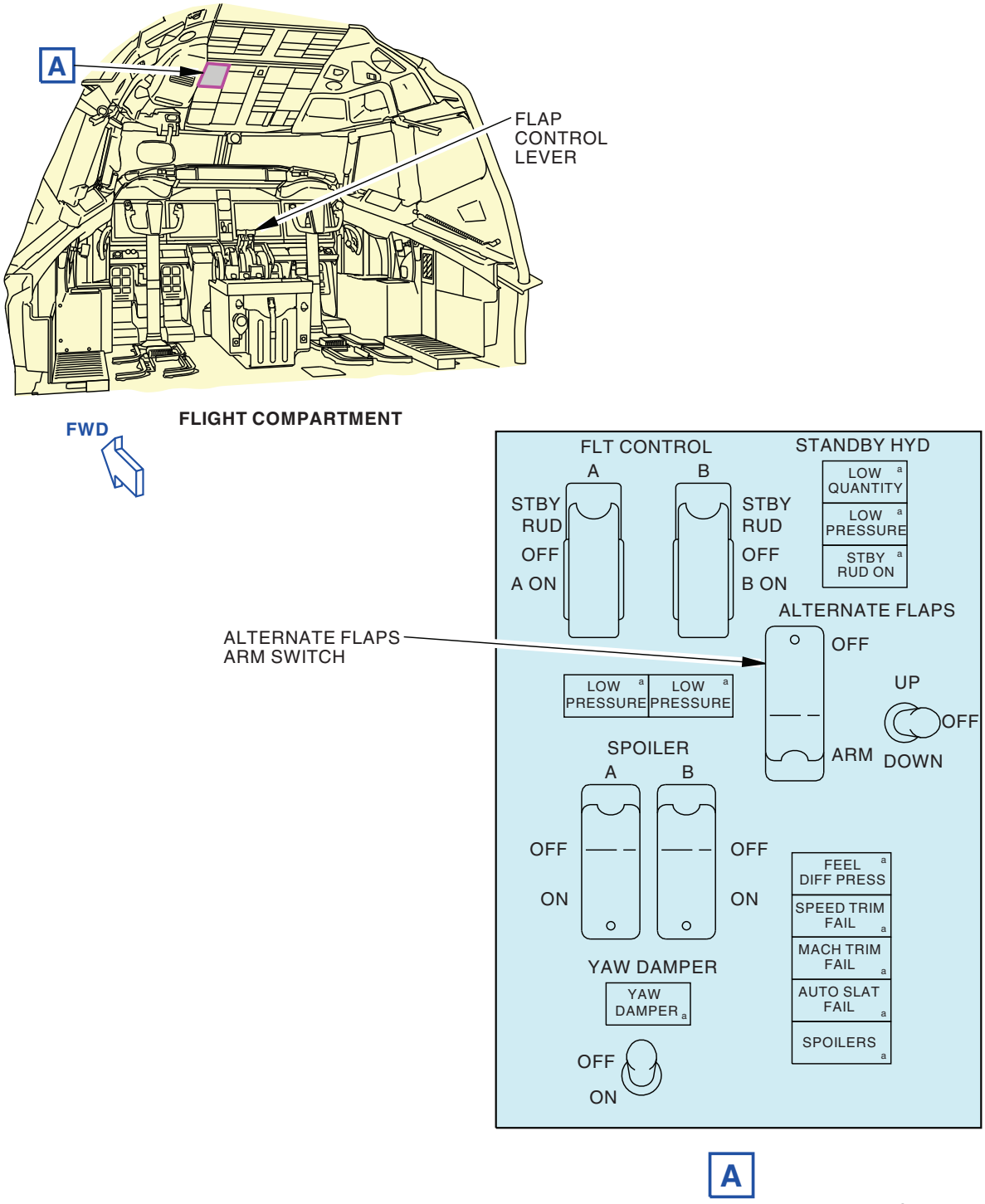
E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-02-860-015

(1) Remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

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Trailing Edge Flap Bypass Valve Test
Figure 501/27-51-02-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLAP PRIORITY VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Flap Priority Valve Removal

(2) Flap Priority Valve Installation

TASK 27-51-04-000-801

2. Flap Priority Valve Removal

(Figure 401)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-1054	Container - Fuel Resistant, 5-Gallon (19-Liter)

C. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 27-51-04-480-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-04-860-001

(2) Remove the pressure from the system B hydraulic reservoir (TASK 29-09-00-860-802).

E. Flap Priority Valve Removal

SUBTASK 27-51-04-020-001

(1) Do these steps to remove the flap priority valve [1]:

(a) Disconnect the flap flow limiter [6]

(b) Disconnect hydraulic line [2]and hydraulic line [7] from the flap priority valve [1]:

1) Put a 5-gallon (19-liter) fuel resistant container, STD-1054, below the flap priority valve [1] to hold hydraulic fluid.

2) Disconnect hydraulic line [2] and hydraulic line [7] from the flap priority valve [1].

3) Install plugs in hydraulic line [2] and hydraulic line [7].

(c) Remove the nuts [11] and washers [12], and remove the flap priority valve [1] from the bracket [10].

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SUBTASK 27-51-04-020-002

(2) If you will install a new flap priority valve [1], do these steps:

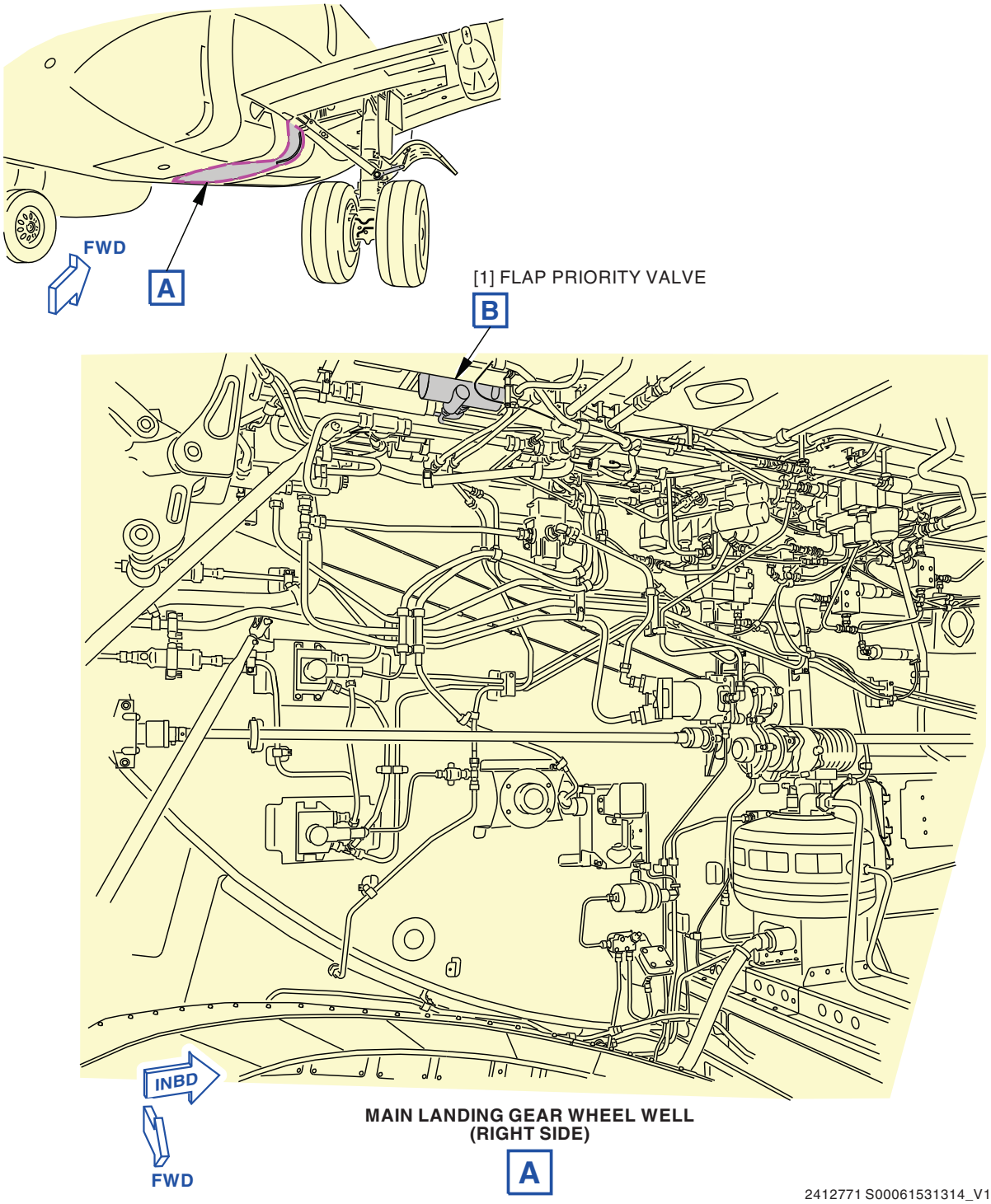
(a) Remove the union [5] and packings [4] from the flap priority valve [1].

(b) Remove the union [8] and packing [9] from the flap priority valve [1].

(c) Remove the reducer [3] and packings [4] from the flap priority valve [1].

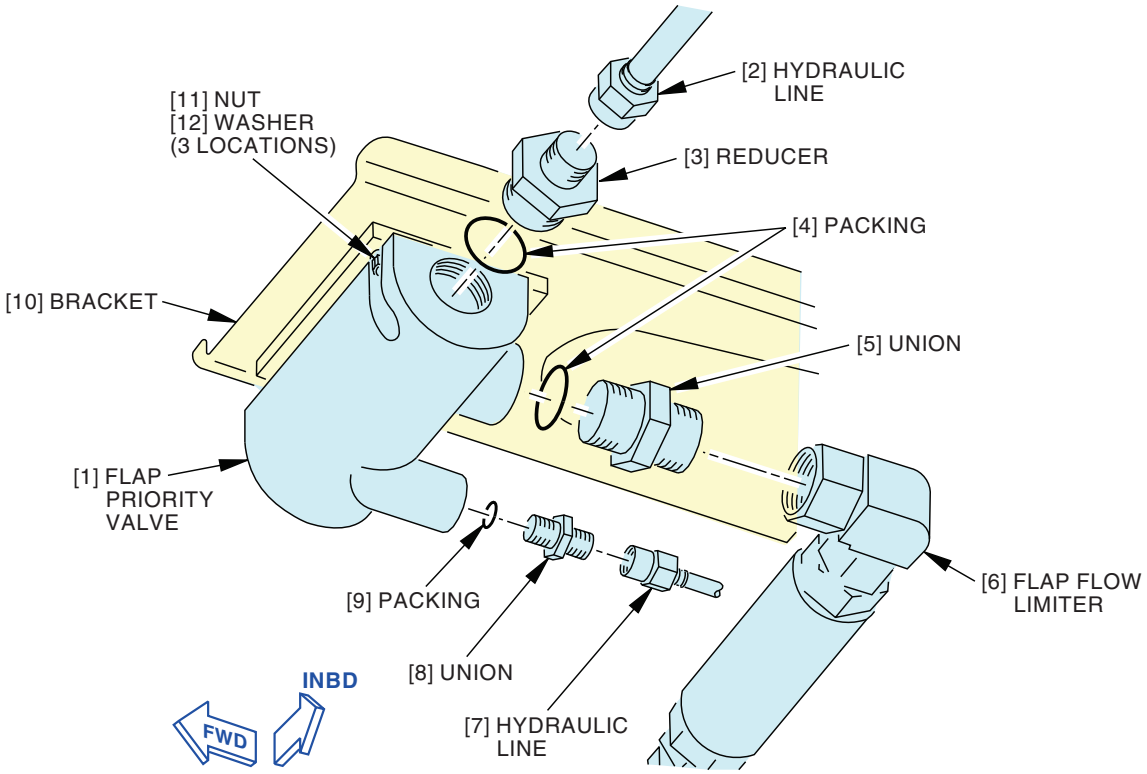
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap Priority Valve Installation
Figure 401/27-51-04-990-801 (Sheet 1 of 2)

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FLAP PRIORITY VALVE

B

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Flap Priority Valve Installation
Figure 401/27-51-04-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-04-400-801

3. Flap Priority Valve Installation

(Figure 401)

A. References

Reference	Title
27-51-00-720-801	Flap Priority Valve Test (P/B 501)
27-51-00-860-801	Trailing Edge Flap System Operation With Primary Control (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Flap priority valve	27-51-04-02-005	PTF ALL
4	Packing	27-51-04-02-015	PTF ALL
9	Packing	27-51-04-02-030	PTF ALL

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Flap Priority Valve Installation

SUBTASK 27-51-04-420-001

(1) If you will install a new flap priority valve [1], do these steps to install the union [5], union [8], and reducer [3]:

(a) Apply MCS 352B fluid, D00054, or hydraulic fluid, D00153, to the new packings [4] and new packing [9].

(b) Install the packing [4] and union [5] into the flap priority valve [1].

1) Tighten the union [5] to 450 ±23 in-lb (50.8 ±2.6 N·m).

(c) Install the packing [9] and union [8] into the flap priority valve [1].

1) Tighten the union [8] to 110 ±6 in-lb (12.4 ±0.7 N·m).

(d) Install the packing [4] and reducer [3] into the flap priority valve [1].

1) Tighten the reducer [3] to 450 ±23 in-lb (50.8 ±2.6 N·m).

SUBTASK 27-51-04-420-002

(2) Do these steps to install the flap priority valve [1]:

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WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the bolts on the bracket [10].

(b) Put the flap priority valve [1] in its location on the bracket [10].

(c) Install the nuts [11] and washers [12] to hold the flap priority valve [1].

(d) Remove the plugs from hydraulic line [2] and hydraulic line [7].

(e) Connect hydraulic line [2] and hydraulic line [7] to the flap priority valve [1].

(f) Tighten the B-nut on the hydraulic line [2] to 500 ±25 in-lb (56.5 ±2.8 N·m).

(g) Tighten the B-nut on the hydraulic line [7] to 140 ±7 in-lb (15.8 ±0.8 N·m).

(h) Connect the flap flow limiter [6] to the flap priority flap priority valve [1].

(i) Tighten the nut of the flap flow limiter [6] to 900 ±45 in-lb (101.7 ±5.1 N·m).

SUBTASK 27-51-04-860-002

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Supply system B hydraulic power (TASK 29-11-00-860-801).

SUBTASK 27-51-04-860-003

(4) Use the primary system to extend and retract the flaps (TASK 27-51-00-860-801).

SUBTASK 27-51-04-210-001

(5) Examine the hydraulic fittings for leaks.

SUBTASK 27-51-04-720-001

(6) Do this task: Flap Priority Valve Test, TASK 27-51-00-720-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLAP CONTROL LEVER POSITION SENSOR - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the flap control lever position sensor

(2) An installation of the flap control lever position sensor

(3) An adjustment of the flap control lever position sensor

(4) An system test of the flap control lever position sensor

(5) A detailed visual inspection of the flap control lever sensor, linkage and rod ends.

B. The flap control lever position sensor is installed in the control stand and operates from the flap control lever.

TASK 27-51-06-000-801

2. Flap Control Lever Position Sensor Removal

(Figure 201)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-51-06-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-51-06-010-001

(2) Remove the lower right side cover [3] from the lower right side of the control stand [7]:

(a) Remove the screws [5] and the pocket assembly [6] from the lower right side cover [3].

(b) Loosen the two lower screws [8] on the right side of the control stand [7].

(c) Remove the screws [4] that attach the lower right side cover [3] to the control stand [7].

(d) Remove the lower right side cover [3] from the control stand [7].

1) Move the lower right side cover [3] down, away from the upper right side cover [1].

C. Flap Control Lever Position Sensor Removal

SUBTASK 27-51-06-020-001

(1) Disconnect the electrical connector [9] for the flap control lever position sensor [20].

SUBTASK 27-51-06-020-002

(2) Remove the flap control lever position sensor assembly [2]:

NOTE: The position sensor assembly [2] includes the position sensor [20], bracket assembly [23], bracket [24] and the crank [18].

(a) Remove the bolt [15], washers [16] and clamp [14] that attach the wire for the electrical connector [9].

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(b) Remove the bolts [11], washers [12], nuts [13] and clamps [10] that attach the wire for the electrical connector [9].

(c) Remove the bolt [22] and nut [21] that attach the link [17] to the crank [18].

(d) Remove the bolts [19] and washers [30] that attach the flap control lever position sensor assembly [2] to the control stand [7].

SUBTASK 27-51-06-020-003

(3) Remove the flap control lever position sensor [20] from the position sensor assembly [2]:

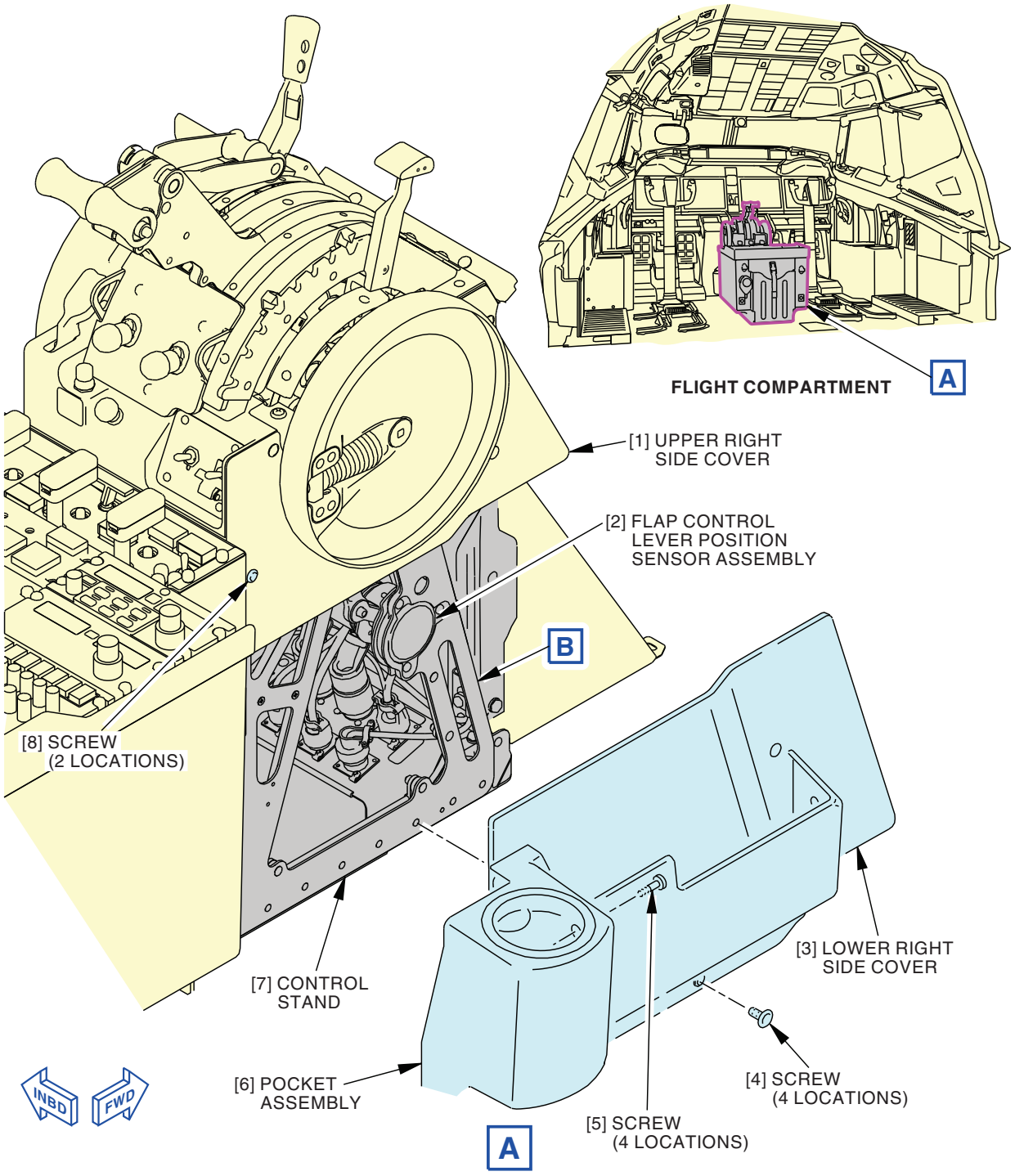
(a) Remove the screw [27], washers [28] and nut [29] that attach the crank [18] to the shaft on the position sensor [20].

(b) Remove the screws [25] and washers [26] that attach the position sensor [20] to the bracket [24].

(c) Remove the flap control lever position sensor [20] from the bracket [24].

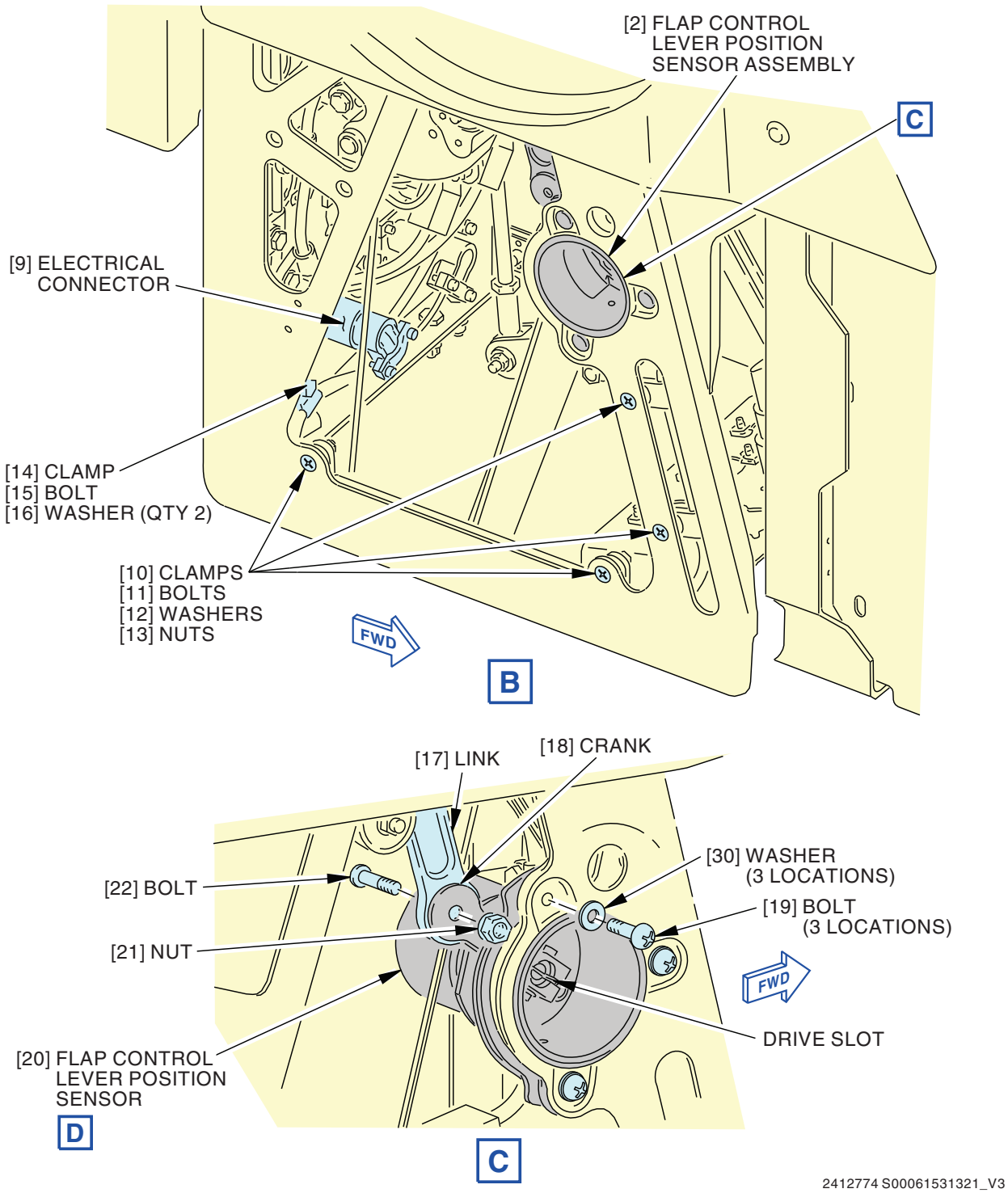
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



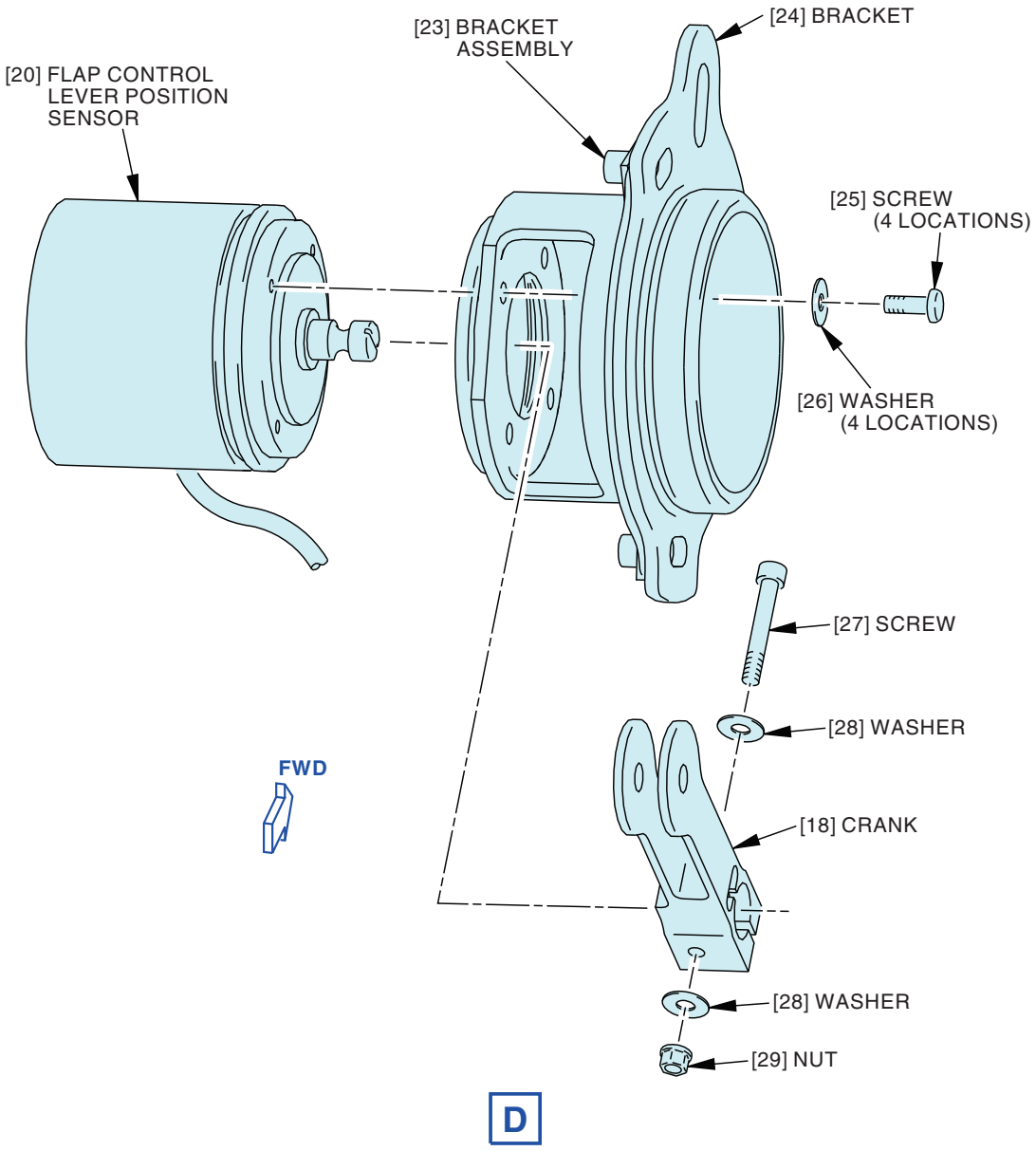
Flap Control Lever Position Sensor Installation
Figure 201/27-51-06-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap Control Lever Position Sensor Installation
Figure 201/27-51-06-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap Control Lever Position Sensor Installation
Figure 201/27-51-06-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-06-400-801

3. Flap Control Lever Position Sensor Installation

(Figure 201)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Upper right side cover	25-11-21-25-080	PTF ALL
2	Sensor assembly	27-51-06-01-085	PTF ALL
3	Lower right side cover	25-11-21-25-150	PTF ALL

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Flap Control Lever Position Sensor Installation

SUBTASK 27-51-06-860-002

(1) Make sure that this circuit breaker is open and has safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-51-06-010-002

(2) Install the flap control lever position sensor [20] on the position sensor assembly [2]:

NOTE: The position sensor assembly [2] includes the position sensor [20], bracket assembly [23], bracket [24] and the crank [18].

(a) Put the bracket assembly [23] in its position on the bracket [24].

(b) Put the crank [18] in its position through the slot in the bracket [24].

(c) Put the crank [18] on the shaft of the position sensor [20] and put the position sensor [20] in its position on the bracket [24].

(d) Install the screw [27], washers [28] and nut [29] to attach the crank [18] to the position sensor [20].

(e) Install the screws [25] and washers [26] to attach the position sensor [20] to the bracket [24].

(f) Make sure that the wire for the electrical connector [9] is approximately 180° from the crank [18].

SUBTASK 27-51-06-420-001

(3) Install the flap control lever position sensor assembly [2]:

(a) Put the flap control lever position sensor assembly [2] in its position on the control stand [7].

(b) Install the bolts [19] and washers [30] that attach the flap control lever position sensor assembly [2].

(c) Install the bolt [22]and nut [21] to attach the link [17] to the crank [18].

SUBTASK 27-51-06-420-002

(4) Connect the electrical connector [9] for the flap control lever position sensor [20].

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SUBTASK 27-51-06-420-003

(5) Install the clamps [10] with the bolts [11], washers [12], and nuts [13] to attach the wire for the electrical connector [9] to the control stand [7].

SUBTASK 27-51-06-420-004

(6) Install the clamp [14] with the bolt [15] and washers [16] to attach the wire for the electrical connector [9] to the control stand [7].

SUBTASK 27-51-06-860-003

(7) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	12	C00213	FC TE FLAP POS/SKEW SNSR & IND LEFT

SUBTASK 27-51-06-820-001

(8) Do this task: Flap Control Lever Position Sensor Adjustment and Test, TASK 27-51-06-820-801.

SUBTASK 27-51-06-730-001

(9) Do this task: Flap Control Lever Position Sensor System Test, TASK 27-51-06-730-801.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-06-410-001

(1) Install the lower right side cover [3] on the lower right side of the control stand [7]:

(a) Put the lower right side cover [3] in its position on the control stand [7], under the upper right side cover [1].

(b) Install the screws [4] that attach the lower right side cover [3] to the control stand [7].

(c) Tighten the lower screws [8] that attach the upper right side cover [1].

(d) Install the pocket assembly [6] with the screws [5] on the lower right side cover [3].

——— END OF TASK ———

TASK 27-51-06-820-801

4. Flap Control Lever Position Sensor Adjustment and Test

(Figure 201, Figure 202)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

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D. Prepare for the Adjustment

SUBTASK 27-51-06-860-005

(1) If it is supplied, remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

NOTE: If system B hydraulic power is supplied, the flaps and slats will operate during the adjustment.

SUBTASK 27-51-06-010-003

(2) Remove the lower right side cover [3] from the lower right side of the control stand [7] (Figure 201):

(a) Remove the screws [5] and the pocket assembly [6] from the lower right side cover [3].

(b) Loosen the two lower screws [8] for the upper right side cover [1] on the right side of the control stand [7].

(c) Remove the screws [4] that attach the lower right side cover [3] to the control stand [7].

(d) Remove the lower right side cover [3] from the control stand [7].

1) Move the lower right side cover [3] down, away from the upper right side cover [1].

SUBTASK 27-51-06-010-004

(3) To get access to the Flap/Slat Electronics Unit (FSEU) on the E1-1 shelf in the main equipment center, do this task (Figure 202).

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Flap Control Lever Position Sensor Adjustment

SUBTASK 27-51-06-860-006

(1) Move the flap control lever to the 10-unit position.

SUBTASK 27-51-06-740-001

(2) Use the FSEU to find the output of the flap control lever position sensor [20]:

(a) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

(b) Push the down arrow until OTHER FUNCTNS? shows on the display.

(c) Push the YES button to select OTHER FUNCTNS? on the FSEU.

(d) Push the down arrow until I/O MONITOR? shows on the display.

(e) Push the YES button to select I/O MONITOR? from the menu.

(f) Push the down arrow until SYNCHRO INPUTS? shows on the display.

(g) Push the YES button to select SYNCHRO INPUTS? from the menu.

(h) Make sure that the LVR POS shows in the limit of 52.3 to 52.9 degrees.

SUBTASK 27-51-06-820-002

(3) If the output of the position sensor [20] shown on the FSEU is not in the limit, adjust the position sensor [20] (Figure 201):

(a) Loosen the bolts [19] that attach the flap control lever position sensor assembly [2] to the control stand [7].

(b) Put a 1/4 inch flat head screwdriver in the drive slot on the position flap control lever sensor assembly [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Carefully turn the position sensor assembly [2] until the FSEU shows the LVR POS in the limit of 52.3 to 52.9 degrees.

(d) Tighten the bolts [19].

(e) Remove the 1/4 inch flat head screwdriver from the drive slot.

(f) Make sure that the LVR POS shows in the limit of 52.37 to 52.87 degrees.

SUBTASK 27-51-06-730-002

(4) Do this task: Flap Control Lever Position Sensor System Test, TASK 27-51-06-730-801.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-06-860-007

(1) Push the ON/OFF button on the front panel of the FSEU to turn it off.

SUBTASK 27-51-06-410-002

(2) For the main equipment center, do this task.

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-51-06-410-003

(3) Install the lower right side cover [3] on the lower right side of the control stand [7]:

(a) Put the lower right side cover [3] in its position on the control stand [7], under the upper right side cover [1].

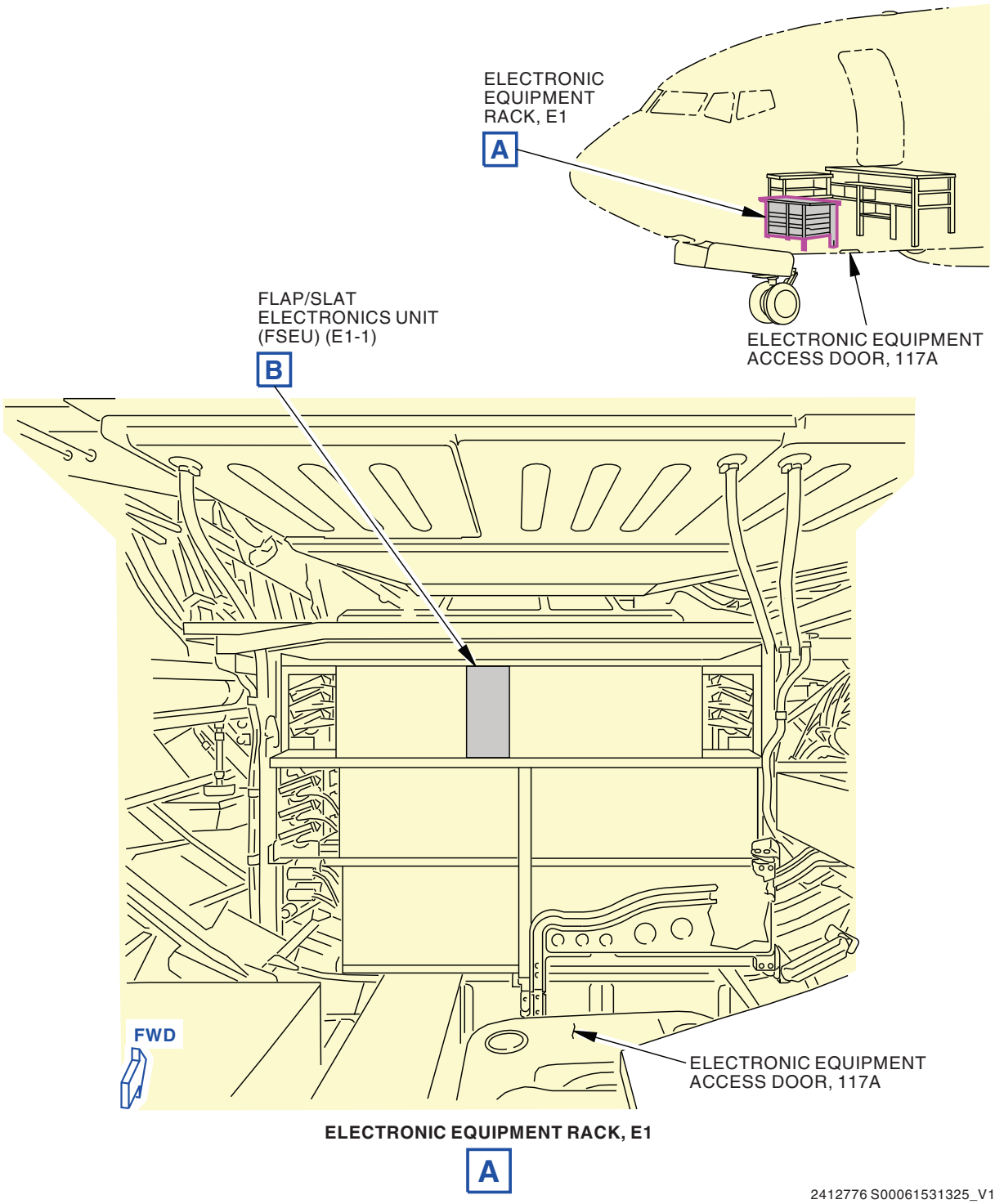
(b) Install the screws [4] that attach the lower right side cover [3] to the control stand [7].

(c) Tighten the lower screws [8] that attach the upper right side cover [1]

(d) Install the pocket assembly [6] with the screws [5] on the lower right side cover [3].

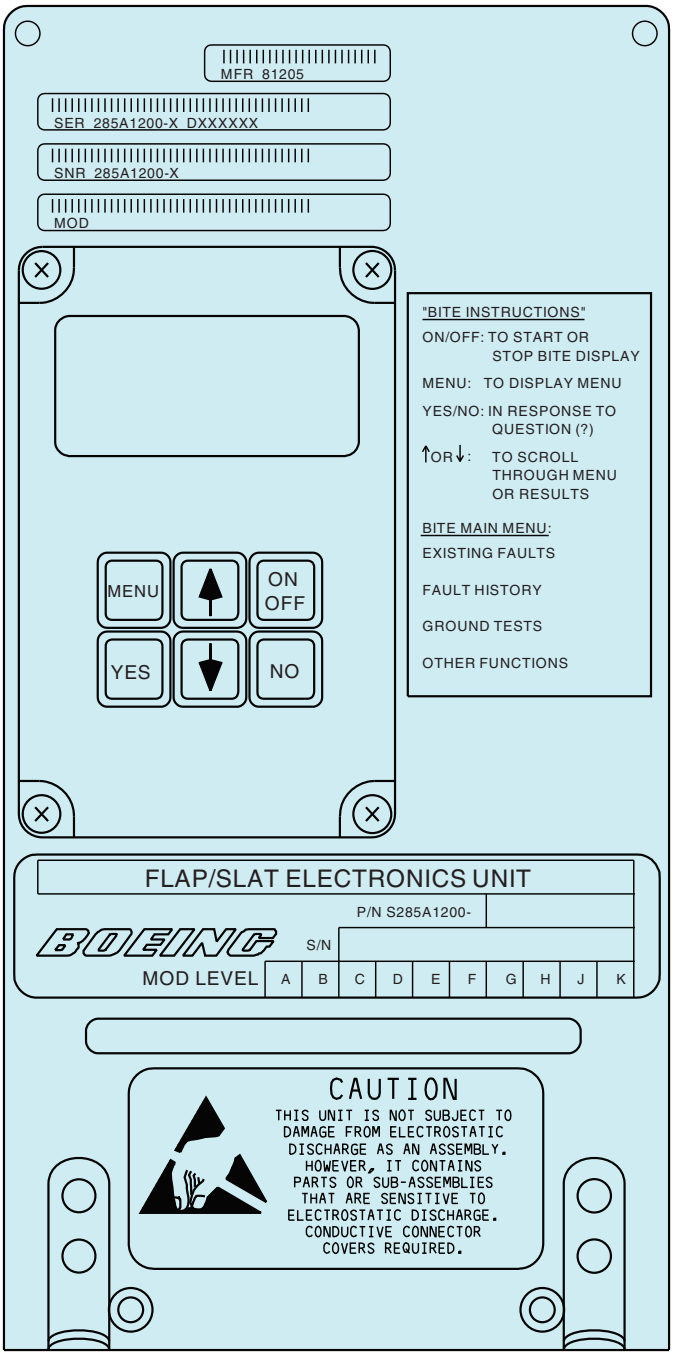
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap/Slat Electronics Unit Installation
Figure 202/27-51-06-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLAP/SLAT ELECTRONICS UNIT (FSEU)

B

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Flap/Slat Electronics Unit Installation
Figure 202/27-51-06-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-06-730-801

5. Flap Control Lever Position Sensor System Test

(Figure 202)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 27-51-06-860-009

(1) If it is supplied, remove system B hydraulic power, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

NOTE: If system B hydraulic power is supplied, the flaps and slats will operate during the system test.

SUBTASK 27-51-06-010-005

(2) To get access to the Flap/Slat Electronics Unit (FSEU) on the E1-1 shelf in the main equipment center, do this task.

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Flap Control Lever Position Sensor Test

SUBTASK 27-51-06-730-003

(1) Move the flap control lever to the UP position.

SUBTASK 27-51-06-740-002

(2) Use the FSEU to find the output of the flap control lever position sensor [20] in the UP position:

(a) Push the ON/OFF button on the front panel of the FSEU to turn on the display.

(b) Push the down arrow until OTHER FUNCTNS? shows on the display.

(c) Push the YES button to select OTHER FUNCTNS? on the FSEU.

(d) Push the down arrow until I/O MONITOR? shows on the display.

(e) Push the YES button to select I/O MONITOR? from the menu.

(f) Push the down arrow until SYNCRO INPUTS? shows on the display.

(g) Push the YES button to select SYNCHRO INPUTS? from the menu.

(h) Make sure that the LVR POS shows on the FSEU in the limit of -2.8 to +2.8 degrees.

SUBTASK 27-51-06-730-004

(3) Move the flap control lever to the 10-unit position.

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(a) Make sure that the LVR POS shows in the limit of 50.6 to 54.6 degrees.

SUBTASK 27-51-06-730-005

(4) Move the flap control lever to the 40-unit position.

(a) Make sure that the LVR POS shows in the limit of 105.4 to 110.9 degrees.

SUBTASK 27-51-06-730-006

(5) Move the flap control lever to the UP position.

(a) Make sure that the LVR POS shows in the limit of -2.8 to +2.8 degrees.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-06-860-010

(1) Push the ON/OFF button on the front panel of the FSEU to turn it off.

SUBTASK 27-51-06-410-004

(2) For the main equipment center, do this task.

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 27-51-06-210-801

6. Flap Control Lever Position Sensor and Mechanism Detailed Visual Inspection

(Figure 203)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Prepare for the Inspection

SUBTASK 27-51-06-010-008

(1) Remove the upper right side cover [1] and lower right side cover [3] from the right side of the control stand [7].

(a) Remove the screws [5] and pocket assembly [6] from the lower right side cover [3].

(b) Remove the screws [8] that attach the upper right side cover [1] and lower right side cover [3] to the control stand [7].

(c) Remove the upper right side cover [1] and lower right side cover [3] from the control stand [7].

1) If it is necessary, remove the stabilizer trim wheel [50] from the right side of the control stand [7].

a) Remove the bolt [51] and spacer [52] that attaches the stabilizer trim wheel [50] to the shaft.

b) Remove the stabilizer trim wheel [50] from the shaft.

C. Flap Control Lever Position Sensor Detailed Visual Inspection

SUBTASK 27-51-06-020-005

(1) Do a detailed visual inspection of the flap control lever sensor assembly [2] and crank [18].

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SUBTASK 27-51-06-210-001

(2) Do a detailed visual inspection of the ends of the link [17].

D. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-06-410-005

(1) Install the upper right side cover [1] and lower right side cover [3] on the right side of the control stand [7].

(a) Put the upper right side cover [1] and lower right side cover [3] in their position on the control stand [7].

NOTE: The lower right side cover goes under the upper right side cover.

(b) Install the screws [8] that attach the upper right side cover [1] and lower right side cover [3] to the control stand [7].

(c) Install the pocket assembly [6] with the screws [5] on the lower right side cover [3].

SUBTASK 27-51-06-420-005

(2) If removed, install the stabilizer trim wheel [50] on the right side of the control stand [7].

(a) Install the stabilizer trim wheel [50] on the shaft with the handle 90 +/- 15 degrees apart from the handle on the other stabilizer trim wheel [50].

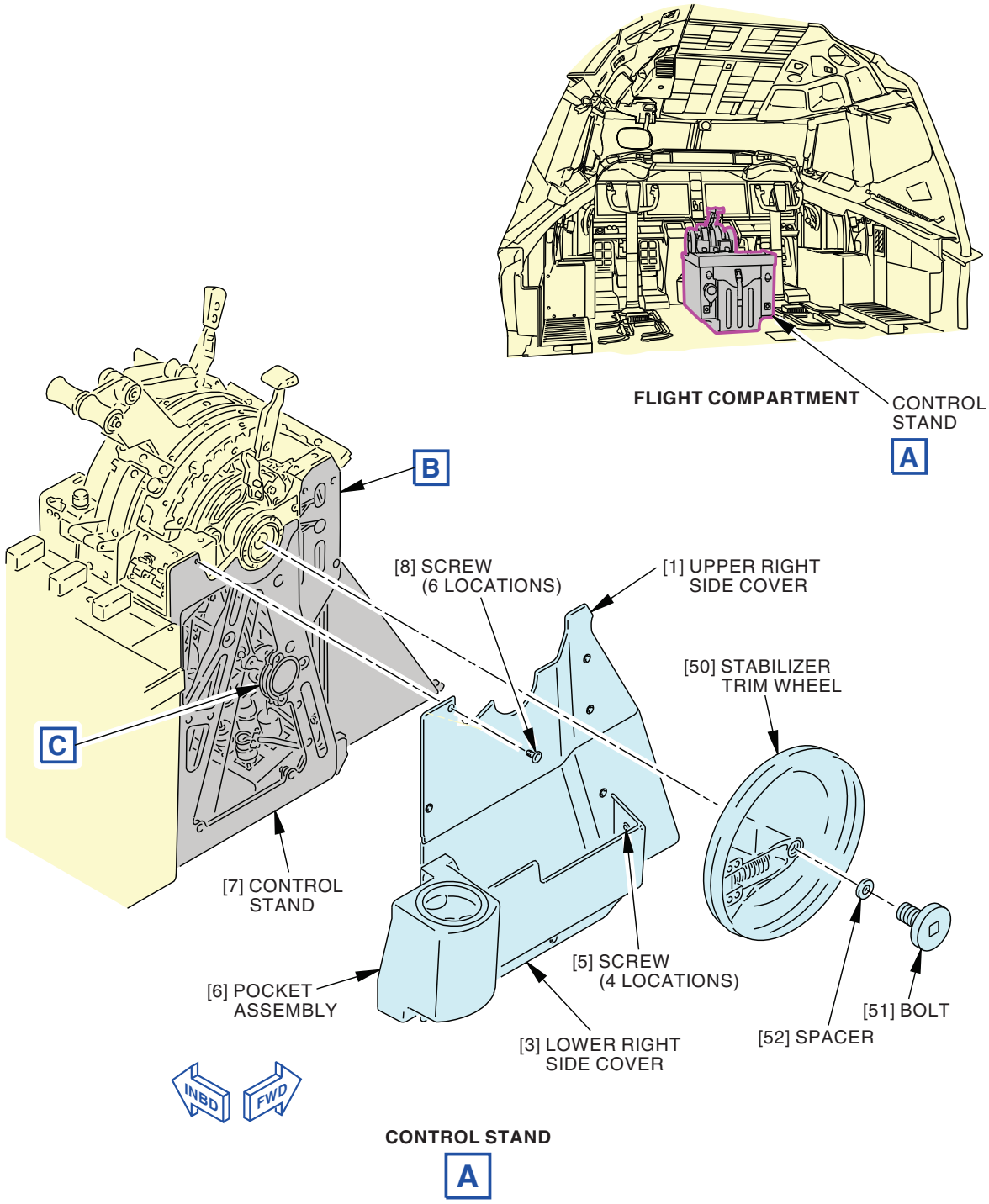
1) Make sure that an equal amount of the shaft is visible on both sides of the pedestal.

(b) Install the spacer [52] and bolt [51] to attach the stabilizer trim wheel [50] to the shaft.

(c) Tighten the bolt [51] to 150 in-lb (16.9 N·m) - 160 in-lb (18.1 N·m) plus run-on torque.

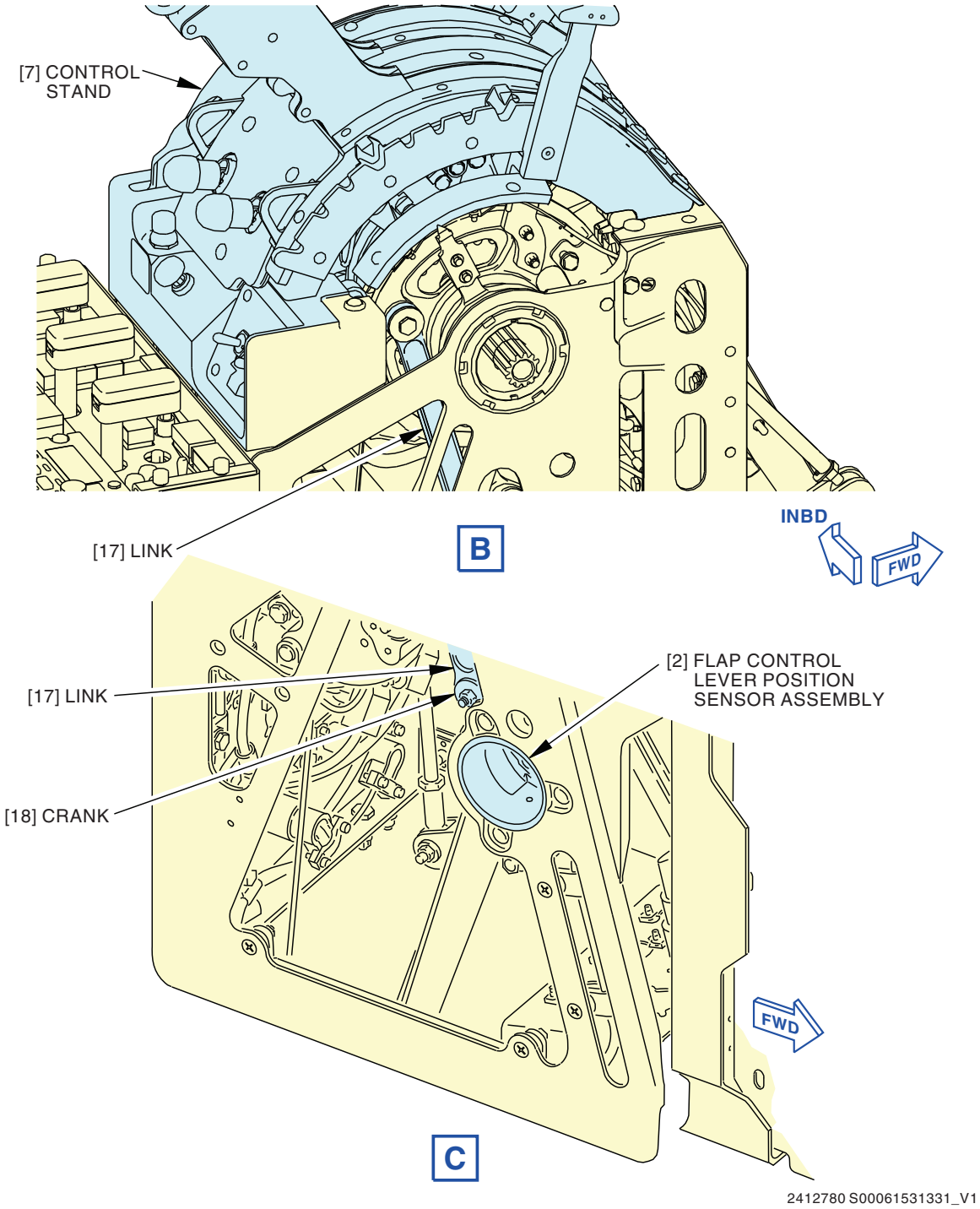
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap Control Lever Position Sensor Inspection
Figure 203/27-51-06-990-803 (Sheet 1 of 2)

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Flap Control Lever Position Sensor Inspection
Figure 203/27-51-06-990-803 (Sheet 2 of 2)

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FLAP LEVER POSITION SWITCHES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Flap lever position switch removal

(2) Flap lever position switch installation.

B. There are two flap lever position switches installed on the control stand:

(1) The flap lever 10 to 40-unit position switch

(2) The flap lever UP position switch.

C. The flap lever position switches are installed on the control stand and operate with the flap control lever.

D. There are two options for removal and installation of the flap lever position switches:

(1) You can remove and install the two flap lever position switches as an assembly.

(2) You can remove and install the two flap lever position switches independently by cutting and splicing the wires.

TASK 27-51-08-000-801

2. Flap Lever Position Switch Removal

(Figure 401)

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Access Panels

Number	Name/Location
112A	Forward Access Door

C. Prepare for the Removal

SUBTASK 27-51-08-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	10	C00284	PANEL & INSTR ELEX PANEL

SUBTASK 27-51-08-010-002

(2) Remove the stabilizer trim wheel [8] from the right side of the control stand [13]:

(a) Remove the bolt [31] and spacer [32] that attaches the stabilizer trim wheel [8].

(b) Remove the stabilizer trim wheel [8] from the shaft.

SUBTASK 27-51-08-010-003

(3) Remove the upper side cover [7] and lower side cover [11] from the right side of the control stand [13]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the screws [10] and the pocket assembly [12] from the lower side cover [11].

(b) Remove the screws [6] that attach the upper side cover [7] and the lower side cover [11] to the control stand [13].

(c) Remove the upper side cover [7] and the lower side cover [11] from the control stand [13].

SUBTASK 27-51-08-010-004

(4) Remove the stabilizer trim lightplate [3] from the control stand [13]:

(a) Remove the screws [4] and remove the stabilizer trim lightplate [3].

(b) Remove the nuts [17], washers [16] and washers [15] that attach the wires [14] to the stabilizer trim lightplate [3].

SUBTASK 27-51-08-010-005

(5) Remove the screws [2] and bolt [5] and remove the control stand cover [1] from the control stand [13].

D. Flap Lever Position Switch Removal

SUBTASK 27-51-08-020-001

(1) Disconnect the electrical connector [28] for the flap lever position flap lever position switches [24]:

(a) To get access to the electrical connector [28] from below the control stand [13], do this task.

Open this access panel:

Number	Name/Location
112A	Forward Access Door

(b) Disconnect the electrical connector [28] from below.

(c) Remove the screws [29] and washers [30] that attach the electrical connector [28].

(d) Remove the electrical connector [18] from the control stand [13].

SUBTASK 27-51-08-020-002

(2) Remove the flap lever position switches [24] as an assembly:

(a) Remove the bolts [22], bolt [27] and washers [23] that attach the flap lever position switches [24] and the clamp [21] to the control stand [13].

(b) Remove the bolt [25], washer [26] and clamp [21].

(c) Remove the bolts [19], washers [20] and clamps [21] that attach the wires to the control stand [13].

SUBTASK 27-51-08-020-003

(3) Remove a flap lever position switch [24] independently:

(a) For the 10 to 40-unit flap lever position switch [24], remove the bolts [22] and washers [23] that attach the flap lever position switch [24] to the control stand [13].

1) Remove the bolt [25], washer [26] and clamp [21].

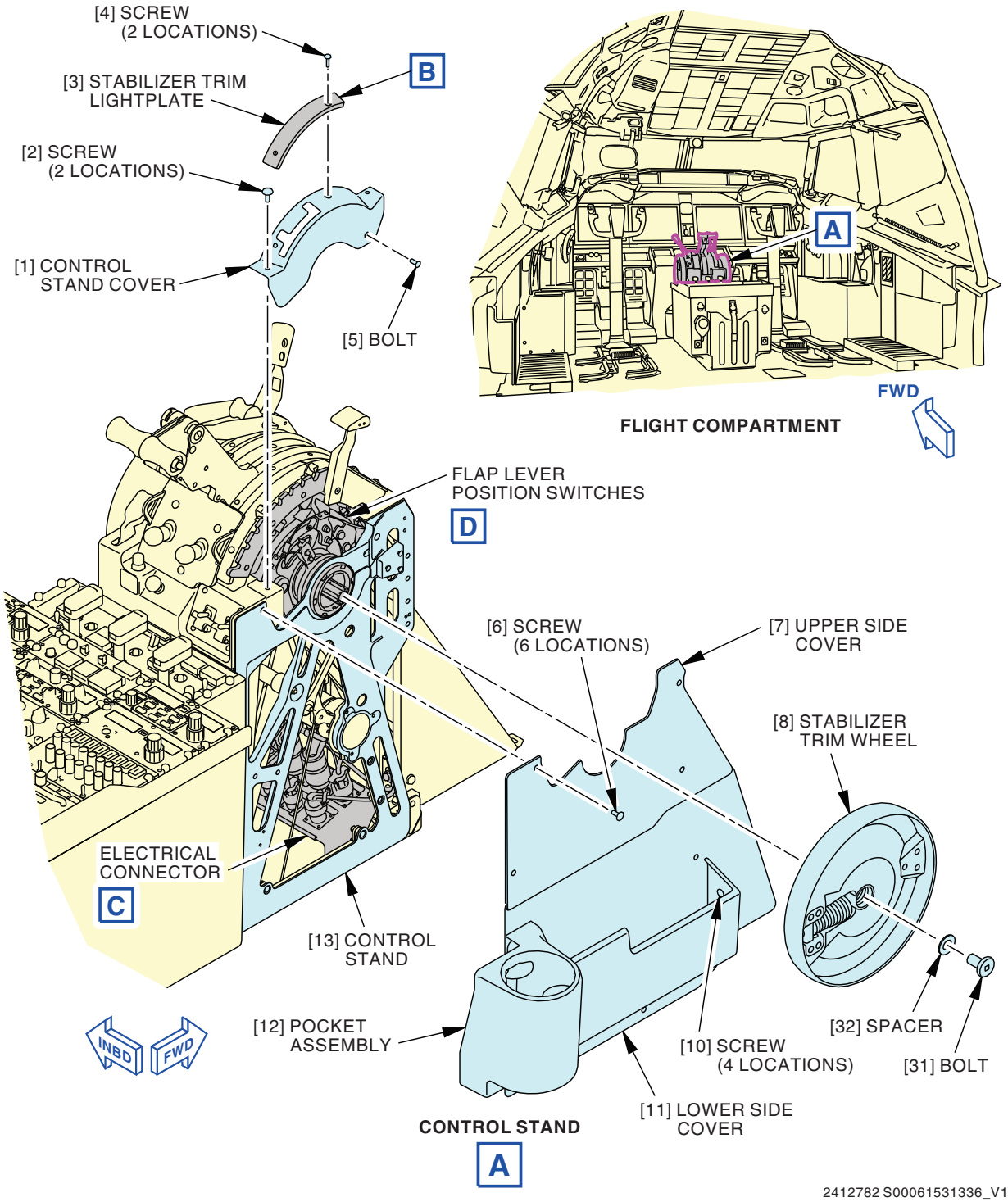
(b) For the UP flap lever position switch [24], remove the bolt [22], bolt [27] and washers [23] that attach the flap lever position switch [24] and the clamp [21] to the control stand [13].

(c) Remove the bolt [19] and washer [20] from a sufficient number of clamps [21] to get access to cut the wires.

(d) Cut the wires for the applicable flap lever position switch [24].

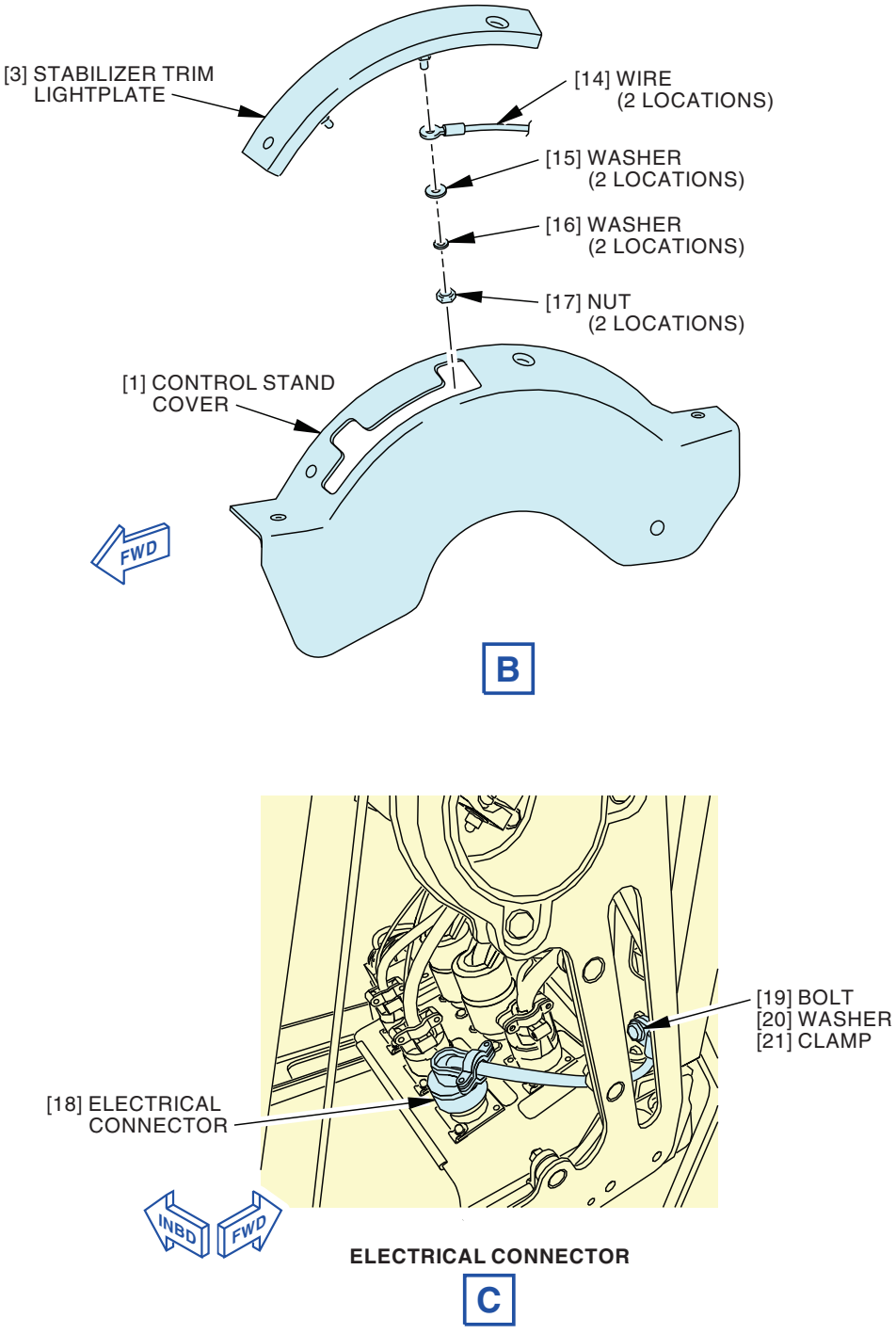
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



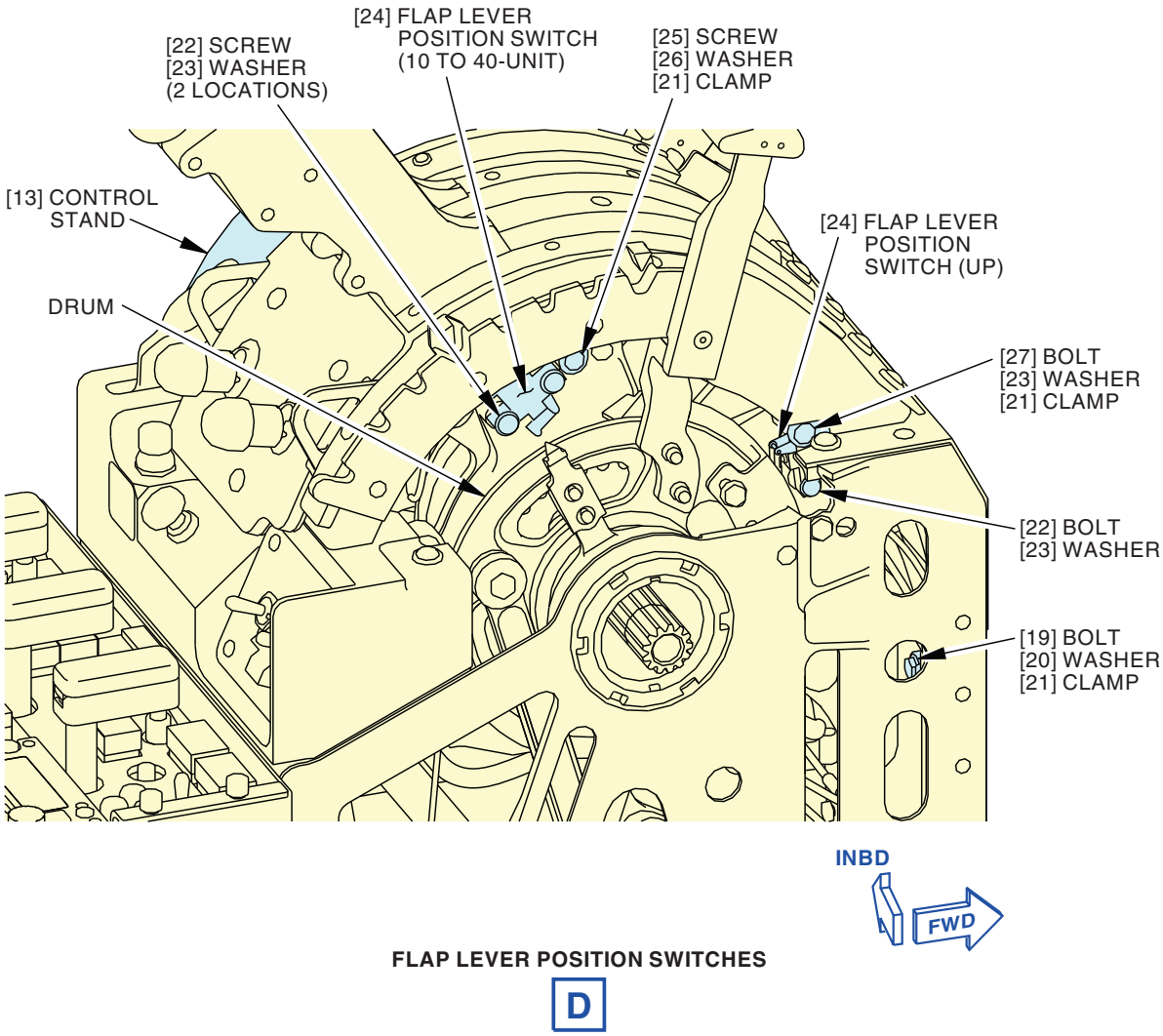
Flap Lever Position Switch Installation
Figure 401/27-51-08-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap Lever Position Switch Installation
Figure 401/27-51-08-990-801 (Sheet 2 of 5)

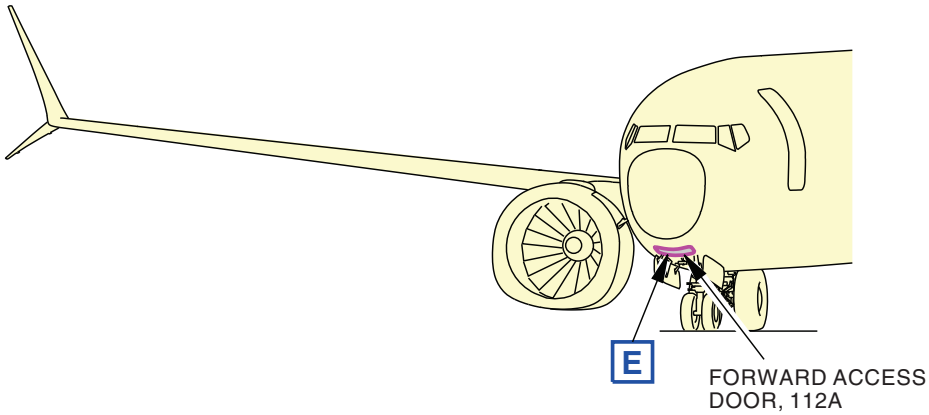
ECCN 9E991 BOEING PROPRIETARY - See title page for details

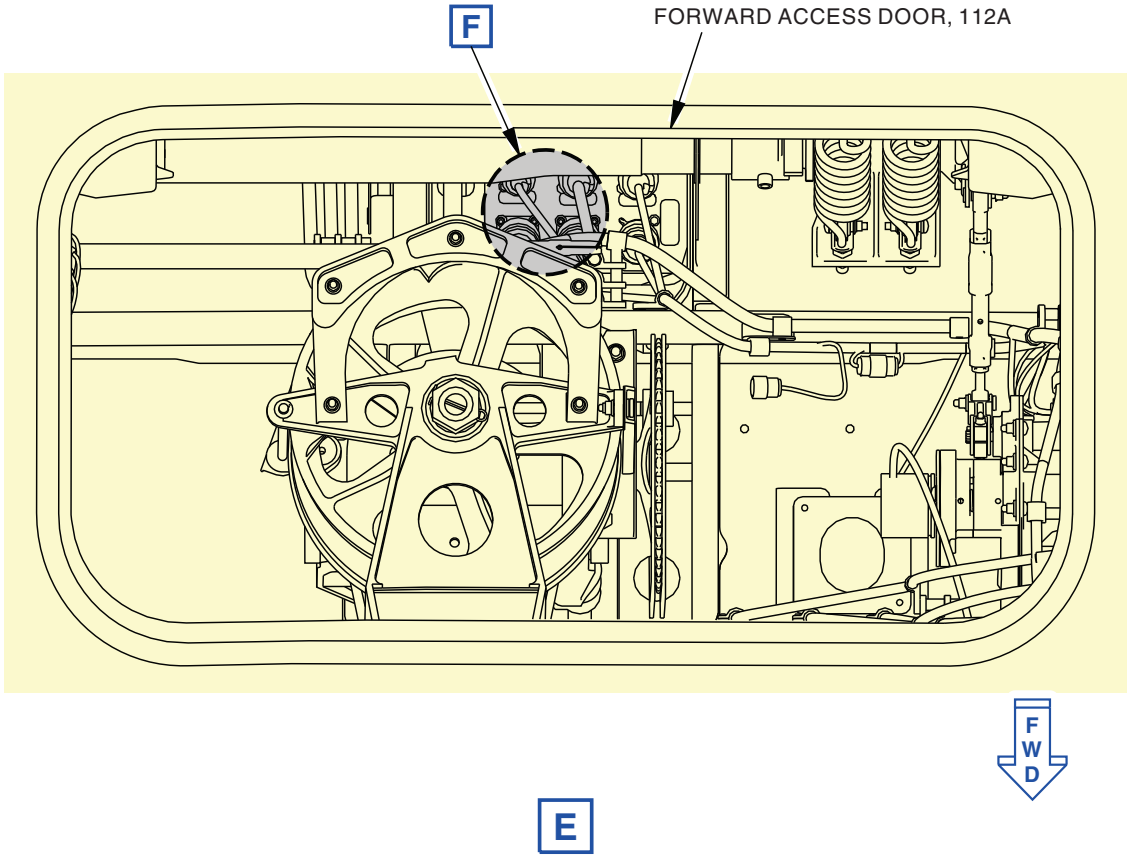


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Flap Lever Position Switch Installation
Figure 401/27-51-08-990-801 (Sheet 3 of 5)

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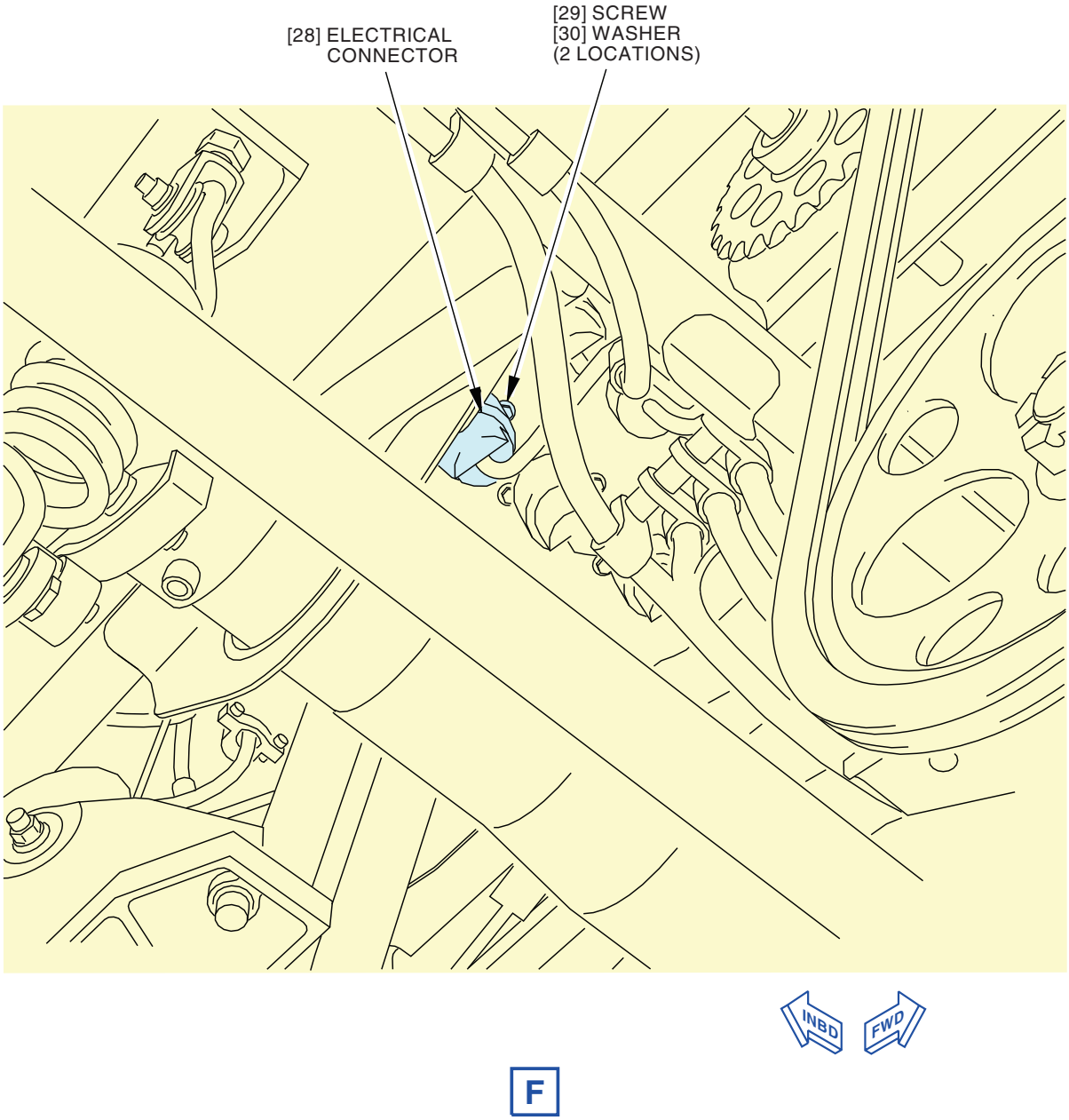




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Flap Lever Position Switch Installation
Figure 401/27-51-08-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap Lever Position Switch Installation
Figure 401/27-51-08-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-08-400-801

3. Flap Lever Position Switch Installation

(Figure 401)

A. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-51-00-740-803	Trailing Edge Flap Load Relief System BITE Test (P/B 501)
27-51-08-820-801	Flap Lever Position Switch Adjustment (P/B 501)
27-81-00-730-801	Leading Edge Cruise Depressurization Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Lower side cover	25-11-21-25-210	PTF ALL
12	Pocket assembly	25-11-21-25-065	PTF ALL
24	Flap lever position switch	27-51-08-01-100	PTF ALL

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Flap Lever Position Switch Installation

SUBTASK 27-51-08-860-002

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	10	C00284	PANEL & INSTR ELEX PANEL

SUBTASK 27-51-08-420-001

(2) Install the flap lever position switches [24] as an assembly:

(a) Put the flap lever position switches [24] with the wires and electrical connector [18], in their position in the control stand [13].

(b) Install the bolts [22], bolt [27] and washers [23] that attach the flap lever position switches [24] and the clamp [21] to the control stand [13].

(c) Install the bolt [25], washer [26] and clamps [21] that attach the wires to the control stand [13].

(d) Install the bolts [19], washers [20] and clamps [21] that attach the wires to the control stand [13].

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SUBTASK 27-51-08-420-002

(3) Install a flap lever position switch [24] independently:

(a) Splice the wires from the new flap lever position flap lever position switch [24] with the wires in the control stand [13] (SWPM 20-10-11).

(b) For the 10 to 40-unit flap lever position switch [24], install the bolts [22] and washers [23] that attach the flap lever position switch [24] to the control stand [13].

1) Install the bolt [25], washer [26] and clamps [21] that attach the wires to the control stand [13].

(c) For the UP flap lever position switch [24], install the bolt [22], bolt [27] and washers [23] that attach the flap lever position flap lever position switch [24] and the clamp [21] to the control stand [13].

(d) Install the applicable bolts [19], washers [20] and clamps [21] that attach the wires to the control stand [13].

SUBTASK 27-51-08-820-001

(4) Do this task: Flap Lever Position Switch Adjustment, TASK 27-51-08-820-801.

SUBTASK 27-51-08-420-003

(5) Connect the electrical connector [18]

(a) Do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

(b) Put the electrical connector [18] in its position.

(c) Install the screws [29] and washers [30] to attach the electrical connector [18].

(d) Connect the electrical connector [28] from below the control stand [13].

(e) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-51-08-860-003

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

SUBTASK 27-51-08-740-001

(7) Do this task: Trailing Edge Flap Load Relief System BITE Test, TASK 27-51-00-740-803.

SUBTASK 27-51-08-730-001

(8) Do this task: Leading Edge Cruise Depressurization Test, TASK 27-81-00-730-801.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-08-410-001

(1) Install the control stand cover [1] on the control stand [13] with the screws [2] and the bolt [5].

SUBTASK 27-51-08-010-006

(2) Install the stabilizer trim lightplate [3] on the control stand [13]:

(a) Connect the wires [14] to the stabilizer trim lightplate [3]:

1) Put the wires [14] for in their position on the stabilizer trim lightplate [3].

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2) Install the washers [15], washers [16] and nuts [17] to attach the wires [14].

(b) Put the stabilizer trim lightplate [3] in its location on the control stand [13].

(c) Loosely install the screws [4].

(d) Adjust the position of the stabilizer trim lightplate [3] to align with the stabilizer trim indicator.

(e) Tighten the screws [4].

SUBTASK 27-51-08-410-002

(3) Install the upper side cover [7] and the lower side cover [11] on the right side of the control stand [13]:

(a) Put the upper side cover [7] and the lower side cover [11] in their position on the control stand [13], with the lower side cover [11] under the upper side cover [7]

(b) Install the screws [6] that attach the upper side cover [7] and lower side cover [11] to the control stand [13].

(c) Install the pocket assembly [12] with the screws [10] on the lower side cover [11].

SUBTASK 27-51-08-410-004

(4) Install the stabilizer trim wheel [8] on the right side of the control stand [13]:

(a) Put the stabilizer trim wheel [8] on the shaft.

(b) Make sure that the handle on the stabilizer trim wheel [8] is 75 to 105 degrees from the handle on the left-side stabilizer trim wheel [8].

(c) Install the bolt [31] and spacer [32] to attach the stabilizer trim wheel [8].

(d) Tighten the bolt [31] to 150 in-lb (16.9 N·m) – 160 in-lb (18.1 N·m) above run-on torque.

SUBTASK 27-51-08-860-004

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	10	C00284	PANEL & INSTR ELEX PANEL

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLAP LEVER POSITION SWITCHES - ADJUSTMENT/TEST

1. General

A. This procedure has an adjustment of the flap lever position switches.

B. The flap lever position switches are installed on the control stand and operate with the flap control lever.

TASK 27-51-08-820-801

2. Flap Lever Position Switch Adjustment

(Figure 501)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Adjustment

SUBTASK 27-51-08-860-005

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	10	C00284	PANEL & INSTR ELEX PANEL

SUBTASK 27-51-08-010-007

(2) Remove the stabilizer trim wheel [8] from the right side of the control stand [13]:

(a) Remove the bolt [9] and spacer [26] that attach the stabilizer trim wheel [8].

(b) Remove the stabilizer trim wheel [8] from the shaft.

SUBTASK 27-51-08-010-008

(3) Remove the upper side cover [7] and lower side cover [11] from the right side of the control stand [13]:

(a) Remove the screws [10] and the pocket assembly [12] from the lower side cover [11].

(b) Remove the screws [6] that attach the upper side cover [7] and the lower side cover [11] to the control stand [13].

(c) Remove the upper side cover [7] and the lower side cover [11] from the control stand [13].

SUBTASK 27-51-08-010-009

(4) Remove the stabilizer trim lightplate [3] from the control stand [13]:

(a) Remove the screws [4] and remove the stabilizer trim lightplate [3].

(b) Remove the nuts [17], washers [16] and washers [15] that attach the wires [14] to the stabilizer trim lightplate [3].

SUBTASK 27-51-08-010-010

(5) Remove the screws [2] and bolt [5] and remove the control stand cover [1] from the control stand [13].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Flap Lever Position Switch Adjustment

NOTE: The adjustment of the flap lever position switch [19] is changed by moving the cam [21] on the flap control lever drum [20].

SUBTASK 27-51-08-020-004

(1) Disconnect the electrical connector [18] for the flap lever position switches [19]:

SUBTASK 27-51-08-860-006

(2) Move the flap control lever to the 5-unit position.

SUBTASK 27-51-08-020-005

(3) Loosen the screws [22] that attach the cam [21] to the flap control lever drum [20].

SUBTASK 27-51-08-820-002

(4) Move the cam [21] fully forward.

SUBTASK 27-51-08-820-003

(5) Adjust the cam [21] on the flap control lever drum [20]:

(a) Do a continuity check between these pins on the electrical connector [18] D5150:

1) Make sure there is continuity between pins 8 and 9.

2) Make sure there is no continuity between pins 9 and 11.

(b) Move the flap control lever to the middle position between the 5-unit and 10-unit detents.

(c) Do a continuity check between these pins on the electrical connector [18] D5150:

1) Make sure there is continuity between pins 8 and 9.

2) Make sure there is no continuity between pins 9 and 11.

(d) Hold the flap control lever in the middle position between the 5-unit and 10-unit detents.

(e) Slowly move the cam [21] on the flap control lever drum [20] aft until the flap lever 10-unit switch operates.

1) Make sure there is no continuity between pins 8 and 9.

2) Make sure there is continuity between pins 9 and 11.

(f) Tighten the screws [22] that attach the cam [21] to the flap control lever drum [20].

1) Make sure there is no continuity between pins 8 and 9.

2) Make sure there is continuity between pins 9 and 11.

SUBTASK 27-51-08-820-004

(6) Move the flap control lever to the 5-unit position.

(a) Make sure there is continuity between pins 8 and 9.

(b) Make sure there is no continuity between pins 9 and 11.

SUBTASK 27-51-08-820-005

(7) Move the flap control lever to the 10-unit position.

(a) Make sure there is no continuity between pins 8 and 9.

(b) Make sure there is continuity between pins 9 and 11.

SUBTASK 27-51-08-820-008

(8) Move the flap control lever to the 40-unit position and verify full travel is accomplished without interference or binding.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-08-820-006

(9) Do a check of the adjustment of the UP flap lever position switch [19]:

NOTE: There is no independent adjustment of the UP flap lever position switch [19].

(a) Move the flap control lever to the UP position.

1) Make sure there is no continuity between pins 5 and 6.

2) Make sure there is continuity between pins 5 and 7.

(b) Move the flap control lever to the 1-unit position.

1) Make sure there is continuity between pins 5 and 6.

2) Make sure there is no continuity between pins 5 and 7.

(c) Move the flap control lever to the UP position.

1) Make sure there is no continuity between pins 5 and 6.

2) Make sure there is continuity between pins 5 and 7.

(d) If the cam [21] is not adjusted correctly for the flap lever position switch [19], then adjust the cam [21] again.

SUBTASK 27-51-08-420-004

(10) Connect the electrical connector [18].

SUBTASK 27-51-08-860-007

(11) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES

D. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-08-410-005

(1) Install the control stand cover [1] on the control stand [13] with the screws [2] and the bolt [5].

SUBTASK 27-51-08-010-011

(2) Install the stabilizer trim lightplate [3] on the control stand [13]:

(a) Connect the wires [14] to the stabilizer trim lightplate [3]:

1) Put the wires [14] in their position on the stabilizer trim lightplate [3].

2) Install the washers [15], washers [16] and nuts [17] to attach the wires [14].

(b) Put the stabilizer trim lightplate [3] in its location on the control stand [13].

(c) Loosely install the screws [4].

(d) Adjust the position of the stabilizer trim lightplate [3] to align with the stabilizer trim indicator.

(e) Tighten the screws [4].

SUBTASK 27-51-08-410-006

(3) Install the upper side cover [7] and the lower side cover [11] on the right side of the control stand [13]:

(a) Put the upper side cover [7] and the lower side cover [11] in their position on the control stand [13], with the lower side cover [11] under the upper side cover [7].

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(b) Install the screws [6] that attach the upper side cover [7] and lower side cover [11] to the control stand [13].

(c) Install the pocket assembly [12] with the screws [10] on the lower side cover [11].

SUBTASK 27-51-08-410-007

(4) Install the stabilizer trim wheel [8] on the right side of the control stand [13]:

(a) Put the stabilizer trim wheel [8] on the shaft.

(b) Make sure the handle on the stabilizer trim wheel [8] is 75 to 105 degrees from the handle of the left-side stabilizer trim wheel [8].

(c) Install the bolt [9] and spacer [26] to attach the stabilizer trim wheel [8].

SUBTASK 27-51-08-860-008

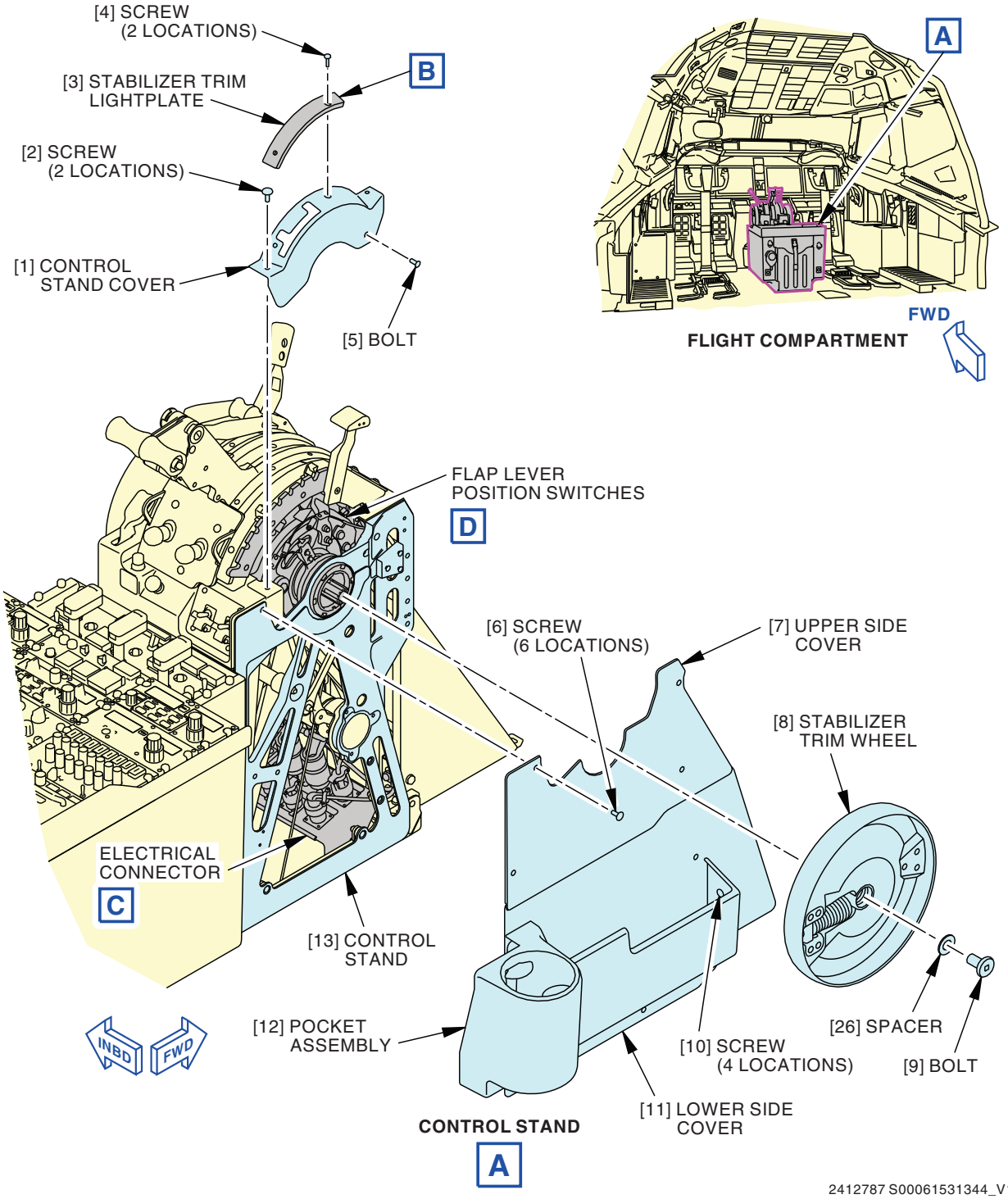
(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	10	C00284	PANEL & INSTR ELEX PANEL

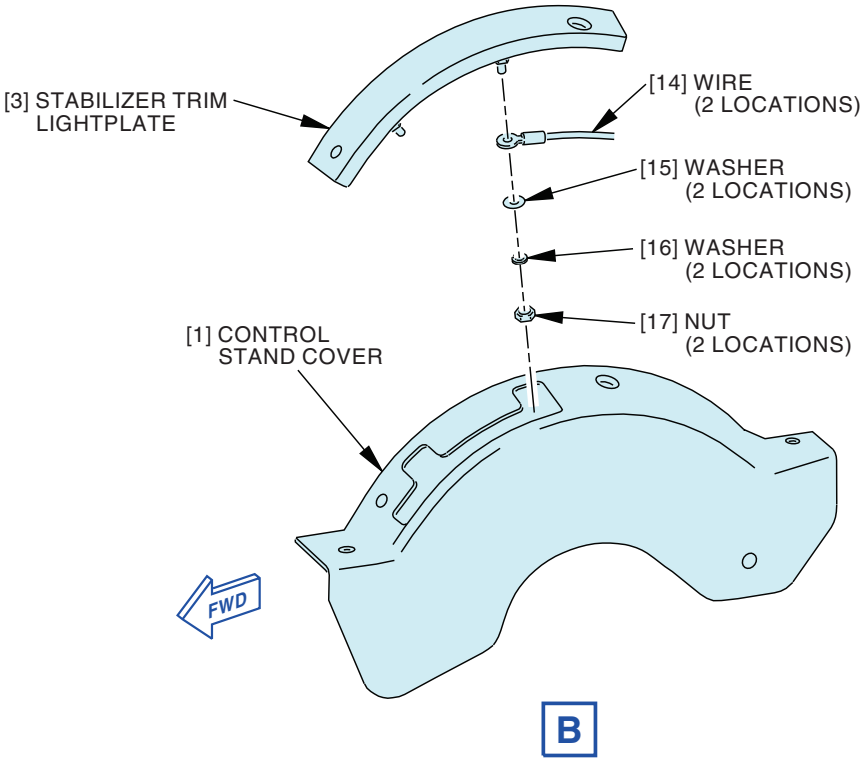
——— END OF TASK ———

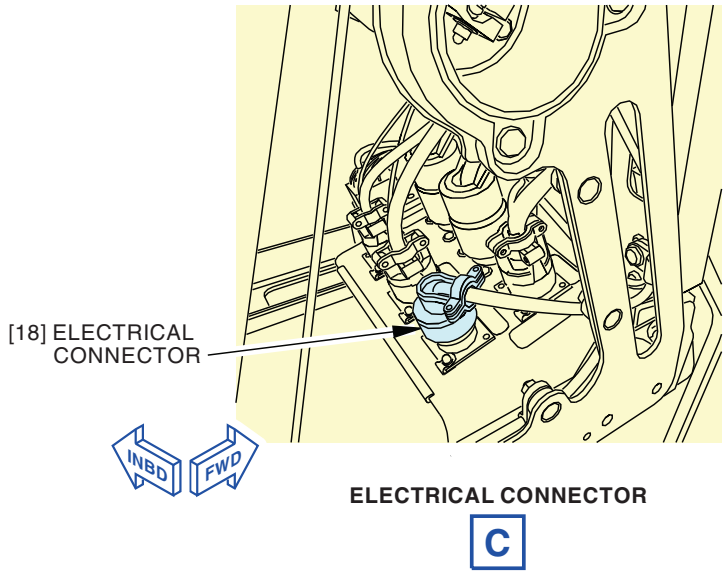
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flap Lever Position Switch Adjustment
Figure 501/27-51-08-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

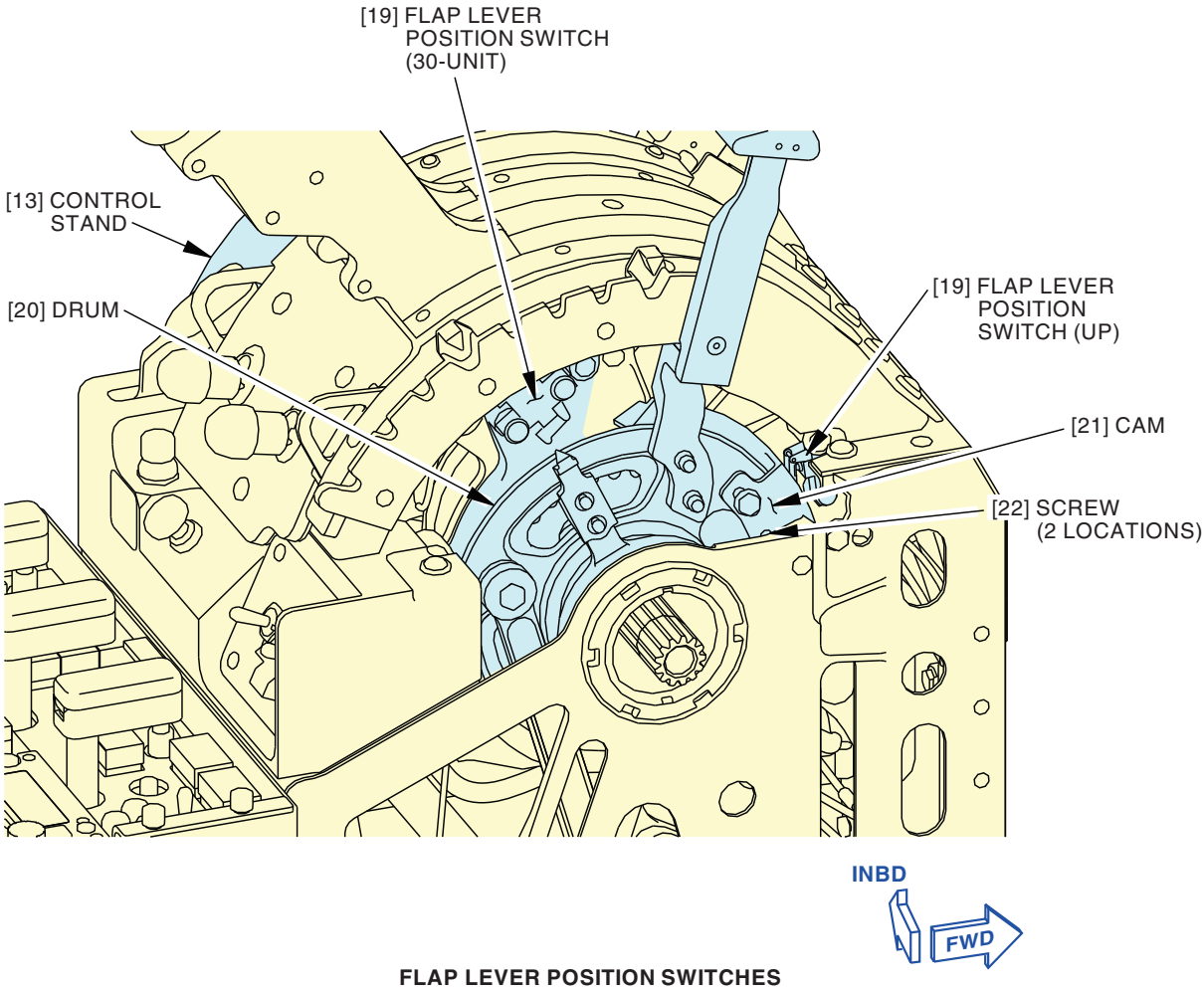




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Flap Lever Position Switch Adjustment
Figure 501/27-51-08-990-802 (Sheet 2 of 3)

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FLAP LEVER POSITION SWITCHES

D

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Flap Lever Position Switch Adjustment
Figure 501/27-51-08-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD TRAILING EDGE FLAP - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of inboard trailing edge flap

(2) An installation of inboard trailing edge flap

(3) A removal of inboard trailing edge flap bullnose seal

(4) An installation of inboard trailing edge flap bullnose seal

(5) A removal of inboard trailing edge flap outer seal

(6) An installation of inboard trailing edge flap outer seal

B. This procedure includes the removal and installation of the inboard main flap with the aft flap. You can remove and install the main flap with the aft flap either removed or installed. To remove and install the aft flap only, do this tasks: Inboard Aft Flap Removal, TASK 27-51-12-000-801 and Inboard Aft Flap Installation, TASK 27-51-12-400-801.

C. The inboard main flap and aft flap together weigh approximately 333 lb (151 kg). The main flap alone weighs approximately 300 lb (136 kg).

TASK 27-51-11-000-801

2. Inboard Trailing Edge Flap Removal

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-980-801	Trailing Edge Flap System Manual Operation (P/B 201)
27-51-18-000-802	Inboard Flap Support Aft Fairing Removal (P/B 401)
53-51-21-000-803	Aft Wing-To-Body Fairing Panel Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1729	Sling Equipment - Trailing Edge Flaps Part #: C27052-39 Supplier: 81205

C. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

D. Access Panels

Number	Name/Location
194AL	Aft Wing To Body Fairing Panel
194AR	Aft Wing To Body Fairing Panel
194BL	Flap Track Lubrication Panel - Aft
194BR	Flap Track Lubrication Panel - Aft

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(Continued)

Number	Name/Location
194CL	Aft Wing To Body Fairing Panel
194CR	Aft Wing To Body Fairing Panel

E. Prepare for the Removal

SUBTASK 27-51-11-860-017

(1) Extend the trailing edge flaps to the 40-unit position (TASK 27-51-00-860-803).

SUBTASK 27-51-11-040-001

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-11-010-001

(3) Do this task: Inboard Flap Support Aft Fairing Removal, TASK 27-51-18-000-802.

SUBTASK 27-51-11-010-002

(4) Remove these access panels from the wing to body fairing (TASK 53-51-21-000-803):

(a) On the left wing, open these access panels:

Number	Name/Location
194AL	Aft Wing To Body Fairing Panel
194BL	Flap Track Lubrication Panel - Aft
194CL	Aft Wing To Body Fairing Panel

(b) On the right wing, open these access panels:

Number	Name/Location
194AR	Aft Wing To Body Fairing Panel
194BR	Flap Track Lubrication Panel - Aft
194CR	Aft Wing To Body Fairing Panel

F. Inboard Trailing Edge Flap Removal

SUBTASK 27-51-11-020-001

(1) Remove the support assembly [131] (Figure 401):

(a) Remove the three bolts [130].

1) Keep the three bolts [130] for the installation.

(b) Remove the support assembly [131].

SUBTASK 27-51-11-020-002

(2) Disconnect the skew control rods for the skew mechanisms from the carriages (Figure 401):

(a) At the outboard carriage [13], remove the nut [8], washer [7], washer [5], bolt [4], and bushing [6] that attaches the skew control rod [3] to the outboard carriage [13].

(b) At the inboard carriage [52], remove the nut [47], washer [46], washer [44], bolt [43], and bushing [45] that attaches the skew control rod [41] to the inboard carriage [52].

SUBTASK 27-51-11-020-003

(3) Disconnect the aft flap drive rod [25] from the crankshaft [18]:

(a) Remove the pin [24], nut [23], washer [22], washer [20], bolt [19], and bushing [21] from the crankshaft [18].

(b) Move the aft flap drive rod [25] away from the inboard flap [1].

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SUBTASK 27-51-11-980-001

(4) If the aft flap [2] is installed, move the aft flap [2] to the retracted position on the inboard flap [1]:

(a) Move the crankshaft [18] aft until the aft flap [2] is in the retracted position.

CAUTION DO NOT USE A RIG PIN IN THE CRANKSHAFT TO HOLD THE AFT FLAP IN THE RETRACTED POSITION. THIS CAN CAUSE DAMAGE TO THE CRANKSHAFT OR THE FITTING ON THE MAIN FLAP.

(b) Install the restraint strap [83] (part of the sling equipment, SPL-1729) to hold the aft flap [2] in the retracted position.

SUBTASK 27-51-11-480-001

(5) Install the sling [82] on the inboard flap [1] (Figure 402):

(a) Remove the existing bolts [85] at the inboard and outboard ends of the front spar on the inboard flap [1].

1) Keep these bolts [85] for the installation.

(b) Install the sling fittings [81] in the location of the bolts [85].

(c) Remove the existing bolts [86] at the rear spar at the aft end of the inboard flap [1].

1) Keep these bolts [86] for re-installation.

(d) Install the sling fitting [84] in the location of the bolts [86].

(e) Connect the sling [82] to the sling fittings [81] and sling fitting [84].

(f) Connect the sling [82] to an overhead lifting device.

(g) Use the sling [82] to hold the weight of the inboard flap [1].

SUBTASK 27-51-11-020-004

(6) Remove the stop fittings from the flap tracks (Figure 401):

(a) On the flap track [39], remove the nut [56], washer [55], bushings [54], and bolt [53] from the end of the flap track [39].

(b) On the outboard flap track [14], remove the roller stop [30], roller stop [37], and the aft fairing support fitting [35]:

1) Remove the nuts [29], washers [28], bolt [27], and bolts [36] that attach the aft fairing support fitting [35] to the flap track [14].

2) Remove the nuts [34], washers [33], washers [32], and bolts [31] that attach the aft fairing support fitting [35], roller stop [30], and roller stop [37] to the flap track [14].

3) Remove the roller stop [30], roller stop [37], and the aft fairing support fitting [35] from the flap track [14].

SUBTASK 27-51-11-860-002

(7) Use the manual drive system to extend the trailing edge flaps to the stops on the ballscrews (TASK 27-51-00-980-801).

NOTE: During the manual operation of the trailing edge flap, it is not necessary to do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-11-020-005

(8) Remove the inboard flap [1]:

(a) Hold the weight of the inboard flap [1] with the sling [82] and the overhead lifting device.

(b) Disconnect the inboard carriage [52] from the gimbal [40]:

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1) Install a support to hold the ballscrew [42] when it is disconnected from the inboard carriage [52].
2) Remove the nuts [50], washers [49], and bolts [48] that attach the carriage bushings [51] to the inboard carriage [52].
3) Remove the carriage bushings [51] from the inboard carriage [52].
4) Move the gimbal [40] away from the inboard carriage [52].

(c) Disconnect the outboard carriage [13] from the gimbal [16]:
1) Install a support to hold the ballscrew [17] when it is disconnected from the outboard carriage [13].
2) Remove the nut [11], washer [10], and bolt [9] that attach the carriage bushing [15] and carriage carriage bushing [26] to the outboard carriage [13].
3) Remove the carriage bushing [15] and carriage bushing [26] from the outboard carriage [13].
4) Move the gimbal [16] away from the outboard carriage [13].

(d) Use the sling [82] to move the inboard flap [1] aft and down away from the flap tracks.

CAUTION

BE CAREFUL WHEN YOU LOWER THE FLAP TO THE GROUND. YOU CAN CAUSE DAMAGE TO THE TRAILING EDGE OF THE FLAP.

(e) Use the sling [82] to move the inboard flap [1] outboard and aft away from the airplane.

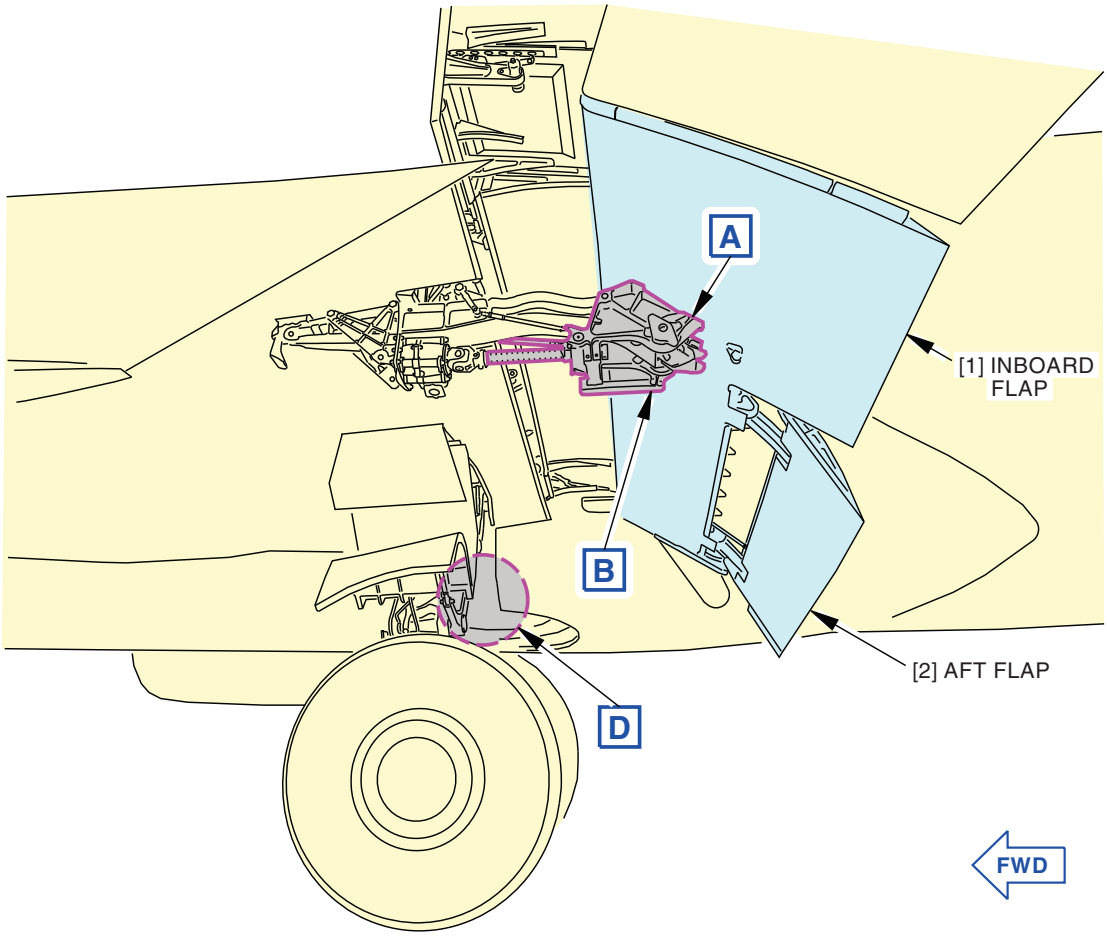
SUBTASK 27-51-11-210-001

(9) Do a check of the configuration of the inboard flap carriage roller cartridges, and inboard seals:
(a) Do a check of the position of the index marks [120] for the carriage roller cartridges on the inboard and outboard carriages.
1) Make a record of the position to use during the installation.
(b) If you will install a different flap, remove and keep the shim and bullnose seal for the installation.
1) Keep and install the same shim and bullnose seal you removed from the airplane.

NOTE: If you install a different bullnose seal or shim, it can cause unsatisfactory flight performance.

END OF TASK

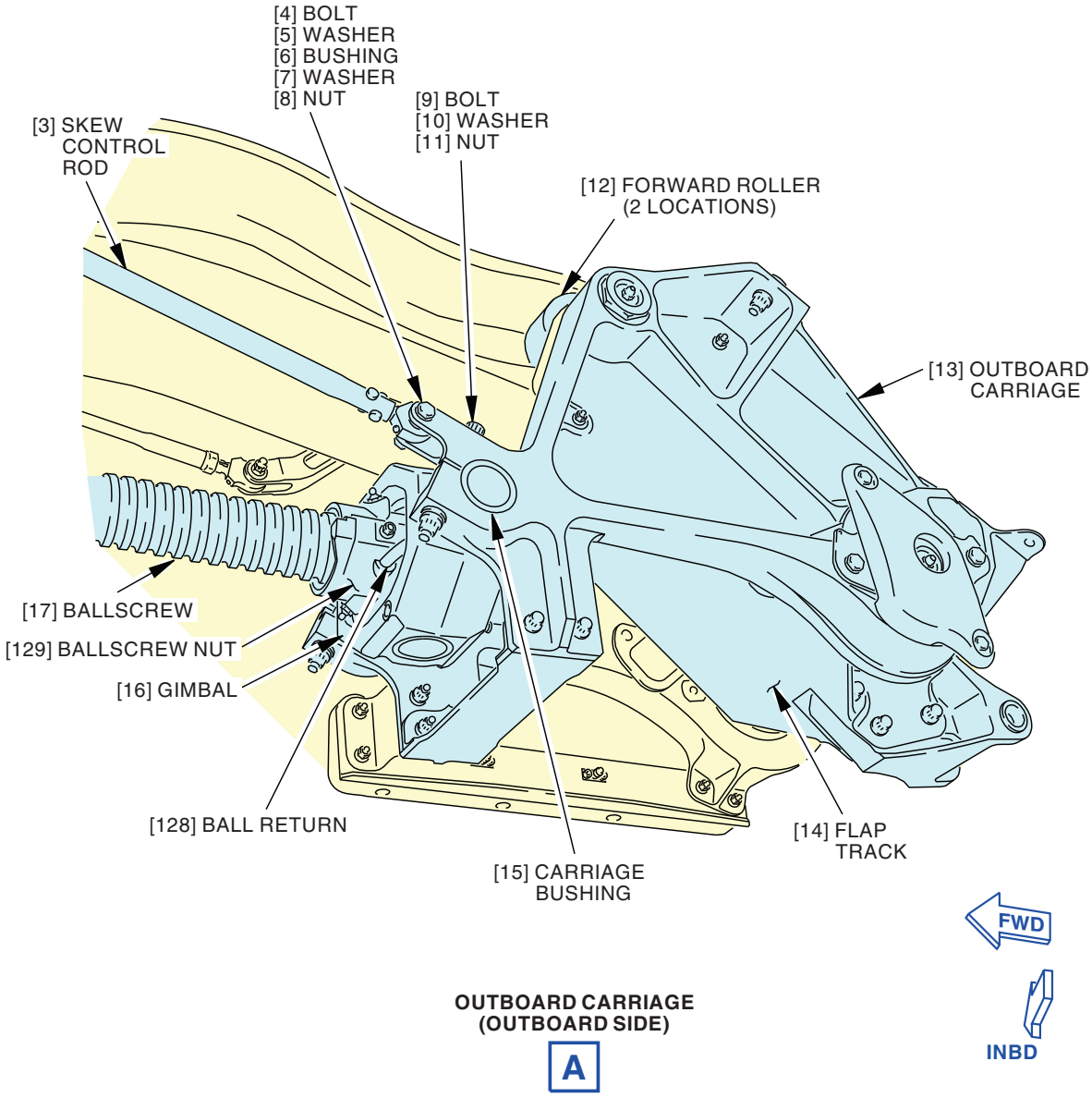
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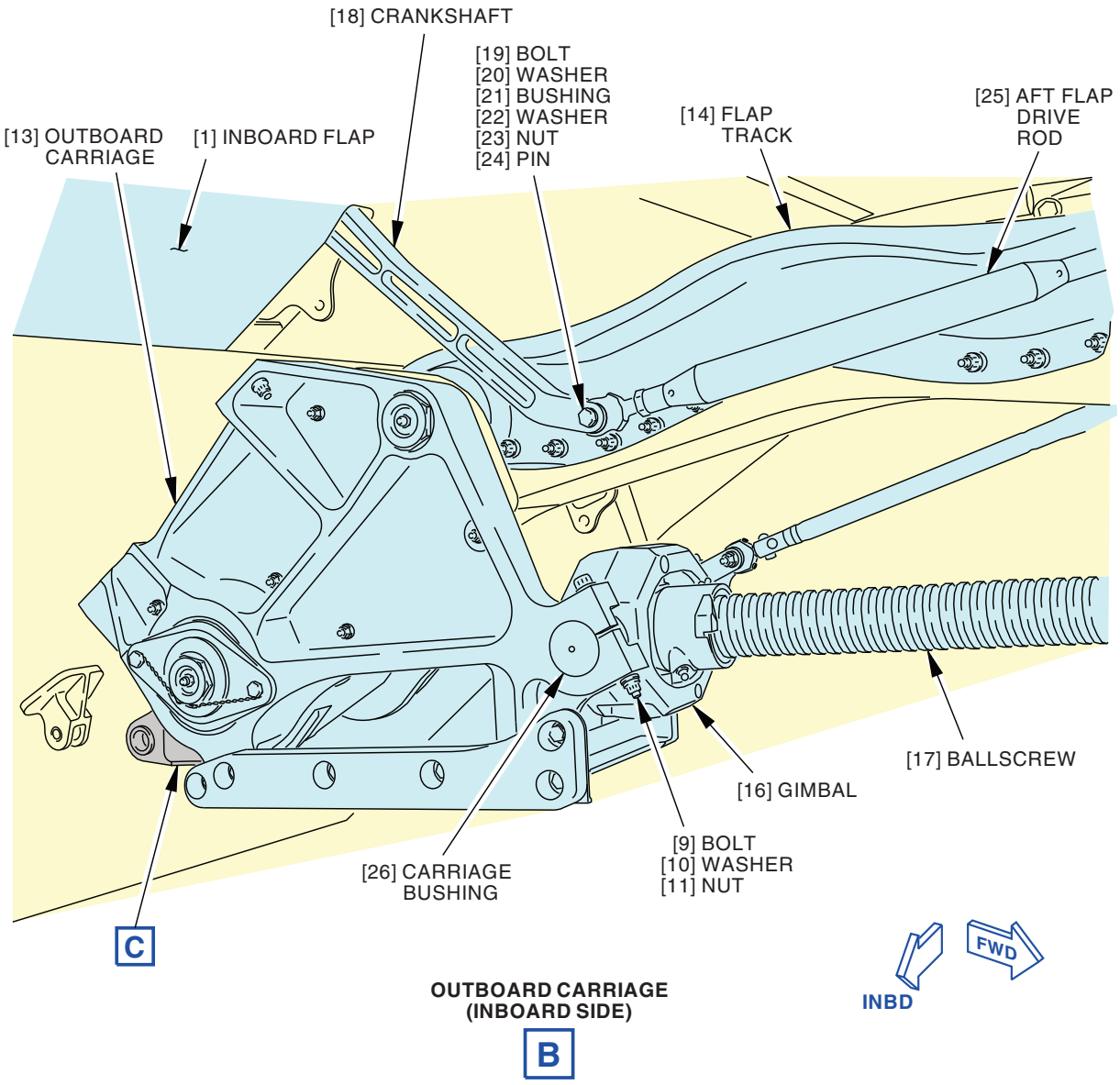
Inboard Trailing Edge Flap Installation
Figure 401/27-51-11-990-801 (Sheet 1 of 7)

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Inboard Trailing Edge Flap Installation
Figure 401/27-51-11-990-801 (Sheet 2 of 7)

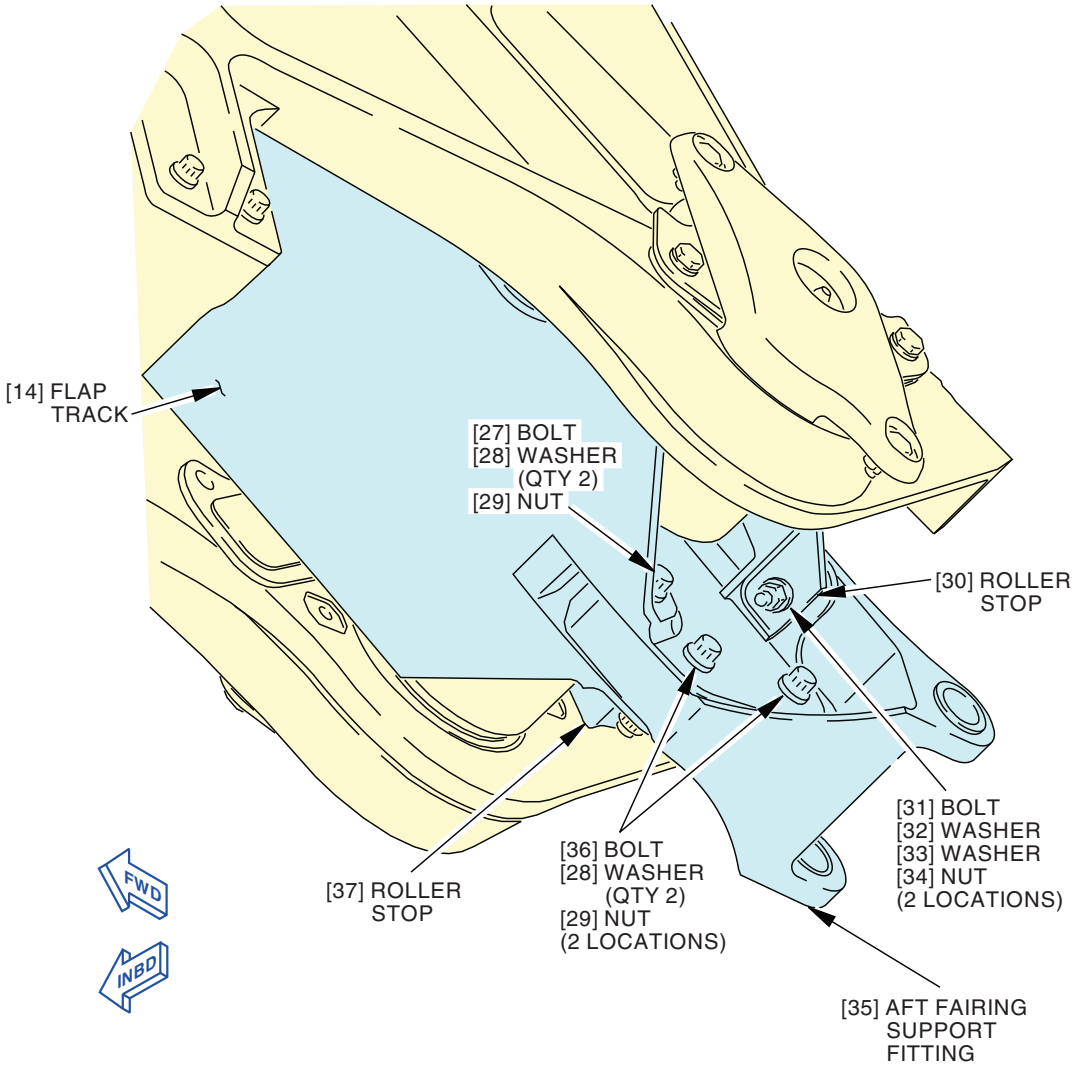
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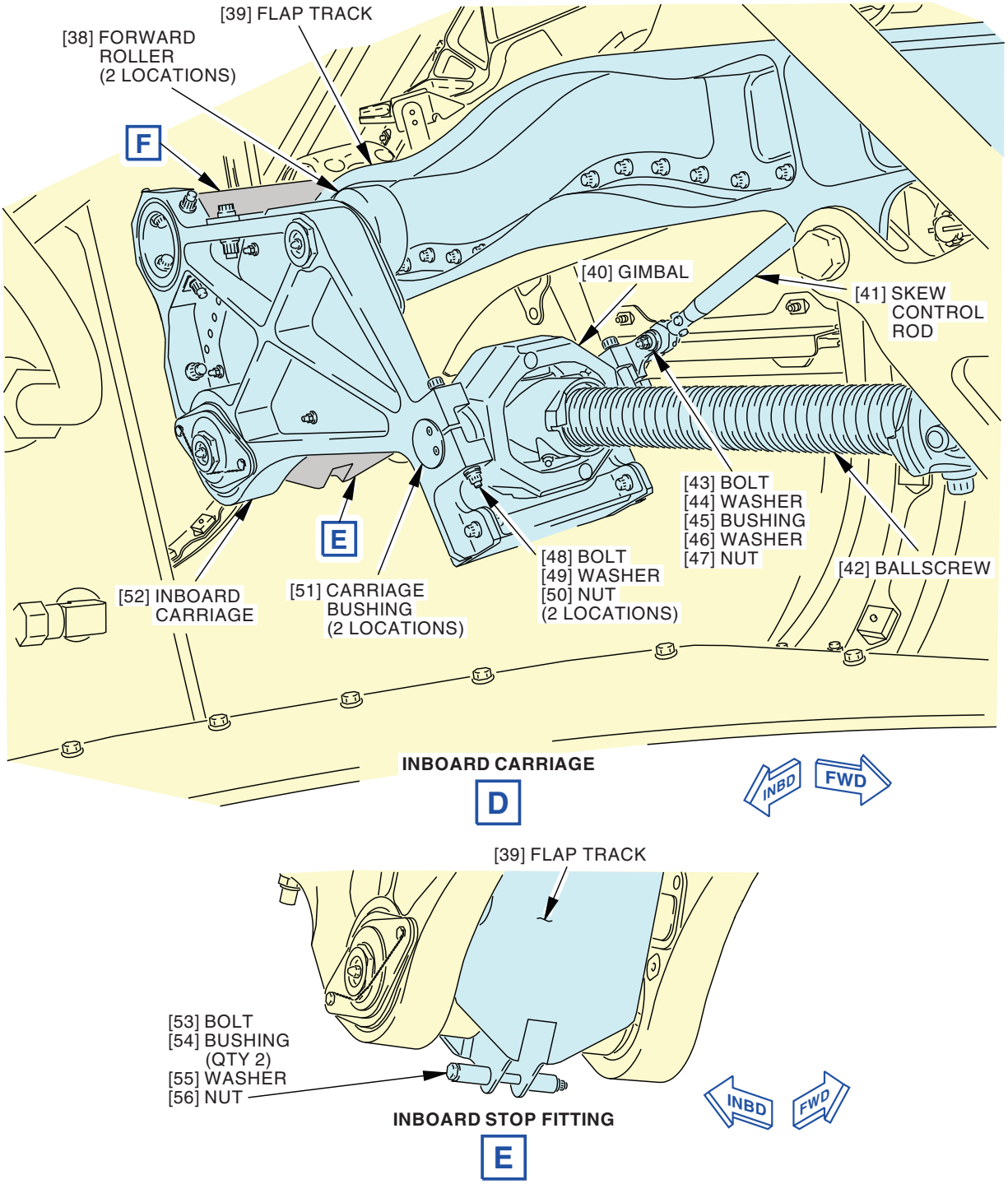
Inboard Trailing Edge Flap Installation
Figure 401/27-51-11-990-801 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



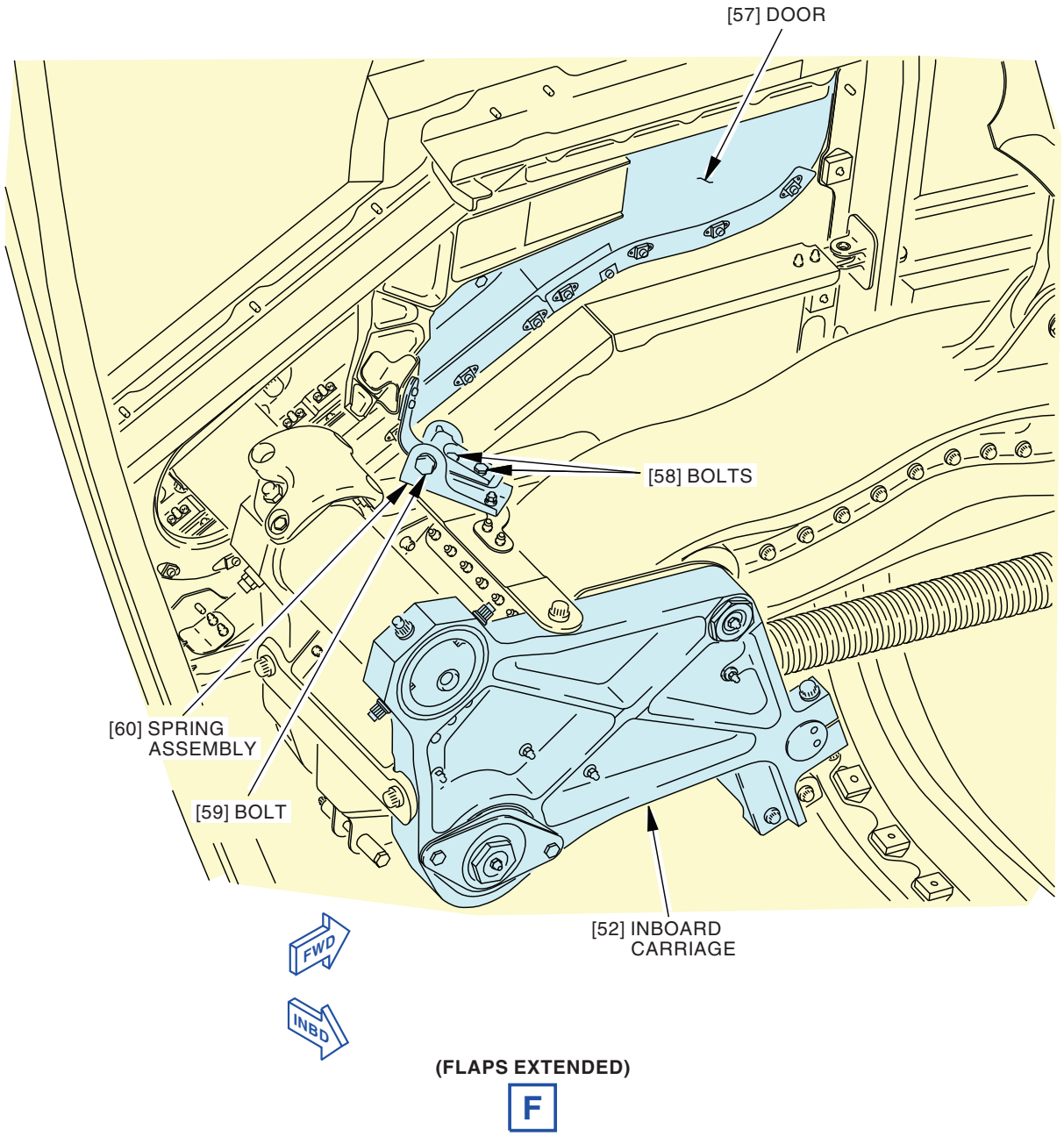
Inboard Trailing Edge Flap Installation
Figure 401/27-51-11-990-801 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



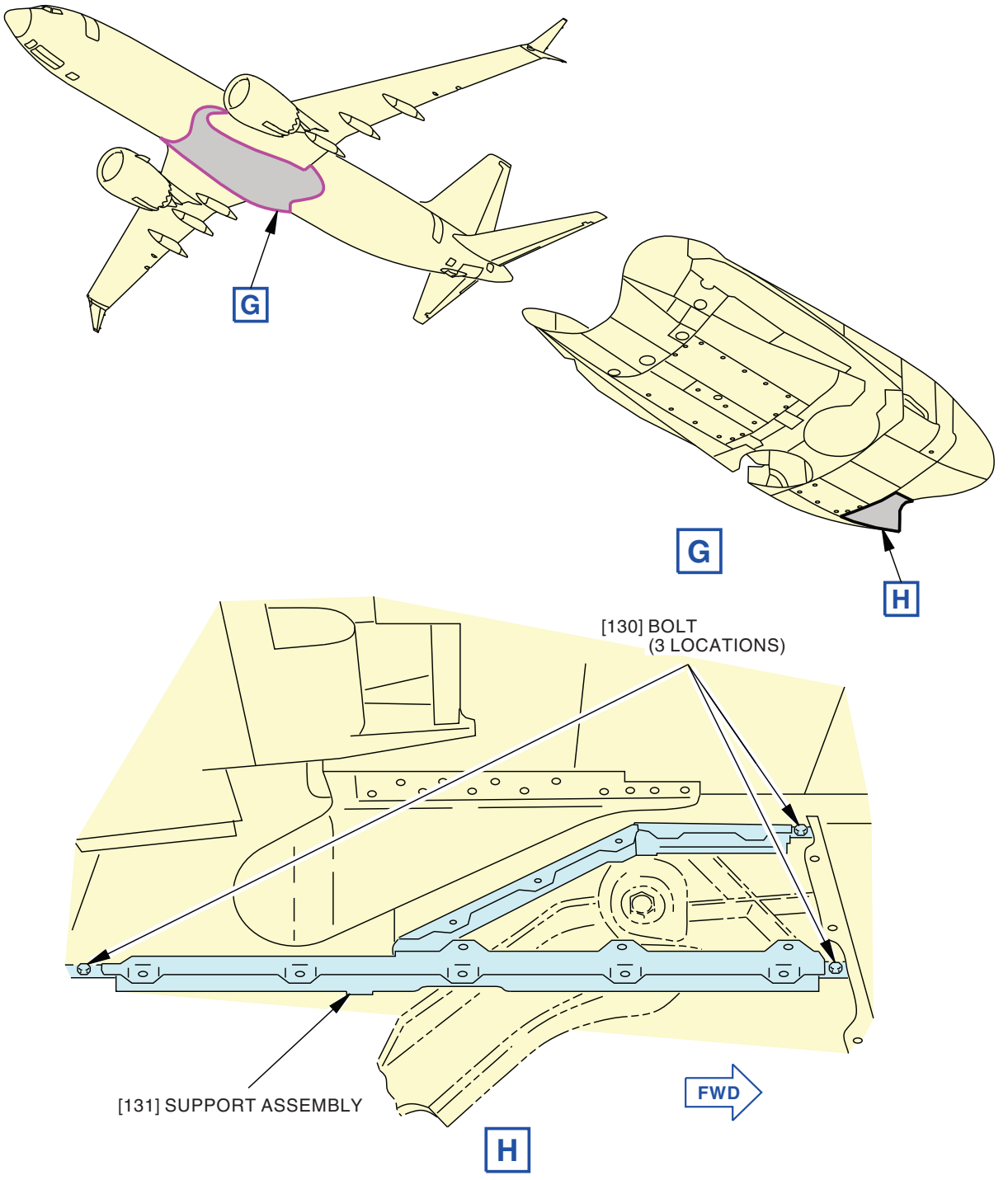
Inboard Trailing Edge Flap Installation
Figure 401/27-51-11-990-801 (Sheet 5 of 7)

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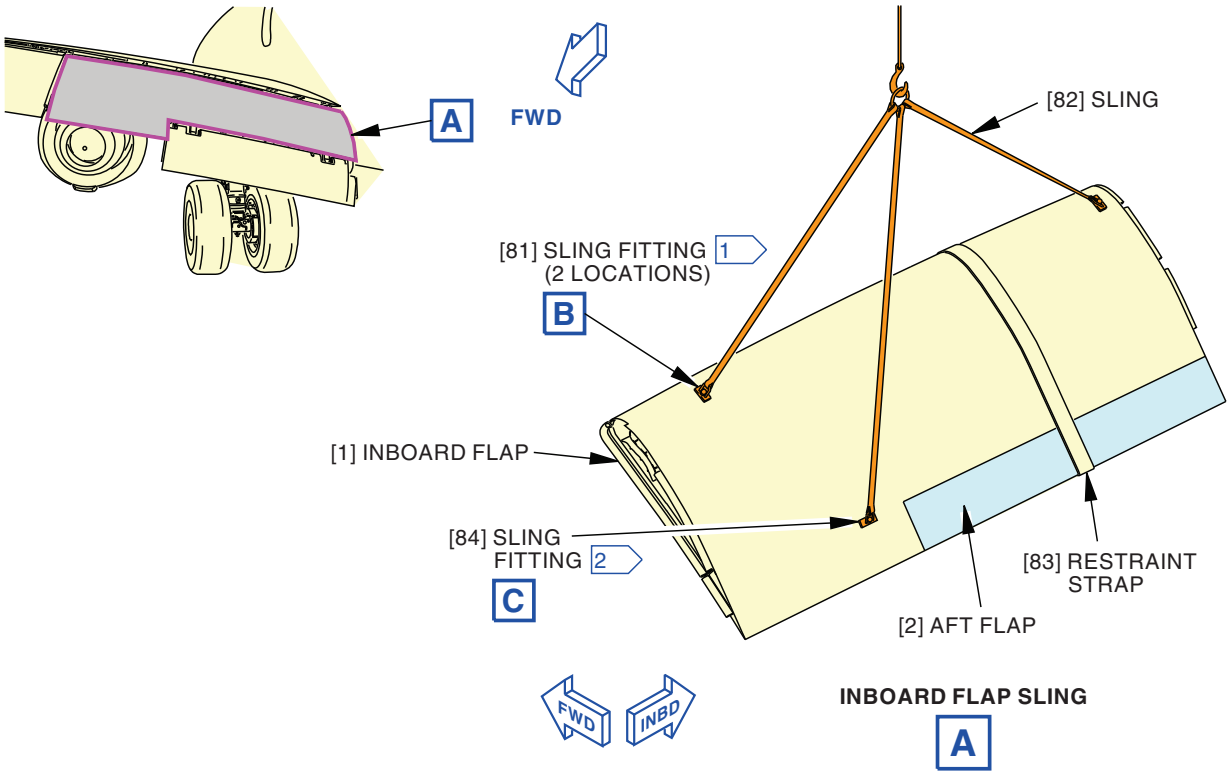
Inboard Trailing Edge Flap Installation
Figure 401/27-51-11-990-801 (Sheet 6 of 7)

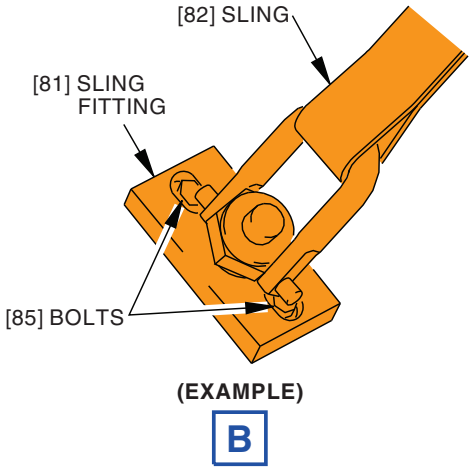
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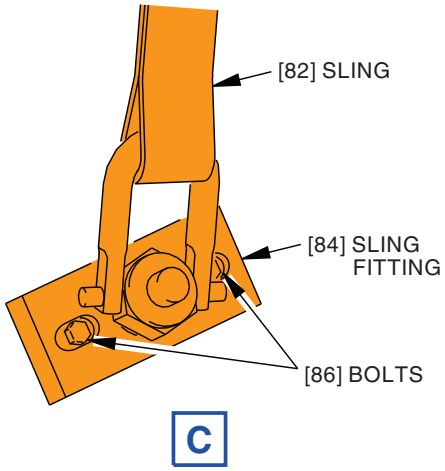


Inboard Trailing Edge Flap Installation
Figure 401/27-51-11-990-801 (Sheet 7 of 7)

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1 ITEF MFSTA 88.4, 175.5 FORWARD OF FLAP FRONT SPAR PLANE.

2 ITEF MFSTA 163.4, COMMON TO FLAP REAR SPAR PLANE.

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Inboard Flap Sling
Figure 402/27-51-11-990-802

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TASK 27-51-11-400-801

3. Inboard Trailing Edge Flap Installation

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-51-11-820-801	Inboard Trailing Edge Flap Adjustment (P/B 501)
27-51-18-400-802	Inboard Flap Support Aft Fairing Installation (P/B 401)
53-51-21-400-803	Aft Wing-To-Body Fairing Panel - Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1729	Sling Equipment - Trailing Edge Flaps Part #: C27052-39 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
D00633	Grease - Aircraft General Purpose	BMS3-33
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Inboard flap	27-51-11-01-100	PTF ALL
		27-51-11-01-105	PTF ALL
2	Aft flap	27-51-12-01A-073	PTF ALL
		27-51-12-01A-076	PTF ALL

E. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

F. Access Panels

Number	Name/Location
194AL	Aft Wing To Body Fairing Panel
194AR	Aft Wing To Body Fairing Panel
194BL	Flap Track Lubrication Panel - Aft
194BR	Flap Track Lubrication Panel - Aft
194CL	Aft Wing To Body Fairing Panel
194CR	Aft Wing To Body Fairing Panel

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G. Prepare for the Installation

SUBTASK 27-51-11-820-001

(1) If the eccentric bushing [115] are installed on the flap assembly, make sure that the carriage roller cartridges are in the correct position (Figure 403):

NOTE: The eccentric bushings are not installed on every flap assembly. If they are not installed, adjustment is not necessary.

(a) If you install a different inboard flap [1] than the one you removed, make sure that the cartridge [116] and cartridge [125] are in the same position as the flap you removed.

(b) If you did not record the position of the carriage roller from the flap you removed, then adjust the cartridge [116] and cartridge [125] to the nominal position.

NOTE: The nominal position for the cartridges is -45 degrees for the left wing, and +30 degrees for the right wing.

NOTE: There is no mismatch or alignment dimension to the fixed structure or the outboard flap that the flap is adjusted to meet. The cartridges can be changed from the nominal position to meet lateral trim tolerances.

(c) Make sure that the cartridge [116] and cartridge [125] for the inboard flap [1] are in the same position.

(d) If it is necessary, adjust the cartridge [125] on the inboard carriage [52] as follows:

1) Loosen the nut [127] on the roller [123].
2) Turn the eccentric bushing [115] until the index mark [120] is in the correct position.
3) Tighten the nut [127] to 160 in-lb (18 N·m) - 975 in-lb (110 N·m).
4) Install the pin [126] in the nut [127].
5) Install the lockwire between the bolts that attach the cartridge [125].

(e) If it is necessary, adjust the cartridge [116] or cartridge [125] on the outboard carriage [13]:

1) Adjust the outboard cartridge [116]:
   a) Remove the bolts [113], washers [122], and rub block [121] from the retainer [114].
   b) Loosen the nut [118] on the roller [117].
   c) Turn the eccentric bushing [115] until the index marks [120] are in the correct position.
   d) Tighten the nut [118] on the roller [117] to 160 in-lb (18 N·m) - 975 in-lb (110 N·m).
   e) Install the pin [119] in the nut [118].
   f) Put the rub block [121] in its position on the retainer [114] and install the bolts [113] and washers [122].
2) Adjust the cartridge [125] as follows:
   a) Loosen the nut [127] on the roller [123].
   b) Turn the eccentric bushing [115] until the index marks [120] are in the correct position.
   c) Tighten the nut [127] on the roller [123] to 160 in-lb (18 N·m) - 975 in-lb (110 N·m).
   d) Install the pin [126] in the nut [127].

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e) Install lockwire between the bolts that attach the cartridge [125].

SUBTASK 27-51-11-210-002

(2) If you will install a different flap, install the shim and bullnose seal you removed from the original flap.

NOTE: Install the same shim and bullnose seal you removed from the airplane. If you install a different bullnose seal or shim, it can cause unsatisfactory flight performance.

SUBTASK 27-51-11-020-006

(3) Make sure that the stop fittings are not installed on the flap tracks.

SUBTASK 27-51-11-480-002

(4) If it is not installed, install the sling [82] on the inboard flap [1] (Figure 402):

(a) Remove the bolts [85] at the inboard and outboard ends of the front spar on the inboard flap [1].

(b) Install the sling fittings [81] in the location of the bolts [85].

(c) Remove the bolts [86] at the rear spar at the aft end of the inboard flap [1].

(d) Install the sling fitting [84] in the location of the bolts [86].

(e) Connect the sling [82] to the sling fittings [81] and sling fitting [84].

SUBTASK 27-51-11-980-002

(5) If it is installed, make sure that the aft flap [2] is restrained in the retracted position on the inboard flap [1]:

CAUTION DO NOT USE A RIG PIN IN THE CRANKSHAFT TO HOLD THE AFT FLAP IN THE RETRACTED POSITION. THIS CAN CAUSE DAMAGE TO THE CRANKSHAFT OR THE FITTING ON THE MAIN FLAP.

(a) Install the restraint strap [83] (part of the sling equipment, SPL-1729) to hold the aft flap [2] in the retracted position.

H. Inboard Trailing Edge Flap Installation

SUBTASK 27-51-11-420-001

(1) Install the inboard flap [1] (Figure 401) as follows:

CAUTION BE VERY CAREFUL WHEN YOU LIFT THE FLAP FROM THE GROUND. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE TRAILING EDGE OF THE FLAP.

(a) Use the sling [82] and the overhead lifting device to move the inboard flap [1] up into its position aft of the flap track [14] and flap track [39].

(b) Align the forward rollers [12] for the outboard carriage [13] with the flap track [14].

(c) Move the inboard flap track [39] to align with the forward rollers [38] for the inboard carriage [52].

CAUTION DO NOT USE THE SLING TO MOVE THE FLAP ON THE FLAP TRACK AFTER THE AFT ROLLERS ENGAGE. THIS CAN CAUSE DAMAGE TO THE FLAP.

(d) Move the inboard flap [1] up and forward on the flap tracks until the carriages align with the gimbal [16] and gimbal [40].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Connect the outboard carriage [13] to the gimbal [16]:

1) Remove the support from the ballscrew [17] and move the gimbal [16] to its position in the outboard carriage [13].

2) On the left wing, make sure that the ball return [128] on the ballscrew nut [129] are outboard.

3) On the right wing, make sure that the ball return [128] on the ballscrew nut [129] are inboard.

NOTE: This is the recommended position for maintenance.

(f) Install the carriage bushing [15] and carriage bushing [26] to attach the gimbal [16] to the outboard carriage [13].

(g) Install the bolts [9], washers [10], and nuts [11] that attach the carriage bushing [15] and carriage bushing [26] to the outboard carriage [13].

(h) Tighten the nuts [11] to 65 in-lb (7.3 N·m) - 75 in-lb (8.5 N·m) more than the run-on torque.

(i) Connect the inboard carriage [52] to the gimbal [40]:

1) Remove the support from the ballscrew [42] and move the gimbal [40] to its position in the inboard carriage [52].

2) Apply grease, D00633, to the inside and outside of the carriage bushings [51].

3) Install the carriage bushings [51] to attach the gimbal [40] to the inboard carriage [52].

4) Install the bolts [48], washers [49], and nuts [50] that attach the carriage bushings [51] to the inboard carriage [52].

5) Tighten the nuts [50] to 65 in-lb (7.3 N·m) - 75 in-lb (8.5 N·m).

SUBTASK 27-51-11-420-002

(2) Install the stop fittings for the flap tracks:

(a) On the inboard flap track [39], install the bolt [53], bushings [54], washer [55], and nut [56] on the end of the flap track [39].

(b) On the flap track [14], install the roller stop [30], roller stop [37], and the aft fairing support fitting [35]:

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Apply Cor-Ban 27L Compound, G50237, to the mating surfaces of the aft fairing support fitting [35], roller stop [30], roller stop [37], and the flap track [14].

2) Put the roller stop [30], roller stop [37], and the aft fairing support fitting [35] in their positions on the flap track [14].

a) Make sure that the roller stop [30] and roller stop [37] touch the web of the flap track [14].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

3) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the bolt [27], bolts [36], and bolts [31].

4) Install the bolts [31], washers [32], washers [33], and nuts [34] that attach the aft fairing support fitting [35], roller stop [30], and roller stop [37] to the flap track [14].

5) Install the bolts [36], bolt [27], washers [28], and nuts [29] that attach the aft fairing support fitting [35] to the flap track [14].

SUBTASK 27-51-11-080-001

(3) Remove the sling [82] from the inboard flap [1] (Figure 402):

(a) Disconnect the sling [82] from the overhead lifting device.

(b) Disconnect the sling [82] from the sling fittings [81] and sling fitting [84].

(c) Remove the sling fittings [81] and sling fitting [84] from the inboard flap [1].

(d) Apply compound, C00528, to the holes for the bolts [85] and bolts [86] in the inboard flap [1].

(e) Install the bolts [85] at the inboard and outboard ends of the front spar on the inboard flap [1].

(f) Install the bolts [86] at the aft end of the inboard flap [1].

SUBTASK 27-51-11-980-003

(4) If the aft flap [2] is installed, move the aft flap [2] to the extended position on the inboard flap [1] as follows:

(a) Remove the restraint strap [83] from the aft flap [2].

(b) Move the crankshaft [18] aft until the aft flap [2] is in the extended position.

SUBTASK 27-51-11-420-003

(5) Connect the aft flap drive rod [25] to the crankshaft [18] (Figure 401) as follows:

(a) Move the aft flap drive rod [25] to its position on the crankshaft [18].

(b) Install the bolt [19], washer [20], bushing [21], washer [22], and nut [23] that attach the aft flap drive rod [25] to the crankshaft [18].

1) Make sure that the head of the bolt [19] is on the inboard side.

(c) Tighten the nut [23] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

(d) Install the pin [24] through the nut [23].

SUBTASK 27-51-11-420-004

(6) Connect the skew control rods for the skew mechanisms to the carriages:

(a) Connect the skew control rod [41] to the inboard carriage [52]:

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WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the bolt [43].
2) Install the bolt [43], washer [44], bushing [45], washer [46], and nut [47] to attach the skew control rod [41] to the inboard carriage [52].
   a) Install the bolt [43] with the head away from the center of the flap track [39].

(b) Connect the skew control rod [3] to the outboard carriage [13]:

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

1) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the bolt [4].
2) Install the bolt [4], washer [5], bushing [6], washer [7], and nut [8] to attach the skew control rod [3] to the outboard carriage [13].
   a) Install the bolt [4] with the head away from the center of the flap track [14].

SUBTASK 27-51-11-420-005

(7) Install the support assembly [131] (Figure 401):
   (a) Install the support assembly [131].
   (b) Install the three bolts [130].

SUBTASK 27-51-11-410-001

(8) Install these access panels for the wing to body fairing (TASK 53-51-21-400-803):
   (a) On the left wing, close these access panels:

Number	Name/Location
194AL	Aft Wing To Body Fairing Panel
194BL	Flap Track Lubrication Panel - Aft
194CL	Aft Wing To Body Fairing Panel

   (b) On the right wing, close these access panels:

Number	Name/Location
194AR	Aft Wing To Body Fairing Panel
194BR	Flap Track Lubrication Panel - Aft
194CR	Aft Wing To Body Fairing Panel

SUBTASK 27-51-11-860-003

(9) Extend the trailing edge flaps:
   (a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.
   (b) Extend the trailing edge flaps to the 40-unit position (TASK 27-51-00-860-803).
   (c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-11-210-003

(10) With the flaps fully extended, make sure that the full surface of the spring assembly [60] on the inboard carriage [52] touches the door [57] on the wing to body fairing and that it holds the door closed (Figure 401).

SUBTASK 27-51-11-820-002

(11) If the spring assembly [60] does not touch the door [57], or hold it closed, adjust the position of the spring assembly [60]:

(a) Loosen the bolts [58] and bolt [59] that hold the spring assembly [60] in its position.

(b) Move the spring assembly [60] until it touches the door [57] with its face parallel to the door.

(c) Tighten the bolts [58] and bolt [59].

SUBTASK 27-51-11-820-003

(12) Do this task: Inboard Trailing Edge Flap Adjustment, TASK 27-51-11-820-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-11-410-002

(1) Do this task: Inboard Flap Support Aft Fairing Installation, TASK 27-51-18-400-802.

SUBTASK 27-51-11-440-001

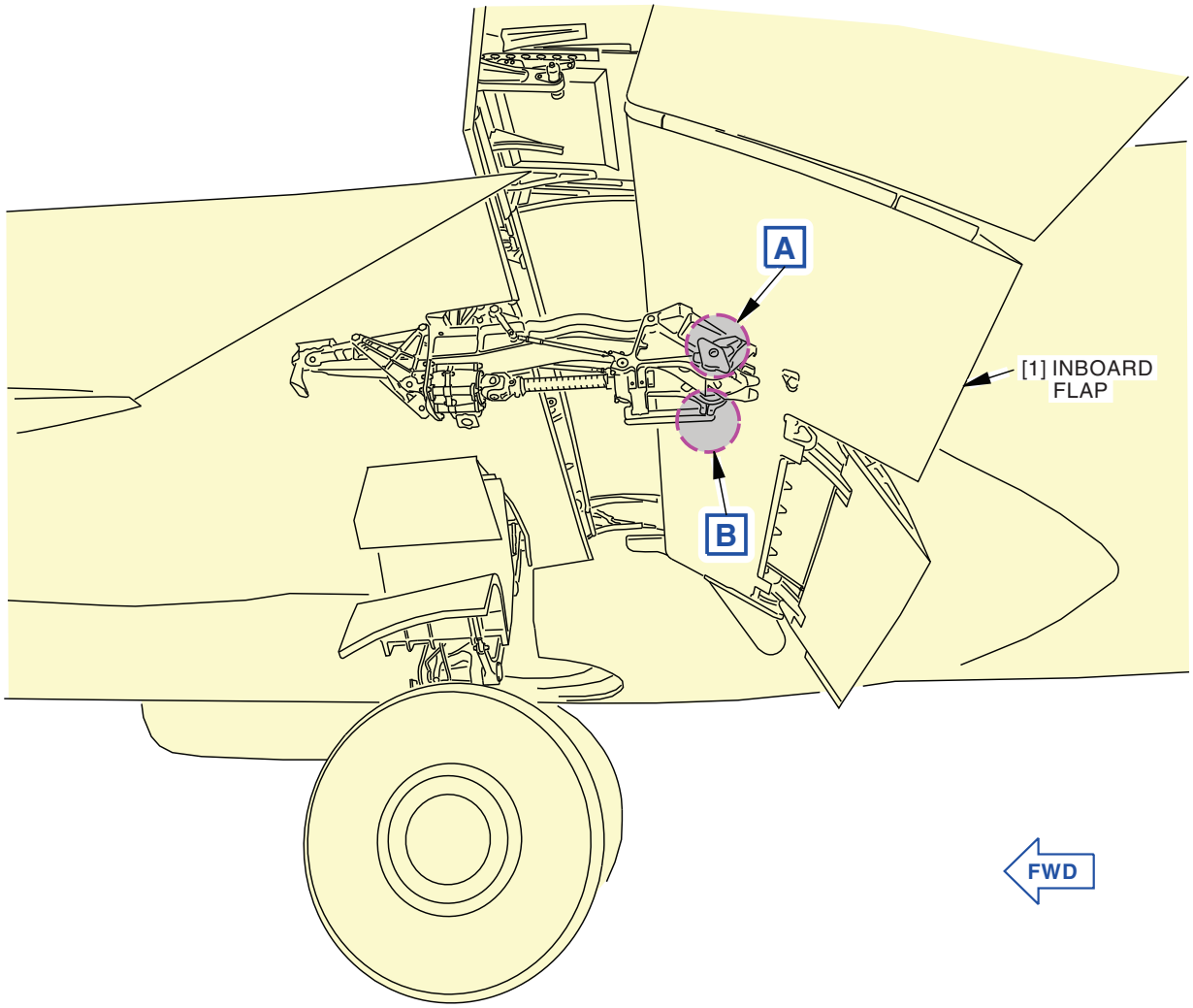
(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-11-860-004

(3) If the trailing edge flaps are not in the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

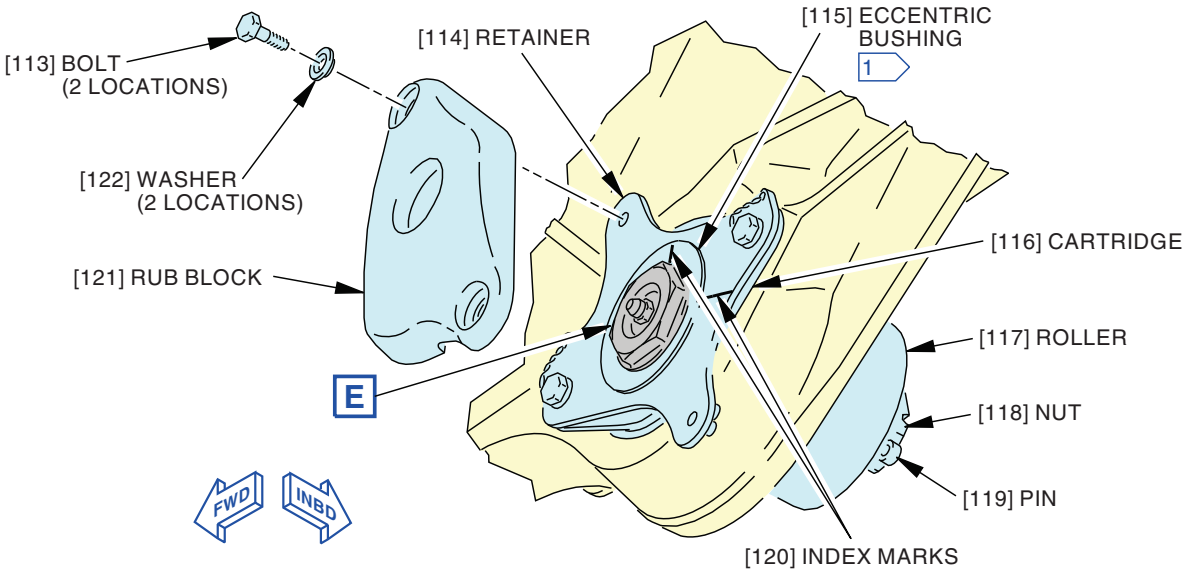


1 IF INSTALLED
2 LEFT WING NOMINAL POSITION
3 RIGHT WING NOMINAL POSITION

2412800 S00061531361_V1

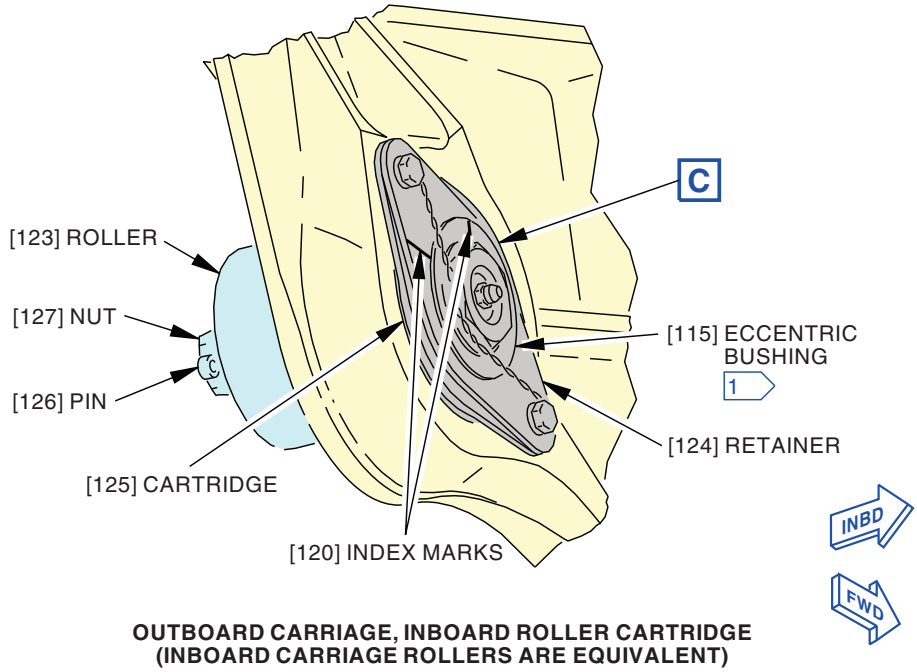
Inboard Flap Roller Adjustment
Figure 403/27-51-11-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD CARRIAGE, OUTBOARD ROLLER CARTRIDGE

A



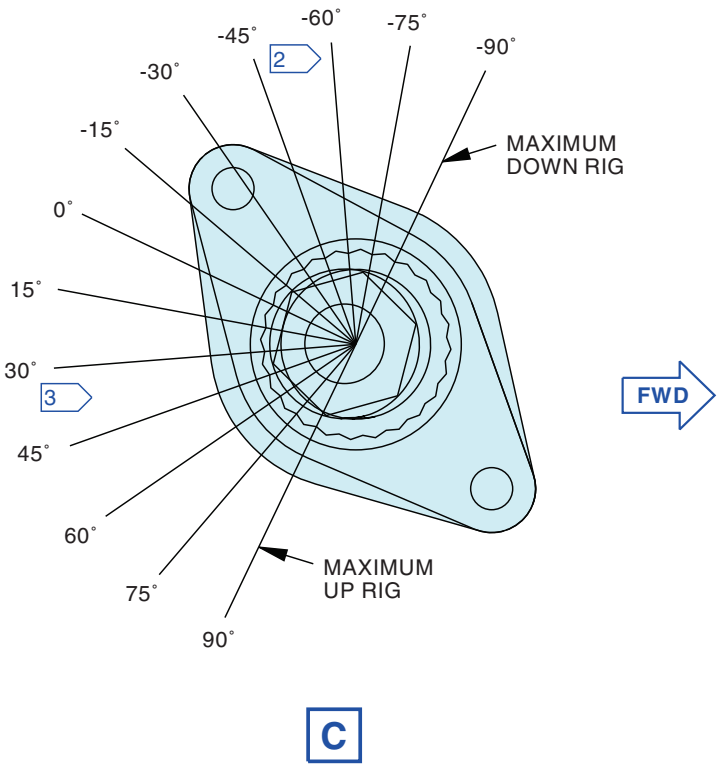
OUTBOARD CARRIAGE, INBOARD ROLLER CARTRIDGE
(INBOARD CARRIAGE ROLLERS ARE EQUIVALENT)

B

2412801 S00061531362_V1

Inboard Flap Roller Adjustment
Figure 403/27-51-11-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412802 S00061531363_V1

Inboard Flap Roller Adjustment
Figure 403/27-51-11-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-11-000-802

4. Inboard Trailing Edge Flap Bullnose Seal Removal

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

C. Prepare for the Removal

SUBTASK 27-51-11-860-005

(1) Extend the trailing edge flaps to the 15-unit position (TASK 27-51-00-860-803).

SUBTASK 27-51-11-040-002

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-11-480-003

WARNING MAKE SURE THAT THE GROUND LOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE GROUND LOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(3) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-11-020-007

(4) Remove the bullnose seal [66] (Figure 404):

(a) Move the torque tube door up and inboard to get access to the bullnose seal [66].

(b) Remove the bolts [67] and washers [68] that attach the bullnose seal [66] to the inboard flap [1].

(c) Remove the bullnose seal [66] and shim [69] from the inboard flap [1].

(d) If you will install a different flap, remove and keep the shim [69] and bullnose seal [66] for the installation.

NOTE: Keep and install the same shim [69] and bullnose seal [66] you removed from the airplane. If you install a different bullnose seal or shim, it can cause unsatisfactory flight performance.

END OF TASK

TASK 27-51-11-400-802

5. Inboard Trailing Edge Flap Bullnose Seal Installation

(Figure 404)

A. References

Reference	Title
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
66	Bullnose seal	27-51-11-01-060	PTF ALL
		27-51-11-01-065	PTF ALL

C. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

D. Inboard Trailing Edge Flap Bullnose Seal Installation

SUBTASK 27-51-11-420-006

(1) Install the bullnose seal [66]:

(a) Move the torque tube door up and inboard to get access to the bullnose seal [66].

(b) Put the bullnose seal [66] and the shim [69] in their positions on the inboard flap [1].

(c) Install the bolts [67] and the washers [68] to attach the bullnose seal [66].

(d) Tighten the bolts [67].

(e) Release the torque tube door and let it close.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-11-860-006

(1) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-11-860-007

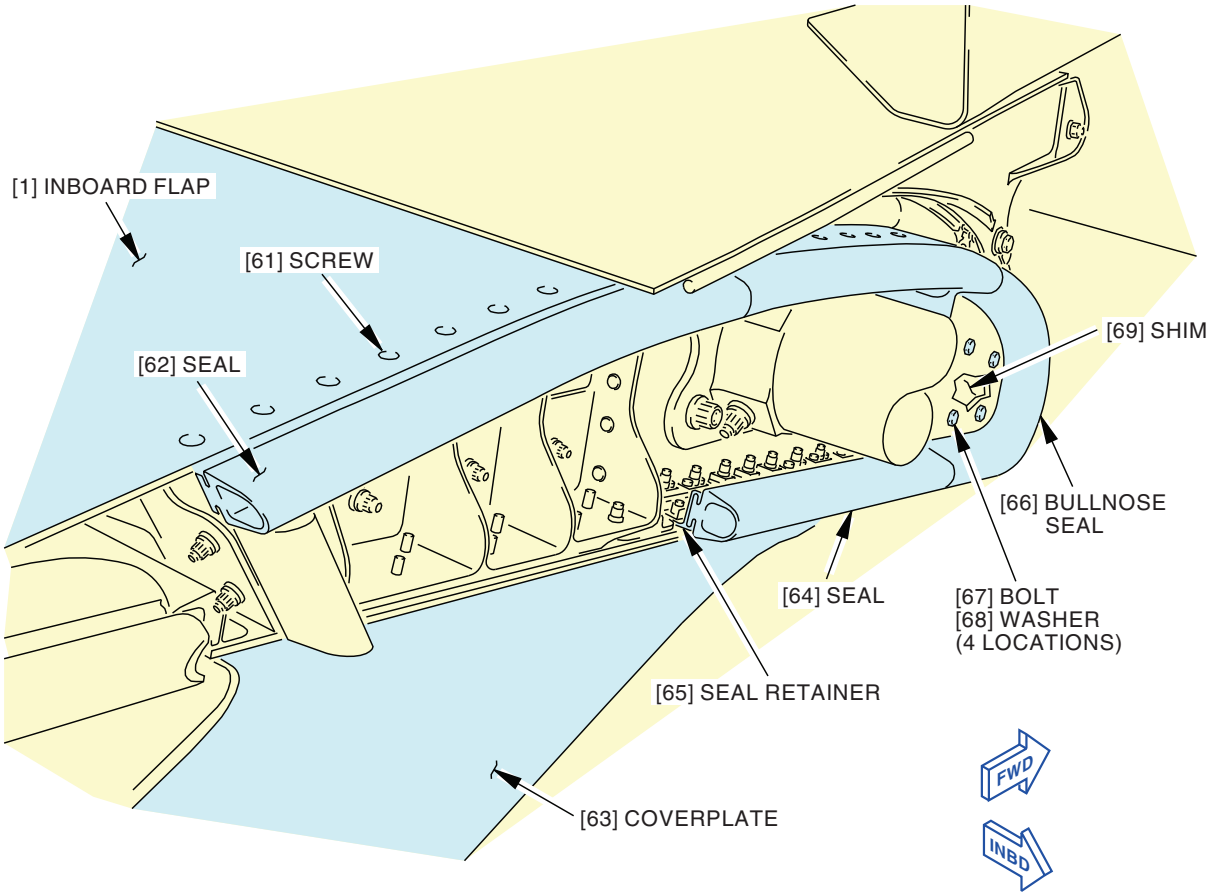
(2) Retract the flaps to the retracted position (TASK 27-51-00-860-804).

SUBTASK 27-51-11-860-008

(3) Do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



H

NOTE:
WING-TO-BODY FAIRING NOT SHOWN FOR CLARITY.

2412803 S00061531366_V2

Inboard Trailing Edge Flap Bullnose Seal Installation
Figure 404/27-51-11-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-11-000-803

6. Inboard Trailing Edge Flap Outer Seal Removal

(Figure 405)

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-61-00-040-801	Spoiler Control System - Deactivation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

C. Prepare for the Removal

SUBTASK 27-51-11-000-001

(1) Extend the trailing edge flaps to the 30-unit position (TASK 27-51-00-860-803).

SUBTASK 27-51-11-860-021

(2) If necessary, deploy and deactivate the spoilers (TASK 27-61-00-040-801).

SUBTASK 27-51-11-000-002

(3) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-11-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(4) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

D. Inboard Trailing Edge Flap Outer Seal Removal

SUBTASK 27-51-11-000-004

(1) Do these steps to remove the seal [2]:

(a) Remove the screws [3], self align washers [6], washers [4], and nuts [5] that attach the seal [2] to the flap assembly [1].

(b) Remove the seal [2] from the flap assembly [1].

(c) Remove the plug from the seal [2].

SUBTASK 27-51-11-000-005

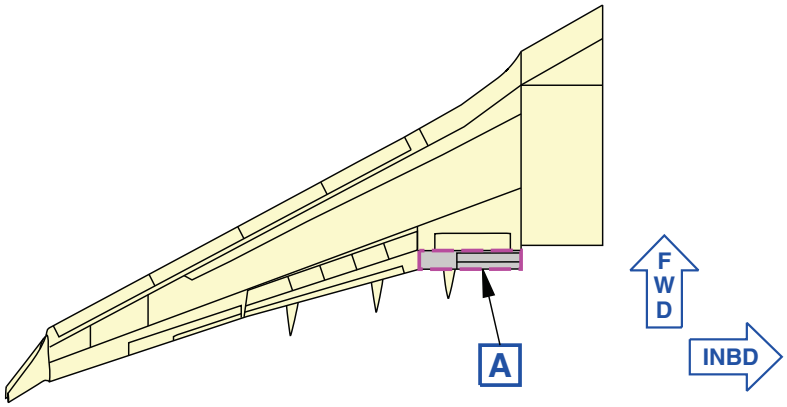
(2) Do these steps to remove the seal [141] and seal [142]:

(a) Remove the screw [3], self align washer [6], washer [4], and nut [5] that attach the seal [141] to the flap assembly [1].

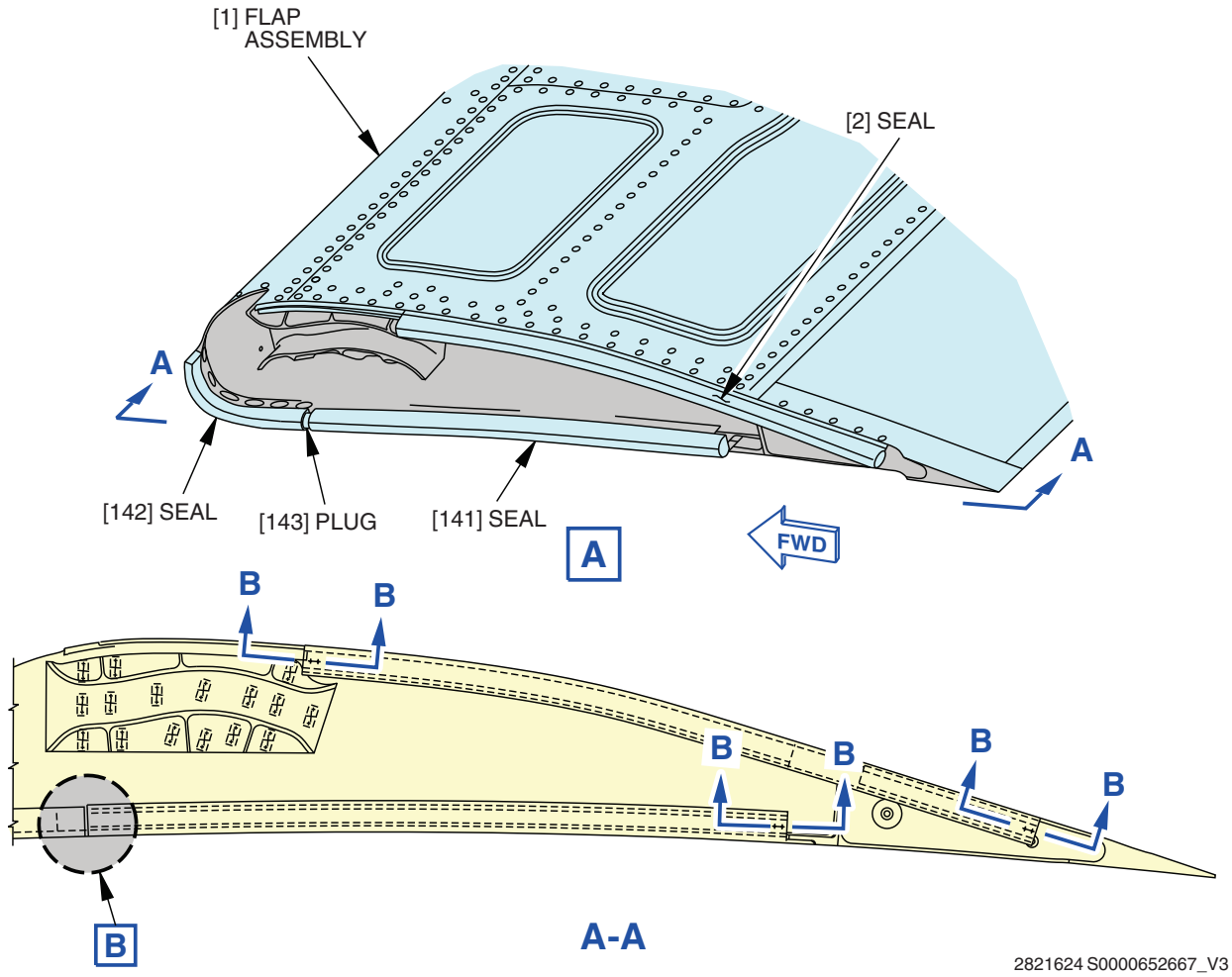
(b) Remove the seal [141], seal [142], and plug [143] from the flap assembly [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

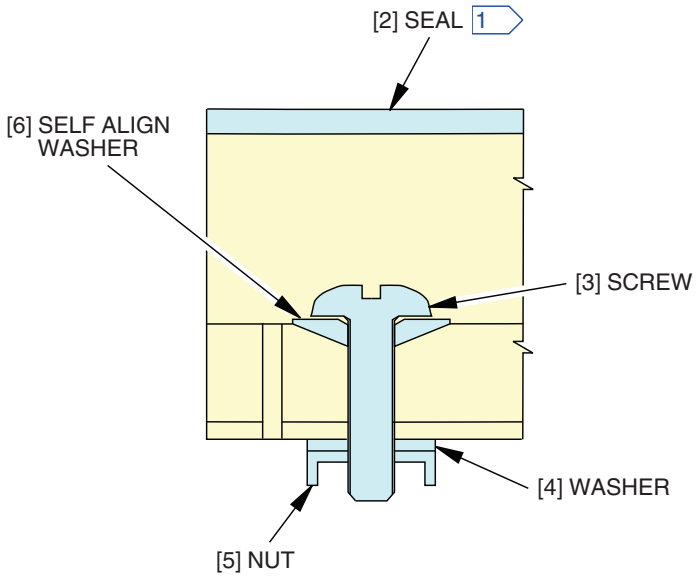


(LEFT WING IS SHOWN, RIGHT WING IS OPPOSITE)



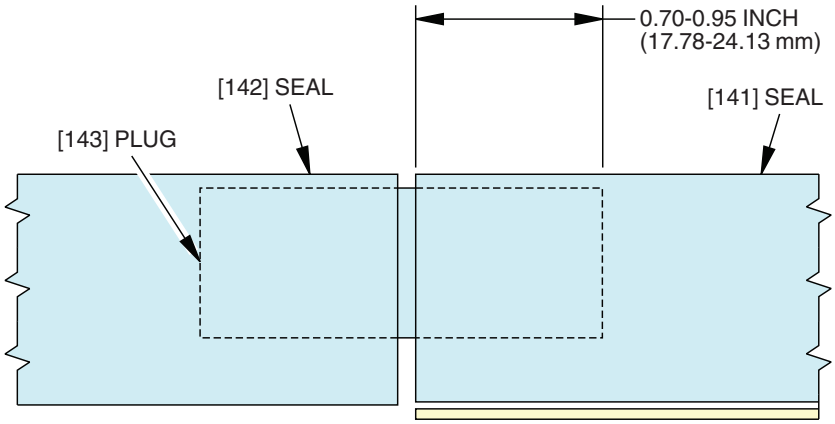
Inboard Trailing Edge Flap Outer Seal Installation
Figure 405/27-51-11-990-810 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

B-B



1 SEAL [2] IS SHOWN, SEAL [141] IS EQUIVALENT

2821625 S0000652668_V4

Inboard Trailing Edge Flap Outer Seal Installation
Figure 405/27-51-11-990-810 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-11-400-803

7. Inboard Trailing Edge Flap Outer Seal Installation

(Figure 405)

A. References

Reference	Title
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-00-440-801	Spoiler Control System - Activation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00027	Adhesive - Silicone Rubber, 1 Part, RTV	BAC5010 Type 60
G00110	Sponge - Silicone Rubber, Closed-Cell, General Purpose, Non-Solvent Use	BMS1-23

C. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

D. Inboard Trailing Edge Flap Outer Seal Installation

SUBTASK 27-51-11-000-007

(1) If necessary, remove the inboard trailing edge flap (TASK 27-51-11-000-801).

SUBTASK 27-51-11-400-001

(2) Do these steps to install the seal [2].

(a) Install the plug at the forward end of the seal [2].

1) Cut the plug 0.5 in. (12.7 mm) - 1 in. (25.4 mm) long from sponge, G00110.

2) Apply adhesive, A00027, to the mating surface of the plug.

a) Do not cover the ends of the plug with adhesive, A00027.

(b) Put the seal [2] on the flap assembly [1].

(c) Install the screws [3], self align washers [6], washers [4], and nuts [5] that attach the seal [2] to the flap assembly [1].

(d) Tighten the screws [3].

NOTE: Tighten the fastener until the seal [2] starts to visibly compress and clamps to the seal retainer.

SUBTASK 27-51-11-400-003

(3) Do these steps to install the seal [141] and seal [142].

(a) Put the seal [141] and seal [142] on the flap assembly [1].

(b) Apply adhesive, A00027, to the mating surfaces of the plug [143].

(c) Put the plug [143] between the seal [141] and seal [142].

1) Install the plug [143] 0.7 in. (17.78 mm) - 0.95 in. (24.13 mm) inside the seal [141] and seal [142].

(d) Install the screw [3], self align washer [6], washer [4], and nut [5] that attach the seal [141] to the flap assembly [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Tighten the screw [3].

NOTE: Tighten the fastener until the seal [141] starts to visibly compress and clamps to the seal retainer.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-11-400-005

(1) If it was removed, install the inboard trailing edge flap (TASK 27-51-11-400-801).

SUBTASK 27-51-11-440-005

(2) If they were deactivated, reactivate the spoilers (TASK 27-61-00-440-801).

SUBTASK 27-51-11-400-002

(3) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-11-860-019

(4) Retract the flaps to the retracted position (TASK 27-51-00-860-804).

——— END OF TASK ———

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INBOARD TRAILING EDGE FLAP - ADJUSTMENT/TEST

1. General

A. This procedure has an adjustment of the inboard trailing edge flap.

B. This procedure includes the adjustment of the main flap with the aft flap either removed or installed.

C. The adjustment instructions included in this procedure are included in the Rigging Diagram - Inbd and Outbd Flaps, Wing Trailing Edge 113A3017.

TASK 27-51-11-820-801

2. Inboard Trailing Edge Flap Adjustment

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-51-00-980-801	Trailing Edge Flap System Manual Operation (P/B 201)
27-51-18-000-802	Inboard Flap Support Aft Fairing Removal (P/B 401)
27-51-18-400-802	Inboard Flap Support Aft Fairing Installation (P/B 401)
27-51-21-820-801	Outboard Trailing Edge Flap Adjustment (P/B 501)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-755	Scale - Spring, Push/Pull Type, 0 - 150lbs (0 - 68kg) Capacity

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50375	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule)	BACC13AT3K, AMS 5689

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
541	Left Wing - Fairing Flap Support No. 4
542	Left Wing - Fairing Flap Support No. 3
553	Left Wing - Inboard Flap
641	Right Wing - Fairing Flap Support No. 5
642	Right Wing - Fairing Flap Support No. 6

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(Continued)

Zone	Area
653	Right Wing - Inboard Flap

E. Prepare For the Adjustment

SUBTASK 27-51-11-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 27-51-11-040-003

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-11-010-003

(3) Do this task: Inboard Flap Support Aft Fairing Removal, TASK 27-51-18-000-802 (if required).

SUBTASK 27-51-11-820-004

(4) Make sure the outboard flap [1] is adjusted correctly.

(a) To adjust the outboard flap [1], do this task: Outboard Trailing Edge Flap Adjustment, TASK 27-51-21-820-801.

F. Inboard Trailing Edge Flap Adjustment

NOTE: There is no mismatch, or alignment dimension to the outboard flap, or to fixed structure that the inboard flap is adjusted to meet. The eccentric carriage rollers can be changed from the nominal position to meet lateral trim tolerances.

NOTE: The eccentric bushing is not installed on every inboard flap.

SUBTASK 27-51-11-860-009

(1) Do a check of the adjustment of the deflection roller [12] on the outboard end of the inboard flap [2] (Figure 501):

(a) Manually extend and retract the inboard flap [2] through its full travel (TASK 27-51-00-980-801).

1) While you retract the inboard flap [2], monitor the roller [12] to see if there is preload when it engages the deflection control track [3].

(b) If there is preload when the roller [12] engages the deflection control track [3], then adjust the roller [12]:

1) Manually extend the inboard flap [2] to get access to the roller [12] (TASK 27-51-00-980-801).

2) Remove the bolt [6] and washer [5] that hold the retainer [4] and retainer [13] to the deflection control rib [14].

3) Remove the retainer [13] for the inboard eccentric bushing [10].

4) Remove the pin [7] and loosen the nut [8] on the bolt [11].

5) Turn the inboard eccentric bushing [10] to change the position of the roller [12].

a) Make sure that you do not change the position of the roller [9].

6) Install the retainer [13] for the inboard eccentric bushing [10].

7) Install the bolt [6] and washer [5] that hold the retainer [4] and retainer [13].

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8) Tighten the nut [8] finger tight and install the pin [7].

a) If it is necessary, tighten the nut [8] to the minimum amount necessary to install the pin [7].

SUBTASK 27-51-11-860-010

(2) Retract the trailing edge flaps to the retracted position (TASK 27-51-00-860-804).

SUBTASK 27-51-11-820-006

(3) Do a check of the adjustment of the aft flap drive rod [17]:

(a) Make sure the trailing edge flaps are in the retracted position.

(b) Install rig pin A/S-8 from the rigging pin kit, SPL-1585, in the rig pin hole [18] in the crankshaft [16].

SUBTASK 27-51-11-820-007

(4) If you cannot install the rig pin, adjust the length of the aft flap drive rod [17]:

(a) Loosen the nuts [15] on the aft flap drive rod [17].

(b) Turn the aft flap drive rod [17] to change the length until you can install the rig pin in the crankshaft [16].

(c) Tighten the nuts [15] on the aft flap drive rod [17].

(d) Install lockwire on the nuts [15].

SUBTASK 27-51-11-080-002

(5) Remove the rig pin A/S-8 rig pin from the crankshaft [16].

SUBTASK 27-51-11-210-004

(6) Do a check of the adjustment of the seals [55] on the inboard end of the inboard flap [2] (Figure 502):

NOTE: These seal compression requirements are not necessary on airplanes with undersize seals [55] installed on the inboard end of the inboard flap [2].

(a) With the trailing edge flaps retracted, make sure the seals [55] on the inboard end of the inboard flap [2] do not contact the wing-to-body fairing [56].

(b) Extend the trailing edge flaps to the 40-unit position:

1) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

2) Extend the trailing edge flaps to the 40-unit position (TASK 27-51-00-860-803).

3) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(c) With the trailing edge flaps extended, make sure the seals [55] on the inboard end of the inboard flap [2] do not contact the wing-to-body fairing [56].

SUBTASK 27-51-11-820-008

(7) Do this procedure to adjust the applicable seal [55] (Figure 502):

(a) Loosen the screws [67] that attach the seal [55] and the seal retainer [68] to the inboard inboard flap [2].

(b) Adjust the position of the seal [55] and the seal retainer [68] outboard, relative to the airplane.

1) Make sure you move the seal [55] and the seal retainer [68] together as a unit.

(c) Take gap measurements on both inboard flap inboard ends, at these locations:

1) Aft end of the upper forward seal retainer.

2) Forward end of the upper forward seal retainer.

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3) Forward end of the lower seal retainer.

(d) Use these measurements to find which side of the airplane has the smallest gaps at the measured locations from the wing-to-body fairing.

(e) Tighten the screws [67] for the retainer assemblies on the flap with the smallest measured gaps.

(f) If the measurement between a given left-right pair is not within 0.10 in. (2.54 mm), or the gaps between the seals and wing-to-body fairing are less than 0.3 in. (7.6 mm), adjust the seals until the symmetry and gap requirements are met, or until the maximum adjustment limit is reached.

(g) Tighten the screws [67].

(h) Apply sealant, A00247, to the inboard side of the seal retainer [68] and the threads of the screws [67].

SUBTASK 27-51-11-860-011

(8) Retract the trailing edge flaps:

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Retract the flaps to the retracted position (TASK 27-51-00-860-804).

(c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-11-820-009

(9) Do this procedure to adjust the coverplate [63] to the wing-to-body fairing panels (Figure 502):

(a) Make sure that the clearance between the coverplate [63] and the wing-to-body fairing is 0.1000 +0.1200 / -0.0000 in. (2.5400 +3.0480 / -0.0000 mm).

NOTE: This distance is referred to as dimension C.

(b) If the measurement between the coverplate [63] and the wing-to-body fairing is not within the tolerance, then adjust the coverplate [63]:

1) Loosen the screws that attach the coverplate [63] to the lower surface of the inboard flap.

2) Apply sealant, A00247, to both sides of the coverplate [63] in the area of the attach screws.

3) Adjust the position of the coverplate [63].

4) Apply the sealant, A00247, in the slots before you install the screws.

5) Tighten the screws.

6) Fill and fair the exposed slots if and as required with the sealant, A00247.

NOTE: The gap between the coverplate [63] and the wing-to-body fairing in the flaps retracted position should not be less than 0.1000 in. (2.54 mm), but can be increased up to 0.2200 in. (5.59 mm) if required for the coverplate [63] to clear the wing-to-body fairing panels by at least 0.0200 in. (0.508 mm) in the flaps extended position.

SUBTASK 27-51-11-210-005

(10) On airplanes with the seal [62] on the wing to body fairing, do these steps:

(a) Make sure the seals [62] on the wing-to-body fairing [56] touch the coverplate [63] on the inboard end of the inboard flap [2].

1) If the seal [62] does not touch the coverplate [63], then adjust the applicable seal [62]:

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a) Loosen the screws [61] that attach the seal [62] and the seal retainer [60] to the wing-to-body fairing [56].

b) Adjust the position of the seal [62] and the seal retainer [60].

<1> Make sure that you move the seal [62] and the seal retainer [60] together as a unit.

c) Tighten the screws [61].

d) Apply sealant, A00247, to smooth the area around the retainer.

SUBTASK 27-51-11-820-010

(11) Do a check of the clearance at the aft flap seal [69]:

(a) At the aft end of the seal retainer [70], measure the distance between the base of the seal retainer [70] and the wing-to-body fairing [56].

NOTE: This distance is referred to as dimension A.

(b) Make sure the aft flap seal [69] is the correct part for the actual dimension A (Table 501).

Table 501/27-51-11-993-801

Dimension A Inches (mm)	Aft Flap Seal Part Number
Less Than 1.05 in. (26.7 mm)	113A2700-29
1.05 in. (26.7 mm) to 1.45 in. (36.8 mm)	113A3172-56
More Than 1.45 in. (36.8 mm)	113A3172-57

(c) If the aft flap seal [69] is not the correct part number, then replace the aft flap seal with the correct part:

1) Remove the lockwire from the ends of the aft flap seal [69].

2) Remove the aft flap seal [69] from the seal retainer [70].

3) Put the new aft flap seal [69] in the seal retainer [70].

4) Install lockwire on the ends of the aft flap seal [69].

SUBTASK 27-51-11-860-012

(12) Extend the trailing edge flaps:

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Extend the trailing edge flaps to the 40-unit position (TASK 27-51-00-860-803).

(c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-11-210-006

(13) With the flaps fully extended, make sure the full surface of the spring assembly [66] on the inboard carriage [59] touches the door [58] on the wing-to-body fairing [56] and that it holds the door closed.

SUBTASK 27-51-11-820-011

(14) If the spring assembly [66] does not touch the door [58], or hold it closed, adjust the position of the spring assembly [66]:

(a) Loosen the bolts [64] and bolt [65] that hold the spring assembly in its position.

(b) Move the spring assembly [66] until it touches the door [58].

(c) Apply a 5 lbf (22.24 N) force with a spring scale, STD-755, against the end of the spring assembly [66] to push it up against the door [58].

(d) Make sure the spring assembly [66] is flat against the adjacent surface of the door [58].

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(e) Tighten the bolts [64] and bolt [65].

SUBTASK 27-51-11-860-013

(15) Retract the trailing edge flaps:

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Retract the flaps to the retracted position (TASK 27-51-00-860-804).

(c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-11-220-001

(16) Do a check of the adjustment of the seals between the inboard and outboard flaps (View A-A, Figure 502):

(a) Measure the distance between the lower skin of the outboard flap [1] and the inboard flap [2].

NOTE: This distance is referred to as Dimension B.

(b) Extend the trailing edge flaps:

1) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

2) Extend the trailing edge flaps to the 40-unit position (TASK 27-51-00-860-803).

3) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(c) Measure the distance the outboard flap seal [51] extends past the skin of the outboard flap [1].

1) The distance the outboard flap seal extends should be Dimension B minus 0.75 in. (19.05 mm).

(d) Measure the distance the inboard flap seal [52] extends past the skin of the inboard flap [2].

1) The distance the outboard flap seal extends should be Dimension B minus 0.70 in. (17.78 mm).

SUBTASK 27-51-11-820-012

(17) If the inboard flap seal [52] and the outboard flap seal [51] are not in the correct position, then adjust the seals (View A-A, Figure 502):

(a) Adjust the outboard flap seal [51]:

1) Loosen the screws [54] that attach the seal retainer for the outboard flap seal [51] to the outboard flap [1].

2) Adjust the position of the seal retainer for the outboard flap seal [51].

3) Tighten the screws [54].

4) Make sure the outboard flap seal [51] extends Dimension B minus 0.75 in. (19.05 mm) from the edge of the outboard flap skin.

(b) Adjust the inboard flap seal [52]:

1) Loosen the screws [53] that attach the seal retainer for the inboard flap seal [52] to the inboard flap [2].

2) Adjust the position of the seal retainer for the inboard flap seal [52].

3) Tighten the screws [53].

4) Make sure the inboard flap seal [52] extends Dimension B minus 0.70 in. (17.78 mm) from the edge of the inboard flap skin.

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SUBTASK 27-51-11-860-014

(18) Retract the trailing edge flaps:

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Retract the flaps to the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

(c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

G. Inboard Fixed Panel Adjustment

SUBTASK 27-51-11-860-015

(1) Make sure the flap is in the retracted position.

SUBTASK 27-51-11-220-002

(2) Measure the mismatch between the fixed lower panels [85] and the lower surface of the inboard flap [2] (Figure 503).

SUBTASK 27-51-11-820-013

(3) If the mismatch is not in the limit of 0.00 ±0.06 in. (0.00 ±1.52 mm), adjust the applicable fixed lower panel [85] (Figure 504):

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Extend the trailing edge flaps to get access to the rods for the fixed lower panels [85] (TASK 27-51-00-860-803).

(c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(d) Adjust the position of the outboard fixed lower panel [93]:

1) Loosen the nuts [91] on the rod [92], rod [94], rod [95] and rod [98].

2) Adjust the length of the rod [92], rod [94], rod [95] and rod [98].

3) Tighten the nuts [91] on the rod [92], rod [94], rod [95] and rod [98].

(e) Adjust the position of the fixed lower panel [107]:

1) Remove the nuts [105], washers [104], washers [103] and bolts [102] that attach the rod [106] to the fixed lower panel [107].

2) Adjust the length of the rods [106].

3) Install the bolts [102], washers [103], washers [104] and nuts [105] to attach the rods [106] to the fixed lower panel [107].

4) Install lockwire on the nuts [91].

(f) Adjust the position of the beam assembly [100]:

NOTE: The beam assembly [100] is between panel [107] and panel [101].

1) Loosen the nuts [91] on the rod [99].

2) Adjust the length of the rod [99].

3) Tighten the nuts [91] on the rod [99].

4) Install lockwire on the nuts [91].

(g) Adjust the position of the inboard fixed lower panel [101]:

1) Adjust the rods [111] on the inboard end of the fixed lower panel [101]:

a) Remove the nut [115], washer [114], washer [113] and bolt [112] that attach the rods [111] to the fixed lower panel [101].

b) Adjust the length of the rods [111].

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c) Install the bolt [112], washer [113], washer [114] and nut [115] to attach the rods [111] to the fixed lower panel [101].

d) Install lockwire on the nut [91].

2) Adjust the rod [108], rod [109] and rod [110] for the fixed lower panel [101]:

a) Loosen the nuts [91] on the rod [108], rod [109] and rod [110].

b) Adjust the length of the rod [108], rod [109] and rod [110].

c) Tighten the nuts [91] on the rod [108], rod [109] and rod [110].

d) Install lockwire on the nuts [91].

(h) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(i) Retract the trailing edge flaps to the retracted position (TASK 27-51-00-860-804).

(j) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(k) Make sure the mismatch between the fixed lower panels and the lower surface of the inboard flap [2] is in the limits.

SUBTASK 27-51-11-220-003

(4) Do a check of the adjustment of the blade seals [84] on the fixed lower panels [85] (Figure 503):

(a) Measure the distance between the lower surface of the inboard flap [2] and the blade seal [84].

SUBTASK 27-51-11-820-014

(5) If the distance between the inboard flap [2] and the blade seal [84] is not in the limit of 0.00 ±0.03 in. (0.00 ±0.76 mm), then adjust the position of the blade seal [84]:

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Extend the trailing edge flaps to get access to the fasteners for the seal retainers [83] (TASK 27-51-00-860-803).

(c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(d) Loosen the screws [82] that attach the seal retainers [83] to the fixed lower panel [85].

(e) Move the seal retainers [83] to the correct position.

1) Make sure that you move the blade seal [84] and the seal retainer [83] together as a unit.

(f) Torque the screws [82] that attach the seal retainers [83] to15 in-lb (1.69 N·m).

(g) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(h) Retract the trailing edge flaps to the retracted position (TASK 27-51-00-860-804).

(i) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(j) Make sure the distance between the inboard flap [2] and the blade seal [84] is in the limit.

SUBTASK 27-51-11-820-015

(6) Do a check of the adjustment of the bulb seals [81] on the fixed lower panels [85] (Figure 504):

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Extend the trailing edge flaps to get access bulb seals [81] on the fixed lower panels [85] (TASK 27-51-00-860-803).

(c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(d) Measure the distance from the aft end of the blade seal [84] to the aft end of the bulb seal [81].

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(e) If the distance between the blade seal [84] and the bulb seal [81] is not in the limit of 1.428 ±0.060 in. (36.271 ±1.524 mm), adjust the position of the bulb seal [81]:

1) Loosen the screws [82] that attach the bulb seal retainer [83].

2) Adjust the position of the bulb seal [81] and the seal retainer [83].

a) Make sure that you move the bulb seal [81] and the seal retainer [83] together as a unit.

3) Torque the screws [82] to 15 in-lb (1.69 N·m).

SUBTASK 27-51-11-410-003

(7) Measure the gap between the fixed upper panel, outboard of spoiler #6 on the left wing or spoiler #7 on the right wing, and the upper surface of the inboard flap [2].

(8) If the gap between the panel trailing edge and the flap upper surface is not within 0.005 in. (0.127 mm) to 0.100 in. (2.540 mm), adjust the upper panel (Figure 503):

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

1) Extend the inboard trailing edge flaps to get access to the turnbuckle rod [92], rod [94], rod [95], rod [98] and rod [99] for the fixed upper panels (TASK 27-51-00-860-803).

2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(b) Loosen the two fasteners common to the serrated plates.

(c) Remove the existing safety device wires.

(d) Loosen the lower panel turnbuckle rods common to the upper panel support beam.

(e) Loosen the bolt [118], washer [119], washers [120], washer [121] and nut [122] at each end of the upper panel turnbuckle rod assembly.

1) Add washers [120] as required to achieve a maximum gap of 0.03 in. (0.76 mm) before clamp-up.

2) Make sure that there is a minimum of one washer per side.

(f) Adjust the turnbuckle rods and serrated plates to set the gap between the panel trailing edge and the flap upper surface to within 0.005 in. (0.127 mm) to 0.100 in. (2.540 mm) (Figure 503).

1) Make sure that the threads are visible through the turnbuckle rod inspection hole.

(g) Retighten all fasteners loosened or removed.

(h) Install the safety cable kit, G50375, to the turnbuckle rods as necessary.

SUBTASK 27-51-11-410-004

(9) Do the following steps to adjust the upper fixed trailing edge panel inboard of spoiler #6 on the left wing or spoiler #7 on the right wing:

(a) Loosen the fasteners at each end of the upper turnbuckle rod assembly.

(b) Loosen the fasteners at the panel attach lug.

(c) Remove the existing safety device wires.

(d) Loosen the jam nuts on either end of the turnbuckles.

(e) Adjust the turnbuckle rods to set the gap between the panel trailing edge and the flap upper surface to within 0.005 in. (0.127 mm) to 0.100 in. (2.540 mm) (Figure 503).

(f) Tighten the jam nuts on either end of the turnbuckle rods.

1) Install safety cable kit, G50375, on both ends of the turnbuckle rod.

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(g) Tighten the fasteners at the panel attach lugs and at each end of the upper panel turnbuckle rod assemblies.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-11-410-005

(1) Do this task: Inboard Flap Support Aft Fairing Installation, TASK 27-51-18-400-802.

SUBTASK 27-51-11-440-002

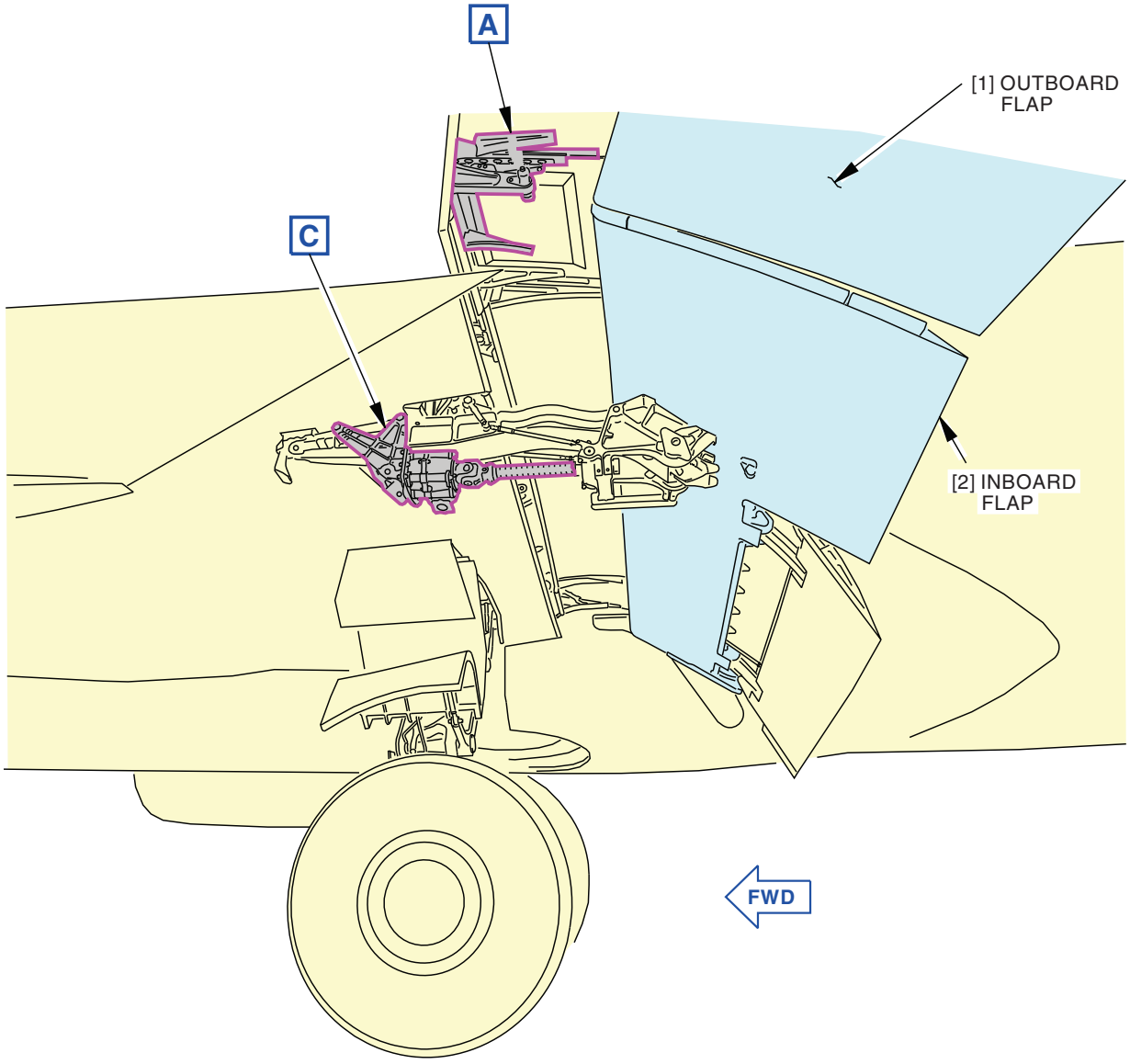
(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-11-860-016

(3) Retract the trailing edge flaps to the retracted position (TASK 27-51-00-860-804).

——— END OF TASK ———

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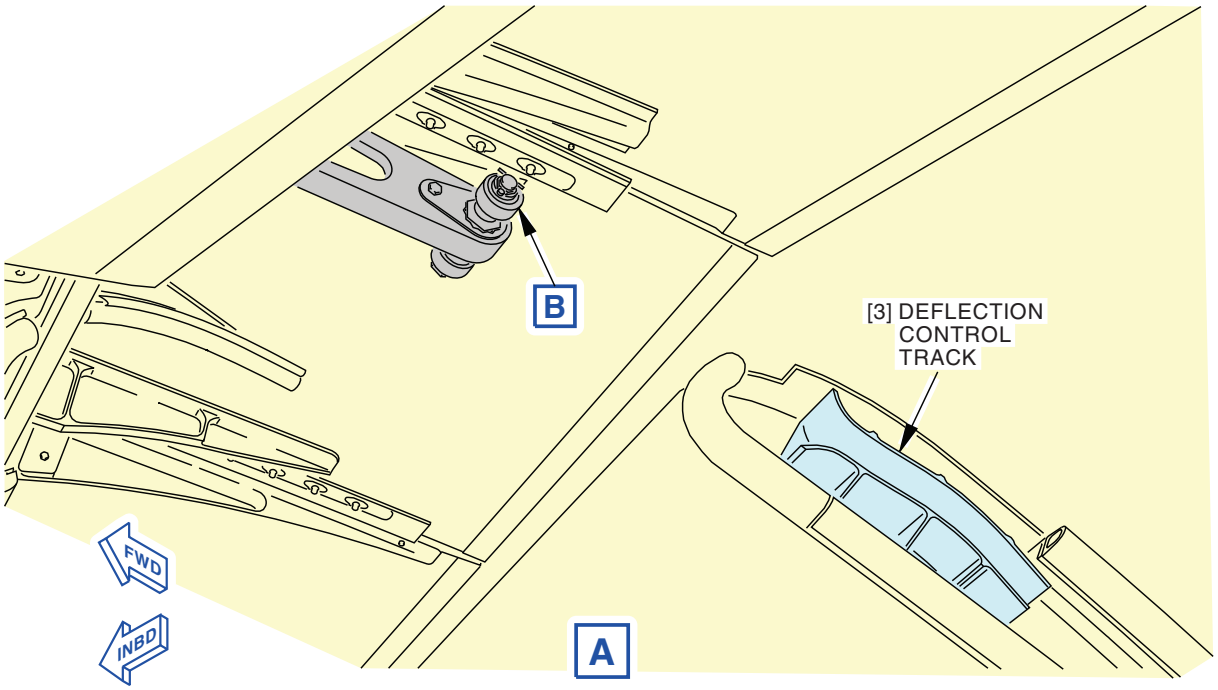


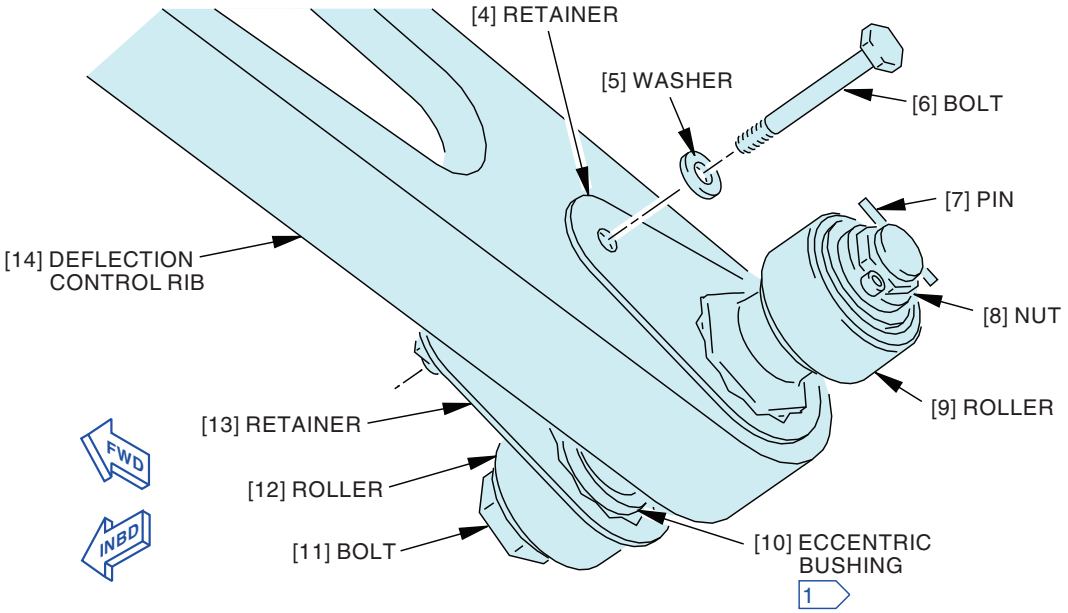
1 IF INSTALLED

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Inboard Trailing Edge Flap Adjustment
Figure 501/27-51-11-990-805 (Sheet 1 of 3)

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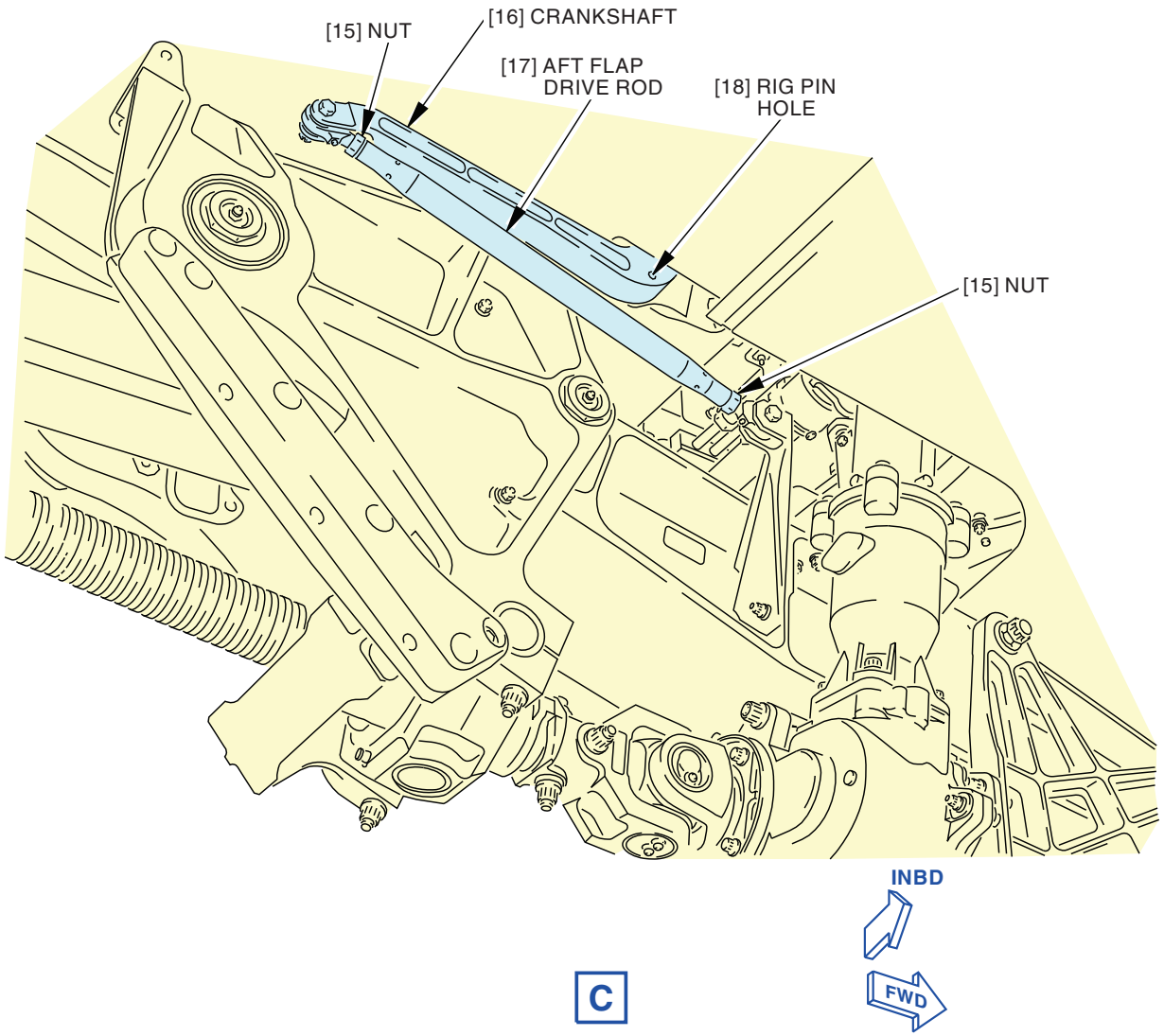




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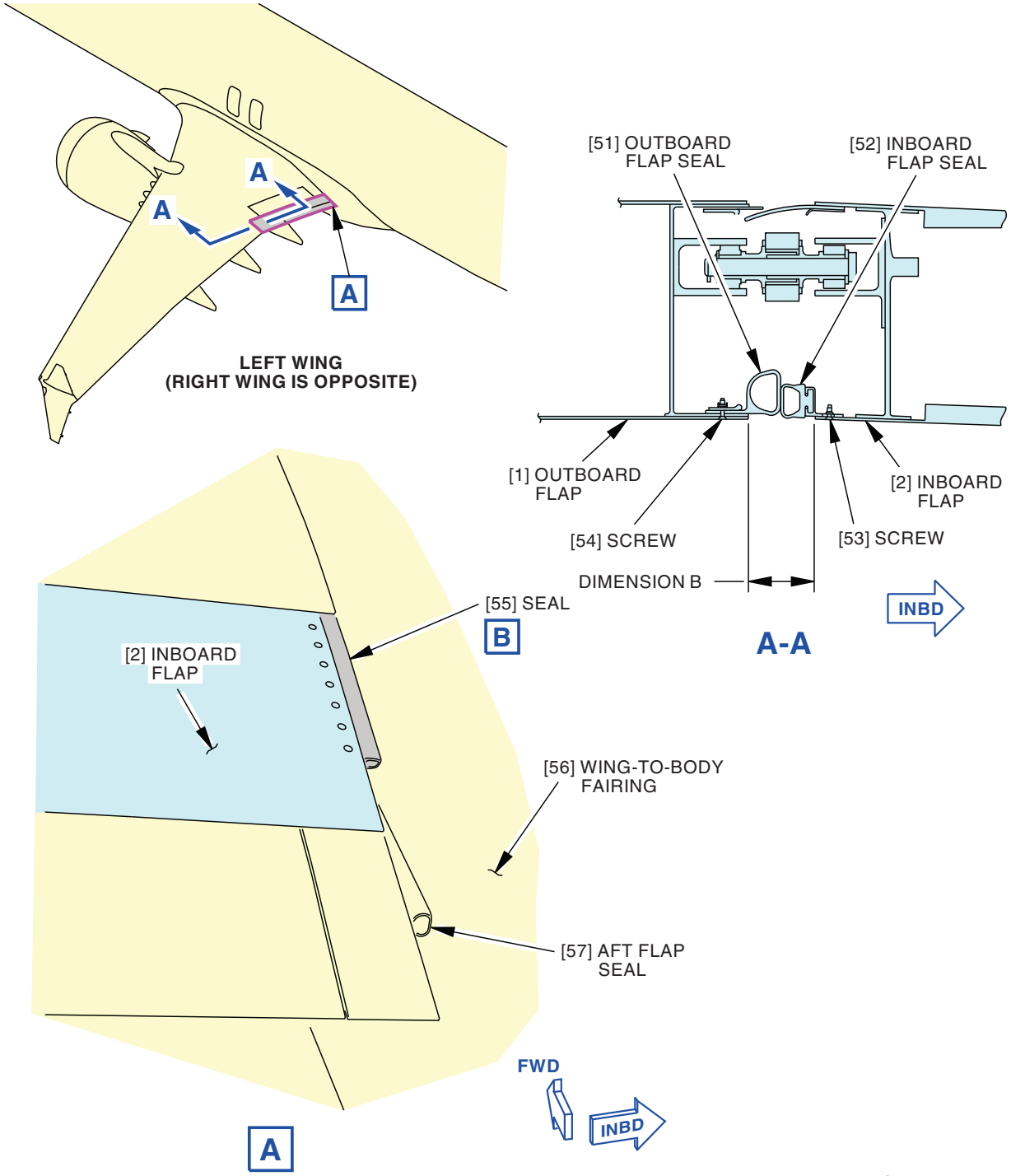
Inboard Trailing Edge Flap Adjustment
Figure 501/27-51-11-990-805 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



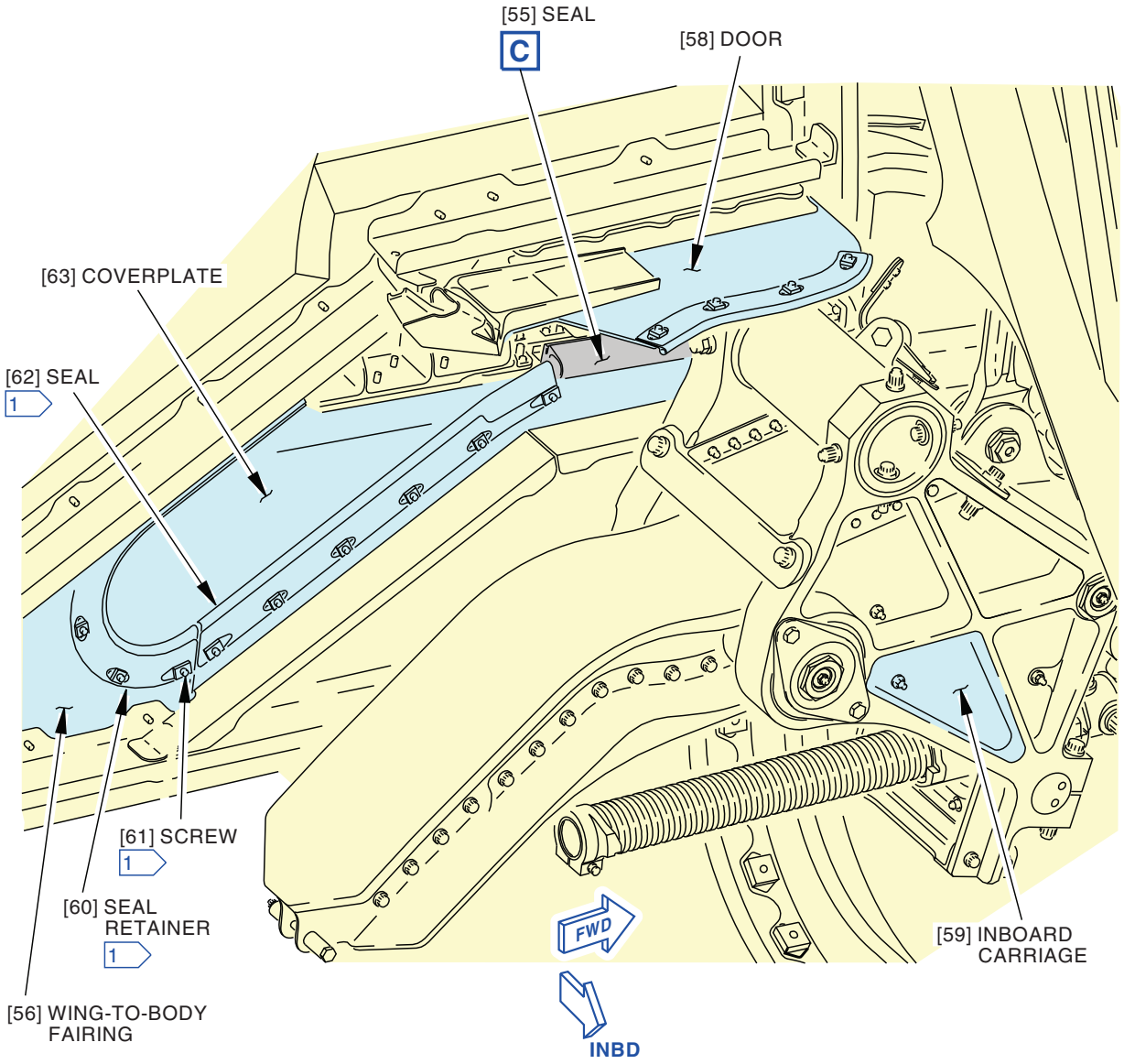
Inboard Trailing Edge Flap Adjustment
Figure 501/27-51-11-990-805 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



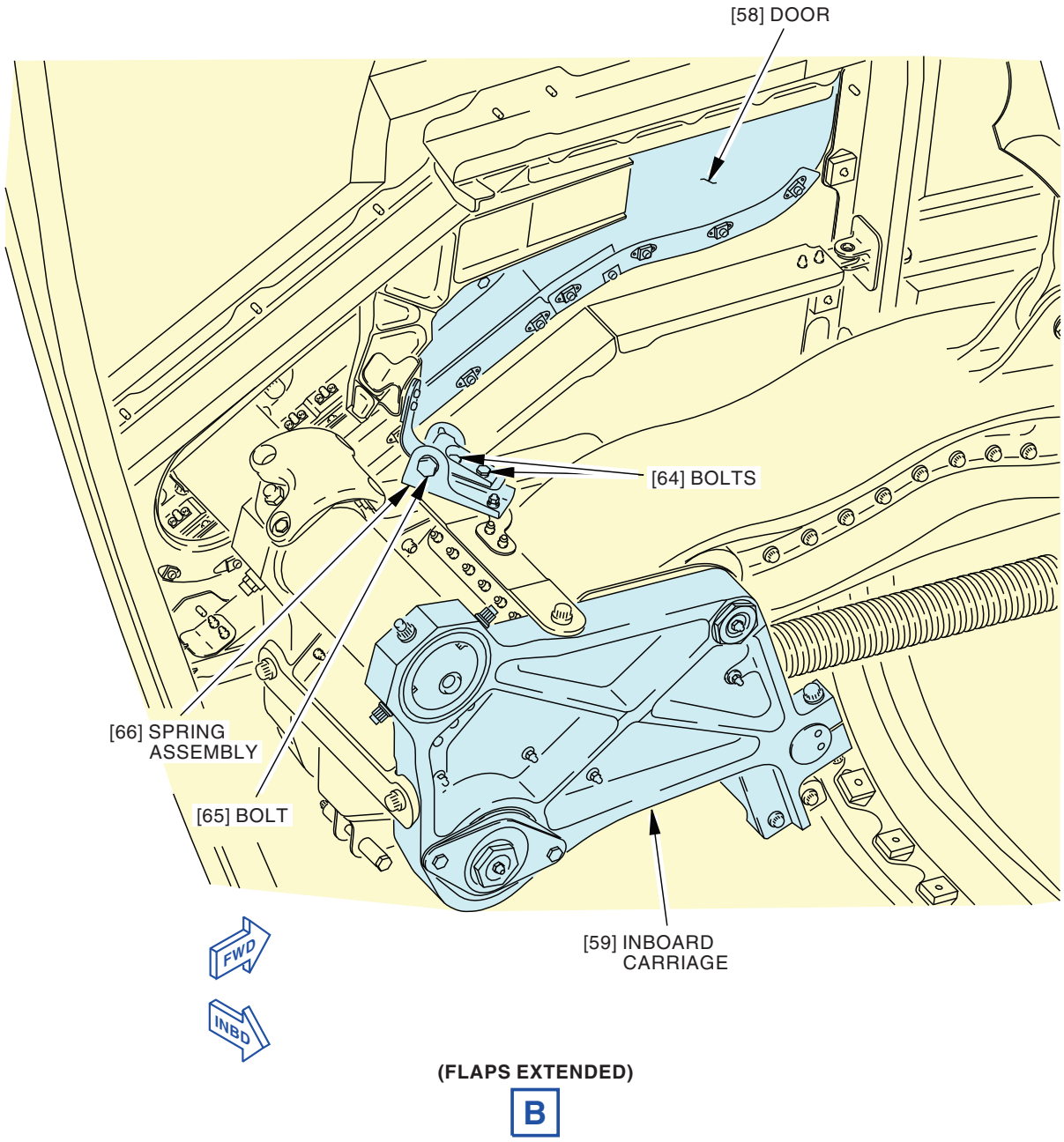
Inboard Flap Seal Adjustment
Figure 502/27-51-11-990-806 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



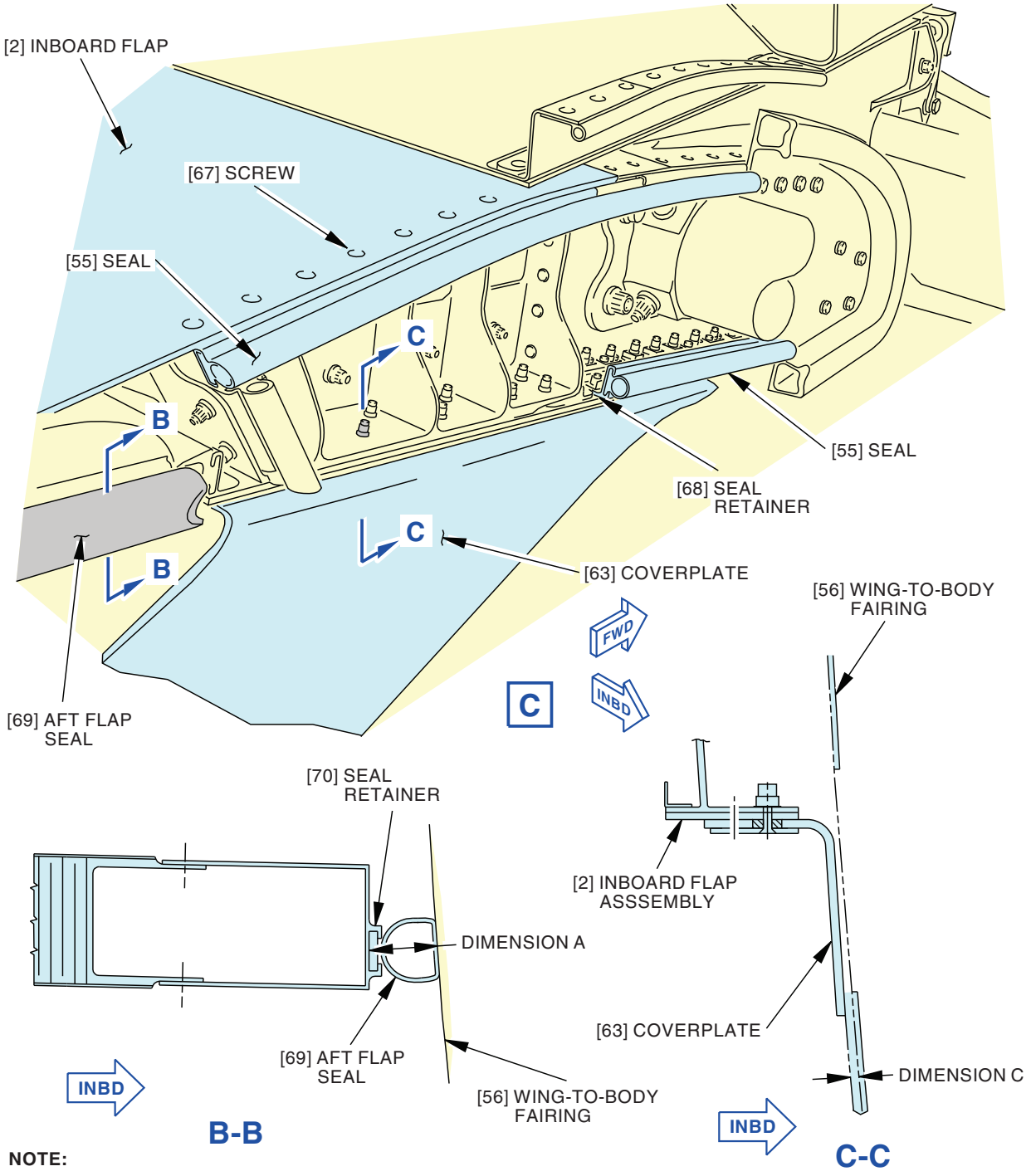
Inboard Flap Seal Adjustment
Figure 502/27-51-11-990-806 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Flap Seal Adjustment
Figure 502/27-51-11-990-806 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



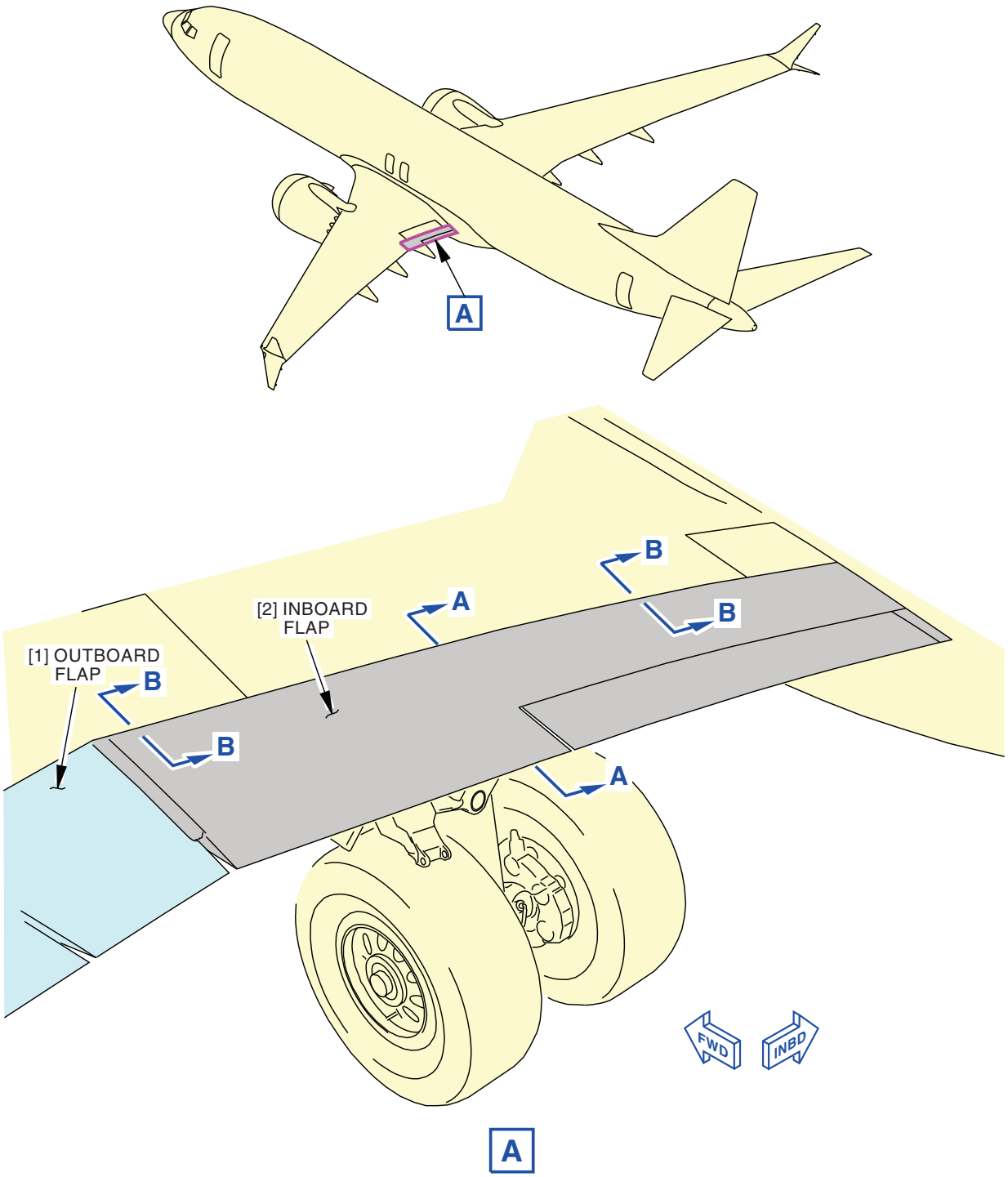
NOTE:

WING-TO-BODY FAIRING NOT SHOWN FOR CLARITY.

2412812 S00061531377_V1

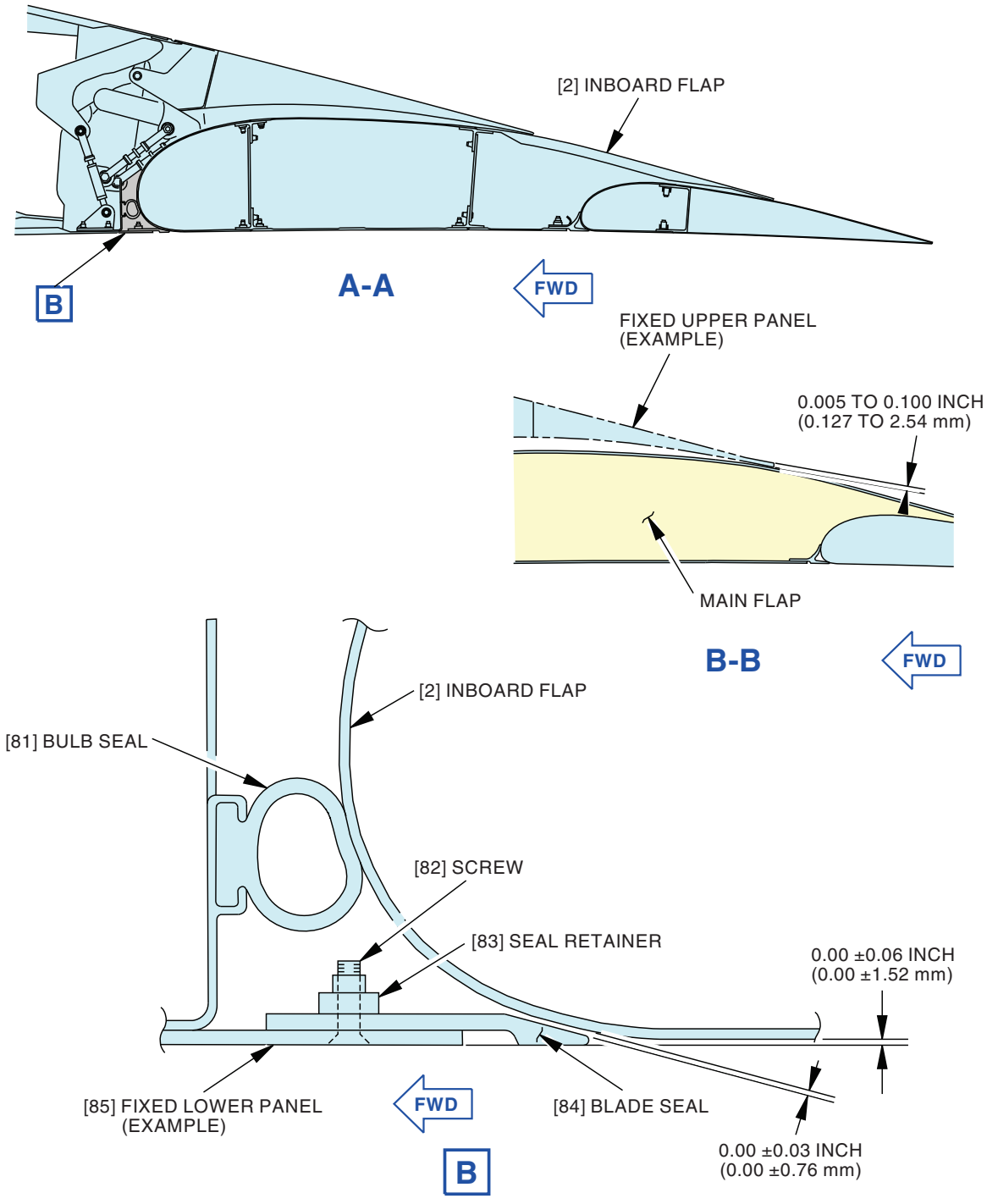
Inboard Flap Seal Adjustment
Figure 502/27-51-11-990-806 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



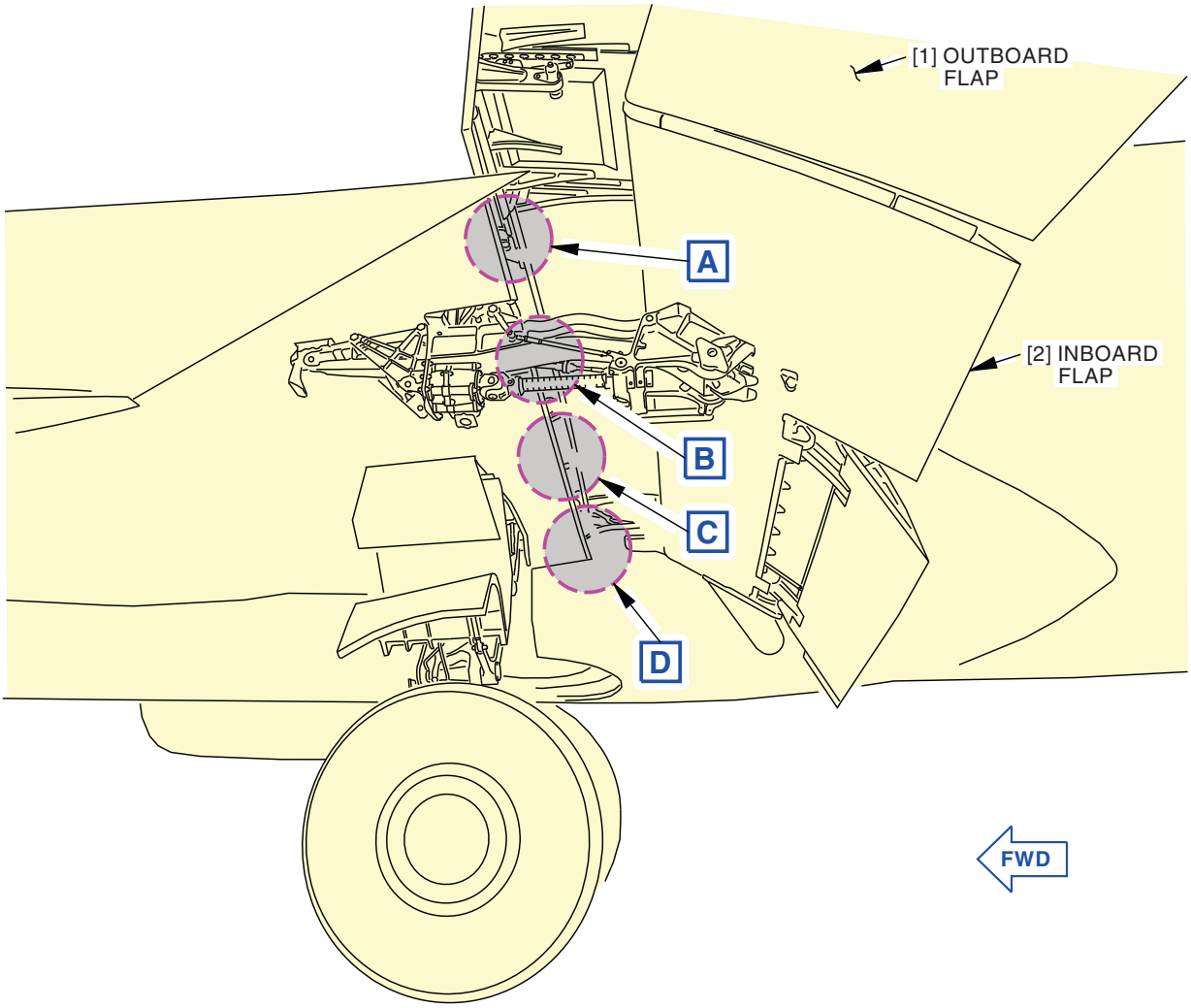
Inboard Trailing Edge Flap Clearances
Figure 503/27-51-11-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Trailing Edge Flap Clearances
Figure 503/27-51-11-990-807 (Sheet 2 of 2)

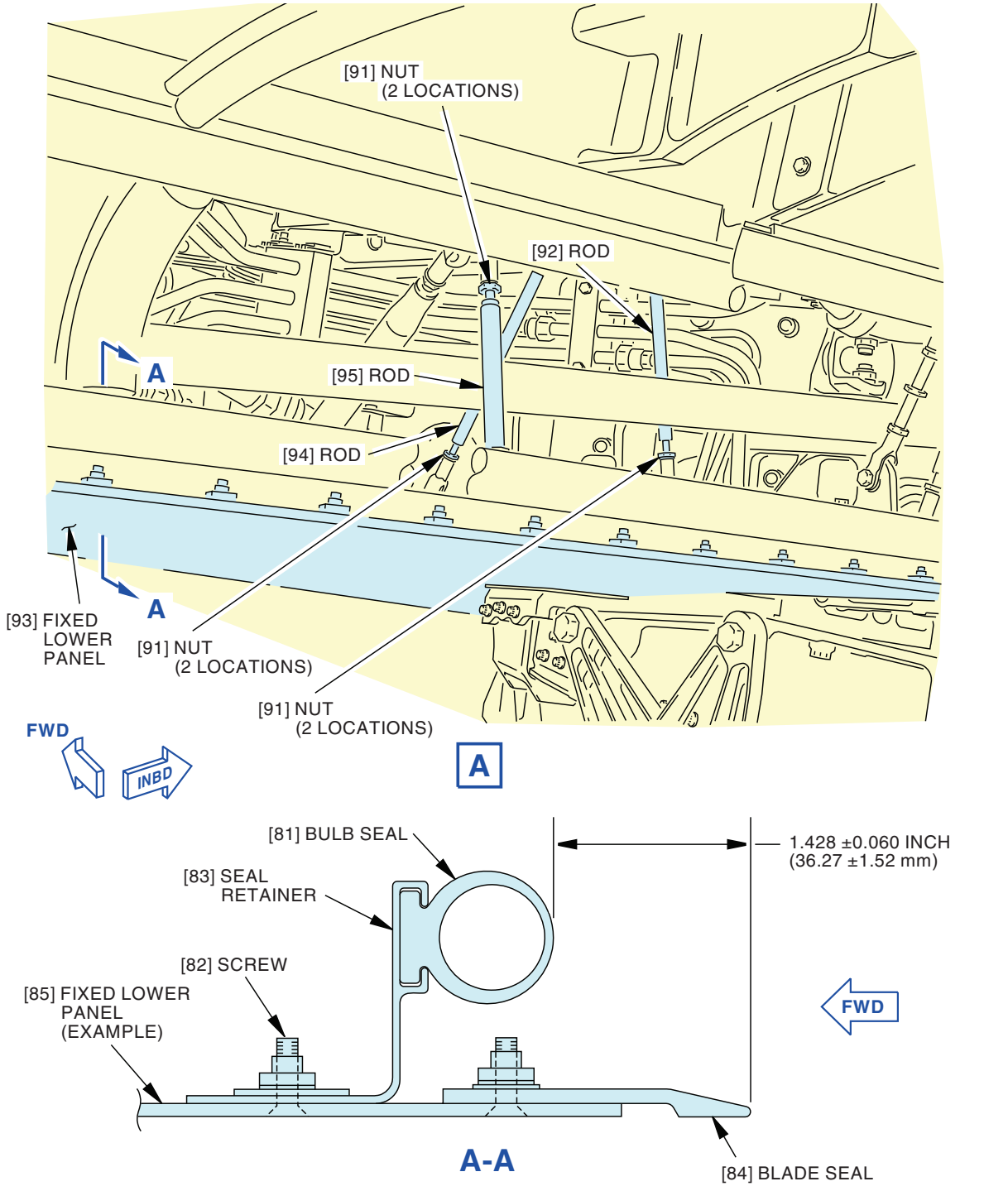
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412815 S00061531380_V1

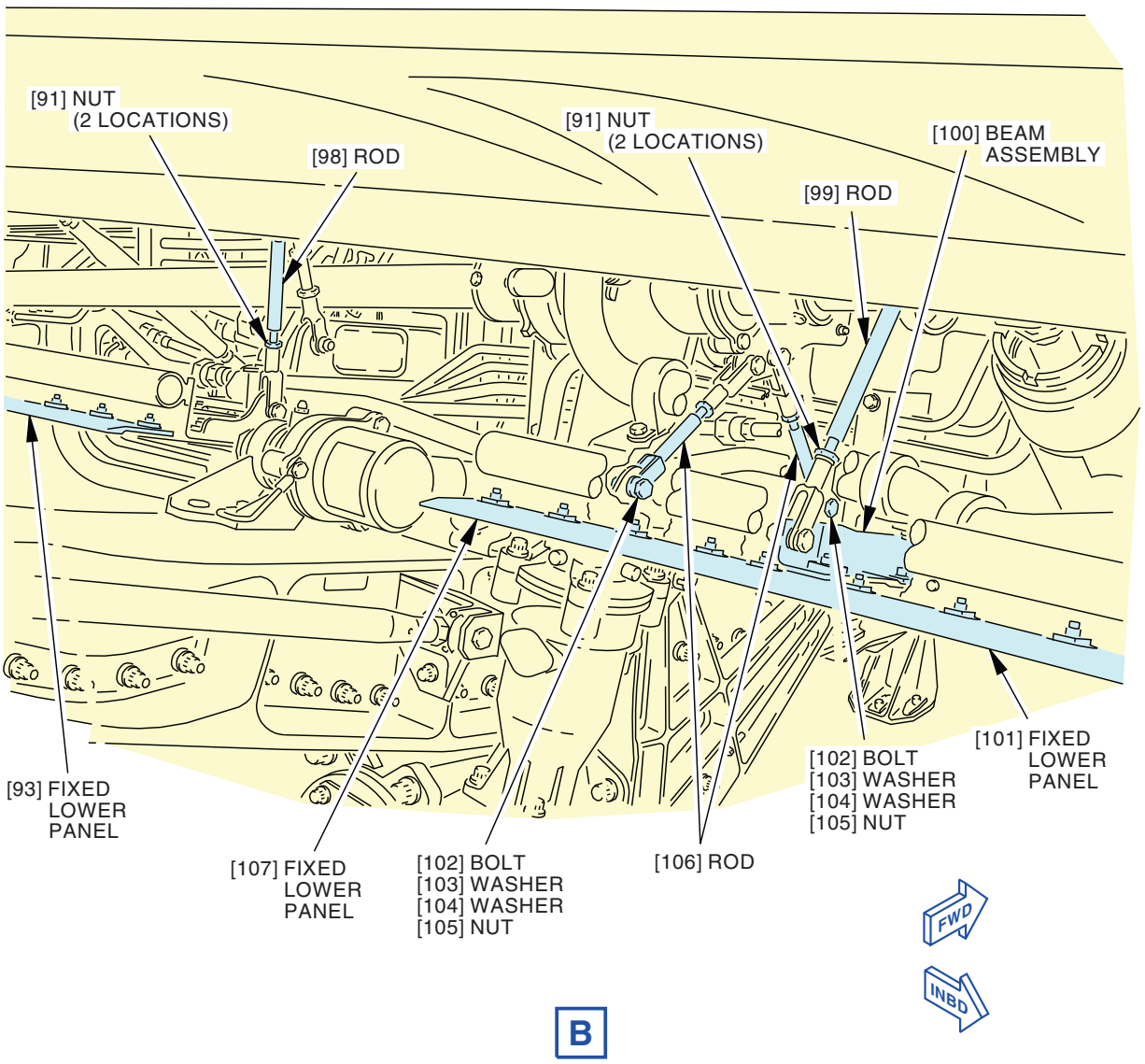
Inboard Fixed Panel Adjustment
Figure 504/27-51-11-990-808 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



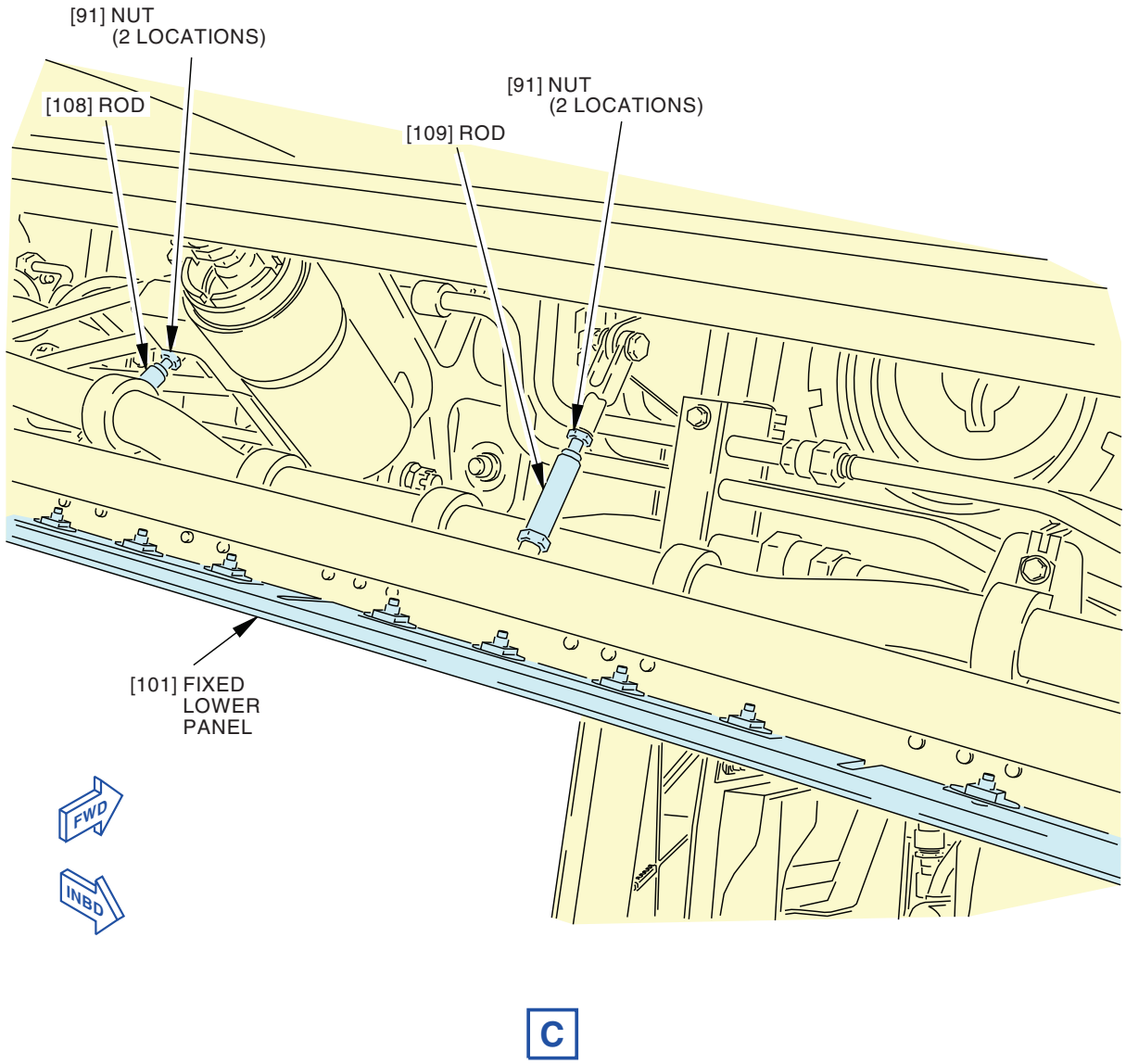
Inboard Fixed Panel Adjustment
Figure 504/27-51-11-990-808 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Fixed Panel Adjustment
Figure 504/27-51-11-990-808 (Sheet 3 of 5)

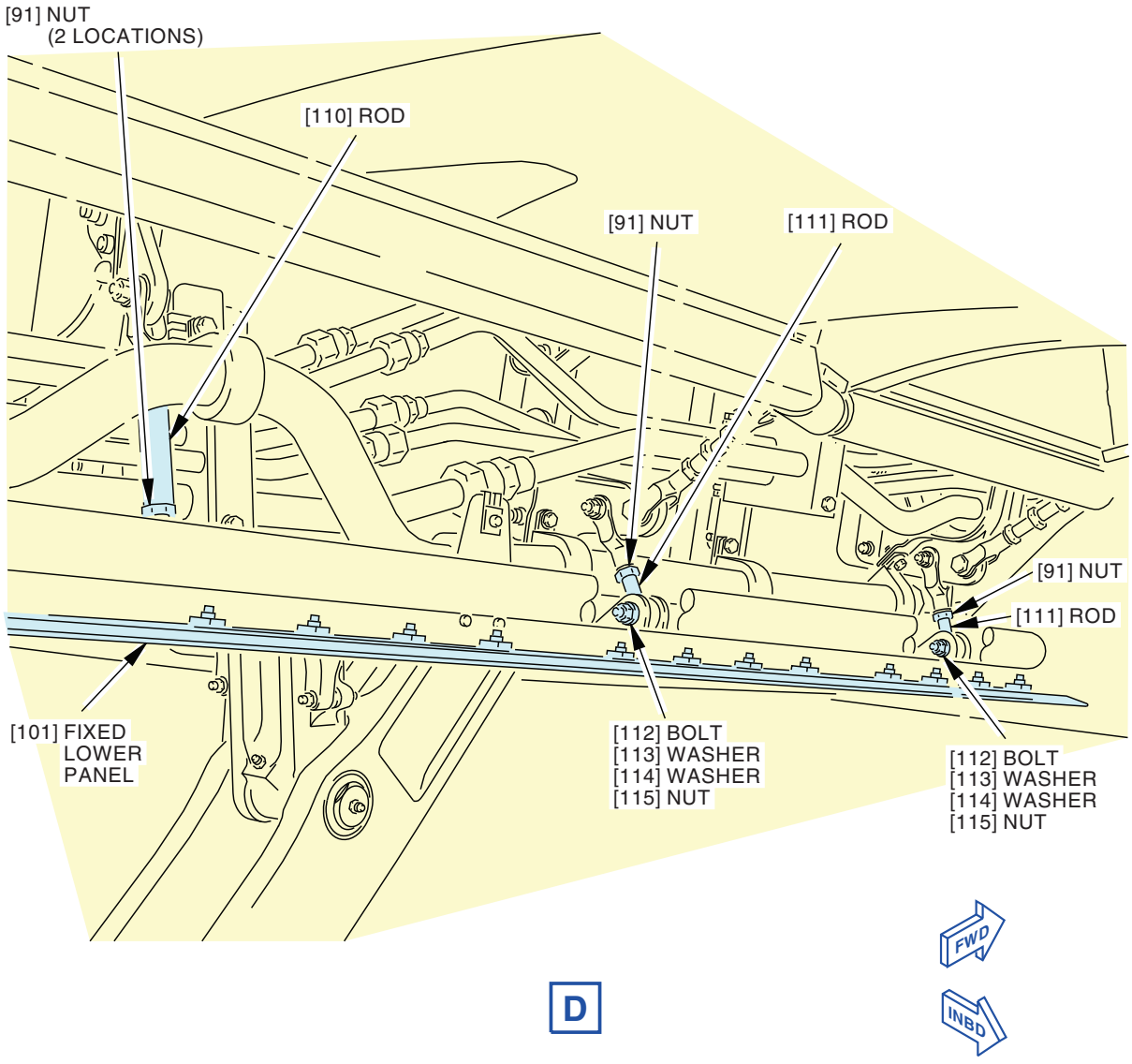
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412818 S00061531383_V1

Inboard Fixed Panel Adjustment
Figure 504/27-51-11-990-808 (Sheet 4 of 5)

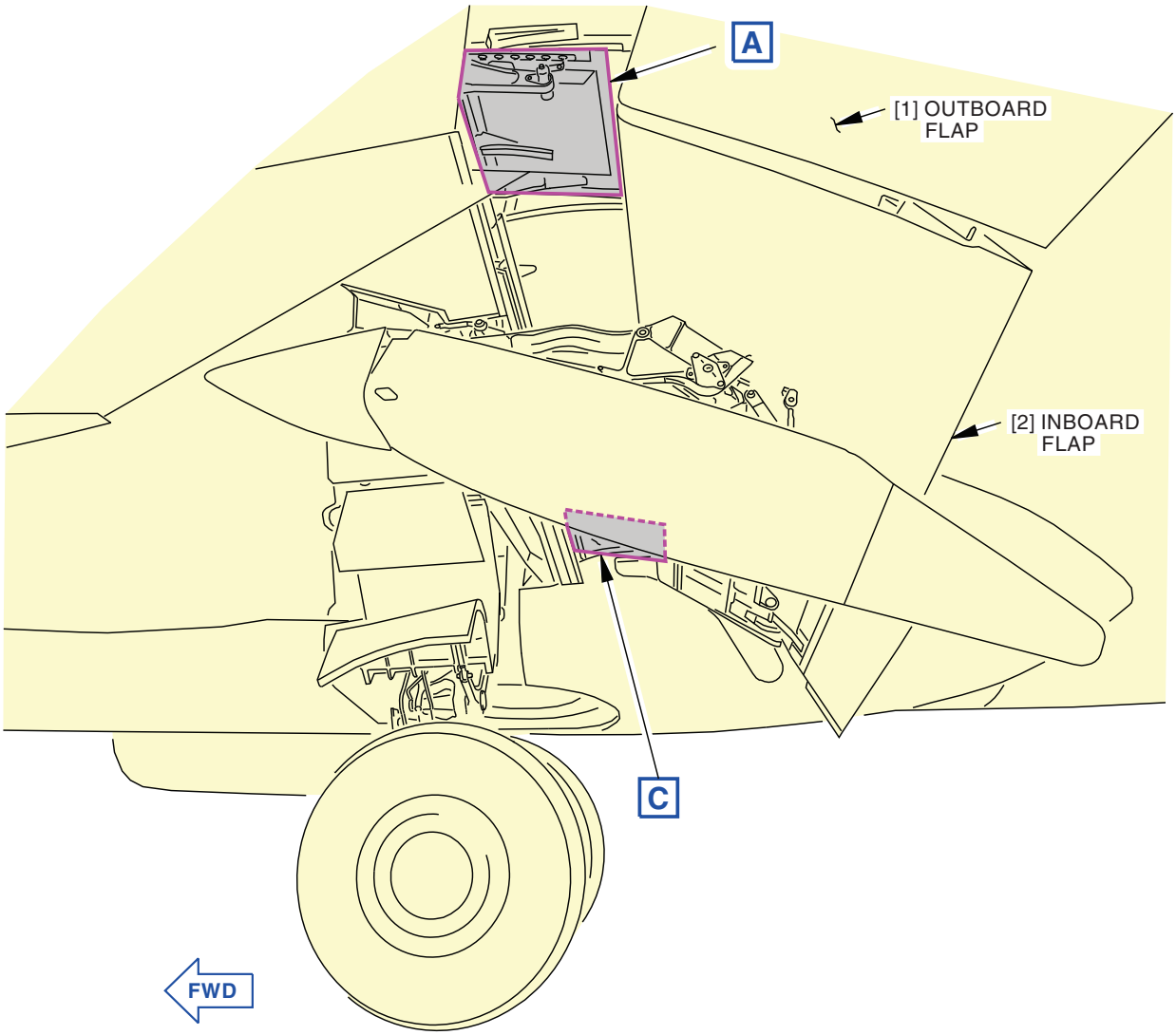
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412819 S00061531384_V1

Inboard Fixed Panel Adjustment
Figure 504/27-51-11-990-808 (Sheet 5 of 5)

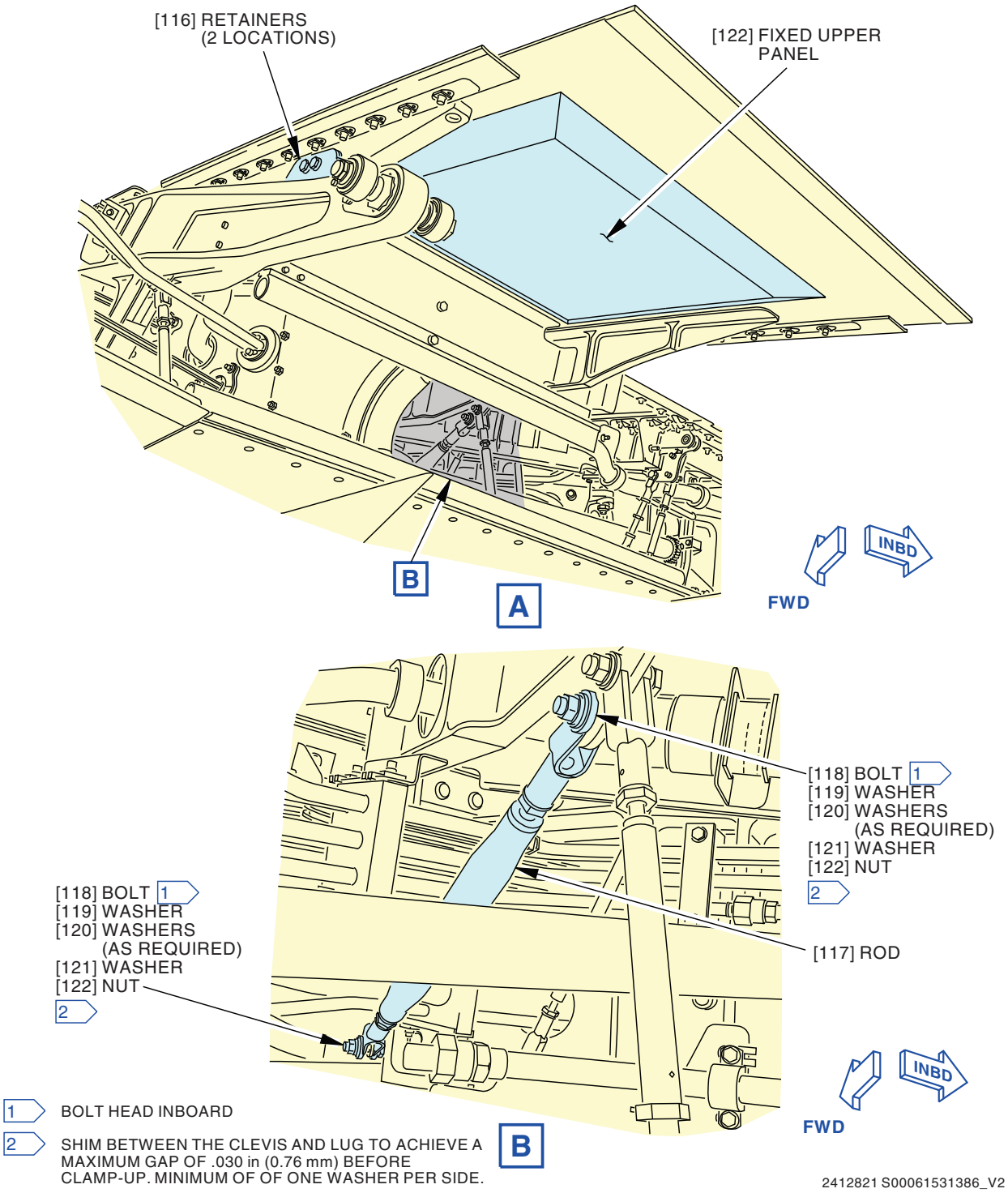
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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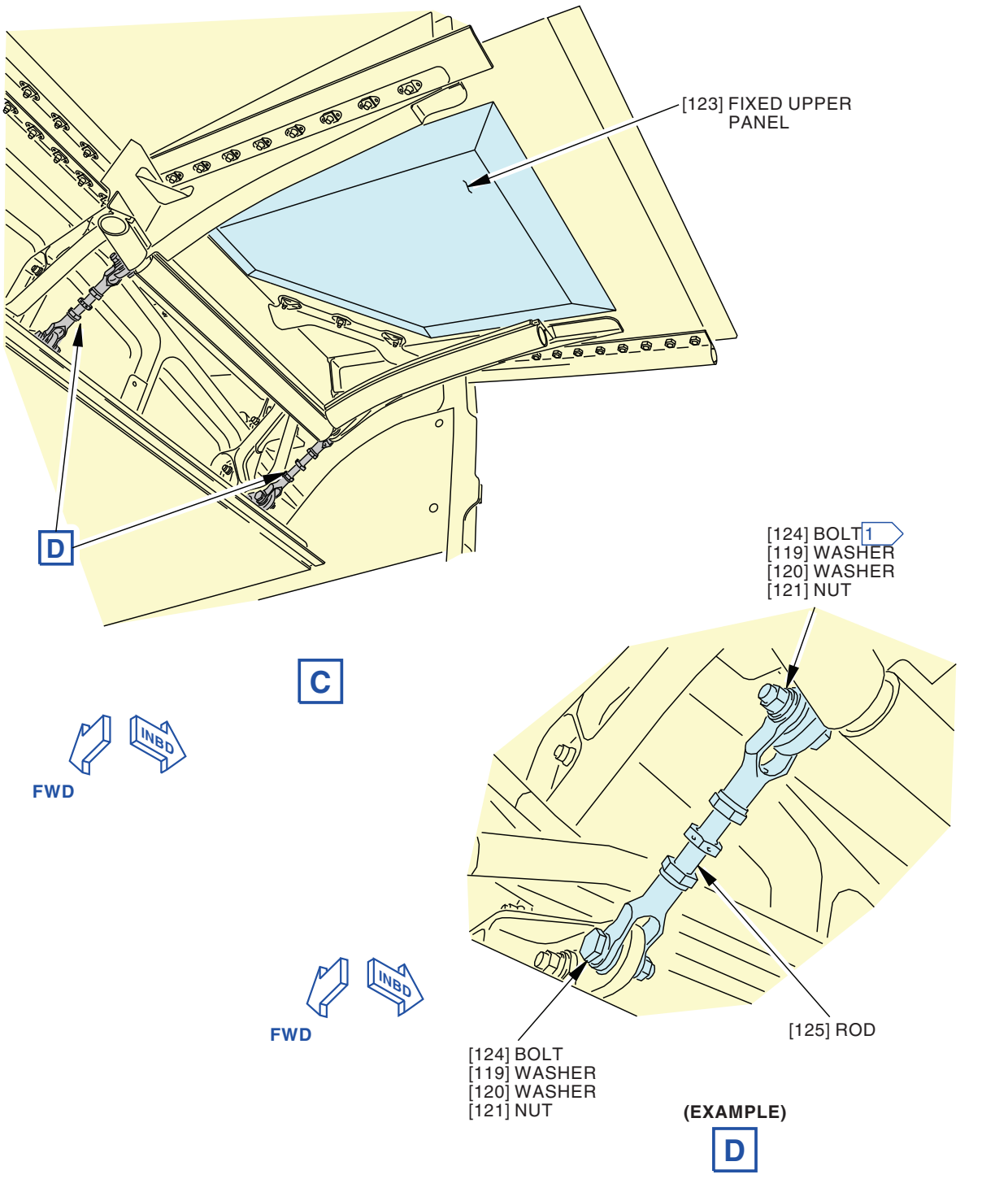
Inboard Fixed Upper Panel Adjustment
Figure 505/27-51-11-990-809 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Fixed Upper Panel Adjustment
Figure 505/27-51-11-990-809 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Fixed Upper Panel Adjustment
Figure 505/27-51-11-990-809 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD AFT FLAP - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the inboard aft flap

(2) Installation of the inboard aft flap.

B. The aft flap weighs approximately 60 lb (27 kg).

TASK 27-51-12-000-801

2. Inboard Aft Flap Removal

(Figure 401)

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)

B. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

C. Prepare for the Removal

SUBTASK 27-51-12-860-001

(1) Extend the trailing edge flaps to the 40-unit position (TASK 27-51-00-860-803).

SUBTASK 27-51-12-040-001

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

D. Inboard Aft Flap Removal

SUBTASK 27-51-12-020-001

(1) At the outboard flap support, disconnect the crankshaft [3] from the aft flap drive rod [10]:

(a) Remove the pin [9], nut [8], washer [7], bushing [6], washer [5] and bolt [4] that attach the crankshaft [3] to the aft flap drive rod [10].

SUBTASK 27-51-12-980-001

(2) Use the crankshaft [3] to manually move the aft flap assembly [2] to the extended position.

SUBTASK 27-51-12-020-002

(3) Disconnect the inboard and outboard aft flap pushrods [11]:

(a) Remove the pins [19], nuts [18], washers [17], washers [15], bolts [14] and bushings [13] that attach the aft flap pushrods [11] to the aft flap assembly [2].

(b) Move the aft flap pushrods [11] away from the aft flap assembly [2].

SUBTASK 27-51-12-020-003

(4) Remove the aft flap assembly [2]:

NOTE: The aft flap assembly [2] weighs approximately 60 lb (27 kg).

(a) Support the aft flap assembly [2].

(b) Disconnect the aft flap tracks [21] from the aft flap assembly [2]:

1) Remove the pins [22], nuts [23], washers [24], washers [26], bolts [27] and bushings [25] that attach the aft flap tracks [21] to the aft flap assembly [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Move the aft flap tracks [21] forward, away from the aft flap assembly [2].

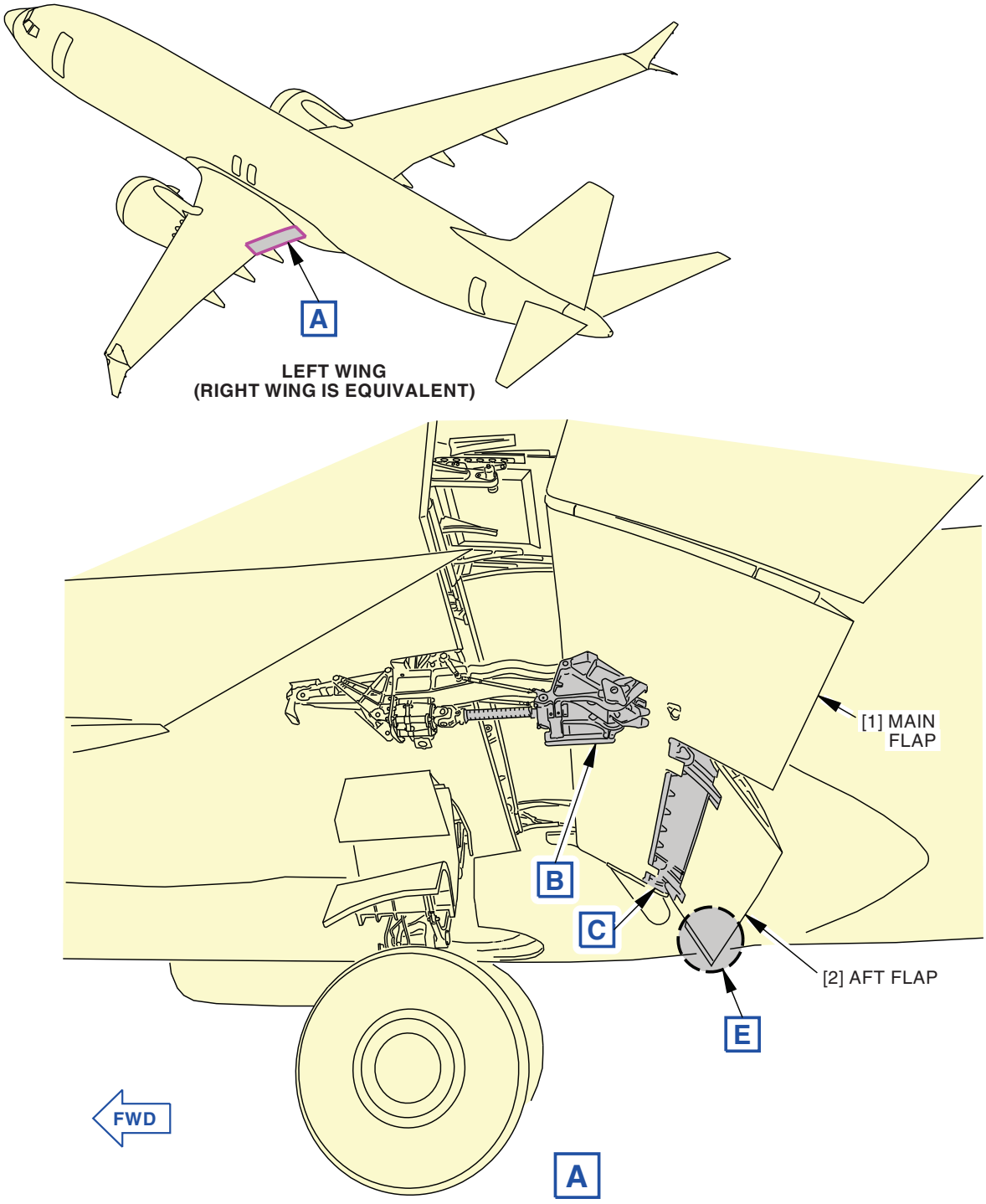
(c) Move the trailing edge of the aft flap assembly [2] down, until it is perpendicular to the main flap [1].

(d) Disengage the inboard and outboard rollers [28] from the programming tracks [38].

1) Disengage the inboard roller [28]first.

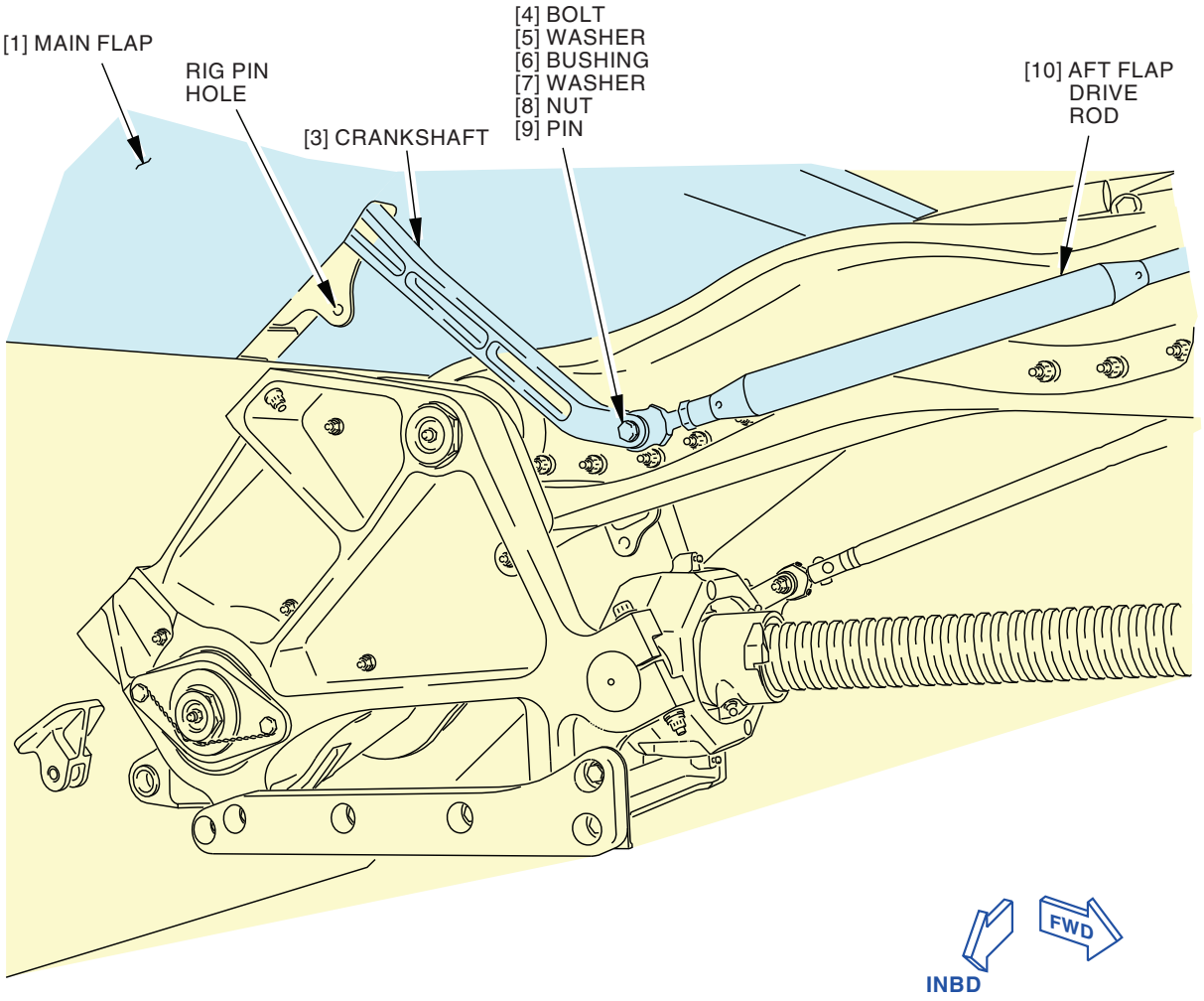
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



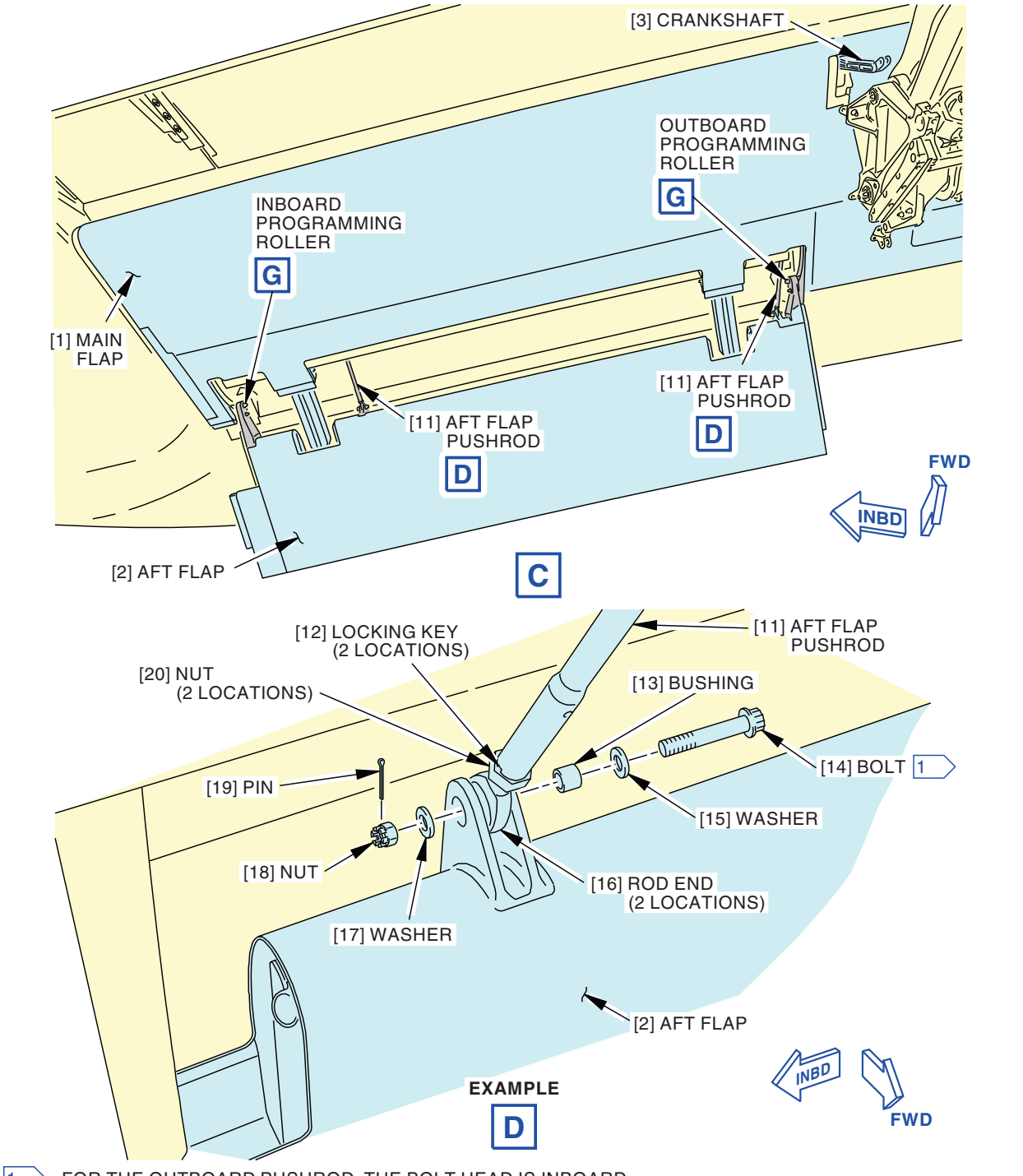
Inboard Aft Flap Installation
Figure 401/27-51-12-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Aft Flap Installation
Figure 401/27-51-12-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

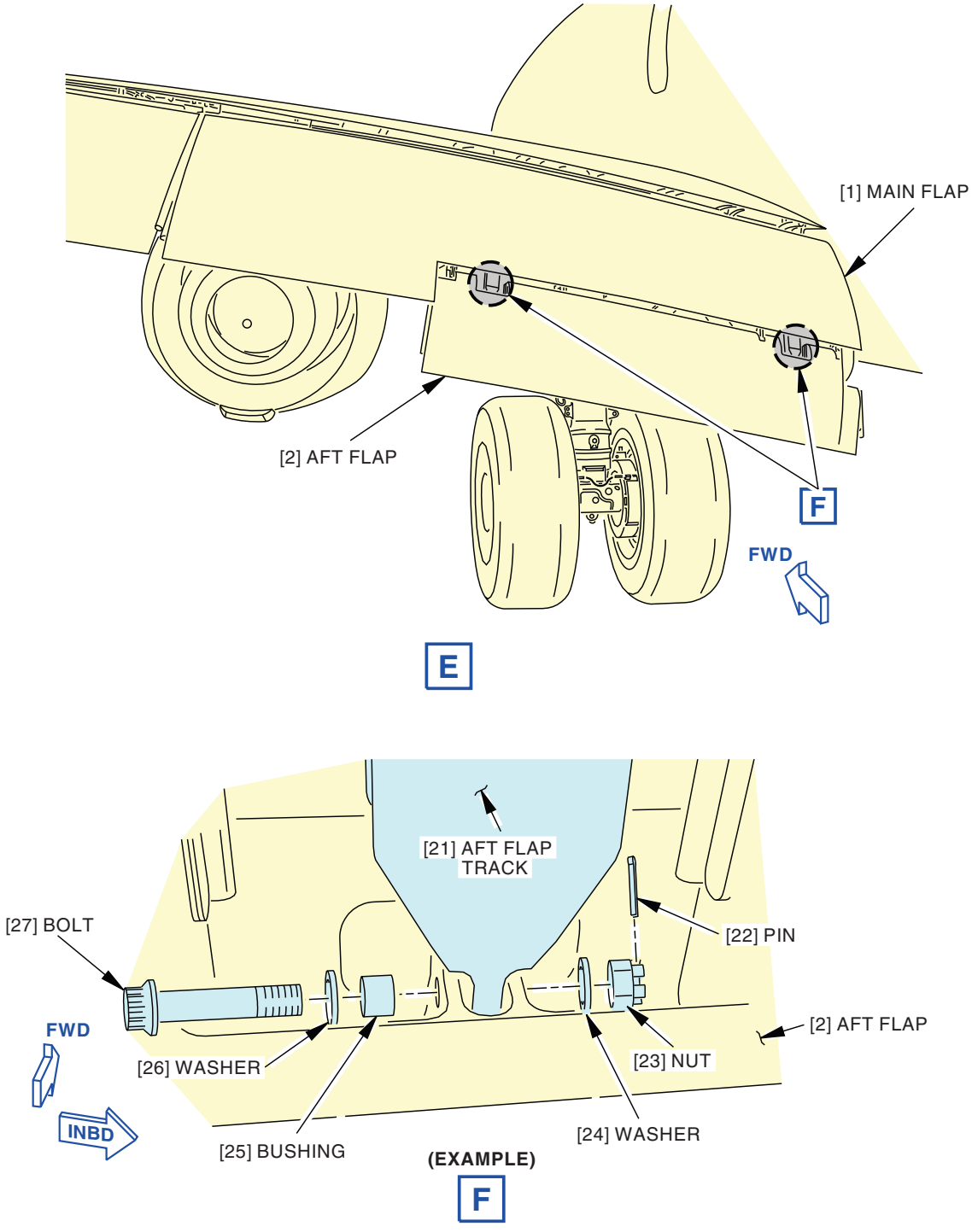


1 FOR THE OUTBOARD PUSHROD, THE BOLT HEAD IS INBOARD

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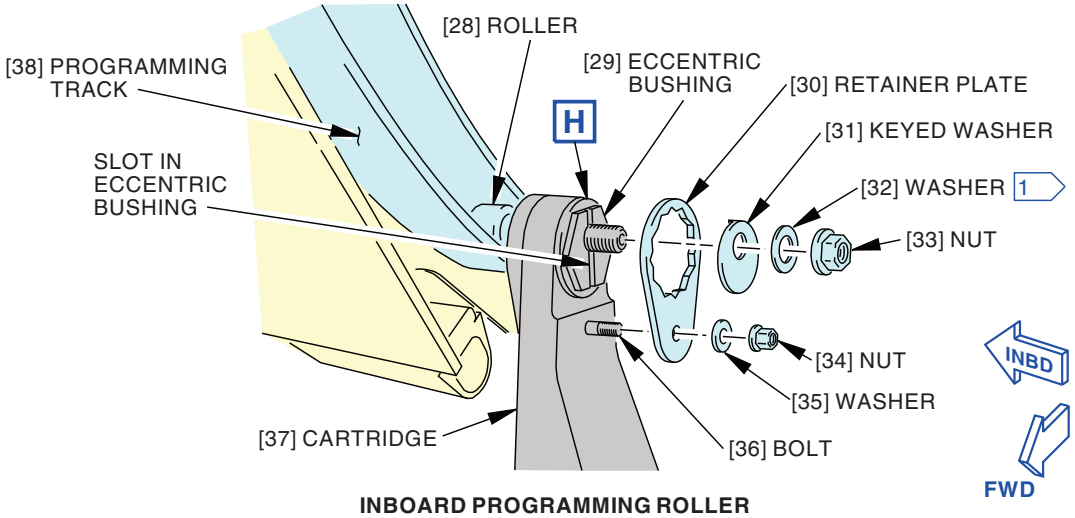
Inboard Aft Flap Installation
Figure 401/27-51-12-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



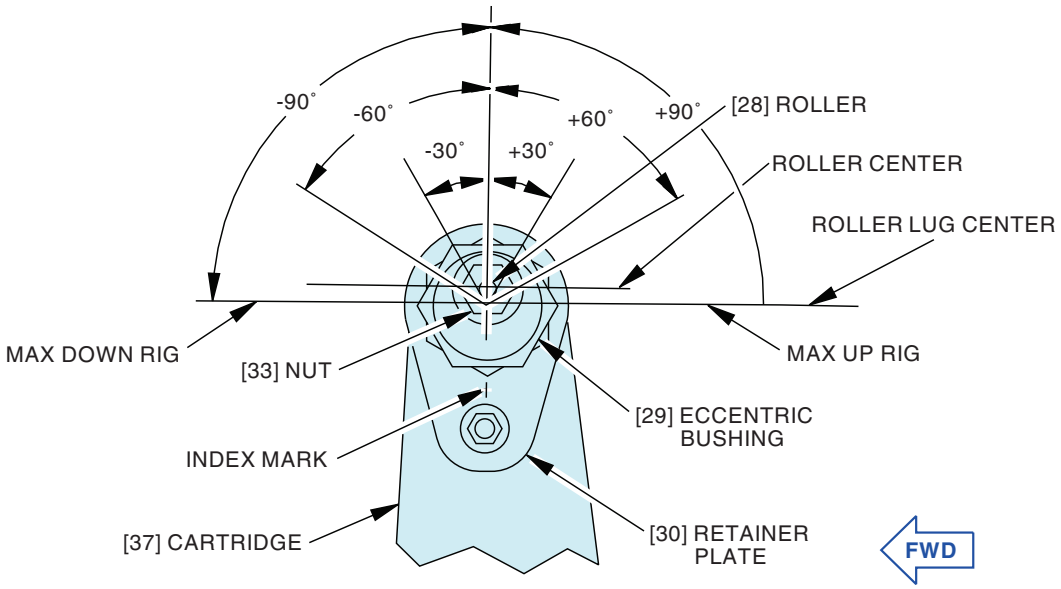
Inboard Aft Flap Installation
Figure 401/27-51-12-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



INBOARD PROGRAMMING ROLLER
(OUTBOARD PROGRAMMING ROLLER IS EQUIVALENT)

G



NOMINAL RIG POSITION

H

1 SOME AIRPLANE CONFIGURATIONS HAVE A TABBED WASHER.

2412827 S00061531395_V2

Inboard Aft Flap Installation
Figure 401/27-51-12-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-12-400-801

3. Inboard Aft Flap Installation

NOTE: The sum of the lengths of external repairs to the inboard aft flap trailing edge wedge and the outboard aft flap trailing edge wedge assemblies must be less than or equal to 24 inches. A single repair that includes both the upper and lower surfaces is counted only once against the spanwise total.

A. References

Reference	Title
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-51-16-400-801	Inboard Aft Flap Track Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-600	Mirror - Inspection
STD-7028	Equipment - Borescope Inspection

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00633	Grease - Aircraft General Purpose	BMS3-33
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Flap assembly	27-51-12-01A-073	PTF ALL
		27-51-12-01A-076	PTF ALL
9	Pin	27-51-14-01-020	PTF ALL
19	Pin	27-51-12-01A-020	PTF ALL
22	Pin	27-51-12-01A-020	PTF ALL

E. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

F. Prepare for the Installation

SUBTASK 27-51-12-820-001

(1) If you will not install the same aft flap assembly [2] and main flap [1], make sure that the inboard and outboard rollers [28] are in the most forward position (Figure 401):

NOTE: With the rollers [28] in the most forward position, the slot in the eccentric bushing [29] will align with the index mark on the retainer plate [30].

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(a) Remove the nut [34], washer [35] and retainer plate [30] that hold the eccentric bushing [29].

(b) Loosen the nut [33] on the roller [28].

(c) Turn the eccentric bushing [29] until the roller [28] is in the forward position and the slot aligns with the bolt [36].

NOTE: The bolt [36] will stay in its position.

(d) Apply sealant, A00247, to the threads of the bolt [36].

(e) Install the retainer plate [30] with the washer [35] and nut [34].

(f) Torque the nut [33] on the roller [28] to 102.5 ±7.5 in-lb (11.6 ±0.9 N·m).

(g) If a tabbed washer is installed, bend a minimum of two tabs of the washer [32] over the nut [33].

SUBTASK 27-51-12-420-001

(2) Make sure that the aft flap tracks are installed on the main flap [1] and are in the retracted position.

(a) If they are not installed, do this task: Inboard Aft Flap Track Installation, TASK 27-51-16-400-801.

G. Inboard Aft Flap Installation

SUBTASK 27-51-12-420-002

(1) Install the aft flap assembly [2]:

(a) Hold the aft flap assembly [2] below and approximately perpendicular to the main flap [1].

(b) Engage the outboard roller [28] with the outboard programming track [38].

(c) Engage the inboard roller [28] with the inboard programming track [38].

(d) Lift the trailing edge of the aft flap assembly [2] and hold it in its usual position.

(e) Connect the aft flap assembly [2] to the aft flap tracks [21]:

1) Move the aft flap tracks [21] aft until they align with the aft flap assembly [2].

2) Apply a thin layer of grease, D00633, to the outside diameter of the bushings [25].

3) Install the bushings [25], bolts [27], washers [26], washers [24] and nuts [23] that attach the aft flap tracks [21] to the aft flap assembly [2].

4) Torque the nuts [23] to 125 ±25 in-lb (14 ±3 N·m).

5) Install the pins [22] in the nuts [23].

SUBTASK 27-51-12-020-004

(2) Connect the inboard and outboard aft flap pushrod [11]:

(a) Carefully pull the aft flap pushrod [11] through the cutouts to move them to the extended position.

NOTE: The crankshaft [3] will move when you pull on the aft flap pushrod [11].

(b) Apply a thin layer of grease, D00633, to the outside diameter of the bushings [13].

(c) Install the bushings [13], bolts [14], washers [15], washers [17], and nuts [18] that attach the aft flap pushrod [11] to the aft flap assembly [2].

(d) Torque the nuts [18] to 65 ±15 in-lb (7.3 ±1.7 N·m).

(e) Install the pins [19] in the nuts [18].

SUBTASK 27-51-12-980-002

(3) Use the crankshaft [3] to manually move the aft flap assembly [2] to the retracted position.

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H. Inboard Aft Flap Adjustment

SUBTASK 27-51-12-480-001

(1) Install the rig pin (no. F70207-113) from the rigging pin kit, SPL-1585, in the crankshaft [3].

SUBTASK 27-51-12-220-001

(2) Measure the distance between the trailing edge of the aft flap assembly [2] and the trailing edge of the upper cove panel [61] on the main flap [1] (Figure 402).

NOTE: Measure the distance aft of the inboard and outboard aft flap pushrods [11].

(a) At the outboard aft flap pushrod [11], make sure that the distance is 9.2745 ±0.0600 in. (235.57 ±1.53 mm).

(b) At the inboard aft flap pushrod [11], make sure that the distance is 9.2319 ±0.0600 in. (234.49 ±1.52 mm).

SUBTASK 27-51-12-820-002

(3) If the distance at the inboard or outboard aft flap pushrods [11] are not in the limit, then adjust the length of the applicable aft flap pushrod [11] (Figure 401):

(a) Remove the rig pin from the crankshaft [3].

(b) Use the crankshaft [3] to manually extend the aft flap assembly [2] to the extended position.

(c) Adjust the length of the applicable aft flap pushrod [11]:

NOTE: For each 90 degree turn of the aft flap pushrod [11], the trailing edge of the aft flap will move 0.04 in. (1.02 mm).

1) Loosen the nuts [20] on the aft flap pushrod [11].

2) Adjust the length of the aft flap pushrod [11].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

3) Apply a thin layer of Cor-Ban 27L Compound, G50237 to the threads of the rod ends [16].

4) Tighten the nuts [20].

5) Install lockwire on the nuts [20] and the locking keys [12].

(d) Use the crankshaft [3] to manually retract the aft flap assembly [2] to the retracted position.

(e) Install the rig pin in the crankshaft [3].

(f) Make sure that the distance from the trailing edge of the aft flap assembly [2] to the trailing edge of the upper cove panel [61] is in the limit.

SUBTASK 27-51-12-980-003

(4) Give support to the aft flap assembly [2].

NOTE: This is the position the rollers [28] will be in with airloads applied to the aft flap assembly [2].

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(a) Do a visual inspection to make sure that the programming rollers [28] touch the bottom surface of the programming tracks [38], and that the aft rollers on the aft flap tracks [21] touch the top surface of the tracks.

NOTE: If necessary, use an inspection mirror, STD-600 or a borescope, STD-7028 to aid with the inspection.

SUBTASK 27-51-12-220-002

(5) At the outboard end of the aft flap assembly [2], measure the vertical alignment at the trailing edge of the aft flap assembly [2] and the main flap [1] (Figure 402).

SUBTASK 27-51-12-820-003

(6) If the vertical alignment of the trailing edge is not in the limit of 0.000 ±0.100 in. (0.00 ±2.54 mm), then adjust the rollers [28] for the inboard and outboard programming tracks [38] (Figure 401):

(a) Make sure that you keep the two rollers [28] adjusted to the same position.

(b) Remove the rig pin from the crankshaft [3].

(c) Use the crankshaft [3] to manually extend the aft flap assembly [2] to the extended position.

(d) Adjust the position of the rollers [28]:

1) Remove the nut [34], washer [35] and retainer plate [30] that hold the eccentric bushing [29].

2) Loosen the nut [33] on the roller [28].

3) Turn the eccentric bushing [29] to adjust the position of the roller [28].

NOTE: Refer to Table 401 to find how far you must turn the eccentric bushing [29].

Table 401/27-51-12-993-801 Programming Roller Eccentric Bushing Adjustment

Eccentric Bushing Position	Aft Flap Trailing Edge Movement Inches (Millimeters)
Nominal to +30	+0.046 (+1.168)
+30 to +60	+0.034 (+0.864)
+60 to +90	+0.013 (+0.330)
Nominal to -30	-0.046 (-1.168)
-30 to -60	-0.034 (-0.864)
-60 to -90	-0.013 (-0.330)

4) Apply sealant, A00247, to the threads of the bolt [36].

5) Install the retainer plate [30] with the washer [35] and nut [34].

6) Torque the nut [33] on the roller [28] to 102.5 ±7.5 in-lb (11.6 ±0.9 N·m).

(e) Use the crankshaft [3] to manually retract the aft flap assembly [2] to the retracted position.

(f) Install the rig pin in the crankshaft [3].

(g) Make sure that the vertical alignment of the trailing edge of the aft flap assembly [2] and the main flap [1] is in the limit.

SUBTASK 27-51-12-220-003

(7) Measure the clearance between the upper cove panel [61] on the main flap [1] and the upper surface of the aft flap assembly [2] (Figure 402).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Measure the clearance at all of the cove panel supports [62] and at the aft flap roller fittings [76].

SUBTASK 27-51-12-820-004

(8) If the clearance between the main flap [1] and the aft flap assembly [2] is not in the limit of 0.010 +0.100 / -0.005 in. (0.254 +2.540 / -0.127 mm), then change the thickness of the shims [75] on the cove panel support [62] and the aft flap roller fittings [76]:

(a) Remove the rig pin from the crankshaft [3].

(b) Use the crankshaft [3] to manually extend the aft flap assembly [2] until you can get access to the fasteners for the shims [75] and shims [77].

(c) If it is necessary, change the thickness of the shims [75] on the cove panel supports [62]:

1) Remove the nuts [74], washers [73] and bolts [72] that attach the upper cove panel [61] to the cove panel support [62].

2) Remove the shim [75] from the cove panel support [62] and the upper cove panel [61].

3) Change the thickness of the shim [75].

NOTE: The maximum permitted thickness of the shim [75] is 0.10 in. (2.54 mm).

4) Apply sealant, A00247, to the mating surfaces of the shim [75].

5) Put the shim [75] in its position between the cove panel support [62] and the upper cove panel [61].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

6) Apply a layer of Cor-Ban 27L Compound, G50237 to the shank and threads of the bolts [72].

7) Install the bolts [72], washers [73] and nuts [74].

(d) If it is necessary, change the thickness of the shims [77] on the aft flap roller fittings [76]:

1) Remove the nuts [80], washers [79] and bolts [78] that attach the upper cove panel [61] to the aft flap roller fitting [76].

2) Remove the shims [77] from the aft flap roller fitting [76] and the upper cove panel [61].

3) Change the thickness of the shim [77].

NOTE: The maximum permitted thickness of the shim [77] is 0.10 in. (2.54 mm).

4) Apply sealant, A00247 to the mating surfaces of the shim [77].

5) Put the shim [77] in its position between the aft flap roller fitting [76] and the upper cove panel [61].

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WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

6) Apply a layer of Cor-Ban 27L Compound, G50237 to the shank and threads of the bolts [78].

7) Install the bolts [78], washers [79] and nuts [80].

(e) Use the crankshaft [3] to manually retract the aft flap assembly [2] to the retracted position.

(f) Install the rig pin in the crankshaft [3].

(g) Make sure that the clearance between the main flap [1] and the aft flap assembly [2] is in the limit.

SUBTASK 27-51-12-220-004

(9) Measure the clearance between the lower cove panel [68] on the main flap [1] and the lower surface of the aft flap assembly [2].

(a) Measure the clearance at the two lower cove panel supports [62].

SUBTASK 27-51-12-820-005

(10) If the clearance between the main flap [1] and the aft flap assembly [2] is not in the limit of 0.000 ±0.060 in. (0.00 ±1.52 mm), then change the thickness of the shims [81] on the lower cove panel [68] support:

(a) Remove the rig pin from the crankshaft [3].

(b) Use the crankshaft [3] to manually extend the aft flap assembly [2] until you can get access to the fasteners for the shims [81].

(c) Change the thickness of the shim [81] on the cove panel supports [62]:

1) Remove the nuts [71], washers [70], and bolts [69] that attach the lower cove panel [68] to the cove panel support [62].

2) Remove the shim [81] from the cove panel support [62] and the lower cove panel [68].

3) Change the thickness of the shim [81].

NOTE: The maximum permitted thickness of the shim [81] is 0.10 in. (2.54 mm) millimeters).

4) Apply sealant, A00247, to the mating surfaces of the shim [81].

5) Put the shim [81] in its position between the cove panel support [62] and the lower cove panel [68].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

6) Apply a layer of Cor-Ban 27L Compound, G50237 to the shank and threads of the bolts [69].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

7) Install the bolts [69], washers [70] and nuts [71].

(d) Use the crankshaft [3] to manually retract the aft flap assembly [2] to the retracted position.

(e) Install the rig pin in the crankshaft [3].

(f) Make sure that the clearance between the main flap [1] and the aft flap assembly [2] is in the limit.

SUBTASK 27-51-12-220-005

(11) At the inboard end of the aft flap assembly [2], measure the compression of the seal [82] on the aft side of the main flap [1]:

(a) Measure and record the distance between the trailing edge of the aft flap assembly [2] and the lower cove panel [68].

NOTE: This distance is referred to as Dimension A.

(b) Remove the rig pin from the crankshaft [3].

(c) Use the crankshaft [3] to manually extend the aft flap assembly [2] until the seal [82] is not compressed against the aft flap assembly [2].

(d) Measure and record the distance between the aft edge of the seal [82] and the lower cove panel [68].

NOTE: This distance is referred to as Dimension B.

SUBTASK 27-51-12-820-006

(12) If Dimension A minus Dimension B is not in the limit of 12.8393 ±0.1000 in. (326.12 ±2.54 mm), then adjust the position of the seal [82]:

(a) Use the crankshaft [3] to manually extend the aft flap assembly [2] until you can get access to the fasteners for the seal [82].

(b) Loosen the nuts [86], washers [85] and bolts [84] that attach the retainer [83].

(c) Move the retainer [83] with the seal [82] to change the distance to the compression of the seal [82].

NOTE: This will change the Dimension B.

(d) Tighten the bolts [84], washers [85] and nuts [86] that attach the retainer [83].

(e) Use the crankshaft [3] to manually retract the aft flap assembly [2] to the retracted position.

(f) Install the rig pin in the crankshaft [3].

SUBTASK 27-51-12-220-006

(13) Outboard of the inboard aft flap track [21], measure the distance between the aft edge of the seal [63] and the seam on the lower surface of the aft flap assembly [2].

(a) If the distance between the seam on the aft flap assembly [2] and the seal [63] is not in the limit of 4.7091 ±0.1000 in. (119.61 ±2.54 mm), then adjust the position of the seal [63]:

1) Remove the rig pin from the crankshaft [3].

2) Use the crankshaft [3] to manually extend the aft flap assembly [2] until you can get access to the fasteners for the seal [63].

3) Loosen the nuts [66], washers [65] and bolts [64] that attach the retainer [67].

4) Move the retainer [67] with the seal [63] to change the distance to the seam on the aft flap assembly [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) Tighten the nuts [66], washers [65] and bolts [64] that attach the retainer [67].

6) Use the crankshaft [3] to manually retract the aft flap assembly [2] to the retracted position.

7) Install the rig pin in the crankshaft [3].

(b) Make sure that the distance between the trailing edge of the aft flap assembly [2] and the seal [63] is in the limit.

SUBTASK 27-51-12-220-007

(14) Do a check of the adjustment of the seals at the inboard and outboard ends of the aft flap assembly [2]:

(a) On the inboard and outboard ends of the aft flap assembly [2], measure the distance between the aft flap assembly [2] and the main flap [1].

(b) Remove the rig pin from the crankshaft [3].

(c) Use the crankshaft [3] to manually extend the aft flap assembly [2] until the seal [88] and seal [93] are clear of the main flap [1].

(d) Measure the distance the seal [88] and seal [93] extends away from the aft flap assembly [2].

(e) If the distance the seal [88] and seal [93] extends is not in the limit of the distance you measured plus 0.150 ±0.100 in. (3.81 ±2.54 mm), then adjust the position of the applicable seal [88] or seal [93]:

1) Loosen the nuts [87], washers [91] and bolts [89] that attach the retainer [90].

2) Move the retainer [90] and the seal [88] or seal [93] to change the distance the seal extends past the aft flap assembly [2].

3) Tighten the nuts [87], washers [91] and bolts [92] that attach the retainer [90].

(f) Use the crankshaft [3] to manually retract the aft flap assembly [2] to the retracted position.

NOTE: The seal [88] and seal [93] can extend a maximum of 0.5 in. (12.7 mm) below the lower surface of the main flap [1] when the seals are compressed.

SUBTASK 27-51-12-080-001

(15) Remove the rig pin from the crankshaft [3] if it is necessary.

SUBTASK 27-51-12-420-003

(16) At the outboard flap support, connect the crankshaft [3] to the aft flap drive rod [10] (Figure 401):

(a) Use the crankshaft [3] to manually extend the aft flap assembly [2] until the crankshaft [3] aligns with the aft flap drive rod [10].

(b) Install the bolt [4], washer [5], bushing [6], washer [7] and nut [8] that attach the crankshaft [3] to the aft flap drive rod [10].

1) Make sure that the head of the bolt [4] is on the inboard side.

(c) Torque the nut [8] to 40 ±10 in-lb (4.5 ±1.1 N·m).

(d) Install the pin [9] through the nut [8].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-12-440-001

(1) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

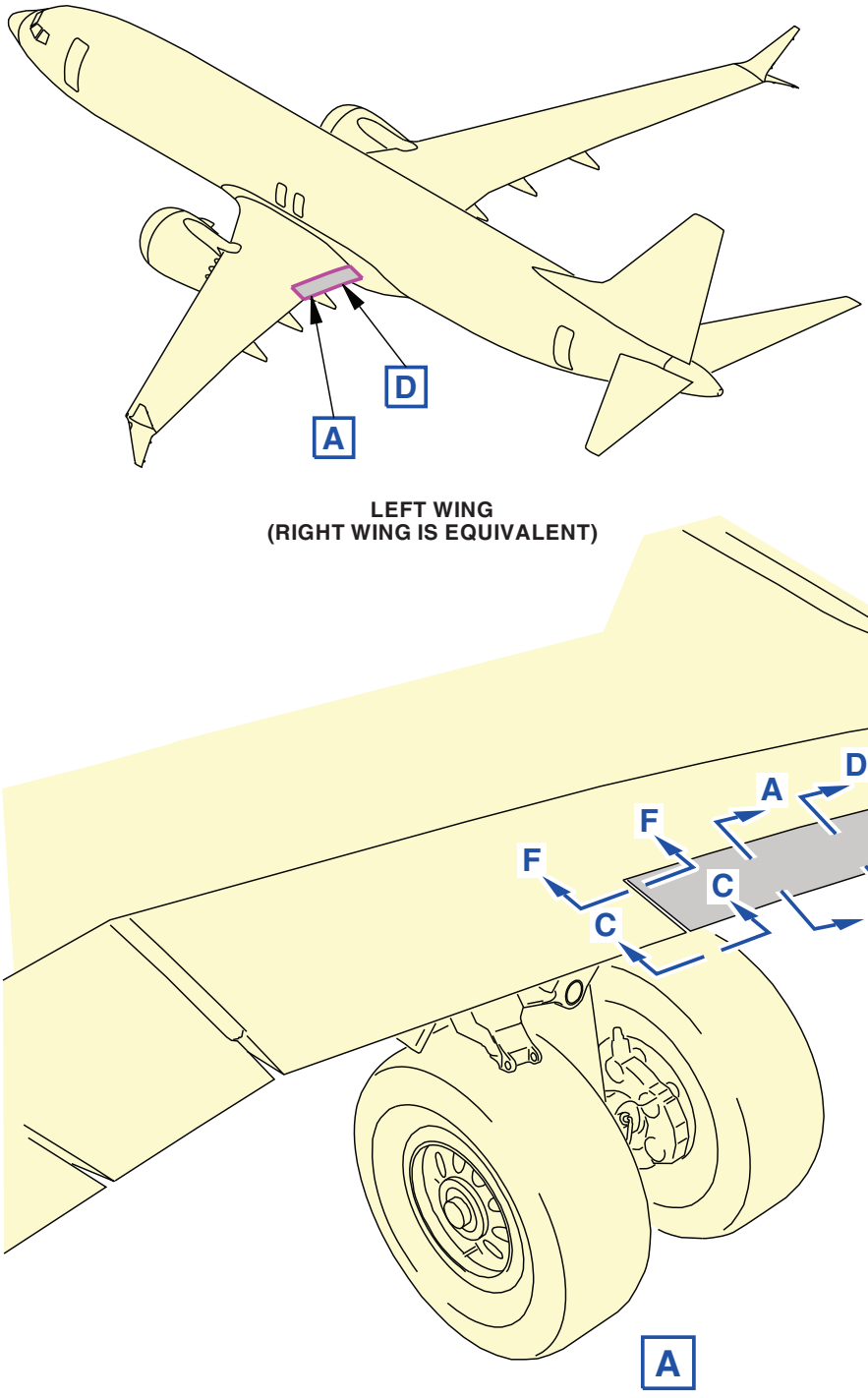
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-12-860-002

(2) Retract the trailing edge flaps to the retracted position (TASK 27-51-00-860-804).

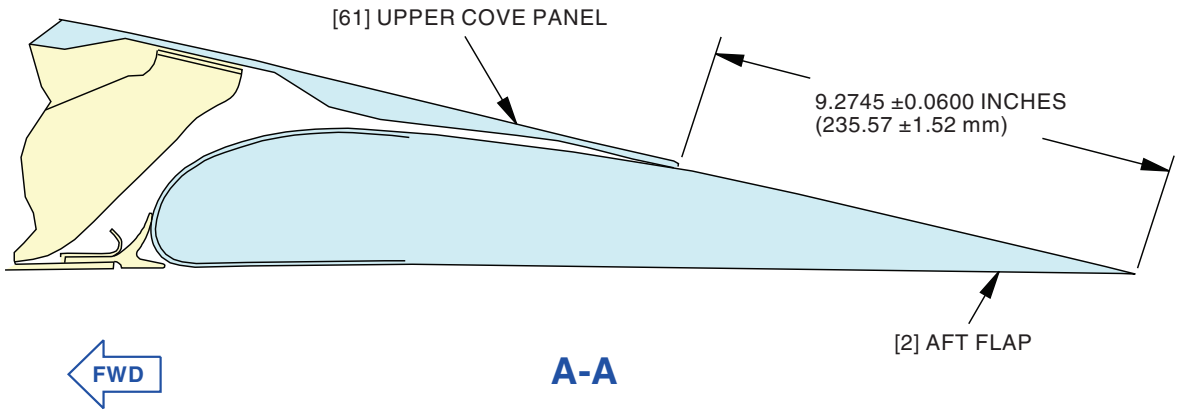
END OF TASK

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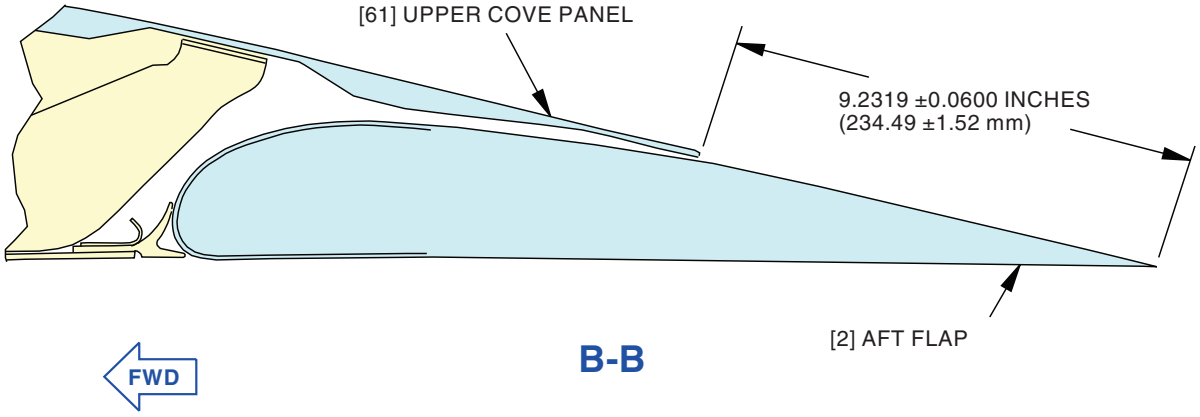


Inboard Aft Flap Adjustment
Figure 402/27-51-12-990-802 (Sheet 1 of 6)

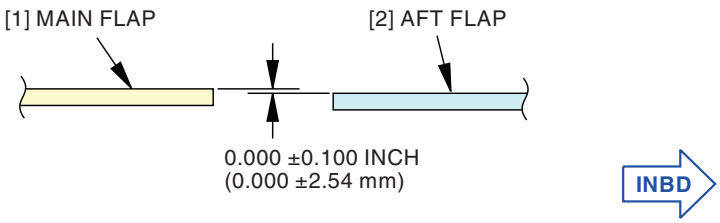
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A-A



B-B



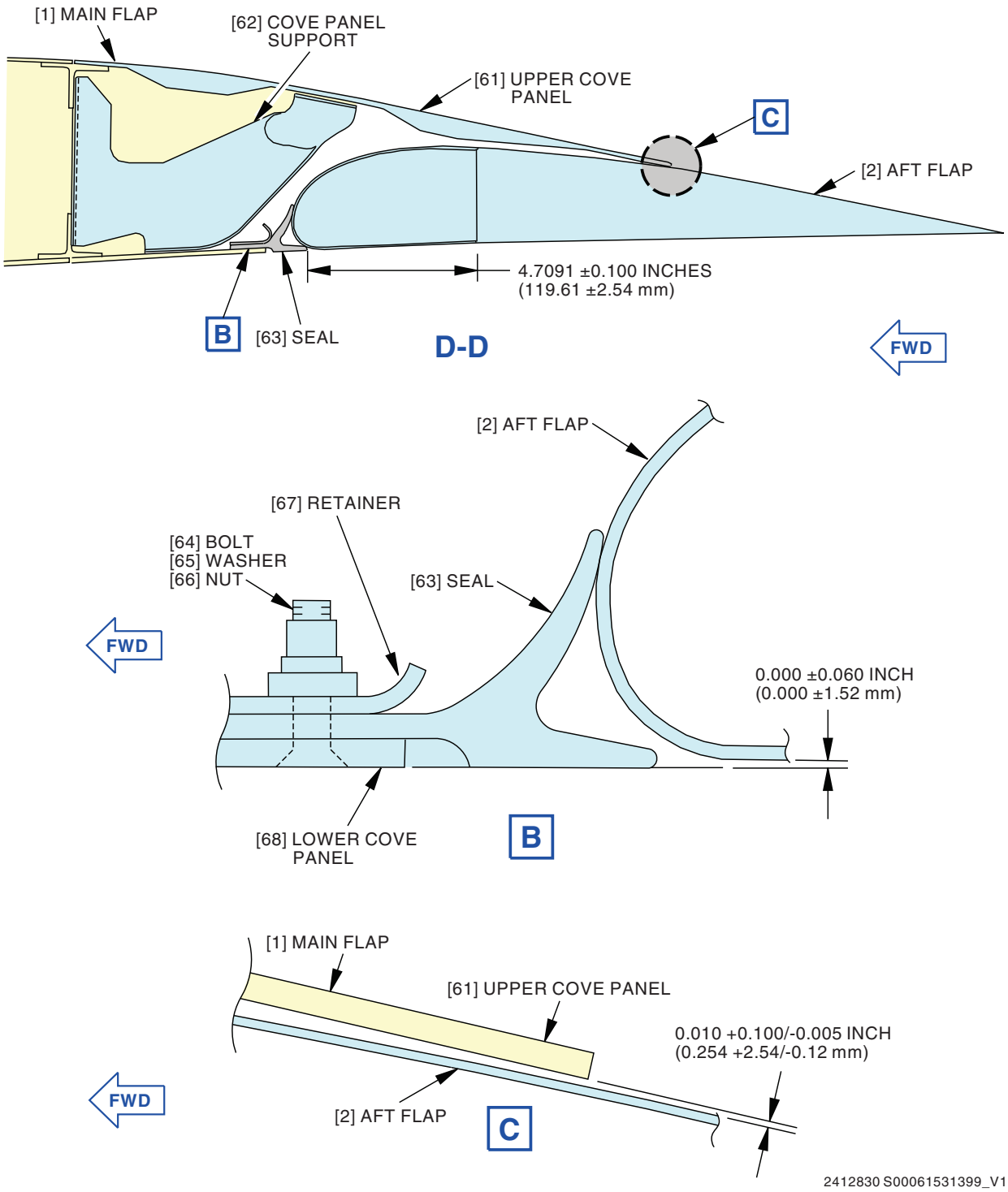
VERTICAL ALIGNMENT

C-C

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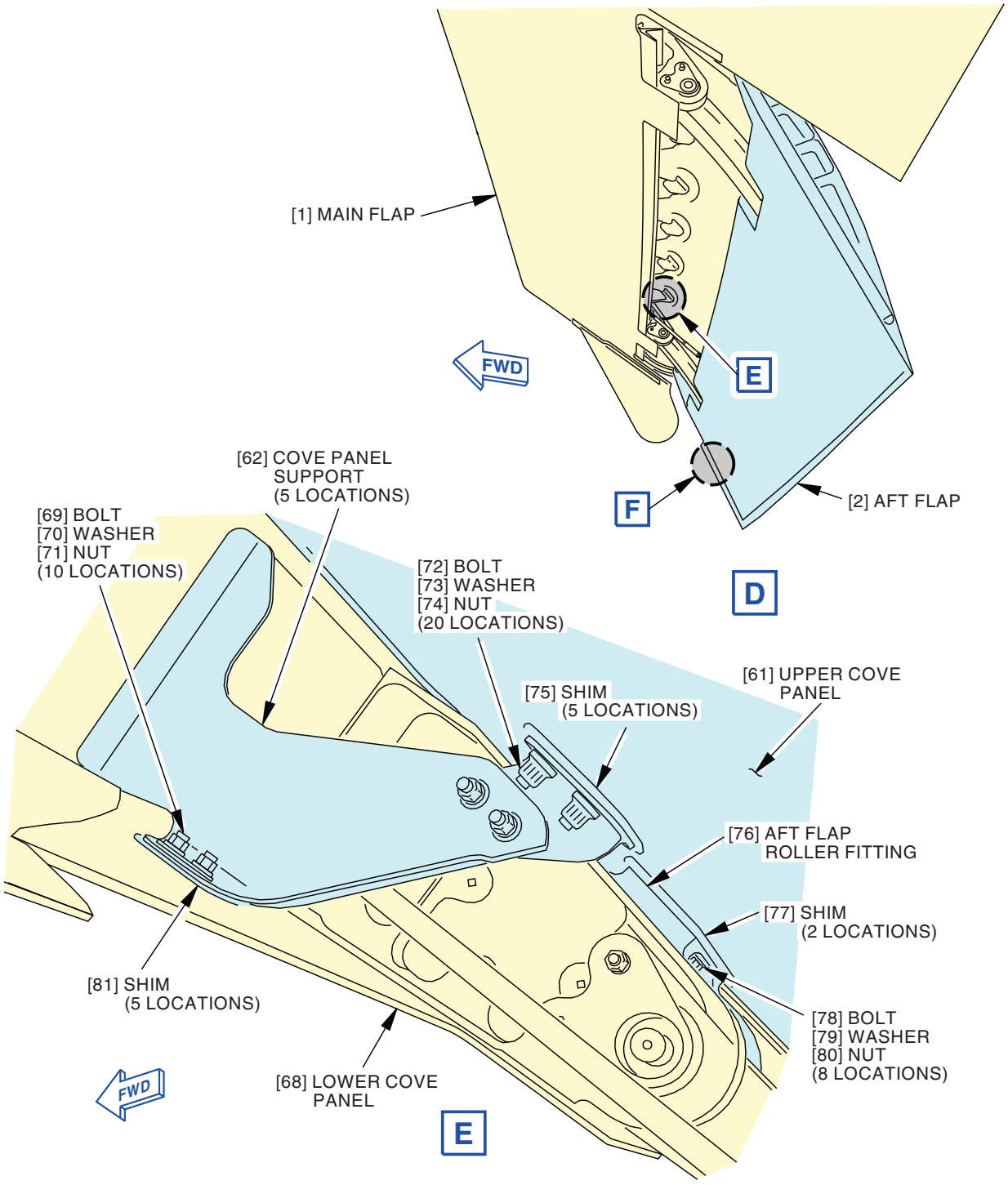
Inboard Aft Flap Adjustment
Figure 402/27-51-12-990-802 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



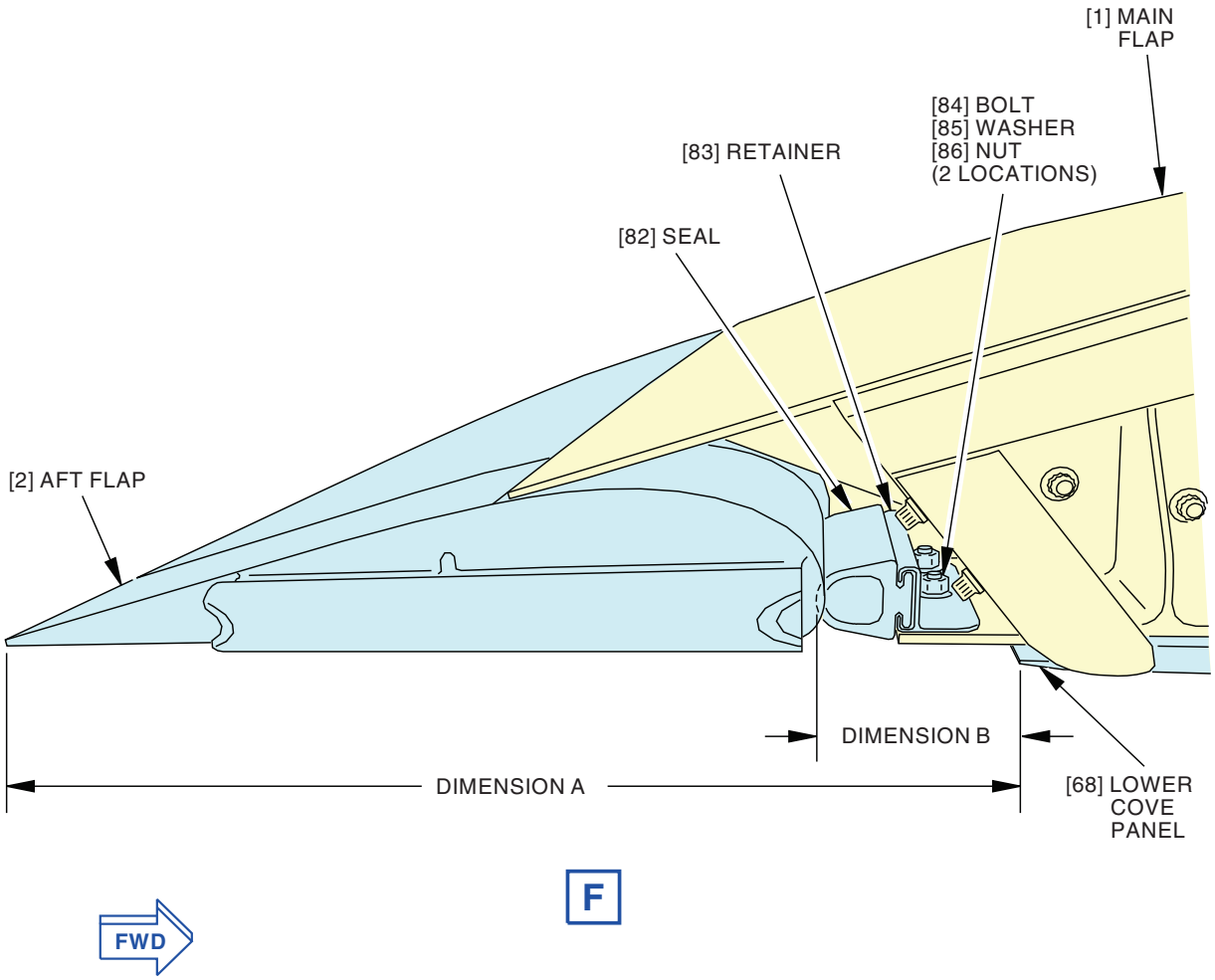
Inboard Aft Flap Adjustment
Figure 402/27-51-12-990-802 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Aft Flap Adjustment
Figure 402/27-51-12-990-802 (Sheet 4 of 6)

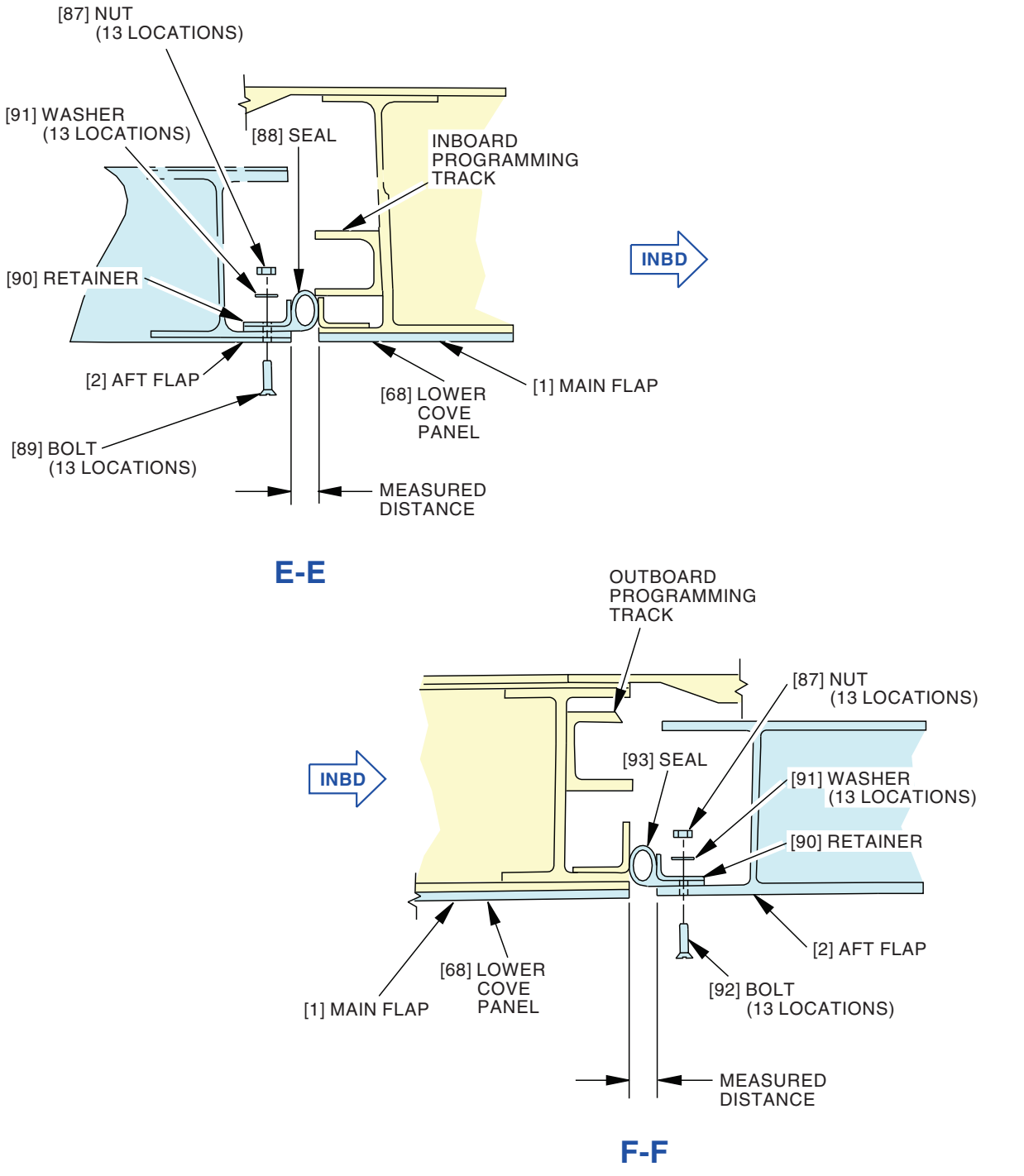
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Inboard Aft Flap Adjustment
Figure 402/27-51-12-990-802 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Aft Flap Adjustment
Figure 402/27-51-12-990-802 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD AFT FLAP - INSPECTION/CHECK

1. General

A. This procedure contains a task to do an inspection of the attachment points for the inboard aft flap. The attachment points are the aft flap drive rods and the aft flap tracks.

TASK 27-51-12-200-801

2. Inboard Aft Flap Inspection

(Figure 601)

A. References

Reference	Title
27-51-12-000-801	Inboard Aft Flap Removal (P/B 401)
27-51-12-400-801	Inboard Aft Flap Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

C. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

D. Procedure

SUBTASK 27-51-12-020-005

(1) Do this task: Inboard Aft Flap Removal, TASK 27-51-12-000-801.

SUBTASK 27-51-12-020-007

(2) Examine the parts of the inboard aft flap (Table 601):

(a) Remove the parts that you will measure.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are not in tolerance, replace or repair the parts.

Table 601/27-51-12-993-802

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE CLEARANCE inches/mm
#1 BUSHING	I.D.	0.2500 in. (6.3500 mm)	0.2505 in. (6.3627 mm)	0.2545 in. (6.4643 mm)	0.0050 in. (0.1270 mm)
#1 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2455 in. (6.2357 mm)	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-51-12-993-802 (Continued)

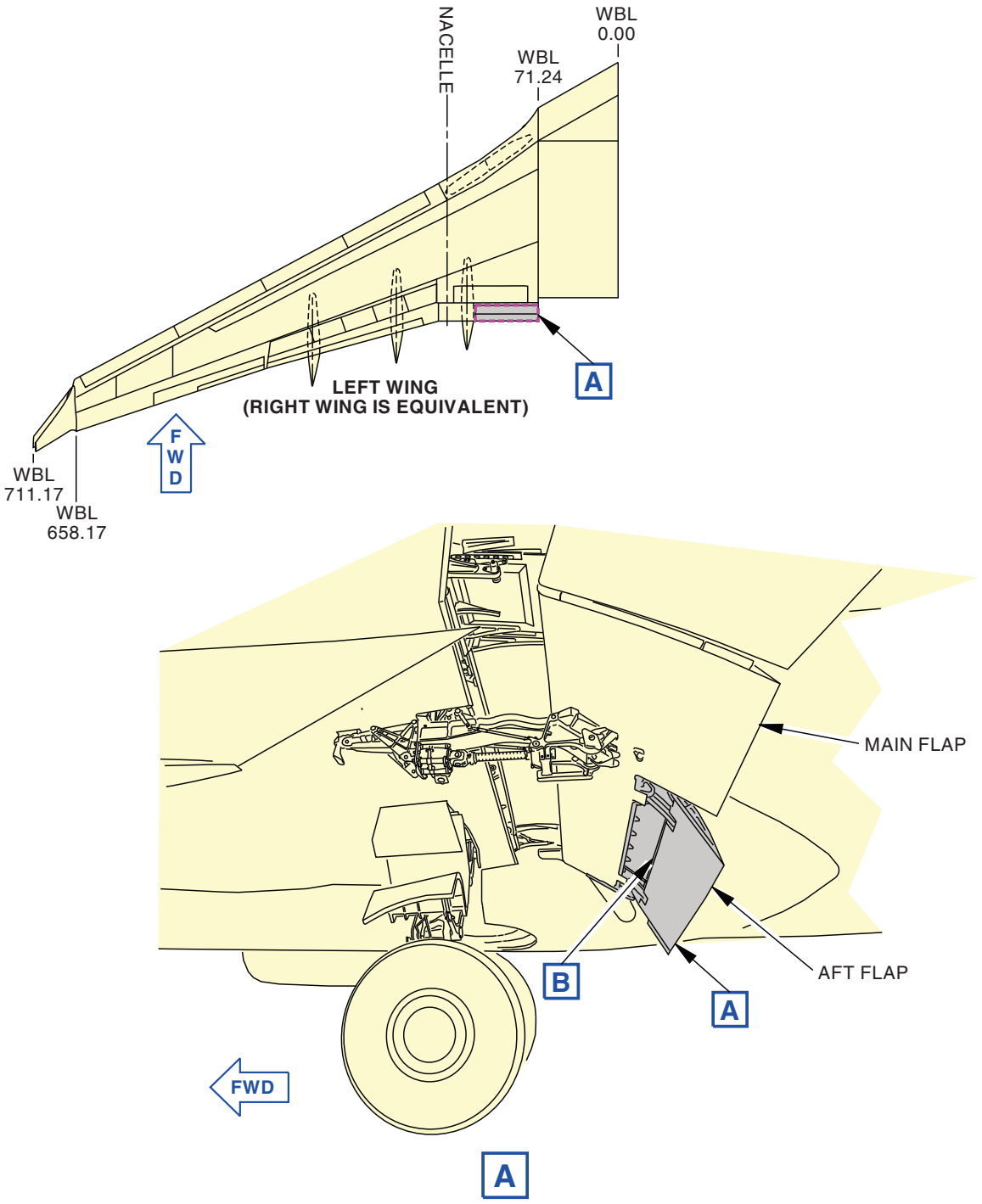
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE CLEARANCE inches/mm
#2 ROD END	I.D.	0.2500 in. (6.3500 mm)	0.2500 in. (6.3500 mm)	0.2500 in. (6.3500 mm)	0.0015 in. (0.0381 mm)
#2 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2485 in. (6.3119 mm)	
#3 BUSHING	I.D.	0.3750 in. (9.5250 mm)	0.3756 in. (9.5402 mm)	0.3756 in. (9.5402 mm)	0.0016 in. (0.0406 mm)
#3 BUSHING	O.D.	0.3740 in. (9.4996 mm)	0.3745 in. (9.5123 mm)	0.3740 in. (9.4996 mm)	
#4 BUSHING	I.D.	0.2500 in. (6.3500 mm)	0.2505 in. (6.3627 mm)	0.2545 in. (6.4643 mm)	0.0050 in. (0.1270 mm)
#4 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2455 in. (6.2357 mm)	
#5 BUSHING	I.D.	0.3125 in. (7.9375 mm)	0.3130 in. (7.9502 mm)	0.3170 in. (8.0518 mm)	0.0050 in. (0.1270 mm)
#5 BOLT	O.D.	0.3110 in. (7.8994 mm)	0.3120 in. (7.9248 mm)	0.3080 in. (7.8232 mm)	
#6 BUSHING	I.D.	0.4375 in. (11.1125 mm)	0.4382 in. (11.1303 mm)	0.4382 in. (11.1303 mm)	0.0017 in. (0.0432 mm)
#6 BUSHING	O.D.	0.4365 in. (11.0871 mm)	0.4370 in. (11.0998 mm)	0.4365 in. (11.0871 mm)	
#7 TRACK	I.D.	0.3125 in. (7.9375 mm)	0.3130 in. (7.9502 mm)	0.3170 in. (8.0518 mm)	0.0050 in. (0.1270 mm)
#7 BOLT	O.D.	0.3110 in. (7.8994 mm)	0.3120 in. (7.9248 mm)	0.3080 in. (7.8232 mm)	
#8 BUSHING	I.D.	0.3125 in. (7.9375 mm)	0.3131 in. (7.9527 mm)	0.3170 in. (8.0518 mm)	0.0050 in. (0.1270 mm)
#8 BOLT	O.D.	0.3110 in. (7.8994 mm)	0.3120 in. (7.9248 mm)	0.3081 in. (7.8257 mm)	

SUBTASK 27-51-12-420-004

(3) Do this task: Inboard Aft Flap Installation, TASK 27-51-12-400-801.

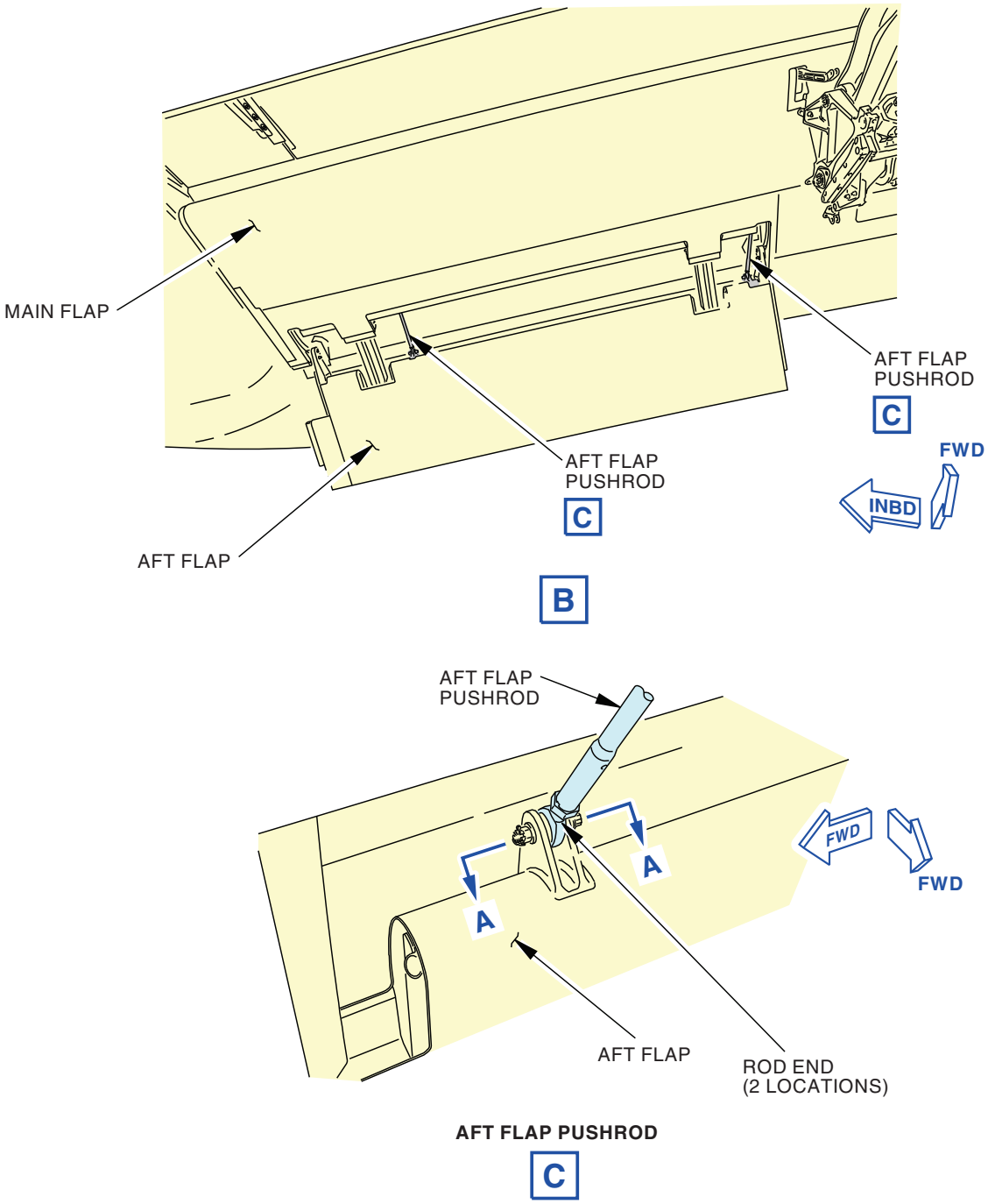
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Aft Flap Inspection
Figure 601/27-51-12-990-803 (Sheet 1 of 4)

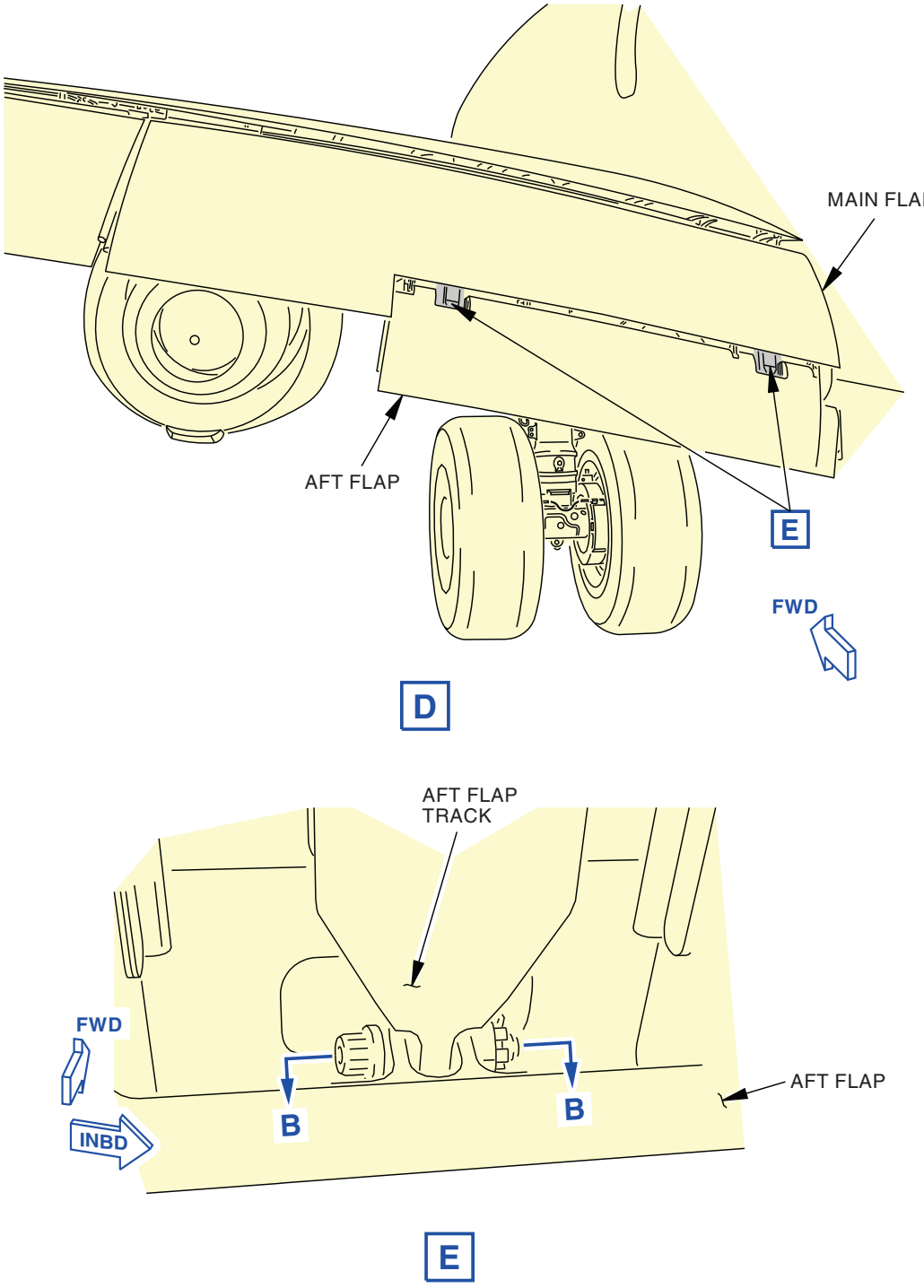
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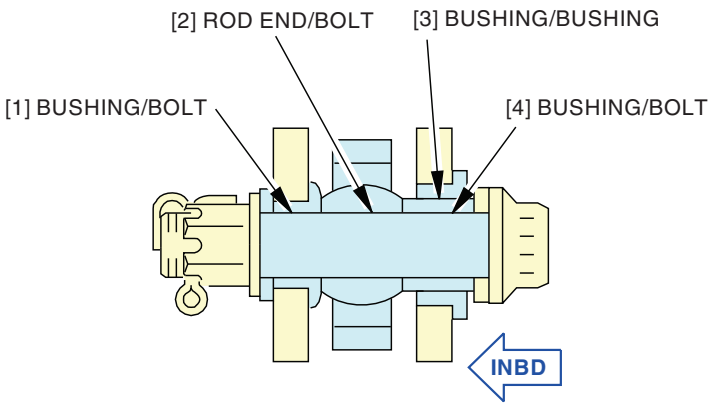
Inboard Aft Flap Inspection
Figure 601/27-51-12-990-803 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

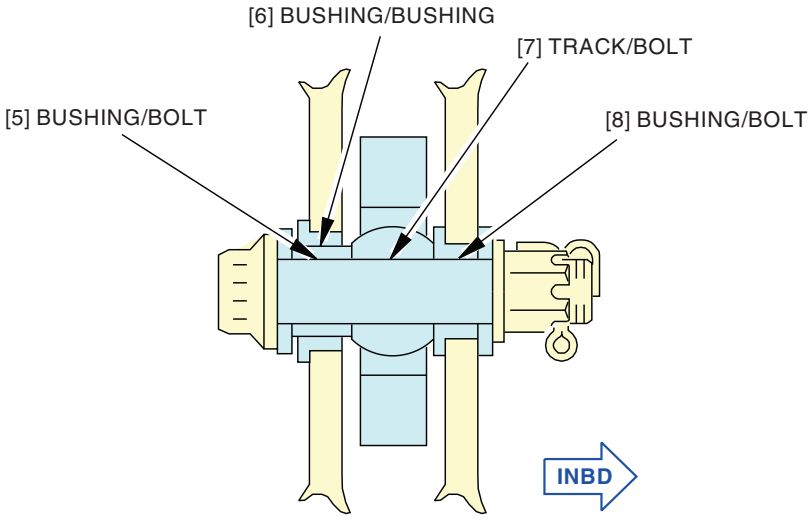


Inboard Aft Flap Inspection
Figure 601/27-51-12-990-803 (Sheet 3 of 4)

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A-A



B-B

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Inboard Aft Flap Inspection
Figure 601/27-51-12-990-803 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD FLAP CARRIAGES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Inboard flap carriage removal (selection)

(2) Inboard flap carriage installation (selection)

(3) Removal of the outboard carriage for the inboard flap

(4) Installation of the outboard carriage for the inboard flap

(5) Removal of the inboard carriage for the inboard flap

(6) Installation of the inboard carriage for the inboard flap.

TASK 27-51-13-000-801

2. Inboard Flap Carriage Removal (Selection)

A. Location Zones

Zone	Area
541	Left Wing - Fairing Flap Support No. 4
542	Left Wing - Fairing Flap Support No. 3
543	Left Wing - Fairing Flap Support No. 2
553	Left Wing - Inboard Flap
641	Right Wing - Fairing Flap Support No. 5
642	Right Wing - Fairing Flap Support No. 6
643	Right Wing - Fairing Flap Support No. 7
653	Right Wing - Inboard Flap

B. Inboard Flap Carriage Removal (Selection)

SUBTASK 27-51-13-000-001

(1) Do one of these tasks to remove the applicable inboard flap carriage:

(a) Do this task: Inboard Flap Outboard Carriage Removal, TASK 27-51-13-000-802.

(b) Do this task: Inboard Flap Inboard Carriage Removal, TASK 27-51-13-000-803.

——— END OF TASK ———

TASK 27-51-13-400-801

3. Inboard Flap Carriage Installation (Selection)

A. Location Zones

Zone	Area
541	Left Wing - Fairing Flap Support No. 4
542	Left Wing - Fairing Flap Support No. 3
543	Left Wing - Fairing Flap Support No. 2
553	Left Wing - Inboard Flap
641	Right Wing - Fairing Flap Support No. 5
642	Right Wing - Fairing Flap Support No. 6
643	Right Wing - Fairing Flap Support No. 7
653	Right Wing - Inboard Flap

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Inboard Flap Carriage Installation (Selection)

SUBTASK 27-51-13-400-001

(1) Do one of these tasks to install the applicable inboard flap carriage:

(a) Do this task: Inboard Flap Outboard Carriage Installation, TASK 27-51-13-400-802.

(b) Do this task: Inboard Flap Inboard Carriage Installation, TASK 27-51-13-400-803.

——— END OF TASK ———

TASK 27-51-13-000-802

4. Inboard Flap Outboard Carriage Removal

(Figure 401)

A. General

(1) This task is applicable to the flap carriages at flap supports No. 3 and 6.

B. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-11-000-801	Inboard Trailing Edge Flap Removal (P/B 401)

C. Location Zones

Zone	Area
542	Left Wing - Fairing Flap Support No. 3
553	Left Wing - Inboard Flap
642	Right Wing - Fairing Flap Support No. 6
653	Right Wing - Inboard Flap

D. Prepare for the Removal

SUBTASK 27-51-13-040-001

(1) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-13-020-001

(2) Do this task: Inboard Trailing Edge Flap Removal, TASK 27-51-11-000-801.

E. Inboard Flap Outboard Carriage Removal

SUBTASK 27-51-13-010-001

(1) Remove the upper skin panel [3] from the main flap [1]:

(a) Remove the bolts [4] that attach the upper skin panel [3] to the main flap [1].

(b) Remove the upper skin panel [3].

SUBTASK 27-51-13-020-002

(2) Remove the carriage [2] from the main flap [1]:

(a) Remove the nuts [15], washers [14], washers [13], bolts [12] and bolts [16] that attach the aft end of the carriage [2].

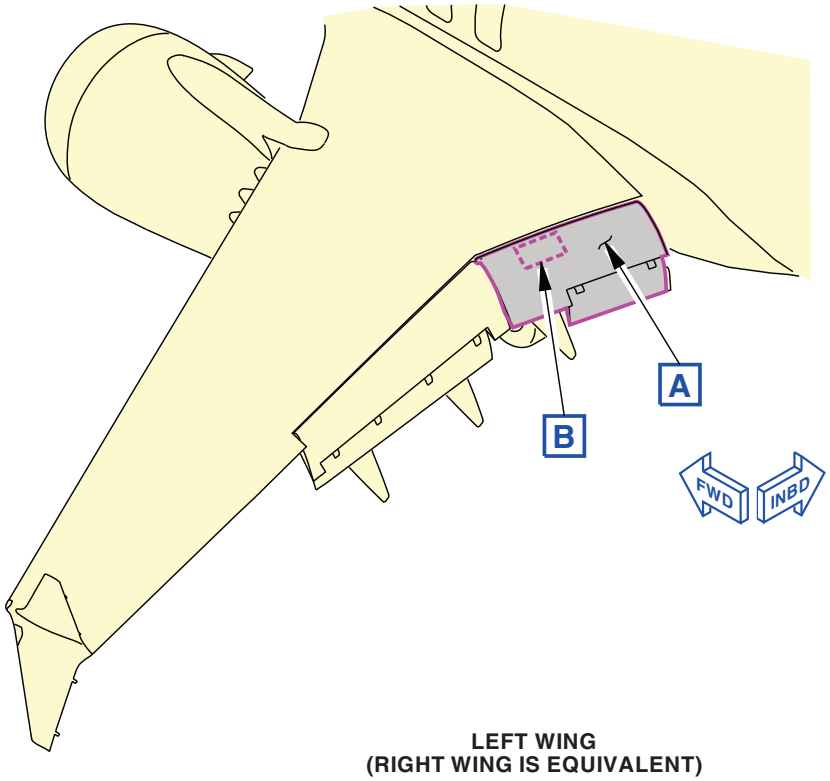
(b) Remove the nuts [11], washers [10] and bolts [9] from the forward end of the carriage [2].

(c) Remove the nuts [8], washers [7], bolts [5] and fillers [6] from the forward end of the carriage [2].

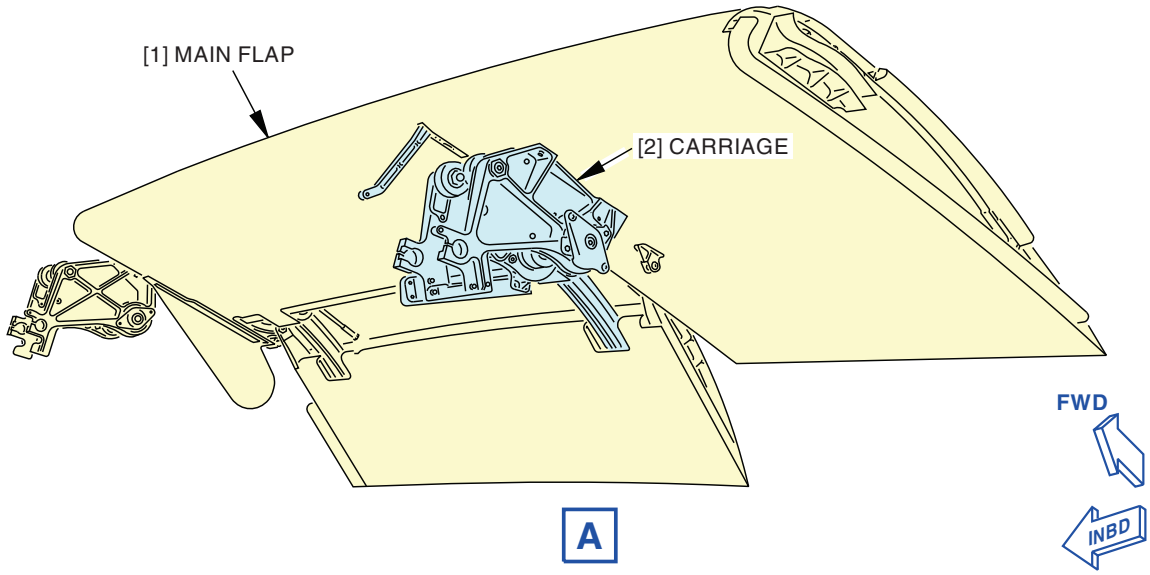
(d) Remove the carriage [2] from the main flap [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



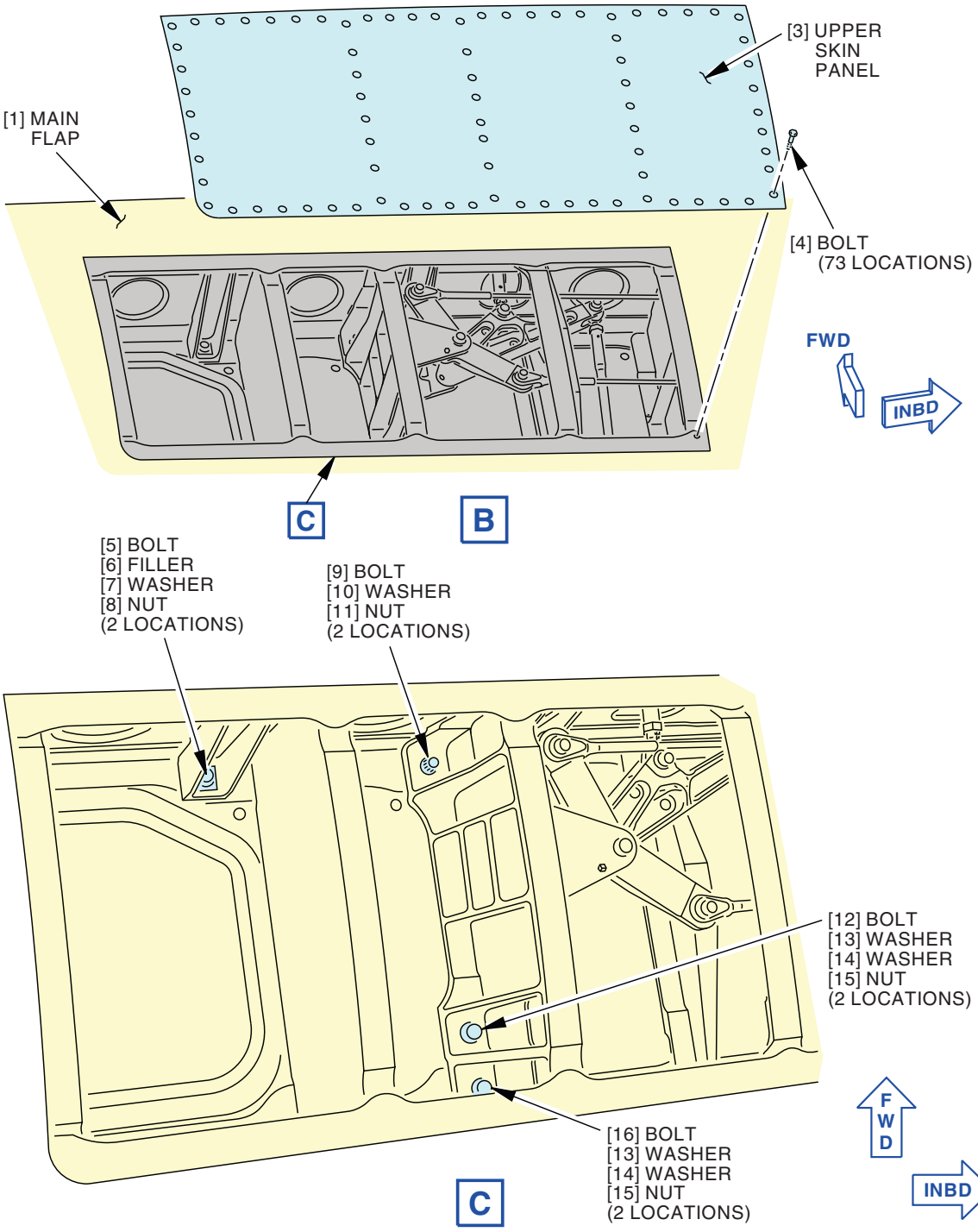
LEFT WING
(RIGHT WING IS EQUIVALENT)



2412838 S00061531412_V1

Inboard Flap Outboard Carriage Installation
Figure 401/27-51-13-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Flap Outboard Carriage Installation
Figure 401/27-51-13-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-13-400-802

5. Inboard Flap Outboard Carriage Installation

(Figure 401)

A. General

(1) This task is applicable to the flap carriages at flap supports No. 3 and 6.

B. References

Reference	Title
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-11-400-801	Inboard Trailing Edge Flap Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Carriage	27-51-13-01A-080	PTF ALL
		27-51-13-01A-085	PTF ALL

E. Location Zones

Zone	Area
542	Left Wing - Fairing Flap Support No. 3
553	Left Wing - Inboard Flap
642	Right Wing - Fairing Flap Support No. 6
653	Right Wing - Inboard Flap

F. Inboard Flap Outboard Carriage Installation

SUBTASK 27-51-13-020-012

(1) Inspect the rub pads [55] for wear.

(a) If the rub pads [55] are worn greater than 50%, they will need to be replaced.

SUBTASK 27-51-13-420-001

(2) Install the carriage [2] on the main flap [1]:

(a) Make sure the upper skin panel [3] is removed from the main flap [1].

(b) Apply sealant, A00247, to the mating surface of the carriage [2] and the main flap [1].

(c) Put the carriage [2] in its position on the main flap [1].

(d) Apply sealant, A00247, to the mating surface of the fillers [6] and the main flap [1].

(e) Put the fillers [6] in their location on the main flap [1].

(f) Apply sealant, A00247, to the shank and threads of the bolts [5], bolts [9], bolts [12] and bolts [16].

(g) Install the bolts [5], washers [7] and nuts [8] to attach the forward end of the carriage [2] to the main flap [1].

(h) Install the bolts [9], washers [10] and nuts [11] at the forward end of the carriage [2].

1) Make sure that the bolts [9] are installed with the bolt heads down.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Install the bolts [12], bolts [16], washers [13], washers [14] and nuts [15] to attach the aft end of the carriage [2] to the main flap [1].

SUBTASK 27-51-13-410-001

(3) Install the upper skin panel [3] on the main flap [1]:

(a) Apply sealant, A00247, to the mating surface of the upper skin panel [3] and the main flap [1].

(b) Put the upper skin panel [3] in its position.

(c) Install the bolts [4]:

1) Apply compound, C00528 to the forward and aft rows of holes in the upper skin panel [3].

2) Install the bolts [4] in the forward and aft rows of holes.

3) Apply sealant, A00247, to the bolts [4] that go in the ribs and stiffeners in the main flap [1].

4) Install the bolts [4] in the ribs and stiffeners.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-13-420-002

(1) Do this task: Inboard Trailing Edge Flap Installation, TASK 27-51-11-400-801.

SUBTASK 27-51-13-440-001

(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

——— END OF TASK ———

TASK 27-51-13-000-803

6. Inboard Flap Inboard Carriage Removal

(Figure 402)

A. General

(1) This task is applicable to the flap carriages at flap supports No. 4 and 5.

B. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-11-000-801	Inboard Trailing Edge Flap Removal (P/B 401)

C. Location Zones

Zone	Area
541	Left Wing - Fairing Flap Support No. 4
553	Left Wing - Inboard Flap
641	Right Wing - Fairing Flap Support No. 5
653	Right Wing - Inboard Flap

D. Prepare for the Removal

SUBTASK 27-51-13-040-002

(1) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-13-020-003

(2) Do this task: Inboard Trailing Edge Flap Removal, TASK 27-51-11-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Inboard Flap Inboard Carriage Removal

SUBTASK 27-51-13-010-002

(1) Remove the lower skin panel [31] from the main flap [1]:

(a) Remove the bolts [32] that attach the lower skin panel [31] to the main flap [1].

(b) Remove the lower skin panel [31].

SUBTASK 27-51-13-020-004

(2) Remove the secondary torque tube [40] from the main flap [1]:

(a) Remove the retainer [39] from the secondary torque tube [40]:

1) Remove the bolt [41], washer [42], nut [44] and filler [43] that attach the secondary torque tube [40] to the main flap [1].

2) Remove the retainer [39] from the secondary torque tube [40].

(b) Remove the nuts [49], washers [48], washers [51] and bolts [50] that attach the inboard end of the carriage [34] and the secondary torque tube [40] to the primary torque tube [38].

(c) Remove the secondary torque tube [40].

SUBTASK 27-51-13-020-005

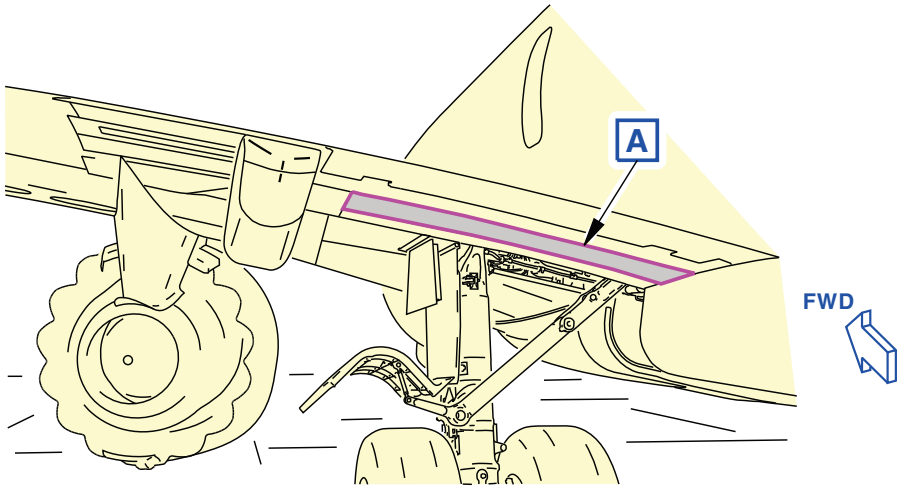
(3) Remove the carriage [34] from the main flap [1]:

(a) Remove the nuts [46], washers [45], bolts [47], bolt [54] and spacers [52] that attach the carriage [34], attach plate [37] and rub block [33] to the primary torque tube [38].

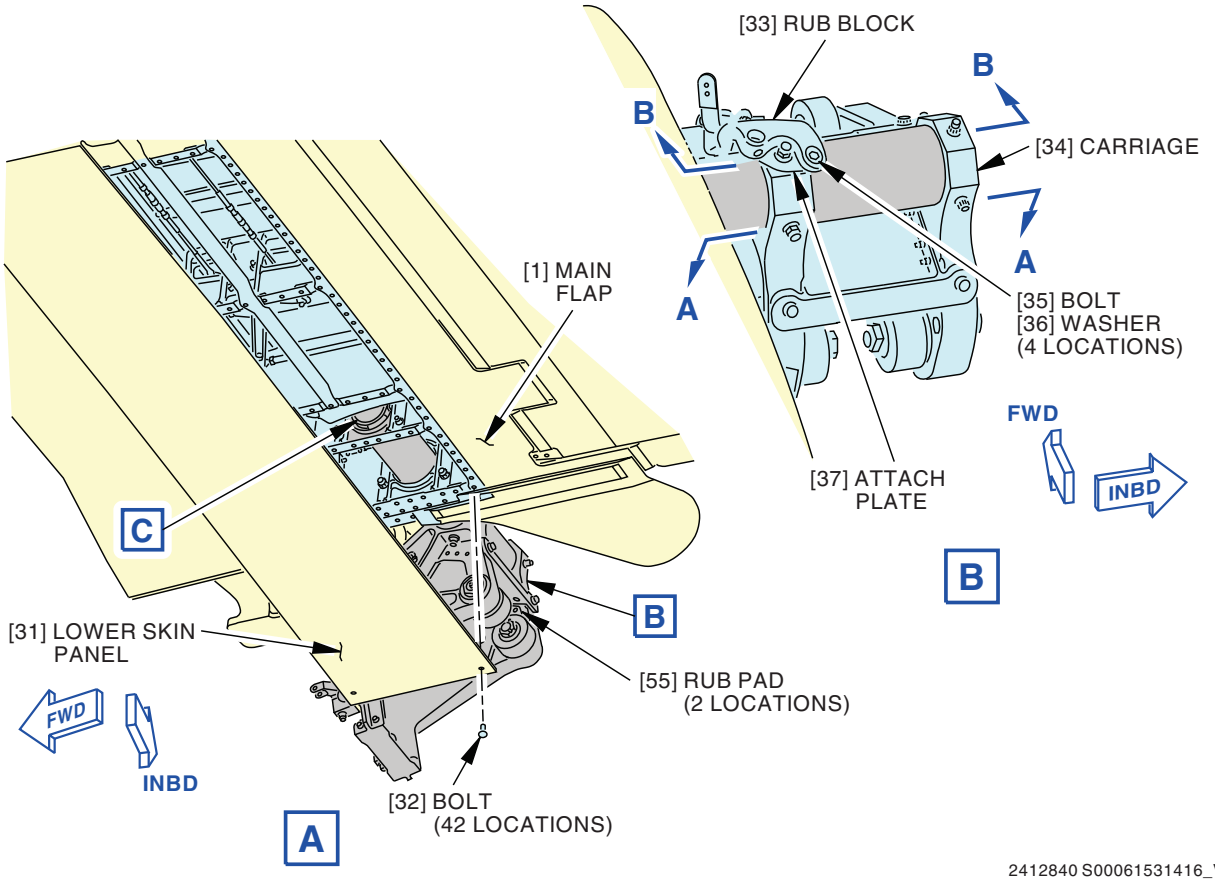
(b) Slide the carriage [34] inboard, off of the primary torque tube [38].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



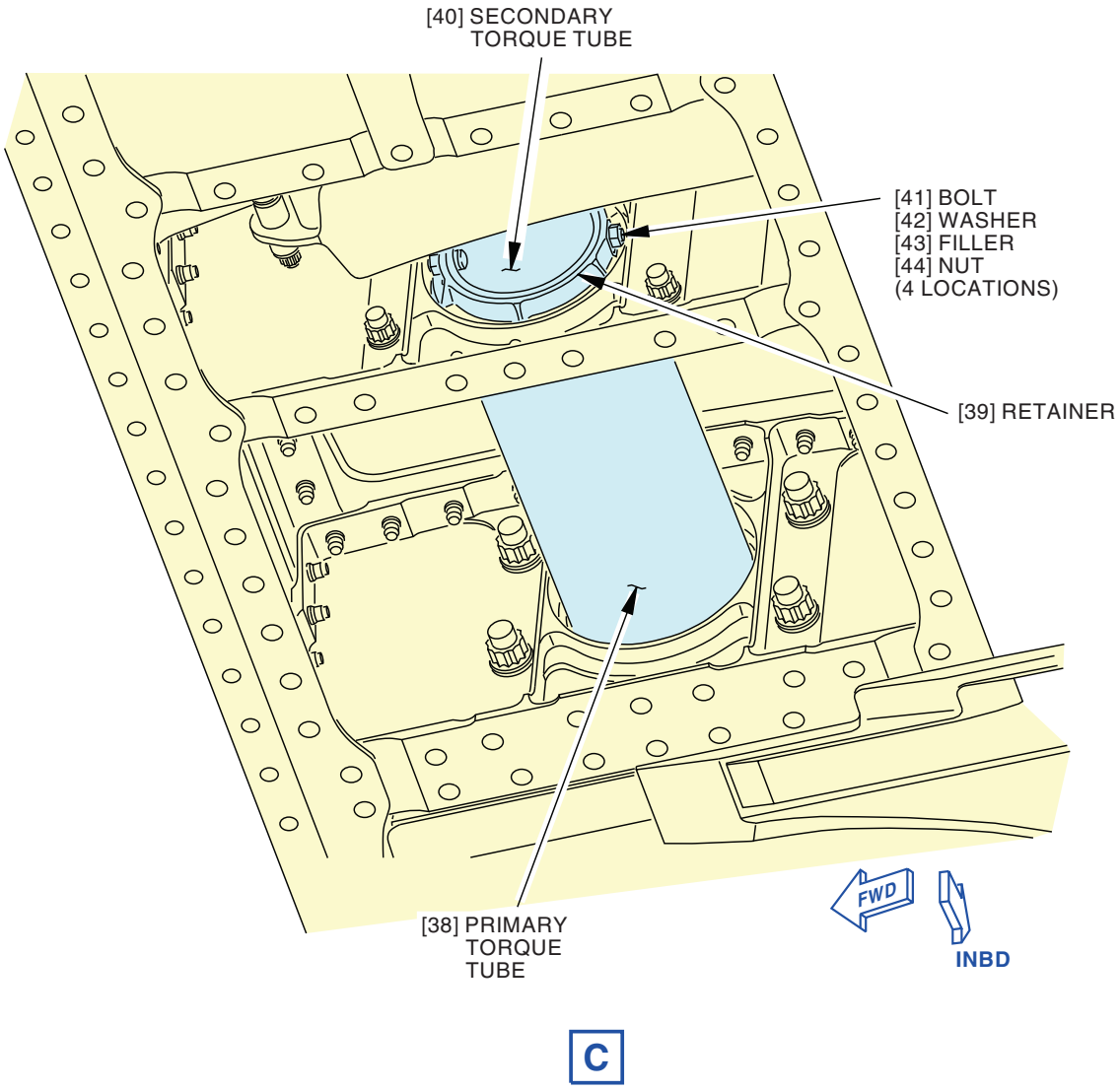
LEFT WING
(RIGHT WING IS EQUIVALENT)



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Inboard Flap Inboard Carriage Installation
Figure 402/27-51-13-990-802 (Sheet 1 of 3)

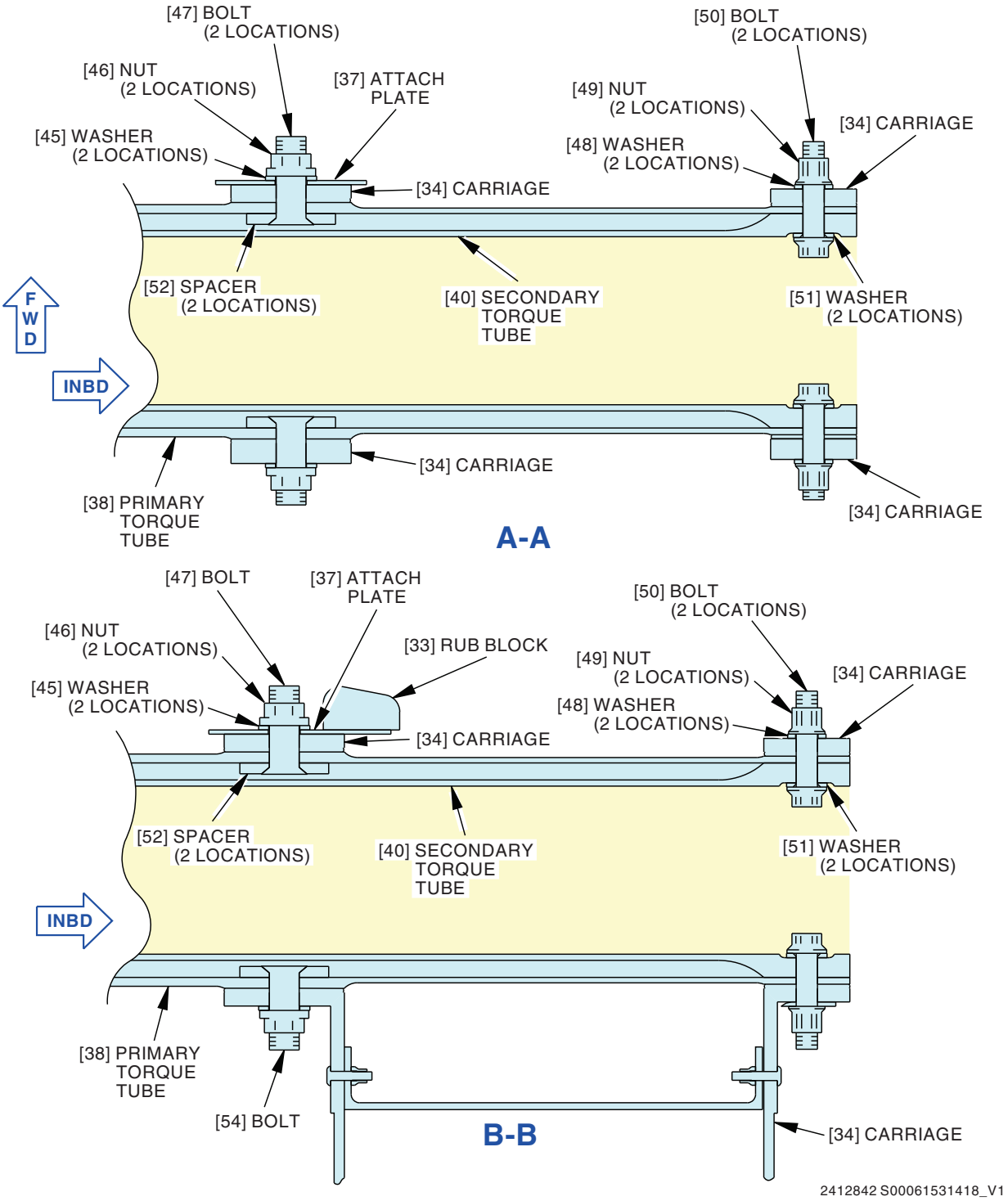
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Inboard Flap Inboard Carriage Installation
Figure 402/27-51-13-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Flap Inboard Carriage Installation
Figure 402/27-51-13-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-13-400-803

7. Inboard Flap Inboard Carriage Installation

(Figure 402)

A. General

(1) This task is applicable to the flap carriages at flap supports No. 4 and 5.

B. References

Reference	Title
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-11-400-801	Inboard Trailing Edge Flap Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
34	Carriage	27-51-13-01A-435	PTF ALL
		27-51-13-01A-440	PTF ALL

E. Location Zones

Zone	Area
541	Left Wing - Fairing Flap Support No. 4
553	Left Wing - Inboard Flap
641	Right Wing - Fairing Flap Support No. 5
653	Right Wing - Inboard Flap

F. Inboard Flap Inboard Carriage Installation

SUBTASK 27-51-13-020-013

(1) Inspect the rub pads [55] for wear.

(a) If the resin material of the rub pads [55] is worn more than 40%, replace the rub pads [55].

SUBTASK 27-51-13-420-003

(2) Install the carriage [34] on the main flap [1]:

(a) Put the carriage [34] in its position on the primary torque tube [38].

(b) Apply sealant, A00247, to the mating surface of the spacers [52].

(c) Apply sealant, A00247, to the shank and threads of the bolts [47] and bolt [54].

(d) Install the bolts [47] and bolt [54] with the spacers [52] through the primary torque tube [38] and the carriage [34].

(e) Put the attach plate [37] and rub block [33] in their position on the bolts [47].

(f) Install the washers [45] and nuts [46] on the bolts [47] and bolt [54].

1) Tighten the nuts [46] to torque tool set point of 470 in-lb (53.10 N·m).

a) Tighten the nuts [46] again before the sealant becomes hard.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) If they are not installed, install the bolts [35] and washers [36] that attach the rub block [33] to the attach plate [37].

SUBTASK 27-51-13-420-004

(3) Install the secondary torque tube [40] in the main flap [1]:

(a) Put the secondary torque tube [40] in its location in the primary torque tube [38].

(b) Install the bolts [50] and washers [51] through the secondary torque tube [40], primary torque tube [38] and the carriage [34].

(c) Install the washers [48] and nuts [49] on the bolts [50].

(d) Tighten the nuts [49] to torque tool set point of 320 in-lb (36.2 N·m).

1) Tighten the nuts [49] again before the sealant becomes hard.

(e) Install the retainer [39] on the secondary torque tube [40].

1) Turn the retainer [39] until the holes for the bolts [41] align.

2) Apply sealant, A00247, to the fillers [43].

3) Install the bolts [41], washers [42], fillers [43] and nuts [44].

SUBTASK 27-51-13-410-002

(4) Install the lower skin panel [31] on the main flap [1]:

(a) Apply sealant, A00247, to the mating surface of the lower skin panel [31] and the main flap [1].

(b) Put the lower skin panel [31] in its position.

(c) Install the bolts [32]:

1) Apply compound, C00528, to the forward and aft rows of holes in the lower skin panel [31].

2) Install the bolts [32] in the forward and aft rows of holes.

3) Apply sealant, A00247, to the bolts [32] that go in the ribs and stiffeners in the main flap [1].

4) Install the bolts [32] in the ribs and stiffeners.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-13-420-005

(1) Do this task: Inboard Trailing Edge Flap Installation, TASK 27-51-11-400-801.

SUBTASK 27-51-13-440-002

(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD FLAP CARRIAGES - INSPECTION/CHECK

1. General

A. This procedure contains the tasks to do inspections of the carriages for the inboard trailing edge flaps.

TASK 27-51-13-200-801

2. Inboard Flap Carriage Inspection (Selection)

A. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
541	Left Wing - Fairing Flap Support No. 4
542	Left Wing - Fairing Flap Support No. 3
641	Right Wing - Fairing Flap Support No. 5
642	Right Wing - Fairing Flap Support No. 6

B. Inboard Flap Carriage Inspection (Selection)

SUBTASK 27-51-13-200-001

(1) Do one of these tasks to inspect the applicable inboard flap carriage:

(a) Do this task: Inboard Flap Outboard Carriage Inspection, TASK 27-51-13-200-802.

(b) Do this task: Inboard Flap Inboard Carriage Inspection, TASK 27-51-13-200-803.

——— END OF TASK ———

TASK 27-51-13-200-802

3. Inboard Flap Outboard Carriage Inspection

(Figure 601)

A. General

(1) This task gives wear limits for the outboard carriage on the inboard flap.

(2) This task is applicable to the carriages at flap supports No. 3 and 6.

B. References

Reference	Title
27-51-13-000-802	Inboard Flap Outboard Carriage Removal (P/B 401)
27-51-13-400-802	Inboard Flap Outboard Carriage Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
542	Left Wing - Fairing Flap Support No. 3
642	Right Wing - Fairing Flap Support No. 6

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Inboard Flap Outboard Carriage Inspection

SUBTASK 27-51-13-020-006

(1) Do this task: Inboard Flap Outboard Carriage Removal, TASK 27-51-13-000-802.

SUBTASK 27-51-13-020-014

(2) Inspect the rub pads [8] for wear.

(a) If the rub pads [8] are worn greater than 50%, they will need to be replaced.

SUBTASK 27-51-13-020-007

(3) Use these tools to do a check for wear on the part of the inboard flap outboard carriage:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-51-13-020-008

(4) Examine the parts of the inboard flap outboard carriage (Table 601):

(a) Remove the parts that you will measure.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are not in tolerance, replace or repair the parts.

Table 601/27-51-13-993-801

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE CLEARANCE inches/mm
#1 BUSHING	I.D.	1.1870 in 30.150 mm	1.1878 in 30.170 mm	1.1915 in 30.264 mm	0.0050 in
#1 PIN	O.D.	1.1855 in 30.112 mm	1.1865 in 30.137 mm	1.1828 in 30.043 mm	0.1270 mm
#2 ROLLER	I.D.	1.1868 in 30.145 mm	1.1875 in 30.163 mm	1.1915 in 30.264 mm	0.0050 in
#2 PIN	O.D.	1.1855 in 30.112 mm	1.1865 in 30.137 mm	1.1825 in 30.036 mm	0.1270 mm
#3 ROLLER	O.D.	2.9990 in 76.175 mm	3.0010 in 76.225 mm	2.9970 in 76.124 mm	
#4 BUSHING	I.D.	1.2506 in 31.765 mm	1.2514 in 31.786 mm	1.2540 in 31.852 mm	0.0050 in
#4 PIN	O.D.	1.2480 in	1.2490 in	1.2470 in	0.1270 mm

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-51-13-993-801 (Continued)

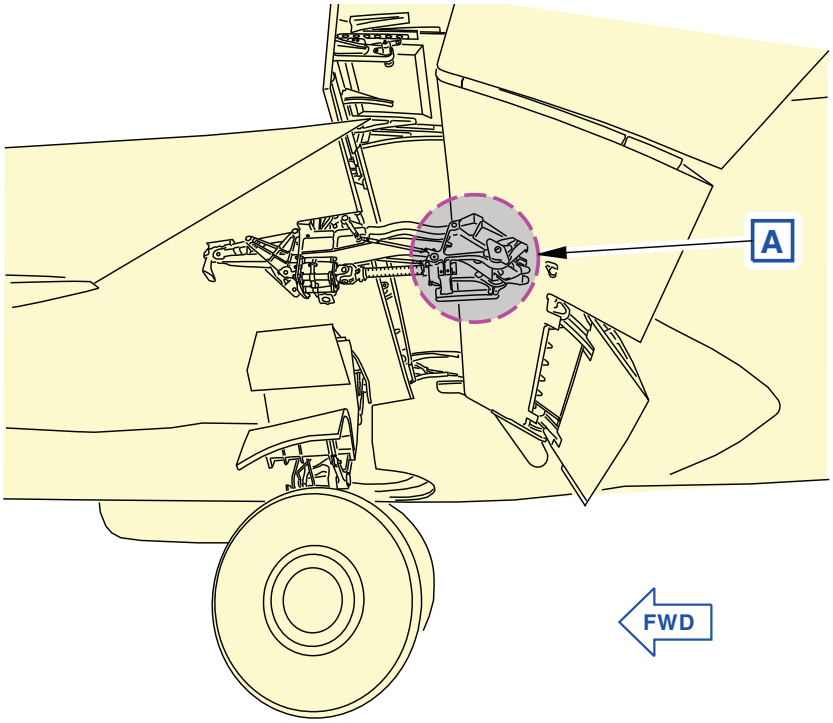
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE CLEARANCE inches/mm
		31.699 mm	31.725 mm	31.674 mm	
#5 BUSHING	I.D.	3.3000 in 88.820 mm	3.3020 in 83.871 mm	3.3040 in 83.922 mm	0.0050 in 0.1270 mm
#5 CARTRIDGE	O.D.	3.2980 in 83.769 mm	3.2990 in 83.795 mm	3.2970 in 83.744 mm	
#6 ROLLER	I.D.	1.2493 in 31.732 mm	1.2500 in 31.750 mm	1.2540 in 31.852 mm	0.0050 in 0.1270 mm
#6 PIN	O.D.	1.2480 in 31.699 mm	1.2490 in 31.725 mm	1.2450 in 31.623 mm	
#7 ROLLER	O.D.	2.9990 in 76.175 mm	3.0010 in 76.225 mm	2.9970 in 76.124 mm	

SUBTASK 27-51-13-420-006

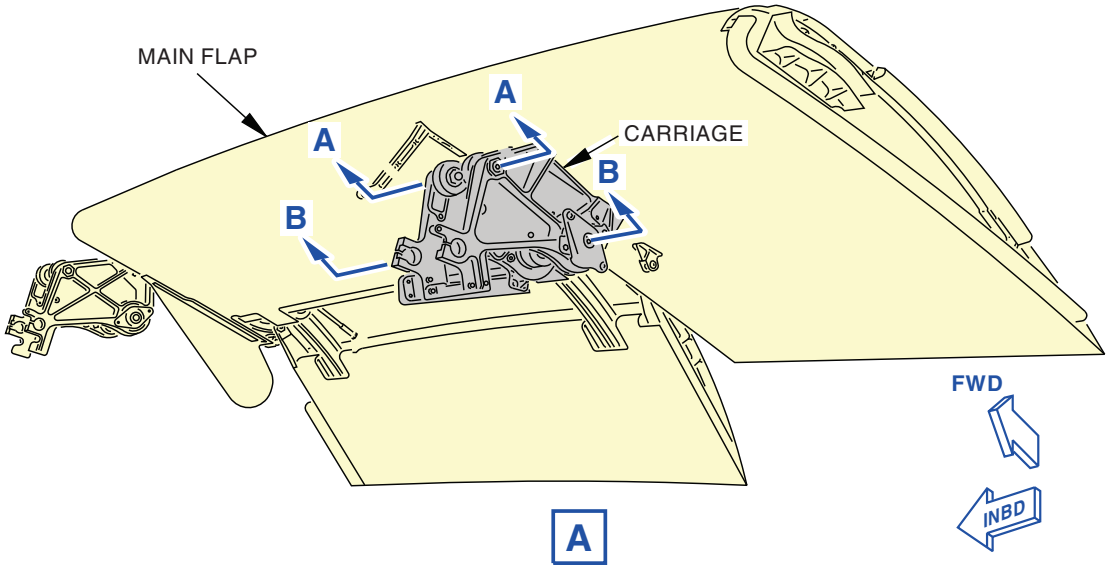
(5) Do this task: Inboard Flap Outboard Carriage Installation, TASK 27-51-13-400-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

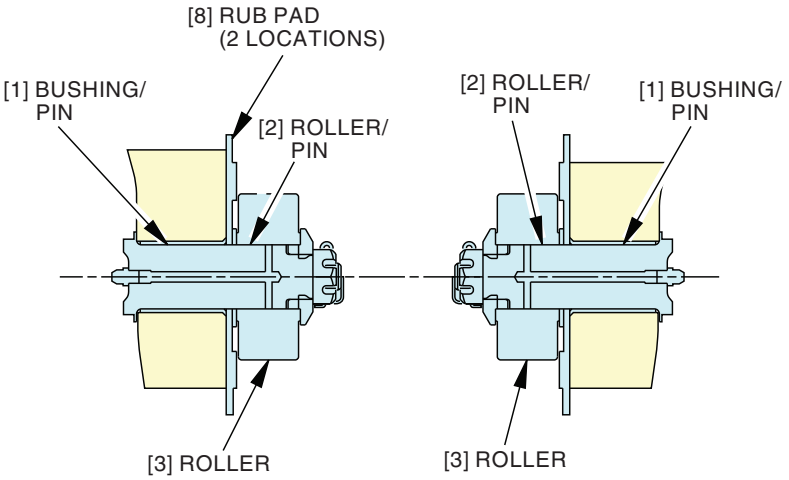


LEFT WING
(RIGHT WING IS EQUIVALENT)

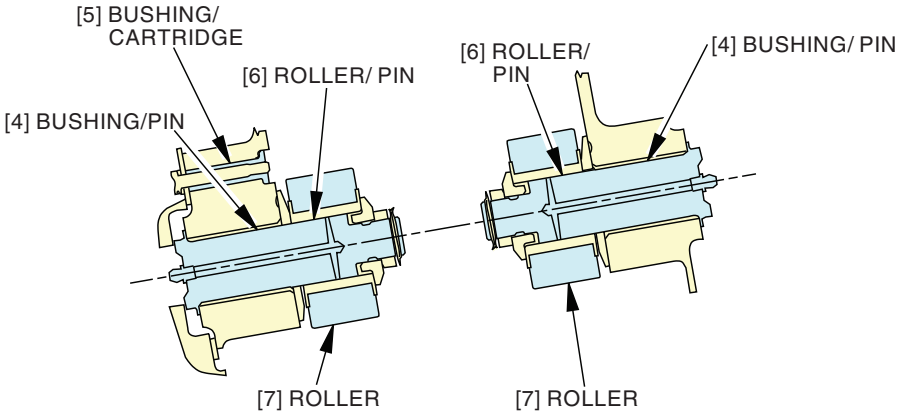


Inboard Flap Outboard Carriage Inspection
Figure 601/27-51-13-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

2412844 S00061531423_V3

Inboard Flap Outboard Carriage Inspection
Figure 601/27-51-13-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-13-200-803

4. Inboard Flap Inboard Carriage Inspection

(Figure 602)

A. General

(1) This task gives wear limits for the inboard carriage on the inboard flap.

(2) This task is applicable to the carriages at flap supports No. 4 and 5.

B. References

Reference	Title
27-51-13-000-803	Inboard Flap Inboard Carriage Removal (P/B 401)
27-51-13-400-803	Inboard Flap Inboard Carriage Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
541	Left Wing - Fairing Flap Support No. 4
641	Right Wing - Fairing Flap Support No. 5

E. Inboard Flap Inboard Carriage Inspection

SUBTASK 27-51-13-020-009

(1) Do this task: Inboard Flap Inboard Carriage Removal, TASK 27-51-13-000-803.

SUBTASK 27-51-13-020-015

(2) inspect the rub pads [8] for wear.

(a) If the rub pads [8] are worn greater than 40%, they will need to be replaced.

SUBTASK 27-51-13-020-010

(3) Use these tools to do a check for wear on the part of the inboard flap inboard carriage:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-51-13-020-011

(4) Examine the parts of the inboard flap inboard carriage (Table 602):

(a) Remove the parts that you will measure.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are not in tolerance, replace or repair the parts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 602/27-51-13-993-802

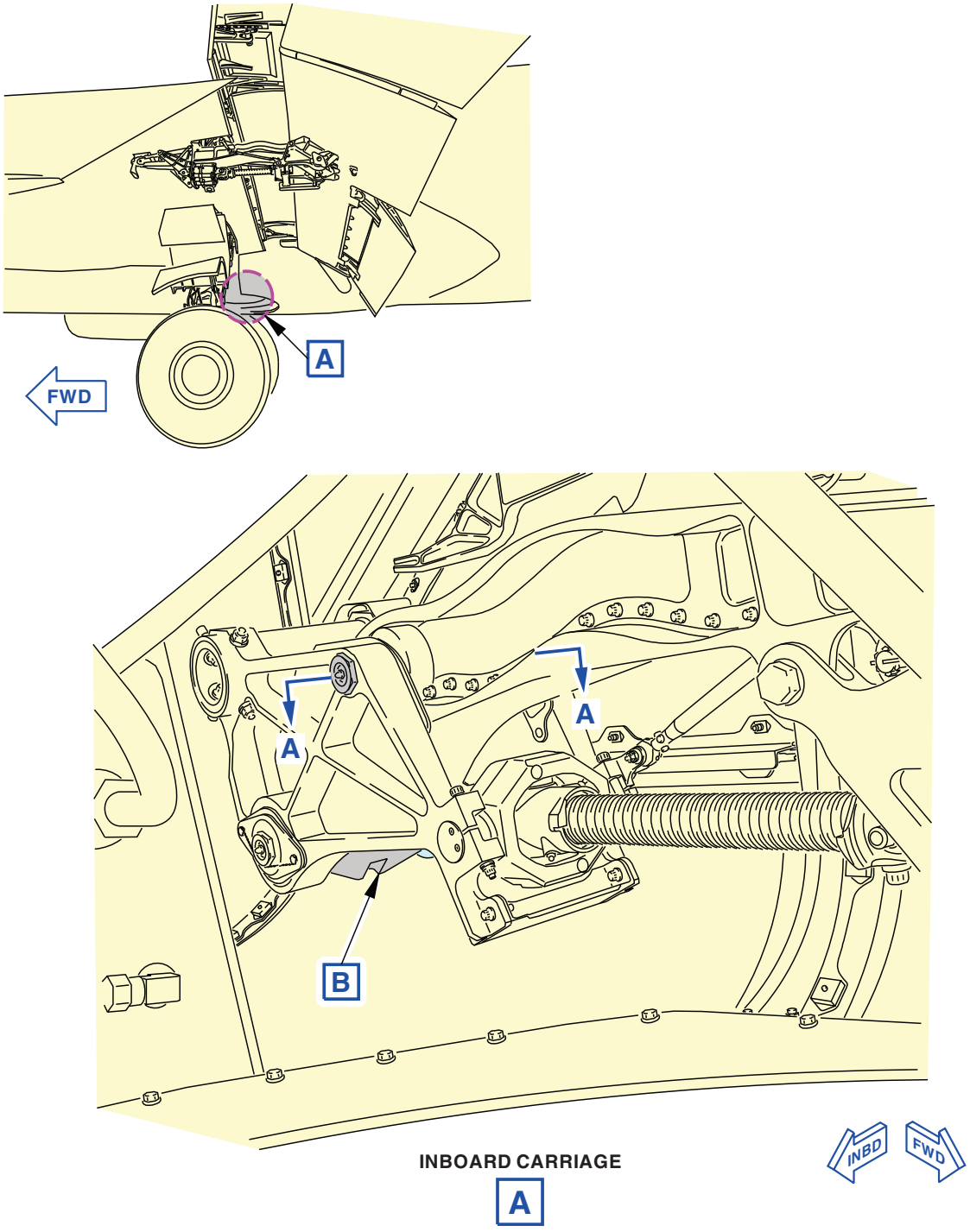
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE CLEARANCE inches/mm
#1 BUSHING	I.D.	1.1870 in 30.150 mm	1.1878 in 30.170 mm	1.1915 in 30.264 mm	0.0050 in 0.1270 mm
#1 PIN	O.D.	1.1855 in 30.112 mm	1.1865 in 30.137 mm	1.1828 in 30.043 mm	
#2 ROLLER	I.D.	1.1868 in 30.145 mm	1.1875 in 30.163 mm	1.1915 in 30.264 mm	0.0050 in 0.1270 mm
#2 PIN	O.D.	1.1855 in 30.112 mm	1.1865 in 30.137 mm	1.1825 in 30.036 mm	
#3 ROLLER	O.D.	2.9990 in 76.175 mm	3.0010 in 76.225 mm	2.9970 in 76.124 mm	
#4 ECCENTRIC BUSHING	I.D.	1.2506 in 31.765 mm	1.2514 in 31.786 mm	1.2540 in 31.852 mm	0.0050 in 0.1270 mm
#4 PIN	O.D.	1.2480 in 31.699 mm	1.2490 in 31.725 mm	1.2470 in 31.674 mm	
#5 ROLLER	I.D.	1.2493 in 31.732 mm	1.2500 in 31.750 mm	1.2540 in 31.852 mm	0.0050 in 0.1270 mm
#5 PIN	O.D.	1.2480 in 31.699 mm	1.2490 in 31.725 mm	1.2450 in 31.623 mm	
#6 ROLLER	O.D.	2.9990 in 76.175 mm	3.0010 in 76.225 mm	2.9970 in 76.124 mm	

SUBTASK 27-51-13-420-007

(5) Do this task: Inboard Flap Inboard Carriage Installation, TASK 27-51-13-400-803.

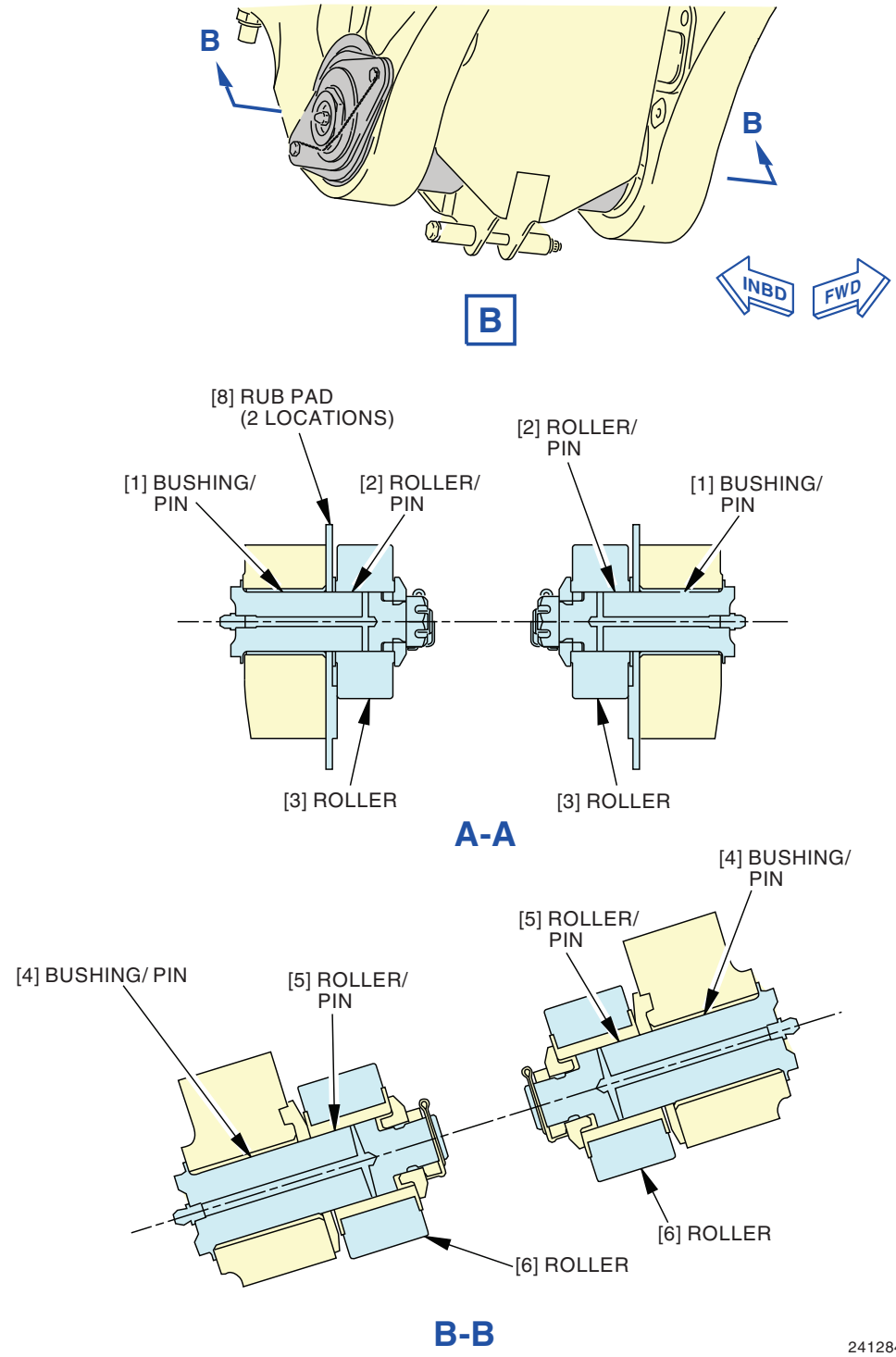
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Flap Inboard Carriage Inspection
Figure 602/27-51-13-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Flap Inboard Carriage Inspection
Figure 602/27-51-13-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD AFT FLAP ACTUATING MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Inboard aft flap actuating mechanism removal

(2) Inboard aft flap actuating mechanism installation.

B. The aft flap actuating mechanism includes these parts:

(1) The aft flap drive rod

(2) The interconnect pushrod

(3) The crankshaft

(4) The aft flap pushrods

(5) The cables

(6) The outboard bellcrank

(7) The inboard bellcrank

TASK 27-51-14-000-801

2. Inboard Aft Flap Actuating Mechanism Removal

(Figure 401)

A. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-18-000-802	Inboard Flap Support Aft Fairing Removal (P/B 401)

B. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

C. Prepare for the Removal

SUBTASK 27-51-14-860-001

(1) Extend the trailing edge flaps to the 40-unit position. To extend them, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-51-14-040-001

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-14-010-001

(3) Do this task: Inboard Flap Support Aft Fairing Removal, TASK 27-51-18-000-802.

SUBTASK 27-51-14-010-002

(4) Remove the upper skin panel [4] from the outboard end of the main flap [1]:

(a) Remove the bolts [3] that attach the upper skin panel [4] to the main flap [1].

(b) Remove the upper skin panel [4].

SUBTASK 27-51-14-010-003

(5) Remove the lower skin panel [58] from the inboard end of the main flap [1]:

(a) Remove the bolts [57] that attach the lower skin panel [58] to the main flap [1].

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(b) Remove the lower skin panel [58].

SUBTASK 27-51-14-010-004

(6) Remove the leading edge panel [40] from the outboard end of the main flap [1]:

(a) Remove the bolts [42] that attach the leading edge panel [40] to the main flap [1].

(b) Remove the leading edge panel [40].

D. Inboard Aft Flap Mechanism Removal

SUBTASK 27-51-14-020-001

(1) Remove the aft flap drive rod assembly [43]:

(a) Remove the pins [50], nuts [49], washers [48], bushings [47], washers [46] and bolts [45] that attach the aft flap drive rod assembly [43] to the drive rod fitting [44] and the crankshaft assembly [17].

(b) Remove the aft flap drive rod assembly [43].

SUBTASK 27-51-14-020-002

(2) Remove the interconnect push rod assembly [18]:

(a) Remove the pin [26], nut [25], washer [24], washer [22], bolt [21] and bushing [23] that attach the interconnect push rod assembly [18] to the crankshaft assembly [17].

(b) Remove the pin [16], nut [15], washer [14], washer [12], bolt [11] and bushing [13] that attach the interconnect push rod assembly [18] to the outboard bellcrank assembly [33].

(c) Remove the interconnect push rod assembly [18].

SUBTASK 27-51-14-020-003

(3) Remove the crankshaft assembly [17]:

(a) Remove the pin [56], nut [55], washer [54], washers [52], bolts [51] and bushings [53] that attach the crankshaft assembly [17] to the ribs [41].

(b) Remove the crankshaft assembly [17].

SUBTASK 27-51-14-020-004

(4) Remove the aft flap push rod assemblies [34]:

(a) Remove the pin [32], nut [31], washer [30], washer [28], bolt [27] and bushing [29] that attach the outboard aft flap push rod assembly [34] to the outboard bellcrank assembly [33].

(b) Remove the pin [32], nut [31], washer [30], bushing [29], washer [28] and bolt [27] that attach the inboard aft flap push rod assembly [34] to the inboard bellcrank assembly [67].

(c) Remove the pin [80], nut [79], washer [78], washer [76], bolt [75] and bushing [77] that attach the aft flap push rod assemblies [34] to the aft flap [2].

(d) Remove the aft flap push rod assemblies [34].

SUBTASK 27-51-14-020-005

(5) Remove the cable assemblies [19] and cable assemblies [64]:

(a) Remove the locking clips [66] from the turnbuckles [65].

(b) Turn the turnbuckles [65] to loosen the cable assemblies [19] and cable assemblies [64].

(c) Remove the pins [63], nuts [62], washers [61], washers [60], teflon washers [90] and bolts [59] that attach the cable assemblies [64] to the inboard bellcrank assembly [67].

(d) Remove the pins [39], nuts [38], washer [37], washers [36], teflon washers [90] and bolts [35] that attach the cable assemblies [19] to the outboard bellcrank assembly [33].

(e) Remove the cable assemblies [19] and cable assemblies [64].

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SUBTASK 27-51-14-020-006

(6) Remove the outboard bellcrank assembly [33]:

(a) Remove the pin [10], nut [9], washer [8], washer [6], bolt [5] and bushings [7] that attach the outboard bellcrank assembly [33] to the main flap [1].

(b) Remove the outboard bellcrank assembly [33].

SUBTASK 27-51-14-020-007

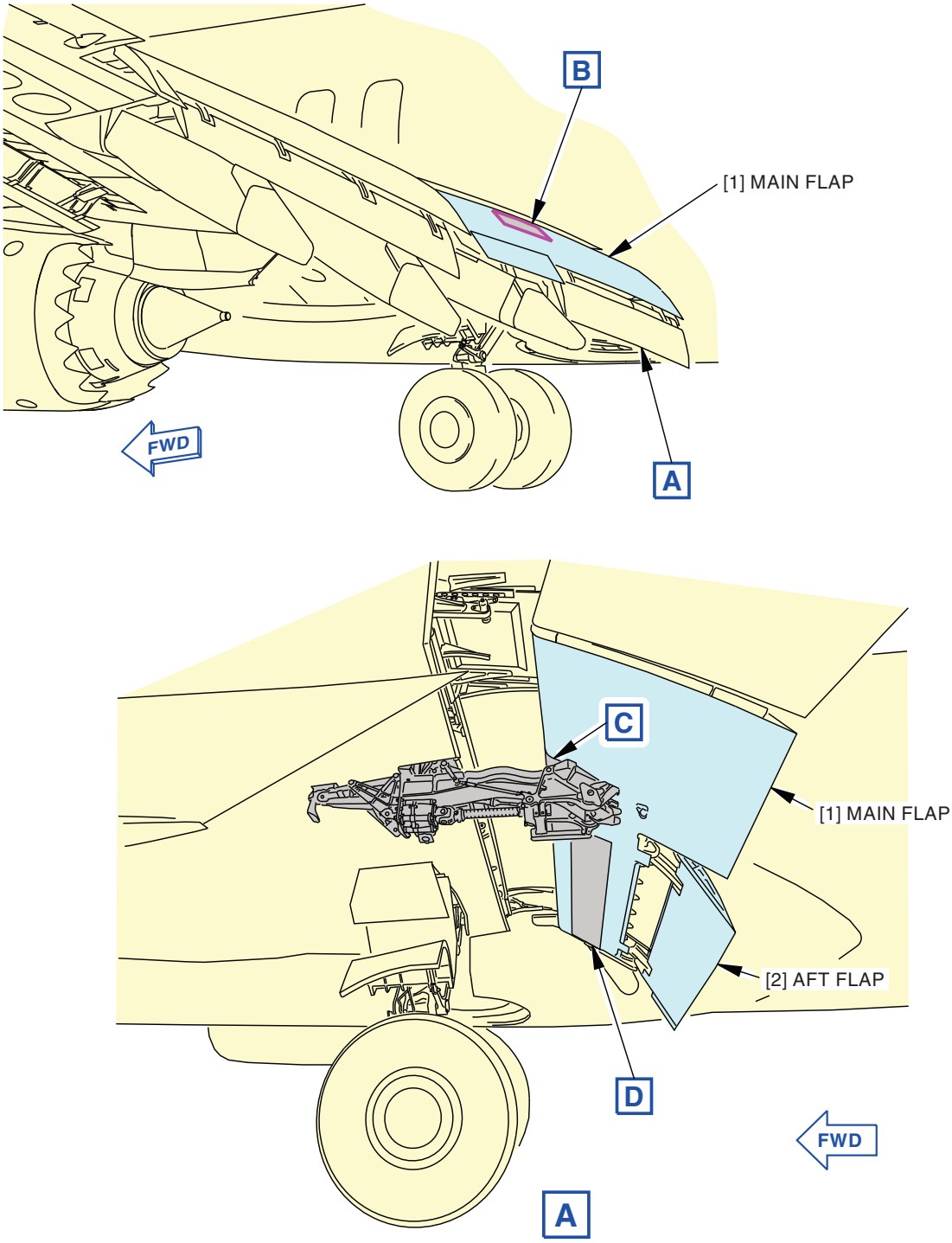
(7) Remove the inboard bellcrank assembly [67]:

(a) Remove the pin [73], nut [72], washer [71], washer [69], bolt [68] and bushings [70] that attach the inboard bellcrank assembly [67] to the main flap [1].

(b) Remove the inboard bellcrank assembly [67].

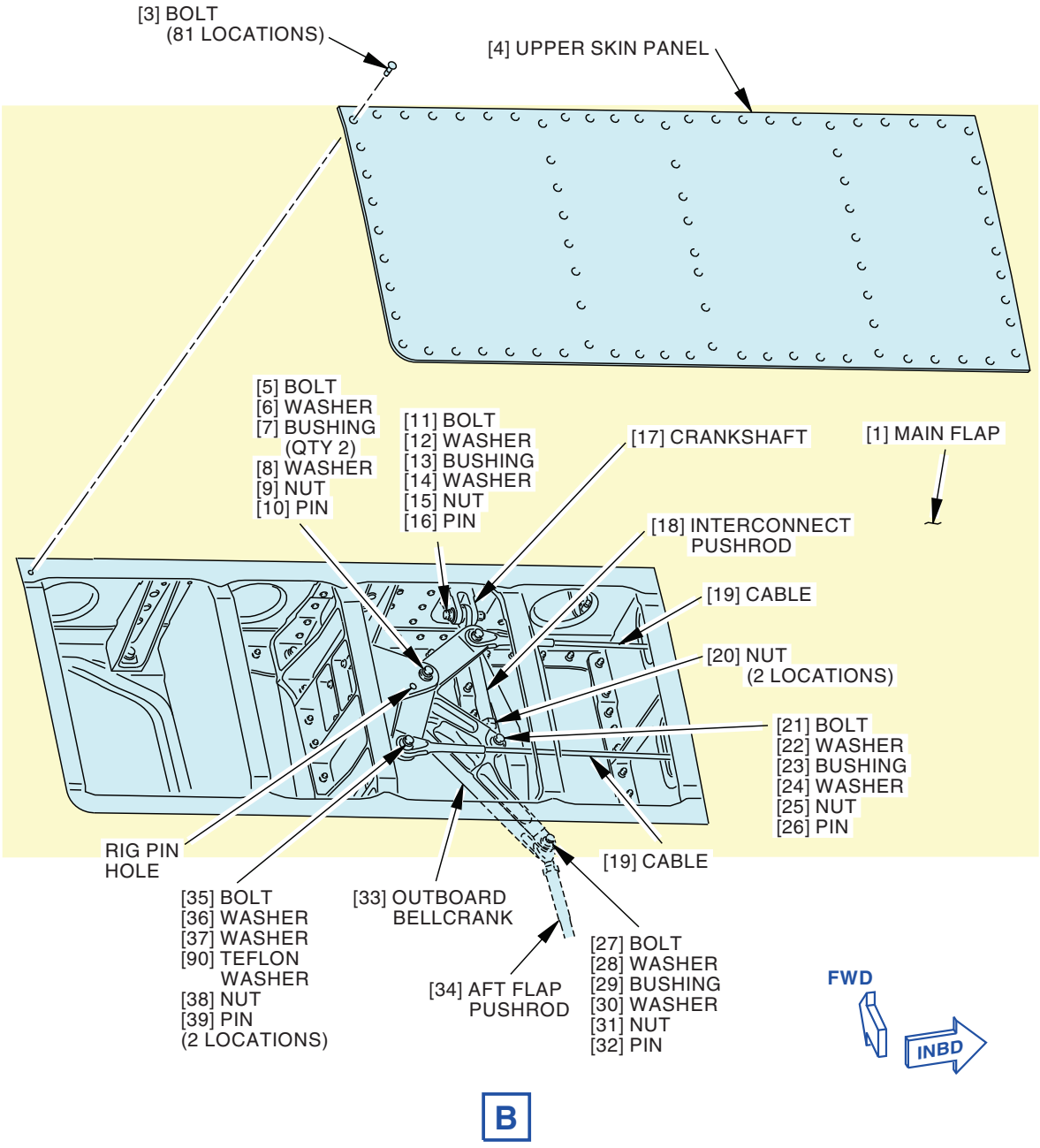
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Aft Flap Actuating Mechanism Installation
Figure 401/27-51-14-990-801 (Sheet 1 of 5)

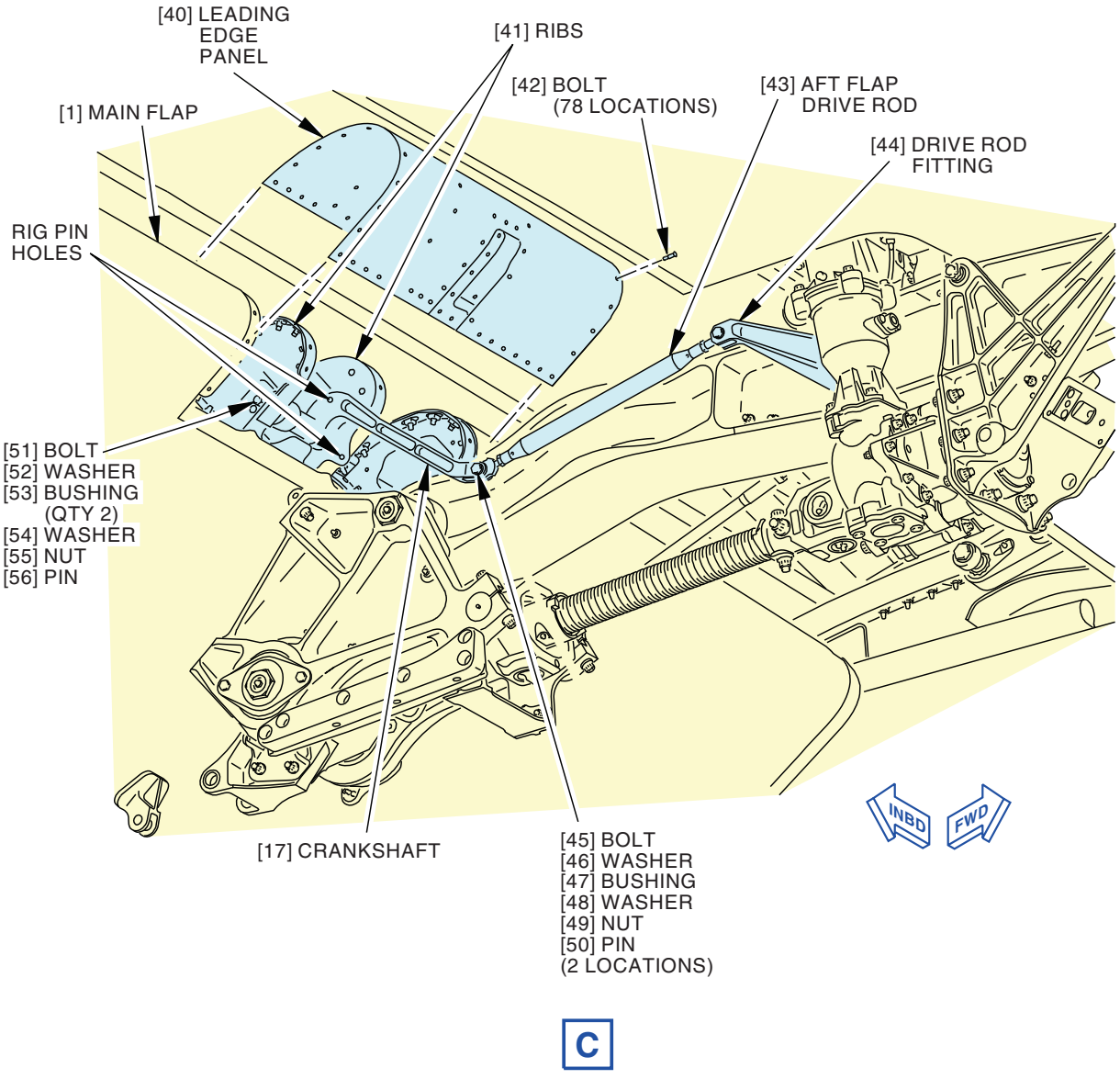
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Inboard Aft Flap Actuating Mechanism Installation
Figure 401/27-51-14-990-801 (Sheet 2 of 5)

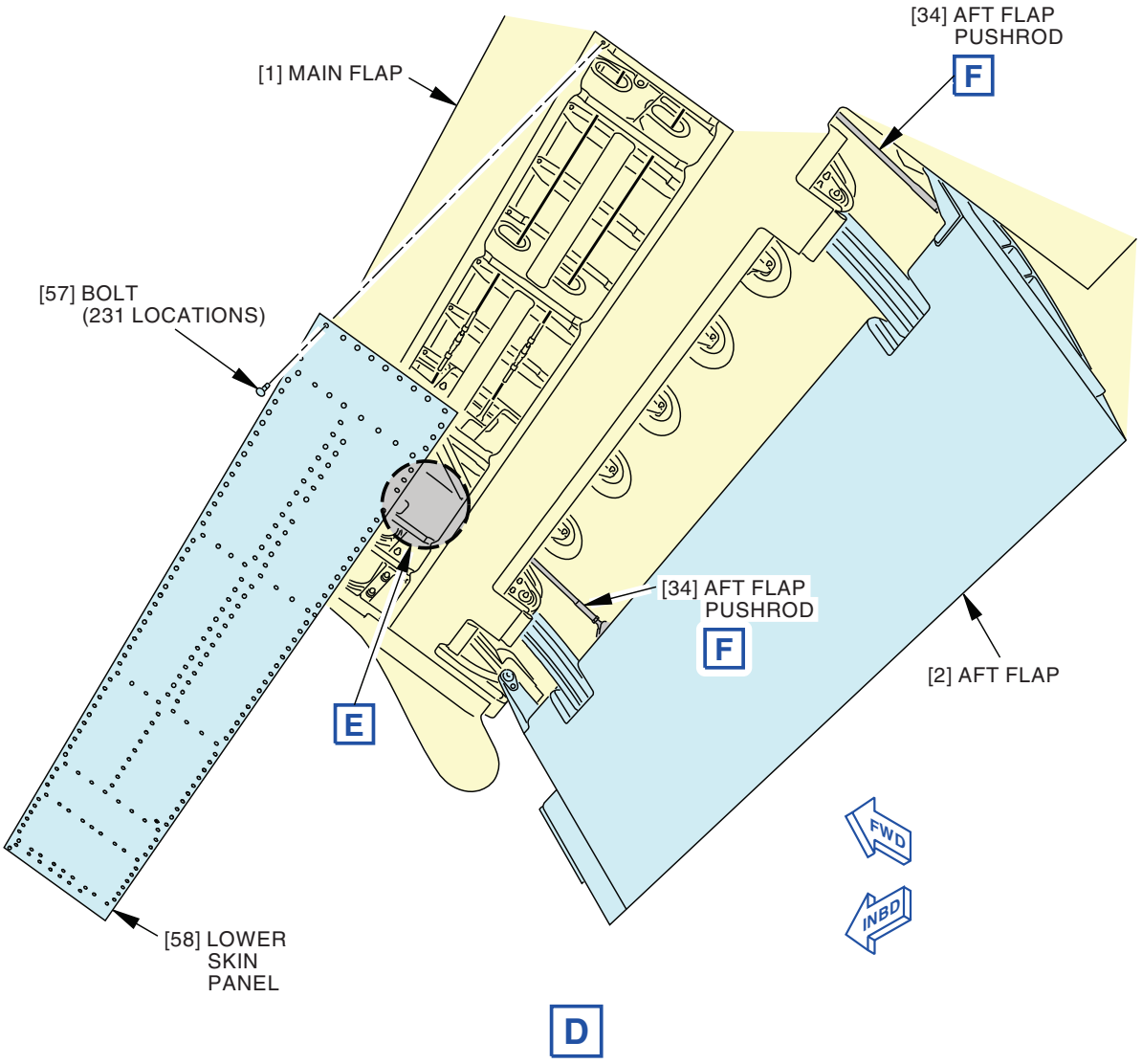
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Inboard Aft Flap Actuating Mechanism Installation
Figure 401/27-51-14-990-801 (Sheet 3 of 5)

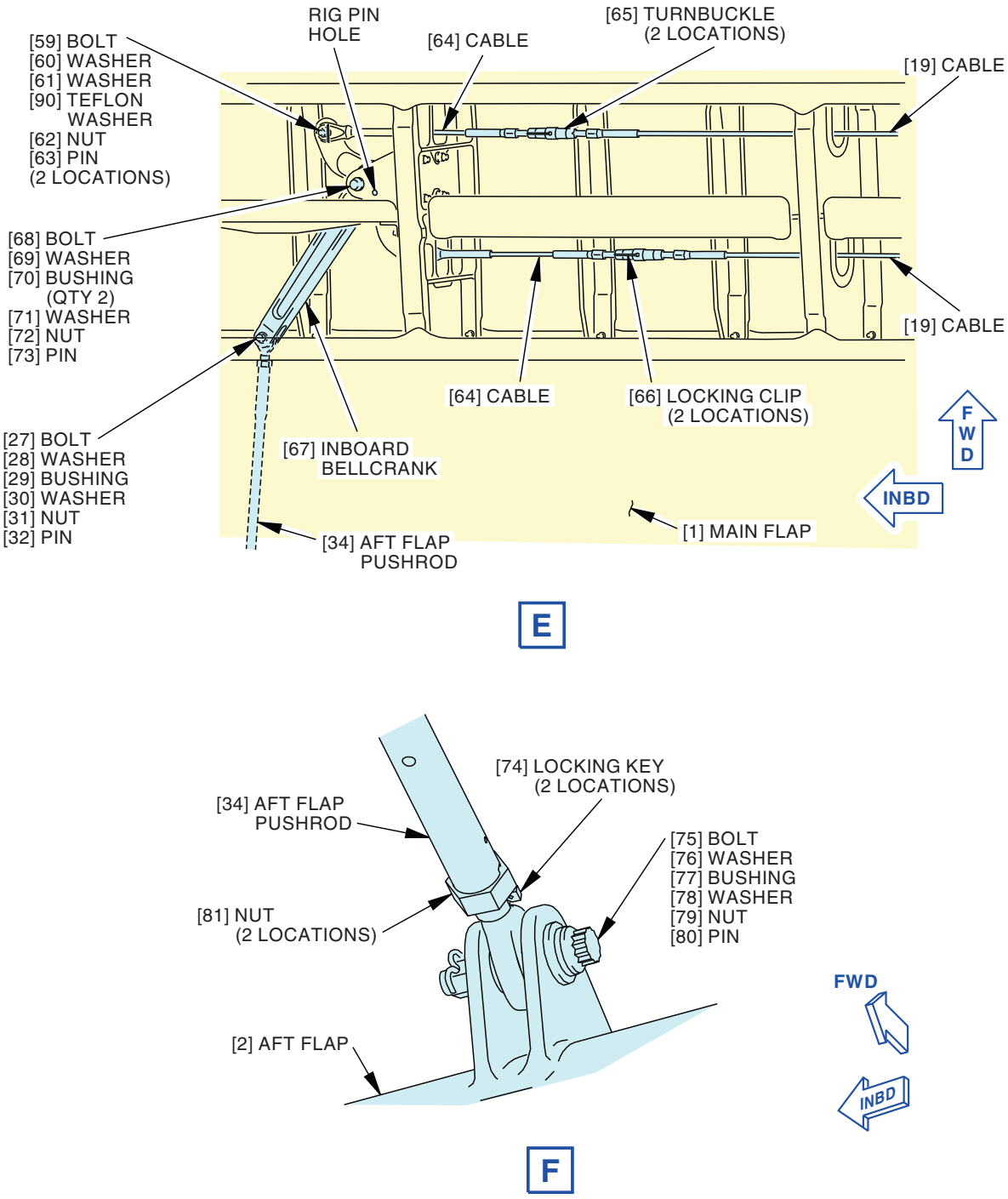
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Inboard Aft Flap Actuating Mechanism Installation
Figure 401/27-51-14-990-801 (Sheet 4 of 5)

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Inboard Aft Flap Actuating Mechanism Installation
Figure 401/27-51-14-990-801 (Sheet 5 of 5)

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TASK 27-51-14-400-801

3. Inboard Aft Flap Actuating Mechanism Installation

(Figure 401)

A. References

Reference	Title
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-51-18-400-802	Inboard Flap Support Aft Fairing Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1554	Tensiometer - Cable, High Tension (200 lbs and above) Part #: 102-03108 Supplier: 21844 Part #: 102-03120 Supplier: 21844 Part #: ACM-300 Supplier: 13331 Part #: ACM-600 Supplier: 13331 Part #: ACX-500-1 Supplier: 13331 Part #: T5-8008-106-00 Supplier: 0N8U4 Part #: T60-1001-C9-1A Supplier: 0N8U4
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	Pin	27-51-14-01A-135	PTF ALL
16	Pin	27-51-14-01A-030	PTF ALL
17	Crankshaft assembly	27-51-14-01A-125	PTF ALL
		27-51-14-01A-130	PTF ALL
18	Rod assembly	27-51-14-01A-210	PTF ALL
19	Cable assembly	27-51-14-01A-290	PTF ALL
		27-51-14-01A-300	PTF ALL
26	Pin	27-51-14-01A-135	PTF ALL
32	Pin	27-51-14-01A-135	PTF ALL
33	Bellcrank assembly	27-51-14-01A-310	PTF ALL
34	Rod assembly	27-51-14-01A-245	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
39	Pin	27-51-14-01A-135	PTF ALL
43	Rod assembly	27-51-14-01-050	PTF ALL
50	Pin	27-51-14-01-020	PTF ALL
56	Pin	27-51-14-01A-060	PTF ALL
63	Pin	27-51-14-01A-135	PTF ALL
64	Cable assembly	27-51-14-01A-300	PTF ALL
66	Clip	27-51-14-01A-285	PTF ALL
67	Bellcrank assembly	27-51-14-01A-355	PTF ALL
73	Pin	27-51-14-01A-135	PTF ALL
80	Pin	27-51-12-01A-020	PTF ALL

E. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

F. Inboard Aft Flap Actuating Mechanism Installation

SUBTASK 27-51-14-420-001

(1) Install the inboard bellcrank assembly [67]:

(a) Put the inboard bellcrank assembly [67] in its position in the main flap [1].

(b) Install the bushings [70], bolt [68], washer [69], washer [71] and nut [72] to attach the inboard bellcrank assembly [67].

NOTE: Install the bolt [68] with the head down.

(c) Tighten the nut [72] to 100 in-lb (11.3 N·m) to 150 in-lb (16.9 N·m).

(d) Install the pin [73] in the nut [72].

SUBTASK 27-51-14-420-002

(2) Install the outboard bellcrank assembly [33]:

(a) Put the outboard bellcrank assembly [33] in its position in the main flap [1].

(b) Install the bushings [7], bolt [5], washer [6], washer [8] and nut [9] to attach the outboard bellcrank assembly [33].

NOTE: Install the bolt [5] with the head up.

(c) Tighten the nut [9] to 100 in-lb (11.3 N·m) to 150 in-lb (16.9 N·m).

(d) Install the pin [10] in the nut [9].

SUBTASK 27-51-14-420-003

(3) Install the cable assemblies [19] and cable assemblies [64]:

(a) Put the cable assemblies [19] and cable assemblies [64] in their position in the main flap [1].

(b) Install the bolts [35], washers [36], washers [37], teflon washers [90] and nuts [38] to attach the cable assemblies [19] to the outboard bellcrank assembly [33].

(c) Install the bolts [59], washers [60], washers [61], teflon washers [90] and nuts [62] to attach the cable assemblies [64] to the inboard bellcrank assembly [67].

(d) Tighten the nuts [38] and nuts [62] to 50 in-lb (5.6 N·m) to 80 in-lb (9.0 N·m).

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(e) Install the pins [39] in the nuts [38].

(f) Install the pins [63] in the nuts [62].

(g) Connect the cable assemblies [19] and cable assemblies [64] together at the turnbuckles [65].

NOTE: Do not tighten the turnbuckles [65] or install the locking clips [66] at this time.

SUBTASK 27-51-14-420-004

(4) Install the crankshaft assembly [17]:

(a) Put the crankshaft assembly [17] in its position between the ribs [41] on the leading edge of the main flap [1].

(b) Install the bushings [53], bolt [51], washer [52], washer [54], and nut [55] that attach the crankshaft assembly [17] to the ribs [41].

(c) Tighten the nut [55] to 160 in-lb (18.1 N·m) to 240 in-lb (27.1 N·m).

(d) Install the pin [56] in the nut [55].

SUBTASK 27-51-14-420-005

(5) Install the interconnect push rod assembly [18]:

(a) Put the interconnect push rod assembly [18] in its position between the crankshaft assembly [17] and the outboard bellcrank assembly [33].

(b) Install the bushing [13], bolt [11], washer [12], washer [14] and nut [15] that attach the interconnect push rod assembly [18] to the crankshaft assembly [17].

NOTE: Install the bolt [11] with the head outboard.

(c) Install the bushing [23], bolt [21], washer [22], washer [24] and nut [25] that attach the interconnect push rod assembly [18] to the outboard bellcrank assembly [33].

NOTE: Install the bolt [21] with the head up.

(d) Tighten the nut [15] and nut [25] to 50 in-lb (5.6 N·m) to 80 in-lb (9.0 N·m).

(e) Install the pin [16] in the nut [15].

(f) Install the pin [26] in the nut [25].

SUBTASK 27-51-14-820-001

(6) Adjust the interconnect push rod assembly [18], cable assemblies [19] and cable assemblies [64]:

(a) Move the crankshaft assembly [17] to the retracted position.

(b) Install the rig pin from the rigging pin kit, SPL-1585 (F70207–113 or F70207–114) in the crankshaft assembly [17].

(c) Install the rig pin from the rigging pin kit, SPL-1585 (F70207–113 or F70207–114) in the outboard bellcrank assembly [33].

(d) If you cannot install the rig pin in the outboard bellcrank assembly [33], adjust the length of the interconnect push rod assembly [18]:

1) Loosen the nuts [20] on the interconnect push rod assembly [18].

2) Turn the interconnect push rod assembly [18] until you can install the rig pin in the outboard bellcrank assembly [33].

3) Tighten the nuts [20] on the interconnect push rod assembly [18].

4) Install MS20995C32 lockwire, G01048 on the nuts [20].

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(e) Install the rig pin from the rigging pin kit, SPL-1585 (F70207–113 or F70207–114) in the inboard bellcrank assembly [67].

(f) If you cannot install the rig pin in the inboard bellcrank assembly [67], adjust the cables:

1) Remove the rig pin from the rigging pin kit, SPL-1585 from the crankshaft assembly [17].

2) If it is necessary, remove the tension from the forward cable assembly [19] and cable assembly [64]:

a) Remove the locking clips [66] from the turnbuckles [65].

b) Turn the turnbuckles [65].

3) Install the rig pin from the rigging pin kit, SPL-1585 (F70207–113 or F70207–114) in the inboard bellcrank assembly [67] and the outboard bellcrank assembly [33].

4) Use a high tension cable tensiometer, COM-1554 to adjust the tension in the aft cable assembly [19] until it is in the limit of 200 ±25 lbf (889.6 ±111.2 N).

NOTE: The cable is 3/16” (0.1875 in. (4.7625 mm)) diameter.

5) Use a high tension cable tensiometer, COM-1554 to adjust the tension in the forward cable assembly [19] until you can easily remove and install the rig pins in the bellcranks.

a) Make sure that the tension in the aft cable assembly [19] stays in the limit.

NOTE: The cable is 3/16” (0.1875 in. (4.7625 mm)) diameter.

(g) Install the locking clips [66] in the turnbuckles [65] for the cable assemblies [19] and cable assemblies [64].

(h) Remove the rig pins from the inboard bellcrank assembly [67], outboard bellcrank assembly [33] and the crankshaft assembly [17].

SUBTASK 27-51-14-420-006

(7) Install the aft flap push rod assemblies [34]:

(a) Put the aft flap push rod assemblies [34] in their location between the aft flap [2] and the bellcranks.

(b) Install the bolt [27], washer [28], bushing [29], washer [30] and nut [31] that attach the inboard aft flap push rod assembly [34] to the inboard bellcrank assembly [67].

(c) Install the bushing [29], bolt [27], washer [28], washer [30] and nut [31] that attach the outboard aft flap push rod assembly [34] to the outboard bellcrank assembly [33].

(d) Install the bushings [77],bolts [75], washers [76], washers [78] and nuts [79] that attach the aft flap push rod assemblies [34] to the aft flap [2].

(e) Tighten the nuts [31] and nuts [79] to 50 in-lb (5.6 N·m) to 80 in-lb (9.0 N·m).

(f) Install pins [32] in the nuts [31].

(g) Install pins [80] in the nuts [79].

SUBTASK 27-51-14-820-002

(8) If the aft flap [2] is installed, do an adjustment of the aft flap push rod assemblies [34]:

(a) Use the crankshaft assembly [17] to manually move the aft flap [2] to the retracted position.

(b) Install the rig pin from the rigging pin kit, SPL-1585 (F70207–113 or F70207–114) in the crankshaft assembly [17].

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(c) Measure the distance between the trailing edge of the aft flap [2] and the trailing edge of the upper cove panel [101] on the main flap [1] (Figure 402).

NOTE: Measure the distance aft of the inboard and outboard aft flap push rod assemblies [34].

1) At the outboard push rod assembly [34], make sure that the distance is 9.2745 ±0.0600 in. (235.57 ±1.53 mm).

2) At the inboard push rod assembly [34], make sure that the distance is 9.2319 ±0.0600 in. (234.49 ±1.52 mm).

(d) If the distance at the inboard or outboard pushrod is not in the limit, then adjust the length of the applicable aft flap push rod assembly [34] (Figure 401):

1) Remove the rig pin from the crankshaft assembly [17].

2) Use the crankshaft assembly [17] to manually extend the aft flap [2] to the extended position.

3) Adjust the length of the applicable aft flap push rod assembly [34]:

NOTE: For each 90 degree turn of the aft flap push rod assembly [34], the trailing edge of the aft flap will move 0.04 in. (1.02 mm).

a) Loosen the nuts [81] on the aft flap push rod assembly [34].

b) Adjust the length of the aft flap push rod assembly [34].

c) Tighten the nuts [81].

d) Install MS20995C32 lockwire, G01048 on the nuts [81] and the locking keys [74].

4) Use the crankshaft assembly [17] to manually retract the aft flap [2] to the retracted position.

5) Install the rig pin from the rigging pin kit, SPL-1585 (F70207–113 or F70207–114) in the crankshaft assembly [17].

6) Make sure that the distance from the trailing edge of the aft flap [2] to the trailing edge of the upper cove panel [101] is in the limit.

(e) Remove the rig pin from the crankshaft assembly [17].

(f) Use the crankshaft assembly [17] to manually extend the aft flap [2] to the extended position.

SUBTASK 27-51-14-420-007

(9) Install the aft flap drive rod assembly [43]:

(a) Put the aft flap drive rod assembly [43] in its position between the drive rod fitting [44] and the crankshaft assembly [17].

(b) Install the bolts [45], washers [46], bushings [47], washers [48] and nuts [49] that attach the aft flap drive rod assembly [43] to the crankshaft assembly [17] and the drive rod fitting [44].

1) Make sure that the heads of the bolts [45] are on the inboard side.

(c) Tighten the nuts [49] to 30 in-lb (3.4 N·m) to 50 in-lb (5.6 N·m).

(d) Install the pins [50] through the nuts [49].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-14-410-001

(1) Install the leading edge panel [40] on the outboard end of the main flap [1]:

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(a) Apply sealant, A00247 to the mating surface of the leading edge panel [40] and the main flap [1].

(b) Put the leading edge panel [40] in its position.

(c) Install the bolts [42]:

1) Apply compound, C00528 to the forward and aft rows of holes in the leading edge panel [40].

2) Install the bolts [42] in the forward and aft rows of holes.

3) Apply sealant, A00247 to the bolts [42]b that go in the ribs and stiffeners in the main flap [1].

4) Install the bolts [42] in the ribs and stiffeners.

SUBTASK 27-51-14-410-002

(2) Install the upper skin panel [4] on the outboard end of the main flap [1]:

(a) Apply sealant, A00247 to the mating surface of the upper skin panel [4] and the main flap [1].

(b) Put the upper skin panel [4] in its position.

(c) Install the bolts [3]:

1) Apply compound, C00528 to the forward and aft rows of holes in the upper skin panel [4].

2) Install the bolts [3] in the forward and aft rows of holes.

3) Apply sealant, A00247 to the bolts [3] that go in the ribs and stiffeners in the main flap [1].

4) Install the bolts [3] in the ribs and stiffeners.

SUBTASK 27-51-14-410-003

(3) Install the lower skin panel [58] on the inboard end of the main flap [1]:

(a) Apply sealant, A00247 to the mating surface of the lower skin panel [58] and the main flap [1].

(b) Put the lower skin panel [58] in its position.

(c) Install the bolts [57]:

1) Apply compound, C00528 to the forward and aft rows of holes in the lower skin panel [58].

2) Install the bolts [57] in the forward and aft rows of holes.

3) Apply sealant, A00247 to the bolts [57] that go in the ribs and stiffeners in the main flap [1].

4) Install the bolts [57] in the ribs and stiffeners.

SUBTASK 27-51-14-410-004

(4) Do this task: Inboard Flap Support Aft Fairing Installation, TASK 27-51-18-400-802.

SUBTASK 27-51-14-440-001

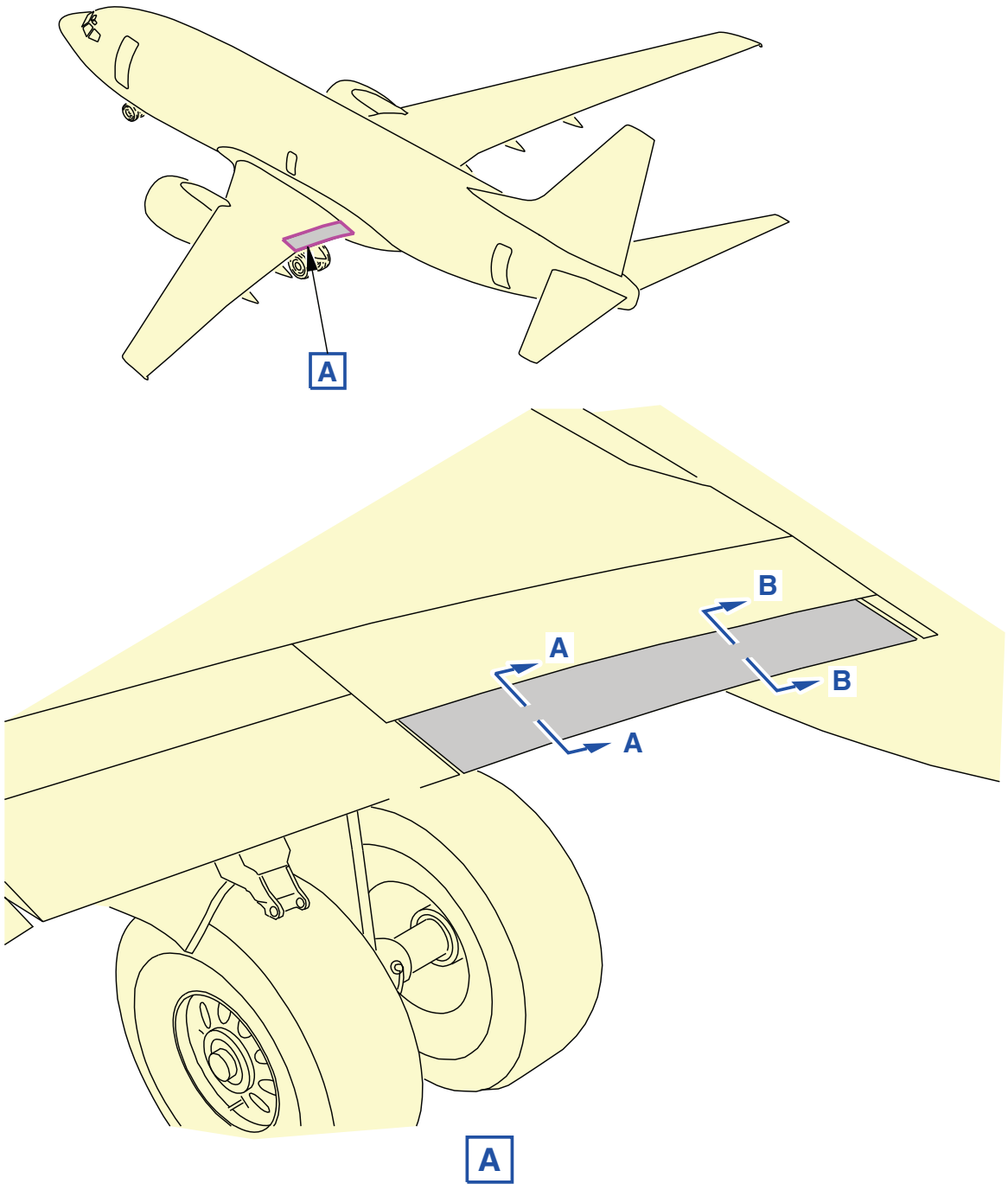
(5) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-14-860-002

(6) If the trailing edge flaps are not in the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

——— END OF TASK ———

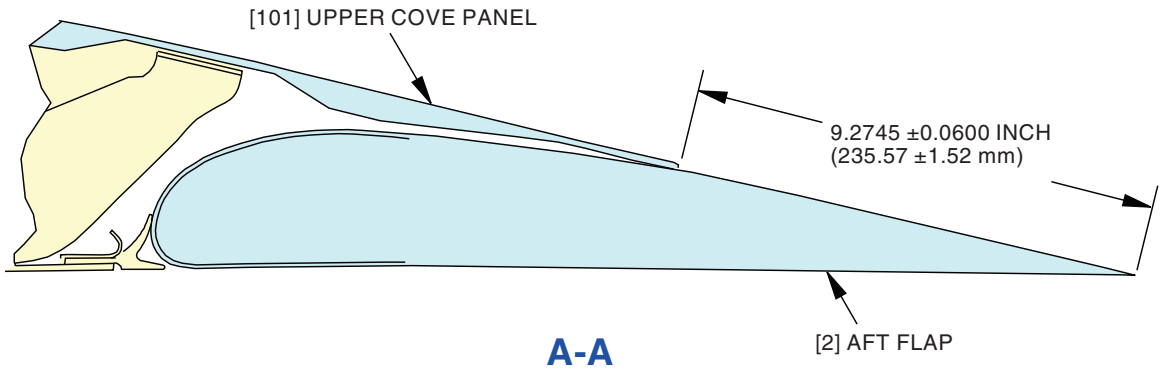
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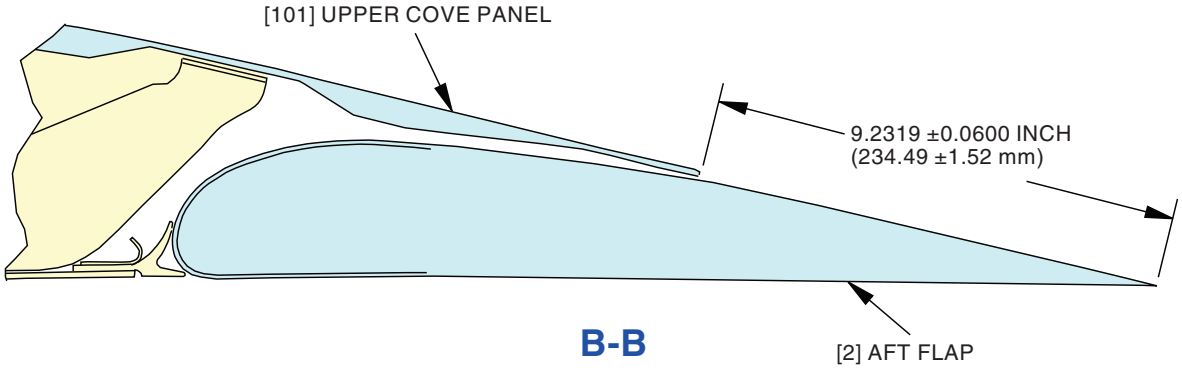


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Inboard Aft Flap Pushrod Adjustment
Figure 402/27-51-14-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2412853 S00061531437_V1

Inboard Aft Flap Pushrod Adjustment
Figure 402/27-51-14-990-802 (Sheet 2 of 2)

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INBOARD FLAP TRACKS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Inboard Flap Track Removal (Selection)

(2) Inboard Flap Track Installation (Selection)

(3) Inboard Flap Outboard Track Removal

(4) Inboard Flap Outboard Track Installation

(5) Inboard Flap Inboard Track Removal

(6) Inboard Flap Inboard Track Installation.

TASK 27-51-15-000-801

2. Inboard Flap Track Removal (Selection)

A. Procedure

SUBTASK 27-51-15-000-001

(1) Do one of these tasks to remove the applicable inboard flap track:

(a) Do this task: Inboard Flap Outboard Track Removal, TASK 27-51-15-000-802.

(b) Do this task: Inboard Flap Inboard Track Removal, TASK 27-51-15-000-803.

——— END OF TASK ———

TASK 27-51-15-400-801

3. Inboard Flap Track Installation (Selection)

A. Procedure

SUBTASK 27-51-15-400-001

(1) Do one of these tasks to install the applicable inboard flap track:

(a) Do this task: Inboard Flap Outboard Track Installation, TASK 27-51-15-400-802.

(b) Do this task: Inboard Flap Inboard Track Installation, TASK 27-51-15-400-803.

——— END OF TASK ———

TASK 27-51-15-000-802

4. Inboard Flap Outboard Track Removal

(Figure 401)

A. General

(1) This task is applicable to the flap tracks at flap supports No. 3 and 6.

B. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-11-000-801	Inboard Trailing Edge Flap Removal (P/B 401)
27-51-18-000-801	Inboard Flap Support Forward Fairing Removal (P/B 401)
27-51-18-000-802	Inboard Flap Support Aft Fairing Removal (P/B 401)
27-51-31-000-802	Inboard Flap Outboard Transmission Removal (P/B 401)

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(Continued)

Reference	Title
27-59-02-000-803	Inboard Flap Outboard Skew Mechanism Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1568	Jack - Hydraulic, General Low Profile Part #: HW93718 Supplier: 28047 Opt Part #: B67563 Supplier: 36251 Opt Part #: W93718 Supplier: 36251
SPL-1733	Adapter - Hoist, Removal/Installation, Flap Track Part #: C27040-29 Supplier: 81205 Opt Part #: C27040-1 Supplier: 81205

D. Location Zones

Zone	Area
542	Left Wing - Fairing Flap Support No. 3
642	Right Wing - Fairing Flap Support No. 6

E. Prepare for the Removal

SUBTASK 27-51-15-860-001

(1) Extend the trailing edge flaps to the 40-unit position (TASK 27-51-00-860-803).

SUBTASK 27-51-15-040-001

(2) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-15-010-001

(3) Remove the fairings for the outboard track on the inboard flap, do these tasks:

- Inboard Flap Support Forward Fairing Removal, TASK 27-51-18-000-801
- Inboard Flap Support Aft Fairing Removal, TASK 27-51-18-000-802

SUBTASK 27-51-15-020-001

(4) Do this task: Inboard Trailing Edge Flap Removal, TASK 27-51-11-000-801.

SUBTASK 27-51-15-020-002

(5) Do this task: Inboard Flap Outboard Skew Mechanism Removal, TASK 27-59-02-000-803.

SUBTASK 27-51-15-020-003

(6) Do this task: Inboard Flap Outboard Transmission Removal, TASK 27-51-31-000-802.

F. Inboard Flap Outboard Track Removal

SUBTASK 27-51-15-480-001

(1) If you use the adapter, SPL-1733, install it on the flap track [1]:

(a) Install the adapter, SPL-1733 on the flap track [1] approximately 35 in. (89 cm) aft of the track attach fitting.

(b) Put a low profile hydraulic jack, COM-1568 below the adapter, SPL-1733 to support the flap track [1].

(c) If you do not use the adapter, SPL-1733, make sure that you support the flap track while you disconnect it.

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SUBTASK 27-51-15-020-020

(2) Remove the nut [28], washers [29], and bolt [30] that attach the bonding jumper [27] to the structure.

SUBTASK 27-51-15-020-021

(3) Remove the clamp [34] that attaches the electrical wire to the bracket as follows:

(a) Remove the bolt [31], washer [32], and clipnut [33].

SUBTASK 27-51-15-020-004

(4) Disconnect the forward end of the flap track [1] from the track attach fitting:

(a) Remove the bolts [13] and washers [12] that attach the failsafe fitting [14] to the track attach fitting.

(b) Remove the pin [18], nut [17], washer [16], and fuse pin [15] from the forward end of the flap track [1].

SUBTASK 27-51-15-020-005

(5) Disconnect the aft end of the flap track [1] from the structure:

(a) Remove the bolt [8], washer [9], washer [10], and nut [11] from the outboard side of the flap track [1].

(b) Remove the bolt [7], washer [6], washer [4], and nut [3] from the inboard side of the flap track [1].

(c) Measure the thickness of the shim [5].

1) Make sure to mark the track with this measurement data.

NOTE: This measurement data for the removal and replacement of the flap tracks is an important step to obtain ample free movement in the fuse pin joint at the forward end of the flap track and return of the flaps to former operating condition.

SUBTASK 27-51-15-020-006

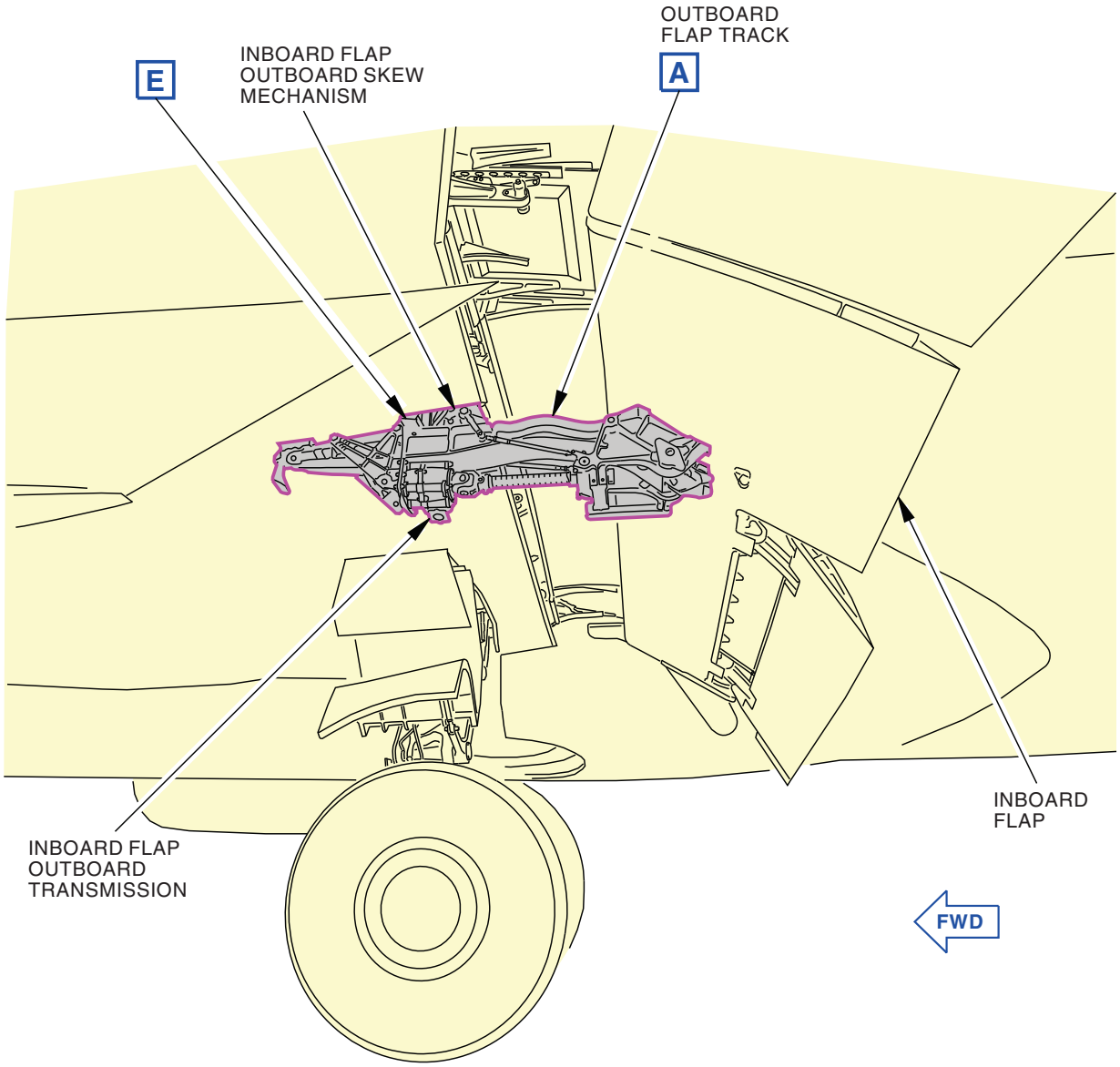
(6) If you install a new flap track [1], remove the transmission attach fitting [2] from the flap track [1]:

(a) Remove the pin [25], nut [24], washer [26], and bolt [19] that attach the forward end of the transmission attach fitting [2].

(b) Remove the pin [23], nut [22], washer [21], and bolt [20] that attach the aft end of the transmission attach fitting [2].

——— END OF TASK ———

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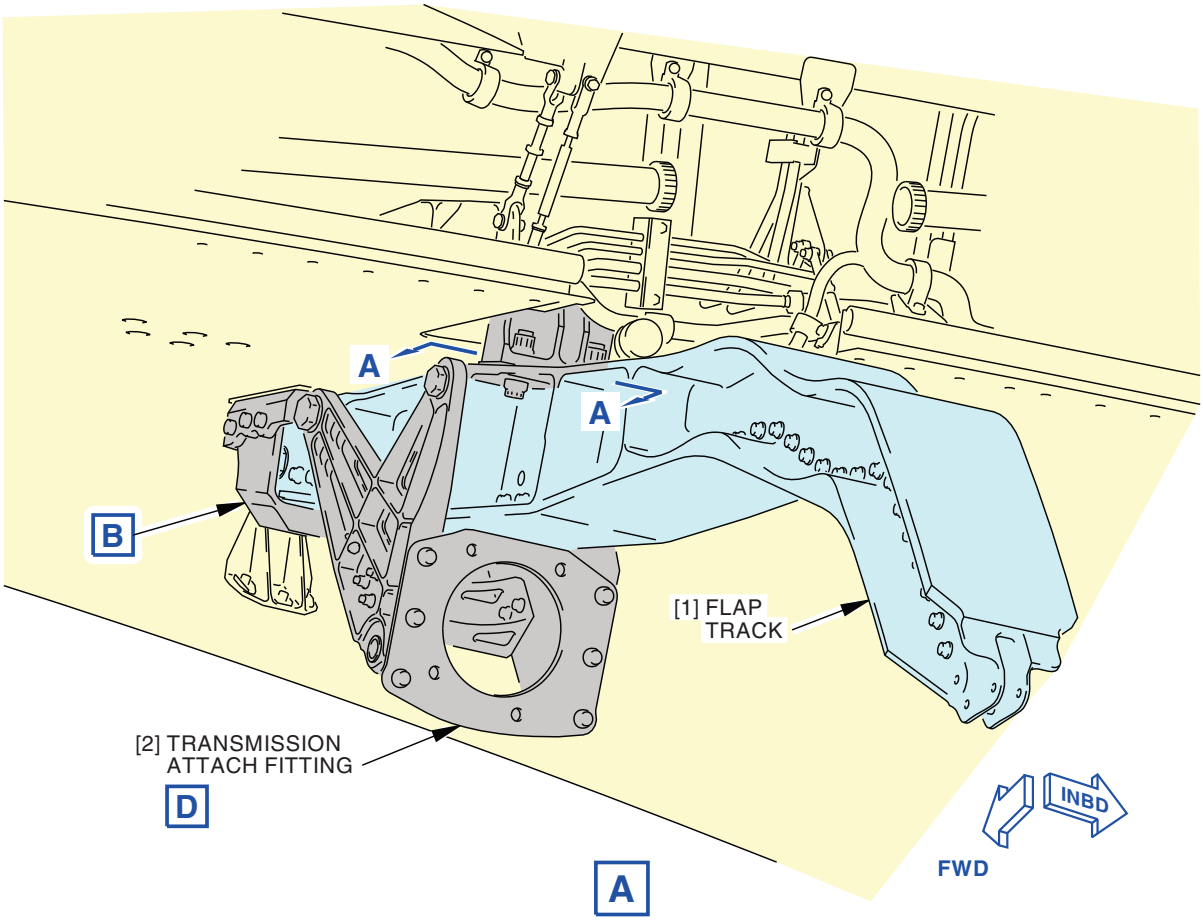


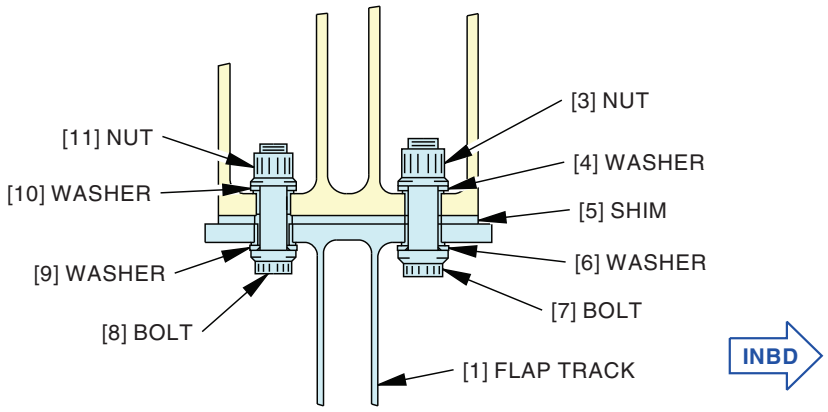
(EXAMPLE)

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Inboard Flap Outboard Flap Track Installation
Figure 401/27-51-15-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



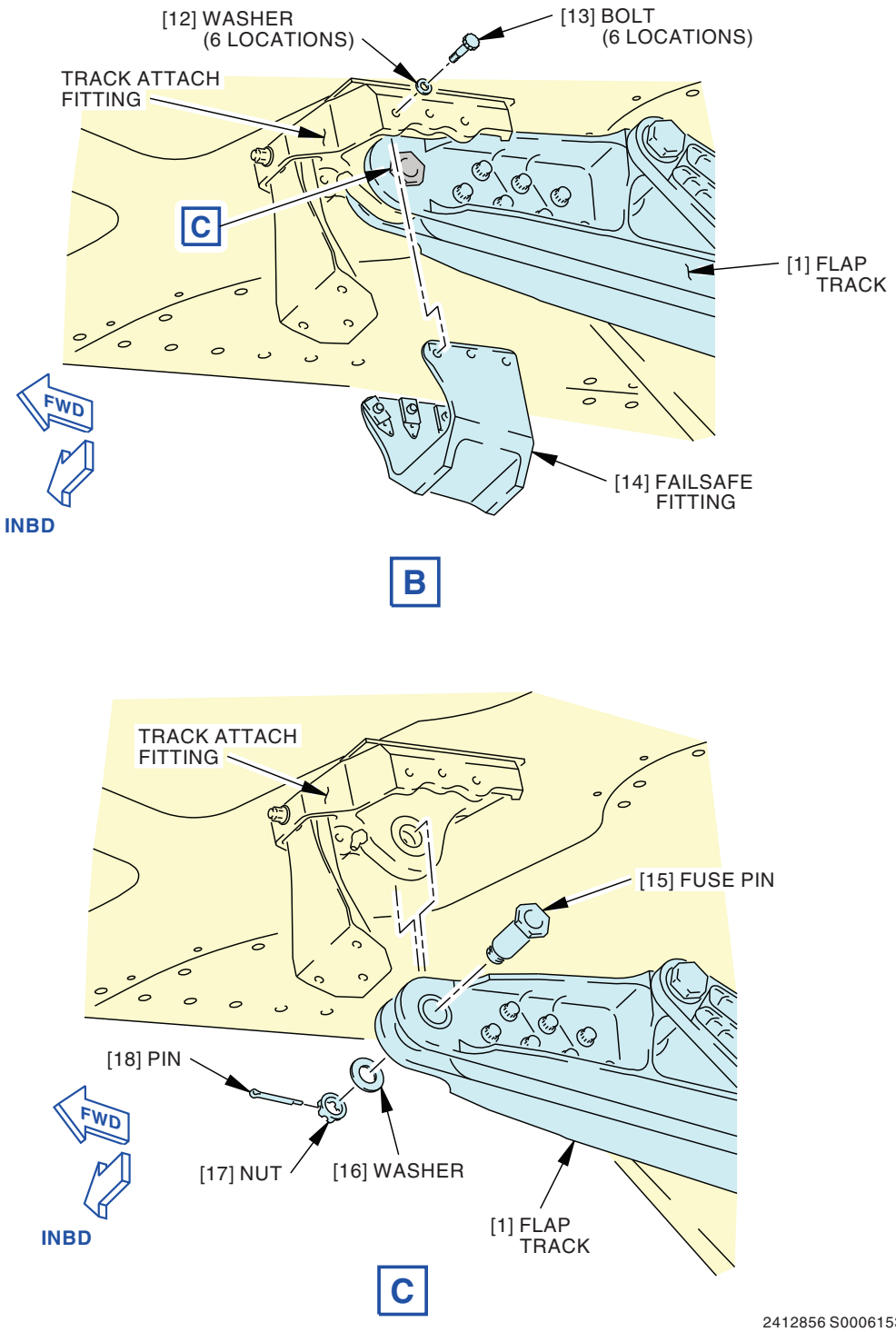


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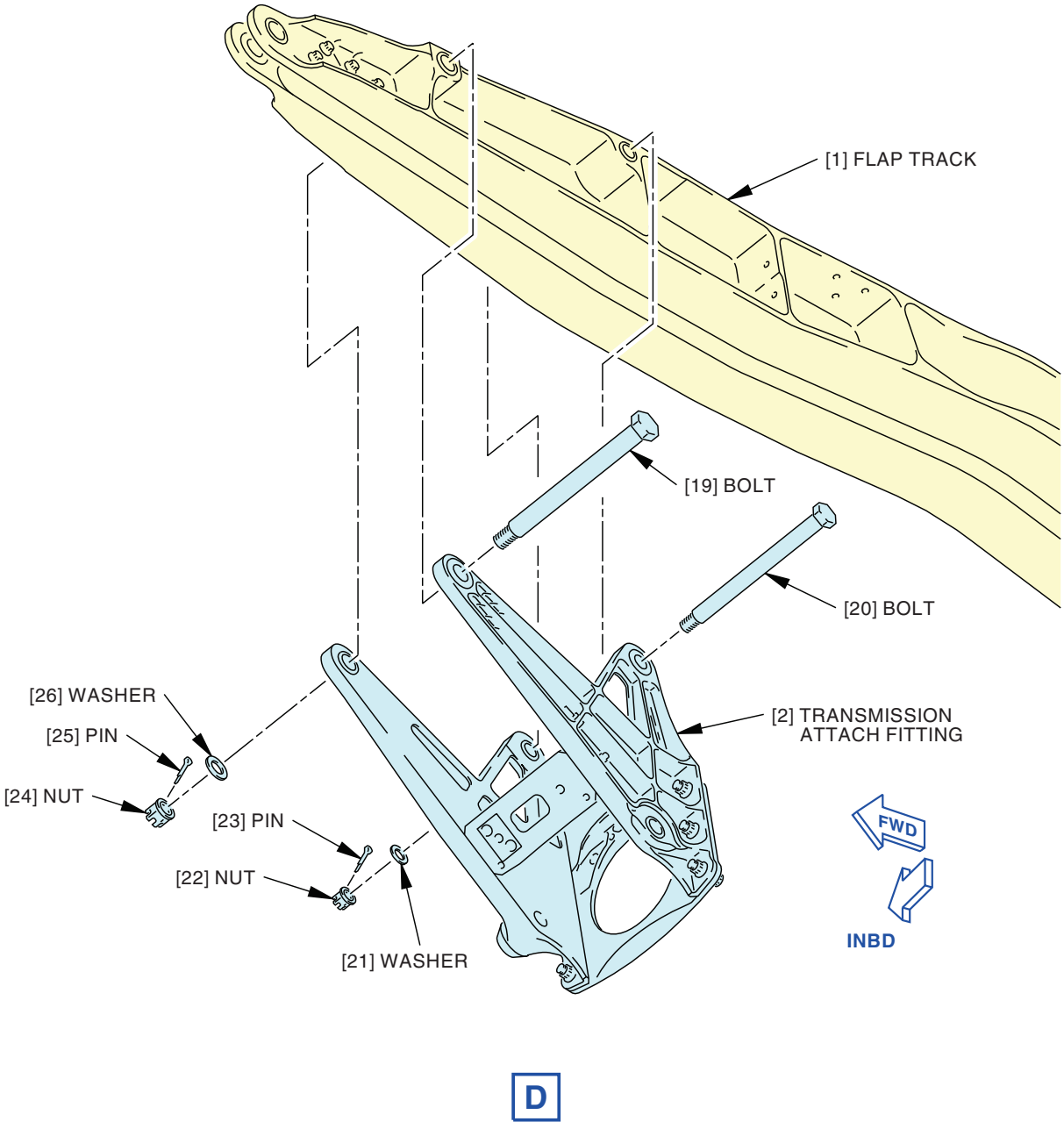
Inboard Flap Outboard Flap Track Installation
Figure 401/27-51-15-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Flap Outboard Flap Track Installation
Figure 401/27-51-15-990-801 (Sheet 3 of 5)

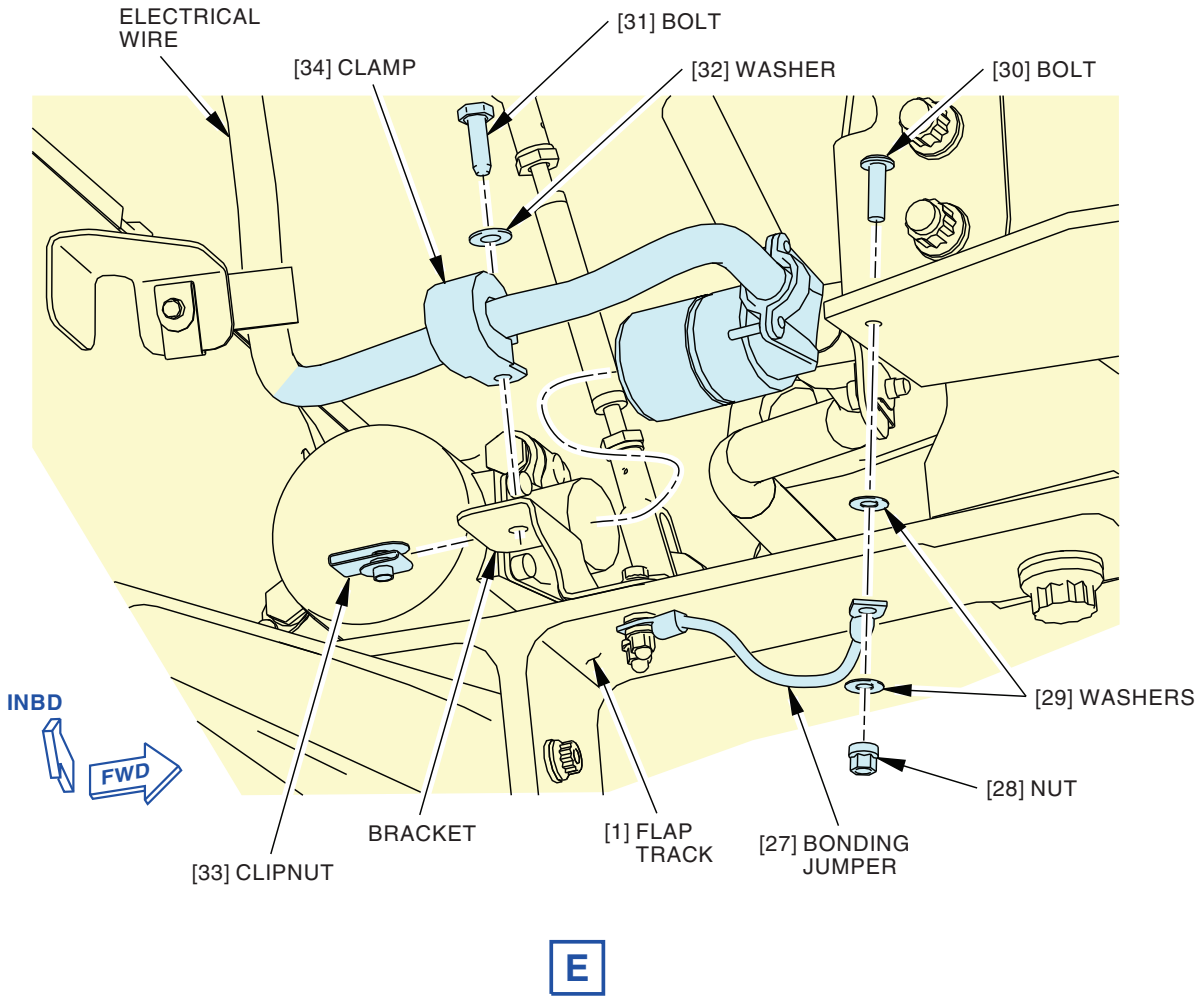
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Inboard Flap Outboard Flap Track Installation
Figure 401/27-51-15-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Inboard Flap Outboard Flap Track Installation
Figure 401/27-51-15-990-801 (Sheet 5 of 5)

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TASK 27-51-15-400-802

5. Inboard Flap Outboard Track Installation

(Figure 401)

A. General

(1) This task is applicable to the flap tracks at flap supports No. 3 and 6.

B. References

Reference	Title
12-22-51-640-814	Inboard Flap Outboard Flap Track Lubrication (P/B 301)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-51-11-400-801	Inboard Trailing Edge Flap Installation (P/B 401)
27-51-11-820-801	Inboard Trailing Edge Flap Adjustment (P/B 501)
27-51-18-400-801	Inboard Flap Support Forward Fairing Installation (P/B 401)
27-51-18-400-802	Inboard Flap Support Aft Fairing Installation (P/B 401)
27-51-31-400-802	Inboard Flap Outboard Transmission Installation (P/B 401)
27-59-00-820-801	Flap Skew Detection System Adjustment (P/B 501)
27-59-02-400-803	Inboard Flap Outboard Skew Mechanism Installation (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
COM-1568	Jack - Hydraulic, General Low Profile Part #: HW93718 Supplier: 28047 Opt Part #: B67563 Supplier: 36251 Opt Part #: W93718 Supplier: 36251
SPL-1733	Adapter - Hoist, Removal/Installation, Flap Track Part #: C27040-29 Supplier: 81205 Opt Part #: C27040-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2

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(Continued)

Reference	Description	Specification
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
B00130	Alcohol - Isopropyl	TT-I-735
D00633	Grease - Aircraft General Purpose	BMS3-33
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50262	Wiper - Cleaning	BMS15-5
G51464	Abrasive - Aluminum Oxide paper, 60 grit or finer	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Flap track	27-51-15-02-290	PTF ALL
18	Pin	27-51-15-02-105	PTF ALL
23	Pin	27-51-15-02-230	PTF ALL
25	Pin	27-51-15-02-235	PTF ALL

F. Location Zones

Zone	Area
542	Left Wing - Fairing Flap Support No. 3
642	Right Wing - Fairing Flap Support No. 6

G. Inboard Flap Outboard Track Installation

SUBTASK 27-51-15-420-001

(1) If it is not installed, install the transmission attach fitting [2]:

(a) Put the transmission attach fitting [2] in its position on the flap track [1].

(b) Install the bolt [20], washer [21], and nut [22] that attach the aft end of the transmission attach fitting [2] to the flap track [1].

(c) Install the pin [23] in the nut [22].

(d) Install the bolt [19], washer [26], and nut [24] that attach the forward end of the transmission attach fitting [2] to the flap track [1].

(e) Install the pin [25] in the nut [24].

SUBTASK 27-51-15-480-002

(2) If you will use the adapter, SPL-1733, install it on the flap track [1]:

(a) Install the adapter, SPL-1733 on the flap track [1] approximately 35 in. (89 cm) aft of the track attach fitting.

(b) Put a low profile hydraulic jack, COM-1568, below the adapter, SPL-1733, to support the flap track [1].

SUBTASK 27-51-15-420-002

(3) Install the flap track [1].

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(a) If it is not installed, install the shim [5] on the structure.

1) Make sure that the shim [5] is the same thickness as the shim [5] that you removed.

NOTE: The original thickness of the shim [5] is on a marker on the structure.

NOTE: If it is necessary to replace the flap track with a new unit, a new shim [5] may need to be made and should be marked and checked after the nuts are torqued that attach the flap track to the structure.

2) Apply sealant, A00247, to the mating surface between shim [5] and flap track [1], and mating surface between shim [5] and structure.

3) Put the shim [5] in its location.

(b) Attach the aft end of the flap track [1]:

1) Lift the flap track [1] to its location on the structure.

2) Install the bolt [7], washer [6], washer [4], and nut [3] that attach the inboard end of the flap track [1].

3) Install the bolt [8], washer [9], washer [10], and nut [11] that attach the outboard end of the flap track [1].

(c) Attach the forward end of the flap track [1]:

1) Apply grease, D00633, to the outside of the fuse pin [15] and the inside of the bushings.

2) Install the fuse pin [15] to attach the forward end of the flap track [1] to the track attach fitting.

(d) Tighten the nut [3] to a torque set point of 1800 in-lb (203 N·m).

(e) Tighten the nut [11] to a torque set point of 1300 in-lb (147 N·m).

(f) Make sure that you can turn the fuse pin [15] in the joint with a maximum of 10 in-lb (1 N·m).

NOTE: You can push up on the forward end of the flap track [1] to take the weight off the fuse pin [15] when you measure the torque.

(g) Install the washer [16] and nut [17] on the fuse pin [15].

(h) Tighten the nut [17] to 330 in-lb (37 N·m) - 350 in-lb (40 N·m).

(i) Loosen the nut [17] to align the hole for the pin [18].

(j) Install the pin [18] in the nut [17].

(k) Apply a fillet seal with sealant, A00247, around the joint where the shim [5], flap track [1], and airplane structure are attached.

SUBTASK 27-51-15-420-003

(4) Install the failsafe fitting [14] on the track attach fitting:

(a) Put the failsafe fitting [14] in its location on the track attach fitting.

(b) Install the bolts [13] and washers [12] to attach the failsafe fitting [14].

SUBTASK 27-51-15-420-014

(5) Install the clamp [34] that attaches the electrical wire to the bracket as follows:

(a) Install the bolt [31], washer [32], and clipnut [33] that attach the clamp [34] and electrical wire to the bracket.

SUBTASK 27-51-15-420-015

(6) Install the bonding jumper [27] as follows:

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(a) Use cotton wiper, G00034, moist in alcohol, B00130, to clean the mating surfaces of the washers [29] and bonding jumper [27].

(b) Clean the mating surface of the structure with 60 grit or finer wet/dry aluminum oxide sandpaper, G51464.

1) Clean the mating surface of the structure with wiper, G50262, moist in alcohol, B00130.

2) If it is necessary, use wiper, G50262, to remove remaining solvent.

(c) Install the bolt [30], washers [29], and nut [28] that attach the bonding jumper [27] to the structure.

1) Apply sealant, A50155, sealant, A50110, sealant, A50231, or sealant, A50296, to the mating surfaces of the bolt [30], washers [29], nut [28], bonding jumper [27], and airplane structure.

2) Tighten the nut [28] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

3) Make sure that there is squeeze out around all the interfaces of each component.

(d) Measure the electrical bonding resistance between the bonding jumper [27] and airplane structure (SWPM 20-20-00).

1) Use the non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

SUBTASK 27-51-15-480-003

(7) If it is necessary, remove the adapter, SPL-1733, from the flap track [1].

SUBTASK 27-51-15-420-004

(8) Do this task: Inboard Flap Outboard Transmission Installation, TASK 27-51-31-400-802.

SUBTASK 27-51-15-420-005

(9) Do this task: Inboard Flap Outboard Skew Mechanism Installation, TASK 27-59-02-400-803.

SUBTASK 27-51-15-420-006

(10) Do this task: Inboard Trailing Edge Flap Installation, TASK 27-51-11-400-801.

SUBTASK 27-51-15-820-001

(11) Do this task: Inboard Trailing Edge Flap Adjustment, TASK 27-51-11-820-801.

SUBTASK 27-51-15-820-002

(12) Do this task: Flap Skew Detection System Adjustment, TASK 27-59-00-820-801.

SUBTASK 27-51-15-640-001

(13) Do this task: Inboard Flap Outboard Flap Track Lubrication, TASK 12-22-51-640-814.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-15-410-001

(1) Install the fairings for the outboard track on the inboard flap, do these tasks:

- Inboard Flap Support Aft Fairing Installation, TASK 27-51-18-400-802
- Inboard Flap Support Forward Fairing Installation, TASK 27-51-18-400-801

SUBTASK 27-51-15-440-001

(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

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SUBTASK 27-51-15-860-002

(3) If the trailing edge flaps are not in the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

——— END OF TASK ———

TASK 27-51-15-000-803

6. Inboard Flap Inboard Track Removal

(Figure 402)

A. General

(1) This task is applicable to the flap tracks at flap supports No. 4 and 5.

B. References

Reference	Title
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-11-000-801	Inboard Trailing Edge Flap Removal (P/B 401)
27-51-31-000-803	Inboard Flap Inboard Transmission Removal (P/B 401)
27-59-02-000-804	Inboard Flap Inboard Skew Mechanism Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1568	Jack - Hydraulic, General Low Profile Part #: HW93718 Supplier: 28047 Opt Part #: B67563 Supplier: 36251 Opt Part #: W93718 Supplier: 36251
SPL-1733	Adapter - Hoist, Removal/Installation, Flap Track Part #: C27040-29 Supplier: 81205 Opt Part #: C27040-1 Supplier: 81205
SPL-9469	Hold Equipment - T.E. Inboard Flap Part #: C27079-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
541	Left Wing - Fairing Flap Support No. 4
641	Right Wing - Fairing Flap Support No. 5

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F. Access Panels

Number	Name/Location
193CL	Wheel Well Panel - Aft Outboard
193CR	Wheel Well Panel - Aft Outboard
194CL	Aft Wing To Body Fairing Panel
194CR	Aft Wing To Body Fairing Panel

G. Prepare for the Removal

SUBTASK 27-51-15-860-003

(1) Extend the trailing edge flaps to the 40-unit position. To extend them, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-51-15-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-51-15-860-004

(3) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-51-15-860-005

(4) For the left inboard flap, if applicable:

Open these access panels:

Number	Name/Location
193CL	Wheel Well Panel - Aft Outboard
194CL	Aft Wing To Body Fairing Panel

SUBTASK 27-51-15-860-006

(5) For the right inboard flap, if applicable:

Open these access panels:

Number	Name/Location
193CR	Wheel Well Panel - Aft Outboard
194CR	Aft Wing To Body Fairing Panel

SUBTASK 27-51-15-860-007

(6) If the inboard flap hold equipment, SPL-9469 is not available, then do this task: Inboard Trailing Edge Flap Removal, TASK 27-51-11-000-801.

SUBTASK 27-51-15-480-005

(7) Do the following steps to install the trailing edge inboard flap hold equipment, SPL-9469:

NOTE: The inboard flap hold equipment, SPL-9469 will support the inboard flap so that inboard flap removal is not necessary.

NOTE: It is not necessary that the aft flap be installed in order to install the inboard flap hold equipment, SPL-9469.

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(a) Inflate the shock strut with a sufficient amount of nitrogen, G00018 to provide clearance for the mounting bracket of the lower brace assembly from hold equipment, SPL-9469 (Main Landing Gear Shock Strut Servicing, Airplane on the Ground, TASK 12-15-31-610-802).

(b) Attach the lower brace assembly to the bottom of the outer cylinder of the main landing gear.

(c) Adjust the length of the lower brace assembly until you can attach the lower brace assembly to the aft flap track assembly.

(d) Attach the upper brace assembly to the upper end of the outer cylinder.

(e) Adjust the length of the upper brace assembly until you can attach the upper brace assembly to the aft most fitting on the lower brace assembly.

(f) Attach the side brace assembly to the side brace of the main landing gear.

NOTE: The flange of the backing plate is oriented in the downward direction.

(g) Adjust the length of the side brace assembly until you can attach it to the forward most fitting on the lower brace assembly.

(h) Extend the three brace assemblies until the weight of the flap is supported.

1) If necessary, push up on the inboard flap while extending the brace assemblies.

SUBTASK 27-51-15-020-007

(8) Do this task: Inboard Flap Inboard Skew Mechanism Removal, TASK 27-59-02-000-804.

SUBTASK 27-51-15-020-008

(9) Do this task: Inboard Flap Inboard Transmission Removal, TASK 27-51-31-000-803.

H. Inboard Flap Inboard Track Removal

NOTE: Make sure you support the flap track [41] while you disconnect it.

SUBTASK 27-51-15-480-006

(1) If you will use the adapter, SPL-1733, install it on the flap track [41]:

(a) Install the adapter, SPL-1733 on the flap track [41] approximately 20 in. (51 cm) aft of the track forward attach fitting.

(b) Put a low profile hydraulic jack, COM-1568 below the adapter, SPL-1733 to support the flap track [41].

SUBTASK 27-51-15-020-010

(2) Disconnect the forward end of the flap track [41]:

(a) Remove the nut [56], washer [55] and bolt [52] that attach the flap track [41] to the forward attach assembly [50].

(b) Remove the nut [58], washer [54] and bolt [51] that attach the flap track [41] to the forward attach assembly [50].

SUBTASK 27-51-15-020-019

(3) Disconnect the link [45] from the structure:

(a) Remove the nut [43], washer [42] and bolt [46] from the bolt [49].

(b) Remove the lockwire from the nut [48] and washer [47].

(c) Remove the nut [48] and washer [47] from the bolt [49].

(d) Remove the bolt [49] that attaches the link [45] to the structure.

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SUBTASK 27-51-15-020-011

(4) If you will install a new flap track [41], remove the transmission attach fitting [53] and the link [45] from the flap track [41]:

(a) Remove the bolts [60], bolt [64], washers [61] and nuts [62] that attach the transmission attach fitting [53] to the flap track [41].

(b) Disconnect the link [45] from the flap track [41]:

1) Remove the pin [66], nut [67], washer [65] and bolt [68] from the bolt [72].

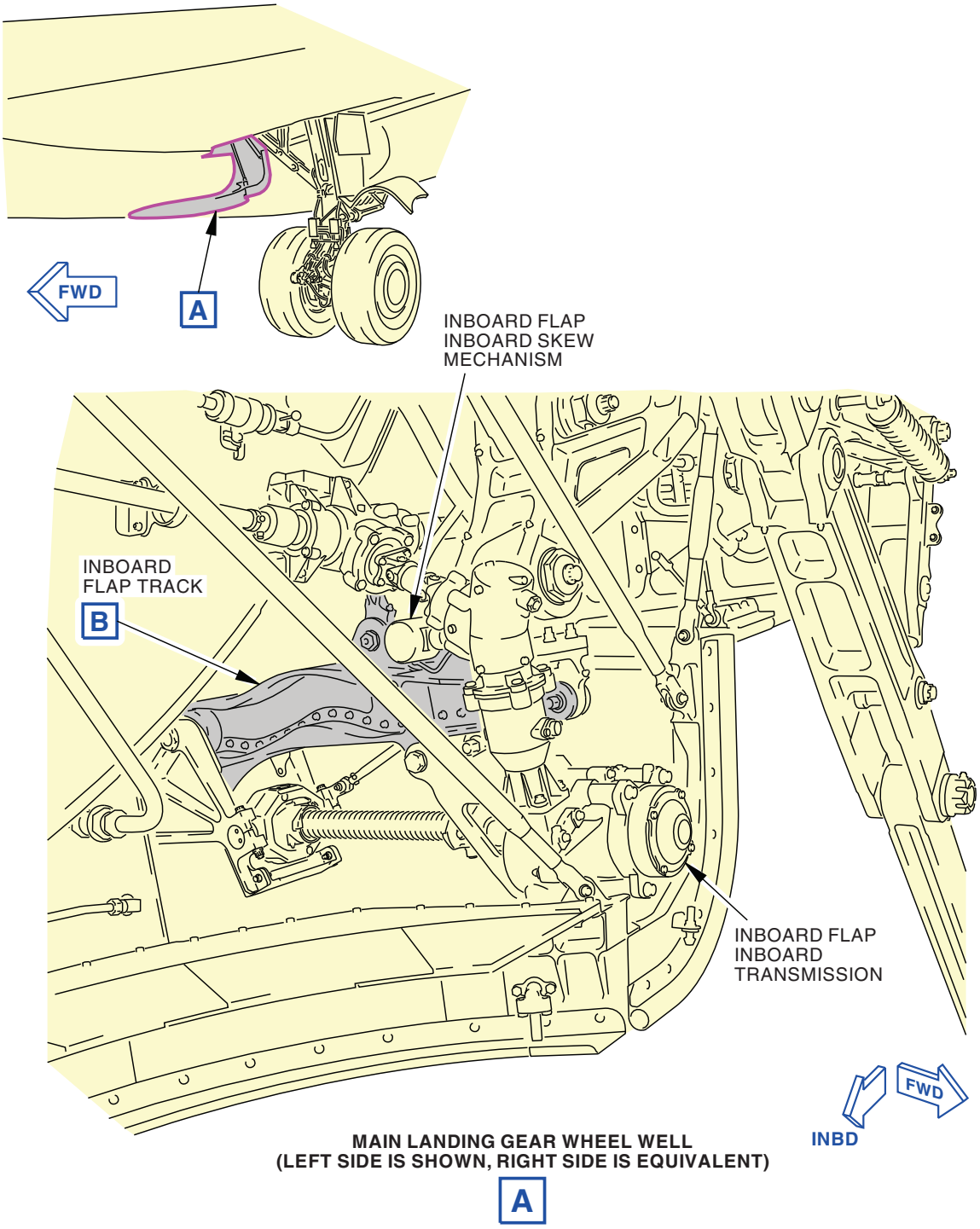
2) Remove the lockwire from the nut [71] and washer [70].

3) Remove the nut [71], washer [70] and washer [69] from the bolt [72].

4) Remove the bolt [72], bushing [73] and washer [69] that attaches the link [45] to the flap track [41].

END OF TASK

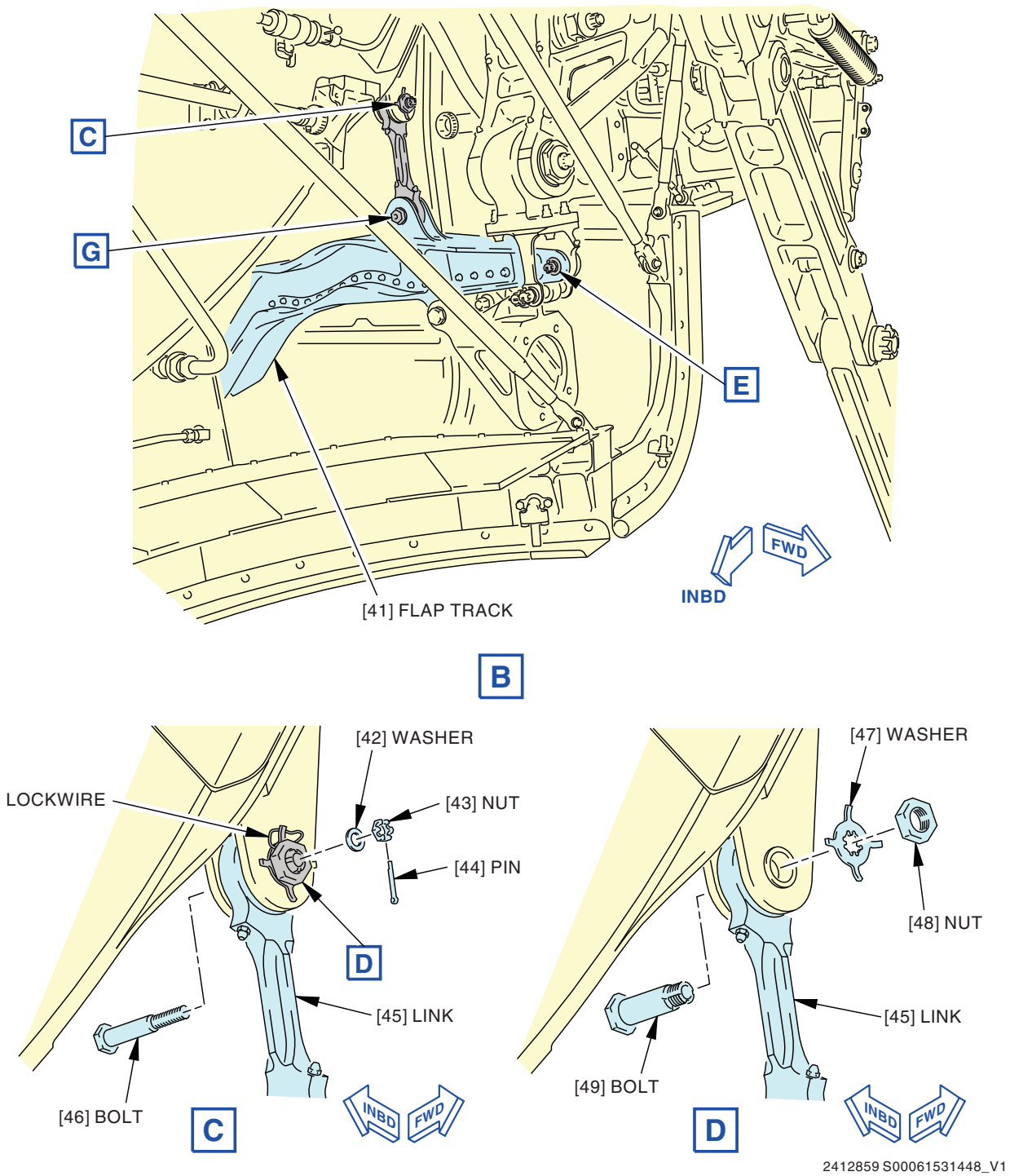
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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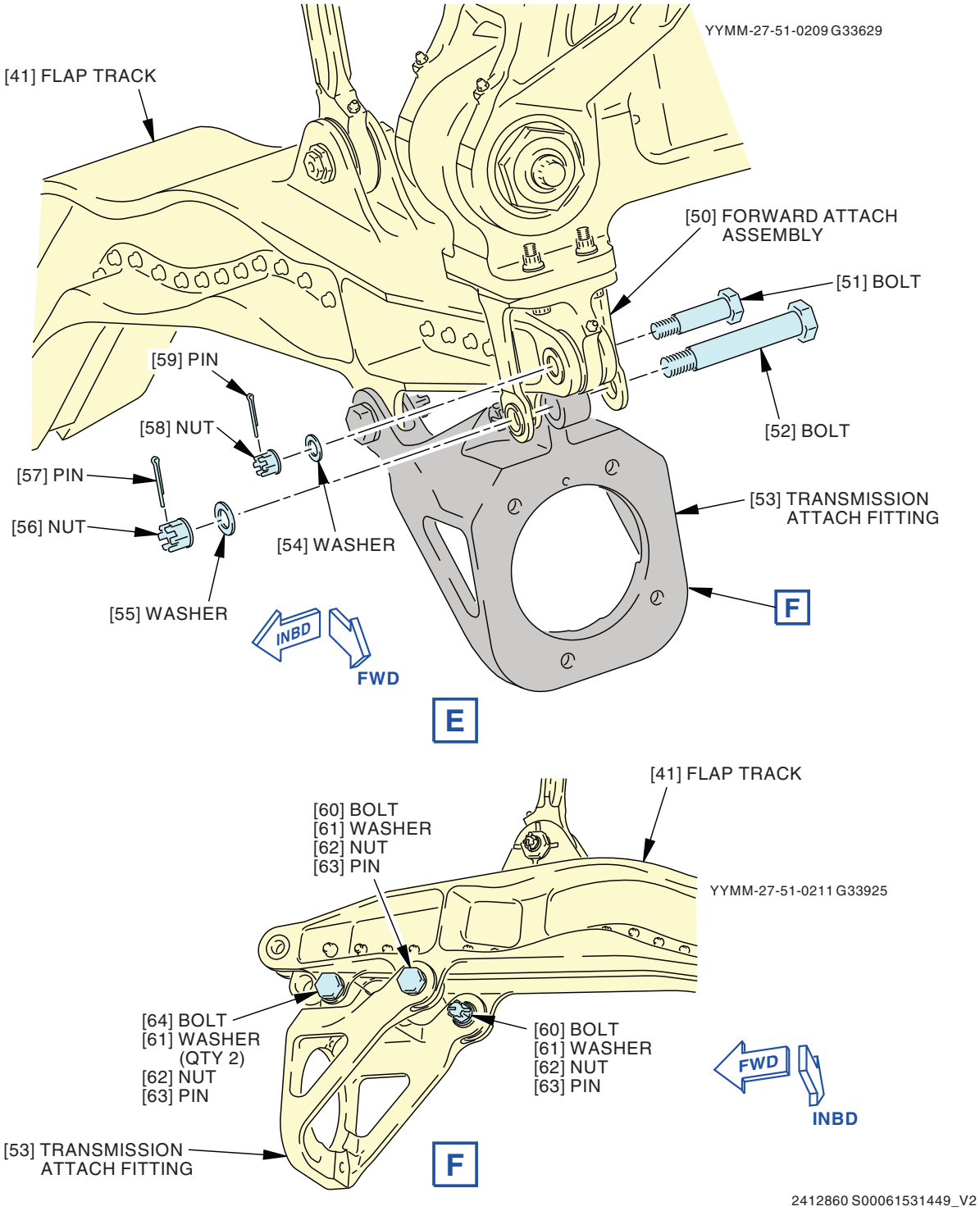
Inboard Flap Inboard Flap Track Installation
Figure 402/27-51-15-990-802 (Sheet 1 of 4)

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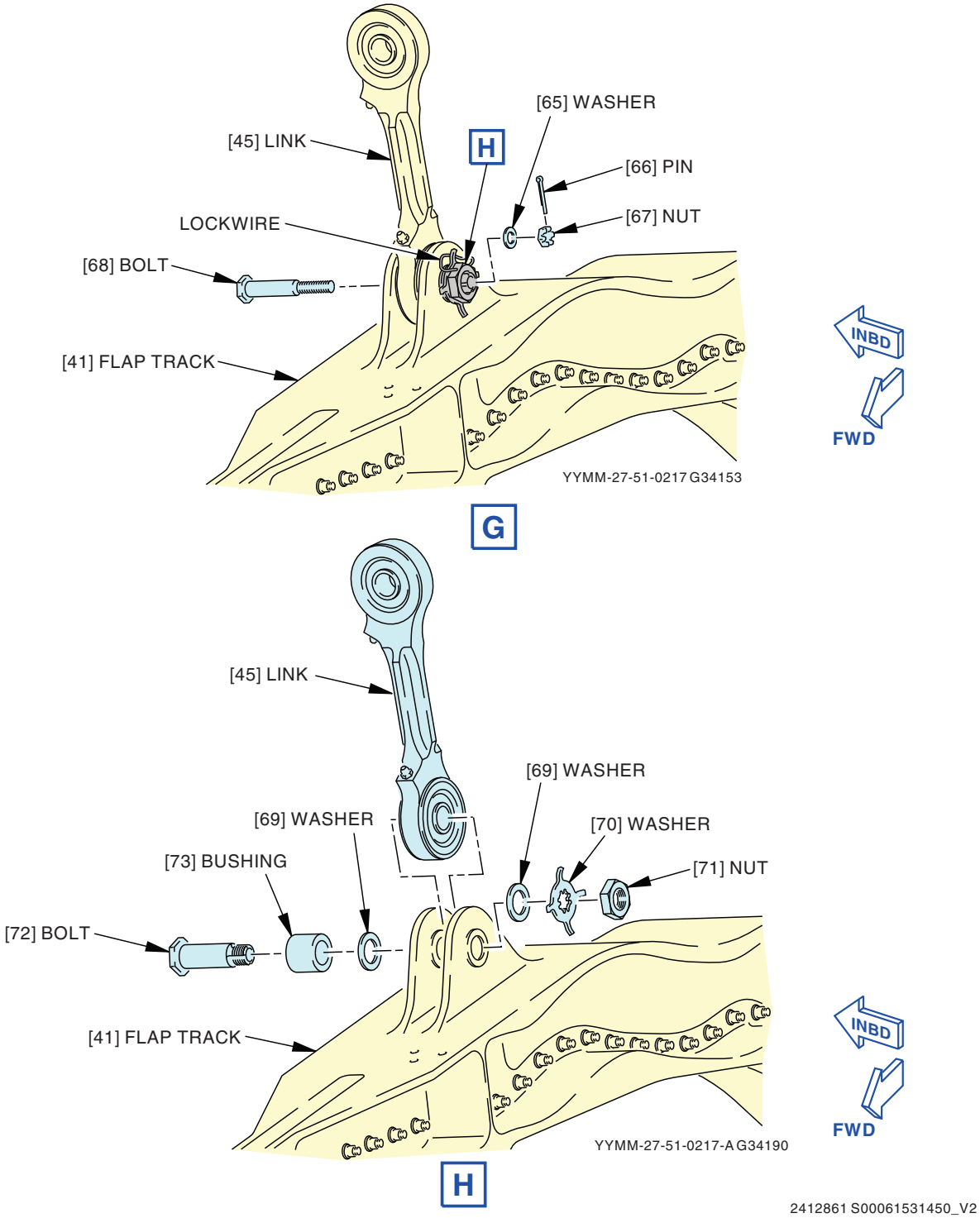
Inboard Flap Inboard Flap Track Installation
Figure 402/27-51-15-990-802 (Sheet 2 of 4)

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Inboard Flap Inboard Flap Track Installation
Figure 402/27-51-15-990-802 (Sheet 3 of 4)

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Inboard Flap Inboard Flap Track Installation
Figure 402/27-51-15-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-15-400-803

7. Inboard Flap Inboard Track Installation

(Figure 402)

A. General

(1) This task is applicable to the flap tracks at flap supports No. 4 and 5.

B. References

Reference	Title
12-15-31-610-802	Main Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
12-22-51-640-813	Inboard Flap Inboard Flap Track Lubrication (P/B 301)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-51-11-400-801	Inboard Trailing Edge Flap Installation (P/B 401)
27-51-11-820-801	Inboard Trailing Edge Flap Adjustment (P/B 501)
27-51-31-400-803	Inboard Flap Inboard Transmission Installation (P/B 401)
27-59-00-820-801	Flap Skew Detection System Adjustment (P/B 501)
27-59-02-400-804	Inboard Flap Inboard Skew Mechanism Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1568	Jack - Hydraulic, General Low Profile Part #: HW93718 Supplier: 28047 Opt Part #: B67563 Supplier: 36251 Opt Part #: W93718 Supplier: 36251
SPL-1733	Adapter - Hoist, Removal/Installation, Flap Track Part #: C27040-29 Supplier: 81205 Opt Part #: C27040-1 Supplier: 81205
SPL-9469	Hold Equipment - T.E. Inboard Flap Part #: C27079-1 Supplier: 81205

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
41	Flap track	27-51-15-01-330	PTF ALL
		27-51-15-01-332	PTF ALL
44	Pin	27-51-15-01-140	PTF ALL
57	Pin	27-51-15-01-065	PTF ALL
59	Pin	27-51-15-01-060	PTF ALL
63	Pin	27-51-15-01-270	PTF ALL
66	Pin	27-51-15-01-270	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

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(Continued)

Zone	Area
541	Left Wing - Fairing Flap Support No. 4
641	Right Wing - Fairing Flap Support No. 5

F. Access Panels

Number	Name/Location
193CL	Wheel Well Panel - Aft Outboard
193CR	Wheel Well Panel - Aft Outboard
194CL	Aft Wing To Body Fairing Panel
194CR	Aft Wing To Body Fairing Panel

G. Inboard Flap Inboard Track Installation

SUBTASK 27-51-15-420-007

(1) If they are not installed, install the transmission attach fitting [53] and link [45] on the flap track [41], do these steps:

(a) Install the transmission attach fitting [53], do these steps:

1) Put the transmission attach fitting [53] in its position on the flap track [41].

CAUTION

MAKE SURE THAT THE HEAD OF THE OUTBOARD BOLT POINTS TO OUTBOARD OF THE TRANSMISSION ATTACH FITTING. THE NUT AND COTTER PIN CAN TOUCH THE FLAP-SKEW-CONTROL ROD AND CAUSE DAMAGE. IF YOU DO NOT OBEY, DAMAGE TO THE FLAP-SKEW-CONTROL ROD CAN OCCUR.

2) Install the bolts [60], bolt [64], washers [61], and nuts [62] that attach the transmission attach fitting [53] to the flap track [41].

a) Make sure that the heads of the bolts [60] are oriented on outside surface of the transmission attach fitting [53] (View F, Figure 402).

3) Tighten the nuts [62] to 600 in-lb (67.8 N·m) - 800 in-lb (90.4 N·m).

4) Install the pins [63] in the nuts [62].

(b) Install the link [45] on the flap track [41], do these steps:

1) Put the link [45] in its position on the flap track [41].

2) Install the bolt [72], bushing [73], washers [69], washer [70], and nut [71] to attach the link [45] to the flap track [41].

3) Tighten the nut [71] to 290 in-lb (32.8 N·m) - 510 in-lb (57.6 N·m).

4) Install lockwire between the nut [71] and washer [70].

5) Install the bolt [68], washer [65], and nut [67] in the bolt [72].

6) Tighten the nut [67] to 30 in-lb (3.4 N·m) - 60 in-lb (6.8 N·m).

7) Install the pin [66] in the nut [67].

SUBTASK 27-51-15-480-007

(2) If you use the adapter, SPL-1733, install it on the flap track [41], do these steps:

(a) Install the adapter, SPL-1733, on the flap track [41] approximately 20 in. (51 cm) aft of the track forward attach fitting.

(b) Put a low profile hydraulic jack, COM-1568, below the adapter, SPL-1733, to support the flap track [41].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-15-020-012

(3) Connect the link [45] to the structure, do these steps:

(a) Put the link [45] in its position on the structure.

(b) Install the bolt [49], washer [47], and nut [48] to attach the link [45] to the structure.

(c) Tighten the nut [48] to 290 in-lb (32.8 N·m) - 510 in-lb (57.6 N·m).

(d) Install lockwire between the nut [48] and washer [47].

(e) Install the bolt [46], washer [42], and nut [43] in the bolt [49].

(f) Tighten the nut [43] to 30 in-lb (3.4 N·m) - 60 in-lb (6.8 N·m).

(g) Install the pin [44] in the nut [43].

SUBTASK 27-51-15-420-008

(4) Connect the forward end of the flap track [41], do these steps:

(a) Put the flap track [41] in its position on the forward attach assembly [50].

(b) Install the bolt [51], washer [54], and nut [58] that attach the flap track [41] to the forward attach assembly [50].

(c) Install the pin [59] in the nut [58].

(d) Install the bolt [52], washer [55], and nut [56] that attach the flap track [41] to the forward attach assembly [50].

(e) Install the pin [57] in the nut [56].

SUBTASK 27-51-15-480-008

(5) If it is necessary, remove the adapter, SPL-1733, from the flap track [41].

SUBTASK 27-51-15-420-009

(6) Do this task: Inboard Flap Inboard Transmission Installation, TASK 27-51-31-400-803.

SUBTASK 27-51-15-420-010

(7) Do this task: Inboard Flap Inboard Skew Mechanism Installation, TASK 27-59-02-400-804.

SUBTASK 27-51-15-420-011

(8) If the inboard flap hold equipment, SPL-9469, was not used, do this task: Inboard Trailing Edge Flap Installation, TASK 27-51-11-400-801.

SUBTASK 27-51-15-080-001

(9) Remove the inboard flap hold equipment, SPL-9469.

(a) Retract the three brace assemblies to remove the weight of the inboard flap from the hold equipment, SPL-9469.

(b) Remove the side brace assembly.

(c) Remove the upper brace assembly.

(d) Remove the lower brace assembly.

SUBTASK 27-51-15-820-003

(10) Do this task: Inboard Trailing Edge Flap Adjustment, TASK 27-51-11-820-801.

SUBTASK 27-51-15-820-004

(11) Do this task: Flap Skew Detection System Adjustment, TASK 27-59-00-820-801.

SUBTASK 27-51-15-640-002

(12) Do this task: Inboard Flap Inboard Flap Track Lubrication, TASK 12-22-51-640-813.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-15-840-001

(1) If the hold equipment, SPL-9469, was used, do these steps:

(a) Inflate the shock strut to the proper servicing height (TASK 12-15-31-610-802).

(b) For the left inboard flap, close these access panels:

Number	Name/Location
193CL	Wheel Well Panel - Aft Outboard
194CL	Aft Wing To Body Fairing Panel

(c) For the right inboard flap, close these access panels:

Number	Name/Location
193CR	Wheel Well Panel - Aft Outboard
194CR	Aft Wing To Body Fairing Panel

SUBTASK 27-51-15-440-002

(2) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

SUBTASK 27-51-15-860-008

(3) If the trailing edge flaps are not in the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INBOARD FLAP TRACKS - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Inboard Flap Track Inspection (Selection)

(2) Inboard Flap Outboard Track Inspection

(3) Inboard Flap Inboard Track Inspection.

TASK 27-51-15-200-801

2. Inboard Flap Track Inspection (Selection)

A. References

Reference	Title
27-51-13-200-802	Inboard Flap Outboard Carriage Inspection (P/B 601)
27-51-13-200-803	Inboard Flap Inboard Carriage Inspection (P/B 601)

B. Procedure

SUBTASK 27-51-15-200-001

(1) Do one of these tasks to inspect the applicable inboard flap track:

(a) Do this task: Inboard Flap Outboard Carriage Inspection, TASK 27-51-13-200-802.

(b) Do this task: Inboard Flap Inboard Carriage Inspection, TASK 27-51-13-200-803.

——— END OF TASK ———

TASK 27-51-15-200-802

3. Inboard Flap Outboard Track Inspection

(Figure 601)

A. General

(1) This task gives wear limits for the outboard track on the inboard flap.

(2) This task is applicable to the flap tracks at flap supports No. 3 and 6.

B. References

Reference	Title
27-51-15-000-802	Inboard Flap Outboard Track Removal (P/B 401)
27-51-15-400-802	Inboard Flap Outboard Track Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
542	Left Wing - Fairing Flap Support No. 3
642	Right Wing - Fairing Flap Support No. 6

E. Procedure

SUBTASK 27-51-15-020-013

(1) Do this task: Inboard Flap Outboard Track Removal, TASK 27-51-15-000-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-51-15-020-014

(2) Use these tools to do a check for wear on the part of the inboard flap outboard track:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-51-15-020-015

(3) Examine the parts of the inboard flap outboard track (Table 601):

(a) Remove the parts that you will measure.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are not in tolerance, replace or repair the parts.0.070 in. (1.778 mm).

Table 601/27-51-15-993-801

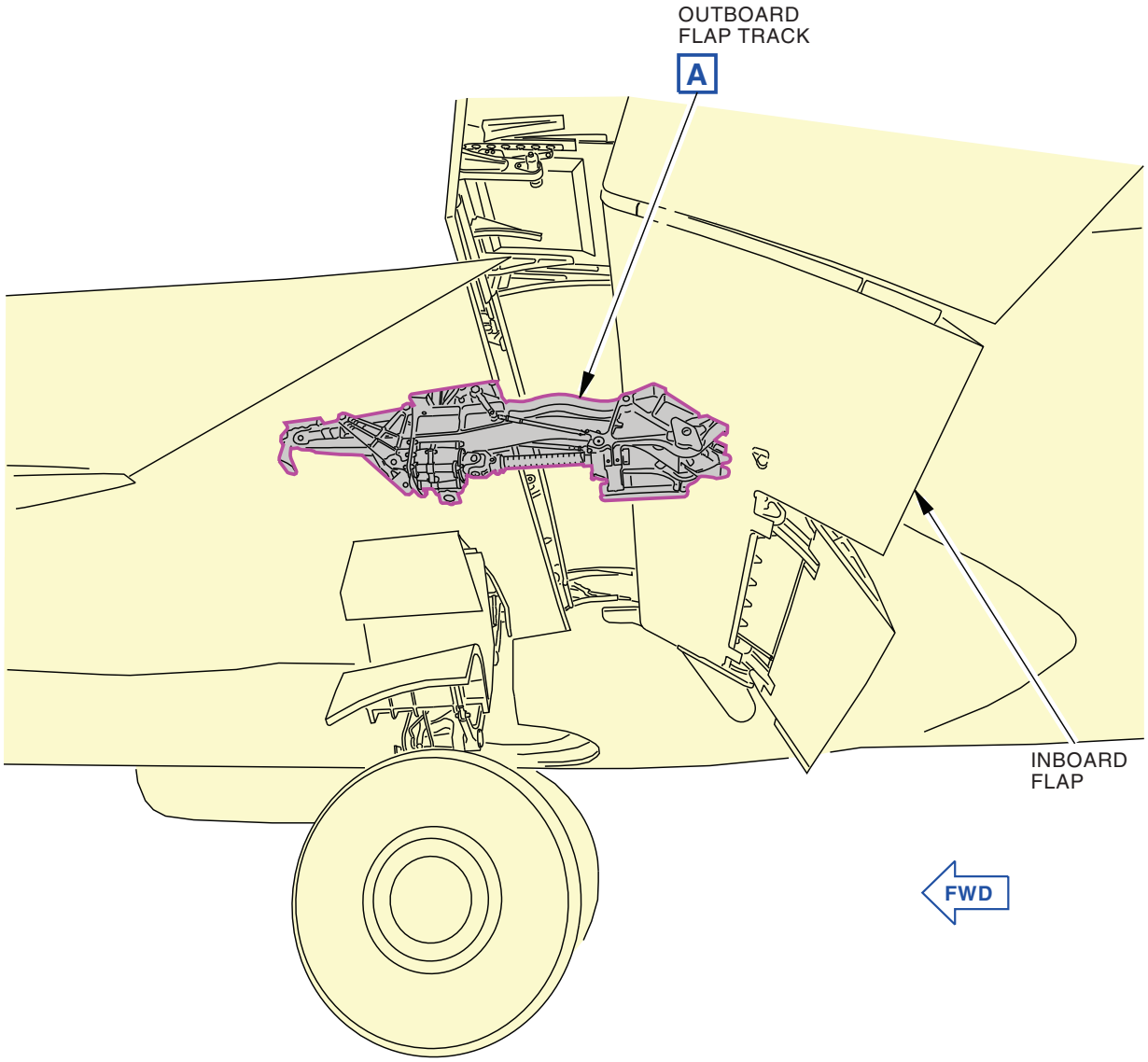
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE CLEARANCE inches/mm
BUSHING [1]	I.D.	0.7000 in. (17.7800 mm)	0.7095 in. (18.0213 mm)	0.7118 in. (18.0797 mm)	0.0336 in. (0.8534 mm)
FUSE PIN [5]	O.D.	0.6800 in. (17.2720 mm)	0.6805 in. (17.2847 mm)	0.6782 in. (17.2263 mm)	
BUSHING [2]	I.D.	0.7000 in. (17.7800 mm)	0.7095 in. (18.0213 mm)	0.7118 in. (18.0797 mm)	0.0336 in. (0.8534 mm)
FUSE PIN [5]	O.D.	0.6800 in. (17.2720 mm)	0.6805 in. (17.2847 mm)	0.6782 in. (17.2263 mm)	
BUSHING [3]	I.D.	0.5620 in. (14.2748 mm)	0.5670 in. (14.4018 mm)	0.5695 in. (14.4653 mm)	0.0080 in. (0.2032 mm)
BOLT [7]	O.D.	0.5605 in. (14.2367 mm)	0.5615 in. (14.2621 mm)	0.5590 in. (14.1986 mm)	
BUSHING [4]	I.D.	0.5620 in. (14.2748 mm)	0.5670 in. (14.4018 mm)	0.5695 in. (14.4653 mm)	0.0700 in. (1.7780 mm)
BOLT [6]	O.D.	0.4985 in. (12.6619 mm)	0.4995 in. (12.6873 mm)	0.4970 in. (12.6238 mm)	

SUBTASK 27-51-15-420-012

(4) Do this task: Inboard Flap Outboard Track Installation, TASK 27-51-15-400-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

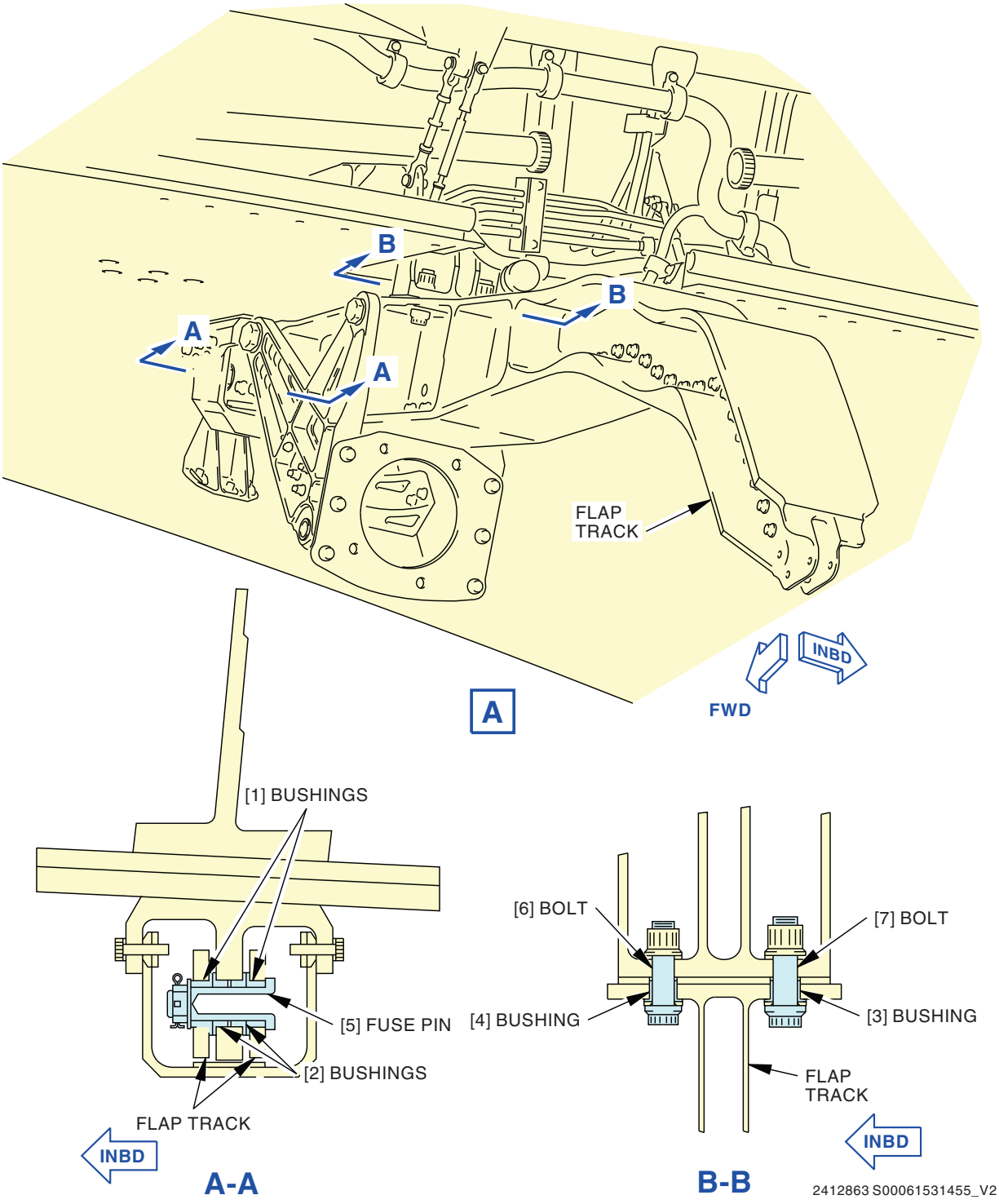


(EXAMPLE)

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Inboard Flap Outboard Flap Track Wear Limits
Figure 601/27-51-15-990-803 (Sheet 1 of 2)

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Inboard Flap Outboard Flap Track Wear Limits
Figure 601/27-51-15-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-15-200-803

4. Inboard Flap Inboard Track Inspection

(Figure 602)

A. General

(1) This task gives wear limits for the inboard track on the inboard flap.

(2) This task is applicable to the flap tracks at flap supports No. 4 and 5.

B. References

Reference	Title
27-51-15-000-803	Inboard Flap Inboard Track Removal (P/B 401)
27-51-15-400-803	Inboard Flap Inboard Track Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
541	Left Wing - Fairing Flap Support No. 4
641	Right Wing - Fairing Flap Support No. 5

E. Procedure

SUBTASK 27-51-15-020-016

(1) Do this task: Inboard Flap Inboard Track Removal, TASK 27-51-15-000-803.

SUBTASK 27-51-15-020-017

(2) Use these tools to do a check for wear on the part of the inboard flap inboard track:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-51-15-020-018

(3) Examine the parts of the inboard flap inboard track (Table 602):

(a) Remove the parts that you will measure.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are not in tolerance, replace or repair the parts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 602/27-51-15-993-802

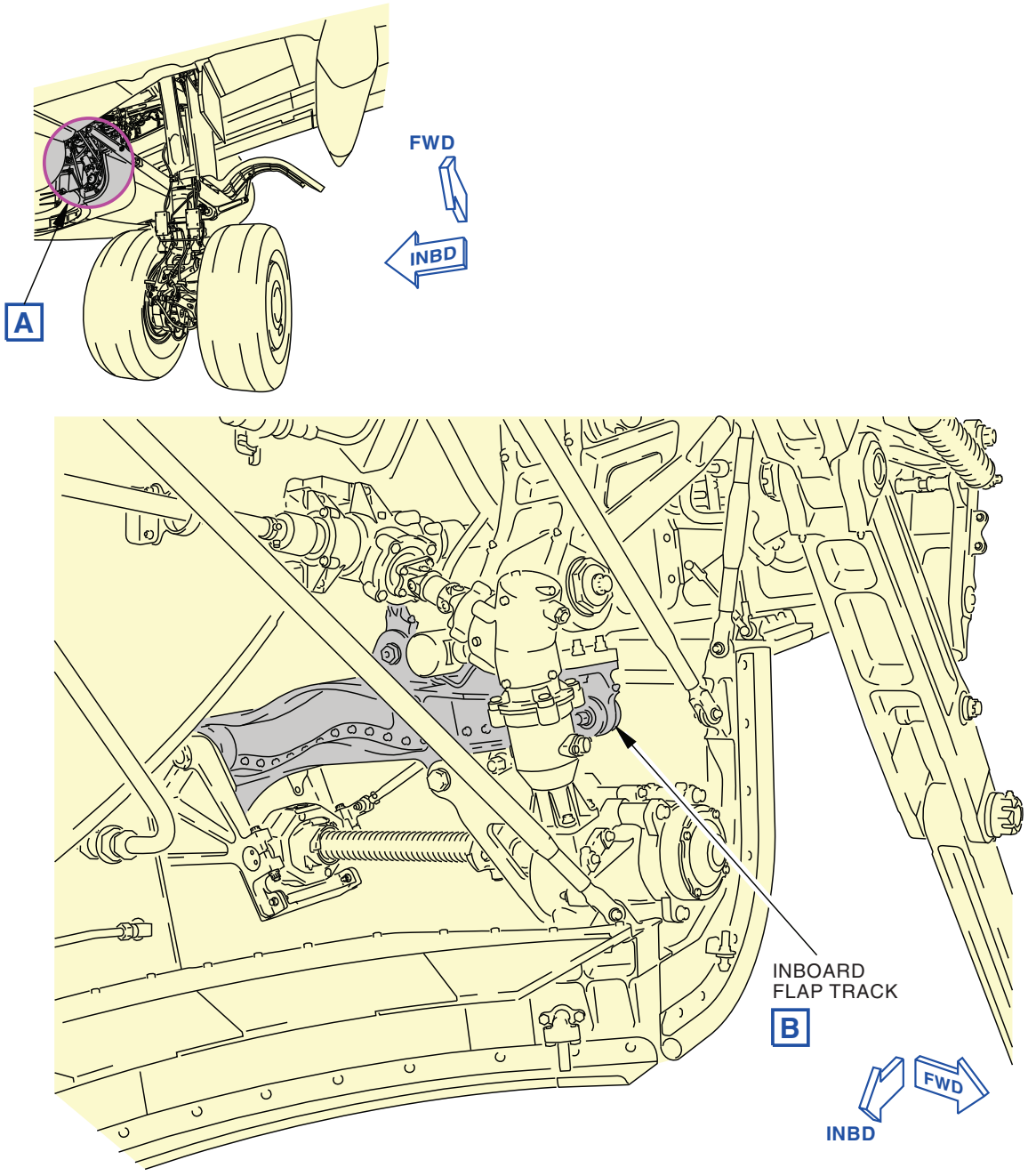
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE CLEARANCE inches/mm
PIN (OUTER) [1]	O.D.	0.7485 in. (19.0119 mm)	0.7490 in. (19.0246 mm)	0.7468 in. (18.9687 mm)	
PIN (OUTER) [1]	I.D.	0.4960 in. (12.5984 mm)	0.4990 in. (12.6746 mm)	0.4990 in. (12.6746 mm)	0.0090 in. (0.2286 mm)
PIN (INNER) [2]	O.D.	0.4900 in. (12.4460 mm)	0.4930 in. (12.5222 mm)	0.4900 in. (12.4460 mm)	
BUSHING [3]	I.D.	0.6245 in. (15.8623 mm)	0.6252 in. (15.8801 mm)	0.6278 in. (15.9461 mm)	0.0064 in. (0.1626 mm)
BOLT [4]	O.D.	0.6230 in. (15.8242 mm)	0.6240 in. (15.8496 mm)	0.6214 in. (15.7836 mm)	
BEARING [5]	I.D.	0.6250 in. (15.8750 mm)	0.6255 in. (15.8877 mm)	0.6255 in. (15.8877 mm)	0.0025 in. (0.0635 mm)
BOLT [4]	O.D.	0.6230 in. (15.8242 mm)	0.6240 in. (15.8496 mm)	0.6230 in. (15.8242 mm)	
BUSHING [6]	I.D.	0.7495 in. (19.0373 mm)	0.7503 in. (19.0576 mm)	0.7525 in. (19.1135 mm)	0.0035 in. (0.0889 mm)
BOLT [7]	O.D.	0.7485 in. (19.0119 mm)	0.7490 in. (19.0246 mm)	0.7468 in. (18.9687 mm)	

SUBTASK 27-51-15-420-013

(4) Do this task: Inboard Flap Inboard Track Installation, TASK 27-51-15-400-803.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



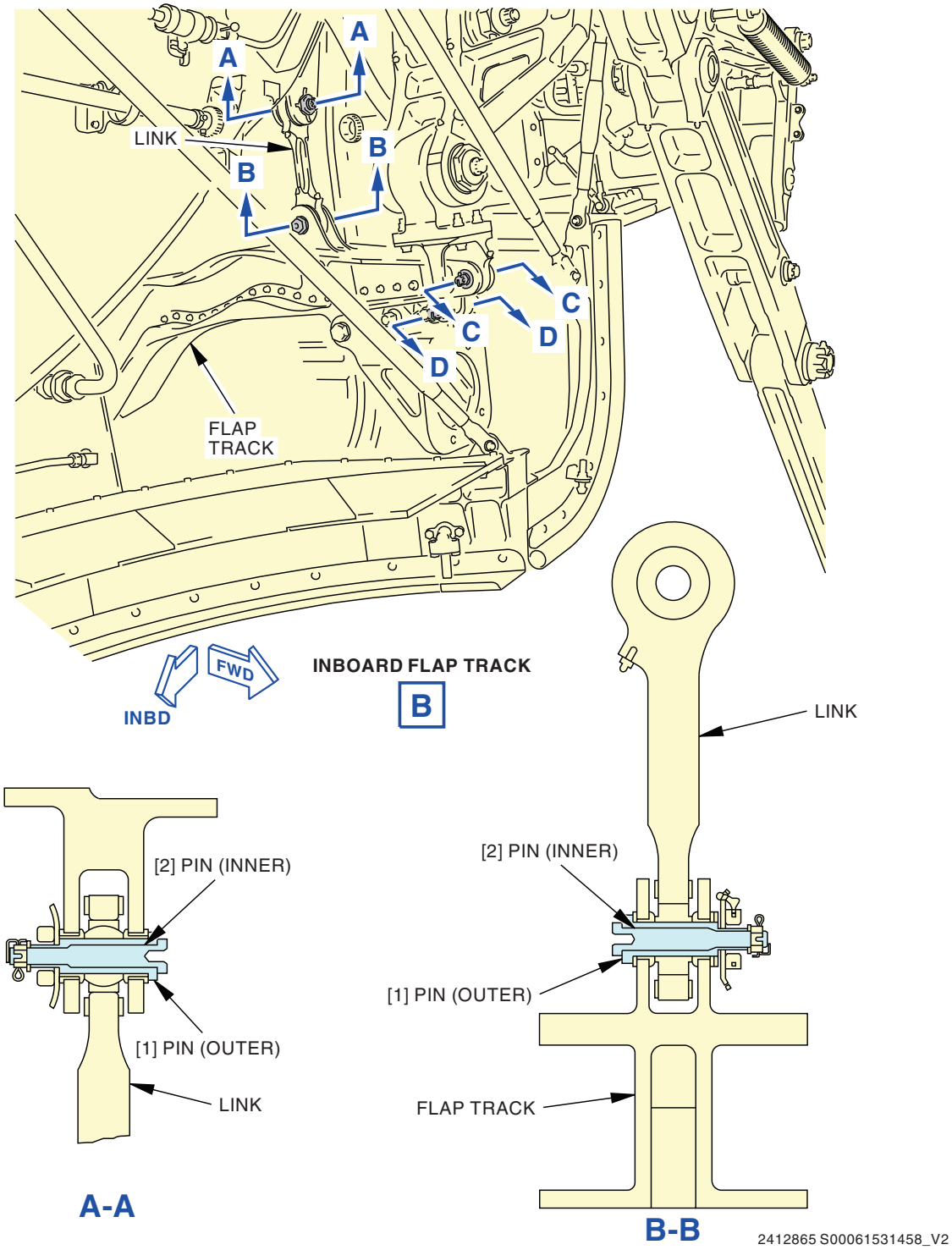
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2412864 S00061531457_V1

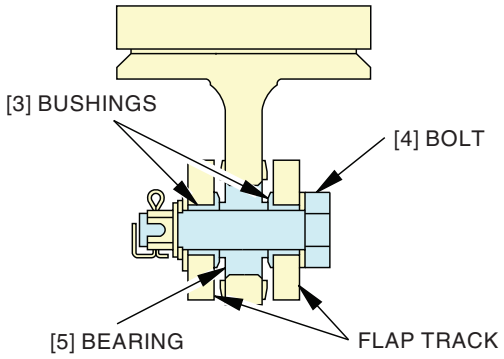
Inboard Flap Inboard Flap Track Inspection
Figure 602/27-51-15-990-804 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

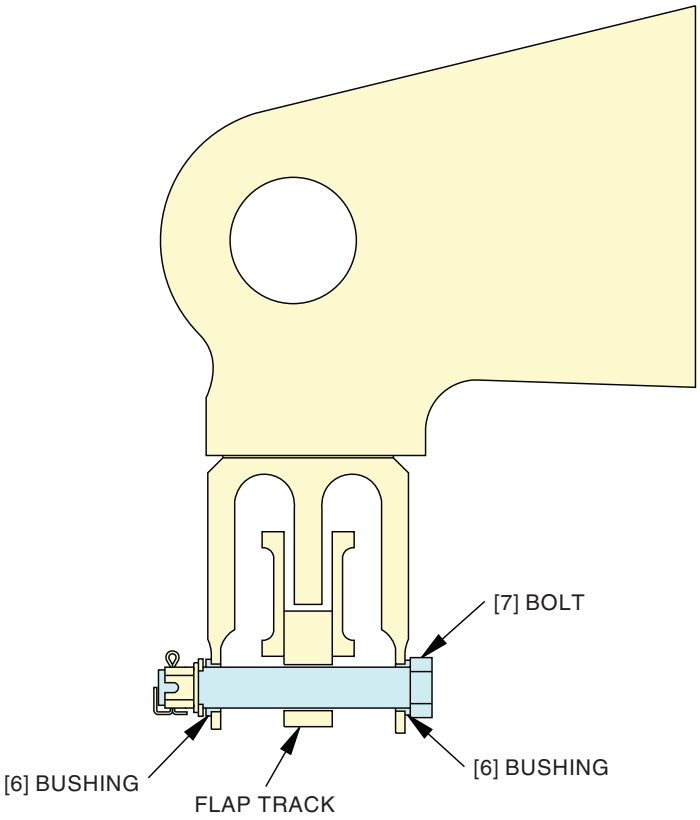


Inboard Flap Inboard Flap Track Inspection
Figure 602/27-51-15-990-804 (Sheet 2 of 3)

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C-C



D-D

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Inboard Flap Inboard Flap Track Inspection
Figure 602/27-51-15-990-804 (Sheet 3 of 3)

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INBOARD AFT FLAP TRACKS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Inboard Aft Flap Track Removal

(2) Inboard Aft Flap Track Installation.

B. This procedure is applicable to the two aft flap tracks on the inboard flap.

TASK 27-51-16-000-801

2. Inboard Aft Flap Track Removal

(Figure 401)

A. References

Reference	Title
27-51-12-000-801	Inboard Aft Flap Removal (P/B 401)

B. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

C. Prepare for the Removal

SUBTASK 27-51-16-020-001

(1) Do this task: Inboard Aft Flap Removal, TASK 27-51-12-000-801.

D. Inboard Aft Flap Track Removal

SUBTASK 27-51-16-020-002

(1) Remove aft flap track [2] from the main flap [1]:

(a) Remove the nuts [4] that attach the cartridge assemblies [3] and cartridge assemblies [5] to the aft flap track support [6] on the main flap [1].

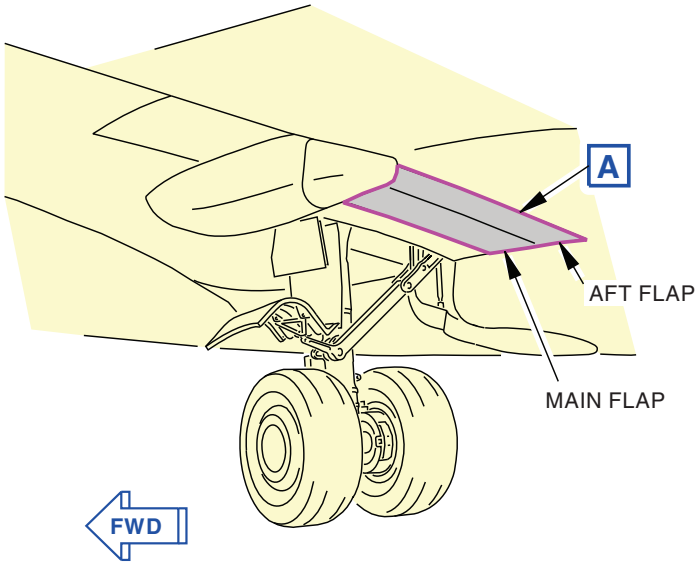
NOTE: There are two cartridge assemblies [3] and two cartridge assemblies [5] on the inboard and the outboard aft flap track supports.

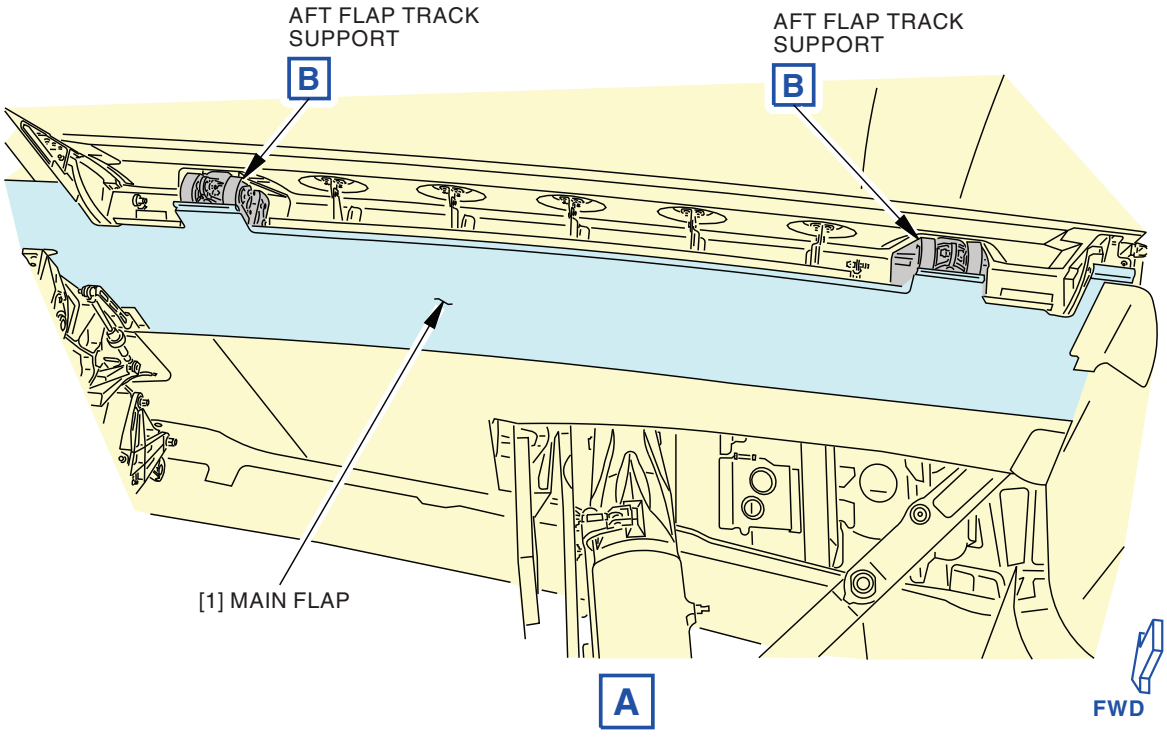
(b) Remove the cartridge assemblies [3] and cartridge assemblies [5] from the aft flap track support [6].

(c) Carefully remove the aft flap track [2] from the aft flap track support [6].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

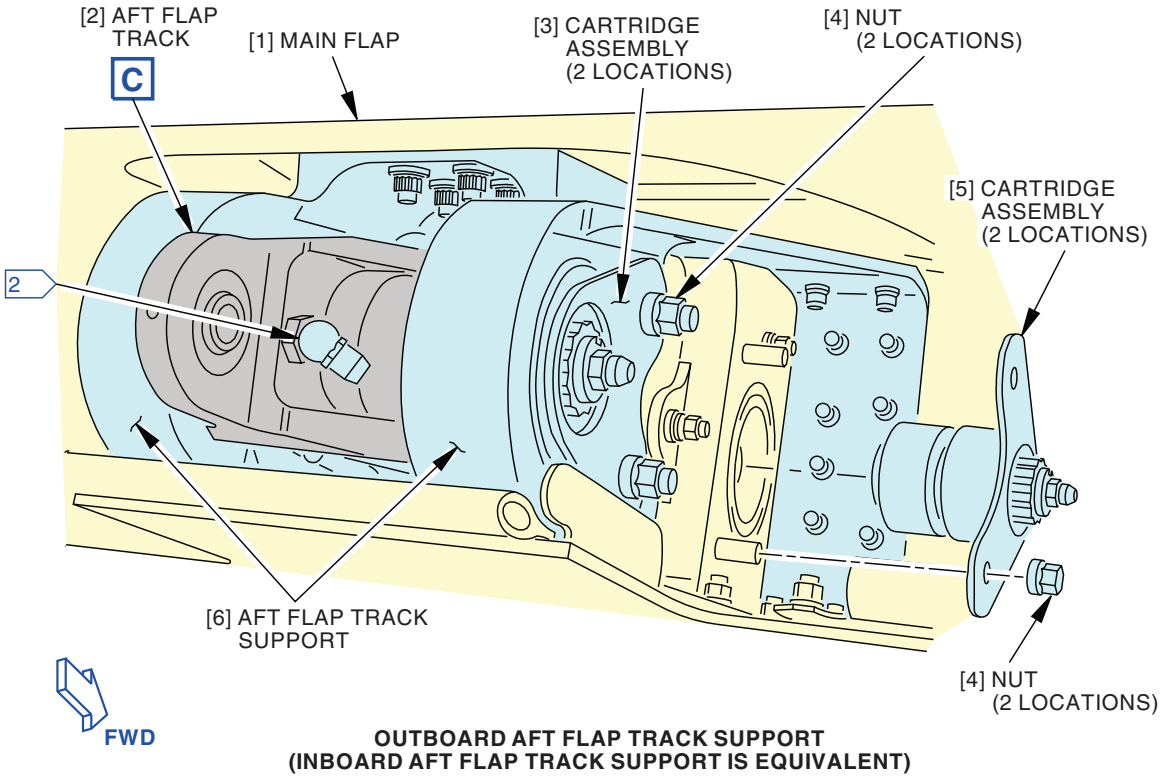


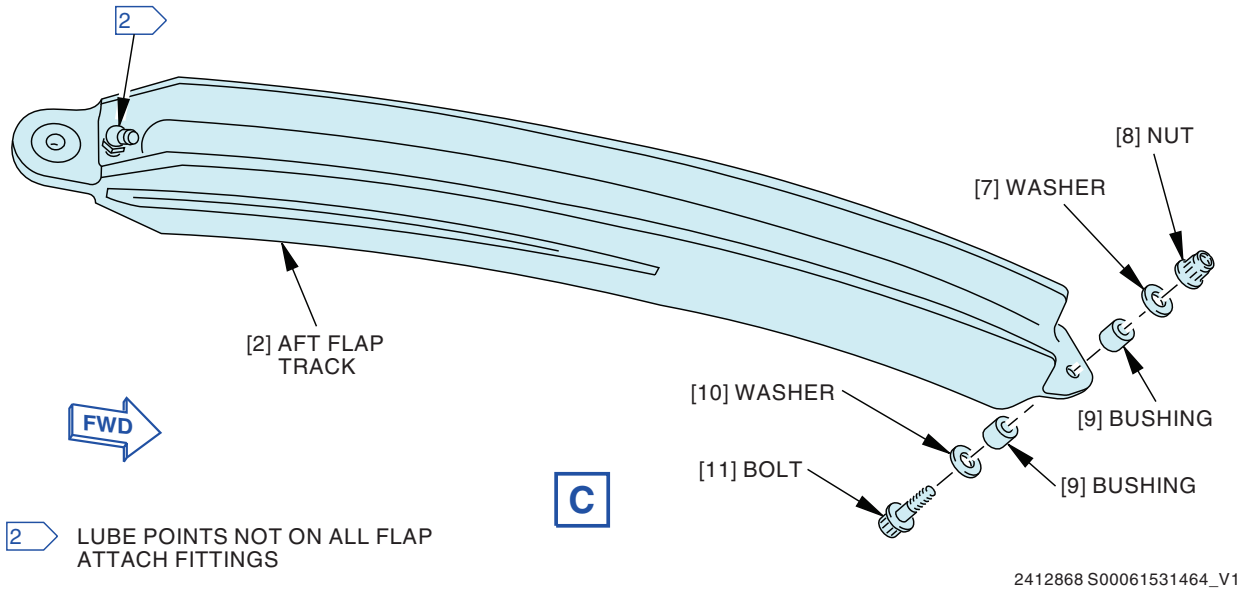


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Inboard Aft Flap Track Installation
Figure 401/27-51-16-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





Inboard Aft Flap Track Installation
Figure 401/27-51-16-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-51-16-400-801

3. Inboard Aft Flap Track Installation

(Figure 401)

A. References

Reference	Title
27-51-12-400-801	Inboard Aft Flap Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Track	27-51-16-01A-065	PTF ALL

D. Location Zones

Zone	Area
553	Left Wing - Inboard Flap
653	Right Wing - Inboard Flap

E. Inboard Aft Flap Track Installation

SUBTASK 27-51-16-420-001

(1) If the downstop is not installed on the aft flap track [2], do these steps:

(a) Install the bolt [11], bushings [9], washer [10], washer [7], and nut [8] on the aft flap track [2].

(b) Tighten the nut [8] to 50 in-lb (5.6 N·m) - 75 in-lb (8.5 N·m).

SUBTASK 27-51-16-420-002

(2) Install the aft flap track [2] in the main flap [1]:

(a) Carefully put the aft flap track [2] in its position in the aft flap track support [6] in the main flap [1].

1) Make sure that the downstop is forward of the forward cartridge assemblies [5].

(b) Apply a layer of sealant, A00247, to the mating surface of the cartridge assemblies [3] , cartridge assemblies [5], and aft flap track support [6].

(c) Put the cartridge assemblies [3] and cartridge assemblies [5] in their position in the aft flap track support [6].

(d) Install the nuts [4] that attach the cartridge assemblies [3] and cartridge assemblies [5] to the aft flap track support [6].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-51-16-420-003

(1) Do this task: Inboard Aft Flap Installation, TASK 27-51-12-400-801.

NOTE: It is necessary to do the aft flap adjustment after you install the aft flap.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details